WLAN PROS TOOLBOX 1.13

WLAN Pros Toolbox: Field Manual

Every tool documented: what it does, how to drive it, worked examples

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WLAN Pros Toolbox · Field Manual

Compiled 2026-07-02 · Field & Trade Reference added 2026-07-05 · reordered to the app's sections and Wi-Fi Classroom added 2026-09-26 · 14 Classroom tools added 2026-09-27 · Find My, Explained added 2026-09-27 · Public Wi-Fi and Wi-Fi Privacy Myths added 2026-09-27 · renamed Wireless Classroom 2026-09-28 · Polarization, OFDMA vs MU-MIMO and One Talker per Channel added 2026-09-29 · covers 264 tools · app v1.13.0

This field manual documents every tool in the WLAN Pros Toolbox, drawn directly from the help text that ships inside the app. Each entry states what the tool does, why it is in the kit, how to drive it, the inputs it takes, the formula or method behind it where one applies, a worked example where one helps, and the field notes that keep you out of trouble. Tools are grouped and ordered the same way they appear in the app, so you can navigate the manual and the Toolbox the same way. Every figure and method is the one the app actually runs.

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Test Network (5 tools)

Live Wi-Fi and internet diagnostics. These tools read the device's real connection, namely the associated AP, link rates, signal, and throughput, and answer the everyday question of whether a slowdown is the Wi-Fi or the internet.

Test My Connection

Answer the everyday question "is the slowdown my Wi-Fi or my internet?" in plain English, and tell the user what to say to support.

Why it's here. The consumer-facing front door. Reached from the "Check My Connection" button on the home screen. The pro "Wi-Fi vs Internet" tool was folded into this one screen; the old `/tools/wifi-vs-internet` deep link now redirects here. Same backends and verdict engine, re-skinned for a non-technical user. Reach for it when you want a one-tap answer plus vetted self-help steps, not engineer numbers.

How to use

1. Tap "Check My Connection" on the home screen. On Wi-Fi the check starts on its own. When the app can't confirm Wi-Fi, it stops before spending data and the button reads "Check My Connection (uses data)" or "(may use data)"; tapping it is the choice to run the speed test.
2. On iOS, install the companion Shortcut first if prompted; the Wi-Fi link read comes from it (see Wi-Fi Information). Without it, the tool still measures the internet and degrades honestly.
3. On macOS, granting Location is optional; it only affects whether the SSID name shows. The verdict never needs Location (it is rate-based). On macOS the app can proactively surface the native "Allow Location" prompt so the name resolves without a settings trip.

4. Read the two status chips (Wi-Fi / Internet) and the headline.

Formula or method. It runs two things in one pass and feeds both into the shared verdict engine: a connected-link read via WifiInfoSourceResolver → MacWifiInfoAdapter (macOS CoreWLAN) or WiFiDetailsBridge → ConnectedAp.fromWifiDetails (iOS Shortcut), AndroidWifiInfoAdapter (Android WifiManager), or WindowsWifiInfoAdapter (Windows Native Wifi); a full net_quality run via OwnEngineQualityClient.forHost('one.one.one.one'). It translates the net_quality grades into the engine's internet-health flag via the shared ConnectionCheck.internetHealth: GOOD only when download AND upload AND latency AND loss all grade good or excellent; otherwise marginal. The engine's six engineer verdicts (WifiVsInternetVerdict: wifiLimiter, upstream, bothContributing, bothHealthy, wifiUnknown, onlineUnmeasured) are re-skinned by ConsumerVerdictMapper into seven consumer outcomes plus a two-axis chip status. Each chip (Wi-Fi / Internet) reads on an absolute 3-tier data-rate scale: "Strong" (100 Mbps and up, the Excellent download band), "Moderate" (25 to under 100 Mbps, the Good band), "Weak" (under 25 Mbps, the Fair and Poor bands; usable Wi-Fi is graded on this same download scale), or "Couldn't check" when that side was not measured (a neutral token, never a fault color). The seven outcomes and their headlines: A "Your Wi-Fi is the slow part"; A-lead "Your Wi-Fi is mostly the slow part"; B "Your internet is the slow part"; C "Your Wi-Fi and internet both look fine"; D1 "We checked your internet, but not your Wi-Fi" (internet measured, Wi-Fi not); D2 "We could not finish the check" (neither measured); E "You are online" (the speed test stalled, but DNS resolved, a public IP was obtained, and cloud apps were reachable, so the honest read is reachable-but-unmeasured, not offline).

Field notes

- Platform differences: macOS reads the link via CoreWLAN (Tx rate, RSSI/SNR, no Rx). iOS reads it via the Shortcut (Tx and Rx). Android and Windows read it natively, with no noise floor or SNR. Served from a WLAN Pi, the check runs on the Pi. If the link can't be read (wired, or iOS without the Shortcut), it falls to the D1/D2 honest path; it does not guess a side. D1 keeps the real internet figure it did measure; D2 admits neither side was read.
- The headline is a hedge by design ("Looks like…", not "your Wi-Fi is broken"). The two chips teach the model that Wi-Fi and internet are two separate things. "Both look fine" is the most useful real-world answer; it points the user at the specific app/site instead of the connection.
- Cloud Apps reachability panel: the result column also mounts a Cloud Apps panel that TCP-connects (port 443) to the recurated kCloudApps named-service list and reports each as reachable or "unreachable" (a word + glyph, never color alone). It proves the service edge/CDN is reachable and times that hop; it is NOT a measure of in-app call or stream quality, and the caption says so. It reuses the same shared ReachabilityProbe the Network Quality tool runs.

- Analyze Results sits beside Copy on the result screen: Copy saves the raw report, Analyze explains the verdict and recommends next steps, entirely on-device (see Analyze Results).
- The verdict copy is deliberately non-diagnostic for a layperson. The underlying numbers and bands come from the same Wi-Fi-vs-Internet engine, which now lives inside this screen. There is no longer a separate tool to switch to for the numbers.
- How it picks a side: it compares how much internet speed you're actually getting against how much your Wi-Fi link could realistically carry. When the internet measures fine on its own, it says both look fine; otherwise it leans toward Wi-Fi or internet depending on which one has the headroom to spare.
- D2 ("Make sure you're on Wi-Fi and try again") is the honest answer when nothing measured, never a fake zero.

Analyze Results

Turn a finished Test My Connection result into a plain-language, conclusion-first report: what your connection is telling you and what to do about it, ordered by importance.

Why it's here. Test My Connection gives the headline verdict and the raw numbers; Analyze Results explains them. Reach for it when you want the why behind the verdict and the recommended next steps, evaluated entirely on the device. It sits alongside Copy on the result screen: Copy saves the raw report for support, Analyze explains it.

How to use

1. Run Test My Connection first so there is a measured result.
2. Tap "Analyze my results" beneath the verdict.
3. Read the findings top to bottom: the most important (the verdict, an open network, packet loss, a dead path) lead, then signal/band/internet-quality context. Each carries a severity word (ACTION / WORTH A LOOK / CONTEXT) plus a color, never color alone.
4. Use the Copy action (top right) to save the whole analysis as plain text.

Formula or method. A pure, local rule engine (AnalyzeEngine) evaluates a structured rule library against the same result data already on screen, read directly from the in-memory models (ConnectedAp, the net_quality result, the Wi-Fi-vs-Internet verdict, the DNS probe, and the cloud

reachability tally), never by re-parsing the copied report text. Every threshold is imported from the app constants, not duplicated: RSSI/SNR from WifiGradingBands, latency/jitter/loss/responsiveness/download from QualityScoring, the verdict thresholds from the Wi-Fi-vs-Internet engine, and the security labels from WifiSecurity. Fired rules are sorted by severity (P1 critical, P2 important, P3 context) then by the rule library's declaration order, so the verdict always leads. Context-only reassurances ("no problem here") are suppressed unless a real finding also fired, so the report never narrates a non-issue. Runs entirely on the device: no network call, nothing stored, nothing leaves the device.

Field notes

- Doctrine guardrails: the "you're on 2.4 GHz" (R-20) and "narrow channel width" (R-23) rules deliberately carry the honest trade-off and never tell a user to blanket-switch bands or force 160 MHz: wider is not automatically better, and 2.4 GHz is the right choice for range and IoT.
- Null discipline: a field the platform did not measure fires no rule. The honesty rules (channel width not captured on iPhone; Wi-Fi signal details not captured) explain the gap instead of guessing.

Network Quality

A one-shot transport-quality measurement covering latency, jitter, loss, download, upload, and responsiveness, plus a reachability check against a set of popular cloud apps. Each dimension is graded on its own; there is deliberately no single composite "score".

Why it's here. When you want to characterize a connection's transport behavior the way Apple's networkQuality or an Orb does, measured by the app's own engine. Reach for it to see whether the internet path is healthy across multiple axes, not just "how many Mbps".

How to use

1. Open the tool. A live latency trend starts sampling immediately (every 30s) while the screen is open.
2. Tap "Run test" for the full one-shot measurement (download/upload/responsiveness run only on a full run). When the app can't confirm Wi-Fi, the button reads "Run test (uses data)" or "Run test (may use data)".
3. Read the six graded rows and the popular-sites reachability table.

Formula or method. All this app's own engine. Latency / jitter / loss: 10 sequential TCP-connect RTTs to one.one.one.one:443 (not ICMP, the sandbox blocks raw sockets). Jitter is RFC-3550-style mean deviation between consecutive samples; loss is failed-connects ÷ attempts × 100. With zero successful samples, latency and jitter report "Unavailable" but loss is a real 100%. Download: parallel-summed, multi-server. Several concurrent streams (5 by default) share one window of about 15 seconds (ThroughputProbe maxDuration 15 s, the first 3 s of warm-up discarded), each against a different independent public server or CDN from a diverse pool (Cloudflare, OVH, Hetzner, Cachefly, ThinkBroadband). The first few seconds of TCP slow-start ramp are discarded, so the number is sustained steady-state throughput, not the ramp. Each server's own rate is measured; any server that comes back below roughly half the median (a throttled or slow outlier) is dropped, and the survivors' rates are summed into the aggregate download figure, so one slow server cannot drag the result down. If every stream fails, it raises an honest "couldn't measure", never a fake 0 Mbps. Upload: single stream with multi-CDN fallback (only Cloudflare __up is a verified large-POST sink, honest single-stream, not faked parallelism). A non-2xx or empty transfer is an honest failure, not 0. Responsiveness (RPM): a simplified single-flow loaded-latency estimate inspired by the IETF responsiveness draft (draft-ietf-ippm-responsiveness) / Apple networkQuality, NOT the full multi-flow RPM method. It samples loaded RTT while a download flow runs, then RPM = 60000 / loadedAvgMs. Reachability: TCP-connect (port 443) to 14 well-known cloud-app hosts, a mix the public and a WLAN pro both recognize: social/consumer (Facebook, Instagram, TikTok, YouTube, Netflix) plus pro/infra (Google, iCloud, Microsoft 365, Cloudflare, AWS, Zoom, Slack, GitHub) and a Cloudflare DNS anchor. Grade bands: Latency ms: Excellent <20, Good <50, Fair <100, Poor ≥100 (grounded in ITU-T G.114, our cut points). Jitter ms: <5/<15/<30. Loss %: 0/<1/<2.5. Responsiveness RPM: ≥1000/≥500/≥100. Download Mbps: ≥100/≥25/≥5 (explicitly a heuristic). Upload Mbps: ≥20/≥5/≥1 (heuristic).

Field notes

- Platform differences: runs on macOS, Windows, Linux, Android, iOS over dart:io sockets/HTTP. Served from a WLAN Pi, the test runs on the Pi (/toolboxapi/conntest): download and upload are the Pi's uplink, the reachability rows are the Pi's connection hops, and a separate row times the Wi-Fi hop between this device and the Pi. Any other browser routes to the download-the-app fallback (no sockets).
- Read each grade word on its own; a connection can be Excellent on latency and Poor on upload at the same time, which is the point. The download/upload Mbps grades are "good enough for a household" heuristics, not standards. RPM is directional only.
- Latency uses TCP round-trip time, so it includes the full SYN/SYN-ACK round trip to a real host (a faithful proxy, slightly higher than ICMP). Only a completed handshake counts as an answered sample; any failure (a timeout, an unreachable, or an actively-refused RST) is caught as a lost sample (LatencyProbe swallows every connect exception as loss). In practice the target is hardcoded to one.one.one.one:443, which

completes normally and never refuses, so the RST case does not arise here; with zero successful samples latency and jitter report "Unavailable" while loss is a real 100%.

- The on-screen footnote states plainly: "these are this app's own measurements, not a third-party score." Download is summed-parallel (so it reflects aggregate link capacity, not a single flow). A failed measurement is "Unavailable" with a note, never 0.

Wi-Fi Information

Show the live connected-AP link details: SSID, BSSID, RSSI, noise, SNR, Tx/Rx rate, channel, width, band, 802.11 standard.

Why it's here. The "what is my radio actually doing right now" read. Reach for it to confirm band/channel/width, check signal and SNR against design targets, or sanity-check a client's negotiated rate.

How to use

1. macOS: open the tool; it reads CoreWLAN, then re-reads it every 2 s while the screen is open (macPollInterval). Tap Refresh for an immediate re-read. SSID and BSSID are gated behind macOS Location Services; tap Grant Location (or open the Location settings pane) to reveal them. The radio metrics do not need Location.
2. iOS: this is live streaming only. Install the combined "WLAN Pros Live" companion Shortcut (one-time, via the iCloud link), then tap Start. The Shortcut loops, harvesting the connected AP each cycle via the stock "Get Network Details" action and handing JSON back to the app; Stop freezes the last values. There is no one-tap snapshot on iOS.

Formula or method. All sources normalize into one ConnectedAp model. macOS comes from CoreWLAN via the com.wlanpros.toolbox/wifi_info native channel; iOS comes from the Shortcut JSON parsed by WiFiDetails (case-insensitive keys, tolerant numeric parse). In iOS Live mode, RSSI and SNR get hard grades against the bands below; Tx/Rx rates get a trend (Rising/Falling/Steady) rather than a hard grade, because a "good" data rate is entirely relative to band/width/MCS. RSSI bands (dBm): Excellent ≥-59 (>-60), Good ≥-67, Fair ≥-72, Poor below. SNR bands (dB): Excellent ≥36 (>35), Good ≥25, Fair ≥15, Poor below.

Field notes

- macOS (CoreWLAN): exposes SSID/BSSID (Location-gated), RSSI, noise, SNR (reported directly), Tx rate, PHY mode → standard label, channel, channel width, band (reported), country code, interface name, hardware MAC. It does NOT expose the Rx rate or Tx power (public CoreWLAN limitation); those render "Not exposed by macOS CoreWLAN", never estimated.
- iOS (Shortcut): exposes SSID, BSSID, channel, RSSI, noise, standard, Rx rate AND Tx rate. SNR is derived app-side (rssi – noise) and band is derived from the channel number (both labeled "derived"). The harvest does NOT return channel width, so width renders "Unavailable" with the note "Not reported for this network", never fabricated. Channel→band derivation: 1 to 14 → 2.4 GHz, 36 to 177 → 5 GHz, 181 to 233 → 6 GHz; the ambiguous low 6 GHz range (1 to 93) is read as 2.4/5 GHz and never silently claimed as 6 GHz.
- Android and Windows are both built and live: Android reads the link via WifiManager (AndroidWifiInfoAdapter), Windows via Native Wifi (WindowsWifiInfoAdapter), each as a snapshot (pull-to-refresh) source like macOS. Neither platform exposes a noise floor or SNR, so those fields render their honest "Unavailable" row rather than a derived value. Web: served from a WLAN Pi, the tool shows the Pi's own radio (/toolboxapi/wifi); any other browser gets the download-the-app fallback.
- RSSI and SNR carry hard grades; read the rate as a trend, not a pass/fail. The standard label combines the 802.11 designation and Wi-Fi generation (e.g. "802.11be (Wi-Fi 7)"; 802.11ax on 6 GHz shows Wi-Fi 6E). Anything marked "derived" was computed, not read from the radio.
- macOS Tx-only and iOS no-width are real platform ceilings, not bugs. On macOS, if the SSID is blank, it's almost always a missing Location grant, not a hidden network. The macOS read has a 5s hang-safety: a stalled CoreWLAN read surfaces an honest "No Wi-Fi reading" rather than freezing.

BSS Load

Read BSS Load, element 11 of the beacon, off the access point you are connected to: associated station count, channel utilization, and available admission capacity.

Why it's here. These are the access point's own numbers about itself, which is a different question from what your client radio is doing. Reach for it when you want the AP's view of the channel alongside your own, or when you are deciding whether a cell is carrying more clients than its coverage suggests. The element is optional, so plenty of access points never send it, and knowing that is itself a finding.

How to use

1. macOS: open the tool and it takes a snapshot; tap Read again to re-read. macOS will not hand beacon information elements to an app without Location access, so if the screen reports no information elements, grant Location and read again.
2. Every other native platform: the screen opens and reports that this device gave it no information elements. No beacon information elements reach this app on iOS, and the connected-AP information-element channel is wired for macOS only today.
3. When there is no reading, read the small label above the sentence first: it says whether the reason is about this access point, about this read, or about what we were told.

Formula or method. The raw information elements are walked by the shared bounds-checked TLV walker (`ie_parser.dart`) and element 11 is decoded by `bss_load_decoder.dart` per IEEE 802.11-2012 section 8.4.2.30, cross-checked field for field against Wireshark's `packet-ieee80211.c` dissector. Station Count is 2 octets little-endian. Channel Utilization is one octet on a linear 0 to 255 scale, so percent = raw × 100 ÷ 255, and the raw octet prints beside it. Available Admission Capacity is 2 octets little-endian in units of 32 microseconds of medium time per second; 1 000 000 ÷ 32 = 31 250 raw represents one full second, which is 100 percent. Nothing is graded. The decoder returns three numbers and one out-of-range flag, and the screen renders three numbers and one out-of-range flag.

Field notes

- An all-zero reading is a reading. An idle access point on a quiet channel really does advertise zeros, and they print as zeros, never as a blank or a dash.
- The screen never collapses the not-available cases into one gray state, and the distinction is the point. "This access point does not advertise BSS Load" is a claim about somebody's network and is shown only when the beacon was walked end to end with every octet accounted for. A capture cut short, a device that exposed no information elements at all, an element this build refuses to decode, and a source that never said whether the capture was whole are all findings about our own read or about what we were told, and each says so under its own label.
- The Cisco "QBSS Version 1, non CCA" variant rides the same element ID 11 with a one-octet admission-capacity field carrying different semantics. It is recognized and deliberately not decoded: the standard layout would produce a plausible wrong number. Do not add support for it without a live Cisco capture pinning what that octet means.
- Channel Utilization is the access point's own sense of the medium over its own measurement window, by physical or virtual carrier sense. It is not a measurement this device made, so two access points on one channel can honestly report different numbers.
- Station Count is an association count, not a traffic measure. It says nothing about how much air those stations use.

- A capture clipped exactly on an element boundary is byte for byte identical to a complete one. There is no evidence in the bytes, so such a blob reads as absent. That limit is stated rather than papered over, and it lives in the layer that capped the buffer, not in the decoder.
- Web, including a WLAN Pi, routes to the download-the-app fallback: there is no method channel in a browser.

Cellular Information

Show the iPhone's mobile-network details: carrier, radio technology, signal bars, country code, roaming.

Why it's here. A quick read of what the cellular radio is doing, useful when comparing Wi-Fi offload behavior or confirming a device fell back to LTE/5G.

How to use

1. iOS only. Install the same combined "WLAN Pros Live" companion Shortcut, then Start; the Shortcut harvests cellular details via "Get Network Details" and hands them over the App Group bridge.

Formula or method. Parses the Shortcut JSON into CellularInfo (case-insensitive keys, with whitespace-trim tolerance after a real on-device bug). Fields: carrier name, radio technology (mapped from raw CTRadioAccessTechnology* constants to friendly labels like "5G (NSA)", "LTE"; unknown values pass through, never blanked), signal bars (coarse 0–4 status-bar indicator), country code, roaming bool.

Field notes

- Platform differences: iOS is the only source. There is deliberately no native CoreTelephony path: CTCarrier is deprecated since iOS 16.4 and returns placeholder junk, and cellular signal strength (RSRP/RSRQ/dBm) is private-API-only and an App Store rejection. So data comes only via the Shortcut. macOS (no radio), Android, Windows show an honest "not available on this platform"; web shows the download fallback.
- Signal bars are bars (0 to 4), never relabeled dBm/RSRP/RSRQ; the app does not have a raw signal value and must not imply it does. Bars are clamped to 0 to 4; an out-of-range value is never trusted.

- This is the one tool where iOS exposes more than a native app could (the Shortcut runs in Apple's privacy context). A missing field renders "Unavailable", never a fabricated value.

Roaming Log

Record each time your device roams from one access point (BSSID) to another on the same network (SSID) during an open monitoring session, with the timestamp, the from→to BSSID pair, and the signal at the roam.

Why it's here. A roam is a BSSID change while the SSID stays the same. Watching the live Wi-Fi signal shows a roam happen, but nothing recorded it. Reach for this when you want a timestamped list of roams from a walk-around, e.g. to see where the device clings to a far AP or hops too often.

How to use

1. macOS, Android, and Windows: recording starts automatically and polls the Wi-Fi link continuously while the screen is open. On macOS the BSSID needs Location access; on Android it needs the Location permission. Walk your space; each roam is logged as it happens.
2. iOS: install the "WLAN Pros Live" companion Shortcut (same one Wi-Fi Information uses), tap Start, then walk with the screen open. Firing the Shortcut switches to the Shortcuts app, so it waits for your deliberate Start tap rather than auto-firing.
3. Read the list newest-first: time · network, the from→to BSSID pair, and the signal (RSSI/SNR) read at the roam.

Formula or method. The shared WifiSignalSampler feeds every fresh connected-AP sample to a pure RoamDetector. The detector records a roam only when the current BSSID differs from the prior non-null BSSID AND the SSID is unchanged. A null/blank BSSID is skipped and keeps the last known anchor (it neither advances nor breaks the chain), so a single unreadable sample mid-walk does not split one roam into two or drop a real one; a roam is still only ever recorded between two real BSSIDs, so a gap can never fabricate one. The first known BSSID seeds the anchor (no roam); a same-BSSID sample is ignored; a changed SSID is a network switch, not a roam, and is excluded. On macOS the detector is fed on every poll (before the sparkline's unchanged-RF guard) so a roam at identical signal still registers; on iOS it is fed from each new streamed payload and reset on a fresh Stop→Start.

Field notes

- Foreground session only on iOS. There is no public iOS API for background Wi-Fi or BSSID-change callbacks, so no app can log roams with the app closed or the phone in a pocket: the same ceiling Wi-Fi Check shares. macOS, Android, and Windows record continuously while the screen is open. Web, including a WLAN Pi, routes to the download-the-app fallback.
- BSSID is the AP radio's MAC address. A roam between two radios in the SAME physical AP (e.g. 2.4 GHz → 5 GHz) is still a BSSID change and is logged; that is correct: it is a real roam.
- The signal shown is the reading at the moment the new BSSID first appeared, not an average. "Signal unavailable" prints honestly when the platform omitted RSSI for that sample, never a fake value.
- Stopping and starting (iOS) clears the session log so a new walk does not inherit the prior walk's roams.
- Copy-to-save: the §8.16 Copy action in the app bar exports the whole recorded roam session as paste-ready plain text (each roam with its time, network, from→to BSSID pair, and signal), built by the pure buildRoamLogCopyText so a walk can be saved to a ticket or note.

How the Toolbox Measures Throughput

The Toolbox reports two throughput numbers, and they answer two different questions. One measures your path to the internet. The other measures the Wi-Fi hop between your device and the Access Point. Read each as its own answer and both make sense. Compare them to each other, or to a headline speed test, and you will only confuse yourself.

Keep them straight from the start.

Two numbers, two questions

- **Internet throughput** is shown by *Test My Connection* and *Network Quality*. The headline figure is your download, what people mean by internet speed, measured against the wider internet right now. Upload is measured too and shown as its own separate number.
- **Usable Wi-Fi capacity** comes from the Wi-Fi-vs-Internet engine (now part of *Test My Connection*). This is roughly how much throughput the Wi-Fi hop itself can carry, based on the rate your radios negotiated.

Same app, two measurements, two purposes. Everything below explains what each one actually does and why a number you see somewhere else may not match.

How the internet number is measured

The internet download figure comes from an aggregate-capacity measurement, the same class of test the well-known speed tests run. It is built to be comparable to them, not built to read low.

Here is what the app actually does:

- **Download opens several streams at once, to a diverse set of independent public servers.** Five streams by default, each against a different provider or CDN, never a single provider. Running parallel streams across independent networks is how you fill a fast connection, and how you keep one provider's rate-limiting from collapsing the whole measurement.
- **The window runs about 15 seconds, and the ramp-up is thrown away.** The first few seconds are TCP slow-start, when the connection has not yet reached full speed. The app discards them and reports the sustained, steady-state rate, not the ramp.
- **Each server's own rate is measured, and slow outliers are dropped.** Any server that comes back far below the pack, below roughly half the median rate, is treated as throttled and excluded. The surviving servers' rates are summed into the aggregate download figure, so one bad or throttled server cannot drag the result down.
- **The reported internet number is the download.** That is what people mean by internet speed. Upload is measured too, over its own window, and shown as its own separate number.
- **When it cannot get a clean measurement, it says so.** You get an honest "couldn't measure," never a fake 0. A zero on a working connection would be a lie, and the app will not tell it.

Responsiveness, the RPM figure, is a single-flow loaded-latency reading taken while a download runs. It holds up well across servers, so it stays steady even when the throughput number moves.

Privacy stays simple. These are anonymous downloads from public servers. Nothing about you is sent or published, and the whole measurement runs on your device.

Why your speed test may read differently

No single number is "the" speed of a connection. Throughput is a measurement, and every measurement depends on how it was taken. Run two honest tests on the same connection and they can land on different numbers for reasons that have nothing to do with either one being broken.

What moves the number:

- **Which server you hit, and how well it is peered to your ISP.** A single nearby edge cache that your provider peers with directly has a shorter, cleaner path than a spread of independent public servers out on the wider internet.
- **How many streams the test opens, and to how many destinations.** More streams across more independent networks measure real aggregate capacity; a single stream to one close cache often posts a higher-looking peak.
- **Whether the ramp-up counts.** A test that includes the first seconds of TCP slow-start can report a different number than one that throws the ramp away and measures only the sustained rate, the way this app does.
- **Time of day and congestion.** The same connection is not the same connection at 3 a.m. and at 8 p.m.

Put those together and the pattern is easy to see. A one-number tool that saturates the single nearest edge cache can post a higher figure than a steady-state measurement summed across several independent public servers. That gap is a difference in method. Both tests measured something real. This app measures the honest, sustained throughput to real internet destinations, which is the number that matches what your connection actually delivers day to day.

What the Wi-Fi number is, and what it is not

The Wi-Fi-vs-Internet engine (folded into *Test My Connection*) produces a different figure entirely: usable Wi-Fi capacity, which the app estimates at about 55% of the negotiated Wi-Fi link rate.

Two things to understand about that 55%.

First, it is deliberately below the headline link rate, and that is correct. The rate your device and the Access Point negotiate is a ceiling, not a delivery. Real-world throughput over Wi-Fi runs roughly 50 to 60% of that negotiated rate once you account for protocol overhead,

acknowledgments, contention, and retries. Taking 55% of the link rate is a reasonable estimate of what the hop can actually carry.

Second, this number describes the Wi-Fi hop only. It is the road between your device and the Access Point, and nothing past it. It says nothing about your internet speed.

Do not compare the Wi-Fi capacity number to an internet speed test. Different road, different question. One tells you how much the local Wi-Fi hop can move; the other tells you how much your connection to the internet can move. The whole point of putting them side by side in the tool is to show you which of the two is the limiting factor, not to make them agree.

What our internet number represents

When the Toolbox shows you an internet download figure, it is real, sustained throughput to a diverse set of neutral public servers, measured over about a 15-second window with the slow-start ramp discarded and throttled outliers dropped. It is not the highest number the app could have coaxed out of a single hand-picked nearby cache, and it is not padded to look good next to anyone else's result.

If another test on the same connection reads higher, that is the method talking: a single nearby edge cache, or a run that counts its ramp-up. If the Toolbox cannot get a clean read, it tells you it couldn't rather than inventing a number.

Two throughput numbers, two questions. The internet figure is your honest path to the wider internet. The Wi-Fi figure is the capacity of the hop to your Access Point. Keep them straight and each one tells you exactly what it means.

Networking Tools (25 tools)

Socket, lookup, and scan utilities for working a network from the device in hand. Ping, traceroute, port and host discovery, DNS and registry lookups, and packet-level senders and inspectors.

This Device (5)

Current Location

Reads the device's GPS location on open and shows latitude, longitude, altitude, and horizontal accuracy directly, no conversion step.

Why it's here. The fast "where am I right now" answer. The same fix is available inside the Lat / Long converter, but people expect a dedicated location tool rather than a coordinate calculator.

How to use

1. Open the tool; it requests Location permission (if needed) and reads one fix automatically.
2. Read latitude / longitude (decimal degrees), altitude (meters), and horizontal accuracy (meters).
3. Tap Update to read a fresh fix. Grant Location or open Settings if prompted.

Formula or method. Reuses the DeviceLocationService seam behind the Lat / Long tool (geolocator, LocationAccuracy.best, 15 s cap). Resolves Location permission/services first and renders the matching sealed state (granted -> fix; needs-permission -> request; blocked -> Settings deep-link; no-fix -> honest unavailable). A genuine no-fix on a granted/enabled device falls back to an IP-derived approximate location, clearly labeled approximate (never presented as GPS). Coordinates render in Roboto Mono as identifier values.

Example. Granted + outdoors: 40.712800, -74.006000, altitude 12 m, accuracy 5 m.

Field notes

- Altitude and accuracy can read "Not reported" on hardware without GPS (e.g. a Mac on a coarse Wi-Fi fix).
- An approximate fix derived from your public IP is labeled as such and is city-level, not GPS-precise.
- No coordinates are transmitted; the IP-approximate fallback derives location from the public IP only.

Device Info

Show the device's own system facts: model (marketing name plus the raw identifier), total physical memory (RAM), system uptime since the last boot, and the cellular IP address where the device has a cellular interface.

Why it's here. The companion to Interface Information: that tool answers "what's my IP, gateway, and Wi-Fi link"; this one answers "what device is this, how much RAM, how long since boot, and what's my cellular IP". Reach for it to confirm the hardware model or to read the cellular-side address that the Wi-Fi-centric interface view doesn't surface.

How to use

1. Open the tool; it reads a snapshot. Each field a platform can't provide renders its honest unavailable state rather than a fabricated value.
2. Tap Refresh in the top bar to re-read (uptime advances; the rest is stable).
3. Use Copy to put the whole snapshot on the clipboard as labeled text.

Formula or method. Model and total memory come from the device_info_plus package (BSD-3): on iOS, modelName (the package maps utsname.machine, e.g. iPhone16,2, to a marketing name) plus physicalRamSize; on macOS, modelName/model plus memorySize; on Windows, productName (falling back to the machine name) plus systemMemoryInMegabytes. Uptime comes from a tiny native MethodChannel (com.wlanpros.toolbox/system_info → systemUptime) reading ProcessInfo.processInfo.systemUptime; that channel is wired on macOS and iOS, so uptime renders its honest unavailable state where the platform does not provide it. The cellular IP comes from dart:io NetworkInterface.list, matching the conventional iOS cellular interface name pdp_ip0.

Field notes

- Cellular IP uses a heuristic: Apple does not treat interface names as a stable API, so detection matches the conventional iOS cellular name pdp_ip0. "No cellular interface" is the normal, honest result on a Wi-Fi-only iPhone, in airplane mode, or on a Mac (which has no cellular interface).
- Total memory is shown in binary units (an 8 GiB device reads "8 GB") to match how RAM is physically sized; the label stays the familiar GB/MB.

- Uptime is seconds since boot, formatted as "3d 4h 12m"; it always shows at least minutes (a just-booted device reads "0m").
- Nulls are honest "not available", never 0 or "". On Android, model is shown but total RAM is not surfaced by the package, so it reads unavailable.
- On Windows, the "Model" line shows the OS product name (e.g. "Windows 11 Pro"), not a hardware model. The package does not expose a distinct machine model on Windows, so this field is deliberately the OS name rather than a fabricated device name. RAM is surfaced; uptime, which rides the macOS/iOS-only native channel, reads unavailable.

Interface Information

Show the device's own network state: per-interface IPv4/IPv6 addresses, interface name and inferred type, plus gateway, subnet mask, Wi-Fi SSID/BSSID, and the device's primary IP where the platform exposes them. (It does not read the device's configured DNS servers. Only Test My Connection's report does, and only on Android.)

Why it's here. The "what's my IP, gateway, and link" foundation. Reach for it first when you need the device's own address (e.g. before a ping/sweep) or to confirm which interface is active.

How to use

1. Open the tool; it reads a snapshot. Each field that a platform can't provide returns null and renders "Not available on this platform". The Wi-Fi identity (SSID/BSSID) may be served from a recent shared reading rather than a fresh live one; when it is, the screen stamps it "as of HH:MM".

Formula or method. Built on dart:io NetworkInterface.list for the address/interface table. The Wi-Fi identity fields (SSID, BSSID, interface name, hardware MAC) come from the shared native ConnectedAp subsystem (WifiInfoSourceResolver → MacWifiInfoAdapter on macOS, the WLAN Pros Live Shortcut payload on iOS, WifiManager on Android, Native Wifi on Windows), the same source the Wi-Fi Information tool uses, because network_info_plus returned a null SSID/BSSID on iOS/macOS that surfaced as a misleading "not available". Gateway, subnet mask, and the Wi-Fi IPv4/IPv6 addresses still come from network_info_plus. Each sub-read is independently guarded so one denied call (e.g. SSID) never blanks the

whole screen. Interface kind (Wi-Fi/Ethernet/Cellular/Loopback/VPN/Other) is a heuristic from the OS interface name, e.g. macOS en0 is guessed Wi-Fi, en1+ Ethernet; explicitly conservative. primaryIPv4 prefers the Wi-Fi link IP, else the first non-loopback IPv4.

Field notes

- The Wi-Fi identity read has a warm-cache path: rather than a fresh native read every time, a shared ConnectedAp reading up to 5 minutes old can be served, stamped with an "as of HH:MM" time so a remembered value is never shown as if it were live. IP/gateway/mask are read fresh.
- Platform differences: SSID/BSSID/gateway exposure depends on OS permission state. On iOS the SSID/BSSID arrive through the WLAN Pros Live Shortcut payload (not a native wifi-info entitlement); macOS needs Location for the SSID name; Android/Windows vary. Served from a WLAN Pi it shows the Pi's own interfaces (/toolboxapi/interfaces); any other browser is gated off. SSID is cleaned of wrapping quotes and the placeholder.
- The interface "type" is a name-based guess; on macOS the en0/en1 split is heuristic, so trust the addresses over the label on unusual hardware.
- Nulls are honest "not available", never 0 or "". Gateway and SSID/BSSID exposure is platform-and-permission dependent. The device's configured DNS resolvers are not among the fields this tool reads.

Join a Network

Picks a Wi-Fi network from a scan and associates to it, then reports what you actually landed on. On macOS and Windows the join runs on this device's own radio. With a WLAN Pi attached, the Pi joins instead and reports its full link record.

Why it's here. Associating is the test. A scan tells you a network is on the air; only a join tells you whether the credentials work, whether the AP will take you, and which BSSID you ended up on when several advertise the same SSID. Doing it from the tool means you see the outcome rather than reading it back out of the OS afterwards.

How to use

1. Pick a radio if more than one is offered, then run a scan. Networks are listed with their SSID, BSSID, channel, signal, and the security they advertise.
2. Tap a network. If it needs a passphrase you are asked for one; open and OWE (Enhanced Open) networks need nothing, because neither takes credentials.
3. Read the result. A native join reports that you are connected and to which SSID. A WLAN Pi join reports its full link record, including the BSSID it landed on.
4. Disconnect from the same screen when you are done.

Example. Two APs advertise the same SSID in a corridor. You join, and the result names the BSSID you associated with, which tells you which of the two actually took you, rather than leaving you to infer it from signal strength.

Field notes

- WHICH RADIO JOINS DEPENDS ON THE PLATFORM, and the screen says so rather than making you guess. macOS and Windows join with their own radio. iOS and Android do not: neither lets a third-party app associate to a network it chooses, so there the tool needs a WLAN Pi and says so.
- JOINING CHANGES THE NETWORK THIS DEVICE IS ON. That is the point of the tool, but it also means a join can move you to a different subnet, and anything depending on the old one is dropped. On a native join you are moving the machine you are standing at.
- OPEN AND OWE NEED NO PASSWORD. OWE, which the Wi-Fi Alliance calls Enhanced Open, encrypts without authenticating, so there is nothing to type. A network advertising OWE transition mode is a marker that BOTH an open and an OWE BSS exist, not a security scheme of its own.
- A FAILED JOIN USUALLY TELLS YOU WHICH KIND OF FAILURE IT WAS. A network that never appears is a different problem from one that appears and then refuses the passphrase, and the two are worth separating before you change anything.
- 802.1X IS NOT SUPPORTED. Enterprise networks need a credential exchange this tool does not carry; it handles open, OWE, WPA2 Personal, and WPA3 Personal.

Link Info

Shows every network interface on this device by what it actually is: whether it has carrier, what speed and duplex it negotiated, its MAC, its driver, its MTU, and which one is carrying your traffic right now.

Why it's here. Most tools tell you about the network. This one tells you about the machine you are standing at, and that is where a surprising number of faults live. A laptop on Ethernet with the Wi-Fi radio still associated will happily report Wi-Fi details that have nothing to do with where its traffic is going. The interface holding the default route is marked, so the question of which link is actually in use is answered rather than assumed.

How to use

1. Open it and read the list. Every interface the device exposes appears, including ones with no carrier.
2. Find the row marked DEFAULT ROUTE. That is the interface carrying your traffic.
3. Read the negotiated speed and duplex. A gigabit port that negotiated 100 Mbps half duplex is a cabling fault, and it looks fine until you look here.
4. Use the transport rows underneath to see which paths this device could use, and why any of them is unavailable.
5. Where a transport is untested, tap Try it now to find out rather than guess.

Inputs

Input	Unit	Range
None	-	Read-only. The screen reports what the operating system says about the hardware

How it works. Interfaces are read from the operating system's own interface table. The default route is resolved by asking the routing table which interface holds it rather than by inferring it from an address, because a device can hold several addresses at once and the one that looks

primary is often not the one in use. Speed and duplex are the negotiated values reported by the driver, not the port's rated capability. Where a value is not reported, the screen says so instead of printing a zero.

Example. A Mac plugged into Ethernet while still associated to Wi-Fi shows both interfaces with carrier, and the DEFAULT ROUTE badge sits on the Ethernet one. That is the difference between a machine that is on Ethernet and a machine that merely has a cable in it.

Field notes

- Not reported is a real answer and is shown as one. Some drivers, some virtual interfaces and some platforms do not expose speed or duplex at all, and printing a zero there would be a fabricated number.
- On a sandboxed Mac App Store build the routing table cannot be read by shelling out, so the default-route badge may be absent. The direct-download build has no such limit.
- A speed figure here is what the link negotiated, which is a ceiling rather than a measurement of throughput. Use Test My Connection for that.

Reachability & Path (5)

Ping (ICMP)

Real ICMP echo round-trip on mobile: live RTT, min/avg/max, loss.

Why it's here. When you specifically want true ICMP echo (the classic ping), available where the platform genuinely supports it.

How to use

1. Enter a host, run (iOS or Android). Streams replies with running stats, same UI shape as TCP Ping. Served from a WLAN Pi, the Pi sends the echoes (1 to 20) and the screen shows the final aggregate.

Formula or method. Real ICMP echo request/reply via the native backend (dart_ping_ios SimplePing/GBPing on iOS; dart_ping spawning system ping on Android). The method label is "ICMP echo", never relabeled from a TCP probe.

Field notes

- Platform matrix (honest): iOS, real ICMP echo available. Android, real ICMP echo via system ping, available. macOS App Store build: the ICMP path spawns the OS ping binary, which the macOS App Sandbox blocks, so it gives an honest "not available in the sandboxed build" and points the user at TCP Ping instead. Windows: ICMP ping is not wired into the Windows build, and the screen says so and points at TCP Ping (defaultIcmpBackend() returns null on all desktop). WLAN Pi: /toolboxapi/ping runs the echo on the Pi. Any other browser: no sockets → download fallback.
- Where available, this is genuine ICMP RTT/loss, the real thing, not a TCP proxy.
- On macOS and Windows, use TCP Ping.

Ping (TCP)

A reachability + round-trip-latency probe that works on every platform, including the sandboxed desktop, by timing a TCP handshake, not ICMP echo.

Why it's here. The portable ping. ICMP is often filtered while a TCP port (443) answers; this is the tcping/paping approach pros already use. It is also the desktop path where real ICMP can't run.

How to use

1. Enter a host, optionally pick a probe port (443 default; presets 443/80/53/22/7) and count, run. Live min/avg/max/loss and a sparkline build as replies land.

Formula or method. Each "ping" is a timed Socket.connect to host:port (default 443). A completed handshake OR an actively-refused RST both count as a successful round trip for latency (exactly how tcping treats it); only a genuine timeout or lookup failure is a loss. Probes are spaced by

the requested interval minus how long the probe took, so cadence stays steady under latency. On a lost probe there is no round-trip time to subtract, so the full timeout is counted as the elapsed time; a burst of fast failures (e.g. an unresolvable host) can therefore fire with little gap between probes rather than waiting out the interval.

Field notes

- Platform differences: identical everywhere native. Served from a WLAN Pi, the Pi runs the ping as an ICMP echo and the screen labels it so. Any other browser is gated off. The screen labels the metric "TCP RTT" and shows the target port so it's never mistaken for ICMP.
- RTT is the TCP handshake time to a port, slightly higher than ICMP and dependent on the chosen port answering. A "refused" target still gives you a valid latency number (the host answered).
- This is not ICMP echo (see Ping (ICMP) for the mobile real-ICMP path). If a host filters the probe port, it reads as loss even if the host is up on another port; try a different probe port.

Ping Plotter

Runs a sustained ping to a target and charts round-trip latency over time, instead of the single-shot result the Ping and ICMP Ping tools give. The live trend, jitter, and visible dropped probes show how stable a path is, not just whether it answers once.

Why it's here. The live performance graph. A single ping says "reachable now"; a trend says "steady, spiky, or dropping packets." Reach for this to watch a flaky link over seconds or minutes, the view the single-shot ping tools can't give.

How to use

1. Enter a host, optionally pick a probe port (443 default; presets 443/80/53/22/7) and a sample interval (0.5s / 1s / 2s / 5s), then Start plot.
2. The chart fills left-to-right as replies land: a lime line for RTT and a red dot on the axis for any lost probe. The readout above shows current / min / avg / max / jitter and loss%.
3. It runs until you tap Stop; the chart keeps the most recent samples (a bounded window) so a long run stays fixed-size. Copy exports the summary plus a per-sample table.

Formula or method. Drives the shipped TCP-handshake Ping engine (PingService) in continuous mode (count = 0), one probe per chosen interval. Each reply folds into a bounded rolling window (default last 60 samples); min/avg/max are over the landed RTTs in that window, jitter is the mean absolute difference between consecutive landed RTTs (a lost probe breaks the chain, so jitter never pairs across a gap), and loss% is lost/sent. A timed-out / unreachable probe is recorded as an honest gap (no RTT) and drawn as a red axis dot, never as a fabricated 0 ms.

Field notes

- This is a TCP round-trip probe, not ICMP echo. The metric is labeled "TCP RTT" and the probe port is shown, so it is never mistaken for ICMP (see Ping (ICMP) for the mobile real-ICMP path).
- Platform: runs anywhere native, including the sandboxed macOS desktop where real ICMP can't (the ICMP path needs a subprocess the App Sandbox blocks). Served from a WLAN Pi, the Pi runs a bounded ICMP ping series, labeled as such. Any other browser is gated off with the download-the-app prompt.
- Dropped probes are shown, never hidden: a lost sample is a red dot on the axis and counts toward loss%, so a flaky path reads honestly instead of as a smooth line.
- The chart retains a bounded window of recent samples (so memory stays flat on a long run); the copy export notes how many of the total samples are shown.

Traceroute (Mobile)

Hop-by-hop path via an ICMP TTL-walk, on Android and on a WLAN Pi (iOS unsupported).

Why it's here. Extends traceroute to mobile where the platform genuinely supports it, built on the same shared ICMP layer as Ping (ICMP).

How to use

1. Android: enter a host, run; hops fill in via a TTL-walk (one ICMP echo per increasing TTL, surfacing the router that answers each).

Formula or method. A TTL-walk on the ICMP layer: send echoes with TTL 1..maxHops; an intermediate router answering TimeExceeded names that hop; the target answering EchoReply ends the walk.

Field notes

- Platform matrix (the critical honesty point): Android, available; dart_ping's TTL maps to ping -t (outbound TTL) and the system ping prints the responding hop on a "Time to live exceeded" line. iOS, not feasible, honestly unavailable. iOS can echo (via GBPing), but GBPing's receive path only accepts ICMP EchoReply (type 0); it never parses TimeExceeded (type 11), the message a traceroute needs to name each hop. Setting a low TTL just makes the echo time out with no hop IP. So iOS gets an honest "not on this device", never faked hops. Desktop: the system traceroute is the path; this ICMP TTL-walk reports sandboxed-desktop and points there. WLAN Pi: /toolboxapi/traceroute runs the walk on the Pi and returns the final hop list.
- Where available (Android), hops are the same TTL/IP/RTT/* * * shape as the system traceroute.
- The iOS limitation is a real platform ceiling in GBPing, documented at length in the source; it is the reason this tool does not run on iOS. DEVICE-PENDING: the Android path is device-pending verification per the source.

Traceroute (System)

Hop-by-hop path discovery via the OS traceroute/tracert (desktop).

Why it's here. The genuine traceroute, where it can actually run. Reach for it on a Mac/PC to see the routed path and where latency or loss enters.

How to use

1. macOS and Windows, or a WLAN Pi. Enter a host, run; on desktop, hops fill in live as the OS tool emits them and the run is cancellable mid-flight. On a Pi the trace runs on the Pi and the full hop list arrives at the end.

Formula or method. Spawns the system traceroute (Unix: -m maxHops -q 3 -w 2) or tracert (Windows: -d -h maxHops -w 2000) and parses each hop line live from stdout/stderr: TTL, host/IP, per-probe RTTs, and * * * timeouts. A real traceroute needs to read ICMP TIME_EXCEEDED replies, which require either a raw socket or the privileged system binary, so faking hops from TCP timing is explicitly refused.

Field notes

- Platform matrix (honest): macOS / Windows / Linux desktop spawns the OS binary. But under the macOS App Sandbox (the App Store build) spawning is blocked, so the screen runs a live isLaunchable() probe (a side-effect-free no-arg launch) and adapts: a non-sandboxed Developer-ID macOS build and Windows/Linux launch it fine; the sandboxed build shows an explicit "binary unavailable" verdict rather than hanging or pretending. iOS / Android: subprocess execution is sandboxed out entirely → "Traceroute runs on desktop", pointing at Ping (on Android, Traceroute (Mobile) maps the path). WLAN Pi: /toolboxapi/traceroute. Any other browser: gated to the download fallback.
- Each hop shows TTL, the responding router (name + IP), and up to three probe RTTs; * * * is a hop that didn't answer (common and not necessarily a problem). Reaching the target is reported as a terminal "complete".
- On a sandboxed macOS App Store build this tool honestly reports unavailable. For a path read on mobile, see Traceroute (Mobile), but note its iOS limitation.

Discovery & Scanning (5)

Lookup (ARP/NDP)

Discover local-network neighbors: IP, and MAC where the platform exposes it.

Why it's here. A neighbor read of the local segment. Reach for it to map IP↔MAC on platforms that can, or just to list responders.

How to use

1. Open and run; it derives the local subnet and sweeps it, attaching cached MACs where available.

Formula or method. Active discovery: derive the local subnet as a /24 around the device's primary IPv4 (the shipped tool always sweeps a /24; it does not read the interface's real prefix), TCP-connect-probe each of the 254 hosts (curated LAN ports 80/443/22/445/139/53/8080, refused RST counts as up), list the responders, and on macOS/Windows read the system neighbor table to attach the real cached MAC (macOS via a Swift sysctl call, Windows via the Win32 IP Helper API GetIpNetTable in pure-Dart FFI). No subprocess (arp -a is sandbox-blocked, which is exactly why those native reads exist). (The underlying sweep library carries a safety cap that refuses any prefix larger than a /22, but the /24 this tool always requests never approaches it.)

Field notes

- Platform differences: macOS and Windows run the active sweep and attach a real MAC from the system neighbor table. Android runs the sweep without MACs, because the system does not let apps read that table. iOS is unavailable. Served from a WLAN Pi, the Pi reads its own neighbor table (/toolboxapi/neigh). Any other browser falls back to a download. The tool does not keep its own list of what each platform can do: it asks the reader that performs the read, so the app cannot describe a capability it does not have. The Windows read is implemented but not yet confirmed on real hardware; a read that fails is reported as a failed read, not as a platform limit.
- A missing MAC is never an invented value, and it always names WHICH kind of missing it is. "Not exposed on this platform" means no reader here can read the table at all. "MAC read failed" means the read was attempted on this platform and did not work. "Not in the ARP cache" means the read succeeded and simply held no entry for that host. Collapsing those three into one claim is how the tool used to tell macOS users their platform could not do a read it was performing. Incomplete/all-zero ARP entries are skipped.
- On macOS and Windows you get a responder list WITH MACs, read from the system neighbor table. iOS and Android are the ceiling: those sandboxes do not expose the table, so responders list without a hardware address. For richer enrichment (device type and vendor) see Network Discovery.

Nearby AP Scan

Lists the Wi-Fi access points a scan can see around you (SSID, BSSID, channel, band, and signal) and draws a simple channel-occupancy bar per band so you can read how crowded each channel is at a glance. Runs on Android, macOS, and Windows, and on a WLAN Pi's own radio.

Why it's here. A fast read of who else is on the air. When you are picking a channel or chasing co-channel interference, seeing the nearby BSSIDs and where they sit across 2.4, 5, and 6 GHz tells you which channels are busy and which are clear, without carrying a separate analyzer.

How to use

1. Tap Scan to run a Wi-Fi scan. Results list each visible access point with its SSID, BSSID, channel, band, and signal (RSSI).
2. Use the sort control to order the list by signal or by channel, and read the per-band occupancy bars to see which channels are crowded.
3. Re-run the scan to refresh. Android rate-limits background scans, so a rapid re-scan may return the previous results until the throttle window clears.

Example. On a busy 2.4 GHz band the occupancy bars cluster on channels 1, 6, and 11; an access point sitting on channel 3 overlaps both 1 and 6, which the bars make visible at a glance.

Field notes

- RUNS ON ANDROID, macOS, AND WINDOWS, and on a WLAN Pi (/toolboxapi/scan, with a radio picker when the Pi has more than one scan-capable radio). iOS blocks nearby-AP scanning at the operating-system level, so the tool is left out of the catalog there.
- Clean fields only. The Android scan exposes SSID, BSSID, channel, band, and RSSI for a scanned (non-connected) network. It does not expose a per-network noise floor, SNR, or data rate, so those columns do not exist here and are never shown.
- Android throttles Wi-Fi scans. Foreground apps get a limited number of scans per window; when throttled, a re-scan returns the last cached results and the screen notes it rather than faking a fresh scan.
- Location permission and Wi-Fi must both be on for the scan to return results; the screen shows an honest Wi-Fi-off or Location-needed card when either is missing.

Network Discovery

Find live hosts on the local network and enrich each with name, services, inferred device type, and (where exposed) MAC/vendor.

Why it's here. A Fing-style LAN scan. Reach for it for a richer inventory than a bare ping sweep: what each host is, not just that it answered.

How to use

1. Open and run. The engine seeds the local /24, connect-scans it, reverse-DNS resolves, mDNS-browses, then (macOS) reads the ARP cache for MAC/vendor.

Formula or method. Four passes: 1. Subnet seed: derive the local /24 host list from network_info_plus (:181-199). 2. Connect-scan: bounded-concurrency (64) TCP connect across the /24 × a curated port set, run in a background isolate; streams progress (:201-262). 3. Reverse DNS: InternetAddress.reverse() per discovered host (null when no PTR) (:264-275, 499-510). 4. mDNS browse: in-house native NetServiceBrowser/NetService (Apple Bonjour daemon) over a curated DNS-SD service set. Then the neighbor-table read (desktop only, never a subprocess: macOS via Swift sysctl NET_RT_FLAGS/RTF_LLINFO, Windows via the Win32 IP Helper API GetIpNetTable in pure-Dart FFI) runs after the scan warms the kernel cache, attaching MAC + vendor (:303-335). A read that fails is reported as a failed read, not as a platform that cannot do it. Finally a pure device-type heuristic runs on each host's open ports + mDNS services. Device-type rules (first match wins, most specific first): IPP/LPD/9100 or printing mDNS → Printer; RTSP → Camera/NVR; iOS lockdownd (62078) → iOS device; SMB (445) → Windows/SMB; _sonos/_spotify-connect → Speaker; _googlecast → Media streamer; _airplay/_raop/_companion-link → Apple device; then weak signals 80/443/8080 → Web server, 22 → SSH host; any mDNS → mDNS device; else Unknown. MAC→vendor resolves through the full bundled IEEE OUI registry; the resolver owns the honesty contract: null for randomized/local MACs, raw-OUI fallback for unlisted global prefixes, named vendor otherwise.

Field notes

- Platform differences (the heart of it): MAC + vendor: desktop only. macOS via a Swift sysctl ARP read, Windows via the Win32 IP Helper API (GetIpNetTable, pure-Dart FFI, still no subprocess); both feed the same bundled OUI vendor table. On iOS/Android a sandboxed app cannot read the neighbor table, so MAC/vendor stay null. mDNS: iOS + macOS via the native NetServiceBrowser channel. It deliberately does NOT use pure-Dart multicast (iOS 14+ silently drops it without Apple's multicast entitlement) and does NOT use bonsoir (GPL-3.0, incompatible with the app's licensing). Android/other platforms get a clean empty mDNS pass (NsdManager deferred). Served from a WLAN Pi, the whole scan runs on the Pi (/toolboxapi/discovery) over the Pi's own /24. Service types must be declared in Info.plist NSBonjourServices. The connect-scan core is pure-Dart and cross-platform; only mDNS/ARP enrichment are native.

- Device type is a heuristic from ports + mDNS plus, on desktop, the OUI vendor read from the MAC; Unknown is a first-class, non-apologetic outcome. Because APs broadcast no "I am an access point" mDNS service, the only reliable infrastructure signal is the OUI vendor. On desktop (macOS via the sysctl ARP read, Windows via the IP Helper API, both of which supply a MAC) the heuristic classifies networking gear honestly: a recognized networking vendor (Ubiquiti, MikroTik, Aruba, Ruckus, Cisco, Meraki, etc.) is promoted to "Access point / Wi-Fi" ONLY when a Wi-Fi/AP keyword (access point, wifi, wi-fi, wlan, unifi, meraki) also appears; a networking vendor with no such keyword is the generic "Network gear" (a switch, router, or gateway: the category it can prove, not a guessed model). On mobile there is no readable MAC, so `vendorOrHostHas(...)` is always false and an AP falls through to the weak port rules (SSH/Web/Unknown): the documented ceiling.
- Any single pass can fail without aborting the run (a failed mDNS browse just means no mDNS enrichment; nothing is faked). On mobile, expect no MAC/vendor and APs to fall through to SSH/Web/Unknown, a documented ceiling.

Ping Sweep

Discover responsive hosts on a subnet via a TCP-probe sweep (no ICMP).

Why it's here. A quick "who's on this segment" without raw sockets or a subprocess. Reach for it to enumerate live hosts on a /24.

How to use

1. Enter a CIDR (192.168.1.0/24), a range (192.168.1.10-40 or full end address), or a single IP; pick one probe port from the presets (443 default, or 80/22/53); run. A live progress bar and a running responsive count build as hosts settle.

Formula or method. For each candidate, a TCP Socket.connect to the one selected probe port (443 default; the chips are single-select, so the sweep probes a single port per run). A completed handshake or a refused RST both prove the host answered on that port; a timeout means silent on that port. Bounded worker pool (default 32 in flight). Hard cap of 254 hosts (a /24); anything larger is rejected with "that's N hosts, the cap is M", never silently truncated.

Field notes

- Platform differences: identical on every native platform. Served from a WLAN Pi, the Pi runs it. Any other browser is gated off.

- A host is reported "responded", NOT "up"; a host silent on the probed port may still be alive (ICMP-only, firewalled, or just not listening on that port). The tool never claims ICMP-style liveness, and it reports reachability on the one port you chose, not across several.
- This finds hosts that answer TCP on the selected port. To check a different service, re-run with another port. For richer host detail (name, services, type, vendor), use Network Discovery. CIDR /31 and /32 include every address; larger blocks exclude network and broadcast.

Port Scan

TCP connect scan of a host, either a common-ports preset or a custom range, reporting each port open/closed/filtered.

Why it's here. A privilege-free nmap -sT for the field. Reach for it to see what services a host exposes.

How to use

1. Enter a host, pick the common-ports preset or type a custom spec (e.g. 22, 80, 443, 8000-8100), run. Results stream in as ports settle.

Formula or method. Per port, Socket.connect(host, port) with an 800ms default timeout. Open = handshake completes; Closed = actively refused/reset (host reachable, nothing listening); Filtered = no response before timeout (a firewall dropping the SYN). Same open/closed/filtered taxonomy nmap reports for a connect scan, with no raw socket. Connects run in a bounded worker pool (default 64 in flight) and stream incrementally. The common-ports preset is 44 curated ports a network pro actually checks (20–27017, with service labels like 443→HTTPS, 3389→RDP).

Field notes

- Platform differences: works identically on every native platform (no entitlement beyond network-client). Served from a WLAN Pi, the Pi runs it. Any other browser is gated off.
- "Filtered" means the SYN went unanswered (likely a firewall); it does not mean the port is closed. A custom range parser de-dupes and bounds-checks (1 to 65535).

- This is a TCP connect scan, not a SYN/stealth scan; it completes the handshake then tears it down. A host that's all-filtered is usually unreachable or firewalled wholesale.

Names & Ownership (5)

IP Geolocation

Country, region, city, coordinates, timezone, ISP/org, and ASN for an IP (or your own public IP).

Why it's here. Quick geo + ownership context for an address. Reach for it to see where an IP resolves and who runs it.

How to use

1. Enter an IP or hostname, or leave blank to locate your own public IP, look up. A copyable "lat,long" and an OpenStreetMap URL are offered.

Formula or method. Queries ipinfo.io first (free, no key, HTTPS), with get.geojs.io as an automatic fallback (also free, keyless, HTTPS). Both clear iOS App Transport Security and the app's keyless-HTTPS rule. ipwho.is was dropped: it resolves to the ISP's registry/datacenter location (it placed a Utah egress in Virginia), while both ipinfo.io and geojs.io locate the real physical egress and agree on coordinates; ip-api.com is HTTP-only and non-commercial-tier, so it is explicitly not used. Endpoints: an empty query hits each provider's "self" endpoint (https://ipinfo.io/json, else https://get.geojs.io/v1/ip/geo.json), geolocating your public egress IP; a specific IP hits https://ipinfo.io/{ip}/json, else https://get.geojs.io/v1/ip/geo/{ip}.json. Logic: try ipinfo first; if it throws (timeout, rate-limit, transport, bad JSON) OR returns no usable lat/lon, fall back to geojs. If both fail, it returns an honest failure carrying geojs's error (the last word), never a fabricated coordinate. There is no ipwho.is-style {"success": false} in-band flag; each provider is read on its own JSON shape and every field is nullable.

Field notes

- Platform differences: JsonHttpClient → dart:io, native-only; served from a WLAN Pi, the Pi runs the lookup (/toolboxapi/ipgeo); any other browser is gated to the download fallback (ipinfo.io/geojs.io CORS unverified).
- Coordinates are shown as selectable mono data plus a copyable pair and an external maps link (no embedded interactive map). Every field is nullable → missing data renders "Not available".
- Geolocation accuracy is the provider's, especially for mobile/CGNAT IPs. A rate-limit response is surfaced explicitly ("wait a minute and try again").

Lookup (BGP/ASN)

ASN, holder, announced prefix, registry, and peer/upstream counts for an IP or ASN.

Why it's here. Routing-layer context: who announces a prefix, which RIR, how it's connected. Reach for it on upstream/ISP questions.

How to use

1. Enter an IPv4/IPv6 or an ASN (AS15169, 15169, or as15169), look up.

Formula or method. Queries the RIPEstat Data API (free, no key, HTTPS, authoritative, operated by RIPE NCC). IP path: network-info (prefix + ASN) then as-overview (holder, type, registry, announced). ASN path: as-overview then asn-neighbours mapped to upstream/peer/downstream counts. Input is classified IP vs ASN before any call.

Field notes

- Platform differences: built on JsonHttpClient (→ dart:io), so native-only; served from a WLAN Pi, the Pi runs the lookup (/toolboxapi/bgpasn); any other browser is gated to the download fallback (RIPEstat CORS unverified, so no maybe-broken web tool).
- Three states: success, empty (API answered cleanly but resolved no ASN, e.g. a private/bogon IP not in the routing table), and failure. Peer/upstream counts are best-effort enrichment; a neighbors failure leaves them null ("Not available"), not zero.

- Every field is nullable; a datum the API omits renders "Not available", never fabricated. Neighbor "type" mapping: left→upstream, right→downstream, unknown→peer.

Lookup (DNS)

A portable dig/nslookup for the field. Resolve a name as a dig-style all-records view (SOA, NS, A, AAAA, MX, TXT, SRV, CAA at once), as a single record type, or run a one-tap reverse PTR for an IP.

Why it's here. Checks records authoritatively rather than from the local cache, and works through captive portals and corporate firewalls. The all-records view is the fast "show me everything" pass; single-type is the focused look; reverse PTR turns an IP back into a hostname without flipping any selector.

How to use

1. Enter a hostname (or an IP). Leave the mode on All records for the dig-style sweep, then look up.
2. For one record type, switch to Single type and pick A, AAAA, MX, TXT, NS, SOA, PTR, SRV, CAA, or SPF.
3. When the input is an IP, a Reverse lookup (PTR) button appears for a one-tap IP to hostname query.
4. Pick the resolver (Cloudflare default, Google, or Quad9) if one provider is blocked.

Formula or method. Resolves over DNS-over-HTTPS (DoH) via basic_utils DnsUtils.lookupRecord, an HTTPS GET to a JSON resolver, not raw UDP/53. DoH is chosen because raw UDP/53 sits behind iOS local-network gating and is often blocked, while DoH rides HTTPS:443 cleanly and needs no extra socket capability. All records mode fans out one DoH query per type (SOA, NS, A, AAAA, MX, TXT, SRV, CAA) concurrently with Future.wait, then groups the answers in dig order; a per-type failure becomes a per-section note, so the records that did resolve still show. Reverse PTR rewrites the input IP to its in-addr.arpa or ip6.arpa name before querying (with a minimal IPv6 literal parser). SPF is not a separate query: it reads TXT and filters for the v=spf1 policy line (RFC 7208). Three result states per query: success, empty (resolved, no records of this type), and failure, kept distinct so the UI shows the right state.

Field notes

- Records resolve against a public resolver (Cloudflare, Google, or Quad9), not the device configured resolver, which is usually what a pro wants (authoritative, uncached). The app does not read the device's own configured DNS servers on any screen. That value is platform-and-permission gated and is deliberately out of scope.
- Quad9 (9.9.9.9) is a security/malware-filtering resolver. An empty result from Quad9 for a name that resolves fine on Cloudflare or Google is a signal, not a bug: the domain is likely on a threat blocklist Quad9 refuses to answer for, rather than genuinely having no records of that type.
- SRV and CAA are parsed into readable fields; an unparseable form is shown raw. Empty is not an error: a name with no MX simply returns the empty state.
- Platform differences: identical on iOS, Android, macOS, and Windows (HTTPS only). Served from a WLAN Pi, the Pi resolves the name (/toolboxapi/dns). Any other browser is gated off.
- dig +trace (the root to TLD to authoritative delegation walk) is not built. DoH only reaches recursive resolvers that return the final answer, so a true iterative trace would need raw UDP/53 to named authoritative servers plus a hand-rolled DNS wire codec, which is heavier and less reliable in the field than this tool is meant to be.

MAC Vendor OUI Lookup

Turn a MAC address into its registered vendor, fully offline.

Why it's here. Identify a device's manufacturer from its MAC with no network, and, crucially, tell you honestly when a MAC has no real vendor (randomized phones).

How to use

1. Enter a MAC in any common notation (colon, hyphen, Cisco dot aabb.ccdd.eeff, or no separators; any case), look up.

Formula or method. Pure-Dart resolver over a bundled IEEE registry table loaded once and cached, no HTTP, no dart:io. It honors the three IEEE block sizes and matches most-specific-first: MA-S (/36, 9 hex), then MA-M (/28, 7 hex), then MA-L (/24, 6 hex), so a /36 sub-allocation names the real sub-assignee, not the /24 parent.

Field notes

- Platform differences: identical everywhere (offline, pure Dart), including web-safe in principle, though it ships in the gated Networking category. No platform exposes more or less here.
- Honesty bits: U/L bit set (0x02) = locally-administered / randomized address (the common case for modern iOS/Android Wi-Fi) → flagged, no vendor invented (a registry hit would be coincidence). I/G bit set (0x01) = multicast/group address, not a single NIC → flagged, no vendor. A globally-administered MAC not in the bundled snapshot → the raw 24-bit OUI is shown (e.g. B8:27:EB), never invented.
- A randomized phone MAC correctly returns "no vendor"; that's the right answer, not a miss. The bundled table has a documented retrieval date in its asset header; refresh it to pick up new allocations.

WHOIS

Domain/IP registration lookup over WHOIS (TCP port 43), with parsed highlights and the raw record.

Why it's here. Registrar, dates, status, and name servers for a domain or IP, from the field. Reach for it on ownership/expiry questions.

How to use

1. Enter a domain or IP, look up.

Formula or method. Raw WHOIS over TCP/43 via Socket.connect, not a whois subprocess (sandbox-blocked) and not RDAP/HTTPS (uneven coverage, no CORS). It does the hierarchical two-hop dance: query whois.iana.org for the target, parse the refer:/whois: referral to the authoritative registry, re-query that, and follow one optional further hop to a Registrar WHOIS Server: if it returns a fuller record. Highlights (registrar, created/updated/expires, status, name servers) are parsed from the free-form record where reliably present; anything missing is omitted, never faked.

Field notes

- Platform differences: same on every native platform. Served from a WLAN Pi, the Pi runs it. Any other browser is gated off.
- Three states: success (raw record + highlights), empty (server answered but the object is unregistered / a "No match" banner), and failure (connection/timeout/bad input). The servers consulted are listed so the path is transparent.
- WHOIS output is registry-specific and free-form; highlights are best-effort. The raw record is the source of truth.

Services & Protocols (5)

Inspector (HTTP Header)

Issue a HEAD or GET, follow and record the redirect chain hop-by-hop, and return the final status plus all response headers.

Why it's here. Shows the full 301→302→200 story, not just the destination. Reach for it to debug redirects, security headers, caching, or CDN behavior.

How to use

1. Enter a URL (bare host assumes https://), pick HEAD (default) or GET, inspect. Each hop is shown with its status, Location, and headers.

Formula or method. Sets followRedirects = false and follows the chain itself, recording one hop per response until a non-redirect status or the 10-redirect cap. HEAD→GET fallback: if HEAD returns 405 (and the caller didn't demand GET), it transparently retries that hop with GET and notes the fallback. Relative Location values are resolved per RFC 7231.

Field notes

- Platform differences: iOS App Transport Security blocks cleartext http://, and the app deliberately does not add a blanket ATS exception (that would weaken every request). So an http:// target fails at the socket layer on iOS, and the tool surfaces a specific message ("On iOS, cleartext HTTP is blocked by ATS, try the https:// URL") pointing you at the https:// URL. Note this cleartext-HTTP message is shown for ANY

connection failure on an http:// URL (refused, DNS failure, host unreachable) on every native platform, not just iOS. On desktop, where ATS does not apply, treat it as a prompt to retry over https:// rather than a literal diagnosis of the cause. Otherwise identical on native; gated off on web.

- The hop chain reads top (first request) to bottom (final response). headFellBackToGet and redirectLimitHit flags are surfaced. Header names are title-cased and sorted.
- Bodies are drained and discarded; this is a header inspector, not a fetcher. On iOS, use the https:// URL.

Inspector (SSL/TLS)

Connect to host:port over TLS and report the server certificate as inspectable data, including expired, self-signed, and name-mismatch certs.

Why it's here. A field cert inspector. Reach for it to read validity, SANs, fingerprints, key size, and the issuer on any TLS service, especially broken ones.

How to use

1. Enter a host (URLs are accepted and stripped to the host), optional port (443 default), inspect.

Formula or method. SecureSocket.connect with onBadCertificate: (cert) => true: it accepts any certificate at the socket layer so an expired/self-signed/mismatched cert is still captured and shown; the validity verdict is computed from the cert dates, not thrown as an error. A bad cert is a successful inspection, not a failure. Field coverage is split: dart:io X509Certificate gives PEM/DER, subject/issuer DN, notBefore/notAfter, SHA-1; basic_utils X509Utils re-parses the PEM to recover structured DN, SAN list, serial, signature algorithm, public-key algorithm + bits, and SHA-256. Validity is computed against "now": valid / expired / not-yet-valid + days-to-expiry.

Field notes

- Platform differences: same on every native platform. Served from a WLAN Pi, the Pi runs it. Any other browser is gated off.

- The validity state is an icon + text (not color alone). Two honest limits are stated on-screen, not faked: (1) ALPN ≠ TLS version/cipher, since dart:io exposes only the ALPN result, not the negotiated TLS version or cipher suite, so those are reported "not exposed by the platform", never invented. (2) Leaf only, since dart:io hands over only the leaf certificate, not the intermediate/root chain it validated against.
- Connection problems (DNS, refused, timeout) are failures; an invalid cert is not. Don't read the ALPN line as the TLS version; the tool deliberately separates them.

Packet Sender

Send a custom TCP or UDP payload to host:port and read the reply (raw bytes + hex + decoded text).

Why it's here. A field "netcat" for poking a service: banner grabs, custom probes, protocol pokes. Reach for it to send a crafted payload and see what comes back.

How to use

1. Enter host, port, transport (TCP/UDP), and a payload (plain text, or \xNN hex escapes plus \r \n \t \0 \), send. The reply is shown as hex and decoded text.

Formula or method. TCP via Socket, UDP via RawDatagramSocket, no raw sockets, no custom ICMP/IP framing (same sandbox wall as ICMP traceroute), so it ships clean everywhere. TCP: connect → send → read until the peer closes or the read goes idle for the timeout, with a hard total cap (timeout×3) so a chatty stream can't run forever. UDP: bind ephemeral → send one datagram → wait up to the timeout for a reply; no reply is a first-class non-error outcome (timedOut/isNoReply), not an exception, because UDP has no delivery guarantee. Errors are typed: DNS failure, refused, unreachable, timeout, invalid input, other, each mapped to a precise message.

Field notes

- Platform differences: identical on every native platform. Served from a WLAN Pi, the Pi runs it. Any other browser is gated off.

- A UDP "no reply" is honest and expected for many services; it does not mean failure. The payload parser returns null only on a malformed \x (a clear authoring mistake worth surfacing).
- Binary replies decode with the Unicode replacement char rather than throwing, so they still render. This is a single-shot send/receive, not an interactive session.

Time Server (NTP)

Query an NTP server over SNTP and report the server's time, your device clock's offset from it, and the round-trip network delay.

Why it's here. When a log timestamp, a certificate window, or an 802.1X/Kerberos handshake looks wrong, the first question is whether the device clock is off. This answers it against a real time source and tells you which way and by how much, in plain language.

How to use

1. Open the tool. The server field defaults to time.apple.com; tap Check time to query it.
2. If the default is blocked or slow, switch to the public fallback pool.ntp.org.
3. Read the stratum, the server time (UTC and local), the device time, the signed clock offset with its plain-language verdict, and the round-trip delay.

Formula or method. A single unicast SNTP exchange (RFC 4330 / RFC 5905): a 48-byte client request whose first byte is 0x1B (leap 0, version 3, mode 3) is sent over an outbound UDP datagram to port 123, and one reply is read. The four-timestamp math runs on integer microseconds: clock offset = ((t2 − t1) + (t3 − t4)) / 2 and round-trip delay = (t4 − t1) − (t3 − t2), where t1/t4 are the client transmit/receive instants and t2/t3 are the server receive/transmit timestamps read from reply bytes 32–39 and 40–47. NTP timestamps are 64-bit fixed-point seconds since 1900-01-01; the converter subtracts the 2208988800-second NTP-to-Unix epoch offset. A positive offset means the device clock is behind the server; the screen translates the sign so you never have to remember the convention.

Field notes

- Not a subprocess and not an HTTPS time API by design: an HTTPS endpoint can report the server's time but cannot measure the symmetric round-trip the SNTP four-timestamp formula needs to separate offset from delay. The outbound UDP/123 datagram is the unicast cousin of Wake-on-LAN's send, covered by the same network-client entitlement, and works the same on iOS / macOS / Android / Windows.
- Honest failures, never a fabricated offset: a timeout (3 s), a DNS failure, a reply shorter than 48 bytes, or a "kiss-o'-death" stratum 0 each surface as a clear message that suggests the fallback server, rather than a zeroed delay or a made-up offset.
- A tiny negative delay from clock jitter on a fast LAN is clamped to 0 so the reading is never nonsensical. Stratum is reported as-is (1–15 valid, 16 unsynchronized, 0 the kiss-o'-death marker).
- Web has no dart:io UDP socket, so the screen routes to the download-the-app fallback via NetworkSupport.ntpSupported.

Wake-on-LAN

Send a Wake-on-LAN magic packet to wake a host by MAC address.

Why it's here. Wake a sleeping machine on the LAN from your phone or laptop.

How to use

1. Enter the target MAC (colons, hyphens, Cisco dots, or no separators all parse), optionally a subnet-directed broadcast (e.g. 192.168.1.255) and port (9 default, 7 alternative), send.

Formula or method. Builds the 102-byte magic packet (6× 0xFF then the 6-byte MAC repeated 16 times) and sends it as a UDP broadcast via RawDatagramSocket with broadcastEnabled = true, no subprocess, no privileged socket. The MAC is normalized to canonical form; an invalid MAC, bad broadcast IP, or out-of-range port is rejected with a clear message.

Field notes

- Platform differences: works on every native platform (outbound UDP broadcast is covered by the existing entitlements). Served from a WLAN Pi, the Pi runs it. Any other browser is gated off.

- Success means the packet was sent; it makes NO claim the device woke. WoL is unacknowledged, a switch may not forward the all-ones broadcast across subnets, and the target may have WoL disabled. The tool shows the bytes sent and the packet hex.
- If the OS reports 0 bytes sent, it suggests a directed broadcast (e.g. 192.168.1.255) instead of 255.255.255.255. "Sent" ≠ "woke"; verify the host separately.

Calculators & Tools (39 tools)

RF math and field utilities. Link-budget building blocks, antenna and coverage geometry, capacity and power figures, coordinate work, and unit conversions, each computed locally on the device.

RF & Propagation (9)

Cable Loss

Estimates total coax attenuation for a run of a known cable type, length, and frequency, plus the per-100 ft loss coefficient at that frequency.

Why it's here. Cable loss is a direct subtraction in the transmit and receive chains. Pick a cable type and length and the tool tells you how much signal the feedline will eat.

How to use

1. Pick the cable type from the list (five LMR types).
2. Enter frequency and pick its unit (GHz or MHz).

3. Enter run length and pick its unit (ft or m).
4. Read total loss in dB plus the per-100 ft coefficient.

Inputs

Input	Unit	Range
Cable type	one of: LMR-100A, LMR-200, LMR-400 (default), LMR-600, LMR-900	-
Frequency	GHz (default) or MHz	must be > 0
Length	ft (default) or m	must be > 0

How it works. Times Microwave publishes an attenuation equation on every LMR datasheet: dB per 100 ft = k1 × √(F in MHz) + k2 × F in MHz, with k1 and k2 constants for each cable (LMR-100A 0.709140 / 0.001740, LMR-200 0.320900 / 0.000330, LMR-400 0.122290 / 0.000260, LMR-600 0.075550 / 0.000260, LMR-900 0.051770 / 0.000160; Times Microwave LMR-100A p.9, LMR-200, LMR-400 p.23, LMR-600 p.29, LMR-900 p.33). The tool evaluates that equation at your exact frequency, so nothing is interpolated between table rows. Total loss (dB) = per-100 ft loss × length in feet ÷ 100. Frequency in GHz is converted to MHz (× 1000) and length in meters to feet (× 3.28084) as needed. Results are rounded to two decimals. As an anchor, LMR-400 at 2400 MHz is 6.6 dB per 100 ft.

Example. LMR-400, 2.4 GHz, 50 ft → per-100 ft at 2400 MHz = 0.122290 × √2400 + 0.000260 × 2400 = 6.615 dB; total = 6.615 × 50 / 100 = 3.31 dB.

Field notes

- Values are manufacturer figures at the datasheet's stated conditions (VSWR 1.0, 25°C). Real loss rises with heat and age, and connectors add loss not modeled here.
- The equation is Times Microwave's own, so it reproduces their published tables wherever the two overlap, and it stays exact at frequencies they never tabulated (2400 MHz among them).

- There is no RG-58, RG-8, RG-213 or RG-214 here, on purpose. "RG" is not one cable: Belden 8259 and Belden 9201 are both stamped RG-58 and differ by 54 percent at 900 MHz, and no manufacturer publishes RG attenuation at Wi-Fi frequencies. For an RG run, read dB/100 ft off your actual cable's datasheet by part number.
- LMR-1200 is deliberately omitted: Times stops tabulating it at 2500 MHz and the equation under-predicts its loss at Wi-Fi frequencies. The five cables offered are all tabulated to 5800 MHz.
- Above 5800 MHz the tool says plainly that it is extrapolating. Treat a 6 GHz result as a lower bound and confirm against your cable's datasheet.

Earth Curvature

Computes the earth bulge (the height of the earth's curvature at the midpoint of a path) for a given path length and atmospheric K-factor.

Why it's here. On long microwave links the earth's curvature, modified by atmospheric refraction, eats into Fresnel clearance. This sizes how much extra antenna height the curvature demands.

How to use

1. Enter path length and pick its unit (km or mi).
2. Pick the K-factor.
3. Read the bulge in meters and feet.

Inputs

Input	Unit	Range
Path length	km (default) or mi	mi × 1.60934 → km
K-factor	select	4/3 standard (1.333, default), 1.0 geometric, 2/3 worst-case (0.667), 2.0 superrefraction

How it works. The bulge is the height of the earth's curvature at the midpoint of the path, using a mean earth radius of 6371 km scaled by the atmospheric K-factor: bulge in meters = (path length in km)² × 1000 ÷ (8 × 6371 × K). The result is also shown in feet (meters × 3.28084).

Example. 20 km path, K = 1.333 → $20^2 \times 1000 / (8 \times 6371 \times 1.333) = 5.89$ m.

Field notes

- K = 4/3 is the standard-atmosphere assumption; K = 2/3 is the conservative worst case used for availability-critical links (smaller K means more bulge).
- The bulge is the maximum at midpoint; add it to required Fresnel clearance when sizing tower heights.
- Atmospheric K-factor varies with weather and geography. This is a planning value, not a guarantee.

Free Space Path Loss

Computes the loss in dB a signal suffers traveling through free space between transmitter and receiver, given the operating frequency and distance.

Why it's here. The starting point of any link budget. Reach for it to estimate how much signal a point-to-point or outdoor link loses before you factor in antennas and cables.

How to use

1. Enter the frequency and pick its unit (GHz or MHz).
2. Enter the distance and pick its unit (km, mi, or m).
3. Read the path loss in dB. The result blanks to a dash until both fields hold valid positive numbers.

Inputs

Input	Unit	Range
Frequency	GHz (default) or MHz	must be > 0
Distance	km (default), mi, or m	must be > 0

How it works. Free space path loss in dB = 20 × log10(frequency in GHz) + 20 × log10(distance in km) + 92.45. The constant 92.45 is the fixed offset for the km/GHz form of the equation. Inputs are converted to those units first (MHz ÷ 1000 → GHz; mi × 1.60934 → km; m ÷ 1000 → km). The output is rounded to one decimal, and blanks if frequency or distance is zero or negative.

Example. 5 GHz, 1 km → 20 × log10(5) + 20 × log10(1) + 92.45 = 106.4 dB.

Field notes

- Free space only. No obstructions, no ground reflection, no atmospheric effects. Real-world links always lose more, so use this as a floor, not a prediction.
- The reference card lists anchor values (2.4 GHz @ 1 km = 100.1 dB, 6 GHz @ 1 km = 108.0 dB).

Fresnel Zone

Computes the first Fresnel zone radius along a point-to-point path, plus the 60% clearance value planners treat as the "keep it clear" threshold.

Why it's here. Line of sight is not enough for a reliable link; the first Fresnel zone must be mostly clear of obstructions. This sizes how much clearance you need above terrain, rooftops, and trees.

How to use

1. Enter frequency in GHz.

2. Enter total path distance in meters.
3. Optionally enter a point-from-TX distance in meters to get the radius at that specific point; leave blank for the midpoint (maximum) only.
4. Read first-zone radius and 60% clearance (in meters, feet shown alongside).

Inputs

Input	Unit	Range
Frequency	GHz (fixed unit)	must be > 0
Total path distance	meters	must be > 0
Point from TX	meters, optional	must be inside (0, total) to add an at-point result, otherwise ignored

How it works. First the wavelength is found from the frequency: λ in meters = 0.3 ÷ frequency in GHz (the speed-of-light form, with c = 3×10^8 m/s). The first Fresnel zone radius at any point along the path is r = √(λ × d1 × d2 ÷ (d1 + d2)), where d1 and d2 are the distances from that point to each end in meters. The midpoint radius (the maximum, always shown) uses d1 = d2 = total distance ÷ 2. The 60% clearance value is r × 0.6, and feet are r × 3.28084.

Example. 5.8 GHz, 10000 m total, midpoint → λ = 0.3/5.8 = 0.05172 m; r = √(0.05172 × 5000 × 5000 / 10000) = 11.37 m; 60% clearance = 6.82 m.

Field notes

- The midpoint radius is the worst case along the path; clearance there is the binding constraint.
- Frequency unit is fixed to GHz here (unlike FSPL, which offers MHz).
- 60% clearance is the common rule of thumb; some designs require more in heavy-foliage or reflective environments.

ITU Rain Fade

Estimates rain attenuation on a microwave link using ITU-R P.838-3 (specific attenuation) and a simplified ITU-R P.530 (effective path length).

Why it's here. Above ~10 GHz, rain is the dominant fade mechanism on outdoor links. This sizes the fade margin a backhaul link needs to survive heavy rain.

How to use

1. Enter frequency in GHz.
2. Enter rain rate in mm/hr.
3. Enter path length and pick its unit (km or mi).
4. Pick polarization (Horizontal or Vertical).
5. Read total rain attenuation, specific attenuation γ, and effective path length.

Inputs

Input	Unit	Range
Frequency	GHz (fixed)	must be > 0
Rain rate	mm/hr	must be > 0
Path length	km (default) or mi	must be > 0
Polarization	Horizontal or Vertical	-

How it works. The two coefficients k and α come from ITU-R P.838-3 Table 5, all 116 published rows spanning 1 to 1000 GHz, with separate horizontal and vertical values. Between rows the tool interpolates k on a log-log basis and α on a log-linear basis, clamping at the table ends.

Specific attenuation is then γ (dB/km) = k × (rain rate)^α. The effective path length shortens the geometric length to account for non-uniform rain cells: L_eff = L ÷ (1 + L/d0), where d0 = 35 × e^(−0.015 × rain rate). Total rain attenuation (dB) = γ × L_eff. Outputs are shown to two decimals for attenuation, four for γ, and two for L_eff.

Example. 23 GHz, 42 mm/hr, 5 km, Horizontal → Table 5 has a 23 GHz row: k = 0.1286, α = 1.0214; γ = 0.1286 × 42^1.0214 = 5.8510 dB/km; d0 = 35 × e^(−0.63) = 18.64, L_eff = 5/(1 + 5/18.64) = 3.94 km; attenuation = 5.8510 × 3.9425 = 23.07 dB.

Field notes

- The model assumes the rain rate is uniform across the path; real cells are smaller, which the L_eff reduction partly accounts for.
- Pick a rain rate matching your target availability (e.g. the 0.01%-of-time rate for your region). The tool does not supply regional rain statistics.
- Below ~10 GHz rain fade is usually negligible.

Link Budget

Full point-to-point link budget: combines transmit power, antenna gains, cable losses, path loss, and miscellaneous losses into a received signal level, then compares against receiver sensitivity for a fade margin.

Why it's here. The decision tool for whether a link will close, and by how much. Assemble all the gains and losses to see the margin before you deploy.

How to use

1. Enter TX power and pick its unit (dBm, W, or mW).
2. Enter TX antenna gain, TX cable loss, path loss, RX cable loss, RX antenna gain, and RX sensitivity.
3. Optionally enter other/miscellaneous losses (treated as 0 when blank).
4. Read received signal (dBm) and link margin (dB), color-coded by health.

Inputs

Input	Unit	Range
TX power	dBm (default), W, or mW	W/mW must be > 0
TX gain	dBi	-
TX loss	dB	-
Path loss	dB	-
RX loss	dB	-
RX gain	dBi	-
RX sensitivity	dBm	-
Other losses	dB	optional, default 0

How it works. Power in watts is first converted to dBm (10 × log10(W × 1000); milliwatts convert via W = mW/1000). The received signal is the sum of gains minus losses along the chain: received (dBm) = TX power + TX gain − TX loss − path loss − RX loss + RX gain − misc losses. The link margin is received signal minus receiver sensitivity. The margin is color-coded: 10 dB or more is healthy (green), 0 to 10 dB is marginal (amber), and below 0 dB is negative (red).

Example. TX 30 dBm, TX gain 20, TX loss 2, path loss 120, RX loss 2, RX gain 20, misc 0, RX sensitivity −85 → received = 30 + 20 − 2 − 120 − 2 + 20 − 0 = −54 dBm; margin = −54 − (−85) = 31 dB (healthy).

Field notes

- Path loss is supplied by the engineer (compute it with the FSPL tool, then add real-world margins).

- This budget does not include rain fade; use the PtP Link Check for that.
- A healthy margin (≥ 10 dB) is the common design target to ride out fading and multipath.

Noise Floor

Computes the thermal noise floor (kTB) for a channel, the receiver noise floor (kTB plus noise figure), and the quick −174 dBm/Hz rule-of-thumb value.

Why it's here. SNR is signal minus noise floor. Knowing the noise floor for a given channel width sets the minimum usable signal and frames link-margin discussions.

How to use

1. Pick the channel bandwidth.
2. Enter receiver noise figure in dB.
3. Optionally adjust temperature (defaults to 20°C when blank).
4. Read thermal noise floor, receiver noise floor, and the rule-of-thumb value.

Inputs

Input	Unit	Range
Bandwidth	MHz	select of 20, 40, 80, 160, or 320 MHz
Noise figure	dB	must be ≥ 0 (default 7; invalid/negative blanks the outputs)
Temperature	°C	defaults to 20 when blank

How it works. The thermal noise floor is the classic kTB relationship: thermal (dBm) = 10 × log10(k × T × bandwidth in Hz) + 30, where k is Boltzmann's constant (1.380649 × 10^{-23} J/K), T is temperature in kelvin (°C + 273.15), and bandwidth converts from MHz to Hz (× 10^{6}). The receiver noise floor adds the noise figure: rx floor = thermal + noise figure. The rule-of-thumb value uses the standard −174 dBm/Hz reference: kT at the noise reference temperature T0 = 290 K (about 17°C, per ITU-R P.372), 10 × log10(1.380649e-23 × 290) + 30 = −173.98, rounded to −174. rule = −174 + 10 × log10(bandwidth in Hz). Outputs are rounded to one decimal.

Example. 20 MHz, NF 7 dB, 20°C → thermal = 10 × log10(1.380649e-23 × 293.15 × 20e6) + 30 = −100.9 dBm; rx floor = −93.9 dBm; rule = −174 + 10 × log10(20e6) = −101.0 dBm.

Field notes

- This is the theoretical noise floor; real receivers see a higher effective floor from co-channel interference, adjacent-channel leakage, and ambient RF.
- The −174 rule is kT at 290 K (about 17°C) rounded to a whole number, so it differs slightly from the temperature-aware thermal value at any other temperature (at 0°C, kT is −174.24 dBm/Hz).

PtP Link Check

Full point-to-point backhaul link budget end to end, from TX power through antenna gains, cable losses, free-space path loss, and rain fade to the received signal, compared against sensitivity for a fade margin and a PASS / MARGINAL / FAIL verdict.

Why it's here. The all-in-one decision tool for a microwave or Wi-Fi backhaul link, combining FSPL and rain fade with the link budget so you get a single go/no-go answer.

How to use

1. Enter frequency (GHz), distance (km/mi), TX power (dBm), TX gain (dBi), RX gain (dBi), and RX sensitivity (dBm).
2. Optionally enter TX loss, RX loss, rain rate, required margin, and pick polarization.

3. Read EIRP, free-space loss, rain fade, received signal, link margin, and the verdict.

Inputs

Input	Unit	Range
Frequency	GHz (fixed)	must be > 0
Distance	km (default) or mi	must be > 0
TX power	dBm	signed
RX sensitivity	dBm	signed; sensitivity usually negative
TX gain	dBi	-
RX gain	dBi	-
TX loss	dB	default 0
RX loss	dB	default 0
Rain rate	mm/hr	default 0 (no fade)
Required margin	dB	default 10
Polarization	Horizontal or Vertical	-

How it works. This tool chains the link budget, free space path loss, and rain fade into one calculation. EIRP = TX power + TX gain − TX loss. Free space path loss = 20 × log10(distance in km) + 20 × log10(frequency in GHz) + 92.45. Rain fade uses the same ITU-R P.838-3 table and interpolation as the Rain Fade tool (γ = k × R^α, d0 = 35 × e^(−0.015 × R), L_eff = distance ÷ (1 + distance/d0), rain fade = γ × L_eff), returning zero when the rain rate is not greater than zero. The received level = EIRP − free space loss − rain fade + RX gain − RX loss, and the margin =

received level − RX sensitivity. The verdict is PASS when margin is at or above the required margin, MARGINAL when the link still closes (margin 0 up to the required value), and FAIL when the margin is below zero. Outputs are shown to one decimal, except rain fade to two.

Example. 5.8 GHz, 10 km, TX 20 dBm, TX gain 23, RX gain 23, sensitivity −78, TX/RX losses 0, rain 0, required margin 10 → EIRP = 20 + 23 − 0 = 43.0 dBm; FSPL = 20 × log10(10) + 20 × log10(5.8) + 92.45 = 127.7 dB; rxLevel = 43 − 127.7 − 0 + 23 − 0 = −61.7 dBm; margin = −61.7 − (−78) = 16.3 dB → PASS. (With TX loss 1 and RX loss 1: EIRP 42.0, rxLevel −63.7, margin 14.3, still PASS.)

Field notes

- This is the most complete link tool in the suite. Prefer it over the bare Link Budget when the link is outdoors and above ~10 GHz.
- FSPL alone underestimates real loss; the rain fade leg only covers rain, not fog, foliage, or multipath.
- The MARGINAL band (link closes but below your required margin) is the tool's own warning state; the underlying pass/fail is purely margin ≥ required.

RF Attenuation

Estimates total path loss through building materials by summing per-layer attenuation for a chosen Wi-Fi band.

Why it's here. Indoor coverage planning. Count the walls, floors, and obstructions between AP and client and get a quick total dB loss to add to the link budget.

How to use

1. Pick the band (2.4, 5, or 6 GHz).
2. Enter a quantity for each material the signal passes through.
3. Read the total attenuation in dB.

Inputs

Input	Unit	Range
Band	select	2.4 GHz, 5 GHz, or 6 GHz
Per-material quantities	integers	only quantities > 0 contribute

How it works. Each material carries a fixed per-layer loss for each band (for example, drywall is 3/4/5 dB at 2.4/5/6 GHz; poured concrete 13/16/19; metal door 20/26/30; foil insulation 25/30/35). The total is simply the sum of each material's per-layer loss for the chosen band multiplied by how many of that material the signal passes through, added up across every material with a quantity greater than zero.

Example. 5 GHz, 2 sheets of drywall (4 dB each) + 1 concrete block (13 dB) → 2 × 4 + 1 × 13 = 21 dB.

Field notes

- The per-material values are typical attenuation figures, not measurements of your building. Real walls vary widely with construction, moisture, and reinforcement.
- Metal, foil/vapor barriers, and low-E glass are near-total blockers and dominate the total.
- Use this for a first-pass coverage estimate, then validate with a survey.

Antenna & Coverage (4)

Antenna Downtilt

Computes the mechanical downtilt angle that aims an antenna's beam center at a target coverage distance on the ground.

Why it's here. When mounting a sector or directional antenna at height, you need the tilt angle to put the main lobe where the users are.

How to use

1. Enter antenna height above ground (AGL) and pick its unit (ft or m).
2. Enter target coverage distance and pick its unit (km, ft, or m).
3. Read the downtilt angle in degrees.

Inputs

Input	Unit	Range
Antenna height (AGL)	ft or m	-
Target coverage distance	km, ft, or m	-

How it works. The downtilt angle is the arctangent of the antenna height divided by the coverage distance, converted to degrees: downtilt = atan(height ÷ distance). Both values are converted to meters first (feet × 0.3048, kilometers × 1000).

Example. 30 m height, 200 m coverage → atan(30/200) in degrees = 8.53°.

Field notes

- This aims the beam center at the target distance; it does not account for the antenna's vertical beamwidth (use Downtilt Coverage for the near/far edges).
- Pure geometry, no consideration of antenna pattern shape or ground slope.

Downtilt Coverage

Computes the near and far ground coverage edges (and coverage depth) of a downtilted antenna, given height, tilt angle, and vertical beamwidth.

Why it's here. The complement to the Downtilt tool. It shows the footprint the beam actually covers, so you can check for coverage gaps or overshoot.

How to use

1. Enter antenna height (AGL) and pick its unit (ft or m).
2. Enter the downtilt angle in degrees.
3. Enter the antenna's vertical beamwidth in degrees.
4. Read near edge, far edge, and coverage depth.

Inputs

Input	Unit	Range
Antenna height (AGL)	ft or m	must be > 0
Downtilt angle	degrees	-
Vertical beamwidth	degrees	must be in the open range (0, 180)

How it works. The beam's upper and lower edges are the tilt angle minus and plus half the vertical beamwidth. The near edge on the ground is the height divided by the tangent of the near (lower) angle; the far edge is the height divided by the tangent of the far (upper) angle. Coverage depth is the far edge minus the near edge. If the upper beam edge reaches or clears the horizon (its angle drops to zero or below), the far edge is unbounded and reported as beam-above-horizon, with no far edge or depth. The result blanks out if height is not positive or beamwidth falls outside the (0, 180) range.

Example. 30 m height, 10° tilt, 8° beamwidth → far angle = 6°, near angle = 14°; near edge = 30/tan(14°) = 120.3 m; far edge = 30/tan(6°) = 285.4 m; depth = 165.1 m.

Field notes

- When the tilt is shallow relative to beamwidth (e.g. tilt 6°, beamwidth 15° → far angle = −1.5°), the upper edge clears the horizon and the far edge is unbounded. The tool flags this rather than returning a negative distance.
- Uses beam edges at the half-power (or stated) beamwidth; actual coverage tails off gradually beyond those edges.

EIRP Calculator

Computes Effective Isotropic Radiated Power, the actual power radiated from an antenna system after accounting for transmitter power, cable/connector loss, and antenna gain.

Why it's here. EIRP is the regulatory ceiling number. Check it against FCC/ETSI limits and use it as the transmit-side input to a link budget.

How to use

1. Enter TX power and pick its unit (dBm, W, or mW).
2. Enter cable loss in dB and antenna gain in dBi.
3. Read EIRP in dBm, plus a secondary line in watts/milliwatts.

Inputs

Input	Unit	Range
TX power	dBm (default), W, or mW	W/mW must be > 0 (a non-positive value makes the log non-finite and blanks the result)
Cable loss	dB	signed allowed
Antenna gain	dBi	signed allowed

How it works. Power entered in watts is converted to dBm as 10 × log10(watts × 1000); milliwatts convert the same way after dividing by 1000 to watts. EIRP in dBm is then simply transmit power in dBm minus cable loss plus antenna gain. To show the watt equivalent, EIRP in dBm converts back to power: 10^(EIRP_dBm ÷ 10), divided by 1000 for watts. EIRP is shown to one decimal; the watt line shows watts at two decimals when the value is at least 1 W, otherwise milliwatts at one decimal.

Example. TX 20 dBm, cable loss 1.5 dB, gain 14 dBi → 20 − 1.5 + 14 = 32.5 dBm → 10^(3.25)/1000 = 1.78 W.

Field notes

- Cable loss should include all connector and jumper losses on the transmit chain.
- The reference card lists regulatory EIRP ceilings (e.g. 2.4 GHz FCC PtMP = +36 dBm / 4 W); those are planning anchors and vary by sub-band, channel width, and power-control rules. Verify against current local regulations.

Wavelength

Converts a frequency to its wavelength in meters, centimeters, feet, and inches.

Why it's here. Antenna element sizing, ground-plane and spacing rules, and quarter-wave / half-wave rules of thumb all key off wavelength. A quick reference when laying out antennas.

How to use

1. Enter the frequency and pick its unit (MHz or GHz).
2. Read the wavelength in all four units at once.

Inputs

Input	Unit	Range
Frequency	MHz (default) or GHz	must be > 0

How it works. Wavelength in meters is 300 divided by the frequency in MHz: the 300 comes from the speed of light (3×10^8 m/s) scaled for the MHz-to-meter form. From there: centimeters = meters × 100, feet = meters × 3.28084, inches = feet × 12. A frequency entered in GHz is multiplied by 1000 to get MHz first. Display precision: meters to 4 decimals, centimeters to 2, feet to 4, inches to 3.

Example. 2400 MHz → 300/2400 = 0.1250 m, 12.50 cm, 0.4101 ft, 4.921 in.

Field notes

- Uses c = 3×10^8 m/s (the rounded 300 constant), not the exact 299792458. The tiny difference is irrelevant for antenna work.
- This is free-space wavelength; velocity factor inside cable or dielectric is not applied.

Capacity & Power (5)

Capacity Planner

An informational disclaimer, not a calculator. The screen is a single read-only statement: "Capacity planning is a design problem, not a calculation." There are no inputs, no formula, and no computed AP count.

Why it's here. An earlier version squeezed capacity planning into a few input boxes and returned a confident AP count. That math was removed: no single formula can honor the variables that actually drive capacity, and a confidently wrong number is worse than no number. The tile, the `/tools/capacity-planner` route, and the "Capacity Planner" title are kept so it still resolves where users expect, but it now tells the truth instead of computing false precision.

How to use

1. There is nothing to enter. Read the statement.
2. For a plan you can trust, bring in a Wi-Fi professional experienced in your kind of environment; a real plan comes from measuring and modeling your specific space and usage, not a one-size-fits-all estimate.

Field notes

- Read-only disclaimer screen: no inputs, no math, no copyable result. It is deliberately a placeholder, not a stub awaiting a calculator.
- Real Wi-Fi capacity depends on the applications in use and their airtime cost, client device capabilities and roaming, AP and antenna selection, channel reuse and co-channel interference, band steering, airtime fairness, QoS, and how people move through the space. Change any one and the answer changes, which is exactly why the calculator was removed rather than kept with a caveat.
- The copy on screen is verbatim from Keith's approved draft.

PoE Budget

Checks a PoE switch's power budget, port count and port type against everything plugged into it, after a reserve you keep back, and gives you a fits, caution, or does-not-fit verdict.

Why it's here. Before you hang a batch of access points (plus cameras and phones) off a switch, make sure the switch can power them all, has a port for each, and that every port type can actually power the device on it.

How to use

1. Enter the switch's PoE budget in watts, its number of PoE ports, and the reserve to keep back (it starts at 0 percent, so set one if you want budget held back).
2. Pick the PSE type of the switch ports (802.3af, 802.3at, 802.3bt Type 3 or Type 4) to check each device against it.
3. For up to six device rows, pick the device's PoE class, or Watts if you only know its draw, then the quantity (defaults to 1).

4. Read the total PSE allocation, the watts left after the reserve, ports needed against ports available, the port check for each row, and the verdict.

Inputs

Input	Unit	Range
Switch PoE budget	watts	must be greater than 0, or you get no result
PoE ports	ports	optional, up to 4 digits; leave it blank and the port count is not checked
Reserve	percent of the budget	0 to 99.9; defaults to 0; blank counts as 0
PSE type (switch ports)	802.3af, 802.3at, 802.3bt Type 3 or Type 4	optional; Not set skips the per-device check
Device rows (up to 6)	a PoE class (0 to 8) or watts per device, and a quantity	a blank watts field counts as 0 and a watts field may not exceed 71.3 W; quantity defaults to 1 and takes up to 4 digits

What the result shows

Line	What it reads
Total PSE allocation	the rows added up, in watts
Switch budget	the budget you entered
Reserve	the percent and the watts it keeps back, for example 10% (37.0 W)

Line	What it reads
Left after reserve	the usable budget minus the total: "33.0 W remaining", or "−5.0 W over"
Utilization	the total as a percent of the usable budget, for example "90% of usable"
Ports	"10 needed of 24 available", with ", 2 short" added when there are too few; "10 needed (enter the port count to check)" when PoE ports is blank
Port check	"every device fits an 802.3at port", or one line per failing row such as "Device 3: Class 6 needs more than an 802.3at port gives (up to Class 4)"; "PSE type not set, so devices were not checked" when the PSE type is Not set

When any row is a Watts row, a note follows: "Watts rows are the PD watts at the device, converted to the PSE watts of the class that wattage falls in (for example 25.5 W is Class 4, 30.0 W). Pick the device's class to be exact."

The verdict line under the result reads one of:

- **Budget OK**
- **Caution: within 20% of budget limit**
- **Over budget. Reduce devices or upgrade switch**
- **Does not fit.** It names the cause: "See the port checks above", "Not enough PoE ports for these devices", or "Not enough PoE ports, and see the port checks above"

Copy results gives the same lines, with Utilization as "90% of usable budget", then "Verdict:" and the short word (Budget OK, Caution, Over budget or Does not fit), then the full line beneath it.

How it works. A class row is budgeted at that class's PSE allocation, the power the switch port must supply: Class 0 15.4 W, 1 4.0 W, 2 7.0 W, 3 15.4 W, 4 30.0 W, 5 45.0 W, 6 60.0 W, 7 75.0 W, 8 90.0 W. A watts row is the device's PD draw at the device; it is converted to the PSE watts of the lowest class whose PD maximum covers it (Class 1 3.84 W, 2 6.49 W, 3 12.95 W, 4 25.5 W, 5 40 W, 6 51 W, 7 62 W, 8 71.3 W), so 25.5 W

budgets at 30.0 W. No PoE type delivers more than 71.3 W to a device, so a watts row above 71.3 W is refused: the field shows an error and there is no result until you fix it. Each row's figure is multiplied by its quantity and the rows are added for the total PSE allocation. The usable budget is the switch budget minus the reserve, and the watts left is the usable budget minus the total. The percent used is the total over the usable budget (shown capped at 100 percent). Ports needed is one per device. The port check fails a class row above the highest class the port type supports (af Class 3, at Class 4, bt Type 3 Class 6, bt Type 4 Class 8), and a watts row above what that port type delivers at the device. The verdict, checked in this order: if the total is over the usable budget it says Over budget; if there are not enough ports or any row fails the port check it says Does not fit; otherwise if you are past 80 percent of the usable budget it says Caution; otherwise Budget OK.

Example. A 370 W, 24-port 802.3at switch with a 10 percent reserve that you set keeps 37 W back, leaving 333 W usable. Ten Class 4 access points are budgeted at 30.0 W each, 300 W in all, so 33 W is left. 300 ÷ 333 is 90.09 percent, which the screen shows as 90% of usable. The result reads Total PSE allocation 300.0 W, Reserve 10% (37.0 W), Left after reserve 33.0 W remaining, Utilization 90% of usable, Ports 10 needed of 24 available, Port check: every device fits an 802.3at port, and the verdict Caution: within 20% of budget limit. Change one row to a Class 6 access point and the port check fails, because an 802.3at port gives up to Class 4.

Field notes

- Use the device's PoE class from its datasheet when you have it. The class is what the switch allocates for, whatever the device draws at idle.
- Budget in PSE watts, not PD watts. PSE is what the switch port must supply; PD is what survives the cable and reaches the device. Twelve Class-4 APs are 306 W of PD but 360 W of PSE: against a 370 W switch that is the difference between comfortable and nearly exhausted. Both kinds of row are budgeted in PSE watts: a watts row is converted from the PD figure you type to the PSE watts of the class that wattage falls in, and the screen says so. Picking the class from the datasheet is still the exact way.
- 802.3bt Type 4 supplies 90 W (IEEE 802.3-2022, Table 145-11, Class 8 P_Class). The 99.9 W figure sometimes quoted is the PSE power rating band (P_Type, Table 145-16), which is a different quantity.
- The reserve is budget you keep back for growth and for switches that hold back some power themselves. It starts at 0, which compares against the full budget; type a percent to hold some back. A reserve outside 0 to 99.9 blanks the result and the screen asks for one from 0 to 99.9%.
- The percent shown stops at 100 for display, but the Over budget verdict triggers on the real negative watts left, so an over-budget case is never hidden.

- The port check uses one PSE type for the whole switch. It does not model a switch with mixed port types, or power lost over long cable runs.

Shannon Capacity

Works out the theoretical maximum data rate a channel can carry, from its bandwidth and signal-to-noise ratio, and in reverse the SNR a target rate would demand.

Why it's here. Every Wi-Fi generation pushes against this ceiling and none of them breaks it. Bandwidth and signal-to-noise decide the maximum, and everything a vendor sells you is an attempt to get closer to it. Put this next to Noise Floor and the Throughput Calculator and you can see how much of the theoretical maximum your link is actually using.

How to use

1. Choose whether to solve for capacity or for the required SNR.
2. Pick the channel bandwidth.
3. Enter the SNR in dB, or in reverse the capacity you are aiming for in Mbps.
4. Set the number of spatial streams, because the bound applies to each stream separately.
5. Read the capacity, the per-stream figure, and the spectral efficiency in bits per second per Hz.

Inputs

Input	Unit	Range
Bandwidth	MHz	Choose from 20, 40, 80, 160, or 320 MHz
SNR	dB	Any value, including negative. Wi-Fi really does operate near and below 0 dB at the lowest data rates

Input	Unit	Range
Target capacity	Mbps	Greater than zero (required SNR mode only)
Spatial streams	NSS	1 to 8

How it works. Capacity equals the number of spatial streams times the bandwidth times the base-2 logarithm of one plus the signal-to-noise ratio, with that ratio as a linear power ratio rather than dB. Reversing it gives the spectral efficiency a target rate needs, and from that the SNR it demands.

Example. At 80 MHz and 25 dB SNR on one spatial stream the ceiling is about 664.8 Mbps, or 8.31 bits per second per Hz. Asked the other way, 1,000 Mbps on one 80 MHz stream needs 12.5 bits per second per Hz, which needs about 37.6 dB. That is why a gigabit demo only works standing under the access point.

Field notes

- This is a single-channel bound, and Wi-Fi is MIMO. A radio advertising 1,200 Mbps is not beating Shannon: that rate is the sum of parallel spatial streams, each sitting under its own bound. That is why spatial streams are an input here.
- The ratio is linear, not dB. Feeding 25 straight into the formula instead of 316 is the most common mistake with this equation, so the screen shows the conversion.
- An impossible target is not an error. Asking for 10 Gbps on one 20 MHz stream honestly needs about 1505 dB, and printing that number is the point.
- The ceiling assumes perfect coding and no interference, so a real link always lands below it. The gap between the two is the useful part.

SSID Airtime

Works out how much of a channel's time is spent on beacons and probe responses, and what one MORE SSID would cost, across 2.4, 5 and 6 GHz side by side.

Why it's here. Best-practice documents routinely ask for three or four SSIDs without ever costing them. This puts a number on the request, so the conversation stops being a matter of opinion. Airtime is a property of the CHANNEL, not of one access point, so every audible co-channel AP multiplies the figure.

How to use

1. Enter how many SSIDs one access point advertises, and how many co-channel APs you can hear (including that one).
2. Choose whether the APs beacon once per SSID, or use MBSSID to carry several BSSs in one beacon.
3. Set the management basic rate for each band. This is the single biggest lever: a beacon at 6 Mbps takes twice the airtime of the same beacon at 12.
4. Read the large number on each band card: that is what advertising ONE more SSID would add.

Inputs

Input	Unit	Range
SSIDs advertised	count	1 to 64, per access point
Audible co-channel APs	count	1 to 40, including this one
Beaconing	toggle	One beacon per SSID, or MBSSID
Basic rate	Mbps	1 to 54. CCK rates are hidden on 5 and 6 GHz, which have no CCK
Beacon length	bytes	60 to 2300, INCLUDING the 4-byte FCS
Beacon interval	TU	20 to 1000. One TU is 1024 microseconds, not 1000
Amendment	select	802.11ac, 802.11ax (default), or 802.11be. Sets the beacon size estimate

Input	Unit	Range
Airtime counted	toggle	Occupancy (default), or with contention added
Probe exchanges	toggle	Not counted (default), or counted

How it works. Each beacon's time on air is the preamble plus the payload rounded UP to whole 4-microsecond OFDM symbols. Without MBSSID an AP sends one full beacon per SSID, so cost rises linearly with SSID count. With MBSSID it sends one beacon carrying a shared part plus a small profile per extra BSS, which is far cheaper and gets relatively cheaper as the count grows. That total is divided by the beacon interval and multiplied by the number of audible co-channel APs, because they all share the same channel time. Occupancy mode counts only the frames themselves; contention mode adds DIFS plus average backoff, which is the time an AP spends winning the medium.

Example. Four SSIDs, three audible co-channel APs, 331-byte beacons at a 12 Mbps basic rate, 100 TU interval, one beacon per SSID: the next SSID you add costs roughly another third of a percentage point of channel time on 5 GHz. Drop the basic rate to 6 Mbps and that cost doubles.

Field notes

- MANAGEMENT FRAMES ONLY. No data traffic, no RTS/CTS or ERP protection, no retries, no DTIM or buffered multicast. Real channel utilisation is always HIGHER than this figure, never lower.
- Beacon length must INCLUDE the 4-byte FCS. Analysers commonly display the length without it; a 327-byte display means 331 here. Getting this wrong under-reports by about 2%.
- Occupancy is the default on purpose. It makes no assumption about contention, so it is the figure that survives an argument.
- The same SSID count lands far harder on 2.4 GHz, because there are only three non-overlapping channels to spread across. A ship network measured in August 2026 ran 2.4 GHz at 2.27 times the channel utilisation of its own 5 GHz, with identical SSIDs.
- Beacon byte sizes per amendment are ESTIMATES. Replace them with a real length from a capture whenever you have one.

Throughput Calculator

Computes the PHY rate and an estimated real throughput for a Wi-Fi connection from standard, channel width, MCS index, spatial streams, and guard interval.

Why it's here. Translates a client's negotiated rate parameters into expected data-rate numbers, and sets realistic expectations for "what speed should I see."

How to use

1. Pick the Wi-Fi standard (802.11n/ac/ax/be).
2. Pick channel width, MCS index, spatial streams, and guard interval (the options reclamp to what's valid for the chosen standard).
3. Read modulation, PHY rate, and estimated real throughput in Mbps.

Inputs

Input	Unit	Range
Standard	select	HT (802.11n), VHT (802.11ac), HE (802.11ax, default), EHT (802.11be)
Channel width	MHz	HT {20,40}; VHT/HE {20,40,80,160}; EHT {20,40,80,160,320}
MCS index	index	max per standard: HT 7, VHT 9, HE 11, EHT 13
Spatial streams	count	up to: HT 4, VHT/HE/EHT 8
Guard interval	µs	HT/VHT {0.4, 0.8}; HE/EHT {0.8, 1.6, 3.2}

How it works. The PHY rate in Mbps is the number of data subcarriers times the bits carried per symbol times the number of spatial streams, all divided by the OFDM symbol duration in microseconds. The data-subcarrier count depends on the standard and channel width (for example,

802.11ax at 80 MHz uses 980 subcarriers). Bits per symbol comes from the MCS index: its modulation and coding rate (for example, MCS 11 carries about 8.3333 bits per symbol). The symbol duration depends on the standard and the chosen guard interval (for example, 802.11ax at a 0.8 µs guard interval is 13.6 µs). Estimated real throughput is the PHY rate times a flat per-standard efficiency factor: 0.70 for 802.11n, 0.72 for 802.11ac, 0.76 for 802.11ax, and 0.80 for 802.11be. Results round to one decimal, and an invalid width/guard-interval combination or an out-of-range MCS returns a blank.

Example. HE, 80 MHz, MCS 11, 2 streams, GI 0.8 → PHY = (980 · 8.3333 · 2) / 13.6 = 1201.0 Mbps; real = 1201.0 · 0.76 = 912.7 Mbps.

Field notes

- The efficiency factors (0.70 to 0.80) are flat per-standard estimates of MAC/overhead, not measured for your environment; real throughput depends on contention, retries, frame aggregation, and airtime sharing.
- PHY rate is the theoretical peak for a single, clean link. Treat the real-throughput number as an optimistic ceiling.

IP & Addressing (4)

IP Subnet Calculator (IPv4)

Two modes of IPv4 mask arithmetic. Subnet works out the full breakdown (network, broadcast, netmask, wildcard, first and last usable host, total addresses, usable host count, and the same address as an integer, in hex and in binary) from an address plus a CIDR prefix or a dotted mask. Range turns a start and end address into the smallest set of CIDR blocks that covers it exactly, and turns a block back into its range.

Why it's here. The CIDR math you'd otherwise do on paper or in your head at a whiteboard. Reach for Subnet to confirm a network or broadcast boundary, a usable-host range, or a mask-to-prefix conversion. Reach for Range when a customer hands you "10.4.16.0 through 10.4.31.255" and you need blocks for an ACL, a firewall rule, or a DHCP scope.

How to use

1. Pick Subnet or Range with the control at the top.
2. Subnet: enter an IPv4 address (for example, 10.20.0.0). You can add the prefix inline as 10.20.0.0/22; an inline prefix wins and the second field is ignored.
3. Subnet: otherwise enter the prefix or mask in the second field: a CIDR prefix (22 or /22) or a dotted mask (255.255.252.0).
4. Range: enter the first and last address. Or type a whole block into the first field (10.4.16.0/20) and read its range back; the second field is then ignored and says so.
5. Everything updates live on every valid keystroke. Bad input shows a short 'Check your input' note with a specific message rather than a wrong answer.
6. Use the Copy action in the top bar to copy the current mode's result as a labeled block of text.

Inputs

Input	Unit	Range
IPv4 address (Subnet)	dotted-decimal, optionally with an inline /prefix	four numbers each 0-255; inline prefix 0-32
Prefix or mask (Subnet)	CIDR prefix or dotted mask	prefix 0-32, or a valid dotted mask (for example, 255.255.252.0); ignored when the address already carries an inline prefix
First address (Range)	dotted-decimal, or a whole CIDR block	any IPv4 address; a /prefix here derives the range and ignores the second field
Last address (Range)	dotted-decimal	any IPv4 address at or after the first

How it works. It does the standard binary subnet math. The mask keeps the leftmost bits set by your prefix. The network address is the address with the host bits cleared; the broadcast address is the network with those host bits all set; the wildcard is the inverse of the mask. Total addresses is 2 raised to the number of host bits. For prefixes up to /30, the usable hosts are the total minus 2 (you drop the network and

broadcast), the first usable host is the network plus 1, and the last is the broadcast minus 1. A /31 is a special point-to-point case: there's no network or broadcast reservation, so both addresses are usable hosts. A /32 is a single-host route: one address, no range, no broadcast. A dotted mask is only accepted if it's a valid run of 1 bits with no gaps, and an address must be exactly four numbers, each 0 to 255. The number forms are the same 32 bits written differently: the integer is the plain unsigned value, the hex is those four bytes, and the binary lines show all 32 bits with a slash sitting exactly where the prefix ends. The slash replaces the octet dot when the two land in the same place, so there is only ever one separator at any position. The address shown in binary is the one you typed, not the network base, because seeing your host address against the boundary is what shows which bits are host bits. Range works greedily: starting at the first address it takes the largest block that begins there, is aligned to its own size, and does not run past the last address, then repeats from the next free address. That produces the fewest blocks possible, and it is why an unaligned range needs several.

Example. 10.20.0.0/22 gives netmask 255.255.252.0, wildcard 0.0.3.255, network 10.20.0.0, broadcast 10.20.3.255, first host 10.20.0.1, last host 10.20.3.254, 1024 total addresses, and 1022 usable. (This is the example the screen opens with.). In number forms, 10.20.0.0 is 169,082,880, hex 0x0A140000, and binary 00001010.00010100.000000/00.00000000 against a mask of 11111111.11111111.111111/00.00000000. In Range mode, 10.4.16.0 through 10.4.31.255 is one block, 10.4.16.0/20, while 192.168.1.1 through 192.168.1.6 takes four: 192.168.1.1/32, 192.168.1.2/31, 192.168.1.4/31 and 192.168.1.6/32.

Field notes

- A /31 (RFC 3021) is a point-to-point link: there's no network or broadcast to set aside, so both addresses are usable (usable = 2). The screen calls this out.
- A /32 is a single-host route: one address, no range, no broadcast (usable = 1).
- A host address inside the block reports the subnet's network, not the host itself. Every value comes from the masked base.
- A dotted mask has to be a valid contiguous run of 1 bits; something like 255.0.255.0 is rejected as invalid.
- No network needed, just math, so it runs on every platform including the web.
- A range only collapses to a single block when it starts on a CIDR boundary and is a whole power of two long. Most real ranges do not, so several blocks is the correct answer and not a failure. The screen never rounds up to one block, because that would claim addresses you did not ask for.

- The binary lines are where masking stops being a rule you memorized. Put the address and the mask side by side and the boundary is visible, including mid-octet prefixes like /22 that dotted decimal hides completely.
- The integer form is what shows up in database columns, ACL generators, and some log formats. Nothing else in the app produced it.

IP Subnet Calculator (IPv6)

Enter an IPv6 address and prefix length and get back the expanded and compressed forms, the network address, the first and last address in the prefix, the number of hosts, and the address type. It also decodes transition addresses: whether the address you typed carries an IPv4 address inside it, and how a given IPv4 address is written the four ways IPv6 can carry one.

Why it's here. IPv6 subnetting and address-type lookups during design and troubleshooting, where doing 128-bit math by hand invites mistakes. The transition section answers a different question: NAT64, 6to4 and IPv4-mapped addresses genuinely turn up in captures and firewall logs, and "what am I looking at" is the moment they cause.

How to use

1. Enter an IPv6 address (default 2001:db8::1).
2. Enter the prefix length (default 32). A link-local address pasted straight off ifconfig or ip -6 addr works as-is, zone index and all: fe80::1%en0 is accepted, and the zone comes back on its own row.
3. Read the expanded form, compressed form, network and prefix, first and last address, host count, and address type.
4. Read the transition section at the bottom. It decodes the same address you typed above and names the format, the embedded IPv4, and what that IPv4 actually is.
5. The IPv4 field in that section runs the other direction: type an IPv4 address and read it as IPv4-mapped, NAT64, a 6to4 prefix, and the deprecated IPv4-compatible form.
6. Read the solicited-node multicast address and the reverse DNS name at the bottom. Both are derived from the address you already typed.

Inputs

Input	Unit	Range
IPv6 address	any valid IPv6 address	only one :: shortcut allowed; an invalid format shows an inline error
Prefix	whole number	0 to 128 (anything outside that shows an error)
Zone index (optional, on the address)	interface name or ifindex after a %	fe80::1%en0, fe80::1%12, or the %25en0 spelling used inside a URL
IPv4 address (transition section)	dotted-decimal	four octets, each 0 to 255

How it works. It expands the address to its full form, works out which bits the prefix fixes, and from that gives you the network address plus the first and last address in the block. It also shows the shortest valid compressed form by collapsing the longest run of zero groups. Host count is 2 raised to the number of host bits (128 minus the prefix); anything above 2 to the 63rd is reported as "More than 2^63" rather than a full number. The address type is identified by matching the address against the well-known IPv6 ranges (unspecified, loopback, link-local, unique-local, multicast, 6to4, IPv4-mapped, documentation, or global unicast). Transition decoding matches the address against five known formats and reports the first that fits: IPv4-mapped (::ffff:0:0/96, RFC 4291), the NAT64 well-known prefix (64:ff9b::/96, RFC 6052), 6to4 (2002::/16, RFC 3056, where the 32 bits after 2002 are the site endpoint), Teredo (2001:0::/32, RFC 4380, where the client address and port are stored with every bit inverted and are shown flipped back), and the deprecated IPv4-compatible form (::/96, RFC 4291), which excludes :: and ::1 because those are not IPv4 at all. Writing an IPv4 address as IPv6 is the same work in reverse.

Example. 2001:db8::1 with a /64 gives expanded form 2001:0db8:0000:0000:0000:0000:0000:0001, compressed form 2001:db8::1, network 2001:db8::/64, 64 host bits (so the host count shows "More than 2^63"), and type Documentation (2001:db8::/32).

Field notes

- Host counts above 2 to the 63rd are shown as "More than 2^63" rather than a full number.
- The "first address" is the network address itself. IPv6 has no broadcast address and does not reserve the all-zeros host the way IPv4 does.
- Address-type detection covers the common well-known ranges, not every reserved block.
- This tool lives under Networking Tools in the current menu.

- An address literal with a dotted IPv4 tail, such as ::ffff:192.168.1.1, is accepted. That form is legal IPv6 and is how a mapped or NAT64 address is written in a log.
- NAT64 is decoded against the well-known 64:ff9b::/96 prefix only. A network can run its own NAT64 prefix at /32, /40, /48, /56 or /64, and the IPv4 bits sit at a different offset in each. That prefix length is not carried in the address, so decoding one without being told which prefix is in use would be guessing. The screen says so rather than guessing.
- Teredo stores the client address and port with every bit inverted. Read raw, they look like nonsense. The screen flips them back and says it did.
- An address with no IPv4 inside is reported as having none. The screen never invents an embedded address to fill the row.
- IPv4-compatible (::a.b.c.d) is deprecated. It is decoded so you can recognize one in an old configuration, and labeled so you do not build on it.
- The 6to4 output is a PREFIX for a whole site (a /48), not a host address.
- A zone index (the %en0 on fe80::1%en0) names the local interface the scope belongs to, so it sits outside the 128 bits and is set aside before the math runs. The Zone row shows it back, because a character you typed that vanishes with no acknowledgement looks like an input the tool failed to read. A half-typed zone such as fe80::1% is reported as an invalid address.
- The solicited-node multicast address (RFC 4291 section 2.7.1) is FF02::1:FF plus the low-order 24 bits of the address. IPv6 neighbor discovery sends its solicitation to that group rather than to all nodes, which is how it avoids waking every device on the link the way ARP broadcast does. When address resolution is failing, whether the node joined the right group is the question, and working the address out by hand is where it goes wrong.
- Only the low 24 bits are used, so two addresses in completely different prefixes can share a solicited-node group. It is a useful filter, not a unique identifier.
- The reverse DNS name (RFC 3596 section 2.5) is all 32 nibbles in reverse order with the ip6.arpa suffix. No trailing dot is shown, because this is a screen rather than a zone file; add one if you are pasting it somewhere that needs a fully qualified name.

MTU & MSS Calculator

Works out the TCP maximum segment size that survives a given link MTU once IP, TCP and any tunnel headers are paid, and in reverse works out what MTU a path really has from an MSS you have seen working.

Why it's here. This is the tool for the fault where a tunnel comes up, small things work, and big transfers stall. Something in the path is taking bytes the endpoints never accounted for, and the arithmetic here names how many. The reverse direction is the one that earns its place: a packet capture shows you the MSS and never tells you what ate the difference, and a gap of 8 bytes is PPPoE, 24 is GRE, and 60 is WireGuard.

How to use

1. Choose whether to go from an MTU to an MSS, or from an observed MSS back to the MTU it implies.
2. Enter the number in bytes.
3. Pick IPv4 or IPv6, because the IP header is 20 bytes on one and 40 on the other.
4. Turn on TCP timestamps if the connection uses them, which costs another 12 bytes and is the overhead people most often forget.
5. Tap any encapsulation that sits in the path. They stack.
6. Read the result, and in reverse mode read how many bytes are still unaccounted for against a standard 1500-byte Ethernet MTU.

Inputs

Input	Unit	Range
Link MTU	bytes	Any whole number. 1500 is standard Ethernet, 1492 is typical of PPPoE, 9000 is a jumbo frame
Observed MSS	bytes	Any whole number, taken from a capture or from the SYN
IP version	-	IPv4 for a 20-byte header, IPv6 for 40
TCP timestamps	bytes	Off, or on for 12 more bytes
Encapsulation	bytes	Any combination of PPPoE, GRE, WireGuard, VXLAN, IPsec ESP, L2TP, or an 802.1Q tag

How it works. MSS equals the link MTU minus the tunnel overhead, minus the IP header, minus the TCP header, minus the TCP timestamp option if it is in use. Going the other way, the path MTU equals the observed MSS plus all of the same. The IP header is 20 bytes for IPv4 and 40

for IPv6, the TCP header is 20, and the timestamp option costs 12 once padding is counted. Where an overhead is genuinely variable, as IPsec ESP is, the calculator sizes to the high end of the range rather than inventing a single number, because sizing to the low end produces a figure that works until the day the cipher or the padding changes.

Example. A standard 1500-byte Ethernet link over IPv4 leaves an MSS of 1460, which is the number most people already know. The same link over IPv6 leaves 1440. Add PPPoE and it becomes 1452, which is why so many DSL connections run a 1492 MTU. Add WireGuard over IPv4 and it becomes 1400. Going the other way, an MSS of 1452 seen working on a link you believed was standard Ethernet implies a path MTU of 1492, and the 8 missing bytes are the signature of PPPoE.

Field notes

- The 802.11 header is NOT subtracted from MTU, and this is the mistake a Wi-Fi engineer is most likely to make precisely because they know that header well. Wi-Fi's MAC header is larger than Ethernet's, but it sits below the IP layer, so it does not come out of the 1500-byte IP MTU. An 802.11 MSDU is 2304 bytes exactly so a 1500-byte IP packet fits with room over. Wi-Fi does not shrink your MTU; a tunnel running over it does.
- An 802.1Q VLAN tag is on the list for the same reason and also costs nothing: the tag extends the Ethernet header rather than the payload, and compliant switches carry a 1522-byte frame so the MTU stays 1500. It only bites on hardware that cannot handle baby giants.
- Where an overhead is genuinely variable, as IPsec ESP is, the calculator sizes to the high end of its range rather than inventing a single number. Sizing to the low end produces a figure that works until the day the cipher or the padding changes.
- This is arithmetic, not a measurement. It tells you what a stack you describe would cost; it does not probe the path. If several layers could account for the same byte count, the tool names them as candidates and stops short of choosing one, because a capture cannot tell them apart by size alone.

Subnet Planner (VLSM)

Two jobs on IPv4 blocks that span more than one network. **Split** carves a parent block into right-sized subnets from a list of host counts. **Summarize** takes a list of networks and returns the single block that covers them all, plus the smallest set of blocks that covers them exactly.

Why it's here. "Here is your /22 for the site, now give me a VLAN for staff, one for guests, one for the IoT gear and a point-to-point link" is a design question the single-subnet calculator cannot answer, because it works on one network at a time. Summarize is the same math in reverse, for when you are writing a route, an ACL, or a DHCP scope and want one line instead of nine.

How to use

1. Pick Split or Summarize with the segmented control at the top.
2. Split: type the block you are carving (10.20.0.0/22), then list the subnets you need, one per line, as a name and a host count ("Staff 500") or just a count ("500").
3. Summarize: paste your networks, one per line. A prefix, a dotted mask, or a bare address all work.
4. Everything recalculates as you type. Copy takes the whole plan as a labeled text block.

Inputs

Input	Unit	Range
Block to carve up (Split)	CIDR, or address and dotted mask	any IPv4 block, /0 to /32
Subnets you need (Split)	one per line: name then host count, or just the count	1 host or more per line; blank lines and # comments are skipped
Networks to summarize	one per line: CIDR, address and mask, or a bare address read as /32	any IPv4 networks, in any order; overlaps and duplicates are fine

Formula or method. Split sorts the requirements largest-first, then places each block at the next free address. Because every block size is a power of two and the list runs largest to smallest, each block lands on its own boundary with no padding, which is what makes VLSM efficient and why the output is not in the order you typed. The prefix for a host count uses the classic reservation (network + broadcast), so 500 hosts gets a /23 with 510 usable and 2 hosts gets a /30; a request for 1 gets a /32. Leftover space is expressed as the largest aligned CIDR blocks that cover it exactly. Summarize sorts the inputs, merges them into continuous ranges (so an overlap counts once and two touching networks become

one range), then reports two different answers: the smallest single block containing all of them, found by walking the prefix down until the lowest and highest addresses agree under the mask; and the minimal set of aligned CIDR blocks that covers the merged ranges exactly. The difference between those two is reported as an address count and enumerated as the gap blocks.

Example. Split 10.20.0.0/22 for Staff 500, Guest 200, IoT 100 and a 2-host point-to-point link gives 10.20.0.0/23, 10.20.2.0/24, 10.20.3.0/25 and 10.20.3.128/30, with 124 addresses still free in five blocks (10.20.3.132/30, .136/29, .144/28, .160/27, .192/26). Summarize 10.0.0.0/24 and 10.0.3.0/24 and the covering supernet is 10.0.0.0/22, which also claims 512 addresses that are not in your list: 10.0.1.0/24 and 10.0.2.0/24.

Field notes

- The Extra count in Summarize is the number that matters. A covering supernet almost always includes addresses you did not ask for, and advertising it pulls in traffic for networks you may not own. The screen names those gaps as blocks so you can see exactly what you would be claiming.
- A subnet that does not fit is named and told why, and the ones that do fit are still placed. A plan where the last VLAN overflowed is more useful than no plan at all.
- A 2-host request gets a /30, because that works everywhere. If your gear supports RFC 3021, a /31 carries the same two hosts on a point-to-point link in half the space, and the screen says so.
- A 1-host request gets a /32: a single host route, with no room for a gateway. Ask for 2 if you meant a device and its router.
- A bad line is skipped by line number rather than rejecting the whole paste, so a messy list still produces a plan.
- A network typed with host bits set (10.0.0.37/24) is read as 10.0.0.0/24, and the screen says it did that rather than quietly answering a different question.
- No network I/O, just integer math, so it runs on every platform including web.

Coordinates & GPS (4)

Distance and Bearing

Computes the great-circle distance and the initial (forward) and reverse bearings between two latitude/longitude points.

Why it's here. Aiming point-to-point antennas. You need the distance (for FSPL) and the compass bearing (to point the dish) between two sites.

How to use

1. Enter latitude and longitude for point 1 and point 2 (decimal degrees).
2. Read distance (km), forward bearing, and reverse bearing.

Inputs

Input	Unit	Range
Point 1 / Point 2 latitude and longitude	decimal degrees	-

How it works. It treats the Earth as a sphere with a mean radius of 6371 km and uses the haversine method for distance. In plain terms: it measures the angular gap between the two points from their differences in latitude and longitude, then multiplies that angle by the Earth's radius to get the distance along the surface. The forward bearing is the compass heading you'd start out on to follow the shortest (great-circle) path from point 1 to point 2, and the reverse bearing is simply that heading plus 180°.

Example. San Francisco (37.7749, −122.4194) to Los Angeles (34.0522, −118.2437) → distance = 559.12 km; forward bearing = 136.5°; reverse bearing = 316.5°.

Field notes

- Spherical earth model (6371 km mean radius), accurate to a fraction of a percent for terrestrial links, not for survey-grade geodesy.
- The forward bearing is the initial bearing of the great-circle path; it changes along the route on long paths.
- The reverse bearing is the simple +180°, which is exact only on a sphere.

Final Point

Computes the destination latitude/longitude given a starting point, an initial bearing, and a distance (the "direct" geodesic problem).

Why it's here. Projecting where a link points. Given a site, a heading, and a distance, find the coordinates of the far end (e.g. to check a candidate tower location).

How to use

1. Enter the start latitude and longitude.
2. Enter the initial bearing in degrees.
3. Enter the distance and pick its unit (km, mi, or m).
4. Read the destination latitude and longitude.

Inputs

Input	Unit	Range
Start latitude / longitude	decimal degrees	-
Bearing	degrees	-
Distance	km, mi, or m	mi × 1.60934 → km; m ÷ 1000 → km

How it works. It treats the Earth as a sphere with a mean radius of 6371 km. First it converts your distance into an angle by dividing it by the Earth's radius. Then, starting from your point and heading off on the bearing you gave, it walks that angular distance along a great circle and reports where you land. The result longitude is wrapped to the −180° to +180° range so it always reads as a normal coordinate.

Example. From (40, −105), bearing 90° (due east), distance 100 km → destination (39.99, −103.83), about 1.17° of longitude east at that latitude (latitude dips marginally because a constant-90° initial bearing follows a great circle, not a parallel).

Field notes

- Direct great-circle solution on a sphere; the bearing is the initial heading and the path curves on long distances.
- Same spherical-model accuracy caveats as the other geographic tools.

Lat / Long Conversion

Converts a coordinate between decimal degrees (DD), degrees-decimal-minutes (DDM), and degrees-minutes-seconds (DMS).

Why it's here. Different tools and maps use different coordinate notations; this normalizes between them when entering a site location.

How to use

1. Enter a latitude value (and/or longitude) in decimal degrees, or tap "Use my location" to fill the fields from the device's GPS fix (grant Location if prompted).
2. Read the DD, DDM, and DMS forms with the correct hemisphere letter.
3. When a GPS fix is used, an inline map, plus altitude and horizontal accuracy, are shown alongside the coordinate forms.

Inputs

Input	Unit	Range
Latitude	decimal degrees	valid range ±90
Longitude	decimal degrees	valid range ±180

How it works. The sign of the value sets the hemisphere: a latitude of zero or more reads N (otherwise S), and a longitude of zero or more reads E (otherwise W). To break a decimal degree into degrees, minutes, and seconds, it takes the whole-number part as the degrees, multiplies the leftover fraction by 60 to get the minutes, then multiplies that leftover fraction by 60 again to get the seconds. Degrees-decimal-minutes simply keeps the minutes and folds the seconds back in as sixtieths. The three forms are displayed to a fixed precision (DD to six decimals). Any value that is out of range or not a valid number blanks the output.

Example. 40.7128° latitude → DD 40.712800; DMS 40° 42' 46.08" N; DDM 40° 42.7680' N.

Field notes

- Validates ranges (±90 lat, ±180 lon) and blanks on out-of-range input.
- The sign drives the hemisphere letter, so enter west longitudes and south latitudes as negative.
- No datum/projection handling; these are plain WGS84-style decimal degrees in, formatted out.
- Beyond the converter, the screen carries a live-GPS surface: a permission-gated "Use my location" control, an inline map of the fix, and an altitude / horizontal-accuracy readout. This is the same device-location seam the Current Location tool uses; altitude and accuracy read "Not reported" on hardware without GPS.

Midpoint

Computes the great-circle midpoint between two latitude/longitude points.

Why it's here. Finding a relay or repeater site halfway along a long link, or the center point of a coverage area between two known locations.

How to use

1. Enter latitude and longitude for point 1 and point 2.
2. Read the midpoint latitude and longitude.

Inputs

Input	Unit	Range
Point 1 / Point 2 latitude and longitude	decimal degrees	-

How it works. Rather than averaging the two coordinates, it finds the point that sits halfway along the shortest (great-circle) path between them. It does this by combining the two positions as directions on the sphere and locating the heading that lands exactly between them, then converts that back to a latitude and longitude. The result longitude is wrapped to the −180° to +180° range so it always reads as a normal coordinate.

Example. Between (40, 0) and (40, 90) the great-circle midpoint lies well north of the latitude average, at (49.88, 45.00), reflecting that the shortest path bows poleward.

Field notes

- This is the great-circle (spherical) midpoint, not the average of the coordinates. On east-west paths the midpoint is noticeably closer to the pole than the simple lat/lon average.
- Spherical model, same accuracy caveats as Distance and Bearing.

Conversions (7)

Architectural Scale

Converts a named drawing scale to its dimensionless ratio and back, and converts a distance measured on a drawing to its real-world length or the reverse. Covers US architectural fractional-inch scales, engineer's decimal scales, and common metric scales. This is a calculator, not a reference page.

Why it's here. WLAN pros work inside plan sets and scaled PDFs they were never taught to read. Every coverage prediction, wall distance, and mounting height is wrong if the scale is wrong. This is the same calibration step Ekahau, Hamina, and iBwave ask for on import.

How to use

1. Pick the scale family (Architectural, Engineering, or Metric).
2. Pick the specific scale, for example 1/4" = 1'-0". Read its ratio (1:48).
3. In Measure, choose a direction: Drawn to Real, or Real to Drawn.
4. Set the drawing units (in / mm / cm) and real-world units (ft / m).
5. Enter your measurement and read the converted length, with a friendly feet-inches or fractional-inch form beneath it.

How it works. ratio = real / drawn. Imperial: ratio = 12 / inches-per-foot (1/8" = 1'-0" gives 12 / 0.125 = 96, so 1:96). Engineer's: feet-per-inch times 12 (1" = 20' gives 240, so 1:240). Metric scales are already ratios. Drawn to real multiplies the measurement by the ratio; real to drawn divides. The ratio is dimensionless, so the two ends can carry different units.

Example. 3.5" measured on a 1/8" = 1'-0" (1:96) sheet is 3.5 x 96 = 336 inches = 28 ft. In reverse, a 45 ft hallway at 1/4" = 1'-0" (1:48) draws at 45 / 48 = 0.9375 ft = 11-1/4 inches.

Field notes

- The result is only as good as the scale. If a PDF has no embedded scale, confirm which scale it is by measuring a known dimension, a 3'-0" door or a 2x4 ft ceiling tile.
- Printed and PDF plans are often not to true scale (fit-to-page printing, cropped sheets). Trust a dimensioned callout on the sheet over any measurement.
- US-primary: architectural scales use fractional inches, engineer's scales use decimal feet per inch. Metric ratios (1:50, 1:100) are provided for ISO drawing sets.
- Data source: pure on-device math, no network.

Channel / Frequency

Convert a Wi-Fi channel number to its center frequency and back, across 2.4 / 5 / 6 GHz, with the bonded-channel component lists.

Why it's here. Specs, radios, and regulators talk in both channels and megahertz. This converts either direction and shows which 20 MHz channels make up a wide channel, so a "channel 100, 80 MHz wide" entry on a controller stops being a guess.

How to use

1. Pick a band (band is required for channel→frequency because channel numbers collide across 5 and 6 GHz, channel 36 exists in both).
2. Enter a channel to get the center frequency, or a frequency to get the channel and the band it falls in (the frequency ranges are disjoint, so the band comes back for free).
3. Read the bonded-channel breakdown for 40/80/160/320 MHz widths.

How it works. One linear rule covers almost every channel: center frequency in MHz = the band's starting frequency + 5 × the channel number, where the starting frequency is 2407 (2.4 GHz), 5000 (5 GHz), or 5950 (6 GHz). Two special cases bypass it: 2.4 GHz channel 14 = 2484 MHz, and 6 GHz channel 2 = 5935 MHz. The 6 GHz start of 5950 is the verified value (channel 1 = 5955), not the 5940+5n some bad sources use. Channel↔frequency is universal physics, identical everywhere; channel availability (US 1–11 vs world 1–13, DFS, UNII-4, 6 GHz adoption) is regulatory and is surfaced as a caveat, never as a reason to change a computed frequency.

Field notes

- The selectable 20 MHz primaries are the exact allowed sets, not a naive arithmetic range: 2.4 GHz is 1–14; 5 GHz is the verified UNII-1/2A/2C/3/4 list (anything like 37, 49, 51, 145, 181 is rejected); 6 GHz is the sequence 1, 5, 9 … 233 plus the special channel 2.
- DFS channels, UNII-4, and the 6 GHz Preferred Scanning Channels are flagged as caveats, not enforced. Channel 2 (6 GHz) is flagged "Special (5935 MHz, reserved/guard)" and channel 14 (2.4 GHz) "Special (2484 MHz)".
- Channel data is drawn only from primary sources: IEEE 802.11-2020/2024 channelization plus the FCC 6 GHz / 5.9 GHz Reports & Orders. No channel data comes from a secondary source.

dBm / Watt Converter

Live two-way conversion between dBm, Watts, and milliwatts. Type in any one field and the other two update in real time.

Why it's here. RF power lands in your lap in three units depending on the spec sheet, the regulator, or the radio. This converts between them so you don't reach for a calculator at the AP.

How to use

1. Type a value in any of the three fields (dBm, Watts, or Milliwatts); the other two recompute instantly.
2. dBm and Watts accept scientific notation (e.g. 1e-10) plus a sign; Milliwatts is unsigned decimal.
3. Watts renders in scientific notation (5 sig figs) because real Wi-Fi receive levels are tiny (e.g. 1e-10 W); mW shows 4 fixed decimals.
4. Entering Watts or mW that are zero or negative shows a dash in the dBm field, since the logarithm of a non-positive number is undefined.

Inputs

Input	Unit	Range
dBm	dBm (decibel-milliwatts)	any real number; signed + scientific notation accepted
Watts	W	> 0 to convert to dBm; ≤ 0 yields a dash; signed + scientific notation accepted
Milliwatts	mW	> 0 to convert to dBm; ≤ 0 yields a dash; unsigned decimal

How it works. Three conversions do the work: dBm→W = 10^(dBm/10) ÷ 1000; dBm→mW = 10^(dBm/10); and W→dBm = 10 × log10(W × 1000). Converting mW→dBm just divides mW by 1000 to get Watts and reuses the W→dBm math. The factor of 1000 is the Watt-to-milliwatt relationship (1 W = 1000 mW). In plain form, dBm = 10 × log10(mW) and W = 10^(dBm/10) ÷ 1000. Watts display in scientific notation to 5 significant figures; mW shows 4 decimals and dBm shows 2. A zero or negative Watts or mW yields a dash in the dBm field because the logarithm is undefined there.

Example. 0 dBm → 10^(0/10) = 1 mW = 1e-3 W. +30 dBm → 10^(30/10) = 1000 mW = 1 W. 20 mW → 10 · log10(20) = 13.01 dBm.

Field notes

- 0 dBm is exactly 1 mW by definition, the reference point of the scale.
- The on-screen reference card anchors common values: +30 dBm = 1000 mW (1 W, FCC 2.4 GHz max conducted), +13 dBm = 20 mW (common default AP Tx power), 0 dBm = 1 mW, -67 dBm = 0.2 nW (minimum for enterprise data), -80 dBm = 10 pW (typical ambient noise floor).
- Watts uses scientific notation deliberately: Wi-Fi receive power sits around 1e-10 W, which fixed notation renders as an unreadable string of zeros.

Hex / ASCII

A live decimal/hexadecimal/binary integer converter with a fourth single-character ASCII field, plus a printable-ASCII reference table (codes 32 to 126).

Why it's here. Reading packet captures, MAC/OUI bytes, and config hex values. Flip a number between bases or look up an ASCII character without leaving the app.

How to use

1. Type a value into any of the three numeric converter fields (decimal, hexadecimal, binary); the other two update live.
2. Type a single printable character into the fourth (Character) field to round-trip it to its code point in the other three; a character outside the printable range (32 to 126) blanks it honestly rather than guessing.
3. Scroll the ASCII table to look up a printable character's decimal/hex/binary code.

Inputs

Input	Unit	Range
Decimal	digits only	-
Hexadecimal	0-9, a-f, A-F (optional 0x prefix stripped)	-
Binary	0/1 (optional 0b prefix stripped)	Unsigned-integer domain, backed by arbitrary-precision integers so arbitrarily long strings never overflow; a blank/invalid field blanks the mirrors

How it works. Each value is parsed as an arbitrary-precision unsigned integer and re-expressed in base 16, base 2, and base 10. The ASCII table is generated from the character codes rather than hand-transcribed: hex is shown as 2-digit uppercase, binary as 8-bit zero-padded.

Example. Enter decimal 65 → hex 41, binary 1000001. In the table, decimal 65 = char A = hex 41 = binary 01000001.

Field notes

- Unsigned integers only: no negative numbers, no fractions, no two's-complement.

- The table covers printable ASCII (32 to 126) only; control characters and extended/Unicode code points are out of scope.
- The converter is base conversion, not text encoding (it converts a single integer, not an ASCII string to its byte sequence).

Metric Conversion

Converts a length between meters, kilometers, miles, feet, centimeters, inches, and nautical miles.

Why it's here. Field work mixes metric and imperial constantly (datasheets in meters, tape measures in feet, link distances in miles). A quick unit converter avoids arithmetic slips.

How to use

1. Enter a value and pick its "from" unit and "to" unit.
2. Read the single converted result (the tool pivots through meters).

Inputs

Input	Unit	Range
Value, from-unit, to-unit	one of m, km, mi, ft, cm, in, nmi	-

How it works. Every unit is defined by its length in meters: m = 1, km = 1000, mi = 1609.344, ft = 0.3048, cm = 0.01, in = 0.0254, nmi = 1852. To convert, multiply the value by the "from" unit's meters, then divide by the "to" unit's meters. Display decimals vary by unit: m 4, km 6, mi 6, ft 4, cm 2, in 4, nmi 6.

Example. 1 mi (from) → km (to) reads 1.609344; switch the "to" unit to feet and the same mile reads 5280.0.

Field notes

- Uses the international mile (1609.344 m) and international foot (0.3048 m), not the US survey foot.
- Conversions are exact within floating point; the per-unit decimal rounding is for display only.

Transfer Time

Divides a file size by a link speed. Pick which of the three you want out: how long a transfer takes, how fast a link has to be to finish in a given window, or how much data moves in that window.

Why it's here. "How long will this AP firmware push take over this link" is a field question, and so is "we have a ten-minute maintenance window, what speed do we need". The arithmetic is one division. The trap is that link speeds are quoted in bits and file sizes in bytes, and the factor of 8 between them is where the wrong answers come from.

How to use

1. Pick Time, Speed or Size at the top: that is the one you want out, and the other two are the ones you type.
2. Type each value and pick its unit from the list. Size units are byte units; speed units are bit units unless the label says B/s.
3. Read the answer, then read the two lines underneath it that restate what you typed in bits. That is the conversion, shown rather than hidden.

Inputs

Input	Unit	Range
Size	B, KB, MB, GB, TB (decimal) or KiB, MiB, GiB, TiB (binary)	any positive number
Speed	bps, kbps, Mbps, Gbps, Tbps, or KB/s and MB/s	above zero when solving for time
Time	ms, s, min, hr, day	above zero when solving for speed

Formula or method. One division, done entirely in bits. The size is converted to bits and the speed to bits per second using the same unit tables the Unit Converter uses, so there is one definition of a MB in the app and not two. Time is bits divided by bits per second; speed is bits divided by seconds; size is bits per second multiplied by seconds. A divisor of zero returns no answer at all rather than an infinity.

Example. 1 GB at 100 Mbps: 1,000,000,000 bytes is 8,000,000,000 bits, divided by 100,000,000 bits per second is 80 seconds, shown as 1 min 20 s. The same file quoted as 1 GiB is 1,073,741,824 bytes and takes 1 min 25.9 s. That six-second gap on a single gigabyte is why the decimal and binary units stay separate.

Field notes

- Link speeds are in bits per second and file sizes are in bytes. 1 byte = 8 bits, so a 100 Mbps link moves 12.5 MB per second, not 100. That line is printed on the screen itself, not buried in the help sheet.
- A KB is 1000 bytes and a KiB is 1024. Your operating system probably says GB and means GiB, which is why a "1 GB" download can take about 7 percent longer than you expected.
- There is no efficiency slider, on purpose. A default of 85 percent would be a made-up number dressed up as help. The answer is the time at exactly the rate entered. For a realistic number, enter a throughput you actually measured rather than the number on the label.
- Real transfers run slower than this: protocol overhead, disk speed, and everyone else on the link all take a share.
- Fully offline: one division on the device.

Unit Converter

Converts a value between units in one of eight categories: data transfer rate, data storage, length, power, metric prefix, speed, temperature, and time.

Why it's here. Field work mixes units constantly. A datasheet quotes throughput in Mbps, your test rig reports MB/s, an RF reading lands in dBm when you wanted milliwatts. Pick a category, pick the two units, read the answer.

How to use

1. Pick a category (data transfer rate, data storage, length, power, metric prefix, speed, temperature, or time).
2. Type a value, then pick the "from" unit and the "to" unit.
3. Read the converted result. It blanks until the value parses to a valid number.

Inputs

Input	Unit	Range
Category	one of eight categories	-
Value, from-unit, to-unit	the units offered in the chosen category	signed and scientific input accepted (a temperature can be negative)

How it works. Linear categories convert through a single base unit: multiply the value by the "from" unit's factor, then divide by the "to" unit's factor. Power and temperature are not linear and use their own math. Power treats dBm with the same log math as the dBm/Watt converter (W = 10^(dBm/10) ÷ 1000; dBm = 10 × log10(W × 1000)); temperature converts through Kelvin using its affine (offset-and-scale) relationships.

Example. 1 GB → 1000 MB (data storage, decimal). 100 °C → 212 °F (temperature). 1 W → 30 dBm (power).

Field notes

- Decimal and binary are kept separate. A KB is 1000 bytes; a KiB is 1024 bytes. The tool never conflates the two, so a storage answer reads true to the standard you picked.
- Bits and bytes are different units, not a display toggle. 1 byte = 8 bits, and the converter treats them that way across both the storage and the rate categories.
- Temperature and dBm are not simple multiply-by-a-factor conversions. They are handled with the correct affine and logarithmic math, so a negative temperature or a sub-milliwatt power reads correctly.

Ham Radio (2)

The pure-math amateur-radio tools. Channel/frequency physics that a Wi-Fi pro already trusts, applied to antenna dimensions and grid-square locators. The band-dependent amateur references live under Quick Reference → Ham Radio.

Antenna Length

Compute the half-wave dipole and quarter-wave vertical physical lengths from a frequency (or a wavelength), velocity-factor adjusted, shown beside the classic 468/f and 234/f rules of thumb.

Why it's here. Cutting an antenna is the one piece of RF where a wrong number means a saw cut you cannot undo. This gives the physics length and the field rule-of-thumb side by side, so you see the spread before you cut.

How to use

1. Enter a frequency in MHz (or switch to wavelength→frequency for the inverse).
2. Set the velocity factor; it defaults to 0.95 (thin bare wire).
3. Read the half-wave dipole and quarter-wave vertical lengths in meters, feet, and inches, plus the 468/234 rule-of-thumb feet.

How it works. It uses the exact speed of light, c = 299.792458 in MHz·m form, not the rounded 300. Wavelength in meters = 299.792458 ÷ frequency in MHz, and the inverse frequency in MHz = 299.792458 ÷ wavelength in meters. Physical length = electrical length × velocity factor: the half-wave dipole is (wavelength ÷ 2) × velocity factor, and the quarter-wave vertical is (wavelength ÷ 4) × velocity factor. The rules of thumb are the classic feet forms: length in feet ≈ 468 ÷ frequency in MHz for the dipole, and length in feet ≈ 234 ÷ frequency in MHz for the quarter-wave vertical. Those figures already fold in a typical end-effect/velocity factor, which is why they read a touch shorter than the bare physics half/quarter wavelength. Both are shown so you see the spread.

Field notes

- Velocity factor is the knob that matters: the 468/234 rules assume a typical wire, while the physics length × your chosen velocity factor is the honest computed value. Insulated wire, traps, and tubing all shift velocity factor, so confirm it for the element in your hand.
- Worked sanity checks: 14.2 MHz → wavelength ≈ 21.11 m, half-wave dipole ≈ 33 ft (velocity factor 0.95); 146 MHz → quarter-wave vertical ≈ 19–20 in; 2400 MHz → quarter-wave vertical ≈ 3 cm.
- Pure math, no network or platform call; the Copy action exports the full computed report as labeled text.

Maidenhead Grid Square

Encode a latitude/longitude to a Maidenhead (QTH) locator and back at 4, 6, or 8 characters, plus the great-circle distance and bearing between two grid squares.

Why it's here. Hams trade positions as grid squares, not lat/long. This crosses between the two and gives the distance-and-bearing leg, the same spherical math the Distance & Bearing tool uses, so cross-tool results agree.

How to use

1. Enter a lat/lon to encode (choose 4, 6, or 8 characters), or a locator to decode to its cell.
2. Decode returns the cell's south-west corner, its width/height in degrees, and the center point (what a locator→position lookup returns).
3. Enter two locators to get the great-circle distance (km and miles) and the initial bearing between their centers.

How it works. It follows the IARU locator standard: normalize longitude to 0–360 and latitude to 0–180, then subdivide each axis level by level, most-significant pair first, with per-level divisions of 18 / 10 / 24 / 10. Each pair writes the longitude character first, then the latitude character; the field is A–R, the square 0–9, the subsquare a–x, the extended square 0–9, and decoding is case-insensitive. Encoding clamps exactly +180 longitude and +90 latitude a hair inside the range so the top edge does not index past the last cell. The great-circle leg is the spherical haversine with an Earth radius of 6371 km, plus the forward-bearing formula; miles = km × 0.621371.

Field notes

- Output precision is in characters: 4 = field + square, 6 = + subsquare, 8 = + extended square. A 6-character locator is the common amateur exchange.
- Verified anchors: lon −122.4, lat 37.4 → CM87 (San Francisco; note lon −122.0 sits exactly on the CM87/CM97 square boundary and encodes as CM97, so pick a longitude inside the square); Berlin 13.4 E, 52.5 N → JO62.
- Pure math, no I/O or platform API, so it works identically on every platform offline.

Utilities & Generators (4)

DTMF Generator

Plays the Touch-Tone keypad tones (0-9, *, #, and A-D) from a standard 4×4 DTMF grid, one tap at a time or as a continuous loop.

Why it's here. Driving a system that still listens for Touch-Tones over the air or down a line: an IVR menu, a repeater controller, a legacy PBX, or a piece of test gear. Hold the device near the microphone and play the digit.

How to use

1. Tap a key to play its tone for a short burst. The selected key and its two frequencies show at the top.
2. Tap Play to loop the selected key's tone continuously; tap Stop to end it.
3. Tapping a different key during a loop retargets the loop to the new key.
4. To play a whole string, type the digits (e.g. 8675309) into the sequence field and tap Play; the app plays each digit's tone in order (about a 200 ms tone with an 80 ms gap between digits). Stop ends the sequence.

Inputs

Input	Unit	Range
Keypad key	one of 1-9, 0, *, #, A-D	defaults to the center key, 5

How it works. Each DTMF key is the sum of two sine waves, one low-group frequency (the row) and one high-group frequency (the column), per ITU-T Q.23. Low group: 697, 770, 852, 941 Hz. High group: 1209, 1336, 1477, 1633 Hz. A single key tap plays the pair for about 200 ms. The tone is synthesized on the device; nothing is dialed or transmitted by the app itself.

Example. Key 5 is 770 Hz + 1336 Hz. Key 1 is 697 Hz + 1209 Hz. The * key is 941 Hz + 1209 Hz.

Field notes

- Every key is two tones added together, which is what "Dual-Tone Multi-Frequency" means. That is why the readout shows two frequencies, not one.
- The A, B, C, and D keys are real DTMF tones that never appeared on consumer phones. They show up in radio, military, and control systems, which is why the full 4×4 grid is here.
- Tones come out of the device speaker. The app generates audio; it does not place a call or send anything down a phone line. To control remote gear, play the tone into that system's microphone or audio input.
- Reliable detection depends on volume, the speaker, and the receiving system. Hold the device close and keep the level up if a stubborn IVR or controller misses a digit.
- The sequence field plays a typed digit string as tones in order (200 ms per tone, 80 ms between), so you can drive a short IVR path or dial code without tapping each key. It is still played out of the speaker, not sent down any line.

Blue Box (MF)

Plays the R1 Multi-Frequency (MF) trunk-routing tones a 20th-century long-distance switch once used to route a call, plus the 2600 Hz supervisory tone that signaled an idle trunk. A mode of the DTMF Generator.

Why it's here. Telephone signaling history and nostalgia. A blue box reproduced the in-band MF signaling AT&T's switches used internally. This mode lets you hear those tones and see the frequency pairs behind them. It is an educational exhibit that shows off the same tone synthesizer the DTMF tool uses.

How to use

1. Switch to the Blue Box mode with the selector at the top of the DTMF Generator.
2. Tap a digit (1-0), KP, ST, or 2600 to hear that signal.
3. The readout shows the signal's frequency pair and burst timing.

Inputs

Input	Unit	Range
Signal	MF digit 1-0, KP, ST, or 2600 Hz	tap to play; no input required

How it works. Each MF signal is the sum of two sine waves drawn from six frequencies: 700, 900, 1100, 1300, 1500, 1700 Hz. The digit pairs use the canonical ITU-T assignment (for example digit 1 = 700 + 900 Hz, digit 2 = 700 + 1100 Hz). KP (Key Pulse, start of digits) = 1100 + 1700 Hz; ST (Start, end of digits) = 1500 + 1700 Hz; the 2600 Hz supervisory tone is a single frequency. The tone is synthesized on the device at an 8000 Hz sample rate; nothing is dialed or transmitted.

Example. Digit 5 is 900 + 1300 Hz. KP is 1100 + 1700 Hz. The seize tone is a single 2600 Hz tone.

Field notes

- These tones do nothing on any modern phone network. The MF and 2600 Hz signaling a blue box exploited lived in the voice band, where the switch listened for it. Carriers moved call control out of the voice band onto a separate data channel (CCIS, then SS7, still in use) in the 1980s-1990s, so there is nothing in the audio path to seize or route. This is reproduced for history and education only.

- The digit-to-frequency-pair mapping here follows the canonical ITU-T assignment by frequency membership. Some hobbyist references order the pairs differently; both describe the same six-frequency set.
- Tones come out of the device speaker. The app generates audio; it does not place a call or send anything down a line.

Red Box (US coin tones)

Plays the US ACTS payphone coin-deposit acknowledgement bursts for a nickel, dime, and quarter. All three are a 1700 + 2200 Hz dual tone; only the burst count and timing differ. A mode of the DTMF Generator.

Why it's here. Telephone signaling history and nostalgia. A red box reproduced the tones an ACTS payphone sent up the line to report that coins had been deposited. This mode lets you hear those coin patterns and see the timing behind them, using the same tone synthesizer the DTMF tool uses.

How to use

1. Switch to the Red Box mode with the selector at the top of the DTMF Generator.
2. Tap Nickel, Dime, or Quarter to hear that coin's burst pattern.
3. The readout shows the 1700 + 2200 Hz pair and the burst timing for that coin.

Inputs

Input	Unit	Range
Coin	Nickel, Dime, or Quarter	tap to play; no input required

How it works. Every US red box tone is the 1700 + 2200 Hz dual tone, in coin-specific burst patterns: nickel = one 66 ms burst; dime = two 66 ms bursts, 66 ms apart; quarter = five 33 ms bursts, 33 ms apart. The tone is synthesized on the device at an 8000 Hz sample rate (the 2200 Hz top component is well under the 4000 Hz Nyquist limit).

Example. A quarter is five 33 ms bursts of 1700 + 2200 Hz, each 33 ms apart.

Field notes

- These tones do nothing on any modern phone network. Red boxing worked only on Automated Coin Toll System (ACTS) payphones that listened for these tones in the voice path. ACTS was phased out across the US, many payphones added acoustic filters, and payphones themselves are nearly extinct. This is reproduced for history and education only.
- This mode covers the well-documented US nickel / dime / quarter set only. The frequencies and burst timings are the corroborated standard 1700 + 2200 Hz values.
- Tones come out of the device speaker. The app generates audio; it does not place a call or send anything down a line.

Hear the Frequency

Turn a frequency into a sounding tone so the logarithmic instinct behind RF (pitch, octaves, the twelve piano keys, and harmonics) becomes something you hear, with the honest bridge to RF stated plainly.

Why it's here. "An octave is a doubling" and "a dB is a logarithm" are the same instinct, and hearing it lands harder than reading it. This makes frequency audible to build that intuition, then states the limit of the analogy out loud so you do not carry a wrong equivalence into RF work.

How to use

1. Tap Play to sound the current frequency. Use the octave ×2 / ÷2 buttons, the one-octave C4→C5 keyboard, or the octave-ladder presets; the sounding voice retunes live with no click.
2. Pick a waveform (Triangle is the default; sine, square, and saw are available).
3. Read the live nearest-note, cents offset, octaves-from-A4, and the first five harmonics.

How it works. The tool uses pure equal-temperament math: every note frequency is computed, never transcribed. Each note's frequency equals 440 Hz times 2 raised to the power of (n – 49) divided by 12, where n is the piano key number and key 49 = A4 = 440 Hz (ISO 16). The semitone

ratio is 2 raised to the 1/12 power (about 1.0594630943592953), an octave is exactly a doubling (octave up multiplies by 2, octave down divides by 2), and 12 semitones make an octave. The nearest-note readout finds the note by taking 69 plus 12 times the base-2 logarithm of the frequency divided by 440, then reports the cents offset. Harmonics are simple integer multiples (the first is the frequency itself, the second is twice it, and so on). When you change octave, key, or preset, the single sounding voice retunes to the new frequency so it glides smoothly without a stop and restart.

Field notes

- The analogy honesty is baked into the copy: an octave is a base-2 FREQUENCY ratio while a dB is a base-10 POWER ratio: both are logarithmic ratio measures (the real, teachable bridge), but they are NOT the same unit, and the tool never converts an octave to a dB. Audio harmonics are WANTED (timbre); RF harmonics are usually UNWANTED (spurious emissions). The integer-multiple math is identical; the desirability flips.
- Playback is clamped to the audible range 20 Hz–20 kHz; a number above it may still be DISPLAYED for the RF analogy, clearly labeled, but is never driven to the speaker. Above ~5 kHz the waveform is forced to sine because square and triangle harmonics alias near the Nyquist limit.
- Honest no-audio state: if audio cannot initialize (no output device, a muted session), the screen shows a non-fatal "No audio output detected" message instead of pretending a tone played. The tone auto-stops on screen exit so it never runs unattended.

Morse Code

Converts plain text to International Morse code and decodes dots and dashes back to text, live in both directions, with an optional audio tone.

Why it's here. A field utility for the times Wi-Fi and cellular are both gone: signaling with a light, a buzzer, or a key by ear, reading a beacon ID that idents in Morse, or just settling what a string of dots and dashes says. Encode once and play it as a tone the receiving ear or detector can pick up.

How to use

1. Pick a direction: Text to Morse, or Morse to Text.
2. Type or paste into the top field. The converted result updates live in the card below.
3. In Morse to Text, separate letters with a single space and words with a slash, for example ... --- ... for SOS.
4. Tap Play to hear the message as a 600 Hz tone; tap Stop to end it.
5. Open Prosigns to see the named procedural signals (SOS, AR, SK, and so on).

Inputs

Input	Unit	Range
Text or Morse	letters, digits, punctuation / dots and dashes	any length; unmapped characters are skipped

How it works. International Morse code per ITU-R M.1677-1. Each letter, digit, and common punctuation mark maps to a fixed run of dots (dits) and dashes (dahs). Encoding joins the symbols within a character with no gap, characters within a word with a single space, and words with a slash. Decoding reverses the same convention, so encodable text converts back and forth without loss (lowercase folds to uppercase). The audio uses standard Morse timing: a dot is one unit, a dash is three units, the gap inside a character is one unit, between characters three units, and between words seven units, all keyed onto a single tone.

Example. SOS encodes to ... --- ... (S = three dots, O = three dashes). HELLO WORLD encodes to .... . .-.. .-.. --- / .-- --- .-. .-.. -.. with letters spaced and the two words split by a slash.

Field notes

- There is no Morse code for the space character itself; word breaks are carried by the slash separator, which is why a decoded message shows spaces between words and not blanks inside them.
- Prosigns like SOS are run together as one continuous string (...---...) with no internal letter gaps. They are procedural signals, not literal text, so they are listed for reference rather than produced when you encode ordinary words.

- Decoding marks any code it does not recognize with a question mark rather than dropping it, so a mistyped symbol stays visible instead of quietly disappearing.
- The audio is generated and played on the device speaker as a single tone; nothing is transmitted over the air or down a line by the app.

QR Code Generator

Turns any text or URL you type into a scannable QR code, then lets you share or save it as an image.

Why it's here. Handing off a config URL, a guest Wi-Fi link, or a site address without making someone type it. Generate the code on the spot and let them scan it with a phone camera.

How to use

1. Type the text or URL you want to encode, or switch to Wi-Fi mode to build a "scan to join" network code.
2. The QR code renders live as you type. Optionally pick a module shape (Square, Rounded, or Dots) and a size (Small, Medium, or Large).
3. Tap Share / Save to send the image through the system share sheet or save it.

Inputs

Input	Unit	Range
Text or URL	any string	empty input shows a prompt and renders no code
Wi-Fi (scan-to-join)	SSID, password, auth (WPA/WEP/open), hidden flag	builds the standard WIFI: payload iOS Camera and Android scanners honor as a join offer

How it works. The code is generated on the device with no network call. It renders as dark modules on a white background with a 4-module quiet zone, which is what scanners expect. Share / Save captures that white tile to a PNG image.

Example. Type https://wlanpros.com and the matching QR code appears below the field, ready to scan or share.

Field notes

- Dark code on a white background, always. Inverted light-on-dark codes look on-brand but many scanners fail to read them, so the tool does not offer that option. The module-shape (Square / Rounded / Dots) and size (Small / Medium / Large) selectors change only the module rendering and image size, never the dark-on-white contrast a scanner depends on.
- The white border around the code is the quiet zone, and it is part of the code. Do not crop it out when you share the image, or the QR may not scan.
- Everything happens on the device. The text you encode never leaves the app except through the share sheet you trigger.
- Encodes plain text and URLs. It is a generator, not a scanner, so it makes codes rather than reading them.
- Wi-Fi scan-to-join mode builds the de-facto-standard WIFI:T:;S:;P:;H:<true|false>;; payload that Apple and Google honor. Special characters in the SSID or password are backslash-escaped, and hex-looking or space-padded values are double-quoted, so a code that scans on your phone scans on the guest's too. Open networks use nopass and omit the password field.

Quick Reference (103 tools)

Offline lookup tables and the laminated field cards. Channel plans, standards, thresholds, connector and cabling pinouts, protocol references, CLI and capture cheat sheets, checklists, and guides, all available without a connection.

Quick Reference also carries the codes, trades, documents, compliance frameworks, and adjacent radios a WLAN pro meets on a real job and never learned in a Wi-Fi cert. Every one of those entries does one job: it lets you recognize what you are looking at, quote it honestly, and hand the ruling to the right authority. They point you at the AHJ, the licensed electrician, the RCDD, the QSA, the biomed team, or the architect of record. They certify nothing and clear no one.

Most of the field and trade entries carry a reference plate you can download as a PDF from inside the app, so you can save, share, or AirDrop the full-resolution plate for print. Several are text reference only, with no plate and no download: CAD & BIM Formats, Structured Cabling, AEC Process & Glossary, Verticals Index, Data Centers & Wi-Fi, and Architectural Scale. LED Decoder and Vendor Model Decode are interactive drill-downs (they carry selection state) rather than static plates. The download control simply does not appear on the text-only entries.

Radio & Spectrum (10)

Adjacent Radio Systems

The non-Wi-Fi radios a WLAN pro coexists with, designs around, and gets asked about: which five (BLE, Bluetooth Classic, Zigbee, Thread, ANT+) contend for your 2.4 GHz airtime, and which (LoRaWAN, Z-Wave, UWB, NB-IoT, LTE-M, CBRS and private 5G, Wi-Fi HaLow) run coexistence-clean in sub-GHz or licensed spectrum.

Why it's here. The client increasingly hands you the whole smart-building radio stack, not just the Wi-Fi. Knowing what shares the 2.4 GHz air, what runs clean in sub-GHz, and when to talk a client out of a private-cellular pitch that Wi-Fi already covers is both authority and self-defense in a spectrum sweep.

How to use

1. Scan the 2.4 GHz contenders: BLE, Bluetooth Classic, Zigbee, Thread, and ANT+ subtract real airtime at scale, so coordinate a Zigbee lighting mesh with your 1/6/11 plan.
2. Use the coexistence table to route congested-band IoT onto a coexistence-clean radio (LoRaWAN, Z-Wave, cellular IoT) rather than fighting for airtime.
3. Carry the three corrections: Matter is an application layer that can ride Wi-Fi, 802.15.4 is not 2.4 GHz only, and CBRS and private 5G never touch your air.
4. Use the "which radio when" picker to answer the "should this be Wi-Fi or something else" question honestly.

Field notes

- Read every range, rate, and battery figure as a real-world envelope, not a hard spec; they move with power, antenna, spreading factor, channel width, and environment.
- CBRS lives at 3.55 to 3.70 GHz and does not overlap any Wi-Fi band; private cellular competes for the job and the budget, never for your air.
- A Zigbee mesh and a wall of BLE beacons show up in a 2.4 GHz sweep as non-Wi-Fi energy; naming what you see is the difference between a diagnosis and a guess.
- Reference only. Confirm current standards editions and regional band allocations for the specific deployment.
- Data source: bands and topologies per the relevant IEEE 802.15.4, Bluetooth SIG, LoRa Alliance, and 3GPP specifications.

Channel Map

A visual channel-bonding map showing, per band, how 20/40/80/160/320 MHz channels bond together, which primary (center) channel labels each bonded block, and which blocks require DFS.

Why it's here. When planning channel widths and bonding, to see at a glance which 20 MHz primaries fold into a given 80 or 160 MHz channel, and which bonds drag in a DFS sub-channel.

How to use

1. Scroll the 5/6 GHz maps horizontally. A block's number is its primary (center) channel.
2. Color/chip legend: No DFS (neutral, an attribute not a verdict), DFS (amber, radar detection required), Mixed / DFS (danger, a 160 MHz bond spanning DFS and non-DFS sub-bands, where any DFS sub-channel subjects the whole bond to DFS), PSC (lime, the 6 GHz preferred scanning channels).
3. The 6 GHz 320 MHz row shows a primary set of three blocks (centers 31, 95, 159) and an overlapping alternative set (centers 63, 127, 191), each alternate marked as such. A primary and its alternate overlap, so only one of the pair is used at a time.

Field notes

- What it shows: a three-option band toggle. 2.4 GHz: the 11 US channels as 20 MHz blocks, with 1/6/11 emphasized (non-overlapping) and the rest faint. 5 GHz: rows for 20/40/80/160 MHz bonded widths, each block labeled with its primary/center channel and tinted by DFS class. 6 GHz: rows for 20/40/80/160/320 MHz across the full US band, UNII-5 through UNII-8 (the 59 20 MHz primaries ch 1,5,9,...,233), with the PSC channels (5,21,37,...,229) marked.
- US-default. The full US 6 GHz plan (5925 to 7125 MHz) is drawn end to end. An earlier build truncated the map at ch 93, restored to the complete band per a beta report. 6 GHz needs no DFS; indoors it runs low-power (LPI) with no AFC, while standard-power use is AFC-coordinated.
- Reference basis: the US (FCC) channel plan. 5 GHz DFS: No DFS = UNII-1 (36 to 48), UNII-3 (149 to 165), and UNII-4 (169 to 177); DFS = UNII-2A/2C. Colors are for readability; the meaning (DFS/PSC/mixed) is what matters.

dB Reference

A decibel reference card: dB change → power/voltage ratio with rules of thumb, and common dBm anchor values with their power and real-world context.

Why it's here. Quick mental math in the field: "+3 dB is double power," "what power is +30 dBm," "what's the FCC 2.4 GHz limit in dBm."

How to use

1. In the ratio table, positive gains render in lime, losses in red.
2. Key anchors: +3 dB = 2× power; +10 dB = 10×; +30 dBm = 1 W; 0 dBm = 1 mW; −67 dBm = enterprise VoIP minimum; −70 dBm = enterprise data minimum; −80 dBm = typical Wi-Fi receiver sensitivity.

Field notes

- What it shows: dB Power Ratios: dB change (+3 to +30, −3 to −20) → power ratio, voltage ratio, and a rule-of-thumb note. Common dBm Reference Points: dBm anchors (+36 down to −100 dBm) → power (watts/mW/nW/pW) and context (regulatory limits, typical Tx powers, sensitivity floors).

- Footnote: "0 dBd is about 2.15 dBi (dipole reference). dBW = dBm − 30. Regulatory limits shown are US FCC; verify before compliance decisions."
- Mixed US (FCC) and one ETSI anchor; read the context cell for the jurisdiction of each limit.
- The dBm context column names specific regulatory limits: FCC 6 GHz standard-power EIRP (+36 dBm, AFC required), FCC 2.4 GHz max conducted (Part 15.247, +30 dBm), FCC UNII-2A/2C and UNII-1 conducted maxes, and ETSI 5 GHz EIRP (EN 301 893, +23 dBm). These specific limits are verifiable against the cited rules; verify before any compliance decision.

How Strong Is Wi-Fi, Really?

A vendor-neutral comparison of how much energy Wi-Fi puts into your body versus the everyday sun, using verified, stated numbers. No inputs and no runtime math: it is a read-along reference.

Why it's here. Wi-Fi exposure questions come up constantly. This puts the energy in perspective against a thing everyone already lives with: sunlight, using verified numbers instead of hand-waving. Using stated assumptions (ten access points at 30 dBm / 1 watt EIRP, four meters away, free-space spread), the ring of APs delivers about 0.05 W/m². The midday sun delivers about 1,000 W/m², full spectrum, which makes it about 20,000 times stronger per square meter.

How to use

1. Read the short version first: one hour of midday sun puts about as much energy into your body as roughly 2.3 years sitting inside a ring of ten Wi-Fi access points 4 meters (13 feet) away.
2. The energy-parity table shows the same idea as time: 10 seconds of sun ≈ 2.3 days in the ring, 1 minute ≈ two weeks, 1 hour ≈ 2.3 years.
3. The assumptions card states every input behind the numbers, so nothing is hidden: 10 APs at 1 W EIRP, 4 m free-space spread, the sun at 1,000 W/m² full spectrum.
4. The safety-limit card states where this sits against the FCC (47 CFR 1.1310) and ICNIRP 2020 public limit of 10 W/m²: the ring is about 200 times below it.

How it works. Power density falls off with the square of distance: density = EIRP / ($4\pi r^2$), free space at 4 m, stated as approximate.

Example. At 30 dBm (1 W) EIRP and 4 m, each AP is 0.00497 W/m²; ten of them ≈ 0.05 W/m². The sun at 1,000 W/m² is ≈ 20,000× stronger, so 1 hour of sun ≈ 2.3 years in the ring.

Field notes

- Honest note on mechanism: Wi-Fi radio energy is non-ionizing. It can only gently warm tissue. It cannot cause the photochemical or DNA damage that sunburn does, at any Wi-Fi power level. This is a comparison of total energy and warmth, never of sunburn: the UV in sunlight burns; Wi-Fi has no UV and no comparable mechanism.
- The basis is 30 dBm (1 W) EIRP per AP (the US 2.4 GHz maximum) at 4 meters (13 feet). 4 m is realistic for how far a person sits from the access points around them; free-space density falls with r^2, so the figures are conservative for the point.
- Where it sits against the limit: the FCC and ICNIRP cap public RF exposure at 10 W/m². The 10-AP ring at about 0.05 W/m² is roughly 200× below that limit, and measured real-world Wi-Fi runs far lower still.
- Every figure is stated as approximate and traces to a verified source: FCC 47 CFR 1.1310 and OET-65; ICNIRP 2020 RF guidelines; ASTM G173 / NREL reference solar spectrum; WHO and IARC on non-ionizing RF (Group 2B) versus solar UV (Group 1); peer-reviewed Wi-Fi exposure surveys (PMC5927334, PMC8172712).

Non-Wi-Fi Wireless Channels

Look up the channel/frequency plans of the common non-Wi-Fi radios that share or sit beside the bands a Wi-Fi pro works in: LoRaWAN, IEEE 802.15.4, Bluetooth Classic, Bluetooth LE, and Zigbee.

Why it's here. When you're chasing interference or co-existence in the 2.4 GHz ISM band, or sizing a sub-GHz IoT deployment, and need to know where these radios actually sit.

How to use

1. Each technology is its own section.
2. LoRaWAN plans flagged with a "verify" chip are version-dependent or sparsely sourced.

3. BLE rows are ordered by physical frequency (not index) so you can see how the 3 advertising channels interleave; an "Advertising" chip marks channels 37/38/39 (2402/2426/2480 MHz).
4. Zigbee's "common picks" (11, 15, 20, 25, 26) are convention, not a mandate.

Field notes

- What it shows: one card per technology. LoRaWAN: regional plan (EU868, US915, AU915, AS923, IN865, KR920, CN470, CN779, RU864), frequency range in MHz, and a channel-plan description. IEEE 802.15.4: band (868 MHz / 915 MHz / 2.4 GHz), channel-number range, spacing, center summary, region. Bluetooth Classic (BR/EDR): 79 channels, 1 MHz spacing, f = 2402 + k MHz, 2402 to 2480 MHz, ~1600 hops/sec, global. Bluetooth LE: all 40 channels in physical-frequency order with index, frequency in MHz, kind (Advertising / Data). Zigbee: 2.4 GHz 802.15.4 ch 11 to 26, sub-GHz bands, common 2.4 GHz channel picks.
- Bluetooth, BLE, and 802.15.4 use globally-fixed channel grids. LoRaWAN frequency plans are entirely region-defined; there is no global LoRaWAN channel map.
- Plans marked "verify" (CN470 version-dependent; CN779 deprecated/limited; RU864 sparsely sourced) must be confirmed against RP002 §2 and the local regulator.
- BLE channel frequencies follow an explicit per-channel lookup, not a naive linear formula; the simple "2402 + 2×index" shortcut is a common BLE chart error.
- 802.15.4's 868/915 MHz bands are region-restricted; 2.4 GHz ch 11 to 26 are global.
- Data source: cross-checked against LoRa Alliance RP002, IEEE 802.15.4, Bluetooth SIG Core Spec, and CSA/Zigbee spec. 802.15.4 centers verified against the standard formula (ch 0 = 868.3 MHz; ch 1 to 10 = 906 + 2·(k−1); ch 11 to 26 = 2405 + 5·(k−11)).

Regulatory Domains

The radio regulator that governs Wi-Fi in each market: name, official website, governing docs, and 2.4/5/6 GHz band and power notes for 43 jurisdictions, as a dated snapshot.

Why it's here. Working a market you do not know? Find its regulator, the governing rule, and a starting band and power note, then confirm against the official site.

How to use

1. Search by jurisdiction or regulator abbreviation. Shared abbreviations (NCC, TRA, CRA) carry the jurisdiction so they stay unambiguous.
2. The page leads with a snapshot date: the band and power notes are a dated snapshot, not settled constants.
3. Tap a regulator's website to confirm any value against the current rule text before relying on it.

Field notes

- What it shows: a per-jurisdiction directory of 43 jurisdictions, each with its regulator (name, abbreviation, logo), the governing regulation, official website, and a 2.4/5/6 GHz bands note. The structural rules (band edges, DFS/TPC, which power classes exist) are high confidence; the exact dBm figures are a dated snapshot to verify against the regulator.
- Regulatory-volatility caveat: the screen leads with a persistent snapshot banner ("Snapshot verified . Regulations change; confirm against the regulator before relying on a value."). The band and power figures (the 6 GHz dBm values especially) are a dated snapshot, never a settled constant: verify before deploying or certifying.
- Provenance: the entries are compiled and cross-checked from official regulator documents and vendor regulatory white papers, current as of the on-screen snapshot date. Offline, read-only.

RF Bands

A frequency map of where the common wireless technologies live in the spectrum, low to high: RFID, GPS/GNSS, cellular, the 2.4 GHz ISM crowd, and Wi-Fi across all its bands. Five spectrum neighborhoods, each with its band rows, plus a region-variance list for the bands where "what operates where" changes by regulator. A log-scale spectrum-bar plate sits at the top.

Why it's here. You design inside three or four Wi-Fi bands, but those bands have neighbors, and the neighbors are who you fight for airtime and chase for interference. This is the one screen that shows the whole map so you can see what sits just upstairs and downstairs of your channels: C-band 5G right below 6 GHz, the sub-GHz IoT radios under 2.4, the microwave oven leaking at 2.45.

How to use

1. Read top to bottom, low frequency to high. Each neighborhood card carries its band rows and a one-line takeaway on why those radios cluster there.
2. The Wi-Fi rows carry the single lime accent so your home turf stands out inside each crowded band.
3. The warning-toned region-variance list at the bottom is the part that bites: 6 GHz, sub-GHz ISM, Z-Wave center, HaLow, UHF RFID, 2.4 GHz channel count, 60 GHz WiGig, and 5 GHz DFS all change by regulator.

Field notes

- This is a frequency map, not a channel plan and not a security chart. Every band edge is a nominal allocation, not a guaranteed-clear channel. Local power limits, DFS, and licensing constrain real use further.
- The single highest-stakes variance for a Wi-Fi pro is 6 GHz: the US runs the full 5.925 to 7.125 GHz (1.2 GHz), the EU opened only 5.945 to 6.425 GHz (the lower 480 MHz), and some regions have not opened 6 GHz at all. Never assume the upper 6 GHz exists outside the US.
- The 2.4 GHz neighborhood is the most contested on the chart: Wi-Fi, Bluetooth/BLE, Zigbee/Thread, and microwave ovens all sit on top of each other in 83.5 MHz. That is why only 1/6/11 are non-overlapping in North America.
- 5G NR FR1 runs right up to about 7.125 GHz, so its C-band is the immediate downstairs neighbor of 6 GHz Wi-Fi, and FR2 mmWave shares the 24 GHz neighborhood with the 24 GHz ISM band.
- The spectrum-bar plate sits on a dark card; every fact in it is also in the tables, so the screen reads end-to-end without the image.
- Data source: every load-bearing figure cross-verified against at least two independent sources. US/FCC default; region splits are called out explicitly.

Signal Thresholds

RSSI and SNR targets: a quality scale for RSSI, minimum RSSI/SNR by application, and the SNR needed to reach each typical MCS index.

Why it's here. When validating a survey or troubleshooting a complaint: "is −68 dBm good enough for VoIP?" or "what SNR do I need to hold MCS 7?"

How to use

1. RSSI bands carry a colored verdict dot plus the quality word (green = Excellent/Good, amber = Fair, red = Poor); the word always carries the meaning, never color alone.
2. For the application table, read across to the min RSSI and min SNR your target use case needs.
3. The SNR→MCS table tells you the SNR floor to sustain a given rate.

Field notes

- What it shows: three blocks. RSSI quality scale (four bands): Excellent (> −60 dBm) / Good (−60 to −67) / Fair (−68 to −72) / Poor (−73 or weaker). These are Keith Parsons' canonical field thresholds (confirmed 2026-07-12), rendered from the app's single source of truth, `WifiGradingBands.kRssiBands` in `lib/services/network/wifi_grading.dart`, the very same list the Live-mode and Analyze verdict engines grade on, so the number the app shows can never drift from the number it grades. Minimum signal by application: VoIP/real-time, HD video, general browsing, email/basic data, IoT/low-rate, and location/RTLS, each with min RSSI, min SNR, and a note. SNR to MCS (80 MHz, 1 SS): the SNR floor for each MCS index (MCS 0 at 5 dB up to MCS 11 at 35 dB) with an indicative rate.
- The RSSI scale is a convention, not physics (Keith: great connectivity happens at −75 dBm too); treat every value here as a field-planning guideline, not a guarantee. The application and SNR→MCS thresholds vary by client hardware, environment, and AP vendor.
- The SNR→MCS rates are indicative (80 MHz, single stream); MCS 10/11 rates are noted as 802.11ax.
- Provenance: the RSSI quality scale is Keith's reviewed canonical scale (source: `kRssiBands`). The application-minimum and SNR→MCS tables are field-planning reference values ported from the RF Tools PWA; they are widely-used working numbers, not reproduced from a single named standard.

Spectrum Reference

A per-band fact sheet for the three Wi-Fi bands: total usable spectrum, supported standards, channel counts, non-overlapping counts, channel widths, DFS/radar requirements, common co-existing interferers, and key deployment notes.

Why it's here. The one-screen "tell me everything about this band" reference: what lives in it, what interferes with it, what the power/DFS rules are.

How to use

1. Pick a band (2.4 / 5 / 6 GHz); read the eight facts top to bottom.
2. The colored range badge is decorative chrome paired with the band label (2.4 GHz = amber/congested, 5 GHz = blue, 6 GHz = green).
3. The "Co-existence" row names the band's common interferers or managed incumbents.

Field notes

- What it shows: each band shows a range badge plus eight key/value facts: Total spectrum, Standards, Channels (US), Non-overlapping, Channel widths, DFS / Radar, Co-existence, and Key notes. Below the three band sheets sits an additional geography-to-regulator card, listing which body governs Wi-Fi in each region, so the US-default facts above point you at the right regulator elsewhere.
- US-default; the footnote on every band reads "US (FCC) regulatory domain. Verify local rules before deployment."
- Carries useful specifics: UNII-1 indoor-only in some regions; UNII-2A/2C DFS implies a 60-second channel availability check before first use (10 minutes for 5600-5650 MHz in the EU), and on radar the AP leaves within 10 seconds and stays off for 30 minutes; 6 GHz has four US power modes, namely Standard Power (up to 36 dBm EIRP, AFC required indoors and outdoors), LPI (up to 30 dBm, no AFC), VLP (up to 14 dBm EIRP, no AFC, mobile), and Geofenced Variable Power (up to 24 dBm EIRP, U-NII-5 and U-NII-7 only); WPA3 mandatory on 6 GHz.
- Data source: US (FCC) regulatory domain. Values: 2.4 GHz = 83.5 MHz (US); 5 GHz = ~580 MHz usable (UNII-1/2A/2C/3); 6 GHz = 1200 MHz (5925 to 7125 MHz).

Wi-Fi HaLow

A per-section reference for IEEE 802.11ah, Wi-Fi moved down into the sub-1-GHz ISM bands for IoT: what it is, bands by region, channel widths, the headline numbers, a single-stream MCS rate table, power features, PHY/MAC, use cases, a comparison against the other IoT radios, and 2026 maturity. A channel-width plate compares HaLow's 1 to 16 MHz channels against a 20 MHz Wi-Fi channel.

Why it's here. HaLow is the Wi-Fi most Wi-Fi pros have never touched, and it is full of confidently-wrong numbers online. This is the honest read: it trades raw speed for about 1 km range, multi-year battery life, and thousands of devices per AP, while keeping native IP and WPA3. When a client asks "should we use HaLow for the sensor network," you want the real ceiling and the real caveats, not a vendor slide.

How to use

1. Read the region-lock banner first, it is the load-bearing fact. Frequency and channel width are set by each regulator, so a device certified for one region cannot legally run in another.
2. Work down through the cards: headline numbers, bands by region (with a confidence tag on the secondary-source rows), channel widths, the MCS rate table, then power, PHY/MAC, use cases, and the comparison.
3. The MCS table's peak cell (MCS 9, 256-QAM, 16 MHz, short guard interval) carries the lime accent: that is the 86.7 Mbps headline.

How it works. The clean mental model: the 802.11ac PHY clocked at one tenth. Same OFDM machinery, ten times slower clock, so symbols are 10x longer (more resilient over distance and multipath) and rates land at about a tenth of 802.11ac. Capacity comes from a 13-bit Association ID (2^13 minus 1 = 8,191 devices per AP) plus a hierarchical TIM. Power efficiency comes from Target Wake Time, Restricted Access Window, Extended Max Idle, non-TIM mode, and short MAC headers.

Field notes

- The defensible single-stream maximum is 86.7 Mbps (MCS 9, 256-QAM, 16 MHz, SGI), from the Wi-Fi Alliance overview. Use that number, NOT the contested 433.3 Mbps. Wikipedia's 433 figure is a 4-spatial-stream claim, but a 4x scaling of the WFA figure is about 347 Mbps, and first-generation HaLow silicon is single-stream. The far edge of the cell drops to about 150 kbps.
- HaLow does NOT use 2.4, 5, or 6 GHz. Lower frequencies travel farther for the same power, which is the entire reason it exists. Even the widest HaLow channel (16 MHz) is narrower than the minimum 20 MHz 2.4 GHz Wi-Fi channel; narrow channels concentrate energy, which is how it reaches farther.
- Bands by region: US 902 to 928 MHz (widest, full 1 to 16 MHz channels), EU 863 to 868 MHz (narrow, duty-cycle limited, 1/2 MHz only), AU/NZ 915 to 928 MHz. Japan, Korea, China, and Singapore carry a Medium-confidence tag because they come from secondary technical sources, not the WFA doc.

- Where it sits: more range and device count than BLE, Zigbee, and Z-Wave, and more data rate plus native IP than LoRaWAN, Sigfox, and NB-IoT. It does not match LoRa/Sigfox/NB-IoT for multi-kilometer range, and unlike carrier NB-IoT it needs its own AP infrastructure. Its strongest case is replacing short-range mesh radios with longer reach and direct IP, and carrying video where LoRa/Zigbee cannot.
- Maturity (2026): certified and shipping, with Morse Micro the clear silicon leader (MM6108, MM8108) and sub-$130 developer gateways (HaLowLink 1 at $99, HaLowLink 2 at $129). It is early-mainstream, NOT yet mass-deployed like Zigbee or BLE. "HaLow replaced Zigbee" is a roadmap claim, not a 2026 fact.
- The channel-width plate sits on a dark card; every fact is also in the tables, so the screen reads without the image.

Wi-Fi Standards & Terminology (6)

802.11 Feature Matrix

A Wi-Fi 5 to Wi-Fi 7 capability comparison across the features that matter in design: bands, channel width, modulation, OFDMA, MU-MIMO, BSS Coloring, TWT, MLO, preamble puncturing, spatial streams, and the theoretical PHY-max rate.

Why it's here. Sizing what a generation can and cannot do, side by side, without hunting through four standards.

How to use

1. Read each feature row across the four generation columns (Wi-Fi 5 / 802.11ac, Wi-Fi 6 / 802.11ax, Wi-Fi 6E, Wi-Fi 7 / 802.11be).

Field notes

- What it shows: feature presence/absence and the stream/width/QAM ceilings per generation, plus a theoretical PHY-max rate row.
- The Max PHY rate is a theoretical ceiling, not a real-world client rate. It carries a 'ceiling' chip and a footnote with the math (e.g. Wi-Fi 7 ~46 Gbps = 16 streams x 320 MHz x 4096-QAM, with no shipping client above 2-4 streams). Never an achievable speed.

- Data source / standard: IEEE 802.11ac/ax/be; 802.11be (Wi-Fi 7) values are from the draft approaching final approval and may shift (flagged in the footnote). Offline, read-only.

802.11 Standards

A PHY-layer comparison of every major 802.11 amendment from the original 802.11 (1997) through Wi-Fi 7, with year, bands, max PHY rate, MIMO, channel widths, and modulation.

Why it's here. Settling "which generation does what": bands reached, max rate, MIMO ceiling, and modulation per amendment.

How to use

1. Scan by generation badge (Wi-Fi 4 through Wi-Fi 7).
2. The band filter answers "which generations reach 6 GHz" (Wi-Fi 6E and Wi-Fi 7).
3. The original 802.11 shows a dash for generation and MIMO (it predates both).

Field notes

- What it shows: one card per amendment with the IEEE designation, a Wi-Fi generation badge, year, and rows for Bands (GHz), Max PHY rate, MIMO, Channel width (MHz), and Modulation. An optional band filter (All / 2.4 / 5 / 6 GHz) narrows the list.
- Two footnotes: (1) "Official Wi-Fi Alliance generation naming begins at Wi-Fi 4 (802.11n); earlier amendments are shown by their 802.11 names only"; (2) "Wi-Fi 7 certification began 2024; IEEE 802.11be was published 2025."
- Max PHY rate is the theoretical aggregate ceiling; real-world throughput is typically 50 to 60% of it.
- Provenance: the amendment facts (bands, MIMO, channel widths, modulation, max PHY rate) are ported from the IEEE 802.11 amendments; the Year column is the Wi-Fi Alliance certification year, footnoted separately against the IEEE ratification year. Key rows: 802.11ac = Wi-Fi 5 (2013, 6.9 Gbps); 802.11ax = Wi-Fi 6 (2019) and Wi-Fi 6E (2020, adds 6 GHz); 802.11be = Wi-Fi 7 (2024, 23.1 Gbps with MLO, 4K-QAM, up to 320 MHz).

MCS Index

Look up the modulation, coding rate, and PHY data rate for any 802.11 MCS index across channel widths, for 802.11n (HT), 802.11ac (VHT), 802.11ax (HE), and 802.11be (EHT / Wi-Fi 7), scaled by spatial-stream count.

Why it's here. When you see an MCS index in a capture or client stats and want to know the modulation/coding behind it and the rate it should deliver at a given width and stream count.

How to use

1. Choose the standard (n / ac / ax / be) and stream count (1 to 8); rates update (rate = per-stream value × streams).
2. MCS 0 (BPSK 1/2) is the most resilient/slowest; higher MCS indices use denser modulation (up to 1024-QAM for ax and 4096-QAM for be, MCS 12-13) for higher rates needing better signal.
3. Cells shown as "N/A" are genuinely invalid combinations, not zero or fabricated.

Field notes

- What it shows: two selectors, 802.11 standard (n / ac / ax / be) and spatial streams (1 to 8; 802.11n is 1 to 4). Columns per standard: 802.11n = 20 LGI, 20 SGI, 40 LGI, 40 SGI; 802.11ac = 20/40/80/160 SGI; 802.11ax = 20/40/80/160 MHz; 802.11be = 20/40/80/160/320 MHz (ax and be both at an 800 ns guard interval).
- 802.11ac MCS 9 is invalid at 20 and 40 MHz for a single stream (shown "N/A"); some cells are invalid only at certain stream counts. Above 4 spatial streams no rate is shown at all. A documented gap in the published source, stated plainly rather than computed.
- Guard-interval definitions: 802.11n LGI = 800 ns / SGI = 400 ns; 802.11ac uses SGI; 802.11ax and 802.11be define 0.8 / 1.6 / 3.2 microsecond guard intervals (no 400 ns short GI above 802.11ac).
- The notes card states actual throughput is typically 50 to 65% of the PHY rate. Rates are PHY-layer maximums, not delivered throughput.
- Provenance: values come from Keith Parsons' MCS Index Chart and mcsindex.net (by @VergesFrancois, SemFio Networks), cross-checked against the IEEE 802.11n/ac/ax/be PHY rate tables.

Modulation

Teaches what a modulation constellation is on the I/Q plane and why each step up in order (BPSK -> QPSK -> 16/64/256/1024-QAM) carries more bits per symbol but demands a cleaner link (lower EVM, higher SNR).

Why it's here. The visual companion to the MCS Index table. Reach for it when you want to understand WHY a higher MCS needs better signal, or to explain constellations, bits per symbol, and EVM to someone, rather than to look up a spec rate.

How to use

1. Read the six constellation diagrams in order; each step doubles the points and adds one bit per symbol while the points pack closer together.
2. Read the Error Vector Magnitude (EVM) explainer to see what the receiver must overcome to keep each symbol on the correct side of its decision boundary.
3. Read the summary card for the order -> bits/symbol -> SNR/EVM relationship at a glance.
4. Tap any diagram to open a full-screen pinch-zoom view for the detail-dense planes.

Field notes

- The SNR and EVM figures on the cards are REPRESENTATIVE order-of-magnitude demands, not exact 802.11 MCS thresholds. The relationship, not the precise number, is the point. Use the MCS Index tool for the spec rate/modulation table.
- EVM (Error Vector Magnitude) is the distance from the ideal symbol point to where the symbol actually landed, expressed as a percentage (or dB) of the reference amplitude. As order rises, decision boundaries shrink, so higher-order QAM demands a lower EVM and a higher SNR.
- The eight diagrams sit on an always-dark card in both light and dark modes so they never read inverted; every fact is also in the screen's prose and copy text, so the screen reads fully without the images.
- The Copy action exports the order -> bits -> SNR/EVM summary as a tab-separated table plus the representative-numbers caveat.

Wireless Glossary

Plain-language definitions of 123 wireless terms, grouped by topic and searchable live across the term, abbreviation, and definition. 93 are the Wi-Fi terms a working engineer meets, in 8 categories. The other 30 cover the radios and links people mix up with Wi-Fi, in 4 more: Cellular, Short-Range Radio, Location & Satellite, and Home Internet. The same grouped, searchable screen as the authentication glossary, with its own dataset.

Why it's here. When a term in a config screen, a log, or a standards document is the thing standing between you and understanding what is happening. It answers what a term means in Wi-Fi terms, in Keith's voice, without a vendor's slant.

How to use

1. Browse the terms grouped by category, or type in the search box to filter live.
2. Each entry shows the full term, its abbreviation when it has one, and a short plain-language definition that makes sense on its own.
3. Use the copy action to grab the current view (the filtered subset when searching, otherwise the full list).

Field notes

- Multilingual: a language picker (English default, plus Spanish, French, Italian, and German) switches the definition language, so a non-English-first engineer can read the same definition in their own language. The English definitions remain the source of truth: whenever a non-English language is active the screen shows a "Translations in beta" note, because the translations are drafts pending professional review, and that draft flag travels with any copied text.
- Vendor-neutral by design. Definitions describe standards-based behavior, not one vendor's implementation.
- Data source: the curated 123-term Wireless Glossary (93 Wi-Fi terms, then 30 wireless terms beyond Wi-Fi). The tool was called Wi-Fi Glossary until 2026-09-28, and searching for that name still finds it. The Wi-Fi Authentication Glossary below is the security-focused sibling with its own dataset.

Wi-Fi Standards Bodies

Who defines, certifies, and coordinates Wi-Fi and adjacent wireless: IEEE, the Wi-Fi Alliance, ITU-R, and more, grouped by the three layers a pro keeps straight.

Why it's here. People conflate the bodies. This separates who writes the standard from who certifies products from who sets the legal rules, so you cite the right one.

How to use

1. Three layers: a standards body (IEEE 802.11) defines how the radio works; a certification body (Wi-Fi Alliance) verifies interoperability and owns the brand; a regulator sets the legal channel and power rules per country.
2. Tiles are grouped by layer, plus an IoT/adjacent group (Wi-SUN, Connectivity Standards Alliance for Zigbee and Matter, LoRa Alliance).
3. 'Wi-Fi' is a Wi-Fi Alliance trademark, not an acronym and not 'Wireless Fidelity'. The per-country rules live on the Regulatory Domains reference, cross-linked here.

Field notes

- IEEE writes 802.11; the Wi-Fi Alliance certifies and brands it. They are different organizations.
- ETSI appears here as a standards body and also as the EU's referenced harmonizer on the Regulatory Domains page.
- Data source: each body's official site (IEEE, Wi-Fi Alliance, WBA, ITU, IETF, 3GPP, Bluetooth SIG, CWNP, ETSI, Ecma, Wi-SUN, CSA, LoRa Alliance).

Security & Auth (5)

802.1X / EAP Types

The EAP methods a supplicant and authentication server negotiate inside 802.1X, compared across credential type, server-cert requirement, client-cert requirement, and mutual authentication, with a typical-use note.

Why it's here. Choosing or auditing an 802.1X method: this is the side-by-side of what each EAP type needs and protects.

How to use

1. Read the matrix for each EAP method (EAP-TLS, PEAP, EAP-TTLS, EAP-FAST, EAP-PWD, EAP-SIM/AKA, and more) across its four security axes.
2. Read the warning banner: clients that skip server-certificate validation on a tunneled method (PEAP/TTLS/FAST) are exposed to evil-twin credential theft.

Field notes

- What it shows: the EAP methods, their credential model, whether they require a server certificate, a client certificate, and mutual auth, plus a typical-use note.
- Note: EAP-FAST is RFC 4851; RFC 7170 defines TEAP, its standards-track successor (not the same method).
- Data source / standard: each method's defining RFC (5216 EAP-TLS, 5281 EAP-TTLS, 4851 EAP-FAST, 5931 EAP-PWD, 4186 EAP-SIM, 4187 EAP-AKA, 5448 EAP-AKA'). Offline, read-only.

Diffie-Hellman

The Diffie-Hellman key exchange taught by colors: a staged paint-mixing analogy paired with the real modular-exponentiation math, then tied to WPA3 SAE. A paint-mixing diagram sits at the top.

Why it's here. SAE is the heart of WPA3, and SAE is Diffie-Hellman wearing a Wi-Fi hat. If you understand how two parties reach a shared secret over a public channel without ever sending their private values, you understand why WPA3 resists the offline dictionary attacks that broke WPA2-PSK. This is the read-first explainer for that.

How to use

1. Read the paint analogy and the matching math side by side: public base g and modulus p are the common paint; each party's private exponent is their secret color; the mixtures (g^a mod p and g^b mod p) are public; both sides reach the same blend.
2. The eavesdropper row is the point: a passive listener sees the common paint and both mixtures and still cannot un-mix them. Recovering a private exponent is the discrete-logarithm problem, which is hard.

How it works. Public parameters: base g, modulus p. Alice computes A = g^a mod p and sends it; Bob computes B = g^b mod p and sends it. Alice then computes s = B^a mod p, Bob computes s = A^b mod p, and both equal (g^a)^b mod p = (g^b)^a mod p, the shared secret. Mixing is easy (one-way); un-mixing (the discrete log) is hard, which is the security.

Field notes

- The Wi-Fi tie-in is the whole reason this is in the kit: Diffie-Hellman is the basis of SAE (Simultaneous Authentication of Equals), the Dragonfly handshake in WPA3. It replaced the WPA2 pre-shared-key 4-way exchange and resists offline dictionary attacks, because the password is never exposed to a passive listener.
- The diagram sits on a dark card; the analogy, the math, and the WPA3 note all live in the text too, so the screen reads without the image.
- This is a fundamentals explainer, not a calculator. It does not do the modular arithmetic for you; it teaches the shape so SAE stops being a black box.

FreeRADIUS on WLAN Pi

A step-by-step guide to standing up a lab RADIUS server on a WLAN Pi for learning and testing 802.1X (WPA2/WPA3-Enterprise), built around a bundled install script you copy to the Pi and run there.

Why it's here. Reach for it when you want a working RADIUS server to practice 802.1X against without building a full production AAA stack. The screen shows the install script inline, lets you download it, and points at the two files to customize (the shared secret and the user list).

How to use

1. Read the lab caveat: the script ships test accounts (student01-student10) with cleartext passwords and a shared secret named secretwlanpros. Change the secret and use real credentials before anything real.
2. Download install_freeradius.sh (or copy it from the inline view).
3. Copy it to the Pi: scp install_freeradius.sh wlanpi@:~
4. Make it executable: chmod +x install_freeradius.sh
5. Run it on the Pi: ./install_freeradius.sh
6. Verify with the listed radtest / journalctl / tcpdump commands.

Field notes

- The Toolbox runs no shell. This is reference text plus a file you run on your own WLAN Pi, not in the app. The inline script and the download are the exact same bytes.
- Lab use only as shipped: cleartext-password test accounts and a known shared secret. Change the secret in /etc/freeradius/3.0/clients.conf and replace the test users in /etc/freeradius/3.0/users before using it on a real network.
- The script configures PEAP/MSCHAPv2, opens UDP 1812, and runs a radtest against student01 to confirm the server answers.
- Guide and install script by Ferney Munoz; bundled with permission.

Wi-Fi Authentication Glossary

Plain-language definitions of the Wi-Fi authentication terms, 58 of them, that a network or security pro actually meets: the 802.1X / EAP framework, RADIUS and AAA, WPA2 / WPA3 and SAE, PSK and Enterprise modes, certificates, and the supporting acronyms. Each entry pairs the full term and its abbreviation with a definition written for working engineers, grouped by topic.

Why it's here. When a term in a config screen, a log, or a standards document is the thing standing between you and understanding the authentication flow. It answers what AAA, SAE, PMF, or EAP-TLS actually mean in Wi-Fi terms, without a vendor's slant.

How to use

1. Browse the terms grouped by topic (Network Fundamentals, Roaming & Passpoint, Security & Encryption, EAP & Key Exchange, Cellular/3GPP, Core Authentication, and Identity & Credentials).
2. Type in the search box to filter live across the term name, abbreviation, and definition.
3. Each entry shows the full term, its abbreviation when it has one, and a definition in Keith's voice.
4. Use the copy action to grab the current view (the filtered subset when searching, otherwise the full list) as plain text.

Field notes

- Vendor-neutral by design. Definitions describe the standards-based behavior, not one vendor's implementation.
- This is the authentication-focused sibling of the Wireless Glossary; same searchable, grouped layout with its own set of terms.
- Data source: the curated Wi-Fi Authentication Glossary edition, 58 terms.

WPA Security

A matrix of Wi-Fi security modes (WEP through WPA3-Enterprise) with encryption, key method, PMF support, and a deployment verdict, plus a reference of the advanced features that distinguish them.

Why it's here. When choosing or auditing an SSID's security, confirm a mode's cipher, key method, PMF requirement, and whether it's recommended, acceptable, or to be avoided.

How to use

1. Each mode shows a verdict chip whose color reflects the deployment recommendation, always with the word: WEP/WPA1 = "Do not deploy"/"Deprecated" (red); WPA2-Personal = "Acceptable" (amber); WPA3-Personal/WPA3-Enterprise = "Recommended" (green); Enhanced Open = "Open networks", WPA2-Enterprise = "Enterprise std" (blue/info).
2. Below, the feature rows explain the mechanisms (e.g. SAE replaces the PSK 4-way handshake with forward secrecy; OWE encrypts open networks without a password; 6 GHz requires WPA3 or OWE).

Field notes

- What it shows: Security modes: WEP, WPA (WPA1), WPA2-Personal, WPA3-Personal, Enhanced Open, WPA2-Enterprise, WPA3-Enterprise, each with category, encryption suite, key method, PMF (Not supported / Optional / Required), and a status verdict chip. Advanced features: SAE, PMF (802.11w), OWE, Forward Secrecy, 192-bit Security Mode, WPA3-mandatory-on-6 GHz, and 802.1X/RADIUS roles.
- WPA3-Enterprise's 192-bit mode (GCMP-256 + HMAC-SHA-384 + ECDH/ECDSA-384) is noted as required for government/classified deployments. 6 GHz does not permit WPA2 or older.
- The verdicts are reference guidance reflecting current best practice.
- Data source: IEEE 802.11 / Wi-Fi Alliance security standards. Verdict colors meet WCAG contrast.

Clients & Field Wi-Fi (6)

Apple Wi-Fi Support Tips

Apple's own Wi-Fi support guidance distilled into four sections: recommended router/Wi-Fi settings for Apple devices, how to run Wireless Diagnostics on a Mac, the Option-click Wi-Fi menu, and iOS/iPadOS Wi-Fi troubleshooting steps. Three of the four sections carry a link to the

Apple article they came from; the Option-click section instead links to the in-app macOS Menu-Bar Wi-Fi companion, which decodes those fields.

Why it's here. Most of the clients you support carry Apple gear, and Apple publishes specific Wi-Fi guidance that engineers either ignore or never find. This is that guidance in one place: what Apple actually recommends for router settings, how to pull diagnostics off a Mac, and the iOS triage path, with the source articles one tap away.

How to use

1. Read the four sections; three carry a tappable link chip to the Apple support article they were distilled from (the Option-click section links to the in-app companion instead).
2. The Option-click menu section links straight to the macOS Menu-Bar Wi-Fi companion, which owns the per-field "what each RF value means" detail.
3. If a link fails to open, the screen shows the full URL so you can copy it.

Field notes

- Honesty bits carried on-screen, not hidden: Apple is silent on transmit power (so the tool does not invent a figure), and the iOS troubleshooting steps come from a single Apple source, which is flagged. Keith's own domain note is attributed to Keith, not to Apple.
- This is reference guidance, not a live read of the device. For the live association detail on a Mac, use Wi-Fi Information or macOS Menu-Bar Wi-Fi.
- Data source: Apple's Wi-Fi support documentation, footnoted to Apple support URLs.

How Your Devices Access the Internet

A reference plate drawing every hop between a device and the Internet, in order and to scale, so it is visible that Wi-Fi is only the first one. Beside it, one control shows the slowest link winning: pick which hop is slowest (the Wi-Fi link, a 100 Mbps switch port or the ISP (internet service provider) plan) and the end-to-end number equals that hop.

Why it's here. This is the plate for the conversation that starts with the Wi-Fi is down. Wi-Fi is one hop of several, and most of what gets called a Wi-Fi problem happens somewhere past it: the router, the modem, the provider, or the service at the far end. Showing the whole chain in one picture settles the argument faster than explaining it does.

How to use

1. Use it to locate a fault rather than to fix one. Work along the chain and ask which hop the evidence actually points at.
2. Show it to someone who believes Wi-Fi and the Internet are the same thing. That is what it is for.
3. Read the destination row at the bottom: every service shown there sits on the far side of the Internet, not on your Wi-Fi.
4. Beside the plate (below it on a phone), Try it: the slowest link wins has one control, Make this hop the slowest: Wi-Fi, Switch or ISP plan. Each hop shows what it carries, the slowest one is marked in words, and End to end always equals it. The last line says what a faster Wi-Fi link would change.

Inputs

Input	Unit	Range
Make this hop the slowest	choice	Wi-Fi (the device far from the AP, MCS (modulation and coding scheme) 1); Switch (a 100 Mbps switch port; default); ISP plan (every other hop fast)

How it works. No calculation on the plate itself. The plate is a drawing of the path, in connection order, with the local network and the Internet marked as separate things. The panel beside the plate: end to end = the smallest hop. Wi-Fi link = the 802.11ax PHY (physical layer) rate at 5 GHz, 80 MHz, 2 spatial streams and a 0.8 µs guard interval (MCS 9 near the AP, MCS 1 far from it, MCS 11 at best) x 0.6 efficiency (illustrative), the Repeaters and Mesh Backhaul model. Switch port = line rate x 1448 / 1538, the TCP (transmission control protocol) payload of a full-size Ethernet frame: 94.1 Mb/s at 100 Mbps, 941.5 Mb/s at 1 Gbps. ISP plan = 300 Mb/s (illustrative).

Example. A device shows full signal and nothing loads. Full signal means the first hop is healthy, which rules out the hop most people blame and points at everything after it. In the panel: a 100 Mbps switch port carries 94.1 Mb/s, so a Wi-Fi link carrying 576.4 Mb/s and a 300 Mb/s plan still

give 94.1 Mb/s end to end, and the best Wi-Fi link here (720.6 Mb/s) leaves it at 94.1. With the device far from the AP, Wi-Fi is the slowest hop at 86.5 Mb/s; the best Wi-Fi link raises the total to 300 Mb/s and no further, because then the plan is the slowest hop.

Field notes

- The single most useful idea on the plate is that the local network is not the Internet. A device can be perfectly associated, with a strong signal and a valid address, and still reach nothing, because association and Internet access are different achievements.
- Wi-Fi Information and Test My Connection answer those two halves separately.
- The panel's Wi-Fi efficiency and its 300 Mb/s plan are illustrative; the Ethernet figures are arithmetic. CWNA-109 objective 6.6.1 lists LAN port speed and insufficient Internet or WAN bandwidth among the causes of insufficient throughput.

macOS Menu-Bar Wi-Fi

The RF data a Wi-Fi pro can pull from a stock Mac without a third-party app, across four built-in paths: the Option-click Wi-Fi menu, sudo wdutil info, the Wireless Diagnostics app, and the Shortcuts "Get Network Details" action. This screen owns the per-field "what each RF value means" reference.

Why it's here. Before you reach for any third-party Wi-Fi app on a Mac, the OS already exposes most of what you need: live RSSI, noise, channel, width, PHY, Tx rate. Knowing the four built-in paths and what each field means turns any borrowed Mac into a usable Wi-Fi read. This is the reference that names them and decodes the fields.

How to use

1. Start with the four-path overview: what each path gives you and whether it needs sudo.
2. Section A decodes the Option-click Wi-Fi menu fields (what each live value means and why a pro cares). Section B walks the wdutil info Wi-Fi block. Section C lists the Wireless Diagnostics Window-menu tools. Section D covers the Shortcuts "Get Network Details" action.
3. Hold Option and click the Wi-Fi menu-bar icon to see the live association detail inline, no app required.

Field notes

- The load-bearing honesty note: wdutil info needs sudo. Without sudo it prints only its usage text (checked on macOS 26.7.1); run `sudo wdutil info` for the Wi-Fi block with RSSI, noise and MCS. The callout states this plainly.
- Security caveat: the Option-click menu, wdutil and CoreWLAN all reported an Enhanced Open (OWE) network as open or None while the AP beacon advertised OWE (checked 25 October 2026). Check the beacon before calling a link unencrypted.
- The airport CLI is removed on current macOS and is NOT documented as usable. Do not reach for it.
- The Shortcuts "Get Network Details" action is the one path that exposes RF fields an app cannot otherwise read, and it works on iOS too. It is the same bridge the Wi-Fi Information tool uses.
- Data source: distilled from Apple docs plus corroborating sources. Reference text only.

Roaming Parameters

The 802.11k/r/v fast-roaming protocols (what each does, what it requires) plus RSSI/SNR/latency design thresholds for enterprise roaming.

Why it's here. When designing or troubleshooting roaming for VoIP/UC, confirming the protocol roles and the signal-overlap targets that make handoffs work.

How to use

1. The protocol heading shows the designation (lime) and full name.
2. In the thresholds table, the scenario word is status-tinted with a dot: green = good (the two design targets), amber = marginal (the overlap zone), red = bad (sticky-client trigger and unusable).
3. Design rules carry the actionable target (e.g. "≥ 2 APs at −67 dBm everywhere," "second AP at −72 dBm or better across the overlap zone").

Field notes

- What it shows: Protocols block: 802.11r (Fast BSS Transition), 802.11k (Neighbor Report), 802.11v (BSS Transition Management), each with what it does, deployment requirements, and a field note. Thresholds block: five scenarios (VoIP/UC design target, standard data design target, roaming overlap zone, sticky-client trigger, unusable) each with min RSSI, min SNR, roam latency, design rule, and a status verdict.
- The intro states targets vary by client hardware and AP vendor: design guidelines, not guarantees.
- Field notes flag real client behavior: some legacy clients have 802.11r compatibility issues; Android/iOS generally honor 802.11v but some Windows drivers ignore BSS-TM entirely.
- Data source: IEEE 802.11k/r/v. The reference defines exactly these three protocols; OKC is deliberately not included.

Throughput Testing: Where You Test

A reference plate showing how the placement of a test server decides what a throughput number actually measures: the Wi-Fi link, the local network, or the path out to the Internet. Beside it, one control shows the slowest link winning: pick which hop is slowest (the Wi-Fi link, a 100 Mbps switch port or the ISP (internet service provider) plan) and the end-to-end number equals that hop.

Why it's here. Almost every argument about a disappointing speed test is really an argument about where the test server was. A number taken across the Internet includes the WAN, the provider and the far end, none of which are Wi-Fi, and it gets quoted as a Wi-Fi result anyway. Putting the test server in the right place is the difference between measuring the thing you are being asked to fix and measuring everything else.

How to use

1. Decide what you are actually trying to prove before you test.
2. To measure the Wi-Fi link, put the server on the wired side of the same AP, so the radio is the only variable.
3. To measure the local network, put it across the switch fabric on the far side of the routing you care about.
4. To measure the Internet path, use a public server and accept that the result includes everything in between.
5. Say which of the three you measured when you report the number.
6. Beside the plate (below it on a phone), Try it: the slowest link wins has one control, Make this hop the slowest: Wi-Fi, Switch or ISP plan. Each hop shows what it carries, the slowest one is marked in words, and End to end always equals it. The last line says what a faster Wi-Fi

link would change.

Inputs

Input	Unit	Range
Make this hop the slowest	choice	Wi-Fi (the device far from the AP, MCS (modulation and coding scheme) 1); Switch (a 100 Mbps switch port; default); ISP plan (every other hop fast)

How it works. No calculation on the plate itself. The plate lays out the three common server placements against what each one includes and excludes. The panel beside the plate: end to end = the smallest hop. Wi-Fi link = the 802.11ax PHY (physical layer) rate at 5 GHz, 80 MHz, 2 spatial streams and a 0.8 µs guard interval (MCS 9 near the AP, MCS 1 far from it, MCS 11 at best) x 0.6 efficiency (illustrative), the Repeaters and Mesh Backhaul model. Switch port = line rate x 1448 / 1538, the TCP (transmission control protocol) payload of a full-size Ethernet frame: 94.1 Mb/s at 100 Mbps, 941.5 Mb/s at 1 Gbps. ISP plan = 300 Mb/s (illustrative).

Example. A client associated at a 1,200 Mbps PHY rate tests at 94 Mbps against a public server and the Wi-Fi gets blamed. The same client against a server on the AP's own switch returns 700 Mbps. The Wi-Fi was never the limit; the 100 Mbps uplink was. In the panel: a 100 Mbps switch port carries 94.1 Mb/s, so a Wi-Fi link carrying 576.4 Mb/s and a 300 Mb/s plan still give 94.1 Mb/s end to end, and the best Wi-Fi link here (720.6 Mb/s) leaves it at 94.1. With the device far from the AP, Wi-Fi is the slowest hop at 86.5 Mb/s; the best Wi-Fi link raises the total to 300 Mb/s and no further, because then the plan is the slowest hop.

Field notes

- A wired test server needs enough headroom not to become the bottleneck itself, which in practice means the wired path should have roughly an order of magnitude more capacity than the radio can use.
- Test in both directions. Uplink and downlink fail differently and for different reasons.
- A single number from a single spot is a data point, not a survey.
- The panel's Wi-Fi efficiency and its 300 Mb/s plan are illustrative; the Ethernet figures are arithmetic. CWNA-109 objective 6.6.1 lists LAN port speed and insufficient Internet or WAN bandwidth among the causes of insufficient throughput.

Wi-Fi Tools Comparison

A vendor-neutral, offline reference that compares professional Wi-Fi survey, design, spectrum-analysis, and troubleshooting toolkits, grouped by the activity each one serves. Each config lists its vendor, product, license model, up-front cost, and 3-year total cost of ownership, with a neutral note on what the bundle does and does not include. A per-vendor summary and a typical-toolkit roll-up ride alongside.

Why it's here. There is no neutral, capability-level map of the professional Wi-Fi tooling landscape. This is that map, built from Keith's vendor-interviewed workbook, so a leveling-up engineer can see the four activities and which tools serve which job without a vendor sales pitch.

How to use

1. Browse by activity: Design, Validation, Spectrum Analysis, then Troubleshooting. Within each activity, configs are listed alphabetically by vendor.
2. Search by vendor (e.g. Ekahau), product, activity (e.g. spectrum), capability (e.g. survey), or license model (e.g. perpetual). The match is a case-insensitive substring and narrows the activities in place.
3. Read the up-front and 3-year TCO figures as planning estimates, not quotes. Use the vendor Website and Docs links to confirm current pricing and product details before you buy.
4. A query that matches nothing shows an honest "No match" card; it never fabricates a tool. The typical-toolkit roll-up and the per-vendor summaries are always available below the activities.

Field notes

- The figures are still being confirmed with vendors, and a few may change before the final release.
- Pricing is dated. The figures are as of July 2026; confirm current pricing with each vendor, because prices, bundles, and product names change often.
- Cost figures are MODELED ESTIMATES assembled by WLAN Pros from vendor-supplied numbers and list pricing, not vendor-published quotes. Treat every dollar amount as a planning estimate, never a binding price.
- This is NOT a ranking. There is no score and no "best". Tools are listed alphabetically and the same vendor can appear under more than one activity, because real toolkits are assembled across vendors. The set reflects vendors interviewed by WLAN Pros, not every tool that exists,

so an omitted tool is not a snub.

- No vendor logos or product photos appear here. Those are trademarks and copyrighted images that need each vendor's written permission, which is still being gathered. The reference is text and data only for now.
- Fully offline: the comparison is bundled in the app and works with no connection.
- Source: the Wi-Fi Design, Validation, Spectrum Analysis & Troubleshooting Tools V6 workbook (Keith Parsons, vendor-interviewed, last revised 2026-02-16).

Capture & Troubleshooting (9)

802.11 Reason Codes

The 802.11 deauthentication/disassociation reason codes (RC) and association status codes (SC) that appear in captures, with a searchable filter.

Why it's here. When a capture shows a deauth with reason code 15 or an association response with status 17, and you need the plain-language meaning fast.

How to use

1. Type a code number or keyword (e.g. "15" or "handshake") to filter; a "no match" card appears if nothing matches.
2. Reason codes (RC) appear in Deauthentication and Disassociation frames; status codes (SC) appear in Authentication, Association, and Reassociation Response frames.
3. Code 0 in the status group is the success value, rendered green.

Field notes

- What it shows: reason codes grouped by theme: Common (1 to 9), Capability/Channel mismatch (10 to 11), Security frame/element errors (13 to 14, 17 to 22, 24), Security handshake failures (15, 16, 23), QoS/load management (34 to 39), Fast Roaming/802.11r (45 to 48). Plus a separate Association Status Codes group (the most-common subset, 0 to 104), where code 0 ("Successful") is highlighted in green.
- The status-code list is the "most common" subset, not the full table.
- The code numbers and their meanings follow the IEEE 802.11 reason/status code definitions; nothing is invented.
- Data source / standard: IEEE 802.11-2020 §9.4.1.7 (reason codes) and §9.4.1.9 (status codes), cited in the footnote.

Association Sequence

The frame-by-frame 802.11 association and roaming sequences, showing the order and direction of frames between the STA, AP, RADIUS server, and DHCP server.

Why it's here. When analyzing a capture or explaining an association/roam, confirm what frame should come next and which entities exchange it.

How to use

1. Pick a scenario (Open / WPA2-PSK, WPA3-SAE, OWE / Enhanced Open, Passpoint / Hotspot 2.0, WPA2-Enterprise (802.1X/EAP), 802.11r Roam); read the phases top to bottom.
2. Each frame carries a neutral type code: MGMT (management frame), EAP (EAP / EAPOL key), WIRED (RADIUS/DHCP over the wire), DHCP.
3. The legend at the bottom expands each code. Type is carried by the text code, not by color.

Field notes

- What it shows: a scenario selector with six scenarios (Open / WPA2-PSK, WPA3-SAE, OWE / Enhanced Open, Passpoint / Hotspot 2.0, WPA2-Enterprise 802.1X/EAP, and 802.11r FT roam). Each scenario is broken into named phases (e.g. Probe & Auth, Association, 4-Way Handshake, EAP Authentication, DHCP); each frame shows a step number, direction, frame name, a frame-type code, and an explanatory note.

- These are representative sequences, not exhaustive; optional/passive-scan paths and EAP-method round-trips are summarized (e.g. the WPA3 DHCP phase is shown as one combined Discover/Offer/Request/Ack line "identical to WPA2 flow").
- Provenance: the frame sequences are ported from the RF Tools PWA (FX_SCENARIOS) and represent the real 802.11 exchanges (SAE Dragonfly commit/confirm for WPA3, the 4-way handshake, 802.1X/EAP over RADIUS, 802.11r FT over-the-air). They are representative teaching sequences, not reproduced clause-by-clause from the standard. Not region-specific.

Find the Switch and Port (LLDP/CDP)

A how-to reference for reading LLDP (IEEE 802.1AB) or CDP so a switch tells you its own name, the exact port you are on, its management IP, and often the VLAN. Covers the fastest path (switch CLI), the built-in-vs-capture split per OS, the Windows `Get-NetLldpAgent` correction, and the "which port is the AP on" workflow. The app does not capture; it points you at the tools already on your machine.

Why it's here. An AP is dark and you need to know which switch port it lives on, or a device is plugged in somewhere and you need to trace it without walking the patch panel. LLDP and CDP put that answer in a single frame. This tells you how to read it on whatever OS is in front of you.

How to use

1. If you have switch CLI access, start there. It is the fastest and it names the AP directly.
2. Otherwise pick your OS row below and run its built-in path. Linux prints a parsed table; macOS and Windows need a capture with a tool that already ships with the OS.
3. Remember the two hard limits: LLDP and CDP are wired only (a Wi-Fi laptop sees nothing) and single hop (you only see the device on the other end of your cable).

Fastest path: switch CLI. `show lldp neighbors [detail]` or `show cdp neighbors [detail]` lists every neighbor, local and remote port, platform, and management IP. This is where you see the AP and its port, with no host tooling at all. Alongside it: `show interfaces status` (link up/down and speed) and `show power inline` (whether the port is delivering PoE and how much) separate "no link" from "link but no power" from "powered but not booting."

Per-OS one-liners.

- **Linux (built in, cleanest):** `lldpcli show neighbors` from the `lldpd` daemon. Add `details` for the full record. Install first if needed: `sudo apt install lldpd` (or `dnf install lldpd`), then `sudo systemctl enable --now lldpd`.
- **macOS (built-in tcpdump, needs sudo):** `sudo tcpdump -nn -v -i en5 'ether proto 0x88cc'` for LLDP on the wired interface (usually a USB/Thunderbolt adapter like `en5` / `en7`). For CDP, capture the same way but match the CDP MAC `01:00:0C:CC:CC:CC` (CDP has no EtherType).
- **Windows (built-in pktmon, no driver):** `pktmon` filters on `0x88CC` and the CDP MAC, captures, then `pktmon etl2txt` converts the log to readable text. Built into Windows 10 1809 and later, Windows 11, and Server 2019 and later.

The Windows correction. `Get-NetLldpAgent` does NOT show the neighbor. It reads the local agent config only (interface alias, index, scope, MAC), and its module needs the Data Center Bridging feature. "LLDP is enabled" on Windows means the local agent is on, not that you have a neighbor viewer. To read which switch and port, use `pktmon`, not the cmdlet.

The AP-port workflow.

1. Switch CLI access? `show lldp neighbors` / `show cdp neighbors` names the AP against its switch port. Done.
2. No switch access? Read LLDP from the AP itself, via its console or management UI. The AP's own table names the switch and port.
3. Your laptop's LLDP reports the port the LAPTOP is on, NOT the AP's port. Do not confuse the two.
4. Confirm by plugging a laptop into the exact port and cable the AP used, then read LLDP there before you re-patch.

Field notes

- The app runs no capture and no shell. It is reference text; you run these commands on the switch, the WLAN Pi, or the laptop yourself.
- Nothing shows up? Cisco ships LLDP off by default (CDP on), and many non-Cisco switches ship LLDP off too. Check that the switch is sending before you blame the tool.
- Enable on a Cisco switch with `lldp run` (global), `lldp transmit` / `lldp receive` (per interface). For any other vendor (Aruba, Juniper, UniFi), the enable command is not universal: check your vendor.

- Client-Windows (Windows 10/11, not Server) availability of the `NetLldpAgent` module and the Data Center Bridging feature varies by build and NIC driver and is unconfirmed here; on Windows Server the feature install is documented (`Install-WindowsFeature Data-Center-Bridging`).
- Source / basis: fact-checked LLDP/CDP how-to brief (2026-07-15), cross-checked against the Wireshark LLDP wiki, Microsoft Learn (Get-NetLldpAgent, pktmon), lldpd.github.io, Baeldung/tcpdump, and Study-CCNA/Cisco docs.

How to NOT Have a Wireless Problem

A run-through-it AP install checklist organized into Before / Install / After phases, most of which is wired-side verification, on the premise that many "wireless" problems are not wireless problems.

Why it's here. You're mounting and turning up an access point and want to confirm the cabling, PoE, VLAN, DHCP, and routing path are right before you blame the radio, then document and validate after.

How to use

1. Three phases; tap each item as you complete it; the top count tracks progress.
2. Remember the premise: many wireless problems are not wireless problems. Work these checks before, during, and after the install, using a LinkSprinter, LinkRunner AT, EtherScope, or CyberScope, or just a laptop with a command window and the right commands.

Example. Before Installing Access Point: Cable meets/exceeds Cat5e specs · Total cable distance with patch cords < 100 m · PoE meets the AP's specific requirements · Check 802.3 af, at, or bt · Confirm DHCP address & VLAN · Confirm correct VLAN assignment · Confirm access or trunk port as required · Confirm default gateway · Ping default gateway · Confirm target IP addresses reachable · Confirm DNS reachable · Confirm target DNS addresses reachable · Management VLAN assigned & available. Install Access Point: Install access point (kept as its own one-item phase). After Installing Access Point: Document AP's MAC & assigned name · Document AP's location · Document AP's switch / port used · Document AP's IP address · Confirm AP installed in proper orientation · Confirm external antennas installed correctly · Wait for access point to receive configuration · Wait for a 2nd reboot of the AP if needed · Listen in air for all SSIDs being broadcast · Connect client to each SSID · Check each SSID for proper VLAN & IP pool.

Field notes

- Checked state is not saved; it resets when you leave the screen.
- Source / basis: Keith Parsons / WLAN Pros original card (© 2024 WLAN Pros). The "After Installing" list is renumbered to a clean 1-11 (the original card was gap-numbered with no item 2, so the eleven items now read 1 through 11).

Linux / WLAN Commands

A grouped Linux command reference for WLAN work: file/process basics, modern and legacy networking, the wireless-specific tools (iw, iwconfig, airmon-ng, rfkill), tested monitor-mode sequences, and the macOS non-root packet-capture setup.

Why it's here. You're driving a WLAN Pi, a Linux capture box, or a survey laptop and need to put an adapter into monitor mode on a specific channel/width, or recall the exact iw syntax to read the current link.

How to use

1. Commands are grouped (File, Directory, Process, Network, Wireless, Monitor-mode, macOS capture).
2. Filter by command or group name; a group-label match surfaces the whole group.

Example. Commands as shipped. Wireless: iw dev, iw dev wlan0 info (type/channel/mode), iw dev wlan0 link (SSID/signal/rate), iw dev wlan0 scan (sudo), iw dev wlan0 set channel 6, iw phy (PHY caps), iwconfig (legacy), iwlist wlan0 scan (legacy), rfkill list, rfkill unblock wifi. Monitor-mode: sudo airmon-ng start wlan0 (creates wlan0mon), sudo airmon-ng start wlan0 36 (monitor + channel 36), sudo airmon-ng stop wlan0mon, sudo airmon-ng check kill, sudo ifconfig wlan0 down / sudo iwconfig wlan0 mode monitor / sudo ifconfig wlan0 up (3-step), sudo iw dev wlan0 set channel 36 HT40+ (40 MHz secondary above), sudo iw dev wlan0 set channel 40 HT40- (40 MHz secondary below), sudo iwconfig wlan0 mode managed, sudo iw dev wlan0 info, lsusb, sudo dmesg, sudo ethtool -i wlan0, lsmod. macOS capture: sudo dseditgroup -o edit -a USERNAME -t user access_bpf (non-root capture), dscl . read /Groups/access_bpf (verify membership), sudo wdutil info (macOS 14+).

Field notes

- Caveat: iwconfig/iwlist/ifconfig are legacy "wireless extensions" tools; modern distros prefer iw and the iproute2 suite; monitor-mode sequences need sudo and a capable adapter/driver.
- Footnote: wireless extensions are deprecated in favor of iw + nl80211 (both shown because field gear still ships the legacy tools); HT40+/HT40- selects a 40 MHz channel with the secondary 20 MHz channel above (+) or below (-) the control channel; the access_bpf group lets a non-root macOS user capture via libpcap (BPF devices). Replace USERNAME and wlan0 with your actual user and interface.
- Source / basis: targets Linux primarily, with a macOS-capture group. Cross-checked against Linux man-pages, iw/nl80211 docs, and aircrack-ng docs.

Network CLI Commands

A three-column Windows, macOS, and Linux command reference for the everyday network-troubleshooting tasks (reachability, path tracing, DNS, interface config, sockets, ARP, routing, Wi-Fi link state), each with the field-common flags. A trailing Linux-only "shell essentials" group covers the capture-rig / WLAN Pi context.

Why it's here. You're at a client site on whatever laptop is in front of you and need the right command for this OS without looking it up: "what's the macOS equivalent of ipconfig /all," "how do I see the connected SSID/BSSID/RSSI from the CLI on Windows," "what's the Linux version on the WLAN Pi."

How to use

1. One card per task. Each card shows the Windows command, the macOS command, and the Linux command in three separate columns, plus a one-line description and a flag subset. Where macOS and Linux genuinely match, they read identically; where they diverged, each column carries its own command.
2. Filter by command name or task (e.g. "ping" or "DNS"). Where a platform has no native command, the card says so honestly rather than blanking. WLAN-relevant tasks lead.

Example. Commands as shipped (Task | Windows | macOS | Linux | key flags): Connected Wi-Fi interface state | netsh wlan show interfaces | wdutil info (needs sudo) | iw dev wlan0 link | show interfaces (Win), sudo wdutil info (macOS), iw dev wlan0 link (Linux). List visible Wi-Fi

networks | netsh wlan show networks mode=bssid | wdutil info | iw dev wlan0 scan (sudo) | (macOS removed the airport CLI). Reachability/RTT via ICMP echo | ping host | ping host | ping host | -t (Win continuous), -n count (Win), -c count (nix), -i interval. Trace L3 path | tracert host | traceroute host | traceroute host | -d (Win), -m max (nix), -I (nix ICMP), -T (Linux TCP SYN). DNS query full detail | (no native command) | dig name | dig name | +short, -x addr, @server. Interface IP config | ipconfig /all | ifconfig | ip addr | /all, /release, /renew, /flushdns. Active connections | netstat -ano | netstat -an | ss -tunap | -a, -n, -o (Win PID). ARP cache | arp -a | arp -a | ip neigh | -a, -d addr. IP routing table | route print | netstat -rn | ip route | print (Win), -rn (macOS).

Field notes

- The 3-column split is deliberate: macOS and Linux have diverged enough (ifconfig vs ip, netstat vs ss, DHCP renew, flush DNS) that folding them into one "macOS/Linux" column would ship a wrong command on one of the two platforms. Where they are identical, the two columns simply read the same.
- The caveat warns that some commands need administrator/sudo rights and that the flags shown are the field-common subset, not exhaustive.
- The footnote notes that ifconfig, route, arp, iwconfig, and netstat are legacy on Linux (modern distros prefer the iproute2 suite: ip addr, ip route, ip neigh, iw, ss); on macOS use wdutil info (sudo) or the Wireless Diagnostics app for Wi-Fi link details; and netsh wlan is Windows only.
- The reference is broader than the tasks shown above. The Windows column also carries the profile commands `netsh wlan show profiles` and `netsh wlan show profile name="SSID" key=clear`, plus pathping and nbtstat, alongside the everyday reachability/DNS/interface/routing tasks. The latter profile command reveals a saved network's plaintext key and is flagged as sensitive output.
- Source / basis: data consolidated from Keith's Network CLI sheet plus the WLAN Pros Linux cheat sheets, reconciled against current Windows/macOS/Linux docs. The macOS Wi-Fi entry shows only wdutil info; the deprecated airport CLI was removed entirely.

VoIP over Wi-Fi Filters

Wireshark display filters for analyzing voice calls carried over Wi-Fi: the SIP signaling, the RTP and RTCP media, and then the three questions that are specifically about the air rather than about VoIP in general. Did the QoS marking survive the trip onto the wireless side, did the call break

at a roam, and is power save eating it. A seventh group covers the loss and jitter figures Wireshark computes as statistics rather than as filterable fields.

Why it's here. Anyone can list `sip` and `rtp`, and the web is full of enterprise VoIP filter sheets built around firewall and session-border playbooks. A WLAN pro opens a capture for a different reason: a call sounded bad, the wired side looks perfect, and the answer is usually in the 802.11 header. These are the filters that put the IP marking and the 802.11 User Priority on screen at the same time, and the ones that line an RTP gap up against a reassociation. This is the companion to the Wireshark 802.11 Filters card, which carries no VoIP filters at all, and to the DSCP / QoS Markings reference, which explains the marking this card teaches you to check.

How to use

1. Filters are grouped by the question they answer; filter the list by syntax or task, and a group-label match surfaces the whole group.
2. The syntax is selectable for copy.
3. Several groups carry a short note above their filters. That note is the method, not decoration: it says what the filter is for and how to read what comes back.
4. The last group is deliberately not display filters. Those are tshark statistics taps and Wireshark menu paths, and typing them into the filter bar will not work.

Example. Filters as shipped. Find the call, signaling: sip, sip.Method == "INVITE", sip.Method == "BYE", sip.Status-Code >= 400, sip.Status-Code == 486 (Busy Here), sip.Call-ID (present on every message of one dialog), sip.resend == 1 (a retransmitted SIP message), sdp, sdp.media.format, udp.port == 5060, tcp.port == 5060. Find the media: rtp, rtcp, rtp.ssrc == 0x12345678 (one stream, by its synchronization source), rtp.seq, rtp.timestamp, rtp.marker == 1 (talkspurt starts), rtp.p_type, and the payload types rtp.p_type == 0 (G.711 PCMU), == 8 (G.711 PCMA), == 9 (G.722), == 18 (G.729), plus rtp.setup-frame. Did the QoS marking survive: ip.dsfield.dscp == 46 (EF, what voice should be marked on the wire), wlan.qos.priority == 6 (User Priority 6, the Voice access category on the air), ip.dsfield.dscp == 46 && wlan.qos.priority != 6 (the bug: marked EF on the wire, not Voice on the air), ip.dsfield.dscp == 46 && wlan.qos.priority == 5 (the classic signature, EF landing in the Video access category), rtp && wlan.qos.priority == 0 (media riding Best Effort), ip.dsfield.dscp == 0 && rtp, sip && ip.dsfield.dscp == 0, ip.dsfield.dscp == 40 (CS5), ip.dsfield.dscp == 34 (AF41), wlan.qos.priority. Did it break at a roam: wlan.fc.type_subtype == 0x02 (Reassociation request, the roam itself), == 0x03 (Reassociation response), == 0x0b (Authentication), == 0x0c (Deauthentication, a roam that was not the client's idea), wlan.tag.number == 54 (Mobility Domain element, present when 802.11r Fast Transition is in play), rtp.ssrc == 0x12345678 && wlan.fc.retry == 1. Is power save

eating it: wlan.fc.pwrmgt == 1, wlan.fc.pwrmgt == 1 && rtp (a voice client sleeping mid-call), wlan.fc.type_subtype == 0x2c (QoS Null), == 0x1a (PS-Poll), wlan.qos.eosp == 1 (end of a U-APSD service period). What the endpoint itself says: rtcp.pt == 200 (Sender Report), rtcp.pt == 201 (Receiver Report), rtcp.ssrc.fraction (Fraction lost), rtcp.ssrc.fraction > 0, rtcp.ssrc.cum_nr (Cumulative number of packets lost), rtcp.ssrc.high_seq (Highest sequence number received), rtcp.ssrc.jitter (Interarrival jitter), rtcp.ssrc.jitter > 30, rtcp.senderssrc. Loss and jitter as statistics rather than filters: tshark -q -z rtp,streams -r capture.pcapng, tshark -q -z sip,stat -r capture.pcapng, tshark -q -z follow,sip -r capture.pcapng, and Telephony then RTP then RTP Streams in the Wireshark GUI.

Field notes

- The teaching point sits on the card, not only in the help. DSCP lives in the IP header and User Priority lives in the 802.11 header, so something has to map one to the other. When that mapping is missing, wrong, or stripped by a tunnel, the call competes with everything else on the channel and the wired capture still looks perfect.
- Most equipment, with no explicit policy applied, derives User Priority from the top three bits of the DSCP value. EF is 46, which is 101110 in binary, and the top three bits are 101, which is 5. So EF lands in User Priority 5, the Video access category, rather than User Priority 6, which is Voice. That is why the `ip.dsfield.dscp == 46 && wlan.qos.priority != 6` filter usually turns up UP 5 rather than nothing, and it is why the card ships a filter for that exact case. The DSCP / QoS Markings reference carries the full mapping table and the RFC 8325 fix.
- Reading a roam: filter to one rtp.ssrc, note the sequence numbers either side of the gap, then look at what the client did in between. A roam that costs 300 ms is audible. A roam that costs 50 ms is not. An RTP gap that lines up with a reassociation is a roaming problem wearing a VoIP costume.
- Reading power save: a voice client that sleeps between packets sounds exactly like a network with loss, and the far end's own report will call it loss. The Power Management bit rides in the frame control field of every frame, so watch for where it flips rather than looking for a single announcement.
- There are two sources for loss and jitter and they measure different things. The RTCP fields are what the far endpoint reported about what it received. The tshark rtp,streams tap is what this capture actually saw. When the two disagree, that is the result rather than an error: the capture point and the endpoint did not experience the same stream, which on Wi-Fi usually means the loss happened between them.
- Caveat: the filters that combine an IP field with an 802.11 field need both headers visible in the same frame. That means a monitor-mode capture of an open or decrypted network, not a capture taken on the client's own interface, where there is no 802.11 header to filter on. Wireshark also does not label a UDP stream as RTP unless it saw the SIP or SDP exchange that set the call up. A monitor-mode capture

started mid-call usually has not, so every rtp filter here will match nothing until you right-click one of the media packets, choose Decode As, and set it to RTP. The rtcp filters are unaffected.

- Worth flagging (intentional, not a defect): Wireshark has no `rtp.analysis` display filter of any kind. Its per-stream loss and jitter is a statistic, not a filterable field, so there is no field name to type into the filter bar however plausible one sounds. Four names that look right and do not exist are rtp.analysis.lost, rtp.analysis.jitter, rtp.analysis.delta, and rtp.analysis.out_of_seq. All four were compile-tested against Wireshark and rejected before this card shipped.
- Frame type and subtype values are shown in hexadecimal on this card (0x02 is Reassociation request, 0x0c is Deauthentication). The Wireshark 802.11 Filters card shows the same values in decimal. Wireshark accepts either form.
- The RTCP jitter figure is in RTP timestamp units, not milliseconds. At an 8 kHz sampling clock, one unit is 125 microseconds.
- Source / basis: every filter on the card was compiled with dftest, the display-filter compiler Wireshark ships, against Wireshark 4.6.6 on 2026-09-15, before it shipped. RTP and RTCP per RFC 3550, payload type numbers per RFC 3551, DSCP to User Priority per RFC 8325.

Wi-Fi Client Testing Checklist

An ordered, twelve-step client-side connectivity test to run from a client device after an install or when triaging a connectivity complaint.

Why it's here. A user reports "Wi-Fi is broken." This walks you from "can the client even see the SSID" through association, auth, DHCP, gateway/DNS reachability, data rate, and a speed test, in the order failures actually cascade.

How to use

1. One flat ordered list (no phases). Tap each as you complete it; the count tracks progress.

Example. Items in order: Can see all SSIDs being broadcast · Associate to target SSID · Complete SSID authentication · Receive an IP address via DHCP · Receive default gateway & DNS · Ping default gateway · Ping DNS · Ping remote IP address · Ping remote DNS address · Check client MCS · Check client Tx data rate · Complete network speed test.

Field notes

- Per-session only; not saved.
- Source / basis: Keith Parsons / WLAN Pros original card (© 2024 WLAN Pros).

Wireshark 802.11 Filters

Copy-ready Wireshark display filters (typed into the filter bar after capture) and capture filters (BPF syntax, applied during capture) for 802.11 analysis: frame type/subtype, addressing, BSSID/SSID, RadioTap metadata, and RSN cipher/AKM selectors. Now also carries the 802.11 status-code and reason-code lookup tables next to the filters, so the moment a filter surfaces a deauth or a failed assoc, the code's meaning is right there. It now pairs the 802.11 filters with a general TCP/IP display-filter set (IP addressing, TCP/UDP, and the common higher-layer protocols), so a capture that drops below the radio layer stays in one screen.

Why it's here. You have a capture open and need the exact display-filter field to isolate deauths, beacons, a specific BSSID, or a security cipher, without guessing field names from memory. Then, once the deauth is on screen, you need to know what reason code 15 actually means without leaving the tool.

How to use

1. Filters are grouped; filter the list by syntax or task, and a group-label match surfaces the whole group.
2. The syntax is selectable for copy.
3. Below the filters, the status-code and reason-code tables list the highest-frequency 802.11 codes: status codes appear in Auth/Assoc responses; reason codes appear in Deauth/Disassoc frames.

Example. Filters as shipped. Frame type/subtype (display): wlan.fc.type == 0 (all management), == 1 (all control), == 2 (all data); wlan.fc.type_subtype == 0 (Assoc req), 1 (Assoc resp), 2 (Reassoc req), 3 (Reassoc resp), 4 (Probe req), 5 (Probe resp), 8 (Beacon), 9 (ATIM), 10 (Disassoc), 11 (Auth), 12 (Deauth), 13 (Action), 24 (Block Ack Req), 25 (Block Ack), 26 (PS-Poll), 27 (RTS), 28 (CTS), 29 (Ack), 36 (Null data), 40 (QoS data), 44 (QoS Null). Address (display): wlan.addr == aa:bb:cc:dd:ee:ff (any field), wlan.ta, wlan.ra, wlan.sa, wlan.da. BSSID/SSID: wlan.bssid == ..., wlan.ssid == "MyNetwork", wlan.ssid contains "Guest". RadioTap: radiotap.channel.freq == 2412, radiotap.datarate >= 6, radiotap.dbm_antsignal > -70, radiotap.dbm_antnoise < -90, radiotap.channel.freq >= 2400 && < 2500 (2.4 GHz), >= 5000 && < 5900 (5 GHz),

>= 5925 && <= 7125 (6 GHz). Capture filter (BPF): type mgt, type ctl, type data, type mgt subtype beacon, type mgt subtype probe-req, type mgt subtype deauth, type ctl subtype rts, type ctl subtype ack, wlan host aa:bb:cc:dd:ee:ff. RSN cipher (display): wlan.rsn.pcs.type == 4 (CCMP-128, 00-0F-AC:4), == 8 (GCMP-128, 00-0F-AC:8), == 9 (GCMP-256, 00-0F-AC:9), wlan.rsn.gcs.type == 2 (group cipher TKIP, 00-0F-AC:2). RSN AKM (display): wlan.rsn.akms.type == 1 (802.1X, 00-0F-AC:1), == 2 (PSK, 00-0F-AC:2), == 8 (SAE / WPA3-Personal, 00-0F-AC:8), == 18 (OWE, 00-0F-AC:18). TCP/IP display filters. IP addressing: ip.addr == 10.0.0.5 (source or destination), ip.src == 10.0.0.5, ip.dst == 10.0.0.5, ip.addr == 192.168.1.0/24 (any address in a subnet, CIDR), !(ip.addr == 10.0.0.5) (exclude an address), ipv6.addr == 2001:db8::1, ipv6.src == 2001:db8::1, ipv6.dst == 2001:db8::1, ip.ttl < 5 (low TTL, near a routing loop or a traceroute). TCP / UDP: tcp.port == 443 (source or destination port), tcp.dstport == 22 (destination only), udp.port == 53, tcp.flags.syn == 1 && tcp.flags.ack == 0 (connection attempts), tcp.flags.reset == 1 (resets), tcp.flags.fin == 1 (graceful close), tcp.analysis.retransmission, tcp.analysis.zero_window (receiver told the sender to stop), tcp.analysis.flags (all of Wireshark's TCP expert findings), tcp.stream eq 0 (every packet of one conversation), tcp.len > 0 (segments carrying payload, excluding pure ACKs). Higher-layer protocols: icmp, icmpv6, arp, dns, dns.flags.response == 1 (responses only), http, http.request, http.response.code == 404, tls, tls.handshake.type == 1 (Client Hello), dhcp (was bootp in older Wireshark).

Field notes

- The groups above are a representative selection. The tool also ships a Retries / QoS / weak-signal group, an 802.11k / v / r roaming group, a Security / EAPOL (4-way-handshake) group, and an Operators reference group, alongside the frame-type, address, BSSID/SSID, RadioTap, capture-BPF, and RSN cipher/AKM groups shown here. It also ships three general TCP/IP display-filter groups: IP addressing, TCP / UDP, and higher-layer protocols.
- The TCP/IP display filters use Wireshark dfref field names (ip, ipv6, tcp, udp, icmp, arp, dns, http, tls). For the Layer 3-4 header fields those filters match on (the byte offsets and flags behind ip.ttl, tcp.flags, an ICMP type/code), see the Packet Decode reference.
- Caveat: display-filter field names match Wireshark's dfref; capture filters use libpcap/BPF "type/subtype" syntax and only work when capturing with a RadioTap/PPI header.
- Footnote: type_subtype is the combined value (type in the high bits, subtype in the low bits) matching IEEE 802.11 frame type/subtype assignments; capture filters require capturing with a RadioTap header (monitor mode); for the full RSN cipher/AKM number-to-name map, see the RSN groups or the WPA Security reference tool. The bundled status-code and reason-code tables list the highest-frequency 802.11 codes only (the full tables live in the 802.11 Reason Codes reference tool).
- Two deliberate corrections are baked in. (1) The RSN cipher-suite vs AKM tables were rebuilt from IEEE 802.11-2020 Tables 9-149 (cipher = wlan.rsn.pcs.type / wlan.rsn.gcs.type) and 9-151 (AKM = wlan.rsn.akms.type) because the original source card mislabeled cipher values

as AKM. (2) The 5 GHz/2.4 GHz/6 GHz band filters ship a deliberate safe fallback using documented radiotap.channel.freq ranges instead of the unverified radiotap.channel.flags.5ghz child-token. Band-edge detail: the 5 GHz range stops at < 5900 and the 6 GHz range starts at >= 5925, so center frequencies in the 5900 to 5924 MHz gap fall into neither band filter. This is intentional.

- Source / basis: targets Wireshark display-filter (dfref) and libpcap/BPF capture-filter conventions, sourced from the Wireshark dfref, the RadioTap dfref, pcap-filter(7), and IEEE 802.11-2020.

Networking & Protocols (14)

DHCP Options

The DHCPv4 option codes a network or Wi-Fi pro meets, led by Option 138 (CAPWAP-AC), how a lightweight AP learns its WLAN controller address from DHCP, plus the Option 53 message-type table.

Why it's here. Reading a DHCP capture or a scope config: this names the option code and what value it carries, with the controller-discovery option called out first.

How to use

1. Read the option-code table (Option 138 first, then standard options in code order) for code, name, and purpose.
2. Use the Option 53 message-type table to decode DISCOVER / OFFER / REQUEST / ACK and the rest.

Field notes

- What it shows: common DHCPv4 option codes with names and plain-English purpose, plus the Option 53 DHCP message types.
- Option 138 (CAPWAP-AC, RFC 5417) leads because controller discovery is the option most relevant to a Wi-Fi deployment.
- Data source / standard: the IANA BOOTP/DHCP Parameters registry; base options RFC 2132, relay agent info RFC 3046, domain search RFC 3397, message types RFC 2132 section 9.6. Offline, read-only.

DNS Record Types

The DNS resource-record TYPE codes a network or Wi-Fi pro meets: numeric TYPE code, what the record does, and the RFC that defines it.

Why it's here. When a lookup returns an unfamiliar record type, this says what it carries and where it is defined.

How to use

1. Scroll the table. Read by record type (A, AAAA, CNAME, MX, TXT, SRV, CAA, HTTPS/SVCB, and more) or by numeric TYPE code.

Field notes

- What it shows: each DNS resource-record TYPE with its numeric code, a plain-English purpose, and its defining RFC.
- Current registry state reflected: CAA is now governed by RFC 8659 (obsoleting RFC 6844); HTTPS/SVCB by RFC 9460.
- Data source / standard: the IANA DNS Resource Record (RR) TYPEs registry plus the defining RFC cited per row. Offline, read-only.

DSCP / QoS Markings

The Wi-Fi-to-wired QoS mapping: the four WMM Access Categories, the 802.11 User Priority values they carry, and the DSCP markings RFC 8325 recommends so wired and wireless agree.

Why it's here. QoS only works end to end when the wired DSCP markings and the Wi-Fi access categories line up: this is that mapping, with the default-mapping trap flagged.

How to use

1. Read the mapping table for each WMM Access Category, its User Priority values, and the recommended DSCP marking.
2. Read the warning callout beneath the table: the common default mapping demotes voice (EF/DSCP 46) into the video queue.

Field notes

- What it shows: two tables. The WMM-to-DSCP mapping table pairs each of the four WMM Access Categories (Voice, Video, Best Effort, Background), plus a Traffic-type column, with its 802.11 User Priority values, and the DSCP code points (name, decimal, binary) RFC 8325 recommends. Below it, a full DSCP class-point table lists the standard code points (DF/CS0, CS1 to CS7, AF11 to AF43, EF, VA) with name, binary, decimal, and a note.
- The voice-into-video trap is rendered as a warning callout, not buried in a footnote, because it is the misconfiguration this page exists to surface.
- Data source / standard: RFC 8325 (Mapping Diffserv to IEEE 802.11), IEEE 802.11e / 802.11-2020, IEEE 802.1Q, Wi-Fi Alliance WMM; DSCP values per RFC 2474 / 2597 / 3246 / 5865. Offline, read-only.

HTTP Methods & Headers

The HTTP request methods with their safe / idempotent properties, plus the common request and response headers met checking a captive portal, web service, proxy, or API.

Why it's here. Diagnosing a captive portal or an API call: this says whether a method is safe to retry and what a given header means.

How to use

1. Read the methods table for each method and whether it is safe (read-only) and idempotent (repeat = same effect).
2. Read the request- and response-header tables to decode the headers you see in an Inspector (HTTP Header) result.

Field notes

- What it shows: HTTP request methods (GET, HEAD, POST, PUT, PATCH, DELETE, OPTIONS, and more) with safe/idempotent flags, plus common request and response headers.
- Definitions: safe = read-only, no intended state change; idempotent = repeating the request has the same effect as making it once.

- Data source / standard: the IANA HTTP Method and HTTP Field Name registries; RFC 9110 (HTTP Semantics) for method semantics; PATCH per RFC 5789. Offline, read-only.

HTTP Status Codes

The HTTP response status codes, grouped by class, with a plain-English meaning for each. A fast offline lookup when a captive portal, web service, proxy, or API returns a code and you need to know what it means.

Why it's here. When a check returns 403 or 503, or a captive-portal probe comes back 511, and you want the meaning without leaving the toolbox or going online.

How to use

1. Type a code number or keyword (e.g. "404" or "redirect") to filter; a "no match" card appears if nothing matches.
2. The filter matches the code number, the reason phrase, and the plain-English meaning, so "timeout" finds 408 and 504.
3. Use the toolbar copy action to copy the full reference as tab-separated text, one section per class.

Field notes

- What it shows: codes grouped by class: 1xx Informational, 2xx Success, 3xx Redirection, 4xx Client Error, 5xx Server Error. Each row is the code number, its reason phrase, and a short meaning.
- 511 (Network Authentication Required) is the signature of a captive portal: the network blocks access until the client authenticates.
- 418 is registered in the IANA registry as "(Unused)". It is widely known as the "I am a teapot" joke code from RFC 2324; the tool labels it honestly and notes the history.
- Data source: the IANA HTTP Status Code Registry (the authoritative registry), fetched 2026-06-04. Most codes are defined by RFC 9110 (HTTP Semantics). Code numbers and reason phrases are verbatim from the registry; the plain-English meanings are written for this tool. Unassigned and obsoleted codes are omitted; nothing is invented.

IP Address Reference

IANA/IETF special-use address blocks for IPv4 and IPv6 (CIDR prefix, what each block is reserved for, and the defining RFC) plus the IPv6 text-notation rules.

Why it's here. When you see an unexpected address on a link, this tells you fast whether it is private, loopback, link-local, documentation, or multicast, and which RFC governs it.

How to use

1. Scroll the IPv4 and IPv6 special-use tables. Read by CIDR prefix.
2. Check the IPv6 notation card for canonical-form and zero-compression rules before reading or writing an IPv6 address.

Field notes

- What it shows: the IANA IPv4 and IPv6 Special-Purpose Address Registries (private-use, loopback, link-local, documentation, and more), each row carrying its defining RFC.
- The multicast blocks 224.0.0.0/4 (IPv4) and ff00::/8 (IPv6) are not in the special-purpose registries: they live in the separate IANA multicast registries and are sourced to RFC 5771 / RFC 1112 and RFC 4291 section 2.7, flagged with a footnote.
- Data source / standard: IANA IPv4/IPv6 Special-Purpose Address Registries (fetched 2026-06-08) cross-checked against each defining RFC; IPv6 notation per RFC 5952. Offline, read-only.

Naming & Addressing Conventions

Hostname / DNS-label rules, MAC EUI-48 and EUI-64 format, the U/L (universal/local) and I/G (individual/group) bits, and the OUI/CID concept.

Why it's here. When a hostname is rejected or a MAC looks locally administered or multicast, this names the rule and the bit that decides it.

How to use

1. Read the hostname rules for label length, allowed characters, and max FQDN length.
2. Use the MAC-format and U/L, I/G bit tables (and the first-octet bit-field diagram) to tell a universally administered address from a locally administered one, and an individual address from a group/multicast one.

Field notes

- What it shows: hostname/FQDN rules (RFC 952, RFC 1123 section 2.1, RFC 1035), MAC EUI-48/EUI-64 format, the U/L and I/G bit positions in the first octet, and OUI/CID assignment.
- The first-octet bit-field diagram is decorative for screen readers: every fact it depicts (U/L and I/G positions and meanings) is also in the bit table.
- Data source / standard: IEEE Std 802-2014 with RFC 5342 section 2.1 as the freely available IETF restatement, RFC 4291 Appendix A (Modified EUI-64). The IEEE Registration Authority is the sole assigner of OUI/CID. Offline, read-only.

OSI Model

The 7-layer OSI reference model: layer number, name, one-word function, PDU, example modern protocols, and typical hardware.

Why it's here. Localizing a fault: which layer is failing tells you which tool to reach for.

How to use

1. Scroll the table horizontally. Read by layer number.
2. Example mappings: L3 Network = Routing, Packet, IPv4/IPv6/ICMP/IPsec, router/L3 switch; L2 Data Link = Framing, Frame, Ethernet (802.3)/Wi-Fi (802.11)/802.1Q/ARP, switch/AP/bridge/NIC; L1 Physical = Bits, Bit, RF/fiber/copper, cable/radio/hub.

Field notes

- What it shows: the 7 layers, top (7 Application) to bottom (1 Physical), each with: layer number (lime index), name, a one-word function keyword, PDU (Data/Segment/Packet/Frame/Bit), example protocols, and typical hardware. A second card maps the OSI stack onto the TCP/IP (RFC 1122) 4-layer model, showing how the TCP/IP Link layer collapses OSI 1-2 and its Application layer collapses OSI 5-7.
- Footnote: PDU = protocol data unit; layers 5 to 7 are commonly grouped as "data" in TCP/IP practice; ARP is widely placed at Layer 2 (some texts call it L2/L3), and it resolves L3 addresses to L2 addresses.
- The function column is a neutral keyword (no custom mnemonic). Standard reference, not region-specific.
- Data source / standard: ISO/IEC 7498-1:1994 plus standard IETF/IEEE protocol-to-layer mappings.

Packet Decode

A byte-level Layer 3-4 reference for the headers a capture shows beneath 802.11: IPv4, IPv6, TCP, UDP, and ICMP, each as a field / bit-offset / bit-length / meaning table. Adds the TCP control bits, the TCP connection states with the three-way handshake and teardown, a common IP-protocol-numbers table, and the ICMP and ICMPv6 type and code tables.

Why it's here. You have a packet open in Wireshark and need to confirm which field sits at which offset, what a TCP flag combination means, or what an ICMP type and code is telling you, without leaving the app or going online. This is the companion to the Wireshark filters above: those surface the frame, this decodes the bytes inside it.

How to use

1. Scroll to the header you are decoding. Each table lists the field, its bit offset (counted from bit 0, the first / most-significant bit), its bit length, and a plain-language meaning; variable-length fields (Options, padding, payload) read "variable".
2. The IPv4 Type-of-Service octet is shown the modern way, DSCP (6 bits) plus ECN (2 bits); the legacy RFC 791 precedence reading is footnoted, not led with.
3. The IPv4 Protocol and IPv6 Next Header fields point at the IP-protocol-numbers table sitting between the two IP headers. For TCP and UDP port numbers, use the Well-Known Ports reference instead.
4. Copy any table from the toolbar as tab-separated text, one section per header.

Example. Tables as shipped. IPv4 header (field, bit offset, bit length): Version (0, 4), IHL (4, 4), ToS / DiffServ (8, 8), Total Length (16, 16), Identification (32, 16), Flags (48, 3), Fragment Offset (51, 13), Time to Live (64, 8), Protocol (72, 8), Header Checksum (80, 16), Source Address (96, 32), Destination Address (128, 32), Options (160, variable), Padding (variable). IPv6 header: a fixed 320 bits (40 octets) with no header checksum and no in-header fragmentation. TCP control bits (bits 104-111, MSB to LSB): CWR, ECE, URG, ACK, PSH, RST, SYN, FIN. TCP connection states: CLOSED, LISTEN, SYN-SENT, SYN-RECEIVED, ESTABLISHED, FIN-WAIT-1, FIN-WAIT-2, CLOSE-WAIT, CLOSING, LAST-ACK, TIME-WAIT. Common IP protocol numbers: 1 = ICMP, 6 = TCP, 17 = UDP, 47 = GRE, 50 = ESP, 51 = AH, 58 = ICMPv6, 88 = EIGRP, 89 = OSPF, 132 = SCTP. ICMP types: 0 Echo Reply, 3 Destination Unreachable (16-code table, 0-15), 4 Source Quench (deprecated), 5 Redirect, 8 Echo Request, 11 Time Exceeded, 12 Parameter Problem, 13 Timestamp, 14 Timestamp Reply. ICMPv6: error types 1-4, Echo Request / Reply 128 / 129, and Neighbor Discovery 133-137 (Router and Neighbor Solicitation and Advertisement, plus Redirect).

Field notes

- Bit offsets count from bit 0 of each header (bit 0 = first / most-significant bit); bit lengths are exact, never approximate.
- The TCP flags table uses RFC 9293's canonical layout: a 4-bit reserved field plus 8 control bits (CWR through FIN). The bit at position 103 that Wireshark still labels NS is Reserved in RFC 9293. It was NS (Nonce Sum, RFC 3540), which RFC 8311 made Historic, and RFC 9768 reassigns it as AE (Accurate ECN). It ships flagged historic in text, not as a current flag.
- ICMP type 4 (Source Quench) is kept in the table but marked deprecated per RFC 6633.
- Ports are not duplicated here; the app already ships a Well-Known Ports reference.
- Source / basis: every field offset, width, flag, state, type, and code is transcribed from the primary RFC or IANA registry and cited per table: IPv4 (RFC 791, DSCP / ECN per RFC 2474 and RFC 3168), IPv6 (RFC 8200), TCP (RFC 9293), UDP (RFC 768), ICMP (RFC 792 plus the IANA ICMP Parameters registry), and ICMPv6 (RFC 4443, NDP per RFC 4861). Nothing is invented or approximated.

PLMN ID Reference

An offline, searchable, grouped lookup of US Public Land Mobile Network identifiers. Each row pairs a carrier or operator with its MCC, MNC, full PLMN ID, and operational status, covering MCCs 310-316 (US mainland plus Puerto Rico, Guam, the US Virgin Islands, and American Samoa).

Why it's here. When you have a PLMN ID, MCC, or MNC off a cellular scan, a SIM, or a private-LTE / CBRS deployment and need to know which carrier it belongs to, or you need a carrier's code to configure or verify a private cellular network. It works fully offline, so it is reliable in the field where there is no data connection.

How to use

1. Browse the entries grouped by MCC (310 through 316), each group sorted ascending by PLMN ID.
2. Type in the search box to filter live by code (MCC, MNC, or full PLMN ID) or by carrier / operator name.
3. Each row shows the PLMN ID, the MCC/MNC pair, the carrier (and parent operator when different), and the operational status.
4. Use the copy action to grab the current view (the filtered subset when searching, otherwise the full table) as plain text.

Field notes

- MNC and PLMN ID are strings with significant leading zeros (e.g. MNC 004, PLMN ID 310004). A two-digit and a three-digit MNC are different codes; never read them as numbers.
- Status values include operational, not operational, reserved, and unknown. A reserved or not-operational allocation may still appear on the air or in old records, so the status column matters.
- US-only by design. The dataset covers ITU region 3xx (MCCs 310-316); it does not include non-US carriers.
- Data source: US MCC/MNC (PLMN ID) allocations verified against the live Wikipedia 'Mobile country code' tables (2026-06-05).

Speed Test Services

A curated, offline reference to the popular internet speed tests, framed on the two axes that actually change the number: single-stream vs multi-stream, and a nearby CDN edge vs a distant true server.

Why it's here. Two speed tests on the same connection can disagree by a wide margin, and the reason is almost never the connection. It is how many streams the test opens and how far away the server sits. This page lets you pick the right test for the question you are asking, and read a surprising result correctly instead of blaming the Wi-Fi.

How to use

1. Read each service against the two teaching axes: single vs multi-stream, and nearby-edge vs distant-server.
2. Search by name (e.g. Ookla, Fast.com, Cloudflare); a query that matches nothing shows an honest "No match" card.
3. Check the "Runs on" note before trusting a brand as independent: Waveform runs on Cloudflare, Fast.com on Netflix's CDN, ISP tests on Ookla or M-Lab. Tap a website chip to run that test.

Field notes

- Data-per-test figures are the weak column. Each carries a confidence marker ("est.", "rough est.", "measured"), and a persistent band states they are community-measured estimates, not vendor-published numbers.
- Not all of these are independent measurement backends. Where a brand rides on another service's network, the card shows a "Runs on" note.
- Orb is a continuous monitor, not a one-shot test; the Toolbox's own Network Quality tool is the analog and reports no single composite score.
- Fully offline: the service list is a bundled dataset, so the page renders with no network call. Vendor wordmarks render on a neutral chip so they read in both light and dark.

Subnetting / CIDR Table

A /0 through /32 lookup: prefix length, dotted subnet mask, total addresses, usable hosts, and wildcard (inverse) mask.

Why it's here. The fast way to read off a mask, host count, or wildcard for a prefix without doing the powers-of-two in your head.

How to use

1. Find the prefix length (/n) row and read the mask, total addresses, usable hosts, and wildcard across it.
2. For a specific network, broadcast, or host range, use the IPv4 Subnet Calculator tool.

Field notes

- What it shows: every prefix from /0 to /32. Total addresses for a /n is 2 raised to the power (32 minus n); usable hosts is that total minus 2, plus the dotted mask and the inverse (wildcard) mask.
- Exceptions honored: /31 has 2 usable host addresses (point-to-point, RFC 3021), not 0; /32 is 1 host (single-host route, RFC 4632), not -1. Both rows carry an inline note.
- Data source / standard: pure CIDR arithmetic cross-referenced to RFC 4632 section 3.1. Offline, read-only.

Top-Level Domains

A curated reference to the DNS top-level domains a network or IT pro actually meets, grouped by registry type: generic (gTLD), country-code (ccTLD), sponsored/restricted, infrastructure, and notable newer gTLDs. Each entry shows the TLD, its type, and a short managed-by / typical-use note.

Why it's here. Knowing a TLD's type tells you who runs it and what to expect: a sponsored domain like .gov or .edu is eligibility-verified, .arpa is reverse-DNS infrastructure you never register, and .io or .ai are country-code domains used generically rather than true gTLDs.

How to use

1. Scroll the grouped cards, or use the Type filter to narrow to one registry class.
2. Read the TLD (lime, left) then its note. Copy exports the full curated set as a table, regardless of the active filter.

Field notes

- Curated, not exhaustive: the live root zone has roughly 1,500 generic TLDs and about 250 country-code TLDs. This lists the meaningful, field-relevant set.
- Accuracy: .io (British Indian Ocean Territory), .ai (Anguilla), and .co (Colombia) are technically country-code TLDs commonly used generically. They are NOT true generic TLDs, and the per-entry notes say so.

- Data source / standard: the IANA Root Zone Database for registry classification and sponsoring organizations; ICANN for new-gTLD program facts.

Well-Known Ports

Searchable, offline reference of 89 curated TCP/UDP ports a network or Wi-Fi pro meets in the field. Search by port number or by service-name / description substring.

Why it's here. At a packet capture or writing a firewall rule, you ask two questions: "what runs on port N?" and "what port does service X use?" This answers both without leaving the app or going online.

How to use

1. Type a port number (e.g. 443) for an exact-port lookup, or a service name / keyword (e.g. radius, dns, vpn) for a case-insensitive substring match against both the service name and the description.
2. An empty search box lists all 89 curated ports, sorted ascending by port number.
3. Each result shows the service name, the port number with its protocol label (TCP, UDP, or both), and a one-line description.
4. A query that matches nothing shows an honest "No match" card; it never fabricates a port.

Field notes

- What it contains: 89 curated entries spanning ports 1 to 27017 (e.g. 53 dns TCP/UDP, 67 dhcp UDP, 123 ntp UDP, 443 https TCP/UDP where UDP/443 carries HTTP/3 QUIC, 1812 radius UDP and 1813 radius-acct UDP for 802.1X / WPA2-Enterprise, 3389 rdp). Each entry: port number, protocol(s), short service name, one-line description.
- Protocols in the table are TCP and/or UDP only (no SCTP entries present, though the schema supports it). Combinations: 38 TCP-only, 28 UDP-only, 23 TCP+UDP.
- This is a curated subset of the IANA Service Name and Transport Protocol Port Number Registry, trimmed to field-relevant ports, not the full ~49,000-entry registry. Absence means "not in our curated set," which the screen states plainly rather than inventing a row.

- Fully offline: the table is a bundled asset loaded and indexed once at startup; numeric lookups are instant via a port index. Works on every platform.

Cabling, Connectors & Hardware (12)

Antenna Connectors

A 19-connector practical reference for Wi-Fi antenna systems: each connector's full name, reverse-polarity variant, typical Wi-Fi use, indoor/outdoor fit, coupling mechanism, body size, RF signal path, impedance, frequency rating, mating compatibility, and field notes. Plus a polarity-explained diagram, an at-a-glance comparison table, enterprise vendor trends, a to-scale size comparison, and the top 6 a Wi-Fi engineer actually meets.

Why it's here. When you are identifying or mating an antenna lead in the field. It answers the questions a tech actually asks: is this RP-SMA or SMA, will these two mate, does it cover 6 GHz, and which connector does this vendor ship.

How to use

1. Browse the connectors grouped by where they live (enterprise panel/external, outdoor/point-to-point, board-level/internal, test/cellular). Each card shows the connector name, full name, a reverse-polarity marker when it applies, and a photo or line diagram.
2. Each card lists typical use, indoor/outdoor, coupling, size, RF path, impedance, frequency, and mating, then a field-notes line.
3. Type in the search box to filter live across every field (name, vendor, coupling, size, RF path, frequency, or any word in a note).
4. Below the table: a polarity-explained diagram, a Compare-at-a-glance table (connector, size, RF path, typical use) that scrolls sideways on a phone, enterprise vendor trends, the size order largest to smallest with a to-scale diagram, and the top 6 connectors most engineers meet.

Field notes

- The key field gotcha: nearly every connector here is 50 ohm. The one exception is F-Type, which is 75 ohm (a CATV connector folded in from the former RF Connectors reference and not used for Wi-Fi). An RP connector will thread onto its standard counterpart but the center contacts will not connect.
- U.FL intermates with I-PEX MHF I (same footprint); the smaller MHF 4 does NOT mate with either.
- DART is Cisco's Smart Antenna Connector, a proprietary multi-port interface that breaks out to RP-TNC, N-type or RP-N via Cisco adapter cables.
- Size is the connector body width (across-flats for threaded parts, outer diameter for board-level parts): an approximate field-recognition aid, not a precision mechanical spec.
- Some connectors show a photo; where one isn't available (N-Type, TNC, RP-TNC), a line diagram is shown instead.

Source: Keith Parsons.

Bend Radius & Pull Tension

Minimum bend radius and maximum pull tension for copper and fiber, with the TIA-568 numbers kept separate from the rules of thumb that a cable datasheet can override.

Why it's here. Before pulling or dressing a run, confirm how tight you can bend it and how hard you can pull it without degrading the cable.

How to use

1. Installed UTP bends no tighter than 4 times its outer diameter and pulls at no more than 25 lbf (110 N), both per TIA-568.
2. Fiber runs about 10 times its outer diameter installed and 20 times while under tension as rules of thumb; bend-insensitive G.657 goes tighter, and the cable datasheet always wins over any rule of thumb.
3. The 8-times-outer-diameter figure for copper during a pull comes from ISO 11801 and field practice, not TIA.

Field notes

- Over-pulling stretches the conductors and raises return loss and NEXT.
- Each row is labeled Standard or Practice so the distinction survives when you copy it.

Coax Cable

A coaxial cable reference: impedance, velocity factor, outer diameter, maximum usable frequency, and typical use for common RG- and LMR-series cables.

Why it's here. When specifying an antenna run, pick the right LMR size for the length and frequency, and avoid a 75 Ω mismatch.

How to use

1. Each cable is a block: name, then a spec line (impedance / VF% / diameter / max GHz), then the use note.
2. The 75 Ω entry (RG-6) is shown dimmed: it's impedance-mismatched for 50 Ω Wi-Fi and shown for reference only ("CATV/satellite, NOT for Wi-Fi").
3. Higher VF means slightly lower propagation delay and loss.

Field notes

- What it shows: per cable (RG-58, RG-8/U, RG-213, RG-214, LMR-100A through LMR-1200, RG-6): impedance, velocity factor (%), diameter (mm), max frequency (GHz), and a typical-use note.
- A footnote points to the Cable Loss tool for exact attenuation. Wi-Fi is a 50 Ω system, so the dimmed 75 Ω RG-6 is a mismatch. Max frequencies are typical maximums. Standard reference, not region-specific.
- Provenance: the figures are compiled from manufacturer datasheets and common industry specification values for the named cable series; they are widely-used working values, not reproduced from a single named standard.

Ethernet Cable & Connector

The consolidated twisted-pair reference in one tool: the Cat5e-through-Cat8 capability chart (bandwidth, max speed, distance at 1G/10G, PoE support, shielding, typical use) plus the T568A/T568B RJ-45 pinout. This brings the category capability and the pin colors into one place, so you don't have to flip between tools. Twisted-pair side only; coax has its own Coax Cable tool.

Why it's here. When you are choosing cable for a run and then terminating it, you want the category capability and the pin colors in one place, not two. Confirm a category's bandwidth and 10G reach, the PoE++ suitability (notably the Cat6A-for-PoE++ recommendation), and which color goes on which pin, without flipping tools.

How to use

1. Cat capability chart: scroll horizontally. "N/A" in a distance column means that rate isn't supported (e.g. Cat5e has no 10G distance). Key facts: Cat6 hits 10G only to 55 m; Cat6A hits 10G to the full 100 m and supports all 802.3bt; Cat8 carries 1G/10G to 100 m but its 25/40G design rate is limited to about 30 m.
2. T568 pinout: pick the standard (T568B is the default and most common). View is "plug face, clip down, pin 1 on the left." Striped wires show a split swatch (color over white), and the swatch colors are real copper-pair colors. T568A and T568B differ only in swapping the green and orange pairs.

Field notes

- Cat chart: per peer category (Cat5e, Cat6, Cat6A, Cat8) it shows max bandwidth (MHz), max speed, distance at 1 Gbps, distance at 10 Gbps, PoE support, shielding, and a typical-use note. Cat7 and Cat7A are deliberately NOT shown as peer rows. They are ISO/IEC Class F/FA (never ratified by TIA) and use GG45/TERA rather than RJ-45, so they live in a separate warning callout, not the chart. For 10G structured cabling, Cat6A is the TIA-recognized choice. Cells with no applicable value show "N/A".
- Two more reference cards ship below the chart: a BASE-T speed-grade table (1000BASE-T / 2.5GBASE-T / 5GBASE-T / 10GBASE-T, each with its data rate, minimum cabling, and IEEE spec) and an ISO/IEC 11801 shielding-code key decoding the U/UTP, F/UTP, S/FTP, SF/UTP, and U/FTP notation (overall shield / per-pair shield).

- PoE++ tip: bundled Cat6 running PoE++ generates significant heat; Cat6A dissipates it better, so TIA-568 recommends Cat6A for PoE++ in bundles.
- Pinout: each standard's eight pin rows show the pin number, a wire-color swatch and name (e.g. "Orange / White"), the twisted-pair number (1 to 4), and the 100/1000 Base-T function (TX+, RX-, BI-D A+, etc.). A crossover cable uses T568A on one end and T568B on the other, rarely needed today since most switches/NICs auto-MDI-X.
- Reference basis: the TIA-568 / ISO 11801 category and pinout conventions (compiled to reflect those standards, not reproduced verbatim from them); not region-specific.

Fiber Optic

Fiber types (OM1 to OM5, OS1/OS2) with core/cladding, modal bandwidth, jacket color code, and supported distance at 1G/10G/40G/100G.

Why it's here. When specifying fiber, confirm a fiber type's reach at a given rate, its jacket color, and whether it's current or legacy.

How to use

1. Scroll the distance matrix horizontally. A dash in a rate column means that rate isn't supported (OM1/OM2 don't do 40G/100G).
2. OM1/OM2 are shown dimmed (legacy). Jacket colors: OM1/OM2 = Orange, OM3 = Aqua, OM4 = Aqua (violet is a manufacturer convention, not the standard), OM5 = Lime Green, OS1/OS2 = Yellow.
3. Multimode (OM) lists modal bandwidth; singlemode (OS) shows "N/A" bandwidth but reaches 10+ to 80+ km.

Field notes

- What it shows: four headed blocks on one screen. Distance by data rate, per fiber type: core/cladding, modal bandwidth (MHz·km), and supported distance at 1G/10G/40G/100G. Jacket color code & notes, per fiber type: a jacket color swatch and name, and a deployment note. Then the Connectors block and the Polish & endface block (documented below as "Fiber Connectors & Polish", the same screen, not a separate tool), plus a "Two color systems" explainer.

- Footnote: distances are per TIA-568/ISO 11801; actual limits depend on transceiver, splice count, and connector loss. OM3/OM4 are the current deployment standards; OM1/OM2 are legacy (dimmed). OM5's wideband window note (~1,850 to 2,470 MHz·km near 953 nm) is preserved verbatim. Standard reference, not region-specific.
- Standard: TIA-568 / ISO 11801 (cited in the footnote).

Fiber Connectors & Polish (a section within the Fiber Optic tool)

This is not a separate catalog tile. It is the Connectors and Polish/endface portion of the Fiber Optic screen, documented on its own for clarity. The fiber connector form factors (LC, SC, ST, FC, MPO/MTP) with ferrule size, coupling, and use, plus the three endface polishes (PC, UPC, APC) and the two separate color systems: cable jacket color and connector body color.

Why it's here. When you are identifying, specifying, or mating a fiber connector and need to answer the questions that actually cost time in the field: is this LC or SC, will an APC mate a UPC, why is one connector green and another blue, and is that aqua jacket OM3 or OM4. The two single biggest field errors live here: mating APC to UPC, and reading the OM4 jacket color wrong.

How to use

1. Read the connector table for form-factor recognition: ferrule diameter, coupling/latch, simplex or duplex, and typical use.
2. Read the polish rows for endface type, the 8 degree APC angle, the typical return loss, and the connector body color that goes with each polish.
3. Use the two-color-system note to keep cable jacket color and connector body color separate; the same color word can mean two different things depending on which one you are reading.

Connector form factors

Connector	IEC 61754 part	Ferrule	Coupling	Form factor	Typical use
LC	61754-20	1.25 mm	Push-pull latch (RJ-style clip)	Simplex + duplex	Data center / enterprise; SFP/SFP+ transceivers; dominant today
SC	61754-4	2.5 mm	Push-pull snap	Simplex + duplex	FTTH, telecom, enterprise patching; second most common
ST	61754-2	2.5 mm	Bayonet twist-lock	Simplex only	Legacy campus / multimode LANs
FC	61754-13	2.5 mm	Threaded screw nut	Simplex only	Test equipment, precision / high-vibration
MPO / MTP	61754-7	Rectangular multi-fiber (8 / 12 / 24)	Push-pull, keyed	Multi-fiber ribbon	40G/100G/400G parallel optics; data-center trunks

LC and SC dominate new deployments; LC leads the data center because its 1.25 mm ferrule packs more ports into the same SFP density. MTP is US Conec's branded, tighter-tolerance MPO; it is mechanically intermateable with MPO, not a separate standard.

Endface polish

Polish	Endface	Return loss (typical)	Connector body color
PC (Physical Contact)	Slight dome, flat-ish	about -40 dB	Legacy, not separately color-keyed
UPC (Ultra PC)	Finer dome, no angle	about -50 dB	Blue (single-mode)
APC (Angled PC)	8 degree angled ferrule	about -60 dB (best)	Green

The APC ferrule is polished to an 8 degree angle that reflects back-reflection into the cladding, which is why it carries the lowest return loss. APC mates only to APC. Return-loss figures are typical industry numbers, not per-part datasheet guarantees.

Field notes

- HARD RULE: APC and UPC must never be mated. The 8 degree angled ferrule against a flat ferrule causes very high insertion loss and can physically damage both ferrules. Green mates only green.
- Two separate color systems exist and the page's two-color note keeps them distinct. Cable jacket color (TIA-598-D): orange = OM1/OM2, aqua = OM3/OM4, lime green = OM5, yellow = single-mode. Connector body color (TIA-568/598 convention), as the tool presents it: blue = single-mode UPC, green = single-mode APC. (More broadly in the field, the same body-color convention historically used beige for OM1 62.5/125 and black for OM2 50/125 connectors, but the app's note covers the blue/green single-mode pair that actually collides with the jacket colors.)
- The colors collide. Green appears in both systems: lime-green jacket means OM5 multimode, a green connector body means APC angled single-mode. Aqua appears in both: OM3/OM4 jacket and OM3/OM4 connector body. Always check which system you are reading before you trust the color.
- MYTH: "OM4 is violet." TIA-598-D assigns OM4 aqua, the same as OM3. Violet ("Erika Violet") is a manufacturer differentiation convention, not the standard color. Because OM3 and OM4 both default to aqua, the only reliable way to tell them apart is the printed legend on the jacket.
- Field aside (not on the screen): cladding is 125 microns across every fiber type in the distance matrix (9/125 single-mode, 50/125 OM2 to OM5, 62.5/125 OM1), a handy unifying fact when reading the core/cladding column.

Sources: TIA-598-D, IEC 61754 (FOA, Cisco, Fluke corroboration).

LED Decoder

An interactive cross-vendor decoder for an access point's status LED. Pick the vendor, pick the model line when the vendor forks (Cisco Catalyst vs Meraki, Aruba Campus vs Instant On, Extreme IQ Engine vs WiNG), then read that line's own color-and-blink state table: booting, needs adoption, healthy, upgrading, fault, locate, and factory reset. Each state carries a literal colored indicator, a green, amber, red, blue, white, purple, or magenta dot, solid or gently flashing, beside the verbatim signal text. A master cross-vendor comparison chart rides at the top of the vendor picker with the whole color matrix on one plate.

Why it's here. The color on the front of an AP is the fastest read you get before you open a laptop, but the same color means opposite things across vendors, so a flat "green equals healthy" legend is actively wrong. Solid green is healthy-but-no-clients on Meraki and healthy-with-clients on Ruckus; solid white is needs-adoption on UniFi and healthy-on-cloud on Extreme. Resolving the model line first is what keeps the read honest.

How to use

1. Scan the master cross-vendor comparison chart at the top of the picker to see the whole color matrix on one plate; tap it to pinch-zoom, or download it as a PDF.
2. Pick the vendor (enterprise lines first, consumer mesh kept separate).
3. If the vendor forks by management line, pick the line; a single-line vendor jumps straight to its table.
4. Read the state row: the colored indicator (color plus solid or flashing), the verbatim signal text, and what it means in the field.
5. Treat every color as a heuristic and confirm against the exact model's install or getting-started guide.

Field notes

- The colored dot is never the only signal: the color is always named in words beside it, and the verbatim signal text stays the authority for any nuance a dot cannot carry (sequences, alternating patterns, blink-count error codes).
- Undocumented states are marked honestly. A state with no reachable vendor doc renders "Not documented by the vendor, confirm on a lab AP" with a neutral "?" indicator and no invented color. There are exactly six such states.
- Some vendors ship no distinct signal by design (a Meraki factory reset reads as an ordinary reboot); the "reads as X" note is the answer, not a gap.
- MikroTik ships as an honest note, not a table: RouterOS LEDs are user-configurable, so there is no standardized status-LED scheme to decode.
- LED behavior can change with a firmware or dashboard release on cloud-managed lines. Reference only; confirm on the vendor's own documentation.
- Data source: per-line vendor docs, cited on the lines that have a published source (the Cisco Catalyst Getting Started Guides, the Meraki MR46 Installation Guide, the Aruba AP-635 and AP22 installation guides, the Extreme Networks documentation portal, help.ui.com, and

Ruckus KB 000001629). A few lines carry a descriptive read rather than a formal per-line citation, notably the consumer-mesh lines (Orbi, Eero) and Juniper Mist.

Optical Transceivers

Searchable, offline reference of 35 optical Ethernet transceiver variants (1G to 400G) grouped by speed tier, plus the 9-row SFP-to-OSFP form-factor ladder. Each variant lists its designation, data rate, reach, fiber type, wavelength, and connector, with reach as the lead, trust-first column.

Why it's here. Picking an optic, the field's first question is "which module, and how far will it go on this fiber?" This answers it without leaving the app or going online, and keeps the IEEE-vs-vendor line visible so you never quote a vendor reach as a standard guarantee.

How to use

1. Browse by speed tier. The commonly-ordered tiers (10G / 25G / 100G) surface first and are flagged "Commonly ordered"; 1G / 40G / 200G / 400G follow.
2. Search by designation (e.g. LR4), reach, fiber (SMF / MMF / OM4), wavelength (e.g. 850 nm), connector (LC / MPO), or tier (e.g. 100G). The match is a case-insensitive substring across all of those, and it narrows the tiers in place.
3. Read the REACH line first: it is the IEEE maximum on the listed fiber. The fiber type (MMF / SMF) and connector (LC / MPO) give the physical-layer specifics at a glance.
4. A query that matches nothing shows an honest "No match" card; it never fabricates a module. The form-factor table is always available below the variants.

Field notes

- IEEE vs VENDOR: variants standardized in IEEE 802.3 carry a neutral IEEE label. Vendor / coherent variants (1000BASE-EX, 1000BASE-ZX, 10GBASE-ZR, and 400GBASE-ZR) carry a VENDOR label and a "loss-budget dependent" caveat. Their reach is real and widely sold but NOT an IEEE guarantee: it depends on the link's loss budget. Never quote a vendor reach as a standard figure.

- 400GBASE-ZR is coherent DWDM (OIF 400ZR), beyond base IEEE 802.3: a metro / data-center-interconnect optic, not a base Ethernet PHY. Its 80-120 km reach is loss-budget and DWDM-line-system dependent.
- Reach figures are IEEE-standard maximums on the listed fiber; real vendor modules may differ. Multimode reach is grade-dependent (OM3 vs OM4 vs OM5), so those rows give per-grade numbers rather than one figure.
- What it contains: 35 verified variants spanning 1G to 400G across SFP, SFP+, SFP28, QSFP+, QSFP28, QSFP56, and QSFP-DD / OSFP, plus a 9-row form-factor ladder (max rate, lane count, typical power envelope). Power figures are typical vendor ranges, not a single standard: treat as guidance.
- Fully offline: the reference is bundled in the app and works with no connection.
- Verified against IEEE 802.3 standards tables and Cisco / FS.com vendor datasheets. Out-of-scope bleeding edge (800G / 1.6T, CPO / LPO, SFP-DD) was deliberately excluded as not yet stable or field-relevant.

Rack Units

Rack dimensions and mounting hardware: U-to-inches-to-mm conversions, the EIA-310 hole pattern, and 10-32 vs 12-24 vs M6 thread compatibility.

Why it's here. When sizing equipment to a rack or grabbing the right screw, confirm the U height, the real mounting dimensions, and the thread that matches the rack.

How to use

1. 1U is exactly 1.75 in (44.45 mm). Read the conversion table for any U height.
2. '19-inch' describes only the front panel: the mounting holes sit on 18.312 in centers and the opening is about 17.72 in.
3. The vertical holes repeat at 0.5 / 0.625 / 0.625 in within each U, not evenly spaced, which is why an off-by-one mount binds.
4. Match the screw to the rack: 10-32, 12-24, and M6 look alike but will cross-thread and strip if forced.

Field notes

- Many racks ship as bare square holes and need cage nuts; few come tapped.
- Vendor thread guides such as Dell's or HP's are a convention, not a rule.

RJ Connectors

A reference to the registered-jack connector form factors (RJ9/RJ22, RJ11, RJ14, RJ25, RJ45/8P8C, RJ48, RJ48C, RJ48X): positions, conductors, the modular body each uses, and its typical use. This is about the connector body, not the wiring.

Why it's here. In the field you meet the same modular bodies for very different jobs: RJ11 carries one phone line, RJ45 (8P8C) carries Ethernet, and RJ48 reuses the 8P8C body for T1/E1 with a different pin assignment. Telling them apart keeps you from cabling the wrong jack.

How to use

1. Read each connector card: the name, its modular notation (e.g. 8P8C), positions, conductors, and typical use.
2. For T568A / T568B pin-to-pair-color wiring on the RJ45 (8P8C) connector, see the Ethernet Cable & Connector tool.

Field notes

- Accuracy: "RJ45" is the colloquial name for the 8-position 8-conductor (8P8C) modular connector used for Ethernet. Strictly, RJ45 was a telephone wiring standard; the data connector is properly the 8P8C modular jack.
- The PnCm notation means an n-position body with m conductors populated, e.g. 4P4C is the RJ9/RJ22 handset coil-cord body, 6P2C is a 6-position body with 2 conductors (RJ11), 8P8C is fully populated (RJ45/RJ48).
- This table does NOT duplicate the T568A/T568B pin colors: that wiring lives in the Ethernet Cable & Connector tool.
- Data source / standard: the registered-jack (USOC) interface standards and the modular-connector form-factor conventions.

Screw Drives

Recognize the screw drives you meet on access points, enclosures, brackets, racks, and outdoor gear, and carry the matching bit, including the security and tamper drives.

Why it's here. Before a job, know which driver bits you need and whether a fastener is tamper-resistant, so a screw you cannot turn does not stop you.

How to use

1. Match the common drives (slotted, Phillips PH1/PH2, Pozidriv PZ1/PZ2, hex, Torx T10 to T25, Robertson) to the right bit.
2. Pozidriv is easy to mistake for Phillips: the four 45-degree tick marks tell them apart, and the wrong bit strips the head.
3. Security and tamper drives (pin-Torx, pin-hex, one-way, tri-wing, spanner) each need their own bit, so pack the tamper set for outdoor enclosures.

Field notes

- Phillips was not designed to cam out; the 1933 patent sought no cam-out.
- Torx is sized by T-number, not star. Robertson color coding is a trade convention, not an ISO standard.

SD & microSD Cards

How to read every mark on an SD or microSD card face: the capacity standard and the filesystem it commits you to, the bus interface, the three sustained-write class families (C, U, V), the Application Performance Class (A1 and A2), plus endurance ratings, counterfeits, and the write-protect notch.

Why it's here. Because a card face carries up to seven independent marks and only one of them measures two things. The C, U, and V marks each set a minimum sustained sequential write floor and nothing else. A1 and A2 set that same floor, 10 MB/s for both, and add a guarantee the

others never make: random 4 KB IOPS. That is why a V90 card can be slower than an A2 card at booting a Pi and an A2 card can drop frames the V90 handles.

How to use

1. Read the marks in the order the screen lists them: capacity standard first (it is a compatibility gate), then the bus interface (it is the ceiling), then the one class that matches your job.
2. For running an operating system on a Pi or WLAN Pi, read the A class. For packet capture or video, read the V class.
3. Ignore the headline MB/s on the front of the package. That is an unregulated maximum sequential read; the class marks are the only floors on the card.
4. The counterfeit section gives the only test that works. Fill the card completely and read every byte back with h2testw, or f3write and f3read.
5. Use the copy button in the toolbar to copy every table as tab-separated text, one section per topic.

Field notes

- A2 only delivers A2 performance if the host implements Command Queuing. The spec conditions the A2 IOPS figures on the Command Queue scheme with cache enabled, and requires an A2 card to fall back to A1 behavior when they are disabled. That is why real-world Raspberry Pi benchmarks have repeatedly failed to show A2 beating A1. Retail marketing prints A2 unconditionally.
- A1 and A2 both pin sustained write at 10 MB/s, so an A2 card is guaranteed no better than V10 for sustained write unless it separately carries a V mark. An A2 card with no V30 is not a 4K video card, and a V90 card with no A mark says nothing at all about IOPS.
- The capacity mark is a filesystem contract. SDHC means FAT32; SDXC and SDUC mean exFAT. A host that supports SDHC but not SDXC does not fail on capacity, it fails on exFAT. Reformatting an SDXC card as FAT32 makes it work and puts it outside the specification.
- V60 and V90 require UHS-II. A card marked V60 that shows only a Roman I on the face is marked wrongly. A UHS-II card in a UHS-I slot simply runs UHS-I, which is why a fast card in a laptop reader often measures no faster than a cheap one.
- All three sustained-write class families have been OPTIONAL since spec v4.20, so an unmarked card is not necessarily a slow card and a C10 mark is not a modern statement.
- The SD Express maximum is given as ~3.9 GB/s on purpose: the SD Association web table and its own SD 9.1 white paper publish two different figures for the same thing.

- Consumer endurance is quoted in hours at an unstated bitrate and shrinks for 4K, so two cards both labeled high endurance can differ several-fold at the same capacity. TBW is the only comparable endurance number and consumer vendors mostly do not publish it.

Vendor Model Decode

A per-vendor reference for reading an enterprise AP model number. Pick the vendor, then read that vendor's own model-number scheme: what each segment of the SKU encodes (product series, Wi-Fi generation, radio and stream tier, antenna type, regulatory domain), plus a worked example that decodes one real SKU end to end. Covers Cisco (Catalyst/Meraki/CW), HPE Aruba, Ubiquiti UniFi, Ruckus, and Extreme.

Why it's here. Model numbers are position- and suffix-encoded and stable within a vendor generation, so a clean decode is possible offline. But every vendor encodes differently, so this is a per-vendor decoder, never a shared letter dictionary: the same E-style marker means different things across vendors: a regulatory-domain letter on Cisco, an even last digit that flags an external-antenna variant on Aruba, and a product-tier letter on UniFi. One universal letter map would turn all three into one wrong answer.

How to use

1. Pick the vendor.
2. Read the token table left to right: each segment and what it encodes.
3. Read the worked example to see one real SKU decoded segment by segment.
4. Check the confidence-and-caveats note for where a segment needs a per-model datasheet lookup instead of a digit rule.

Field notes

- This is deliberately not a "paste a model number, auto-decode" input. Several vendors (Extreme especially) do not digit-encode Wi-Fi generation, streams, or antenna, so an auto-decoder would fabricate precision the SKU does not carry.
- Aruba's even/odd last-digit antenna rule (even = external, odd = internal) is confirmed back to the Wi-Fi 5 300 series; the 200 series is reported to follow it but is unverified, so decode pre-300-series from a per-model lookup.
- Extreme is Medium confidence: only the first digit (tier) decodes; Wi-Fi generation, stream count, and antenna come from the datasheet.

- Juniper Mist, Fortinet, Cambium, and Omada are flagged for a later pass and are not decoded here; when built, each gets its own module rather than another vendor's rules stretched onto it.
- Reference only. A decode is a heuristic; confirm against the exact model's datasheet or ordering guide before you spec, order, or troubleshoot on it.
- Data source: per-vendor field reference compiled by Keith Parsons / WLAN Pros; sources include the Cisco Catalyst 9130AX datasheet and Getting Started Guide, the Aruba 310 Series datasheet, UniFi Tech Specs, the Ruckus product guide, and Extreme Networks product pages.

Power & Cooling (7)

Batteries

A decode-the-markings reference for cell codes: what the letters and digits on a battery actually mean, and where the pattern stops working.

Why it's here. Survey gear, sensors and test kit run on cells whose codes look systematic and are not. Knowing that CR2032 tells you the size while LR44 tells you nothing about it saves you buying the wrong part.

How to use

1. Start with the hero: CR2032 split into its four groups, chemistry, shape, diameter, height, with a cross-section drawn to the same scale so the code and the object carry the same two numbers.
2. Read the counter-case beside it. LR44 is catalog entry 44, not four by four point four millimetres, and the can it names measures 11.60 by 5.40 mm.
3. Use the size ladder to compare physical sizes at a glance, drawn to scale.

Field notes

- The lithium coin code is dimensional. The aqueous button code is usually a catalog number. Both live in the same clause of the same standard, which is why they are so easily confused.
- Rechargeable cells are not in IEC 60086 at all. That document is titled Primary batteries. Secondary lithium sits under IEC 61960, where the same millimetre rounds in the opposite direction.
- Some sizes on the ladder are not sold at retail and are drawn gray rather than omitted, because the gap is the point.

Cooling & Thermal

Reference for the heat-load conversions you need when sizing cooling for IT spaces: watts to BTU/hr to tons of refrigeration, the IT-load-becomes-heat-becomes-cooling relationship, and the standard sensible-heat airflow (CFM / delta-T) formula.

Why it's here. When you have to size a closet or rack cooling plant from an IT load, convert that load to BTU/hr or tons and read the airflow a given temperature rise needs, so the cooling equipment nameplate matches the heat the gear produces.

How to use

1. The conversion table gives common loads in watts, BTU/hr, and tons, all derived from two anchors: 1 W = 3.412 BTU/hr and 1 ton = 12,000 BTU/hr.
2. The IT-load-to-heat-to-cooling chain shows that essentially all IT power becomes heat, so the cooling load in watts equals the IT load in watts before margin.
3. The airflow note rearranges the standard-air sensible-heat formula so you can find the CFM a target supply-to-return rise needs.

How it works. BTU/hr = watts x 3.412. Tons = BTU/hr / 12,000 (so 1 ton is about 3,517 W). Sensible-heat airflow: BTU/hr = 1.08 x CFM x delta-T (deg F), so CFM = BTU/hr / (1.08 x delta-T). The 1.08 constant is for standard moist air at sea level and shifts with altitude and air density.

Example. A 1,000 W (1 kW) IT load is 3,412 BTU/hr, about 0.284 tons. Removed across a 20 deg F supply-to-return rise it needs about 158 CFM of airflow.

Field notes

- What it shows: a watts/BTU-hr/tons conversion table, the IT-load-to-heat-to-cooling relationship, and the sensible-heat airflow (CFM / delta-T) guidance. No diagram: the relationships are numeric and read cleanly as tables.
- Essentially all electrical power an IT device draws is dissipated as heat into the room, so the cooling load equals the IT load before any headroom for losses and growth.
- The 1.08 airflow constant is the standard-condition approximation, not a universal: verify against site altitude and air density before sizing equipment. "BTU/hr" not "BTUH". Written "Access Point" not "router".
- Data source: the anchor conversions 1 W = 3.412 BTU/hr and 1 ton = 12,000 BTU/hr, plus the standard sensible-heat airflow relationship.

IEC Power Connectors

Reference for the IEC 60320 appliance couplers (C1/C2 through C19/C20, including the C13/C14 PC cord and the C15/C16 kettle cord) and the IEC 60309 industrial pin-and-sleeve connectors, with current ratings, temperature classes, and keying.

Why it's here. When you grab a cord or read an inlet on a PDU, server, or UPS, confirm the coupler pair and its rating so you match cord to inlet, and tell the 70 degC C13/C14 PC cord apart from the 120 degC C15/C16 kettle cord.

How to use

1. The IEC 60320 table lists each coupler pair with its current rating, maximum temperature, nickname, and typical use. Odd number = cord connector (female); even = appliance inlet (male), one greater than its mate.
2. The IEC 60309 table maps connector color to its voltage band; the earth-pin clock position (in 30-degree steps) is mechanical keying so incompatible voltages cannot mate.
3. Use the notes to settle the kettle-cord confusion and to confirm both color and earth-pin hour must match for two IEC 60309 devices to connect.

Example. A C13 connector (female, on the cord) mates with a C14 inlet (male, on the back of a PC or PDU). A C15 cord fits a C14 inlet, but a C13 cord will not fit a C16 inlet because the C15/C16 keying notch blocks it.

Field notes

- What it shows: the IEC 60320 appliance couplers (each face card carries current, max temp, nickname, and a use note) and the IEC 60309 industrial connector (a face card whose specs read color = voltage band; the per-color "use" description is carried in the copy payload rather than shown as an on-screen column), with keying notes rendered as labeled face cards.
- The "kettle cord" nickname properly belongs to C15/C16 (120 degC hot-condition, keyed by a notch), NOT C13/C14 (70 degC cold-condition "PC cord").
- IEC 60309 red spans 380-480V (not a single "415V"): it covers 400V European and 480V US three-phase. Both color AND earth-pin clock hour must match to mate.
- Data source: IEC 60320 and IEC 60309-2.

NEMA Connectors

Reference for North American NEMA straight-blade and locking plug/receptacle configurations: the designation decoder (L prefix, configuration code, current rating, P/R) and the device groups by voltage class with phase, wiring, and amp rating.

Why it's here. When you read a NEMA designation on a plug, receptacle, or PDU, decode it correctly instead of misreading the leading number as a voltage, and confirm the phase and wiring so you do not mistake split-phase for three-phase.

How to use

1. Walk the decoder left to right across a designation like L21-30P: L (locking), 21 (a configuration code, not a voltage), 30 (the amp rating), P (plug; R is receptacle).
2. The device-group tables list types by voltage class (125V, 208/240/250V) with phase, pole/wire count, and amp rating.
3. Use the phase flags to keep the single-phase split 14-series distinct from the three-phase wye L21-series.

Example. L21-30P decodes to L (twist-lock) + 21 (three-phase wye 120/208V, 4-pole 5-wire) + 30 (30A) + P (plug). L21-30R is its receptacle. The leading number is a configuration code, so do no arithmetic on it.

Field notes

- What it shows: the designation decoder with a worked example, plus device groups (125V straight-blade and locking, and 208/240/250V) listing type, voltage, phase, wiring, and amps. The plug and receptacle faces render as labeled face cards above the tables.
- The leading number is a voltage/pole/phase CLASS code, not a literal voltage: 21 = three-phase wye 120/208V, 4-pole 5-wire. Never read it as a voltage.
- The 14-series is single-phase SPLIT (the 4th pin is neutral), NOT three-phase; only the L21-series is three-phase wye. P = plug (male), R = receptacle (female).
- Data source: the NEMA configuration and nomenclature references.

Ohm's Law & Power Wheel

Reference for the V / I / R / P relationships every field tech reaches for: the four core identities (V = I x R, P = V x I, P = I^2 x R, P = V^2 / R), the 12-segment power wheel that expresses each of V, I, R, P in terms of any two of the others, and the single-phase vs three-phase power formulas with the power-factor term.

Why it's here. When you need to solve for a missing electrical quantity at a panel or on a nameplate, find the right form of the wheel and read off the relationship, without re-deriving it. The power-factor caveat keeps you from reading V x I as watts on an AC reactive load.

How to use

1. The core relationships table gives the four identities and what each solves for; everything else derives from V = I x R and P = V x I.
2. The power wheel lists each of V, I, R, and P in three forms, so you can pick the one written in terms of the two quantities you already know.
3. The single-phase vs three-phase table carries the apparent-power (VA) and real-power (W) formulas; real power includes the cos(phi) power-factor term.

How it works. Apparent power S = V x I (single-phase) or square root of 3 x V_LL x I_L (three-phase balanced). Real power P = S x cos(phi). Power factor cos(phi) = 1 only for purely resistive loads and DC; for reactive loads it is less than 1, so V x I gives VA, not W.

Example. A 230V single-phase load drawing 5A has an apparent power of 230 x 5 = 1150 VA. If its power factor is 0.8, real power is 1150 x 0.8 = 920 W, which is why UPS and PDU equipment is rated in both VA and W.

Field notes

- What it shows: the four core identities, the 12-form power wheel (four quantities x three forms), and the single-phase vs three-phase power formulas with a power-factor note.
- Power factor (cos phi) is 1 only for resistive loads and DC, where P = V x I is exact. On reactive loads (motors, compressors, switch-mode supplies) V x I gives apparent power in VA, not real power in watts; do not read P = V x I as universally giving watts in AC.
- Notation: I^2 / V^2 use the caret, the square root is written out, and the power-factor angle is cos(phi). Written "Access Point" not "router".
- Data source: the standard electrical-engineering identity set, the 12-form Ohm's-law wheel, and the AC real/apparent/reactive power relationships.

PoE Reference

Power-over-Ethernet reference: the 802.3 PoE standards (PSE/PD power, powered pairs, class range) and the PD power classes (0 to 8) with max power at the device.

Why it's here. When sizing PoE, confirm what a switch port delivers vs what reaches the device, and which 802.3 standard / class a given AP needs.

How to use

1. PSE power is supplied at the switch; PD power is what's left at the device after cable loss (e.g. 802.3at supplies 30 W PSE, delivers 25.5 W PD).

2. The class table maps a PD's negotiated class to its max draw (e.g. Class 4 = 25.5 W = PoE+ max; Class 8 = 71.3 W = Type 4 max).

Field notes

- What it shows: three tables. (1) PoE standards: 802.3af (PoE), 802.3at (PoE+), 802.3bt Type 3 (PoE++/4PPoE), 802.3bt Type 4 (PoE++ Hi), each with PSE watts, PD watts, powered pairs (2 of 4 or 4 of 4), and supported class range. (2) PD power classes: class 0 to 8 → the PSE output watts, the max power at the PD, the 802.3 standard defining it, and a note (the class ladder carries both the PSE-out and the at-the-PD columns, so you can read the drop across the cable at each class). (3) IEEE 802.3 Types: Type 1 to 4 with PSE voltage range and per-pair current limits, per IEEE Std 802.3-2022 Clause 33 (Type 1/2) and Clause 145 (Type 3/4).
- Footnote points to the PoE Budget tool for sizing a switch against connected devices.
- Written "802.3" (not "802.3x"). Standard reference, not region-specific.
- Data source: the PoE standards and class ladder reflect IEEE 802.3af/at/bt; the Types table and the PSE-output column are cited to IEEE Std 802.3-2022 Clause 33/145.

Power Phasing

Reference for the three power topologies a field tech meets in IT spaces: US single-phase 120V (one hot + neutral), split-phase 120/240V (two hots 180 degrees apart, 240V hot-to-hot, 120V hot-to-neutral), and three-phase wye 208V (three hots 120 degrees apart, 208V line-to-line, 120V line-to-neutral). Renders the 208-vs-240 distinction installers confuse.

Why it's here. When you read a panel or nameplate, confirm which topology you're on and what hot-to-hot voltage to expect, so you don't feed 240V-rated gear from a 208V wye (about 13 percent low) or mislabel split-phase as two-phase.

How to use

1. Each topology card shows its hots, line-to-neutral (L-N), line-to-line (L-L), and phase angle, with a waveform.
2. Split-phase is 240V hot-to-hot because the two legs are 180 degrees apart (2 x 120 in anti-phase); three-phase wye is 208V hot-to-hot because the legs are 120 degrees apart (the square root of 3 times 120, about 208V).

3. Use the 208V-vs-240V table to settle the confusion: same 120V hot-to-neutral, different hot-to-hot, different source topology.

How it works. For a wye system, line-to-line voltage equals the square root of 3 times the line-to-neutral voltage: 120 x 1.732 ≈ 208V. For split-phase, line-to-line is 2 times line-to-neutral in anti-phase: 2 x 120 = 240V.

Example. A 240V single-phase heater fed from a 208V wye panel sees about 13 percent lower voltage and runs at reduced output, because two legs of a 208V wye are 120 degrees apart (208V), not 180 degrees apart (240V).

Field notes

- What it shows: three topology cards (single-phase 120V, split-phase 120/240V, three-phase wye 208V) with hots, L-N, L-L, phase angle, where it's seen, and a clarifying note; plus a 208V-vs-240V comparison table and a why-it-matters note.
- Split-phase is single-phase, NOT two-phase: the two phasors don't define a rotating field. The center conductor is the grounded, current-carrying neutral (center tap), not the equipment ground.
- 208V and 240V are not interchangeable: they differ by about 32V and arise from different topologies (a single-phase center tap vs two legs of a 3-phase wye).
- Written "Access Point" not "router". Nominal voltages per NEC Article 100/210.
- Data source: NEC Article 100/210, split-phase and three-phase electrical-engineering references, and the line-to-line = square root of 3 times line-to-neutral relationship.

Codes, Safety & Compliance (8)

Cloud Tool Trust

How to read the security badges on a cloud Wi-Fi tool before you upload a client's floor plan, AP inventory, or survey data: ISO/IEC 27001 (a certified management system), SOC 2 (an attestation report, not a certificate), GDPR (a law you conform to), the five Trust Services Criteria, and

the adjacent badges (ISO 27017/27018, FedRAMP, CSA STAR).

Why it's here. A compliance badge is a claim about a defined scope and a time window, not a guarantee about your data. You are the one uploading a client's network into someone else's cloud, so the badge tells you which questions to ask, not that the data is safe.

How to use

1. Ask whether it is a certificate or an attestation: ISO 27001 is a certificate; SOC 2 is a report plus an opinion, so "SOC 2 certificate" is a wording tell.
2. Read the scope: does it cover the actual product you will use, or the vendor's corporate IT?
3. For SOC 2, confirm Type 2 (operating effectiveness over a period) over Type 1, and that Confidentiality (and Privacy, if personal data) is in scope.
4. Ask where the data lives and under whose law: residency is not sovereignty, and EU hosting does not by itself solve GDPR.

Field notes

- Third-party-audited claims (ISO 27001 certificate, SOC 2 Type 2, CSA STAR Level 2, FedRAMP) outrank self-asserted ones, and even an audited claim only covers the scope, criteria, and time window printed on it.
- A vendor unwilling to share a SOC 2 report under NDA is itself a flag; read the opinion, scope, period, and the complementary controls you are responsible for.
- Some data, a client's full network design, is sensitive enough that the right home is your own device, not a cloud you never vetted.
- Reference only. Compliance status, scope, and sufficiency are determined by the client's security or compliance officer and a qualified auditor (a CPA firm or counsel), not by this tool.
- Data source: frameworks per ISO/IEC 27001:2022, the AICPA SOC 2 framework, and the EU GDPR.

Credentials & Licenses

The portable IDs and licenses a WLAN pro carries from job to job, and the lead-time landmine in them: why you almost never need an FCC operator license (even for licensed microwave backhaul), and the federal and background-check credentials (TWIC, CAC, DBIDS, SIDA, HAZWOPER-40, background checks) that gate restricted sites with weeks-to-months lead time.

Why it's here. A TWIC, a base credential, or a SIDA badge can each add weeks between the award and touching a cable, so scope the credential before you quote, not after you win. Companion to the Site Access entry.

How to use

1. Settle the license question first: unlicensed Wi-Fi (Part 15) and licensed point-to-point microwave (Part 101) both require no FCC operator license (GROL).
2. For a restricted site, read the credential table for the ID, issuing authority, lead time, and validity you will need.
3. Bucket the lead time: fast and you control it (GROL exam, school check), weeks (SIDA, TWIC, DBIDS), or weeks-to-months (CAC).
4. Confirm what your specific job requires with the issuing authority before you quote and before you go to site.

Field notes

- You would only need a GROL if the work crossed into servicing aviation or marine radios, which is outside a normal WLAN scope of work.
- TWIC gets you the maritime jobs; TSA tells applicants to enroll at least 60 days out and warns processing can exceed 45 days.
- CAC versus DBIDS is the military-base question: the CAC needs a sponsor and an investigation that can run many months, while DBIDS is the lighter get-on-base path.
- Reference only. The issuing authority sets and grants every credential; the Toolbox certifies nothing and clears no one.
- Data source: FCC license scope per Parts 15, 97, and 101; credential detail per TSA, USCG, DoD, and OSHA 29 CFR 1910.120(e).

Enclosure Ratings

The two enclosure-rating systems a WLAN pro reads on outdoor and industrial spec sheets: the IP code (IEC 60529, international) and NEMA types (NEMA 250, US). It decodes each IP digit, lists the NEMA types that matter for Wi-Fi, and shows the one-way NEMA-to-IP translation.

Why it's here. US spec sheets quote NEMA and international sheets quote IP, so you translate between them to compare gear on the same terms and spec an enclosure, AP, antenna, or PoE injector to the real hazard at the mount point.

How to use

1. Read an IP code left to right: first digit is solids and dust (0 to 6), second digit is water (0 to 9K). Each digit maps to a defined lab test.
2. Use the common-ratings table to place a code (IP66 is dust-tight plus powerful jets, the mainstream outdoor AP rating).
3. For a US spec sheet, read the NEMA type, then use the NEMA-to-IP table as a minimum floor. The reverse (IP to NEMA) is not valid.
4. Use the placement table to pick a rating for where the gear mounts.

Field notes

- X means not tested, not zero. IPX7 is water-rated 7 with the solids digit unrated. Read X as no data, never as fails.
- NEMA to IP is valid only as a minimum. NEMA tests corrosion, icing, gasket aging, and oil, which IP never checks, so a NEMA 4X box exceeds IP66 but an IP66 box is not automatically NEMA 4X.
- The water ladder is not a clean superset past 6. An IP67 (immersion) device is not guaranteed to pass IP66 (jets); gear that must survive both is dual-marked, such as IP66/IP67.
- IP tests use fresh water only. Salt, detergents, and solvents need separate chemical-resistance verification.
- This is a field reference, not code or design guidance. Confirm requirements with the AHJ, the architect of record, and a licensed electrician.
- Data source: IP per IEC 60529; NEMA per NEMA 250.

Hazardous Locations

A recognize-and-defer reference for classified (hazardous) areas: the NEC Class and Division system (US, Article 500) and the IEC Zone system (Article 505/506, ATEX, IECEx). It names what the hazard is (Class), how often it is present (Division or Zone), the recognized protection concepts, and the field read, then stops.

Why it's here. A standard commercial AP is a genuine ignition source in a flammable atmosphere. Mounting one in a classified area is illegal, uninsurable, and dangerous. You need to recognize a refinery, grain elevator, fuel depot, or spray booth before quoting, and route the install to rated gear and the AHJ.

How to use

1. Read Class as what the hazard is made of: Class I (gases and vapors), Class II (dust), Class III (fibers and flyings).
2. Read Division as how often it is present: Div 1 during normal operation, Div 2 only under fault. Div 2 is the far larger wireless market.
3. Map to the IEC Zone system where a facility uses it (Div 1 is roughly Zone 0 plus 1 for gas; Div 2 is roughly Zone 2).
4. Match a recognized protection concept (Ex d, Ex i, Ex p, Ex e/nR) to the Division or Zone, or keep the AP out and remote the antenna in.

Field notes

- Never just mount a commercial AP in a classified area. The default move is to keep the AP in the adjacent general-purpose area and remote the antenna into the zone through a rated penetration.
- "Class I Div 2 rated" buys permission to install that specific listed device, in that specific Division or Zone, wired per its control drawing. It is not a blanket safe-anywhere stamp; a Div 2 device is not approved for Div 1.
- You cannot mix Division and Zone classification in the same installation. Which one is in force is set by the facility area-classification drawing and the AHJ, not the installer.
- This is a field reference, not code or design guidance. Confirm requirements with the AHJ, the architect of record, and a licensed electrician.
- Data source: Class/Division per NEC Article 500; Zone per NEC 505/506, ATEX, and IECEx.

NEC Gotchas

A recognize-and-defer reference for the six NEC articles that actually bite a WLAN installer: elevator hoistways (620), plenum air spaces (300.22), PoE bundle heat (725.144), antenna and mast grounding (810), firestopping fire-rated assemblies (300.21), and abandoned cable (800.25). The Article 800 communications-cable rating ladder is deliberately held out of the six and shown as a separate "Supporting reference" band (which jacket fire-rating goes where), not counted as a peer article.

Why it's here. These are the code articles most likely to surface on an everyday install. Recognizing each one on site, then handing it to the AHJ, a licensed electrician, or the equipment listing, keeps you out of trouble. Recognize-and-defer, never how-to-comply.

How to use

1. Use it to recognize a code issue on site: the hoistway you cannot put an AP in, the plenum that dictates your cable jacket, the PoE bundle that can overheat, the antenna that needs grounding, the fire wall you must not breach, and the dead cable you must pull.
2. Read the cable-rating ladder as a jacket fire-rating only (CMP plenum, CMR riser, CM/CMG general, CMX limited); substitution runs downhill only.
3. On the two STOP items (PoE ampacity, firestop assembly), do not eyeball a number or pick an assembly yourself; hand it to the licensed electrician or the listed system.
4. Then defer to the right authority for the actual requirement.

Field notes

- STOP on PoE bundle heat (725.144): the exact ampacity and bundle-count numbers come from the code table and the specific install conditions. Do not eyeball them. Size the bundle with a licensed electrician or the cabling designer against the adopted NEC.
- STOP on firestopping (300.21): the approved firestop is a specific listed assembly matched to the wall type, penetrant, and opening. Never improvise it and never pick the assembly yourself.
- Grounding (810) caveat: nothing survives a direct strike. Bonding and surge protection mitigate nearby strikes and static, not a direct hit. Conductor sizing is an electrician's call.
- This is a field reference, not code or design guidance. Confirm requirements with the AHJ, the architect of record, and a licensed electrician.

- Data source: article numbers per the NEC (NFPA 70); confirm against the locally adopted edition.

Network in Scope

The regulatory frameworks that reach the WLAN itself: PCI DSS (cardholder data), HIPAA (electronic health information), SOX (public-company financial systems), and GDPR (EU personal data). Each entry names the trigger, what the framework generally asks of the network, and the owner to route the specifics to.

Why it's here. When the Wi-Fi carries cardholder data, ePHI, or sits inside a public company's financial systems, the design inherits requirements long before anyone audits it. Recognizing the scope lets you quote honestly and design toward the requirement instead of retrofitting after the auditor shows up.

How to use

1. Recognize the trigger: card-present retail and hospitality (PCI), covered entities and business associates (HIPAA), US public companies (SOX), EU personal data on the network (GDPR).
2. Read the general asks: segmentation and strong crypto for PCI, the section 164.312 technical safeguards for HIPAA, access and change controls for SOX.
3. Route the specifics to the owner: the QSA for PCI, the privacy or security officer for HIPAA and GDPR, internal audit and the external auditor for SOX.
4. Never tell a client the design "meets PCI" or "is HIPAA compliant"; you recognize the framework, and the assessor rules on it.

Field notes

- PCI DSS v4.0.1 makes WPA2-PSK inadequate for the Cardholder Data Environment; WEP and WPA/TKIP are prohibited, and quarterly rogue-AP scanning is required even at a site with no Wi-Fi at all.
- HIPAA encryption is currently "addressable," meaning you implement it or document a defensible reason not to; a January 2025 proposed rule would make it mandatory.

- SOX is the narrowest fit: the network shows up as the access-and-change-control substrate under the financial systems, which is why an AP config change can suddenly need a ticket and an approver.
- Reference only. Whether a network is in scope, and whether a design is sufficient, is determined by the client's compliance officer and a qualified auditor, not by this tool.
- Data source: framework requirements per PCI DSS v4.0.1, HIPAA 45 CFR 164.312, SOX Section 404, and the EU GDPR.

Safety Basics

A field-awareness reference for the personal protective equipment a general contractor expects before badging you onto an active site, a short note on ESD (protecting the electronics you install, not the person), and the recognize-and-STOP hazards you must hand off rather than work.

Why it's here. Half of a WLAN pro's installs happen on a ladder or lift, on an active construction site, around other trades. Knowing the PPE ratings a GC will ask for keeps you from being turned away at the gate, cheap ESD gear prevents latent failures in the gear you install, and the STOP flags keep a routine cable pull from becoming a health or legal incident. Awareness and lookup, not compliance instruction.

How to use

1. Use the PPE ladder to recognize the four common baseline items and the standard each is rated to (hard hat Z89.1, safety-toe F2413, hi-vis ANSI 107, eye protection Z87.1).
2. Read the ESD note as gear protection: a grounded wrist strap, mat, and static bags under ANSI/ESD S20.20 guard against latent damage to APs, optics, and boards.
3. On the four recognize-and-STOP hazards (asbestos or lead, arc flash and energized work, confined spaces, seismic bracing), stop and hand it off. Never run any of them as a procedure yourself.
4. Confirm the site's specific PPE policy with the employer and GC, who own the assessment.

Field notes

- The four PPE items are the common "let me on the site" baseline. On a finished office install street clothes are usually fine; on active construction expect the GC to require all four to badge on.
- ESD risk is highest in data centers and any time you handle bare optics or boards. For a sealed AP coming out of a static bag, mounted and cabled, the risk is lower but not zero.
- The recognize-and-STOP items are named-and-stopped on purpose: the app names the hazard and tells you to hand it off, and deliberately gives no procedure for working any of them.
- This is a field reference, not code or design guidance. Confirm requirements with the AHJ, the architect of record, and a licensed electrician.
- Data source: standards designators ANSI/ISEA Z89.1, ASTM F2413, ANSI/ISEA 107, ANSI Z87.1, ANSI/ESD S20.20, NFPA 70E; confirm the site's PPE policy with the employer and GC.

Site Access

A "Know Before You Go" checklist: eight site types (aerial and man-lifts, rail, hospitals, maritime, warehouse and distribution, schools, data centers, correctional) and the credential, screening, orientation, or escort that can gate you from reaching the work before it even starts.

Why it's here. On many sites you cannot reach the work area without a specific credential, background check, orientation, or escort. That is a quoting and scheduling factor, not just a safety one. Rail screening, a hospital ICRA permit, or lift-operator proof can each add days or weeks between winning a job and touching a cable.

How to use

1. Before you quote, scan the checklist for the site type you are bidding and read what may gate you.
2. Use the "ask about" column as the questions to put to the GC, site owner, or authority: screening programs, orientations, escorts, flagman, PFD and TWIC, ICRA and ILSM, tool control.
3. Budget the lead time (rail screening and hospital ICRA in particular run long) into the schedule you promise.
4. Confirm each requirement with the site, general contractor, and authority before you go to site.

Field notes

- The pattern across every item: the requirement is set by someone other than you, must be satisfied before work starts, and carries real lead time and cost.
- Maritime and over-water work adds PFD and drowning-hazard rules plus heavy salt corrosion, so spec NEMA 4X or high-IP with explicit corrosion resistance.
- There is no OSHA-issued "lift license"; the employer trains and certifies lift operators, and GCs demand documented proof.
- This is a field reference, not code or design guidance. The site, general contractor, and authority set every requirement; the Toolbox certifies nothing and clears no one.

Buildings & Verticals (8)

AEC Process & Glossary

The architecture-engineering-construction workflow you work inside and the shorthand the other trades use: the AIA design phases (Programming, SD, DD, CD, Bidding, CA) and when Wi-Fi should engage, plus the AEC glossary (RFI, submittal, ASI, AHJ, GC, MEP, OAC, AOR/EOR, and more) that trips WLAN pros up. US convention (AIA).

Why it's here. Speak the process fluently and you look like a peer in the room instead of the network person who wandered in. The RFI, the submittal, the punch list, and the OAC are the levers that get your coverage design built the way you drew it. Engaging at Schematic Design is the difference between designing Wi-Fi in and retrofitting it later.

How to use

1. Use the phase table to engage at the right moment: establish RF requirements and reserve IDF and ceiling access at SD, coordinate AP locations against the RCP at DD, finalize plans and telecom sheets at CD.

2. Raise an RFI the moment the Reflected Ceiling Plan and your AP plan disagree, so the conflict is resolved on paper before rough-in.
3. Get your APs, mounts, and cable approved as a submittal (the contractor's proof that what is installed matches the spec).
4. Treat anything about contractual responsibility as "confirm with the architect of record and your contract," never as a ruling.

Field notes

- The AHJ (Authority Having Jurisdiction) is the building official, fire marshal, or inspector who interprets and enforces code locally; the AHJ's word governs.
- An ASI is a minor clarification with no cost or time impact; a Change Order is a formal, signed change to scope, cost, or schedule (what an ASI is not).
- The OAC is the recurring Owner-Architect-Contractor coordination meeting; if Wi-Fi matters, someone has to carry it into the OAC.
- Reference only. Confirm contractual responsibility, code compliance, and phase deliverables with the architect of record, the AHJ, and your contract for the specific project.
- Data source: phase and contract vocabulary per US AIA convention.

CAD & BIM Formats

What the building files an architect hands you actually are: the format decode table (DWG, DXF, DGN, IFC, RVT, NWD/NWC, COBie), the Level of Development (LOD 100 to 500) ladder for how much of a model to trust, and how a building file flows into a Wi-Fi design tool (Ekahau, Hamina, iBwave).

Why it's here. You receive these files on every design job. The difference between a clean import and a wasted afternoon is knowing what the format is, what LOD you were handed, which CAD layers to ask for, and that scale calibration is the single step that makes or breaks the whole design. The Toolbox explains these formats; it does not open, render, or convert them.

How to use

1. Use the decode table to recognize a format on sight: DWG and DXF are Autodesk CAD, IFC is the open BIM standard, RVT is a Revit model, NWD/NWC are Navisworks coordination files, COBie is asset data.
2. Read the LOD you were handed before trusting geometry: LOD 100 to 200 is a massing study (do not derive wall attenuation from it), LOD 300 and up is dimensionally trustworthy.
3. Follow the import flow: import the CAD or PDF, calibrate the scale on a known distance (get this wrong and every downstream distance is wrong), then assign wall materials and run the prediction.
4. Ask the architect for a clean layer set (walls, doors, structure; strip furniture, dimensions, title blocks) instead of an unusable 80-layer dump.

Field notes

- Scale calibration is the step that matters most: draw a line over a known length and enter the real measurement. An error here corrupts every distance, coverage prediction, and attenuation value downstream.
- LOI (Level of Information) is the data-completeness sibling of LOD: LOD is geometry, LOI is data.
- The boundary: the Toolbox explains DWG, IFC, RVT, and the rest and how they reach the design tools. It is not a CAD or BIM viewer, converter, or editor.
- Reference only. Confirm file handling, model reliability, and responsibility with the architect of record and your design-tool documentation.
- Data source: format and LOD conventions per Autodesk, buildingSMART, and the AIA/AGC LOD framework.

Data Centers & Wi-Fi

The read a WLAN pro needs before quoting Wi-Fi in or around a data center: why production Wi-Fi on the floor is usually minimal, why the room is RF-hostile by construction, and how the two resilience frameworks (ANSI/TIA-942 Rated-1 to Rated-4 and the Uptime Institute Tier I to Tier IV) differ.

Why it's here. Walk into a data center expecting office-style coverage and you get two things wrong at once: you fight an RF environment engineered to reflect and contain, and you quote a schedule that ignores the badging and change-control gate.

How to use

1. Clarify the actual use case first: it is rarely production; it is usually out-of-band management, staff mobility, or guest.
2. Survey with the containment in place, because empty-room predictions lie once the racks and aisle barriers are up.
3. Keep the frameworks straight: TIA-942 says Rated, Uptime says Tier, and mixing them ("a Tier 3 TIA rating") is a telling error.
4. Budget the badging and change-control lead time up front, and defer the facility rating to the operator and its design engineer.

Field notes

- Dense metal racks, hot-aisle and cold-aisle containment, and overhead trays chop the space into sealed RF pockets, so the design problem is coverage, not capacity.
- The telecom rooms themselves are small and metal-dense, so APs often go outside or at the doorway rather than inside the rack cage.
- Access is a credentialing exercise: expect badging, mantraps, escort, no-photography rules, NDAs, and change-control windows, the same "know before you go" pattern as any high-security site.
- This is a field reference, not design or facility-rating guidance; confirm resilience ratings, access, and change-control rules with the operator and its design engineer.
- Data source: framework detail per ANSI/TIA-942 and the Uptime Institute Tier Standard.

Healthcare Wi-Fi

Why a hospital is the one building where you cannot design Wi-Fi like an office: the protected WMTS telemetry band, medical-device EMC (IEC 60601-1-2), the voice-and-RTLS roaming grades, shielded rooms, and the four authorities (HIPAA, FDA, FCC, The Joint Commission) plus the in-house biomedical engineering handoff.

Why it's here. The air is shared with life-critical, EMC-regulated devices and a protected telemetry band, a dropped roam can mean a missed alarm, and the "just cover it like an office" instinct is exactly the mistake that gets telemetry stepped on and alarms missed.

How to use

1. Coordinate with clinical and biomedical engineering before you touch a hospital RF environment; that is the single most important handoff.
2. Do not assume all patient monitoring is on Wi-Fi: WMTS is a separate, licensed system on 608 to 614, 1395 to 1400, and 1427 to 1432 MHz, coordinated by a WMTS coordinator.
3. Design to voice-and-RTLS grade, not data grade: continuous facility-wide roaming, overlapping cells, and no dead zones, and design around shielded rooms and the MRI Faraday cage.
4. Recognize each of the four authorities, then defer the rulings to the people who own them.

Field notes

- Medical devices are tested to tolerate a defined RF environment under IEC 60601-1-2, but tested to a limit is not the same as immune to anything; a dense, high-power deployment can push local field strength toward what a nearby monitor was qualified for.
- Coverage grade drives AP count more than floor area does; a hospital quoted at data-grade density fails when it has to carry voice handsets and location services.
- Segmentation is a performance control here, not only a security one: guest, clinical, device, and RTLS traffic get separated for airtime protection as much as for HIPAA.
- This is a field reference, not clinical, code, or compliance guidance; confirm every requirement with biomed, the compliance and security officers, the WMTS coordinator, and the AHJ.
- Data source: standards per the FCC WMTS allocation, IEC 60601-1-2, HIPAA, FDA guidance, and The Joint Commission.

Plan-Set Literacy

A reference for reading an architectural drawing set: how a sheet number is built (discipline letter, sheet-type digit, sequence), what each discipline designator and sheet-type digit means, why the Reflected Ceiling Plan (RCP) is the sheet for AP placement, the rest of a plan set worth knowing, and US drawing scales.

Why it's here. A WLAN designer who can pull the right sheet, read the RCP, and speak in sheet numbers looks like a peer to the other trades instead of a junior. The RCP is where a coverage design meets the physical ceiling, and catching an AP-versus-diffuser conflict on paper is far cheaper than catching it at rough-in.

How to use

1. Read a sheet number left to right: discipline letter, sheet-type digit, sequence. A-201 is Architectural, an elevation, sheet 01.
2. Use the discipline table to find the sheet owner (A architectural, E electrical, T telecommunications; the WLAN and structured cabling usually live on the T sheets).
3. Use the sheet-type digit to find the drawing kind (1 plans, 2 elevations, 6 schedules), so A-1xx is an architectural plan and E-6xx an electrical schedule.
4. For AP placement, open the RCP: overlay your AP locations and resolve conflicts with diffusers, troffers, sprinklers, and soffits before rough-in. Never place APs on the floor plan alone.

Field notes

- The RCP is drawn as if a mirror on the floor reflects the ceiling upward, so it reads in the same left-right orientation as the floor plan; it carries the ceiling grid, mounting heights, and every ceiling-mounted element.
- Always build to the latest revision: the revision cloud marks what changed and the delta triangle carries the revision number.
- US architectural sheets use fractional-inch scales (1/8" = 1'-0" is 1:96); to get the ratio, invert the fraction and multiply by 12.
- Reference only. Confirm drawing conventions, revisions, and responsibility with the architect of record and your contract for the specific project.
- Data source: conventions per the US National CAD Standard; sheet-numbering practice varies on smaller jobs.

Structured Cabling

The TIA and BICSI structured-cabling standards that decide where an AP can go and what feeds it: the TIA-568 family, the 90 m permanent link plus 10 m of cords for a 100 m channel, the cable-category ladder (Cat 5e through Cat 8), and the MDF-IDF topology that ties AP locations to telecom rooms.

Why it's here. The 90 m rule and the Cat 6A bar shape real WLAN decisions: they set the outer edge of where you can place an AP and what uplink it can carry, which quietly drives IDF placement, switch selection, and the whole coverage plan. The physical grounding-as-safety side lives in the NEC codes reference; this is the TIA and BICSI infrastructure side.

How to use

1. Use the TIA family to know who owns what: 568 is cabling, 569 pathways and spaces, 606 labeling and administration, 607 telecom bonding and grounding.
2. Apply the 90 plus 10 m channel rule as an AP-cable-run reality check: an AP more than about 90 m of cable-path from the IDF needs an intermediate closet, a different topology, or fiber.
3. Read the cable-category table for the multi-gig bar: Cat 6A carries 10 Gbps to 100 m and is the practical minimum for Wi-Fi 6, 6E, and 7 APs with multi-gig uplinks.
4. Coordinate with the RCDD (BICSI's cabling-design credential) whose 90 m constraint ties AP locations to IDF locations.

Field notes

- The 90 m permanent link is solid horizontal cable from the telecom room to the outlet; add up to 10 m of stranded patch and equipment cords for a 100 m channel maximum. Exceed it and you are outside TIA.
- T568A and T568B are the two pin-out standards; a jack is wired to one, so be consistent end to end.
- AP count and placement drive IDF count and PoE-switch port budgeting, so the WLAN design and the cabling design have to talk to each other.
- Reference only. Confirm cabling design and standards currency with the RCDD, the architect of record, and your contract for the specific project.

- Data source: standards per ANSI/TIA-568/569/606/607 and BICSI; confirm the current edition.

Telecom Spaces

A decoder for the telecom room names that get used interchangeably and are not interchangeable: Entrance Facility, Equipment Room, Telecommunications Room (the current TIA-569 term), MDF and IDF (legacy distribution-frame terms the field still uses), and "data closet" (slang), plus the hierarchical-star topology that ties them together.

Why it's here. When the electrician says "MDF," the architect's drawing says "TR," and the client says "the data closet," knowing they usually mean the same room keeps you from designing a phantom extra space or missing a real one. The IDF and TR locations are where your APs get their uplink and power.

How to use

1. Use the decode table to map a room name to what it actually is and whether it is a standard, legacy, or slang term.
2. Recognize that MDF and IDF are legacy frame terms and TR is the current TIA-569 word; IDF and TR describe the same space.
3. Read the topology as a hierarchical star: EF in, MDF or Equipment Room at the center, backbone out to the IDF or TR per floor, horizontal out to the AP.
4. On an international job, recognize the ISO/IEC 11801 "distributor" terms (Campus, Building, Floor Distributor) for the same hierarchy.

Field notes

- The cable mechanics (the 90 meter horizontal link, categories, PoE budgeting) live in the Structured Cabling reference and are not repeated here.
- One TR per floor at minimum, more for large floors, because horizontal runs are distance-limited.
- "Data closet" is not a standard term at all; it is an informal catch-all for any TR or IDF.
- Reference only. Confirm the space names, standards currency, and cabling design with the architect of record, the RCDD, and your contract.

- Data source: space vocabulary per ANSI/TIA-569-E, alongside ANSI/TIA-568 and ISO/IEC 11801.

Verticals Index

A plain-language index of the industries you quote and what each one tends to trigger: manufacturing, oil and gas, warehouse, healthcare, hospitality and stadiums, education, retail, data centers, maritime, and correctional or government. It maps the vertical you name to the other reference entries you should read before you quote it.

Why it's here. Nobody standing in a building thinks "this is NAICS 622110"; you think "it's a hospital." Classification codes are back-office filing labels, not design inputs. The vertical you name in the first phone call tells you which reference entry to open before you quote.

How to use

1. Find the vertical you are bidding in the map and read what it tends to trigger.
2. Open the "read first" entry (Hazardous Locations, Enclosure Ratings, Healthcare Wi-Fi, Site Access, Data Centers, or Telecom Spaces) before you quote.
3. For retail, recognize PCI DSS scope when Wi-Fi touches cardholder data, then defer the ruling to the client's QSA.
4. For high-density venues, remember the design axis flips from coverage to capacity, driven by seat count and peak concurrent users.

Field notes

- Miss the hazloc on an oil site, the ICRA gate on a hospital, or the PCI scope in retail, and the timeline and price you promised were wrong before you started.
- Treat a classification code (NAICS, SIC, GICS, ISIC, NACE) as a label a client hands you, never a design input.
- Use the index the other direction too: pick the industry, see the cluster, read the entry that owns the detail.
- This is a field reference, not code, design, or compliance guidance; confirm every requirement with the client, the AHJ, and the relevant assessor.

- Data source: this entry is the index that points at the other Field & Trade Reference entries.

Ham Radio (6)

The band-dependent amateur-radio references, grouped so all the ham material is findable in one place. Every numeric value traces to FCC Part 97 (eCFR Title 47), the FCC Online Table of Frequency Allocations, ARRL, or NCVEC, current as of June 2026, with the corrected figures used throughout: never an older chart. The pure-math ham tools (Antenna Length, Maidenhead Grid) live under Calculators & Tools → Ham Radio.

Band Names & Wavelengths

The two-worlds translation: amateur band names (160 m, 2 m, 13 cm) to their actual frequency ranges, with the λ = 300 / f rule of thumb.

Why it's here. Hams name bands by wavelength in meters and centimeters; Wi-Fi people think in megahertz and gigahertz. This is the lookup that crosses between them.

How to use

1. Find a band by its wavelength name to read its frequency range and ITU region tag.
2. Use the λ = 300 / f rule to sanity-check the relationship in your head.

What it shows

- The band-name → frequency table from 2200 m (135.7–137.8 kHz) up through 3 cm (10.0–10.5 GHz), each with an LF/MF/HF/VHF/UHF/SHF region tag.
- 60 m shown honestly as "~5.332–5.405 MHz (4 ch + segment)" and 13 cm as the split "2300–2310 / 2390–2450 MHz"; 9 cm flagged sunset/out.

Field notes

- The wavelength is the nominal band name; the range is the actual allocation, so the rounded λ = 300 / f relationship is a memory aid, not the exact band edges. (Wavelength in meters is roughly 300 divided by frequency in megahertz.)
- Offline, read-only. Same data source as the band plan.

General License Frequency Chart

A bundled, offline, pinch-zoomable copy of the US amateur General-class HF/VHF/UHF privilege chart: frequency segments, power limits, and the corrected 60 m channels.

Why it's here. You want the General-class privileges as a single printable chart you can zoom on the phone you already have, with no network needed.

How to use

1. Open the card and pinch / double-tap to zoom; the page is fit-to-screen on open.
2. This is a flat print PDF, not an in-app data table, so the content is not screen-reader readable.

Field notes

- Card inner content is not accessible to screen readers (flat rasterized PDF). The card title and "pinch to zoom" gesture are announced.
- Source / basis: Keith's corrected amateur-radio reference, bundled and rendered offline, exactly like the laminated Wi-Fi cards.

Ham Radio General Exam Study Notes

A bundled, offline, pinch-zoomable copy of condensed study notes for the amateur-radio General exam: rules, operating practice, and the RF concepts a Wi-Fi pro already knows.

Why it's here. You want a compact, zoomable General-exam study sheet on the phone, offline, with no network needed.

How to use

1. Open the card and pinch / double-tap to zoom; the page is fit-to-screen on open.
2. This is a flat print PDF, not an in-app data table, so the content is not screen-reader readable.

Field notes

- Card inner content is not accessible to screen readers (flat rasterized PDF). The card title and "pinch to zoom" gesture are announced.
- Source / basis: Keith's amateur-radio study notes, bundled and rendered offline.

Part 15 vs Part 97

Where the amateur 13 cm and 5 cm bands overlap 2.4 and 5 GHz Wi-Fi, and how the unlicensed (Part 15) and amateur (Part 97) rules differ: the rules behind running commodity Wi-Fi hardware as an AREDN mesh node.

Why it's here. The same 802.11 silicon can run under two completely different rule sets. This lays the two side by side so the trade (Part 97's higher power ceiling for Part 15's encryption and commercial freedom) is explicit.

How to use

1. Read the overlap table to see which Wi-Fi channels fall inside an amateur allocation.
2. Read the rule-delta table for how license, power, station ID, encryption, business use, and content differ between Part 15 and Part 97.

What it shows

- The overlapping allocations mapped to the Wi-Fi grid: 13 cm (2390–2450 MHz) overlaps the lower ~50 MHz of the 2.4 GHz ISM band (2400–2450; the 2390–2400 MHz slice sits just below ISM), covering channels 1–6 fully and 7 partially; 5 cm (5650–5925 MHz) covers upper U-NII-2C through U-NII-4; 33 cm (902–928 MHz) is full co-channel with the 900 MHz ISM band.
- The rule deltas: Part 97 requires a licensed control operator and a callsign every 10 minutes (including digital/mesh links), prohibits encryption (mesh traffic must be in the clear), and bars business use and broadcast content; Part 15 needs no license, allows WPA2/WPA3 and commercial use. Part 15 2.4 GHz power is 1 W (30 dBm) conducted / 4 W (36 dBm) EIRP at up to 6 dBi; Part 97 runs to the 1500 W PEP ceiling.
- The AREDN example: AREDN and Broadband-Hamnet run commodity 802.11 a/b/g/n hardware on the overlapping ham channels under Part 97 instead of Part 15.

Field notes

- The 3.3–3.5 GHz amateur band (9 cm) is being sunset and does NOT overlap Wi-Fi 5 GHz; do not confuse it with the 5 cm band (5650–5925 MHz), which does.
- Part 97 (amateur) is secondary on these bands; Part 15 is unlicensed and carries no allocation status. Both carry the same "do not cause harmful interference" ethic. Offline, read-only.

Spectrum Band Designations

The ITU decade-band designations (HF, VHF, UHF, SHF) with their frequency and wavelength spans and what each band's propagation implies, plus the aviation airband and military UHF neighbors a Wi-Fi pro lives next to.

Why it's here. Knowing which decade band you are in tells you how the energy travels: ionospheric skip, line-of-sight, or short-range. This sets that context around the Wi-Fi bands.

How to use

1. Read the four ITU decade bands (each is ×10 the previous) and their propagation character.

2. Check the neighbor list for the non-amateur services that sit beside the bands you work.

What it shows

- The four ITU decade bands a Wi-Fi pro lives near (HF 3–30 MHz sky-wave / ionospheric skip; VHF / UHF / SHF), each with frequency, wavelength, and an operational propagation note. MF/LF below and EHF above are noted (in the intro/copy context), not tabled.
- A neighbors card with five entries that border these bands: the VHF aviation airband, the military UHF airband, and the ISM/U-NII bands a Wi-Fi pro shares the air with (900 MHz ISM, 2.4 GHz ISM, 5 GHz U-NII).

Field notes

- Propagation notes are operational summaries (HF "talk around the world," higher bands line-of-sight), not predictions for a specific path.
- Offline, read-only.

US Amateur Band Plan

The US amateur bands from HF to SHF, by license class: each band's frequency range, the Technician / General / Extra privileges, the power limit, and the mode segments.

Why it's here. A Wi-Fi pro crossing into amateur radio needs to know what they may transmit where and at what power. This is the band plan with the current, corrected numbers, grouped by ITU region (HF / VHF / UHF / SHF).

How to use

1. Browse the bands grouped by region, or type in the search field to filter the list live (a query that matches no band shows an honest "no match" card); read the per-class privileges and the mode/segment notes on each band.
2. Check the power summary and the HF data rule for the rules that cut across bands.

What it shows

- Per-band frequency range, license-class privileges, power, and mode segments.
- The corrected power picture: a 1500 W PEP general ceiling, with the real exceptions (30 m = 200 W PEP; 60 m = 100 W ERP (channels) and 9.15 W ERP (segment); 2200 m = 1 W EIRP; 630 m = 5 W EIRP; a Technician on any HF band = 200 W PEP), always under the "minimum necessary power" rule, 97.313(a). The old blanket "200 W" entries were wrong.
- 60 m as 4 channels plus the 5351.5–5366.5 kHz segment at 9.15 W ERP, effective 13 Feb 2026 (the old 5-channel chart is wrong).
- The HF data rule that replaced the obsolete baud column: no symbol-rate limit on HF, a 2.8 kHz maximum-bandwidth cap, effective 8 Jan 2024 (FCC 23-93). No baud value is shown anywhere, because the VHF/UHF baud status is unconfirmed.
- 9 cm (3300–3500 MHz) is omitted as a band in active sunset that does not overlap 5 GHz Wi-Fi.

Field notes

- Data source: FCC Part 97, the FCC frequency-allocation table, ARRL, and NCVEC. The Toolbox renders its own band plan from FCC data rather than reproducing ARRL's copyrighted chart.
- ASCII hyphen-minus throughout for ranges; a privilege absence reads as a sentence ("No Technician privileges"), never a bare dash. Offline, read-only.

Encoding & Formats (8)

ASCII / Hex / Binary

The full 128-character US-ASCII table with decimal, hex, octal, and binary for each code, plus supplementary quick-reference tables for reading hex dumps and protocol fields.

Why it's here. When decoding a hex dump or a protocol field, look up a byte's character, its four numeric representations, or the meaning of a control code.

How to use

1. The numeric columns (dec/hex/oct/bin) and the glyph render in a monospaced font so octets and look-alike characters (l/I/O/0) read unambiguously.
2. Control rows show a mnemonic (NUL, LF, CR, ESC…); printable rows show the glyph (space is shown as "SP").
3. Filter by typing a decimal, hex (with or without 0x), octal, binary, glyph/mnemonic, or keyword.

Field notes

- What it shows: a "how to read this" card, then Control codes (0 to 31, plus 127) and Printable characters (32 to 126) as tables with Dec / Hex / Oct / Bin / Char / Description. Plus supplementary cards: Range boundaries worth memorizing, Newlines on the wire, The case bit (0x20), Nibble → hex map, Powers of two, Hex place values, High range (128 to 255): no single "extended ASCII", and a Base64 alphabet card (RFC 4648) covering the 3-byte→4-character mapping, the 64-character alphabet, its four contiguous ranges, and the '=' padding rules.
- The high-range card is explicitly honest: ASCII stops at 127; bytes 128 to 255 mean different things depending on the encoding, so there is no single "extended ASCII." It documents UTF-8 (bytes 0 to 127 identical to ASCII; 128 to 255 are part of multi-byte sequences), ISO-8859-1/Latin-1, and Windows-1252 (a common mojibake source), with the rule "bytes 0 to 127 are portable; 128 to 255 are not, so know the encoding before decoding."
- Data source / standard: RFC 20 (US-ASCII) for the 128 values; high-range guidance per ISO-8859-1, Windows-1252, and the Unicode/UTF-8 spec; the Base64 card per RFC 4648. Standard reference, not region-specific.

Data Units

Bit vs byte units, the SI (decimal) vs IEC (binary) prefix ladders, kB vs KiB through EB vs EiB, and how link rates in bits relate to storage in bytes.

Why it's here. The reason a '1 TB' drive shows as ~931 GiB, and the reason a 100 Mbps link is not 100 MB/s, kept in one place.

How to use

1. Read the SI vs IEC ladder for each rank (kB/KiB through EB/EiB), its powers, byte counts, and the percent divergence.
2. Use the bit-vs-byte card and the divide-by-8 rule to convert a bit-rate to byte throughput.

Field notes

- What it shows: the two parallel prefix ladders (SI base-1000 vs IEC base-1024), the per-step divergence (compounding ~2.4 points per rank), and the bit-vs-byte relationship.
- Note: SI uses lowercase k for kilo (kB) but uppercase for M and above; IEC binary prefixes are Ki, Mi, Gi, Ti, Pi, Ei. Ki is a capital K, unlike SI's lowercase k. Network/link speeds are quoted in bits per second; storage in bytes.
- Data source / standard: BIPM SI prefixes and IEC 80000-13 binary prefixes. Offline, read-only.

Hash Lengths

Common hash algorithms by output size (bits / bytes / hex characters), family, and security status, with MD5 and SHA-1 flagged broken / deprecated.

Why it's here. Sizing a digest field or judging an algorithm choice: this is the length and the security verdict at a glance.

How to use

1. Read each algorithm's output length in bits, bytes, and hex characters, plus its family and security-status chip.

Field notes

- What it shows: MD5, the SHA-1, SHA-2, and SHA-3 families, and more, each with output length and a security-status verdict.
- MD5 and SHA-1 carry an explicit broken / deprecated status; the verdict word always accompanies the status color, so color is never the sole carrier of meaning.
- Data source / standard: NIST FIPS 180-4 (SHA-1, SHA-2), FIPS 202 (SHA-3 / Keccak), RFC 1321 (MD5); deprecation per NIST SP 800-131A and the 2017 SHAttered SHA-1 collision. Offline, read-only.

Keyboard Shortcuts

The standard keyboard shortcuts you reach for every day: macOS and Windows system shortcuts, the line-editing keys for the macOS Terminal and Windows PowerShell, how to type special symbols on a Mac with the Option key, and the Greek letters that show up in RF and antenna math.

Why it's here. A fast lookup when you are driving someone else's machine, switching between macOS and Windows, or need a symbol or a Greek letter (lambda for wavelength, mu for micro, ohms) without hunting through a character viewer.

How to use

1. Pick the platform tab (macOS, Windows, Symbols, or Greek). Each tab is a stack of copyable cards.
2. Find the action you want; the key combo is shown as the keys you press together.
3. On the Symbols and Greek tabs, the highlighted glyph is selectable, so you can grab a single symbol directly; the top-bar copy action copies the whole reference.

Field notes

- The true degree sign on a Mac is Option Shift 8; Option 0 gives the similar-looking ordinal, which is flagged on the Symbols tab.
- PowerShell reverse history search (Ctrl R) depends on PSReadLine, which ships with Windows PowerShell 5.1+ and PowerShell 7; a bare console host falls back to F7 / F8.

Markdown Cheatsheet

CommonMark and GitHub Flavored Markdown syntax shown as the literal text you type next to what it renders as, covering headings, emphasis, links, images, lists, task lists, blockquotes, code, and tables.

Why it's here. When writing docs, READMEs, wikis, or notes, confirm the exact Markdown for a structure without guessing, and know which pieces are GitHub extensions that a plain renderer may not support.

How to use

1. Each row pairs the literal Markdown ("You type") with what it renders as. The "You type" column is plain text, not rendered, so the exact syntax is copyable.
2. Rows marked GFM are GitHub Flavored Markdown extensions (tables, task lists, strikethrough, autolinks) and may not work in a plain CommonMark renderer.
3. Gotchas: put a blank line between block elements, escape special characters with a backslash, and end a line with two trailing spaces for a hard line break.

Field notes

- Covers the CommonMark core plus the widely-implemented GFM extensions, with GFM-only rows flagged.
- Data source / standard: the CommonMark specification and the GitHub Flavored Markdown specification.

Phonetic Alphabet

A scannable A to Z reference. Each letter shows its NATO/ICAO spelling word (Alfa, Bravo, Charlie), its international Morse code pattern, its semaphore arm positions, and its international maritime signal flag.

Why it's here. Spell things out clearly over a noisy radio or phone link, read flag and light signals, or settle whether that was M as in Mike or N as in November.

How to use

1. Scan the table for the letter, its NATO spelling word, and its Morse pattern (a dot is a short signal, a dash is a long one).
2. The plate below carries the semaphore arm positions and the colored maritime signal flag for each letter; tap it to zoom.
3. Tap Copy to grab the whole A to Z table as text.

Field notes

- The official NATO/ICAO spelling uses Alfa (not Alpha) and Juliett (double-t) so the words are pronounced correctly across languages.
- This covers A to Z only; there are no digit rows. The maritime flag colors are the real international code of signals colors, not brand colors.

Regex Cheatsheet

Regular-expression syntax (anchors, character classes, quantifiers, groups and references, and alternation / escapes) scoped to the common PCRE2 (Perl-compatible) dialect.

Why it's here. The token you half-remember, looked up fast, with a flag where a dialect would behave differently.

How to use

1. Scroll the syntax tables by section: anchors, character classes, quantifiers, groups/references, and alternation/escapes.
2. Watch for the DIALECT badge: a token marked that way behaves differently outside PCRE2; check the per-row dialect note.

Field notes

- What it shows: the common PCRE2 regex token set with meaning and examples, grouped by function. A 'Dialect: PCRE2 (Perl-compatible)' banner sits at the top.

- Honesty note: there is no single normative regex authority: POSIX (BRE/ERE), PCRE2, ECMAScript, Python, Java, Go (RE2), and .NET differ. No token is presented as universal unless the source data marks it so; non-universal tokens carry a DIALECT badge.
- Data source / standard: the common PCRE2 subset per the PCRE2 syntax reference. Offline, read-only.

Top 30 Emoji

The 30 most-used emoji, ranked 1 to 30, with the official Unicode CLDR name and a descriptive "common use" note.

Why it's here. A lightweight decode reference for what people actually mean by an emoji, useful when meanings drift (e.g. 💀 = "that's hilarious," not death).

How to use

1. Ranked most-used first (😂 face with tears of joy at #1).
2. The official CLDR name is the row's spoken/searchable key; the glyph is excluded from the screen-reader label (readers announce the glyph's own name).
3. The "common use" note describes typical reading, with generational/fandom context where it matters.

Field notes

- What it shows: per emoji: rank (1 to 30), the glyph, the Unicode CLDR official name, and a "common use" note on how people usually read it today.
- The intro is explicit: "common use" is how people usually read each emoji today, not an official Unicode definition; meanings drift by audience, region, and generation.
- The literal/codepoint fields are intentionally omitted. The glyph renders via the platform color-emoji font (Apple Color Emoji on iOS/macOS).
- Data source / standard: Unicode CLDR for the names; ranking is messaging-weighted (private-messaging keyboard frequency, not social-listening-weighted).

Travel & International (4)

Date / Time Standards

A reference for ISO 8601 / RFC 3339 date-time formats, UTC-offset notation, the Unix epoch and the 2038 problem, leap seconds (UTC vs TAI), and NTP stratum levels.

Why it's here. Reading a log timestamp, an API field, or an NTP status: this decodes the format and the time-scale behind it.

How to use

1. Read the format cards for ISO 8601, the strict RFC 3339 Internet profile, and the common format tokens.
2. Read the epoch, leap-second, and NTP-stratum cards for the time-scale context behind a raw timestamp.

Field notes

- What it shows: ISO 8601 and RFC 3339 formats and the difference between them, UTC-offset notation, the Unix epoch and 2038 wraparound, the UTC-vs-TAI leap-second relationship, and NTP stratum levels (1-15 valid; stratum 0 is the kiss-o'-death / unspecified marker).
- Honesty note: the UTC-TAI offset of -37 s is a static educational value current as of the 2017-01-01 leap second, shown with a 'static value' badge and a footnote pointing at the IERS Bulletin C: never presented as a live truth.
- Data source / standard: ISO 8601-1:2019, RFC 3339, RFC 5905 (NTP), POSIX.1-2017, BIPM/IERS. Offline, read-only.

Emergency Phrases

A searchable, offline phrase translator for a Wi-Fi pro working on-site internationally: about 124 travel and emergency phrases in English with Spanish, French, Italian, and German, grouped into seven situations (basics and courtesy, medical and help, directions, food and lodging,

problems and requests, on-site and technical, numbers and time).

Why it's here. When you land on a job in a country whose language you do not speak, a few of the right phrases get you to the server room, to a doctor, or out of trouble. This keeps them one tap away with no signal needed.

How to use

1. Pick how much to show: "One language" (English plus one target you choose) or "All languages" (English plus all four). One language is the default.
2. In one-language mode, choose the target from the language menu (Spanish, French, Italian, or German).
3. Search across all five languages at once. Type a phrase in English or in any target and the list filters live.
4. Tap the copy icon on any phrase to copy just that phrase in the languages currently shown, or use Copy all phrases in the top bar to copy the whole visible list grouped by situation.

Example. Pick German, search "toilet", and the row shows English "Where is the toilet?" with German "Wo ist die Toilette?". Tap copy to put both on the clipboard.

Field notes

- TRANSLATIONS ARE DRAFT. The Spanish, French, Italian, and German wording is draft machine translation and has NOT yet been reviewed by a native or professional translator. A banner at the top of the screen says so, and the caveat is included when you copy the list. Verify any critical phrase locally before relying on it.
- Standard, simple wording was used throughout; no idioms were invented. Phrases are intentionally plain so they travel across dialects.
- Fully offline. The phrase set is bundled in the app, so it works with no signal and no data plan.
- Phrases are grouped by situation in a deliberate, curated order, not alphabetized.

International Power Plugs

Reference for the IEC World Plugs letter system (Types A through M) by region, with the underlying national standard, nominal voltage class and frequency, and current rating, plus the CEE 7 European family breakout.

Why it's here. When you travel or spec gear for a site abroad, confirm the plug type, nominal mains voltage, and frequency you will meet, so power adapters and equipment ratings match what is at the wall.

How to use

1. Search by country at the top (e.g. Germany shows Type C, F at 230V/50Hz). Common names and abbreviations work too, such as USA, UK, or Holland; multi-type countries show all their letters.
2. The plug-type table lists each letter with its standard, voltage class, current rating, and the countries that use it.
3. The CEE 7 family breakout shows how the European C / E / F types and the CEE 7/7 hybrid relate.
4. Read the voltage class as nominal: most of the world is 230V except North America and Japan at 120V. Watch the warnings (for example, Argentine Type I has line and neutral reversed).

Example. A laptop charger rated 100-240V works on both a US Type A/B 120V outlet and a UK Type G 230V outlet with only a mechanical plug adapter; a 120V-only appliance needs a voltage converter, not just an adapter, on a 230V supply.

Field notes

- What it shows: the IEC World Plugs Type A through M table (type, standard, voltage class, current, countries) and the CEE 7 European family breakout. The plug-type faces render as labeled face cards above the tables.
- Voltage classes are nominal: 120V in North America and Japan, 230V across most of the rest of the world. Confirm frequency (50 vs 60 Hz) for motor-driven and timing-sensitive gear.
- Argentine Type I (IRAM 2073) has line and neutral reversed relative to the Australian Type I; the CEE 7/7 plug is a hybrid designed to fit both French (E) and Schuko (F) sockets.
- Data source: the IEC World Plugs letter system and national standards.

Time Zones

A UTC offset reference for coordinating work across sites, surveys, and conference calls: a world map that shows where each one-hour offset band sits, an offset rail with anchor cities for every zone, and a United States table with the standard and daylight abbreviations.

Why it's here. Scheduling a remote survey, a cutover window, or a call across sites. Find each site's offset and the difference in hours is the time difference between you.

How to use

1. Find your site's offset on the rail, then the other site's offset; the difference in hours is the time difference.
2. The world map orients you: vertical bands are one-hour offsets and the prime meridian (UTC 0) runs through Greenwich. Tap it to zoom.
3. The United States table gives the standard and daylight abbreviations (for example EST in winter, EDT in summer) and notes who skips daylight saving.

Field notes

- Offsets shown are standard time. When a region is on daylight saving, its local clock runs one hour ahead of the listed offset; in the US that window runs from the second Sunday in March to the first Sunday in November.
- Most of Arizona stays on Mountain Standard Time all year, and most of Hawaii does not observe daylight saving. A few zones use half-hour offsets (India UTC +5:30, central Australia UTC +9:30).

Educational Resources (1 tool)

The curated directory of places to learn Wi-Fi: blogs, conference talks, video channels, and podcasts, each one a link out to its own site. The directory is a list of outside resources rather than a tool, so it is not documented entry by entry here. This Field Manual also opens from this section. The one in-app tool that lives here is below.

Ham Radio Study Resources

A vetted, offline list of where to study for the amateur-radio exams (hamstudy.org, the ARRL License Manuals, FCC Part 97, and AREDN) each credited, with an "Open website" button and the current exam structure.

Why it's here. The companion to the in-app ham references: when you decide to actually sit an exam, this points you at the strongest current study material instead of a stale chart, with the two currency landmines flagged.

How to use

1. Read the resource cards; tap "Open website" to hand off to the browser (opens in your device's default browser over HTTPS).
2. Read the exam-structure block for the stable "N questions, M to pass" figures.

What it shows

- Four vetted resources: hamstudy.org (the study platform, not its links page), the ARRL License Manuals (the authority reference, cited not reproduced), FCC Part 97 + the frequency-allocation table (the law and the app's data source of truth), and AREDN (the explicit Wi-Fi tie-in: commodity 802.11 hardware run under Part 97).
- The stable exam structure (NCVEC / 97.503): Technician (Element 2) 35 questions / 26 to pass; General (Element 3) 35 / 26; Amateur Extra (Element 4) 50 / 37. No Morse requirement since 23 Feb 2007; the FCC application fee is $35.

Field notes

- The two currency landmines are surfaced as guardrails, never hard-coded around: a new Technician question pool takes effect 1 Jul 2026 (so the pool count is deliberately not memorized, only the stable 35/26 structure is shown), and the 60 m rules changed effective 13 Feb 2026.
- The only network action is a browser hand-off over HTTPS, not an in-app fetch, so no special network exception is needed. A failed hand-off shows an honest inline note with the raw URL, never a silent failure.

Wireless Classroom (91 tools)

Tools for teaching wireless, all of them ours. Guided lessons open the section. Four shelves of simulators follow in teaching order: the wave in free space and through a building, what rides on the wave, who gets the air and when, and the network built from all of it. The WLAN Pros course handouts close it. The simulators compute everything on the device and need no connection, and on a computer or tablet each one has a Present button for a projector. The Teacher's Guide, at the top of the section in the app, covers presenting and lesson sequences; this manual covers what each tool does, how it works, and how to teach it.

Guided Lessons (30)

Antenna Fundamentals

A read-along teaching reference for antenna literacy: what an antenna actually does (it shapes where the radio's energy goes, it does not add power), azimuth vs elevation, why gain trades against beamwidth, polarization and the wall-clock mistake, downtilt, how to read a radiation-pattern polar plot (main lobe, the -3 dB beamwidth points, side lobes, nulls, front-to-back ratio), and which antenna type (omni, patch, sector, Yagi, dish) fits which space. Seven line diagrams are embedded at the points they teach.

Why it's here. It is the antenna-literacy companion to the directional tools in the toolbox (Downtilt, Point-to-Point, Fresnel). When you are choosing or mounting an antenna and want to reason about coverage shape rather than chase a gain number, this is the read-first reference.

How to use

1. Read top to bottom: it builds from one idea (an antenna is a shaper, not a booster) through azimuth/elevation, orientation, reading a pattern chart, and what to use where.
2. Use the embedded diagrams alongside the prose: the polar-plot anatomy diagram labels every feature (main lobe, -3 dB beamwidth, side lobes, nulls, back lobe, front-to-back) so a manufacturer's chart stops being a mystery.
3. The type table and the closing deployment quick-map are the fast lookups: match the space you are covering to the antenna whose pattern fits it.

Field notes

- This is conceptual teaching copy, not a calculator and not a manufacturer spec sheet. It deliberately gives no formula for gain vs beamwidth: "more gain means a narrower beam" is always directionally true, but the exact beamwidth for a given gain depends on the specific antenna's design. Read the published beamwidth on the data sheet; do not try to back it out of the gain number.
- The per-type beamwidth figures (patch about 30-120 degrees, sector about 60-120, Yagi about 15-40, dish about 3-25) are RANGES across real products, not single specs. The antenna in your hand lands somewhere inside its band.
- The cross-polarization penalty is given conceptually: 90 degrees is the worst case (in theory total loss; in practice reflections leave some signal), and a wall-mounted omni costs roughly 6 dB. These are rules of thumb to reason with, not an exact loss figure.
- The floor-plan coverage shapes illustrate how a pattern fills a space; they are not survey predictions or measured coverage.
- The point-to-point note is one line by design: a link needs real Fresnel-zone clearance around the straight line between antennas, not just a visible path. Use the Fresnel and Point-to-Point tools to size that clearance.

Teaching it

A read-along lesson on what an antenna does (it shapes where the energy goes; it does not add power), gain against beamwidth, polarization, downtilt, and how to read a radiation pattern. Pair it with Antenna Pattern.

Spectrum Analysis

A read-along teaching module on RF spectrum analysis for Wi-Fi: why a spectrum analyzer sees energy a Wi-Fi adapter is blind to, how the instrument works (swept versus real-time, and the time-versus-frequency trade), the knobs an operator touches (span, resolution and video bandwidth, reference level, the dBm scale, detectors, max-hold and averaging), the three main views (live trace, waterfall, and density/duty cycle) and how to read each, a nine-card gallery of interferer fingerprints (microwave oven, Bluetooth Classic and BLE, analog video camera, baby monitor, drone/FPV downlink, Zigbee, analog cordless phone, continuous wireless bridge), how to compare captures, the current tool landscape, and the mitigation ladder.

Why it's here. This is the spectrum-analysis companion to the rest of the toolbox: when a Wi-Fi scan shows a clean channel but performance is poor, the cause is usually a non-Wi-Fi interferer that only a spectrum analyzer can see. This module teaches you to recognize those interferers by their waterfall shape and to pick a fix. It is an in-app reference, alongside Antenna Fundamentals.

How to use

1. Open the module from the Wireless Classroom (Guided Lessons), then read the eight topic screens in order from the hub. They build from the why (a spectrum analyzer versus a Wi-Fi adapter) through how it works, the knobs, the three views, fingerprinting, comparing captures, the tools, and mitigation.
2. On Fingerprinting interferers, swipe through the nine signature cards and tap any card to zoom in. The caption under each card is its waterfall fingerprint.
3. On The knobs, tap the Wireless Glossary link to resolve any term in one tap.

Field notes

- This is teaching content, not a working analyzer. A phone's Wi-Fi chipset can join networks but cannot do wideband RF capture, so the app cannot perform spectrum analysis. The module says so in-app.
- The nine signature cards are illustrative teaching diagrams (a representative trace plus a waterfall and a dBm legend drawn to the classic signatures), not captures from your own environment.
- The drone/FPV downlink signature is model-dependent and lower-confidence: analog FPV is a continuous carrier, while digital systems spread a wider, adaptive block. The card flags this.
- Cordless phones split two ways: the 2.4 GHz analog kind interferes, but DECT 6.0 at 1.9 GHz does not touch any Wi-Fi band. Baby monitors vary by model for the same reason.
- Tool details are kept to confirmed capabilities only; pricing is deliberately left out because it moves and varies by region. Oscium now owns MetaGeek, so the Wi-Spy and Chanalyzer line is one product family.
- The 6 GHz band being the cleanest is a current condition, not a permanent property; it will erode as adoption grows.

Teaching it

A read-along lesson on why a spectrum analyzer sees energy a Wi-Fi adapter cannot, how the instrument works, and the signatures of common interferers. Pair it with the Swept vs FFT race in Fourier and FFT.

Find My, Explained

A read-along lesson on how Apple finds your people, your phone and your things, including the AirTag in your suitcase, and what you should turn on before your next trip. Find My is one app that finds three kinds of things: people who choose to share where they are, your Apple devices, and items like AirTags. The lesson follows the free PDF guide of the same name, word for word, with its 11 figures and step-by-step settings for iOS 27 and watchOS 27.

Why it's here. It answers the question Wi-Fi people get asked constantly by people who are not technical: how does Find My work? It also carries the Wi-Fi angle. When you see your bag on the map, you are looking at the location of the phone that heard it, not a reading from the tag. Indoors, and in airports especially, GPS is weak, and the phone leans on the Wi-Fi networks around it. Figure 3 shows how the finder's GPS, Wi-Fi and cell positioning become your dot.

How to use

1. Read top to bottom. It starts with the one idea behind all of it: an AirTag can't tell anyone where it is, and any iPhone that walks past reports its own location, sealed so that only you can read it. Then it covers why it stays private, where the dot comes from, the three radios, the app and sharing, a phone that dies, lost luggage, a tracker that isn't yours, and five things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. The appendix has the step-by-step settings: 13 tasks, one per card.
4. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Written for iOS 27 and watchOS 27. Button names can shift slightly between software updates.
- Figures Apple does not publish, such as Bluetooth range and how often a tag's location updates, are deliberately left out. Figure 4 is not to scale.
- How the finder's GPS, Wi-Fi and cell positioning combine into your dot is our explanation, not Apple's wording.
- Going somewhere public and contacting police before you disable a stranger's tag is our advice, not an Apple instruction.
- The lesson is independent and isn't affiliated with or endorsed by Apple.

Teaching it

How Apple finds your people, your phone and your things, including the AirTag in your suitcase, and what you should turn on before your next trip.

Home Internet, Explained

A read-along lesson on how home internet reaches a house in the US: fiber, cable, DSL, 5G home internet, a local wireless provider, satellite in low and high orbit, and a phone hotspot. It covers speed, lag (latency, the delay before an answer comes back) and data limits, which one to pick, what to read on the label before you sign up, and why your Wi-Fi is a separate question. The lesson follows the free PDF guide of the same name, word for word, with its 18 figures.

Why it's here. It answers the question Wi-Fi people get asked by family, friends and clients: which internet should I get, and why is it slow in the back bedroom? The one idea under it is the Wi-Fi angle. The service you pay for stops at the router, and a speed test on your phone in the back bedroom adds the Wi-Fi on top. Figure 13 shows the two measurements side by side.

How to use

1. Read top to bottom. It starts with the three things you're buying (speed, lag and a data limit), then the six ways in, why every road is shared, wires and radio to your house, why lag is its own number, the service or the Wi-Fi, which one to pick, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 7 has the two-step test that tells the service from the Wi-Fi: a wired laptop at the router, then the same test over Wi-Fi in the room that feels slow.
4. Step 8 has the order to pick in, how to look up your address on the FCC's national broadband map, and the three lines to read on the Broadband Facts label.
5. Two related tools: Why a Busy Line Lags shows latency under load, and The Slowest Link Wins, a panel on the How Your Devices Access the Internet plate, shows why end-to-end speed equals the slowest hop.
6. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Written for US homes, with figures checked on 27 September 2026. Provider plans change often; the Broadband Facts label is the current word for your address. Prices are deliberately left out.
- The wired latency figures are the FCC's own measurements from its 2022 test period, published in 2024.
- Low-orbit figures are Starlink's own published range. The high-orbit floor of about 0.48 seconds is speed-of-light arithmetic.
- The lesson leaves out prices and per-address availability. It is independent and isn't affiliated with or endorsed by any provider named in it.

Teaching it

Fiber, cable, DSL, 5G and satellite: how each one reaches a house, what to check before you sign up, and why the Wi-Fi is a separate question.

Wi-Fi Calling, Explained

A read-along lesson on what happens when your phone makes a call over Wi-Fi instead of the cell tower, when it helps, when it drops, what it means for emergency calls, and what to set up before your next trip. Wi-Fi Calling is not an app. It is your own phone company's calling service, reaching your phone over a Wi-Fi network and the internet instead of over a cell tower. The lesson follows the free PDF guide of the same name, word for word, with its 17 figures, where to turn it on for iPhone and Android, and a checklist for before a trip.

Why it's here. It answers a question Wi-Fi people get asked by people who are not technical: what is Wi-Fi Calling, and why did my call drop? It also carries the network owner's side. When Wi-Fi Calling is on, your phone builds a locked tunnel across the Wi-Fi you are on, through the internet, straight into your phone company. The hotel, the coffee shop and your own router carry the tunnel. They can't see inside it, but they still decide whether it connects and how well the call sounds. The step What your Wi-Fi owes a phone call covers the delay budget, the WMM (Wi-Fi Multimedia) setting that gives voice its own fast lane, and three things to check on your router.

How to use

1. Read top to bottom. It starts with the short answer: same phone, same number, same app, and only the road in the middle changes. Then it covers the two roads to the same phone company, when your phone uses Wi-Fi, walking out the door mid-call, emergency calls, iPad, Mac

and Apple Watch, calling home from abroad, Wi-Fi Calling against app calls, what your Wi-Fi owes a call, and five things people get wrong. Each myth hides its fact until you tap Reveal the fact.

2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. The last step shows where to turn it on, for iPhone and for Android, and a four-step checklist for before a trip.
4. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Carrier rules, prices and country lists change, so read your carrier's current page before you travel. The carrier details in the lesson were checked on 27 September 2026.
- Figures no company publishes, such as battery cost or the exact signal level where a phone switches to Wi-Fi, are deliberately left out.
- What happens to an emergency call over Wi-Fi depends on your carrier and your country. If you ever call for help over Wi-Fi, say where you are.
- The lesson is independent and isn't affiliated with or endorsed by any carrier or device maker.

Teaching it

What happens when your phone makes a call over Wi-Fi instead of the cell tower, when it helps, when it drops, what it means for emergency calls, and what to set up before your next trip.

Starlink, Explained

A read-along lesson on how internet from space reaches your house or RV, how high the satellites are, and why the Wi-Fi inside matters as much as the sky above. A flat dish talks to whichever Starlink satellite is overhead, the satellite passes your data down to a Starlink ground station

called a gateway, and from there it travels over ordinary fiber to the rest of the internet. The lesson follows the free PDF guide of the same name, word for word, with its 13 figures, four tasks in the Starlink app, and what to check before you blame Starlink.

Why it's here. Wi-Fi people get asked about Starlink by friends, family and RV owners who are not technical, and the complaint is usually a slow back bedroom. The lesson carries the Wi-Fi angle: an ordinary speed-test app on a phone measures two links at once, the satellite link and the Wi-Fi, so a slow room is often the Wi-Fi and not the satellites. Figure 11 shows the two links, and Appendix B starts with the link you control.

How to use

1. Read top to bottom. It starts with the one idea behind all of it, then covers how high the satellites fly, the path your data takes, why it feels quick, the flat dish, why speeds change, plans for homes and RVs, trees, snow and rain, the Wi-Fi inside, and six things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode. Figure 1 is drawn to scale.
3. Appendix A has four tasks in the Starlink app, one per card. Appendix B is what to check, in order, when something feels slow.
4. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Plan names and rules are the US ones as of 27 September 2026, from Starlink's own pages. Prices are left out on purpose. Check starlink.com before you buy.
- Figures Starlink does not publish, such as how far away your gateway is, are left out on purpose. Figure 7 is an illustration, not measured speeds.
- The Wi-Fi Calling line is general: some carriers don't support it over satellite internet. T-Mobile's page is the one in the sources; other carriers publish their own.
- The lesson is independent and is not endorsed by SpaceX. Starlink is a trademark of SpaceX.

Teaching it

How internet from space reaches your house or RV, how high the satellites are, and why the Wi-Fi inside matters as much as the sky above.

Tap, Touch and Near, Explained

A read-along lesson on the two short-range radios in a phone: NFC (near field communication), the one that works when you tap to pay, and Ultra Wideband (UWB), the one that measures how far away something is. It covers why a tap has to be a tap, what a tap sends to the store, whether a wallet needs a shield, where the real risk is, and how a phone points the way to your keys or opens a car. The lesson follows the free PDF guide of the same name, word for word, with its 11 figures.

Why it's here. It answers questions Wi-Fi people get asked by people who aren't technical: is tapping to pay safe, and do I need a blocking wallet? The short answers are yes and, for bank cards, no. Each payment sends a code that works only once, and a card has to be within a hand's width of the reader to wake up at all.

How to use

1. Read top to bottom. It starts with the short answer, then the two radios, what a tap sends, the wallet question, the real risks, finding your keys, and four things people get wrong with the habits worth keeping. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 5 has the one Stop box in the lesson: never share a code your bank texts you.
4. Step 6 has Open buttons for Find My, Explained and Bluetooth, Explained when those lessons are in the app.
5. The last row, Take it with you, names the free PDF guide the lesson follows, Tap, Touch and Near, Explained, from WLAN Pros, to read or print away from the app. The PDF isn't inside the app.
6. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Checked on 28 September 2026 against sources that include Visa, EMVCo, Apple, Google, the FTC and the FiRa Consortium. The full source list is the last step of the lesson.
- The lesson is independent and isn't affiliated with or endorsed by any company named in it.

Satellite Texting, Explained

A read-along lesson on how many newer phones send a text through a satellite when there's no cell signal and no Wi-Fi: how the text reaches the satellite, why the phone needs a clear view of the sky, how to hold the phone and wait, why it's a backup and not a promise, and what to set up before you go. The lesson follows the free PDF guide of the same name, word for word, with its 8 figures.

Why it's here. It answers a question Wi-Fi people get asked before a trip into the backcountry: will my phone work out there? The one idea under it is distance. The satellite is far away and the phone's antenna is small, so the phone sends a few words at a time, slowly, and only with open sky above it.

How to use

1. Read top to bottom. It starts with the short answer, then how a text reaches a satellite, why it needs the sky, holding the phone and waiting, the limits, setting it up before you go, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 5 has the one Stop box in the lesson: don't make satellite texting your only way to get help.
4. Step 6 has the setup list and the steps to try the demo on an iPhone and on a Pixel phone while you still have signal.
5. The last step has Open buttons for How GPS Works, Explained, Starlink, Explained and Weak Cell Signal at Home, Explained when those lessons are in the app.
6. The last row, Take it with you, names the free PDF guide the lesson follows, Satellite Texting, Explained, from WLAN Pros, to read or print away from the app. The PDF isn't inside the app.

7. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Checked on 28 September 2026 against Apple, Google, Verizon and T-Mobile support pages. Phones, countries and plans change every few months, so check your carrier and your phone maker before a trip.
- The lesson is independent and isn't affiliated with any phone maker, carrier or satellite company.

Wi-Fi on Planes, Explained

A read-along lesson on Wi-Fi in the air: what airplane mode turns off, why the rule exists, how the internet reaches a plane by ground towers or satellites, why it feels slow when everyone on board shares one link, and how to get online in your seat when the sign-in page won't appear. The lesson follows the free PDF guide of the same name, word for word, with its 11 figures.

Why it's here. It answers a question Wi-Fi people get asked by everyone who flies: why is plane Wi-Fi so slow? Everyone on board shares one link to the ground, and often that link goes through a satellite far out in space, so small jobs fit in your slice and big ones don't.

How to use

1. Read top to bottom. It starts with the short answer, then what airplane mode turns off, why the rule exists, how the internet reaches a plane, why it feels slow, getting online in your seat, four things people get wrong, and tips for your next flight. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 6 covers what to do when the sign-in page won't appear: open a web browser and go to any web page, and on an iPhone don't tap Cancel.

4. The last row, Take it with you, names the free PDF guide the lesson follows, Wi-Fi on Planes, Explained, from WLAN Pros, to read or print away from the app. The PDF isn't inside the app.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Checked on 28 September 2026. Airline Wi-Fi offers and phone menus change often. The full source list is the last step of the lesson.
- When the crew gives an instruction about devices, follow it.

Wi-Fi and Your Health, Explained

A read-along lesson on the numbers behind the question: how much power a Wi-Fi router sends (about a tenth of a watt), how fast the signal fades as it spreads, how far below the safety limits it measures in real homes and schools, how the limits are set, and what health agencies say. The lesson follows the free PDF guide of the same name, word for word, with its 10 figures.

Why it's here. It answers a question Wi-Fi people get asked by family, parents and clients: is Wi-Fi safe to have in the house? The lesson answers with power numbers in plain words. A router sends about a tenth of a watt, the signal fades fast as it spreads, and measurements in homes and schools come out far below the limits.

How to use

1. Read top to bottom. It starts with the short answer, then radio waves against X-rays, a tenth of a watt, why distance matters so much, how far below the limit, how the limits are set, what was measured and what agencies say, four things people get wrong, and what to do if you want less exposure anyway. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.

3. Step 7 quotes the UK Health Security Agency and the World Health Organization in their own words, with the date each page carries.
4. Step 9 has the choices for anyone who wants less exposure anyway, and the one Caution in the lesson: turning Wi-Fi off at night also stops the cameras and doorbells that depend on it.
5. The last row, Take it with you, names the free PDF guide the lesson follows, Wi-Fi and Your Health, Explained, from WLAN Pros, to read or print away from the app. The PDF isn't inside the app.
6. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Checked on 28 September 2026. Agency pages are quoted as they read that day.
- The lesson explains the power numbers in plain words. It isn't medical advice. For a question about your own health, ask your doctor.
- Phones held to your head are a different question from Wi-Fi, and the lesson doesn't cover them.

Travel Routers, Explained

A read-along lesson on the travel router, a small box you pack in your bag that joins the hotel Wi-Fi for you and then makes your own Wi-Fi in the room: its two jobs and two radios, why your devices remember one network, the hotel's device limit, a network of your own, setup at home, joining and signing in at the hotel, a wired jack in the room, when it doesn't help, and a VPN (virtual private network) on the router. The lesson follows the free PDF guide of the same name, word for word, with its 20 figures.

Why it's here. It answers a question Wi-Fi people get asked by travelers: is a travel router worth packing? The one idea under it is that your devices join a network they already know. The box signs in to the hotel once, so the hotel usually counts one device, and every phone, laptop and tablet joins a network it already knows, just like at home.

How to use

1. Read top to bottom. It starts with the short answer, then what it is, why it helps, what you do at home and at the hotel, when it doesn't help, two surprises, a VPN on the router, four things people get wrong, and tips. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 7 has the setup steps to do at home, before the trip.
4. Steps 1 and 8 have an Open button for Captive Portals, Explained, which covers the hotel sign-in page itself.
5. The last row, Take it with you, names the free PDF guide the lesson follows, Travel Routers, Explained, from WLAN Pros, to read or print away from the app. The PDF isn't inside the app.
6. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- The lesson names no brand to buy. It is independent and isn't affiliated with or endorsed by any router maker or hotel company.
- The travel router keeps other guests away from your devices. It doesn't hide what you do online from the hotel.

Mobile Hotspots, Explained

A read-along lesson on the mobile hotspot, which takes the cell signal a phone uses and turns it into Wi-Fi: the two kinds (a pocket box or your phone), how a web page makes the round trip, what decides the speed, a busy network, data and your plan, battery and heat, where to put it, other ways to share, keeping it private, and how it compares with a travel router. The lesson follows the free PDF guide of the same name, word for word, with its 22 figures.

Why it's here. It answers a question Wi-Fi people get asked by travelers and people working away from home: why is my hotspot slow? A laptop's Wi-Fi bars show only the short link to the hotspot. The speed is decided by the cell signal where the hotspot sits and by how many people share the tower.

How to use

1. Read top to bottom. It starts with the short answer, then what it is, how it works, speed, data, battery and heat, where to put it, other ways to share, keeping it private, hotspot or travel router, four good habits, four things people get wrong, and tips. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 9 covers where to put the hotspot: where the cell signal is best, watching the cell bars on the hotspot itself.
4. Open buttons for Travel Routers, Explained, Phone Data Abroad, Explained, Weak Cell Signal at Home, Explained and Home Internet, Explained follow the steps that name them, when those lessons are in the app.
5. The last row, Take it with you, names the free PDF guide the lesson follows, Mobile Hotspots, Explained, from WLAN Pros, to read or print away from the app. The PDF isn't inside the app.
6. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- The lesson names no brand to buy. It is independent and isn't affiliated with or endorsed by any phone, carrier or hotspot maker.

Captive Portals, Explained

A read-along lesson on the Wi-Fi sign-in page, also called a captive portal: why it sits between the Wi-Fi and the internet, how your phone notices it, what to do when the page doesn't show up, why it comes back, what the network can see, how to stay safe on it, and what to do with a device that has no screen. The lesson follows the free PDF guide of the same name, word for word, with its 15 figures.

Why it's here. It answers a question Wi-Fi people get asked in every hotel, airport and cafe: the Wi-Fi says connected, so why won't anything load? Joining the Wi-Fi and reaching the internet are two steps, and the sign-in page sits between them. This lesson is written for the person

holding the phone. For the network side, how a network announces its sign-in page and how a device checks for it, see Connected, No Internet: Captive Portals.

How to use

1. Read top to bottom. It starts with the short answer, then what it is, how your phone notices, what to do when the page doesn't show up, why it comes back, what the network can see, staying safe, devices with no screen, five things people get wrong, and tips for your next trip. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 4 has the steps to try when the page never pops up, and what to do if you use a VPN (virtual private network) or private DNS.
4. Step 7 has the one Stop box in the lesson: never type your email password into a sign-in page.
5. Steps 1 and 8 have Open buttons for Wi-Fi on Planes, Explained and Travel Routers, Explained when those lessons are in the app.
6. The last row, Take it with you, names the free PDF guide the lesson follows, Captive Portals, Explained, from WLAN Pros, to read or print away from the app. The PDF isn't inside the app.
7. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- The lesson is independent and isn't affiliated with or endorsed by any hotel, airline, phone maker or government agency. Companies are named only as sources.
- A real sign-in page is a gate, not a window into your secure pages. Still, type only what the place needs.

Bluetooth, Explained

A read-along lesson on why your earbuds cut out, why the car won't connect, and what to do about each. Bluetooth is a short-range radio that links your phone to earbuds, speakers, watches and your car, and when the sound cuts out, one of three things is usually to blame: something in the way, a crowded band, or a stale pairing. The lesson follows the free PDF guide of the same name, word for word, with its 10 figures.

Why it's here. It answers a question Wi-Fi people get asked by people who are not technical: why do my earbuds keep cutting out, and why won't my car connect? It also carries the Wi-Fi angle. Bluetooth uses the same radio band as a lot of Wi-Fi, so putting the devices that can use it on 5 GHz Wi-Fi leaves more room for your earbuds. Wireless CarPlay uses Bluetooth only to find the car, and Wi-Fi carries the screen and the sound. Figure 8 shows the two radios and their two jobs.

How to use

1. Read top to bottom. It starts with the three usual causes, then pairing and connecting, the band Bluetooth shares with Wi-Fi, your body in the path, crowds, why calls sound thinner than music, one set of earbuds on two devices, your car, wireless CarPlay and Android Auto, what to do when the car won't connect, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 10 has the order to try things in when the car won't connect. Do them while parked.
4. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.
5. Take it with you, at the end of the lesson, names the free PDF guide it follows, Bluetooth, Explained from WLAN Pros, to print or share.

Field notes

- Checked against each company's own pages in September 2026. Menus and car support change with software updates.
- The lesson gives no decibel figure for the human body, because the guide had no sourced number to give. Line thickness in Figure 3 shows the idea, not a measured amount.

- Figure 4 is not to scale.
- The lesson is independent and isn't affiliated with or endorsed by any phone, earbud or car maker.

Weak Cell Signal at Home, Explained

A read-along lesson on why your phone has bars at the curb and drops calls in the kitchen, and the free fix to try first. Most weak signal indoors isn't your phone or the tower. It's the building: a signal that's fine at the curb can lose most of its strength getting through a wall or a window. The lesson follows the free PDF guide of the same name, word for word, with its 13 figures, and compares the three fixes: Wi-Fi Calling, a signal booster and a network extender.

Why it's here. It answers a question Wi-Fi people get asked by family and clients: why do my calls drop inside the house? The physics is the same as Wi-Fi through a wall. Every layer takes a share, metal and coated windows take the most, and higher bands lose more than lower ones. The free fix rides the Wi-Fi: Wi-Fi Calling goes around the walls over your home internet. Figure 7 shows that path, and Figure 13 sets the three fixes side by side.

How to use

1. Read top to bottom. It starts with the short answer and the one idea behind the guide (a booster doesn't make signal, it makes what it hears outside louder), then what a wall costs a signal, what a lab measured, why energy-efficient homes block more, why 5G struggles indoors, how to find where the signal is, Wi-Fi Calling, how a booster works, keeping its two antennas apart, the rules for boosters, network extenders, the three fixes side by side, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 6 is the ten-minute test with your own phone that tells you which fix fits. Judge by whether a call holds up, not by the bars.
4. Where the lesson names another guide in the series, an Open button takes you to that Guided Lesson once it is in the app.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

6. Take it with you, at the end of the lesson, names the free PDF guide it follows, Weak Cell Signal at Home, Explained from WLAN Pros, to print or share.

Field notes

- Checked in September 2026. Carrier products and pages change, so read your carrier's current page before you buy.
- Figure 1 is computed for one layer, head-on, from the ITU-R P.2040 multilayer method. Figure 2 is one UK lab's measurements (Ofcom, 2014). Figure 4 is not to scale.
- The booster rules are the US ones, with a line on the UK and Australia. Rules vary by country, so check with your carrier and your regulator first.
- The Keith's note was written about Wi-Fi. The lesson says so under it: the tower's signal crosses the same materials.
- No booster brand is named or recommended. The lesson is independent and isn't affiliated with any booster maker or carrier.

How GPS Works, Explained

A read-along lesson on how your phone finds itself by listening to satellites, and what to do when the blue dot gets it wrong. Satellites high above the Earth send out the time and where they are. Your phone hears several of them, measures how long each signal took to arrive, and works out where it is. It never sends anything back. The lesson follows the free PDF guide of the same name, word for word, with its 14 figures.

Why it's here. It answers a question Wi-Fi people get asked by people who are not technical: how does my phone know where I am, and does GPS track me? It also sets up the Wi-Fi angle. GPS needs to see the sky, so indoors it mostly gives up and the phone switches to Wi-Fi and cell towers instead. That half of the story is its companion lesson, How Your Phone Knows Where It Is, Explained. Figure 8 shows where GPS works and where it doesn't.

How to use

1. Read top to bottom. It starts with the short answer, then satellites that send the time, why it takes four satellites to find one spot, where GPS works and where it doesn't, when the dot is wrong on purpose (jamming and spoofing), four things people get wrong, and tips for a

better fix. Each myth hides its fact until you tap Reveal the fact.

2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 1 names the indoor half of the story. Its Open button takes you to How Your Phone Knows Where It Is, Explained.
4. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.
5. Take it with you, at the end of the lesson, names the free PDF guide it follows, How GPS Works, Explained from WLAN Pros, to print or share.

Field notes

- Checked on 28 September 2026 against GPS.gov, the US Space Force, the US Coast Guard Navigation Center and the other sources listed at the end of the lesson.
- About 5 m (16 feet) outdoors is typical, not a promise. Near tall buildings or under trees the dot can be off by much more.
- Figures 4, 5, 6, 10 and 13 are not to scale.
- The lesson is independent and isn't affiliated with or endorsed by any government agency, satellite operator or phone maker.

How Your Phone Knows Where It Is, Explained

A read-along lesson on how your phone finds itself indoors and out, and which switches are yours. Your phone listens for five kinds of clues (satellites, Wi-Fi, beacons, cell towers and its own motion sensors) and blends them into one best guess. Outdoors, satellites do much of the work. Indoors, Wi-Fi usually does. The lesson follows the free PDF guide of the same name, word for word, with its 11 figures, the location switches on iPhone and Android, and the _nomap setting for your own router.

Why it's here. It is the Wi-Fi half of the location story, and the question Wi-Fi people get asked most about it: how does my phone know where I am inside? The phone never joins the networks around it. It only hears them and looks them up on a map that phones walking past have already

built. The two steps, building the map and using the map, are David Coleman's, and so is the name Fused Location Positioning (FLP) for the blending. Figure 3 shows the two steps, and Figure 4 shows the dot holding steady through a lobby door.

How to use

1. Read top to bottom. It starts with the short answer, then the five kinds of clues, a map of Wi-Fi, blending it all, your switches, keeping your router off the maps, your location on a 911 call, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 5 has the switches for iPhone and for Android, one card each. Step 6 has the steps to add _nomap to your Wi-Fi name.
4. Step 1 names the outdoor half of the story. Its Open button takes you to How GPS Works, Explained.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.
6. Take it with you, at the end of the lesson, names the free PDF guide it follows, How Your Phone Knows Where It Is, Explained from WLAN Pros, to print or share.

Field notes

- Checked on 28 September 2026. Apple and Google update their help pages often, and phone makers move menus, so check the steps on your own phone.
- The two-step idea and the term Fused Location Positioning come from David Coleman's LinkedIn post on Wi-Fi positioning. The two Keith's notes are from Keith's LinkedIn post of 2026-08-10.
- The 911 step describes calls in the US.
- Figures 1 to 6 and 11 are not to scale.
- Apple, iPhone, Google and Android are trademarks of their owners. The lesson is independent and isn't endorsed by them.

Analog vs Digital, Explained

A read-along lesson on what analog and digital mean, and why your Wi-Fi slows down while your TV just freezes. Analog keeps a smooth copy of something. Digital measures it again and again, and keeps only the numbers. Noise makes an analog signal worse a little at a time, a digital signal ignores noise until it can't and then fails all at once, and Wi-Fi sits in between: it slows down in steps to stay connected. The lesson follows the free PDF guide of the same name, word for word, with its 12 figures.

Why it's here. It answers a question Wi-Fi people get asked by people who are not technical, and carries the idea under every Wi-Fi rate: Wi-Fi sends digital data, but what leaves the antenna is always a smooth wave. Each small change in the wave's height and timing stands for a few bits, and as the signal weakens, Wi-Fi switches to a simpler, slower way of sending. Figure 10 draws the slope, the cliff and the staircase side by side, and the Modulation Simulator shows the same idea on the constellation.

How to use

1. Read top to bottom. It starts with the short answer, then a smooth copy against a list of numbers, why a digital copy doesn't fade, why every radio wave is analog, three ways a signal fades, four things people get wrong, and what this means at home. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 7 has two things to do at home: move closer or clear the path so Wi-Fi climbs back up its staircase, and two tips for TV and music.
4. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.
5. Take it with you, at the end of the lesson, names the free PDF guide it follows, Analog vs Digital, Explained from WLAN Pros, to print or share.

Field notes

- Checked on 28 September 2026. Music app settings change with updates.
- Figure 10 is a sketch, not a measurement.

- The Keith's note quotes Fix Your Own Wi-Fi (WLAN Pros, in preparation, 2026), chapter 5.
- Company names appear only where their own documents are the source. The lesson is independent and isn't affiliated with any maker or broadcaster.

Scales and Ratios, Explained

A read-along lesson on the tiny and huge numbers behind Wi-Fi, turned into things you can hold, see and picture: a foot of wire for a nanosecond, a stack of money for decibels, a ruler for a radio wave, and jars for bytes. No math is needed. The lesson follows the free PDF guide of the same name, word for word, with its 11 figures.

Why it's here. Wi-Fi people use numbers every day that are too small or too big to picture, and the people they teach stop listening at the first one. Each step here gives one picture to keep: light goes about a foot in a nanosecond, adding 3 dB about doubles the power and adding 10 dB makes it ten times bigger, a signal can arrive a billion times weaker than it left and still work, and a 100 Mbps plan moves about 12 megabytes a second. Figure 11 collects four of them.

How to use

1. Read top to bottom. It starts with the short answer, then a nanosecond you can hold, Wi-Fi keeping time in tiny steps, near and far, power and the rule of 3s and 10s, why distance costs so much, how big a radio wave is, bits and bytes, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Figure 8 is drawn at actual size on the printed guide. On a screen it is fitted to the width, so its caption says so.
4. The last step names three companion guides. Each has an Open button that takes you to its Guided Lesson.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

6. Take it with you, at the end of the lesson, names the free PDF guide it follows, Scales and Ratios, Explained from WLAN Pros, to print or share.

Field notes

- Gathered on 28 September 2026.
- Figure 1 is not actual size, and Figure 2 is not to scale. Figure 4 uses three zoom levels, each with its own scale bar.
- Adding 3 dB about doubles the power; the lesson says so, and says it is close enough for any Wi-Fi math.
- Real downloads are slower than the divide-by-8 figure. Other devices share the connection, and every message carries a little extra.

Phone Data Abroad, Explained

A read-along lesson on how to land in another country with your phone working, and no surprise bill when you get home. There are three ways to get data abroad: roaming with your own carrier, a travel eSIM, or a local SIM. Most phones can hold two plans at once, so your home plan can stay on for calls and texts while a second plan carries your data. The lesson follows the free PDF guide of the same name, word for word, with its 13 figures and a four-stop checklist for the trip.

Why it's here. Wi-Fi people get asked about phones abroad by friends, family and colleagues before every trip. The lesson carries the Wi-Fi angle too: hotel, airport and ship Wi-Fi is often the cheapest data on a trip and the least predictable, full Wi-Fi bars only mean the Wi-Fi part worked, and nothing loads until you finish the sign-in page. Figure 11 shows that sign-in step.

How to use

1. Read top to bottom. It starts with the short answer and the one idea behind it, then covers what a SIM does, whether your phone is unlocked, the three ways to get data, roaming with your own carrier, travel eSIMs and local SIMs, keeping your own number, the switches that matter, hotel, airport and ship Wi-Fi, which one to pick for a trip, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.

2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Figure 10 shows which switch stops what, and Figure 13 is the checklist: four stops, two small jobs at each one. Step 7 has an Open button for Wi-Fi Calling, Explained, which the guide names for calls home over hotel Wi-Fi.
4. At the end, Take it with you names the free PDF guide the lesson follows, for printing or reading on a trip.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Carrier prices and plans change often. The lesson's carrier details were checked on each company's own pages in September 2026, so read your carrier's current travel page before you go.
- Prices are left out on purpose. Figure 6 shows how a daily fee adds up, not what any carrier charges.
- Settings paths are given for iPhone and, more loosely, for Android, where menus differ by maker.
- The lesson names no travel eSIM company and recommends none. It is independent and isn't affiliated with any carrier, phone maker or eSIM seller.

Teaching it

Roaming with your own carrier, a travel eSIM and a local SIM, how to keep your own number on while a second plan carries the data, and the switches that stop a surprise bill.

Smart Home Radios, Explained

A read-along lesson on the four names on smart home boxes: Wi-Fi, Thread, Zigbee and Matter. Wi-Fi, Thread and Zigbee are radios. Matter is a shared language that devices speak over Wi-Fi, Thread or a network cable, and Zigbee reaches it through a bridge. The lesson covers why your

smart plug won't join, how to get it to join, what a border router is, and what the Matter logo on the box does and does not promise. It follows the free PDF guide of the same name, word for word, with its 13 figures.

Why it's here. Wi-Fi people get asked why a new smart plug won't join by family, friends and clients. The usual answer is the Wi-Fi: many plugs and bulbs have only a 2.4 GHz radio, and a network that offers only 5 GHz is invisible to them. The lesson also shows why every phone, laptop and TV that moves up to 5 or 6 GHz leaves more room on 2.4 GHz for the small radios that can't leave it. Figure 3 shows the one band they all share.

How to use

1. Read top to bottom. It starts with the short answer and the one idea behind it, then covers three radios and a language, one band shared, why your smart plug won't join, getting it to join, a network for your gadgets, Thread and its border router, one network with several doorways, Zigbee and what Matter changes, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 5 is the fix list, gentlest first, and step 6 is the last fix: a 2.4 GHz network for your gadgets, made a regular network and not a guest one. Figure 13 is four checks before you buy.
4. At the end, Take it with you names the free PDF guide the lesson follows, for printing or handing to someone setting up a smart home.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Checked in September 2026 against the Connectivity Standards Alliance and the Thread Group first, then the platform makers' own help pages. Matter versions and the device types each app supports change often.
- Figure 3 is marked as a simplified picture, not to scale. The lesson gives no channel numbers for Wi-Fi, Thread or Zigbee.
- Some newer smart plugs have a 5 GHz radio too. The lesson's advice is for the ones whose box says 2.4 GHz only.

- The lesson recommends no brand of hub. It is independent and isn't affiliated with any standards body or device maker.

Teaching it

Wi-Fi, Thread, Zigbee and Matter: three radios and a language, why a 2.4 GHz-only smart plug won't join, the gadget network that fixes it, and what a Thread border router does. Figure 3 (one band, shared) is the Wi-Fi hook for a Wi-Fi audience.

Cameras, Doorbells and Baby Monitors, Explained

A read-along lesson on security cameras, video doorbells and Wi-Fi baby monitors. Most gadgets in a home pull things in. A camera pushes video out, often all day, so it is judged by what it can send from the spot where it hangs. The lesson covers upload against download, how many cameras a line can carry, signal at the door, which Wi-Fi lane, wired cameras, where the video goes, baby monitors, keeping strangers out, and what to do when the Wi-Fi is cut on purpose. It follows the free PDF guide of the same name, word for word, with its 11 figures and a half-hour appendix.

Why it's here. Wi-Fi people get asked why a doorbell drops offline or a clip starts late. The answer is usually the Wi-Fi hop or the upload, not the camera. The outside wall is the hardest one for Wi-Fi to cross, a camera with one bar sends slowly and keeps the Wi-Fi busy for everyone else, and a cable to an access point just inside the front wall does most of the good of a cable to the camera. Figure 3 shows the path from a back-of-house router to the front door.

How to use

1. Read top to bottom. It starts with the short answer and the one idea to remember, then covers why the camera sends, how many cameras your line can carry, signal at the door, which Wi-Fi lane, wired cameras and doorbells, where the video goes, baby monitors, keeping strangers out, when the Wi-Fi is cut on purpose, and five things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.

3. Step 3 and the appendix each have an Open button for Network Quality, where the guide says to read your upload row. Step 4 has the fixes for signal at the door, gentlest first.
4. The appendix, Do this now, is four tasks for one camera at a time: check the upload, check the signal where it hangs, lock it down, and test recording without the internet.
5. At the end, Take it with you names the free PDF guide the lesson follows, for printing or handing to a neighbor with a new doorbell.
6. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- The per-camera upload figure, about 3 megabits a second for one maker's best-quality camera, is that maker's published number, not a rating. Figure 2 is marked not to scale.
- Checked in September 2026. Camera plans, apps and rules change, so check the ones you rely on.
- The baby monitor cord distance, at least 1 m (3 feet) from any part of the crib, is the US Consumer Product Safety Commission's.
- Jammer rules are the US ones, from the FCC. The police warnings are as reported by PCWorld and NBC Los Angeles.
- The lesson is independent and isn't affiliated with any camera maker.

Teaching it

A camera is the rare gadget that sends more than it receives: upload against download, signal through the outside wall, what a wired doorbell's wire carries, cloud or home recording, baby monitors, locking a camera down, and jammers. Figure 3 (same house, same doorbell) is the Wi-Fi hook for a Wi-Fi audience.

How Your Devices Access the Internet, Explained

A read-along lesson on the trip everything online takes: from your device to the box in your home, out the line you pay for, across the internet, and to the video or call you wanted. Your Wi-Fi and your internet are two different roads that meet at one box, so they get blamed as one thing. The lesson follows the free PDF guide of the same name, word for word, with its 10 figures, and shows how to tell which road is slow.

Why it's here. It is a conversation Wi-Fi people have all the time: the Wi-Fi is slow, so buy a bigger plan. The lesson separates the two roads. The bars grade only the hop to the box, a faster plan changes only the line to the provider, and cellular skips the box altogether. Figure 9, the hotel with perfect bars and one thin line to the internet, is the case where more Wi-Fi would cost a fortune and fix nothing.

How to use

1. Read top to bottom. It starts with the short answer and the one big idea, then covers the whole trip, two roads and one box, the six stops, what your bars measure, how cellular skips your box, which road is slow, why a bigger plan won't fix slow Wi-Fi, the hotel with perfect bars, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Wherever the guide says to tap Check My Connection, in steps 7, 9 and 11, an Open button takes you to Test My Connection, which grades your Wi-Fi and your internet separately. Step 4 opens Home Internet, Explained and step 6 opens Wi-Fi Calling, Explained, the two guides it names.
4. At the end, Take it with you names the free PDF guide the lesson follows, for printing or sending to whoever asked why the internet is slow.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- The guide is drawn from Keith's book Fix Your Own Wi-Fi (in preparation), chapters 1, 2 and 6, and from the WLAN Pros master graphic How your devices access the Internet, version 3.

- Figure 2 is that master graphic.
- The lesson names no provider and no carrier. It is independent and isn't affiliated with any of them.

Teaching it

Wi-Fi and cellular are ways to reach the internet, not the internet: two roads that meet at one box, six stops, what the bars measure, and why a bigger plan won't fix slow Wi-Fi. Figure 9 (the hotel with perfect bars) is the story to tell a room. Pair it with Test My Connection, which grades the two roads separately.

Mesh, Extenders and Wired Access Points, Explained

A read-along lesson on the three kinds of gear sold to fix a dead zone: extenders, mesh kits and wired access points. Every box you add needs its own link back to your main box, and that link decides how fast the far room will be. A box that uses one radio for the link back and for you has to take turns. The lesson follows the free PDF guide of the same name, word for word, with its 11 figures, and covers where to put the new box, four ways to build the link back, and which one to buy.

Why it's here. Wi-Fi people get asked which extender or mesh kit to buy. The lesson answers with one question: how does the new box link back to the main one? The lesson carries the shared-airtime reason: every piece of your video crosses the air twice on one radio, and the two trips share the same time, so an extender in the dead zone can leave the laptop slower than no extender at all. Figure 6 shows the same far room reached five ways.

How to use

1. Read top to bottom. It starts with the short answer and the one big idea, then covers three ways to fill a dead zone, one radio with two jobs, why it can cut your speed in half, where to put it, five setups for one far room, four ways to build the link back, wired access points, which one to buy, five things people get wrong, and what to test after you set it up. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.

3. The lesson's speeds come from a model in the Wireless Classroom. Steps 4 and 6, and the sources, have an Open button for Repeaters and Mesh Backhaul, that model, so you can move the hops yourself. Step 9 and the related guides open How Your Devices Access the Internet, Explained and Home Internet, Explained.
4. At the end, Take it with you names the free PDF guide the lesson follows, for printing or taking to the store.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- The speeds are from the Classroom model, one long open hallway with typical settings, as the guide says. Your walls and your gear will change every number; the pattern stays the same.
- Google, Ofcom, the MoCA Alliance and the Wi-Fi Alliance are cited from their own published pages, read 27 September 2026.
- The lesson is independent and isn't affiliated with any maker or provider.

Teaching it

Three ways to fix a dead zone, and the one question that decides between them: how does the new box talk to the main one? One radio taking turns, where to put an extender, five setups for one far room, and four ways to build the link back. Pair it with Repeaters and Mesh Backhaul, the model its numbers come from.

Is Your Router Too Old?

A read-along lesson on how to tell in five minutes whether your router still gets security fixes, and what to do if it doesn't. A router's age isn't counted in years. It's counted by whether it still gets fixes. The lesson follows the free PDF guide of the same name, word for word, with its 11 figures: what the router does, the three stages of its life, the five-minute check, who updates it, keep, update or replace, what the 2026 US router rule means for the one you own, and what to look for in the next one.

Why it's here. Wi-Fi people get asked whether an old router needs replacing, and since 2026 whether the new US rule makes it illegal. The lesson gives the test that matters: a router that no longer gets fixes can still stream movies, and the FBI has warned that criminals take over routers like that and use them as a disguise. Figure 6 turns the five-minute check into keep, update or replace.

How to use

1. Read top to bottom. It starts with the short answer and the one idea to remember, then covers what your router does, when a router gets too old, the five-minute check, who updates it, keep, update or replace, the 2026 rule, what to look for in the next one, and four things people get wrong. Each myth hides its fact until you tap Reveal the fact.
2. Tap a figure to open it full screen, then pinch to zoom. With a keyboard, Tab to a figure and press Enter. The figures are the guide's own, on a white card in both light and dark mode.
3. Step 4 is the five-minute check. The appendix, Do this now, is about fifteen minutes in the router's app or settings: update it, lock it down, and write it down. Steps 5 and 8 have an Open button for Mesh, Extenders and Wired Access Points, Explained, where the guide names it.
4. At the end, Take it with you names the free PDF guide the lesson follows, for printing and keeping next to the router.
5. On a computer or tablet, Present opens the lesson full screen for a projector, one step per slide, with the figures and myths on the stage and the rest of the step in the side panel: the Right and Left arrows move between steps, Space reveals the next fact on a myth step, R goes back to the first step with every fact hidden, F switches full screen, ? lists the keys and Esc exits.

Field notes

- Checked on 28 September 2026. The 2026 rule is the US one, from the FCC's own notices; the date makers may keep fixing routers approved before the rule, at least 1 January 2029, is from FCC Public Notice DA 26-454.
- The FBI warning is its Public Service Announcement I-050725-PSA of 7 May 2025, on criminals using end-of-life routers.
- The lesson names no router maker or model on purpose. Menus differ by maker, so a setting may have another name, such as Administration or Advanced.
- The lesson is independent and isn't affiliated with any router maker or provider.

Teaching it

A router is too old when it stops getting security fixes, not when it stops working: the three stages, the five-minute check, who updates the box, keep, update or replace, the 2026 US router rule, and what to look for in the next one.

Why the TV and the Printer Vanish on Guest Wi-Fi

A read-along lesson with one step-through, on why screen casting and printing find nothing from a guest network. A phone finds the TV or the printer with Multicast DNS (mDNS): one question sent to every device on the local network segment at once, addressed to 224.0.0.251 (FF02::FB for IPv6) on UDP port 5353. RFC 6762 defines that address as link-local, so the question never leaves the segment. The step-through follows the question from the phone, with one control: same network, guest network, or client isolation.

Why it's here. It answers a call Wi-Fi people get from hosts and guests: the TV is broken, or the Wi-Fi is broken. Neither is. The guest network is a separate segment, and the discovery question stops at its edge. Client isolation stops it too, even on one network, because the access point will not pass traffic from one wireless device to another.

How to use

1. Read top to bottom. The step-through is in section 2, why the question stays local is in section 3, and what to do about it is in section 5.
2. Pick Same, Guest or Isolation, then press Step. Step 1 shows the question and its address. Step 2 follows it to every device and marks where it stops. Step 3 shows what the phone's list finds. Back takes one step back and Reset returns to step 1.
3. Switch at any step to compare. With Same, the phone is on the main network with the TV and the printer. On the same network the TV and the printer both answer. On the guest network the question stops at the router between the two networks. With client isolation the phone is on the same network as the TV, and the question still stops at the access point.
4. The phone's internet works in all three. Discovery failing does not mean the Wi-Fi is broken.

Field notes

- Some networks run an mDNS gateway, also called a reflector, that repeats discovery questions and answers between segments (VLANs, virtual local area networks) under rules the administrator sets. It is the generic fix when the TV has to sit on a different segment from the

people casting to it.

- Exactly what client isolation blocks is set differently from one product to the next. The lesson shows it stopping the question at the access point.
- A simplified picture: one access point, one router, one TV and one printer, all on Wi-Fi. A wired printer on the same segment, and how each kind of device names and shows what it finds, are left out.
- The lesson names no product. The TV, the printer and screen casting stand in for whatever the room has.

Teaching it

Why screen casting and printing find nothing from a guest network. Pick Same, Guest or Isolation and step the discovery question to the point where it stops. Ask the room first where they think it will stop with isolation on. Pair it with Multicast at the Basic Rate, which covers what multicast costs in airtime.

One Talker per Channel

A read-along lesson with one interactive stage, on why Wi-Fi slows down when everyone is home. On the same channel, only one device can transmit at a time, so the time on a channel is shared: with N devices that all have something to send, each gets about 1/N. Add devices and every bar shrinks; add a second access point on the same channel or on another channel and compare one waiting line with two.

Why it's here. It is the first idea behind every airtime question, told for beginners: a busy channel is a longer line, not a slower access point. The lesson uses a walkie-talkie as the picture: one person talks at a time, and says "over" to hand off. It keeps the rule to one channel and names OFDMA and MU-MIMO as later exceptions without teaching them.

How to use

1. Read top to bottom. The walkie-talkie picture is in section 2, the interactive stage in section 3, two access points in section 4, the slow device in section 5, and what to do about it in section 6.

2. In section 3, use the minus and plus buttons to set 1 to 12 devices on access point 1. Each device has a letter, a bar for its share of the time and a slice of the channel's pie. The headline gives the share as a fraction and a percent.
3. Press Next turn to pass the turn. The device talking now has a ring around its letter, and the line under the room names it: one per channel.
4. Under Second access point, on which channel? pick Same or Other, and set its devices (1 to 6). Same channel is one waiting line, so every device gets 1/(N+M). Other channel is two waiting lines, 1/N on one and 1/M on the other, and two devices can transmit at once.
5. Turn on Make device A slow. A still gets an equal share of the turns, but each of its turns takes 4 times as long, so it uses more of the time and everyone else gets less.
6. On a computer or tablet, Present shows the stage full screen for a projector: Up and Down add or remove a device on access point 1, Space swaps the second access point between same channel and other channel, Right passes the turn, A adds or removes the second access point, S makes device A slow, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Devices on access point 1	devices	1 to 12, 4 at open
Second access point, on which channel?	choice	None, Same, Other
Devices on access point 2	devices	1 to 6, 3 at open
Make device A slow	on or off	off at open

How it works. Share of the time for each device = 1 / (devices on its channel). With device A slow, turns stay 1/N each and A's share of the time = 4 / (N + 3), each other device's = 1 / (N + 3). Access point 1 is on channel 36; a second access point on the other channel is on channel 149.

Example. 4 devices on one access point: 1/4 each, 25%. Add a second access point on the same channel with 3 devices: 1/7 each, 14%. Move it to the other channel: 1/4 (25%) on channel 36 and 1/3 (33%) on channel 149. Make device A slow with 4 devices: A uses 57% of the time and the other three 14% each.

Field notes

- A teaching approximation: every device always has something to send, turns are shared evenly, nothing collides, and two channels do not leak into each other. Channel Planner states the same even split for its contention domains.
- The slow device's 4 times is an illustrative choice. Airtime Fairness models slow clients with real rates and frame sizes.
- OFDMA and MU-MIMO let one access point send to several devices in the same moment. The lesson names them as later exceptions and leaves them out.
- Access point 2 stops at 6 devices so all 18 devices keep a distinct color and letter.
- Next: Channel Planner and Medium Access Simulator, linked from section 7.
- Source: IEEE Std 802.11-2024, 10.2.2 DCF (page 1875), 10.2.3.2 EDCA (page 1876), 10.3.2.1 carrier sense mechanism (page 1885), 10.23.2 EDCA procedure.

Teaching it

Teaches: on the same channel, only one device can transmit at a time, so the time on a channel is shared: N devices get about 1/N each, and a second access point on the same channel is one waiting line, not two.

Control	What it does	Why it's there
Devices on access point 1 (minus, plus)	1 to 12 devices; opens with 4.	The bars shrink as devices are added. That is the whole lesson: the access point did not get slower, the line got longer.
Second access point, on which channel? None / Same / Other	Adds a second access point on channel 36 (Same) or on channel 149 (Other).	Same channel is one waiting line, 1/(N+M) each. Other channel is two lines, 1/N and 1/M, and two devices transmit at once. That is the reason for channel planning.
Devices on access point 2 (minus, plus)	1 to 6; opens with 3. Only shown with a second access point.	Lets the room set N and M separately. It stops at 6 so all 18 devices keep a distinct color and letter.

Control	What it does	Why it's there
Make device A slow	Device A's turns take 4 times as long (an illustrative factor).	Equal turns are not equal time. With 4 devices A still gets 1/4 of the turns but 57% of the time. This is the hook for Airtime Fairness.
Next turn	Passes the turn round the room. A lime ring marks the device talking, and a line names it.	Shows one talker per channel at every turn. With Other selected, two devices talk at once, one on each channel.
Reset	Back to 4 devices, one access point, nothing slow.	A clean start between demos.
Walkie-talkie illustration (section 2)	Two radios: one talking, one listening.	One person talks, then says "over": the class already knows the rule from walkie-talkies.

A demo that works (about 3 minutes)

1. Set up: the opening scene. Four devices, one access point, 25% each.
2. Ask the room: "My neighbor puts in a second access point on my channel. Does my Wi-Fi get faster, slower or stay the same?"
3. Reveal: press A (Same). Seven devices, 14% each: everyone got slower. Press Space (Other): 25% and 33%, and Right shows two devices talking at once.
4. Optional follow-up: press S. Device A takes 57% of the time with the same number of turns. Open Airtime Fairness from section 5.

Pairs with: Medium Access Simulator, Airtime Fairness, Channel Planner.

Left out on purpose: collisions, backoff and priorities (Medium Access Simulator covers them); adjacent-channel leakage and hidden devices; devices that have nothing to send; OFDMA and MU-MIMO, which the lesson names as later exceptions. The even split is the same teaching approximation Channel Planner states.

Public Wi-Fi

A guided lesson on what the person next to you on public Wi-Fi can see. One control picks the network type: Open, Enhanced Open, or Password, and for Password, WPA2-Personal or WPA3-Personal. A drawing shows which stretch of the trip each seal covers, and four color-coded rows, each with a word label, show what a listener on the same network gets: that your device is here, which sites you visit, what you read and type on a secure site, and anything sent without encryption.

Why it's here. It answers the question Wi-Fi people get asked about every cafe and hotel: is public Wi-Fi safe? It corrects both myths, never use public Wi-Fi and no lock means anyone can read my banking, with the Federal Trade Commission's current advice: because of the widespread use of encryption, connecting through a public Wi-Fi network is usually safe. HTTPS (Hypertext Transfer Protocol Secure) protects what you send on any network; Wi-Fi encryption protects the air between your device and the access point (AP).

How to use

1. Read the short answer, then pick a network type in Try it. The rows update at once: Can read it, Can see it, or Sealed, each with a line saying why. On a keyboard, Tab to the control and use the Left and Right arrow keys.
2. Pick Password to turn on the second control, then compare WPA2 and WPA3. The person next to you was given the same password, so WPA2-Personal reads like Open to them once they record you joining, and WPA3-Personal does not.
3. Answer the Predict, then reveal question before you open it, then read the four networks, three myths, and where Wi-Fi encryption stops. Each myth hides its fact until you tap Reveal the fact. The lesson has no Present mode; it reads as a page on any screen.

Field notes

- The listener is fixed: someone on the same network who only records the air, and on a password network was given the same password. A stranger without the password reads nothing on WPA2- or WPA3-Personal.
- Enhanced Open is Opportunistic Wireless Encryption (OWE, RFC 8110). Wi-Fi Alliance expects these networks to show without a lock icon, so the lock in a network list undersells them.
- Wi-Fi encryption ends at the AP. The network owner and its internet provider see site names and anything sent without encryption on all four types.

- Left out: fake networks that copy a familiar name, sign-in (captive) pages, virtual private networks (VPNs), traffic sent to every device at once, and encrypted DNS (Domain Name System) with Encrypted Client Hello, which can hide site names but not the server's address.

Wi-Fi Privacy Myths

A guided lesson in two parts. Part 1 is your phone's private Wi-Fi address: one control picks Off, Fixed or Rotating, and the lesson shows what two routers on the same day, and one cafe on two visits a month apart, record for the same phone, and whether a MAC (media access control) filter set on the first visit lets it back in. Part 2 is the hidden network name: one control hides it, and the name leaves the access point's beacons and appears in the phone's probe requests at home, at the airport and at the cafe.

Why it's here. It corrects three beliefs Wi-Fi people hear every week: that MAC filtering keeps strangers out, that the router's device list shows a phone's real address, and that hiding the network name is security. CWNP lists both MAC filtering and SSID (service set identifier) hiding among the weak security options that should not be used in enterprise networks.

How to use

1. In part 1, pick Off, Fixed or Rotating. Each scene says Can be matched or Can't be matched in words, and the filter row says Let in or Turned away. On a keyboard, Tab to a control and use the Left and Right arrow keys.
2. Answer the Predict, then reveal question before you open it. Show me on Fixed sets the control to the case the answer describes.
3. In part 2, turn Hide the network name on and off and watch the beacons at home and the probe requests in each place. Section 8 has three things people get wrong. Each myth hides its fact until you tap Reveal the fact. The lesson has no Present mode; it reads as a page on any screen.

Field notes

- Every address is illustrative. The hardware address is from the IETF documentation range (RFC 7042), so it belongs to no real device, and the private addresses have the locally administered bit set, as real ones do.

- The two-week rotation is Apple's published figure (Apple 102509, published 5 December 2025): Rotating changes the address every 2 weeks. The on-screen text attributes it to one phone maker and assumes nothing about others, which are not sourced here. The two cafe visits are a month apart so that a two-week rotation has certainly happened.
- Apple's defaults are Fixed on WPA2 or stronger networks and Rotating on open and weak-security ones (Apple Platform Security). The lesson lets you pick either on any network.
- Left out: other ways to recognize a device besides its address, the router's own settings for private addresses, and how networks that need a steady address (reservations, parental controls) cope with Rotating.

Connected, No Internet: Captive Portals

A read-along lesson with one step-through, on why a hotel, airport or airplane network can show full bars and still not reach the internet. Association and reaching the internet are separate steps, and a captive network holds you at its sign-in page, the captive portal, between the two. The step-through follows a device from association to online, with one switch: a network that announces its portal in the address reply (DHCP, the Dynamic Host Configuration Protocol, option 114, defined in RFC 8910) and answers the device's questions through a Captive Portal API (RFC 8908), or a network that announces nothing and intercepts the device's probe.

Why it's here. It answers a question Wi-Fi people get from travelers all the time: the Wi-Fi says connected, so why won't anything load? The bars measure the first step only. It also explains why Wi-Fi Calling does not work in a hotel until the sign-in page is finished: the phone's tunnel to the carrier is held back like everything else.

How to use

1. Read top to bottom. The step-through is in section 2, the two ways a network asks you to sign in are in section 3, and Wi-Fi Calling is in section 4.
2. Press Step to move through the five steps: association, getting an address, the check, the sign-in and online. Back takes one step back and Reset returns to association. Each step shows what the device sends, what the network replies, and what the person holding the device sees.

3. Flip the switch between Announces and Intercepts at any step. Both take the same five steps, so you compare like with like. At step 3, one network answers the device's question straight, and the other intercepts the probe and sends back its sign-in page, tagged Intercepted.
4. The Internet line reads Held and the Wi-Fi Calling line reads Waits for the sign-in until step 4, in both modes, while the Wi-Fi link stays up from step 1.

Field notes

- The sequence is illustrative. It shows the generic probe every device makes and leaves out how often each kind of device probes, which test address it asks, and how long it waits.
- The announcement can also arrive by DHCPv6 (option 103) or an IPv6 Router Advertisement (RFC 8910). The step-through shows the IPv4 DHCP option only.
- RFC 7710 used DHCP option 160 for the same job. RFC 8910 replaced it with 114 because of a conflict.
- RFC 8910 says networks will still need to intercept for older devices, and devices will still probe, for the foreseeable future. So a network that announces its portal still has to intercept for devices that do not read the announcement.
- Left out: paid tiers, session time limits beyond a mention, and networks that block the Wi-Fi Calling tunnel even after the sign-in.

Teaching it

Why a hotel or airplane network shows full bars and still won't load anything. Step through the device's check after association and flip the switch between a network that announces its portal and one that intercepts traffic. Stop at step 3 and ask the room what the device will see. It also explains why Wi-Fi Calling waits for the sign-in. Pair it with Association, Frame by Frame, which plays every frame up to DHCP and DNS.

RF and Propagation (13)

Decibels in Your Head: the Rules of 3 and 10

Shows why a few dB is a big change. One dB (decibel) slider moves a signal from -80 to -60 dBm (decibels relative to one milliwatt) against a fixed -70 dBm reference. Above the slider's dB ruler, a linear bar shows the same power in milliwatts, full length at -60 dBm, and a line joins each whole dB to its place on the bar. The readouts give the power of both, the difference as a ratio, and that ratio built from the rules of 3 and 10 beside the exact figure.

Why it's here. dB is a ratio, so equal steps in dB are equal multiplications of power. +3 dB doubles the power and +10 dB multiplies it by ten. That makes -67 dBm about twice the power of -70 dBm, which the dBm numbers hide. Together the two rules reach any whole number of dB, so a signal difference can be read in your head. This is the dBm-to-milliwatt arithmetic in the CWNA (Certified Wireless Network Administrator) exam objectives.

How to use

1. Predict, then reveal asks: is -67 dBm a little or a lot stronger than -70 dBm? Decide, reveal the answer, then press Show -67 dBm against -70 dBm to see the bar double.
2. Move the Signal level slider. Every 3 dB up doubles the linear bar, and every 10 dB up makes it ten times longer. At -80 dBm the bar is 1% of its length, at -70 dBm 10%, and at -60 dBm all of it.
3. The step buttons add or take away 3 dB or 10 dB, one rule at a time. A step that would leave the -80 to -60 dBm range is turned off.
4. Read the rules path. +1 dB, for example, is +10 -3 -3 -3: ten times, then halved three times, 1.25x against the exact 1.259x.
5. On a computer or tablet, Present opens the tool full screen for a projector: Up and Down move the signal 1 dB, Right adds 3 dB and Left takes 3 dB away, Page Up and Page Down move 10 dB, Space shows -67 dBm against -70 dBm, P reveals the answer, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Signal level	dBm	-80 to -60 in whole dB; default -70, the same as the reference
Reference	dBm	Fixed at -70

How it works. Power in milliwatts = 10^(dBm / 10), the dBm / Watt Converter's formula; picowatts (pW, trillionths of a watt) = milliwatts x 10^9. The power ratio of a difference of D dB = 10^(D / 10), so +3 dB = 10^0.3 = 1.995 and +10 dB = 10 exactly. The rules path writes D as 10a + 3b with the fewest steps (10 and 3 share no factor, so every whole D has one) and estimates the ratio as 10^a x 2^b. The linear bar's length is the signal's milliwatts divided by the milliwatts of -60 dBm.

Example. -70 dBm is 100 pW. -67 dBm is 199.5 pW: +3 dB, 1.995 times the power, about double. -60 dBm is 1,000 pW, ten times the reference. -69 dBm is +1 dB: +10 -3 -3 -3 by the rules, x10 /2 /2 /2 = 1.25x, against the exact 1.259x.

Field notes

- +3 dB is 1.995 times the power, so say about double. +10 dB is exactly ten times.
- The same rules hold for losses: -3 dB halves the power and -10 dB leaves a tenth of it.
- dB compares two powers, and dBm is one power compared with one milliwatt. The difference between two dBm levels is in dB.
- For a level outside -80 to -60 dBm, or a conversion to watts, use the dBm / Watt Converter. The dB Reference lists the common ratios in a table.
- Every number here is arithmetic. Nothing is measured and nothing is illustrative.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: dB is a ratio, so +3 dB doubles the power and +10 dB multiplies it by ten, which makes -67 dBm about twice the power of -70 dBm.

Control	What it does	Why it's there
Signal level (slider, -80 to -60 dBm in whole dB, default -70)	Moves the signal against a fixed -70 dBm reference. The linear milliwatt bar, the lines from the dB ruler and the readouts follow it.	One control, one visible change: a dB slider over a linear bar that doubles every 3 dB. The range was chosen so -80, -70, -67 and -60 dBm fill 1%, 10%, 20% and 100% of the bar; a wider range would have shrunk the -70 vs -67 comparison to a few pixels.
-10 dB, -3 dB, +3 dB, +10 dB (buttons)	Step the signal by one rule. A step that would leave -80 to -60 dBm is turned off.	Lets you act out the rules path the readout prints (+1 dB is +10 -3 -3 -3) one press at a time on the bar.
Reset	Returns the signal to -70 dBm and hides the answer.	Puts the signal back on the reference (1x, the same power) so the next question starts from zero difference.
Reveal the answer / Hide the answer	Shows the answer to "Is -67 dBm a little or a lot stronger than -70 dBm?"	The house predict-then-reveal pattern: the room commits to a guess before the screen answers.
Show -67 dBm against -70 dBm (appears after the reveal)	Sets the signal to -67 dBm.	Puts the prediction on the bar: 10% of the length becomes 20%, the step the dB ruler hides.
Open dBm / Watt Converter, Open dB Reference	Open the two calculator tools.	For any level outside -80 to -60 dBm or a conversion to watts; this tool is the picture, those are the calculators.

A demo that works (about 3 minutes)

1. Set up: open the tool as it starts, signal at -70 dBm, answer hidden.
2. Ask the room: "Is -67 dBm a little or a lot stronger than -70 dBm?" Many will say a little: it is only 3.
3. Reveal: press P, then Space. The bar goes from 10% to 20% of its length: 100 picowatts to 199.5, about double.
4. Optional follow-up: from -70, press Page Up. The bar goes to full length, ten times. Then ask what -69 dBm is and read the rules path: +10 -3 -3 -3, ten times then halved three times, 1.25x against the exact 1.259x.

Pairs with: FSPL Simulator, next on the shelf, where every doubling of distance costs 6 dB. dBm / Watt Converter and dB Reference for the numbers.

Left out on purpose: Nothing here is measured or illustrative; it is arithmetic. +3 dB is 1.995 times, not exactly 2, which the readouts show beside the rule. The range stops at -80 and -60 dBm so the bar stays drawable; outside that, use the dBm / Watt Converter. It says nothing about noise or SNR.

FSPL Simulator

Draws free-space path loss, or the received power it leaves, against distance for 2.4, 5 and 6 GHz on one chart. A cursor reads out each band at any distance, and a Why panel splits each band's loss into the part every band shares and the part that changes with frequency.

Why it's here. Higher bands lose more at the same distance, and the common explanation (the air absorbs 6 GHz more) is wrong. The extra loss comes from the receive antenna: a shorter wavelength makes a smaller antenna that catches less energy. Seeing the two parts side by side makes that stick, and a measured point shows how far a real room sits from the free-space best case.

How to use

1. Turn bands on and off with the chips at the bottom of the screen (on a wide screen, the panel on the right). Each band has its own line style: 2.4 GHz solid with a circle, 5 GHz dashed with a square, 6 GHz dotted with a triangle.
2. Drag across the chart or tap it to move the cursor, or use the Cursor slider under it. The table under the chart shows path loss and received power for each band at that distance, and the difference between bands.
3. Switch the chart between Received and Path loss, and the distance axis between 100 m and 1 km. In the Received view, dashed lines mark design targets. By default they are the -67 dBm voice and -70 dBm HD video targets from the Signal Thresholds tool.
4. Open the controls (the tune button) to pick any 20 MHz channel per band, set Tx power, both antenna gains and other losses, turn on the indoor log-distance model and set its exponent, or enter a measured RSSI and the distance it was taken at. The measured point plots as a diamond with its gap to the free-space curve in dB.

5. Under Design targets in the controls, set each line's level in dBm and a short label, add a line (up to four) or remove one. A line with no label reads as its level, for example -72 dBm. Your lines are saved on this device for every lesson, and Reset to -67 voice / -70 HD video puts the defaults back.
6. On a computer or tablet, Present opens this simulator full screen for a projector: Up and Down double and halve the cursor distance, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Bands	on / off	2.4, 5 and 6 GHz, all on by default
Channel	20 MHz channel number	any 20 MHz channel in the band; defaults 6 (2437 MHz), 100 (5500 MHz), 37 (6135 MHz)
Tx power	dBm	0 to 30; default 20
Tx and Rx antenna gain	dBi	-5 to 15; default 0 (isotropic)
Other losses	dB	0 to 30; default 0
Path-loss exponent n	none	2.0 to 4.0; default 3.0; indoor model only
Measured RSSI	dBm	-120 to 0
Measured distance	m	1 to 1,000
Design targets	dBm and a label	up to 4 lines, each -100 to -30 dBm with a label of up to 24 characters; default -67 dBm voice and -70 dBm HD video

How it works. FSPL (dB) = 20 log10(4 pi d / lambda), with lambda = c / f and c = 299,792,458 m/s. With d in meters and f in MHz this is 20 log10(d) + 20 log10(f) - 27.55. It splits exactly into spreading loss 10 log10(4 pi d^2), the same at every frequency, plus the aperture term -10 log10(lambda^2 / 4 pi), which depends only on frequency. Received power = Tx power + Tx gain + Rx gain - path loss - other losses. The band difference at equal distance is 20 log10(f2 / f1). The indoor model is PL(d) = FSPL(1 m) + 10 n log10(d / 1 m).

Example. At 10 m on channel 6 (2437 MHz): spreading loss 10 log10(4 pi x 100) = 31.0 dB, aperture term 29.2 dB, total 60.2 dB. With 20 dBm Tx and 0 dBi antennas the received power is -40.2 dBm. Channel 100 (5500 MHz) loses 20 log10(5500 / 2437) = 7.1 dB more at every distance, and all of that difference is in the aperture term.

Field notes

- Free space is the best case: no walls, no floor reflection, no people, no fading. Real indoor readings are usually well below the curve.
- Every doubling of distance costs 6 dB, so going from 1 m to 2 m costs as much as going from 50 m to 100 m.
- The indoor overlay is a model with an exponent you choose, not a measurement. n = 2 reproduces free space.
- The aperture term assumes an isotropic (0 dBi) receive antenna. A higher-gain antenna has a larger effective area and wins some of the difference back.
- Measured points are typed in by hand. Nothing is read from the Wi-Fi adapter.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: every doubling of distance costs 6 dB in free space, and higher bands lose more because the receive antenna is smaller at a shorter wavelength, not because the air absorbs more.

Control	What it does	Why it's there
2.4 GHz, 5 GHz, 6 GHz (band chips, all on)	Show or hide each band's curve.	Puts the bands on one chart so the gap between them is visible. Each band has its own line style and marker (solid circle, dashed square, dotted triangle) because the design system has no categorical colors, so the class can tell them apart without color.
Received / Path loss (toggle, default Received)	Plots received power in dBm, or free-space path loss in dB.	Path loss is what the formula gives; received power is the unit a design target is written in, and the target lines only appear in Received.
100 m / 1 km (300 ft / 3,000 ft in imperial)	Sets the end of the distance axis.	Keeps the default view on indoor distances while a point measured past 100 m (330 ft), or an outdoor link, can still land on the chart.
Log / Linear (toggle, default Log)	Switches the distance axis.	Log shows the straight 6 dB-per-doubling lines; linear shows the curve people are used to. Same numbers either way.
Drag or tap the chart; Cursor (slider)	Move the distance cursor. The table under the chart reads path loss and received power per band, and the difference between bands.	The readout makes the second lesson checkable: the difference between two bands is the same at every distance.
2.4 GHz channel, 5 GHz channel, 6 GHz channel (pickers; defaults 6, 100, 37)	Set each band's center frequency.	The band gap is 20 log10(f2/f1), so it depends on the channel, not just the band: 6 GHz channel 37 and channel 117 differ by about 0.5 dB.
Tx power (0 to 30 dBm, default 20)	Raises or lowers every received curve.	Received power = Tx power + gains - path loss - losses; moving it shifts every band equally and leaves the gap alone.
Tx antenna gain, Rx antenna gain (-5 to 15 dBi, default 0)	Add antenna gain at either end.	0 dBi is the isotropic antenna the aperture term assumes. A higher-gain receive antenna has a larger effective area and wins some of the high-band loss back.
Other losses (0 to 30 dB)	Subtracts a fixed loss from every curve.	A stand-in for cable, body or wall loss without modeling any of them.

Control	What it does	Why it's there
Show log-distance curves (switch); Path-loss exponent n (2.0 to 4.0, default 3.0, off until the switch is on)	Overlays the indoor log-distance model.	Free space is the best case; this shows how much faster an indoor path loses. Labeled a model with a chosen exponent, not a measurement; n = 2 reproduces free space.
Measured point: RSSI, Distance, Measured on (band), Clear measured point	Plots a reading you type as a diamond, with its gap to the free-space curve in dB.	The gap between the model and a measurement is where the design work happens. Manual entry only; nothing is read from the Wi-Fi adapter.
Design targets: Level, Label, remove (x), Add line (up to 4), Reset to -67 voice / -70 HD video	Edit the dashed target lines in the Received view. Saved on the device for every lesson.	The lines are editable with a number and a label. The defaults are read from the Signal Thresholds tool so the two tools cannot drift apart.
Show controls / Hide controls (tune button, phone only)	Opens the controls sheet under the chart.	Phone first: the chart owns the screen and the controls collapse to one row of band chips.

A demo that works (about 3 minutes)

1. Set up: all three bands on, Received, Log, cursor at 10 m (33 ft). On channel 6 the path loss is 60.2 dB and, with 20 dBm and 0 dBi antennas, received power is -40.2 dBm.
2. Ask the room: "Move twice as far away, to 20 m (66 ft). How much signal do we lose?"
3. Reveal: press Up. Every band drops 6.0 dB. Press it again at 40 m (131 ft): another 6 dB, the same as the first doubling. Then point at the Why panel: the spreading bar is identical for every band; only the aperture bar changes, and channel 100 sits 7.1 dB under channel 6 at every distance.
4. Optional follow-up: turn on Show log-distance curves at n = 3 and press Up again. A doubling now costs about 9 dB. Or switch to Linear to show the steep drop near the AP is the same 6 dB per doubling, squeezed into the first few meters.

Pairs with: Decibels in Your Head before it. Wi-Fi Through a Wall and 6 GHz Power and PSD after it (the "Why 6 GHz behaves differently" sequence). Rate vs Range and Uplink vs Downlink for the link budget. Lab: the FSPL lab.

Left out on purpose: Free space has no walls, floor bounce, people or fading, so real readings usually sit below the curve. The aperture term assumes a 0 dBi receive antenna. The indoor curve is a model with an exponent you choose. Measured points are typed in by hand.

Wi-Fi Through a Wall

Sends one wave through one wall and shows three things at once: part of the wave reflects off the face, the wave gets smaller as the wall absorbs it, and the same wave continues behind the wall, smaller. The frequency never changes. Pick a real wall, such as an interior stud wall (plasterboard, an air gap, plasterboard), a concrete elevator-shaft wall or a solid wood door, or one material at any thickness. The numbers come from the ITU-R P.2040 building-material model.

Why it's here. Wall loss is usually quoted as one number per material, and 6 GHz is usually said to lose more. This shows where the number comes from and splits it into absorption and reflection. Concrete does lose more at 6 GHz, but a thin panel such as glass, or a layered wall with an air gap, can lose less at a higher frequency, because the echoes from its faces can cancel.

How to use

1. Pick a band and channel and set the Tx power (transmit power). Then pick a wall. The real walls are an interior stud wall, a concrete elevator-shaft wall, a solid wood door, a double-pane window, a single-pane window and a brick wall one brick thick; each lists its layers and their thicknesses. One material, any thickness adds a material list and a thickness you type or drag; that slider is logarithmic from 1 cm to 1 m. The wave reaches the wall at the Tx power: the tool has no distance, so free-space loss is not included.
2. Watch the wave. Its height shows the signal in dB above a -95 dBm noise floor: the incident wave is drawn at the Tx power, and a flat line means the signal is below the noise floor. In front of the wall the incident and reflected waves overlap into a ripple with peaks every half wavelength. Inside, the same wave shrinks in height at a steady rate in dB, at the same wavelength: only the height changes. Behind, the same wave continues, smaller, and even a thick concrete wall still shows a wave while the signal is above the noise floor. The wall is not drawn to scale: it widens with thickness but stays small, a layered wall is split into its layers in equal shares with air left open, and the strip

under the plot names each layer and its true thickness. Every number uses the true thicknesses. Pause stops the animation; with reduced motion on it starts frozen.

3. Read the loss and its two parts, then compare the same wall at 2.4, 5.5 and 6.5 GHz. The measured values card puts published measurements of the wall's main material beside the model, names the source and says how far apart they are; Set the wall switches to one material at the specimen's thickness. Set the angle of arrival and TE or TM (transverse electric or transverse magnetic) to see how a glancing wave reflects more.
4. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses the wave, R returns to the opening wall, Up and Down step through the walls, including One material, any thickness, minus and equals make One material thinner and thicker, F switches full screen, ? lists the keys and Esc exits. In Present the Tx power, angle and polarization fold into one panel.

Inputs

Input	Unit	Range
Band and channel	20 MHz channel	2.4, 5 and 6 GHz channels; defaults 6, 100 and 117
Tx power	dBm	0 to 30; default 20
Wall	real wall or one material	interior stud wall (default), concrete elevator-shaft wall, solid wood door, double-pane window, single-pane window, brick wall one brick thick, one material any thickness
Material (one material only)	ITU-R P.2040 Table 3	concrete, brick, plasterboard, wood, glass, ceiling board, chipboard, plywood, marble, metal
Thickness (one material only)	cm (metric) or in (imperial)	1 to 100 cm (0.39 to 39.4 in)
Angle of arrival	degrees from head on	0 to 80
Polarization	TE or TM	TE: electric field along the wall face; TM: tilted with the angle

How it works. Material properties: relative permittivity e' = a x f^b and conductivity s = c x f^d, with f in GHz and the coefficients from ITU-R P.2040 Table 3; an air gap is the table's air row, e' = 1 and s = 0. Complex permittivity e = e' - j x 17.98 x s / f. Attenuation rate inside a material A = 1636 x s / sqrt(e') dB/m (valid while the loss tangent is under 0.5; the exact value is shown otherwise). The wavelength inside one material, shown as a readout, is the wavelength in air / sqrt(e'). One material: one face reflects R' = (cos t - sqrt(e - sin^2 t)) / (cos t + sqrt(e - sin^2 t)) for TE, with e x cos t in place of cos t for TM. For a wall of thickness d, q = (2 pi d / wavelength) x sqrt(e - sin^2 t), reflection R = R'(1 - e^-j2q) / (1 - R'^2 e^-j2q) and transmission T = (1 - R'^2) e^-jq / (1 - R'^2 e^-j2q). A real wall uses the P.2040 multilayer method (section 2.2.2.1): with u = sqrt(e - sin^2 t) in each layer (cos t in the air on either side) and g = 2 pi u / wavelength, the face between layer n and the next reflects r = (u_n - u_n+1) / (u_n + u_n+1) for TE, or (e_n+1 u_n - e_n u_n+1) / (e_n+1 u_n + e_n u_n+1) for TM. Working from the back face forward, R(n) = (r(n) + R(n+1) e^-j2gd) / (1 + r(n) R(n+1) e^-j2gd), with g and d those of layer n+1, and T is the product over the faces of e^-jgd (1 + r(n)) / (1 + r(n) R(n+1) e^-j2gd). With one layer this gives exactly the one-material result. Transmission loss = -10 log10 |T|^2, split into absorption (the decay along the path inside every layer; air adds none) and reflection (the rest, from the faces and their echoes). Level behind the wall = Tx power - transmission loss. The drawn height is (P - floor) / (Tx power - floor), with P = Tx power + 20 log10 |E| and the floor at -95 dBm: thermal noise in 20 MHz, -174 + 10 log10(20,000,000) = -101 dBm, plus a typical 6 dB receiver noise figure. Inside the wall the drawn level is a straight line in dB between the exact levels at the two outer faces.

Example. Interior stud wall (1.27 cm plasterboard, 8.9 cm air, 1.27 cm plasterboard), head on: 2.7 dB at 2.4 GHz, 2.8 dB at 5.5 GHz and 1.4 dB at 6.5 GHz. The two sheets absorb only 0.5 to 1.2 dB of that; the rest is how the echoes across the gap add or cancel, which is why 6.5 GHz loses less here. Concrete elevator-shaft wall, 20.3 cm: 14.8, 26.9 and 30.4 dB. One material, 102 mm of concrete: 8.1, 14.3 and 16.0 dB, of which about 1.4 to 1.5 dB is reflection. 610 mm (2 ft) of concrete at 5.5 GHz with 20 dBm Tx power: 77.8 dB, so -57.8 dBm behind the wall, 37.2 dB above the noise floor.

Field notes

- These are model values for smooth, uniform layers. Real walls vary: studs, insulation, wiring, reinforcing bars, mortar joints, door frames and window coatings are not modeled. A low emissivity (low-E) window coating can add far more loss than the glass, and P.2040 has no class for it.
- The real walls use common construction sizes, not a measured wall: 1/2 in (1.27 cm) plasterboard on a 2x4 stud's 3 1/2 in (8.9 cm) depth, 8 in (20.3 cm) of concrete, a 1 3/4 in (44.5 mm) door, 1/8 in (0.32 cm) panes with a 1/2 in (1.27 cm) gap, a 6 mm single pane, and a 3 5/8 in (9.2 cm) brick. The losses are whatever P.2040 gives for them; none is tuned to a target.

- P.2040 layered results show frequency-dependent resonance in thin layered walls: the echoes between the layers add or cancel depending on how the gap compares with the wavelength. The model's double-pane window loses 0.5 dB at 2.4 GHz but 8.9 dB at 6.5 GHz, nearly all of it reflection, and its stud wall loses more at 2.4 GHz than at 6.5 GHz. Moving a few MHz can move these numbers.
- Published measurements often disagree with the model. NIST measured 102 mm of lab-cast concrete at roughly twice the model's loss, likely from moisture in fresh specimens, and NYU measured a 45 mm solid-core wood door at nearly three times the model's loss for 45 mm of wood. The measured values card shows each source on its own and never averages them.
- NIST has no data between 2.0 and 3.0 GHz, so its lowest point here is 2.0 GHz, not 2.4.
- Free-space loss is a separate effect: with the same antennas it rises about 7.2 dB from 2.4 to 5.5 GHz and 8.7 dB from 2.4 to 6.5 GHz with no wall at all. The Free Space Path Loss calculator covers it.
- The drawing is slowed to one cycle every 2 seconds and shows the field along a line straight through the wall. It keeps one wavelength everywhere, because the frequency never changes; only the height of the wave changes. The height is in dB, so the ripple in front looks smaller than it would on a linear scale, and the ripple inside the wall from waves bounced off inner faces is not drawn.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: one wave meets one wall, part reflects, the rest shrinks inside and continues behind at the same frequency, and a layered or thin wall can lose less at a higher band because the echoes from its faces cancel.

Control	What it does	Why it's there
Band (2.4 / 5 / 6 GHz, default 5 GHz) and Channel (picker; defaults 6, 100, 117)	Set the frequency.	Channel-accurate wavelength. For thin and layered walls the echoes add or cancel depending on how the gap compares with the wavelength, so moving a few MHz can move the loss.
Tx power (0 to 30 dBm, default 20)	Sets the level at which the wave reaches the wall. The wave height is drawn in dB above a -95 dBm noise floor.	Lets the class see that even thick concrete leaves a wave behind the wall while the signal is above the noise floor. There is no distance in this tool, so free-space loss is not included.

Control	What it does	Why it's there
Wall (picker, default Interior stud wall): Interior stud wall, Concrete elevator-shaft wall, Solid wood door, Double-pane window, Single-pane window, Brick wall, one brick thick, One material, any thickness	Loads a real wall (its layers are listed under the picker and in the strip under the plot) or switches to one material.	Wall loss is usually quoted as one number per material. Real walls computed with ITU-R P.2040 show where the number comes from, and two of them break the "higher band loses more" rule: the stud wall and the double-pane window.
Material (ITU-R P.2040 Table 3) (One material only)	Picks one of ten materials.	Compare materials one at a time on the same model the other wall tools use.
Thickness (field and slider, 1 cm to 1 m (0.4 to 39 in), One material only)	Sets the slab thickness. The slider is logarithmic.	Thickness in cm, and the wall drawn small (not to scale) so a 1 m wall does not swamp the picture. The log scale gives thin glass and thick concrete the same slider travel.
Angle of arrival (0 to 80 degrees, default 0)	Tilts the wave from head on toward skimming along the wall.	A glancing wave reflects more.
Polarization (TE / TM, default TE)	TE: electric field along the wall face. TM: tilted with the angle.	Shows that reflection off a slant depends on polarization; head on the two are the same.
Play / Pause	Animates the wave, one cycle every 2 seconds. It plays when the tool opens; with reduced motion on it starts frozen.	A pause is always offered for accessibility, and a frozen frame is what you point at.
Set the wall to (thickness) (button in the measured values card)	Switches to one material at the published specimen's thickness.	Puts the model and the published measurement side by side at the same thickness. The card names each source and never averages them.

A demo that works (about 3 minutes)

1. Set up: the tool as it opens: Interior stud wall, 5 GHz channel 100, head on, TE, Tx power 20 dBm, wave playing.

2. Ask the room: "This stud wall at 2.4, 5.5 and 6.5 GHz. Which band loses the most?" Many will say 6 GHz.
3. Reveal: read the All three bands card: 2.7, 2.8 and 1.4 dB. The two plasterboard sheets absorb only 0.5 to 1.2 dB of it; the rest is how the echoes across the air gap add or cancel, and at 6.5 GHz they cancel.
4. Optional follow-up: press Up to the Concrete elevator-shaft wall: 14.8, 26.9 and 30.4 dB, so concrete does follow the rule. Then the Double-pane window: 0.5 dB at 2.4 GHz and 8.9 dB at 6.5 GHz, nearly all reflection.

Pairs with: FSPL Simulator (free-space loss is a separate effect), How to Measure Wall Attenuation (same materials), Room Propagation (same wall model on a plan), 6 GHz Power and PSD, and the RF Attenuation calculator.

Left out on purpose: Smooth, uniform layers only: no studs, insulation, wiring, rebar, mortar, frames or low-E coatings (a low-E coating can add far more loss than the glass, and P.2040 has no class for it). The real walls use common construction sizes, not a measured wall. Published measurements often disagree with the model (NIST measured 102 mm of fresh concrete at about twice the model's loss). The drawing is one line straight through the wall, slowed, with the wall not to scale and the ripple inside it not drawn.

How to Measure Wall Attenuation

Walks through measuring a real wall the field way, and shows why the method works. A side view holds an RF (radio frequency) source, such as a hotspot or a small AP (access point), a wall, and a person holding a laptop locked to one channel. Take a series of readings close to the near side of the wall, then close to the far side, and the difference of the two averages is the wall attenuation. The live readout splits that measured number into three parts: the true wall loss, the free-space error from where the readings were taken, and the fading left in the averages.

Why it's here. The far reading is farther from the source than the near reading, so it carries extra free-space path loss (FSPL) on top of the wall. Close to the source the free-space curve is steep, so that extra loss is large: with the source 2 m from the wall and readings 1 m either side, the far reading is 3 m from the source and the near one 1 m, and the measurement reads 20 log10(3 / 1) = 9.5 dB too high before the wall is even counted. With the source 4 m or more away and both readings close to the wall, the two spots are at almost the same distance, and the difference is almost all wall.

How to use

1. Drag the person with the laptop to either side of the wall, or tap a spot in the room. Where the laptop stands on each side is where that side's readings are taken. Go to the far side (or near side) switches sides and keeps each side's spot.
2. Drag the RF source to change its distance from the wall, or use the Source to wall slider. Watch the free-space error grow as the source comes close.
3. Take new readings draws a fresh series on the side the laptop is on. The short ticks on the stage are the readings, the long tick is their average, and the average is what gets subtracted.
4. The scenes set the geometry in one press: Tight (source 4 m away, readings 10 cm from the wall), Loose (readings 1 m from the wall) and Source too close (source 1 m away).
5. Choose the band, the material and the thickness of the wall. The band locks the laptop to one channel, shown under the stage. The materials are the ones in Wi-Fi Through a Wall.
6. Set how many readings each series takes and the fading spread. The spread is illustrative; set it to 0 to see the geometry error alone.
7. Predict, then reveal asks: source 2 m from the wall, readings 1 m either side of it; is the measured wall loss too high, too low, or right? Reveal the answer, then press Show it to put that scene on the stage.
8. On a computer or tablet, Present opens the tool full screen for a projector: Right and Left move the laptop away from and toward the source (crossing the wall at a face), Space takes new readings, S switches sides, Up and Down move the source, P reveals the answer, R resets, F switches full screen, ? lists the keys and Esc exits.
9. Every length follows the metric or imperial switch.

Inputs

Input	Unit	Range
Source to wall	m (metric) or ft (imperial)	0.5 to 15 m (1.6 to 49.2 ft); default 4 m
Laptop side	side	Near side or Far side; default Near

Input	Unit	Range
Near reading, in front of the wall	cm or m (in or ft)	5 cm up to 10 cm short of the source; default 1 m
Far reading, behind the wall	cm or m (in or ft)	5 cm to 5 m; default 1 m
Band	GHz	2.4, 5 or 6; default 5. The laptop is locked to channel 6 (2437 MHz), 100 (5500 MHz) or 117 (6535 MHz)
Material	ITU-R P.2040 Table 3	concrete, brick, plasterboard, wood, glass, ceiling board, chipboard, plywood, marble, metal; default concrete
Thickness	cm (metric) or in (imperial)	1 to 100 cm (0.39 to 39.4 in); default 10.2 cm
Readings per side	readings	1 to 30; default 10
Fading spread, standard deviation	dB	0 to 6; default 2 (illustrative)

How it works. The source radiates 20 dBm with 0 dBi antennas at both ends (decibels over an isotropic antenna). The level at a distance d from the source is 20 dBm - FSPL(d), with FSPL = 20 log10(4 pi d / wavelength), the FSPL Simulator's exact form; behind the wall the true wall loss is subtracted as well. The wall loss is Wi-Fi Through a Wall's ITU-R P.2040 slab model for the chosen material and thickness, head on. With the source D from the near face of a wall t thick, the near reading a_n in front of it and the far reading a_f behind it, the near spot is D - a_n from the source and the far spot D + t + a_f. Each reading is the model level plus a random draw with the chosen standard deviation (illustrative), from a fixed seed, so the same settings give the same readings. Measured wall attenuation = the near average - the far average = true wall loss + 20 log10(far / near) + the fading left in the averages. The middle term is the free-space error; it does not depend on the channel, because the frequency part of FSPL is the same at both spots and cancels. For a wall of no thickness and equal gaps a it is 20 log10((D + a) / (D - a)). The laptop reads nothing below a -95 dBm noise floor (thermal noise in 20 MHz, -101 dBm, plus a 6 dB noise figure); if the far side is below it, the wall cannot be measured. The near reading stays at least 10 cm from the source, where the free-space formula still applies in this model.

Example. At 5 GHz (channel 100, 5500 MHz), with 10.2 cm of concrete and the source 4 m from the wall, the true wall loss is 14.3 dB. With readings 1 m either side, the near average is -36.7 dBm and the far average -55.9 dBm, so the measured wall attenuation is 19.2 dB: 4.6 dB from the geometry, 20 log10(5.1 / 3.0), and 0.3 dB from the fading. Move both readings to 10 cm from the wall and the measurement is 15.2 dB, with the geometry error down to 0.6 dB. With a wall of no thickness, the same 10 cm readings give 20 log10(4.1 / 3.9) = 0.4 dB.

Field notes

- Lock the measuring device to one channel for both series, so every reading is of the same signal.
- Put the source 4 m or more from the wall and take the readings close to it on both sides. The two spots then sit at almost the same distance from the source, and the difference is almost all wall.
- Average a series on each side rather than trusting one reading. The fading spread in this tool is an illustrative setting, not a measurement.
- The wall's own thickness also adds distance between the two spots, so even readings right against both faces keep a small free-space error.
- Left out: the reflection off the near face of the wall, which makes readings close to it rise and fall over a short distance; paths around the wall through doors and other rooms; and the laptop's own body and antenna pattern. Several readings over a small area, averaged, are how a field measurement handles the first.
- For how the wall itself takes the loss, see Wi-Fi Through a Wall. For comparing a predicted wall with a measured one, see Predict, Then Measure.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: measure a wall by averaging a series of readings close to each side with the source 4 m (13 ft) or more away; put the source close and the extra free-space loss on the far side reads as wall.

Control	What it does	Why it's there
Drag the person with the laptop, or tap a spot in the room	Places the laptop on either side of the wall. Where it stands is where that side's readings are taken.	Acts out the field method on the stage: one series in front of the wall, one behind.

Control	What it does	Why it's there
Drag the RF source	Changes the source's distance from the wall.	The free-space curve is steep near the source, so the class watches the error grow as the source comes close.
Scenes: Tight, Loose, Source too close (buttons)	Set the source and both reading distances in one press: Tight is source 4 m, readings 10 cm (4 in); Loose is readings 1 m (3.3 ft); Source too close is source 1 m. The wall and readings settings stay.	Tight is the method to teach. The tool opens on Loose, so the error is on screen from the start. Source too close shows the worst case.
Source to wall (0.5 to 15 m (1.6 to 49 ft), default 4 m)	Same as dragging the source.	Keyboard and exact placement for the lesson's main variable.
Laptop on the: Near side / Far side (toggle); Go to the far side / Go to the near side (button)	Switches sides and keeps each side's spot.	A measurement needs a series on both sides; switching keeps both spots so the difference is always live.
Near reading, in front of the wall; Far reading, behind the wall (sliders)	Set how far from the wall each series is taken. The near reading stays at least 10 cm (4 in) from the source.	The error term is 20 log10(far distance / near distance); these two sliders are what make it large or small.
Band (2.4 / 5 / 6 GHz, default 5)	Locks the laptop to channel 6, 100 or 117 for both series.	Lock the measuring device to one channel so every reading is of the same signal. The free-space error itself does not depend on the channel.
Material (ITU-R P.2040 Table 3), Thickness (1 cm to 1 m, default 10.2 cm (4 in) of concrete)	Set the wall.	The true wall loss comes from the same slab model as Wi-Fi Through a Wall, so the readout can compare the measured number with the truth.
Readings per side (1 to 30, default 10); Fading spread, standard deviation (illustrative) (0 to 6 dB, default 2)	Set how many readings each series takes and how much they scatter.	Average a series rather than trust one reading. Set the spread to 0 to see the geometry error alone.

Control	What it does	Why it's there
Take new readings, near side / far side (button)	Draws a fresh series on the side the laptop is on.	Shows that the fading residual changes every time while the geometry error stays put.
Reveal the answer / Hide the answer; Show it	Answers "Source 2 m from the wall, readings 1 m either side of it. Is the measured wall loss too high, too low, or right?" and puts that scene on the stage.	Predict, then reveal. The far reading is 3 m from the source and the near one 1 m.

A demo that works (about 3 minutes)

1. Set up: the tool opens on Loose: 5 GHz, 10.2 cm (4 in) of concrete, source 4 m (13 ft), readings 1 m (3.3 ft) either side. True wall loss is 14.3 dB and the measurement reads 19.2 dB, 4.6 dB of it geometry.
2. Ask the room the question on screen: source 2 m, readings 1 m either side. Too high, too low, or right?
3. Reveal: press P, then Show it. Too high: 20 log10(3 / 1) = 9.5 dB of free-space loss that is not the wall, about 9.8 dB with this wall's thickness added.
4. Optional follow-up: press Tight. Source 4 m, readings 10 cm (4 in): the measurement is 15.2 dB and the geometry error is down to 0.6 dB. Set the fading spread to 0 to show what is left is geometry, then back to 2 and press Space a few times.

Pairs with: Wi-Fi Through a Wall (how the wall takes the loss) and Predict, Then Measure (a predicted wall against a measured one); the tool has buttons that open both. FSPL Simulator for the free-space term.

Left out on purpose: The reflection off the near face, which makes readings close to it rise and fall over a short distance (averaging several readings over a small area handles it in the field). Paths around the wall through doors and other rooms. The laptop's own body and antenna pattern. The fading spread is illustrative. The source is a fixed 20 dBm with 0 dBi antennas, and nothing below a -95 dBm noise floor can be read.

Multipath Simulator

Shows why the signal changes when you move a few centimeters. A receiver hears the direct signal plus copies that bounced off walls. Each copy travels a different distance, so it arrives with a different phase, and the copies add like arrows placed head to tail. In phase they reinforce; out of phase they cancel.

Why it's here. One RSSI reading taken in one spot can be several dB off from a reading a hand width away, and a survey or a troubleshooting call that trusts it is trusting luck. Seeing the arrows add and cancel explains the dips, why they repeat every half wavelength, and why a second antenna rarely sits in a dip at the same spot as the first.

How to use

1. Pick a scene. One wall shows the direct copy and one reflection, seen from above. Standing wave puts the receiver between a distant access point and a wall. Many paths blocks the direct copy and adds 2 to 30 reflections from random places.
2. Pick a band. The dips repeat every half wavelength: 6.2 cm at 2.4 GHz, 2.7 cm at 5.5 GHz and 2.3 cm at 6.5 GHz.
3. In One wall and Standing wave, pick the wall material (metal, thick concrete, drywall) or set the reflection strength yourself. Drag the receiver in the picture or on the plot, or use the slider, and watch the reflected arrow swing around the direct one.
4. In Many paths, set the number of reflectors and how far away they are, and press New layout for a different room. Move antenna A along the 2 m track and set how far antenna B sits from it, from 0 to 1 wavelength. The readout shows how often each antenna is more than 10 dB below the average and how often both are at once.
5. Still in Many paths, Combine shows what the receiver does with its antennas: A only, Selection (use the stronger antenna at each spot) or MRC (maximal ratio combining: add every antenna, lined up in phase). The combined signal is drawn as a third, bold line. Antennas sets 2 or 4; with 4 they sit one spacing apart. The Diversity gain card gives the textbook gain at 1% for both methods and what this track shows.
6. The delay list shows how late each copy arrives. A copy more than 0.8 microseconds late, about 240 m of extra path, spills past the guard interval into the next symbol.
7. On a computer or tablet, Present opens this simulator full screen for a projector: Up and Down move the receiver a sixteenth of a wavelength, C steps Combine through A only, Selection and MRC in Many paths, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Scene	One wall / Standing wave / Many paths	default One wall
Band	GHz	2.4, 5.5 or 6.5
Wall material	reflection strength |Γ|	Metal 1.0, Thick concrete 0.39, Drywall 0.1, or Custom 0 to 1
Receiver position	cm	0 to 100 along the track (One wall); 0 to 40 from the wall (Standing wave); 0 to 200 (Many paths)
Reflectors	count	2 to 30 (Many paths)
Reflectors up to	m	10, 60 or 300 (Many paths)
Antenna B offset	wavelengths	0 to 1 in steps of 0.05; with 4 antennas, the spacing between neighbors
Antennas	count	2 or 4 (Many paths)
Combine	A only / Selection / MRC	default A only (Many paths)

How it works. Each path adds a copy a · exp(-j k r) / r, where r is the path length, k = 2π/λ, and a is 1 for the direct copy or the reflection coefficient for a reflected one. Every wall reflects with a 180 degree phase flip. Power is 20 log10(|sum| / |direct alone|). One wall uses the image method: the reflected path is as long as a straight line from the access point's mirror image behind the wall. In Standing wave the reflected copy travels 2d further at distance d from the wall, so the copies cancel every λ/2. In Many paths each reflection gets equal power and a random phase, and power is shown against the average; theory (Rayleigh fading) says 1 - e^(-0.1), about 9.5%, of spots sit more than 10 dB below it. Extra delay = extra path / speed of light. Combining works on each antenna's power against its own average: Selection takes the largest, MRC adds them all, so MRC averages 3 dB above one antenna with 2 and 6 dB with 4. For independent Rayleigh antennas the share of the time the combined power is below x is (1 - e^(-x))^M for selection and 1 - e^(-x)(1 + x + … + x^(M-1)/(M-1)!) for MRC. The diversity gain at 1% is the combined level at 1% minus one antenna's level at 1% (-20.0 dB).

Example. At 2.4 GHz, λ = 12.49 cm, so in front of a metal wall the signal drops out every 6.2 cm. Thick concrete with |Γ| = 0.39 gives a swing of 20 log10(1.39 / 0.61) = 7.2 dB instead. A reflection that travels 240 m further arrives 240 / 299,792,458 s = 0.8 µs late, the length of the 802.11 OFDM guard interval. One antenna is below -20.0 dB 1% of the time; two antennas reach the same 1% at -9.8 dB with selection and -8.3 dB with MRC, gains of 10.2 dB and 11.7 dB. Four antennas give 15.8 dB and 19.1 dB.

Field notes

- The wall presets are one reflection strength each. A real wall reflects differently at different angles and bands; thick concrete's 0.39 is for a wave hitting it straight on.
- Many paths blocks the direct copy on purpose. With a strong line-of-sight copy the dips are shallower and rarer than Rayleigh fading predicts.
- A 2 m track at 2.4 GHz holds only a few dozen fades, so the histogram wanders around the Rayleigh curve. Longer tracks or new layouts average it out.
- Layouts are seeded, so the same settings give the same room every time.
- The diversity gains in Selection and MRC assume the antennas fade independently and have equal noise. Antennas half a wavelength apart are close to that but not exactly; the On this track figure is what one 2 m track actually shows, and it wanders with the layout.
- MRC here is the ideal: every antenna perfectly lined up in phase and weighted by its strength. A real receiver estimates those from the preamble and gets close.
- Only the delay list changes much when the reflectors move further away; the pattern along 2 m looks the same at any distance.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: the direct signal and its reflections add like arrows, so moving a few centimeters walks the receiver from a peak into a dip, and the dips repeat every half wavelength; a second antenna, picked or added, fills those dips in.

Control	What it does	Why it's there
Scene (picker): One wall (two-ray), Standing wave (walk to a wall), Many paths (Rayleigh fading); default One wall	Switches between the three models.	The three steps of the lesson: one reflection, the standing wave in front of a wall, then random fading from many paths. It is a picker, not segments, because the labels truncated at phone width.
Band (2.4 / 5.5 / 6.5 GHz, default 2.4)	Changes the wavelength.	The dips repeat every half wavelength: 6.2 cm (2.4 in), 2.7 cm (1.1 in) and 2.3 cm (0.9 in).
Wall material (One wall and Standing wave): Metal (|Γ| 1.00), Thick concrete (|Γ| 0.39), Drywall (|Γ| 0.10), Custom; default Metal	Sets how strongly the wall reflects.	A strong reflector cancels the direct copy completely; thick concrete swings the signal by 7.2 dB; drywall barely ripples it.
Reflection strength |Γ| (slider, 0 to 1)	Sets the reflection yourself; moving it switches the material to Custom.	Lets the class dial the reflection down and watch the nulls fill in.
Receiver position (One wall, 0 to 100 cm) / Distance to the wall (Standing wave, 0 to 40 cm); drag the receiver in the picture or on the plot	Moves the receiver.	Watch the reflected arrow swing around the direct one as the receiver moves.
Reflectors (Many paths, 2 to 30, default 12)	Sets how many reflected copies arrive.	More copies of random phase is Rayleigh fading; the histogram shows it against the theory curve.
Reflectors up to: 10 m / 60 m / 300 m (Many paths; room, large hall, outdoors)	Sets how far away the reflectors sit.	Exists so the guard-interval lesson can fire: only the 300 m layouts put copies more than 0.8 microseconds late.

Control	What it does	Why it's there
New layout (Many paths)	Places the reflectors somewhere else. Layouts are seeded, so the same settings give the same room.	A 2 m track holds only a few dozen fades, so the histogram wanders; new layouts show it averaging toward the Rayleigh curve.
Antenna A position (0 to 2 m); Antenna B offset (λ) (0 to 1 wavelength in steps of 0.05, default 0.5)	Move antenna A along the track and set how far antenna B sits from it.	Selection diversity: two antennas half a wavelength apart rarely fade at the same spot. The readout shows how often each, and both at once, are more than 10 dB below average.
Antennas: 2 / 4 (Many paths, default 2)	Gives the receiver two or four antennas. With four, the offset slider becomes "Antenna spacing (λ)" and they sit one spacing apart.	Four shows diversity gain growing with the antenna count: 15.8 dB (selection) and 19.1 dB (MRC) at 1%, against 10.2 and 11.7 with two. Most client devices have two antennas and many APs have four.
Combine: A only / Selection / MRC (Many paths, default A only)	Adds a third, bold trace: the stronger antenna at each spot (Selection) or all antennas added, lined up in phase (MRC, maximal ratio combining). Antenna A drops to a thin neutral line.	The existing readout only counted how often both antennas fade together. Combine shows what a receiver does with that: the dips fill in, and MRC fills them further because two weak antennas added can clear -10 dB when neither does alone. It defaults to A only so the scene opens as before.
Diversity gain at 1% card: Selection, MRC, On this track	The textbook Rayleigh gain for both methods at the current antenna count, and what this 2 m track shows for the chosen method.	Diversity gain is read at an outage level: one antenna is more than 20 dB below average 1% of the time, and the gain is how much of that the combiner wins back. The track figure wanders on purpose; New layout shows why one short walk is not the textbook number.
Show all N paths / Show the latest N (delay list, Many paths)	Expands or shortens the list of how late each copy arrives.	Keeps the delay list short on screen while the full set is one press away.

A demo that works (about 3 minutes)

1. Set up: Standing wave, 2.4 GHz, Metal.

2. Ask the room: "I move the phone 3 cm (1.2 in). How much can the reading change?" Take guesses in dB.
3. Reveal: drag the receiver onto a peak of the plot, then press Up four times (a quarter wavelength, about 3 cm): it lands in a null, where the two copies cancel. Four more presses reach the next peak, and the next null is 6.2 cm (2.4 in) from the first.
4. Optional follow-up: switch to Thick concrete and the swing drops to 7.2 dB. Or go to Many paths, set Antenna B offset to 0.5 λ, and compare Antenna A, Antenna B and Both at once against the 9.5% Rayleigh figure.

Pairs with: Why Two Devices Disagree (it reuses this fading), Room Propagation (the close-up of the nulls near a wall), Rate vs Range. The "Why the signal jumps when you move" sequence.

Left out on purpose: Each wall preset is one reflection strength; a real wall reflects differently by angle and band, and thick concrete's 0.39 is for a wave hitting it head on. Many paths blocks the direct copy on purpose; with a strong line-of-sight copy the dips are shallower and rarer. The pattern along 2 m looks the same at any reflector distance; only the delay list changes. MRC is the ideal combiner, with every antenna perfectly lined up and weighted; a real radio estimates that from the preamble. The textbook gains assume antennas that fade independently with equal noise; half a wavelength apart is close to that but not exact.

Why Two Devices Disagree

Shows why two devices at the same spot, hearing the same AP, report different signal strength. The true received power at the spot is held fixed, and up to four generic devices (a laptop, a phone, a tablet and a survey adapter by default) each report it their own way.

Why it's here. RSSI (received signal strength indicator) is a number each chipset reports in its own way, so a laptop and a phone side by side commonly disagree by several dB. Comparing raw numbers across devices compares the devices, not the network. This is why survey tools let you enter an offset per adapter, and why a reading taken on one device is not a design target for another.

How to use

1. Start with the question at the top: the laptop says -62, the phone says -68, same spot. Which one is right? Predict, then press Reveal.

2. Each device card shows what that device reports now. The strip chart shows every device's readings over the last 6 seconds, with the true power as a dashed line.
3. Pick a device (A to D) and set its fixed offset, from -10 to +10 dB, and its orientation and grip loss, from 0 to 10 dB. Every offset and loss is illustrative, not a measurement of any real device.
4. For the same device, turn on body loss (the person holding it), choose a 1 dB or 2 dB reporting step, and choose how many readings it averages, from 1 to 10.
5. Turn on Apply offsets to subtract each device's offset, as a survey tool does. With fading off, the fixed spread disappears entirely. With fading on, the fading remains, and so do grip and body loss.
6. Set the spacing between devices. Farther apart than half a wavelength (about 2.7 cm at 5 GHz), each device sees its own fade. Averaging more readings makes the fading spread smaller.
7. Re-sample takes the next few seconds of readings. Reset puts every setting back.
8. On a computer or tablet, Present opens this simulator full screen for a projector: Space re-samples, R resets, Up and Down move the AP 1 m (1 ft when lengths are imperial), F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Band	GHz	2.4, 5 or 6
AP distance	m	1 to 50, default 8
Fading	on / off	default on
Spacing between devices	cm	0 to 30, default 10
Devices	count	2 to 4, default 4
Device offset	dB	-10 to +10, illustrative defaults 0, -4, -2, +1

Input	Unit	Range
Orientation and grip loss	dB	0 to 10, illustrative
Body loss when on	dB	0 to 10, illustrative default 3
Reporting step	dB	1 or 2
Averaging	readings	1 to 10
Apply offsets	on / off	default off

How it works. True power at the spot = effective isotropic radiated power (EIRP) - free-space loss at 1 m - 10 n log10(distance), with an illustrative 14 dBm EIRP and exponent n = 3, the same log-distance model as the Roaming Walk. Each device receives true power + its offset - grip loss - body loss + fading. Fading comes from the Multipath Simulator's Rayleigh model: 12 reflections of equal strength and random phase, direct path blocked, evaluated at each device's own position. The fading pattern moves past the devices at an illustrative 0.1 m/s, sampled 10 times a second. Each device averages its last N samples in milliwatts, converts back to dBm and rounds to its reporting step. Apply offsets subtracts each device's offset from what it reports.

Example. At 5 GHz and 8 m the true power is about -60.4 dBm. With fading off, a device with a -4 dB offset, no grip or body loss and a 1 dB step reports -64 dBm, and one with a +1 dB offset reports -59 dBm: a 5 dB spread. Apply offsets and both show -60 dBm, a spread of 0 dB.

Field notes

- Every device offset, grip and body loss, the AP power and the path loss exponent in this tool are illustrative. Real offsets depend on the device, its antennas and its driver; measure yours against a reference before you rely on them.
- 802.11k defines RCPI (received channel power indicator) as a standardized measurement of received power, which helps comparability where devices report it. This tool shows RCPI as a concept only and does not compute RCPI values.
- An offset calibrates a device. It does not correct how the device is held, the person holding it, or fading at the spot, so a corrected reading still moves.

- Runs are seeded: the same settings and sample set give the same readings every time.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: two devices at the same spot report different RSSI because each adds its own offset, grip, body loss, fading and rounding, and an offset calibrates the device but not how it is held or the fade it sits in.

Control	What it does	Why it's there
Reveal (button beside the question)	Answers "Laptop says -62, phone says -68, same spot. Which one is right?" with this run's numbers.	Predict, then reveal, on the tool's question.
Re-sample	Takes the next few seconds of readings.	Fading changes from one sample set to the next while the fixed offsets do not, so a re-sample shows which part of the spread moves.
Reset	Puts every setting back.	Returns to the four illustrative default devices.
Apply offsets (switch, default off)	Subtracts each device's offset, as a survey tool does once you enter it per adapter.	The fixed spread disappears and fading, grip and body loss remain. This is why survey tools take an offset per adapter.
Fading (switch, default on)	Turns the Rayleigh fading off.	Off shows only the fixed parts of each reading, so Apply offsets can take the spread to zero.
Device A, B, C, D (chips)	Pick the device to edit.	Up to four generic devices, named by letter and type only, with no device or vendor named.
Device X offset (illustrative) (-10 to +10 dB; defaults 0, -4, -2, +1)	Sets that device's fixed offset.	The fixed part of the difference: antenna gain and how the chip calibrates.

Control	What it does	Why it's there
Device X orientation and grip loss (illustrative) (0 to 10 dB)	Sets how much the way it is held costs.	How a device is held changes its reading, and no offset corrects it.
Device X held against a body (switch)	Adds the body loss set under The spot.	The person holding the device is part of the reading (Body Loss covers it in full).
Device X reporting step (1 dB / 2 dB)	Rounds that device's reports.	Two devices hearing the same power can differ by up to a step from rounding alone.
Device X averages the last (1 to 10 readings)	Sets how many readings the device averages, in milliwatts.	Averaging more readings makes the fading spread smaller.
Device X type (Laptop, Phone, Tablet, Survey adapter)	Renames the device. A label only; the numbers are set per device.	Gives the class a familiar kind of device without a brand.
Band (2.4 / 5 / 6 GHz, default 5)	Changes the wavelength and the true power.	Half a wavelength sets how far apart devices must be to fade separately, and it shrinks as the band rises.
AP distance (1 to 50 m, default 8 m (26 ft))	Moves the AP, which changes the true power at the spot.	Shows that the spread between devices does not depend on how strong the signal is.
Spacing between devices (0 to 30 cm (0 to 12 in), default 10 cm)	Sets how far apart the devices sit.	Farther apart than half a wavelength (about 2.7 cm (1.1 in) at 5 GHz), each device sees its own fade.
Devices (Two / Three / Four, default Four)	Sets how many devices are on the spot.	Two devices matches the laptop-and-phone question; four shows the full range.
Body loss when on (illustrative) (0 to 10 dB, default 3)	Sets the loss the body switch adds.	Every body value is illustrative; set it to what you measure.

A demo that works (about 3 minutes)

1. Set up: the defaults: 5 GHz, AP 8 m (26 ft), fading on, four devices.
2. Ask the room the question at the top: laptop -62, phone -68, same spot. Which one is right?
3. Reveal: press Reveal: neither one on its own. Turn Fading off: the traces go flat and the remaining spread is the fixed parts. Turn on Apply offsets: most of it goes, but the phone (Device B) still reads low, because its grip and body loss are not an offset.
4. Optional follow-up: turn Fading back on, then raise the averaging on one device and press Space a few times: its trace steadies. Set spacing to 0 and the devices fade together.

Pairs with: Multipath Simulator (the same fading model), Body Loss, Uplink vs Downlink. The "Why clients misbehave" and "Troubleshooting slow or unstable Wi-Fi" sequences.

Left out on purpose: Every offset, grip and body loss, the 14 dBm AP and the exponent of 3 are illustrative; real offsets depend on the device, antennas and driver. RCPI (802.11k) is shown as a concept card only, with no values computed. Runs are seeded, so the same settings give the same readings. The fading pattern moves past the devices at an illustrative 0.1 m/s.

Room Propagation

Puts an AP, walls and doorways on a floor plan and colors the plan by received power. The signal at any spot is the direct path plus reflections off the walls, each wall both passes and reflects part of the signal, and signal bends past doorways and wall ends into the shadows behind them. A close-up around the client shows the fine ripple that makes RSSI change when you move a few centimeters.

Why it's here. A coverage map is usually a finished picture. This one lets you switch each mechanism on and off and see what it does: turn off reflections, turn off diffraction, change the band. It also splits the question of why 6 GHz loses more into its parts: free-space loss (the smaller antenna aperture at a shorter wavelength), wall absorption, and diffraction.

How to use

1. Pick a floor plan. Drywall offices and Concrete corridor have several walls and doorways; Empty plan has none, so you can draw your own.

2. Drag the AP or the client on the plan, or use the position sliders. The colors are received power in dBm, one shade per 10 dB, with the legend under the plan. The readouts show what the client receives and where the loss comes from: free space, each wall on the straight line, diffraction and reflections.
3. Change the band and watch the same spot on all three bands. Turn Reflections to None and Diffraction off to see the straight-line picture with sharp shadows, then turn them back on. The Fresnel zone overlay shows how wide the path to the client is; the shadow edges show where a doorway's shadow would start if signal traveled only in straight lines.
4. To edit the plan, pick a tool: Draw walls (drag), Add doorways (tap a wall), or Select a wall (tap it), then set its material and thickness, add or close doorways, or delete it. Every edit also works from the wall list, without the plan.
5. On a computer or tablet, Present opens the plan full screen for a projector: R reloads the plan, Up and Down move the AP EIRP, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Floor plan	preset	drywall offices, concrete corridor, open office, glass meeting room, empty plan; up to 20 x 15 m
Band and channel	20 MHz channel	2.4, 5 and 6 GHz channels; defaults 6, 100 and 117
AP EIRP	dBm	0 to 36
AP antenna	orientation	upright (TE) or flat (TM)
Reflections	bounces	none, one or two
Diffraction	on or off	ITU-R P.526 knife edge at wall ends and doorways
Wall material and thickness	ITU-R P.2040 Table 3, mm	concrete, brick, plasterboard, wood, glass, ceiling board, chipboard, plywood, marble, metal; 1 to 500 mm; up to 16 walls

How it works. Field at each point U = sum of a x e^-jkr / r over the direct path and the image reflections (one or two bounces), with 1/r spreading as in three dimensions. Received power = EIRP + 20 log10(wavelength / 4 pi) + 10 log10 |U|^2 for a 0 dBi receiver, so with no walls it is exactly EIRP minus free-space path loss. Every wall's transmission T and reflection R come from the ITU-R P.2040 slab equations at the real angle of incidence, the same model as Wi-Fi Through a Wall. Diffraction uses the ITU-R P.526 knife-edge function: nu = h x sqrt((2 / wavelength) x (1/d1 + 1/d2)), and the exact complex form behind J(nu), so each wall line acts as a screen that is open where there is no wall and passes T where there is. A long wall costs exactly its slab loss, a doorway is the gap in the screen, and an opaque edge right on the line of sight costs 6.0 dB. The map shows the local average (the sum of path powers); the close-up adds every path with its phase to show the ripple. First Fresnel zone radius = sqrt(wavelength x d1 x d2 / (d1 + d2)).

Example. With no walls, 10 m from a 20 dBm EIRP AP on channel 6 the client receives 20 - 60.2 = -40.2 dBm. A metal edge 0.5 m into the path halfway along those 10 m adds 15.5 dB at 2.4 GHz and 19.4 dB at 6.5 GHz. The first Fresnel zone halfway along 10 m is 0.56 m in radius at 2.4 GHz, 0.37 m at 5.5 GHz and 0.34 m at 6.5 GHz. Near a metal wall the nulls repeat every 6.2 cm on channel 6 and every 2.3 cm on channel 117.

Field notes

- This is a 2D plan of a 3D model: signal spreads in three dimensions, but there is no floor or ceiling bounce, no furniture and no people.
- Each wall is one solid, uniform slab. Real walls have studs, cavities, rebar and coatings, and published measurements often disagree with the model. Wi-Fi Through a Wall shows the measured values beside it.
- Reflections stop at two bounces, and reflected paths pass through walls without diffraction. Real rooms add many more weak paths.
- The map is the average over a few centimeters. At one exact spot the signal can sit well above or below it, which the close-up shows: that is why RSSI jumps when you move a phone a little.
- The map is computed on your device, in the background where the platform allows. With two bounces and many walls it takes a moment; the plan says Computing while it works.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: the signal at a spot is the direct path plus reflections and diffraction past edges, and the extra loss at 6 GHz splits into free space, wall absorption and diffraction.

Control	What it does	Why it's there
Floor plan (picker): Drywall offices, Concrete corridor, Open office, Glass meeting room, Empty plan; default Drywall offices	Loads a plan with its walls, doorways, AP and client. Each plan shows its one-line lesson under the picker.	Each preset carries one lesson: a doorway matters little in drywall and a lot in concrete; signal bends past each jamb; metal cabinets cast a shadow; thin glass loses less on channels 100 and 117 than on 6. Empty plan is free space to draw on.
Reset this plan	Reloads the current plan.	Undoes any wall edits in one press.
Band (2.4 / 5 / 6 GHz) and Channel (default 2.4 GHz channel 6)	Set the frequency.	Watch the same spot on all three bands and see why 6 GHz loses more, part by part.
AP EIRP (0 to 36 dBm, default 20)	Raises or lowers the whole map. The receiver is a 0 dBi antenna.	Moves every 10 dB color band together, so the shape of the coverage is a property of the building, not the power.
AP antenna: Upright (TE) / Flat (TM)	Sets the polarization against the walls.	Walls reflect a glancing wave differently for the two polarizations.
Reflections: None / 1 bounce / 2 bounces (default 1)	Sets how many wall bounces are added.	Switch each mechanism off and on to see what it does. None gives the straight-line picture.
Diffraction at wall ends and doorways: On / Off (default On)	Adds the ITU-R P.526 knife-edge diffraction.	Off, every shadow edge is sharp; on, signal bends into the shadow.
Fresnel zone to the client (switch, default on)	Draws the first Fresnel zone and gives its radius at the middle of the path.	Shows how wide the path to the client is; keep it clear of walls and edges.
Doorway shadow edges (switch, default on)	Draws straight lines from the AP past each door jamb.	Where a shadow would start if signal traveled only in straight lines, so the class sees the signal reaching past them.

Control	What it does	Why it's there
Close-up of the ripple (switch, default on)	Shows a close-up around the client with every path added with its phase.	The map is a local average; the close-up shows the peaks and nulls every half wavelength that make RSSI jump when you move a phone a little.
Drag the AP or client on the plan; AP across, AP down, Client across, Client down (sliders)	Move the AP and the client.	The same as dragging, for exact placement and keyboard use.
What a tap or drag on the plan does (picker): Move the AP and client, Draw walls, Add doorways, Select a wall	Chooses what the plan does when touched.	Draw your own plan or change a preset.
Wall (picker), Material (ITU-R P.2040 Table 3), Thickness	Pick a wall and set its material and thickness, or set them for the next wall you draw. Up to 16 walls.	Test a claim live: make one wall concrete, or thicker, and watch the map.
Add a doorway, Close its doorways, Delete wall	Edit the selected wall.	A doorway is a gap in the screen; the class sees how much signal it carries past a lossy wall.

A demo that works (about 3 minutes)

1. Set up: Concrete corridor, 2.4 GHz channel 6, 1 bounce, diffraction on.
2. Ask the room: "The client is in a room off the corridor, behind concrete. Where is its signal coming from?"
3. Reveal: set Diffraction Off: the shadows behind the door jambs go sharp and line up with the doorway shadow edges. Set Reflections to None: the straight-line picture. Turn both back on and read Where the loss comes from: free space, each wall on the line, diffraction and reflections.
4. Optional follow-up: switch to 6 GHz channel 117 and read The same spot on all three bands. Free space alone is about 8.6 dB worse than channel 6, and the diffraction loss grows too.

Pairs with: Wi-Fi Through a Wall (the same wall model), Multipath Simulator (the nulls), FSPL Simulator (the free-space part). The "Why the signal jumps when you move" sequence.

Left out on purpose: It is a 2D plan of a 3D model: no floor or ceiling bounce, no furniture, no people. Each wall is one solid uniform slab. Reflections stop at two bounces, and reflected paths pass through walls without diffraction. The map is the average over a few centimeters; the close-up shows the exact spot.

Antenna Pattern

Shows an antenna's radiation pattern in 3D and as the two 2D cuts a datasheet prints. Pick a dipole, an omni set by its gain, a collinear array, or a directional patch or sector and change it with sliders, or paste an MSI or NSMA pattern file and rotate the 3D estimate built from its two cuts. The Floor coverage view puts the same antenna on a ceiling from 3 to 15 m and shows what the floor under it receives, in both directions.

Why it's here. Antenna gain is easy to misread as more power. Here every antenna radiates the same power, so raising the gain visibly takes signal from somewhere else: a high-gain omni flattens into a pancake with a hole under it, and a directional narrows its beam. It also shows why a 3D view built from a pattern file is an estimate, and how ceiling and wall mounting light different parts of a room. Floor coverage shows why mounting height matters: raise the AP and every spot on the floor moves to a steeper angle of the pattern, so a dipole leaves a hole under it, a higher-gain omni makes the near floor weaker, more power leaves the uplink where it was, and a directional aimed down puts the gain where the clients are.

How to use

1. Pick an antenna, then drag the gain, beamwidth, tilt and floor sliders and watch the 3D surface and the two cuts change. The scale stays fixed, so a gain change reshapes the pattern instead of resizing it.
2. Drag the 3D view to rotate it, pinch or scroll to zoom, and tap Reset view to go back. With a keyboard, focus the view and use the arrow keys and plus or minus.
3. Switch Mounting between Ceiling and Wall to see the same antenna turned 90 degrees against the floor and the mounting surface.

4. To import, choose Imported pattern file, then pick a generated example or paste the text of an MSI (.msi, .pln, .txt) or NSMA (.adf) file and tap Read pattern. There is no file picker: open the file in a text editor and copy all of it. Switch between Summing and Cross-weighted to compare two ways of building 3D from two cuts.
5. On any built-in antenna, tap Rebuild from its two cuts to write it as an MSI file, read it back, and see how far the rebuilt 3D is from the exact one.
6. Set the angle between two antennas' polarizations to see the mismatch loss.
7. Switch the view to Floor coverage to see the antenna from the side on a ceiling, drawn to scale, with the floor colored by what a client 1 m up receives. Pick a starting point (Office, dipole; Warehouse, dipole; Warehouse, 8 dBi omni; Warehouse, +6 dB power; Warehouse, directional), then move the mount height, powers, band, path-loss exponent and the client freely. The antenna card below still changes the antenna. Readouts give both directions directly below and 10, 20 and 30 m out, and how far the floor reaches -67 dBm. Open in Uplink vs Downlink takes the client's link there.
8. On a computer or tablet, Present opens the 3D pattern full screen for a projector: Space spins the view, the Right arrow turns it one step, R resets it, Up and Down move the antenna's gain, elements or shape, F switches full screen, ? lists the keys and Esc exits. In the Floor coverage view, Up and Down move the mount height by 0.5 m, the Right arrow steps to the next starting point and R goes back to the first. V switches between the two views.

Inputs

Input	Unit	Range
Antenna	model	dipole, omni set by gain, collinear array, directional, imported file
Omni gain	dBi	2.15 to 15
Collinear elements and spacing	count, wavelengths	1 to 8 elements, 0.25 to 1.0 wavelength
Downtilt	degrees	0 to 15 electrical (omni, collinear), 0 to 30 mechanical (directional)
Beamwidths	degrees	horizontal 15 to 180, vertical 10 to 120

Input	Unit	Range
Front-to-back and side-lobe floors	dB	10 to 40
Mounting	ceiling or wall	turns the 3D view 90 degrees
Pattern text	MSI or NSMA	pasted file text
Shape what-if	multiplier	0.5 to 2.0 times the file's dB values
Polarization angle	degrees	0 to 90
Mount height (Floor coverage)	m	3 to 15 (10 to 49 ft), in 0.5 m steps
AP and client transmit power (Floor coverage)	dBm	AP 0 to 30, client 0 to 24
Band and path-loss exponent (Floor coverage)	GHz, n	2.4, 5 or 6 GHz; n 2.0 to 4.0
Client position (Floor coverage)	m along the floor	0 to 35

How it works. Dipole: field cos((pi/2) cos theta) / sin theta. Collinear: the dipole times the array factor sin(N psi/2) / (N sin(psi/2)), psi = k d cos theta + k d sin(tilt). Omni by gain: ITU-R F.1336-5, G = G0 - 12 (theta_e / theta3)^2 with theta3 = 107.6 x 10^(-0.1 G0) and its electrical-tilt mapping, floored 30 dB down. Directional: the 3GPP TR 38.901 element, A_V = -min(12 ((theta - 90) / theta3)^2, SLA_V), A_H = -min(12 (phi / phi3)^2, A_max), A = -min(-(A_V + A_H), A_max), with mechanical tilt by rotation. Every built-in shape is scaled so its gain averaged over the sphere is exactly 0 dBi, so the peak shown is the integrated directivity. Imported files: summing, G = G_H(phi) + G_V(theta) in dB with a 30 dB floor (or the deepest value in the file, if deeper), or cross-weighted (Vasiliadis et al., 2005, k = 2); both use the front half of the vertical cut. Polarization loss = -20 log10 |cos d|. Floor coverage: client antenna 1 m up, x along the floor, slant distance r = sqrt(x^2 + (h - 1)^2), angle below the horizon atan((h - 1) / x); AP to client = AP transmit power + G(that direction) - PL(r) + client antenna gain; client to AP = client transmit power + client antenna gain + G - PL(r); PL(r) = FSPL(1 m) + 10 n log10(r). The antenna's gain is the same both ways (reciprocity), so client to AP minus AP to client is the client's transmit power minus the AP's at every point.

Example. A half-wave dipole is 2.15 dBi (0 dBd) with a 78 degree vertical beamwidth. An omni set to 12 dBi has a vertical beamwidth near 6.8 degrees, and 30 degrees above the horizon it is weaker than a 5 dBi omni. A 65 by 65 degree 3GPP element integrates to about 9.8 dBi; the 31,000 rule gives 8.7 dBi and the ideal 41,253 gives 9.9 dBi. Two antennas 45 degrees apart in polarization lose 3.01 dB. On a 9 m ceiling at 5 GHz with n = 3 and the AP at 20 dBm, a dipole leaves the floor under it below -67 dBm out to 2.6 m and reaches -67 dBm out to 17.9 m; an 8 dBi omni there reaches -67 dBm nowhere on the floor; a 65 by 65 degree directional aimed down gives -46.5 dBm directly below.

Field notes

- The 3D is drawn on a fixed dBi scale: radius and color both follow gain, 5 dB per color band, with a floor 35 dB below the top of the scale. The legend gives the numbers, so color is never the only carrier.
- An MSI GAIN with no unit is dBd; reading it as dBi understates the antenna by 2.15 dB. MSI values are losses written without a minus sign, with vertical 90 pointing down. NSMA uses negative values and elevation positive up. Mixing the two turns a pattern upside down with no error.
- A 3D view built from two cuts is an estimate. Summing returns the horizontal cut and the front half of the vertical cut exactly; behind the antenna it can only combine the back of the horizontal cut with the front of the vertical cut. The example files are generated by this app, so the error against the exact 3D can be shown.
- Front-to-back is shown two ways: straight back from the peak, and the worst lobe anywhere in the rear 120 degrees, because the back lobe need not point straight back.
- The F.1336 omni is a planning envelope, not physics: at the dipole's 2.15 dBi it gives a 65.6 degree beamwidth, not 78.
- Mounting turns the 3D view only; the 2D cuts stay in the antenna's own frame. An imported file whose horizontal cut stays within 3 dB is treated as an omni drawn for the ceiling, any other as drawn for a wall.
- NSMA azimuth is read as counterclockwise seen from above. The source used does not state it; it matters only for a horizontal cut that is not symmetric left to right.
- Floor coverage is a model: open floor, no racks, walls or people, a chosen path-loss exponent, and the powers and the client's -2 dBi antenna as illustrative inputs. Straight down, a dipole or collinear has a true null (held 60 dB under the peak here); the omni-by-gain envelope stops 30 dB under its peak, so within a few meters of the point under a high AP it cannot rank two omnis. A real antenna's null depth varies. A level under -95 dBm reads "below -95 dBm", with "(null)" in the pattern's null: a 20 MHz noise floor is about -94 dBm, so no

client or AP reports less, and under a 9 m dipole the model's own figure (-114.2 dBm) is only its 60 dB floor. The exact figure is in the note under Readouts.

- The 25 ft (7.6 m) line is vendor guidance, not a standard: an HPE Aruba Airheads Community thread (Warehouse high ceiling, 2018) quotes a vendor presentation to use downtilt omnis above 25 ft, and a Cisco Community thread (High Ceilings, 2013) has a Cisco employee putting an ordinary omni's limit at about 20 to 25 ft.
- Open in Uplink vs Downlink carries the AP antenna's gain toward the client, but that tool's AP gain runs 0 to 8 dBi; a gain outside that range is carried as the nearest end and the screen says so.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: antenna gain is not more power; the same power is reshaped, so a high-gain omni flattens into a pancake with a hole under it and a directional narrows its beam.

Control	What it does	Why it's there
Antenna (picker): Half-wave dipole, Omni, set by gain, Collinear omni (array), Directional (patch or sector), Imported pattern file; default Omni at 8 dBi	Picks the antenna model.	Change from omni to directional, change the gain, and see the 3D pattern change.
Gain (Omni, 2.15 to 15 dBi)	Changes the omni's gain; the readout gives dBi and dBd.	Every antenna radiates the same power, so raising the gain visibly takes signal from somewhere else. The scale stays fixed, so a gain change reshapes the pattern instead of resizing it.
Electrical downtilt (Omni 0 to 15 degrees; Collinear 0 to 15)	Tilts the cone.	Downtilt tilts the whole cone for an omni, so the ring of strongest signal moves closer on the floor.

Control	What it does	Why it's there
Elements (1 to 8) and Spacing (0.25 to 1.0 wavelength) (Collinear)	Build the array.	Stacking elements makes the nulls between lobes visible; at 1 wavelength spacing a grating lobe appears straight up and down.
Gain (sets both beamwidths) (Directional, 3 to 21 dBi)	Sets both beamwidths together with the practical 31,000 rule.	Gain and beamwidth trade directly. The readouts show what the exact shape integrates to, which is not the rule's number.
Horizontal beamwidth (15 to 180 degrees), Vertical beamwidth (10 to 120 degrees) (Directional)	Set each beamwidth.	Shape the beam one plane at a time.
Front-to-back (A_max), Side-lobe level (SLA_V) (10 to 40 dB); Mechanical downtilt (0 to 30 degrees) (Directional)	Set the rear and side-lobe floors and tilt the beam.	Front-to-back is a worst case, not the value straight back; the readout shows both, because the back lobe need not point straight back.
Rebuild from its two cuts (any built-in antenna)	Writes the antenna as an MSI file, reads it back and rebuilds the 3D from its two cuts.	A 3D view built from two 2D cuts is an estimate; this shows how far off it is against the exact shape.
Generated example (picker), Or paste pattern text (MSI or NSMA), Read pattern (Imported)	Load one of five example files the app generates, or pasted file text. There is no file picker.	Import a two-cut pattern and see it in 3D. The examples are generated here, so the error against the exact 3D can be shown; no manufacturer file is included.
Rebuild the 3D by: Summing / Cross-weighted (Imported)	Switches the reconstruction method.	Compare two ways of building 3D from two cuts, and see that they differ behind the antenna and between the cuts.
What if: shape (x the file's dB) (0.5 to 2.0) (Imported)	Scales the file's dB values.	Change the gain to make it flatter or wider, or make the edges fall off faster.

Control	What it does	Why it's there
Mounting: Ceiling / Wall (default Ceiling)	Turns the 3D view 90 degrees against the floor and the mounting surface.	The same antenna ceiling-mounted and wall-mounted lights different parts of a room. The 2D cuts stay in the antenna's own frame, as a datasheet draws them.
Angle between the two antennas (0 to 90 degrees); See it in 3D: open Polarization	Gives the polarization mismatch loss; the link opens the Polarization tile.	Puts a number on turning one antenna sideways: 45 degrees costs 3.01 dB. The Polarization tile shows the field itself.
Drag, pinch or scroll the 3D view; arrow keys and plus or minus with focus; Reset view; Spin / Stop (presenter bar)	Rotate, zoom and reset the view.	Look underneath a high-gain omni to find the hole, from any angle.
Start from (picker): Office, dipole (3 m); Warehouse, dipole (9 m); Warehouse, 8 dBi omni; Warehouse, +6 dB power; Warehouse, directional (65 by 65 degrees, aimed down)	Sets the antenna, ceiling mounting, height and AP power. Anything moved afterwards reads "Your own settings".	A warehouse story, one change at a time from the 9 m dipole. The presets are illustrative; every slider stays free.
Mount height (3 to 15 m, 0.5 m steps; metric first, feet in parentheses)	Raises or lowers the ceiling and the AP.	The one variable the view is about.
AP transmit power (0 to 30 dBm), Client transmit power (0 to 24 dBm), Band (2.4 / 5 / 6 GHz), Path-loss exponent (2.0 to 4.0)	Set the link.	The same log-distance model and channels as Rate vs Range and Uplink vs Downlink, so the numbers agree across tools.
Client position along the floor (0 to 35 m); drag or tap the floor	Moves the client dot.	Reads both directions at one spot, and sets what Open in Uplink vs Downlink carries.
Readouts: directly below, 10, 20 and 30 m out, both directions; floor cell radius at -67 dBm; hole under the AP	Numbers under the picture.	Client to AP is always AP to client plus (client TX minus AP TX): the antenna's gain is the same both ways.

Control	What it does	Why it's there
Open in Uplink vs Downlink	Opens that tool on this client's link (band, both powers, exponent, slant distance, AP gain toward the client).	Continues the lesson where the asymmetry is the subject. That tool's AP gain runs 0 to 8 dBi, so a gain outside it is carried as the nearest end, and the button's caption says so.

A demo that works (about 3 minutes)

1. Set up: Omni, set by gain at 8 dBi, Ceiling. Press Space to spin it once so the room sees the donut.
2. Ask the room: "I swap this for a 12 dBi omni on the same ceiling. Is the floor right under the AP stronger or weaker?"
3. Reveal: press Up four times to 12 dBi. The donut flattens into a pancake with a hole under it; the vertical beamwidth is near 6.8 degrees, and 30 degrees above the horizon it is weaker than a 5 dBi omni.
4. Optional follow-up: switch Mounting to Wall to see the same antenna light the room differently. Or pick Directional and press Rebuild from its two cuts to show what a planning tool's 3D is built from.

Pairs with: the Antenna Fundamentals guided lesson (read it first), FSPL Simulator (antenna gain in the link), Rate vs Range. The "Antennas" and "RF math and the link budget" sequences.

Left out on purpose: Every built-in shape is normalized to 0 dBi averaged over the sphere, so the peak shown is directivity with no efficiency loss. The omni-by-gain model is the ITU-R F.1336 planning envelope, not physics (it gives the dipole's 2.15 dBi a 65.6 degree beamwidth, not 78). A 3D view from two cuts is an estimate. Mounting turns the 3D only. The NSMA azimuth direction is assumed counterclockwise from above.

Polarization

Draws one radio wave as its electric field along the direction of travel, in a 3D view you can turn. Pick Vertical, Horizontal, Slant 45 degrees, Circular or Elliptical and watch the field: the arrows show which way it points at each place along the wave, and the End-on view square in the corner shows the shape the tip of the field traces, seen from the front end looking back toward the source.

Why it's here. Polarization is usually taught as a word and a number. Seeing the field makes it concrete: vertical and horizontal are the same wave turned 90 degrees, a slant is the same wave turned 45 degrees, and circular is two waves a quarter cycle apart whose sum never shrinks and turns once per wavelength. The component waves show that every polarization is a horizontal wave plus a vertical wave.

How to use

1. Pick a polarization. The readouts name it, say what the tip of the field traces (a line, a circle or an ellipse) and give the axial ratio.
2. Turn on Show the H and V component waves to see the horizontal part (H) and the vertical part (V) that add up to the field. Each is labeled with its letter.
3. Open Adjust to set the two amplitudes and the phase difference yourself. In step (0 or 180 degrees) always gives a line; equal amplitudes 90 degrees apart give a circle; -90 degrees turns the other way.
4. Drag the 3D view to turn it, pinch or scroll to zoom, and tap Reset view to go back. With a keyboard, focus the view and use the arrow keys and plus or minus. Pause stops the wave; with reduced motion on it opens frozen.
5. On a computer or tablet, Present opens the 3D view full screen for a projector: Space plays or pauses, P or the Up arrow moves to the next polarization and Down to the previous, R returns to Vertical and the opening view, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Polarization	preset	Vertical, Horizontal, Slant 45 degrees, Circular, Elliptical, or Custom
H and V amplitude	relative	0 to 1
Phase difference	degrees	-180 to 180, V ahead of H
Show the H and V component waves	on or off	off at open

How it works. E(z, t) = Ax cos(wt - kz) in the horizontal direction plus Ay cos(wt - kz + delta) in the vertical direction, where z is the distance along the direction of travel. The tip of the field traces an ellipse with half-axes sqrt((Ax^2 + Ay^2)/2 +/- R), R = sqrt(Ax^4 + Ay^4 + 2 Ax^2 Ay^2 cos 2delta) / 2. Axial ratio = long half-axis / short half-axis (infinite for a line). The long axis sits at atan2(2 Ax Ay cos delta, Ax^2 - Ay^2) / 2 from horizontal. Every preset has Ax^2 + Ay^2 = 1, so all carry the same power.

Example. Slant 45 degrees is equal horizontal and vertical parts in step (0.71 and 0.71, 0 degrees): a line at 45 degrees, axial ratio infinite. Circular is the same two parts 90 degrees apart: the field stays the same length everywhere, axial ratio 1.0. Elliptical here is 0.89 horizontal and 0.45 vertical, 90 degrees apart: axial ratio 2.0, long axis horizontal.

Field notes

- Circular polarization is shown without a right-hand or left-hand name. Optics and antenna engineering name the direction of turn by different conventions; the tool shows the turn instead.
- The drawing is slowed to one cycle every 3 seconds. The frequency and the wavelength are the same for every polarization.
- The view is dark in both themes, so the lime field and the two component colors are measured against one surface. The legend names each line, so color is never the only carrier.
- For the loss between two antennas at different angles, see the polarization card in Antenna Pattern.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: polarization is the direction the electric field points as the wave passes. Vertical, horizontal and slant are the same wave turned; circular is a horizontal and a vertical wave a quarter cycle apart, whose sum never shrinks and turns once per wavelength.

Control	What it does	Why it's there
Preset (picker): Vertical, Horizontal, Slant 45°, Circular, Elliptical; default Vertical. Reads Custom once the sliders leave every preset.	Sets the two amplitudes and the phase difference. Every preset carries the same power.	The five states a class needs to see.
Show the H and V component waves (default off)	Draws the horizontal part (H) and the vertical part (V) as thinner curves, each labeled with its letter, under the lime field.	Every polarization is two perpendicular waves added together; with this on, the class sees the sum being built.
Adjust: H amplitude, V amplitude (0 to 1), Phase difference, V ahead of H (-180 to 180 degrees, 15 degree steps)	Sets the field directly.	In step always gives a line; equal parts 90 degrees apart give a circle; -90 turns it the other way.
Drag, pinch or scroll the 3D view; arrow keys and plus or minus with focus; Reset view	Rotate, zoom and reset the view. The square in the corner stays face-on: the wave seen end on, from the front end looking back toward the source.	In 3D the traced shape is edge-on from most angles, so it gets its own square.
Play / Pause	Starts and stops the wave (one cycle every 3 seconds). With reduced motion on it opens frozen.	The frequency and wavelength are the same for every preset; only the direction of the field changes.

A demo that works (about 3 minutes)

1. Set up: Vertical, playing, components off. The field goes up and down; the square shows a vertical line.
2. Press P for Horizontal, then Slant 45°: "same wave, turned".
3. Ask the room: "I add a horizontal wave and a vertical wave of the same size. What does the field do?" Turn on the component waves, then press P to Circular.
4. Reveal: the field never shrinks and turns once per wavelength; the square shows a circle, axial ratio 1.0. Press P for Elliptical: axial ratio 2.0.

Pairs with: Antenna Pattern (the polarization mismatch card), the Antenna Fundamentals guided lesson, Wi-Fi Through a Wall (TE and TM). The "Antennas" sequence.

Left out on purpose: No receiving antenna, no mismatch loss and no indoor reflections; Antenna Pattern's polarization card has the loss between two antennas, and a "See it in 3D" link back here. Circular polarization is never named right-hand or left-hand: optics and antenna engineering name the turn differently and the antenna convention was not pinned. The view follows EMANIM Classic by Andras Szilagyi (public domain); no EMANIM code is used.

Rate vs Range

Draws an AP from above with one ring per MCS: inside a ring, a client has enough signal for that MCS. A dashed circle marks the cell edge, where beacons sent at the minimum basic rate can no longer be decoded. Drag the client dot to read its signal, SNR, MCS and data rate, and watch the beacon airtime bar change with the basic rate. A Rate set card turns each of the 12 legacy rates off, supported or basic, and shows which kinds of client can associate, where beacons go, and the rate each ACK goes at.

Why it's here. Rate does not fall smoothly with distance. Each MCS needs a minimum signal, so coverage is a set of rings. Doubling the channel width raises the noise floor 3 dB, so every MCS needs 3 dB more signal and every ring shrinks. Raising the minimum basic rate shrinks the cell edge and cuts the airtime beacons use: 6 to 24 Mbps moves the edge from -82 to -74 dBm at 20 MHz and makes each beacon about 3.6 times shorter. A client must support every basic rate, or the AP refuses it with status 18. And the lowest basic rate is not a floor for ACKs: an ACK goes at the highest basic rate not faster than the frame it answers, so with only 24 Mbps basic, a frame at 12 Mbps is acknowledged at 12.

How to use

1. Drag or tap on the rings to move the client, or use the Client slider under the view. The label next to the dot and the readout below show its received power, noise floor, SNR, MCS and data rate.
2. Pick the band, channel width and spatial streams in the controls (below the view on a phone, in the panel on the right on a wide screen). Change the width and watch every ring shrink while the view stays the same size.

3. Set the AP EIRP, the client antenna gain, the path-loss exponent and an optional margin. The legend lists every ring with its data rate and radius, including rings too thin to label on the view.
4. Build the rate set: tap a rate to cycle it off, supported and basic. The state is written on each chip, and a star marks the rates every device of that kind must support. The four DSSS rates (1, 2, 5.5, 11) exist only at 2.4 GHz. The minimum basic rate is not picked; it is the lowest basic rate, and the dashed cell edge follows it. 9 Mbps and the DSSS rates have no sourced sensitivity, so no cell edge is drawn when one of them is the lowest.
5. Pick the client asking to associate (802.11b only; 802.11g, least it must support; 802.11g; Wi-Fi 6) and, if you like, a PHY the network also requires. The Rate set card says whether it can associate, and if not, why: a missing basic rate means status 18, REFUSED_BASIC_RATES_MISMATCH. It lists every client type, the beacon rate, the Supported Rates element bytes, and an ACK table: for each rate that is on (and HE MCS 0 to 11 for a Wi-Fi 6 client), the rate its ACK goes at.
6. Set the number of SSIDs. The bar shows beacon airtime at the beacon rate against the same beacons at 6 Mbps; DSSS beacons take more, and a notch marks the 6 Mbps figure.
7. On a computer or tablet, Present opens the rings full screen for a projector, with the association verdict beside them: Up and Down move the client out and in, B raises the minimum basic rate (6, 12, 18, 24, 36, 48, 54, then back to 6), C changes the client type, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Band	GHz	2.4, 5 or 6; default 5. Path loss is taken at 2437, 5500 or 6135 MHz
Channel width	MHz	20 or 40 at 2.4 GHz, up to 160 at 5 GHz, up to 320 at 6 GHz; default 20
Spatial streams	streams	1 to 4, for the data-rate labels only; default 2
AP EIRP	dBm	0 to 36; default 20
Client antenna gain	dBi	-5 to 6; default 0

Input	Unit	Range
Path-loss exponent	n	2.0 (free space) to 4.0; default 3.0
Margin	dB	0 to 20; default 0
Rate set	each rate off, supported or basic	1, 2, 5.5, 11 (2.4 GHz only), 6, 9, 12, 18, 24, 36, 48, 54 Mbps; default 6, 12, 24 basic, the rest supported. The minimum basic rate is the lowest basic rate
Client asking to associate	choice	802.11b only; 802.11g, least it must support (1, 2, 6 Mbps); 802.11g; Wi-Fi 6 (default)
Network also requires	choice	Nothing more (default); HT, Wi-Fi 4 (not at 6 GHz); HE, Wi-Fi 6
SSIDs	count	1 to 16; default 4

How it works. Received power = AP EIRP + client antenna gain - path loss, with path loss = free-space loss at 1 m + 10 n log10(d). Each ring radius is the distance where received power equals that MCS's minimum sensitivity plus the margin: d = 10^((EIRP + G - FSPL(1 m) - sensitivity - margin) / 10n). Minimum sensitivity at 20 MHz, MCS 0 to 13: -82, -79, -77, -74, -70, -66, -65, -64, -59, -57, -54, -52, -49, -46 dBm, and 3 dB higher for each doubling of width up to 320 MHz. Noise floor = -174 dBm/Hz + 10 log10(BW in Hz) + a 7 dB noise figure. The cell edge uses the 20 MHz floor of the basic rate. Who can associate: a client must support every basic rate, or it is refused with status 18, REFUSED_BASIC_RATES_MISMATCH (IEEE 802.11-2024 Table 9-80). Beacons go at one of the basic rates (10.6.5.1); the model uses the lowest, a common vendor choice rather than a rule, and with no basic rates, a mandatory rate. The ACK goes at the highest basic rate not faster than the frame it answers, in the same modulation; if none fits, at the highest mandatory rate not faster (10.6.6.5.2). Mandatory rates: 6, 12 and 24 Mbps for OFDM; 1, 2, 5.5, 6, 11, 12 and 24 for 802.11g (ERP). An HT, VHT or HE frame first converts to a non-HT reference rate by its modulation and coding (10.6.11, Table 10-10: 64-QAM 5/6 is 54). In the Supported Rates element a basic rate has its top bit set, in units of 500 kb/s: 6 Mbps basic is 8C (9.4.2.3). Beacon airtime comes from the SSID Airtime tool's calculator (802.11ax beacon size, one AP on the channel, no MBSSID). Data rates come from the MCS Index tool (802.11be, 800 ns guard interval).

Example. At 5 GHz (5500 MHz) with 20 dBm EIRP and n = 3, free-space loss at 1 m is 47.3 dB. The MCS 0 ring reaches 66.8 m at 20 MHz and 42.2 m at 80 MHz. MCS 13 reaches 4.2 m at 20 MHz. A client at 20 m receives -66.3 dBm, just short of MCS 5 (-66 dBm), so it gets MCS 4, with an SNR of 27.7 dB over a -94.0 dBm noise floor. Raising the basic rate from 6 to 24 Mbps pulls the cell edge in from 66.8 m to 36.2 m, and four SSIDs' beacons drop from 2.19% to 0.61% of airtime.

Field notes

- The model's readings, where the standard is silent: when the highest basic rate below a frame is of the other modulation (1, 2, 5.5 or 11 Mbps answering an OFDM frame), the model keeps the modulation and answers from the mandatory OFDM rates; HE frames are answered in OFDM; an 802.11b device, which has no OFDM receiver, cannot decode a beacon sent at an OFDM rate. Rows decided this way are marked.
- A star marks a mandatory rate but does not lock it: the standard says what a device must support, not what an access point must keep on. No status code is shown when a client lacks a required PHY, because none is pinned for that case.
- These sensitivities are conformance floors, the least a radio must manage to pass. Real radios do several dB better, so real rings are larger.
- Only 6 Mbps has a published floor of its own, -82 dBm, equal to MCS 0. 12 to 54 Mbps use the floor of the MCS with the same modulation and coding (24 Mbps is MCS-equivalent to MCS 3, -74 dBm), so they are labeled MCS-equivalent. 9 Mbps and the 2.4 GHz DSSS rates have no sourced floor, so they draw no cell edge.
- The path-loss exponent is a model with a chosen value, not a measurement. Walls, people and furniture make real rings uneven.
- The width is held fixed. A real AP may send a narrower transmission to a far client, which reaches as far as the 20 MHz rings.
- The beacon bar counts beacons only, as time on the air. It leaves out probe responses, the wait for the medium, and other APs on the same channel, all of which the SSID Airtime tool can add.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: rate falls with distance in steps, one ring per MCS; a wider channel shrinks every ring, and a higher minimum basic rate shrinks the cell edge and the beacon airtime. The Rate set card adds: a client must support every basic rate or the AP refuses it with status 18, and the lowest basic rate is not a floor for ACKs.

Control	What it does	Why it's there
Drag or tap on the rings; Client (slider, logarithmic, default 20 m (66 ft))	Moves the client. The label and readout give received power, noise floor, SNR, MCS and data rate.	Read the MCS a client gets at any distance and see the step at each ring edge.
Band (2.4 / 5 / 6 GHz, default 5)	Changes the frequency path loss is taken at (2437, 5500 or 6135 MHz) and the widths on offer.	Higher bands lose more in free space, so every ring is smaller.
Channel width (MHz) (20 or 40 at 2.4 GHz, up to 160 at 5 GHz, up to 320 at 6 GHz; default 20)	Changes every MCS threshold by 3 dB per doubling.	Doubling the width raises the noise floor 3 dB, so every ring shrinks. The view never rescales, so the shrink is visible.
Spatial streams (rate labels only) (1 to 4, default 2)	Changes the data rates on the labels, not the rings.	Streams multiply the rate but do not change the signal each MCS needs, so the rings stay where they are.
AP EIRP (0 to 36 dBm, default 20)	Raises or lowers the downlink level everywhere.	More radiated power pushes every ring out by the same number of dB.
Client antenna gain (-5 to 6 dBi, default 0)	Adds the client's antenna to the link.	A worse client antenna shrinks every ring as surely as a weaker AP.
Path-loss exponent (model) (2.0 to 4.0, default 3.0)	Sets how fast the signal falls with distance.	n = 2 is free space, indoors is usually 3 to 4. Labeled a model with a chosen value, not a measurement.
Margin (0 to 20 dB, default 0)	Adds a safety margin to every MCS threshold.	Shows how much coverage a design margin costs, ring by ring.

Control	What it does	Why it's there
Rate set: 12 chips, 1, 2, 5.5, 11 (2.4 GHz only), 6, 9, 12, 18, 24, 36, 48, 54 Mbps; each tap cycles off, supported, basic; a star marks mandatory rates. Default 6, 12, 24 basic, the rest supported	Sets which rates the AP offers and which every client must have. The minimum basic rate is no longer picked: it is the lowest basic rate, and the dashed cell edge and the beacon bar follow it.	Raising the lowest basic rate shrinks the cell edge and cuts beacon airtime (6 to 24 Mbps moves the edge from -82 to -74 dBm at 20 MHz). Only 6 Mbps has a published floor; when 9 Mbps or a DSSS rate is the lowest basic rate no cell edge is drawn, because neither has a sourced floor. The star marks, it does not lock: the standard says what a device must support, not what an AP must keep on.
Client asking to associate (802.11b only; 802.11g, least it must support; 802.11g; Wi-Fi 6, default)	Picks the client the verdict is about. Every type is also listed.	A missing basic rate means status 18, REFUSED_BASIC_RATES_MISMATCH. An 802.11b device against OFDM basic rates also cannot decode the OFDM beacons, so it never sees the network. The "least it must support" 802.11g device (1, 2, 6 Mbps) is refused by the default 6, 12, 24 basic set.
Network also requires (nothing; HT; HE)	Adds a BSS membership selector.	A client without the PHY cannot associate. No status code is shown, because none is pinned for that case.
ACK table (readout)	For every rate that is on, and HE MCS 0 to 11 for a Wi-Fi 6 client, the rate its ACK goes at, and whether a basic rate or the mandatory fallback answered.	With only 24 basic, a frame at 12 is acknowledged at 12, not 24. Rows marked * are the model's reading where the standard is silent (a DSSS basic rate below an OFDM frame).
SSIDs on this AP (1 to 16, default 4)	Scales the beacon airtime bar.	Every SSID sends its own beacons at the basic rate, so the airtime cost multiplies.

A demo that works (about 3 minutes)

1. Set up: the defaults: 5 GHz, 20 MHz, 2 streams, AP EIRP 20 dBm, n = 3, client at 20 m (66 ft). The client receives -66.3 dBm, just short of MCS 5, so it gets MCS 4.
2. Ask the room: "Widen the channel from 20 to 80 MHz. What happens to the edge of coverage?"
3. Reveal: set 80. The MCS 0 ring pulls in from 66.8 m (219 ft) to 42.2 m (138 ft), and every other ring shrinks with it.

4. Optional follow-up: back to 20 MHz, tap the 6 and 12 chips off (leaving 24 basic). The cell edge pulls in from 66.8 m to 36.2 m (119 ft) and four SSIDs' beacons drop from 2.19% to 0.61% of airtime. Now ask: "A client sends at 12 Mbps. What rate does the ACK go at?" Open the ACK table: 12, because no basic rate is at or below 12, so the mandatory rate answers.

Pairs with: FSPL Simulator, Uplink vs Downlink (the other direction of the same rings), 6 GHz Power and PSD, the SSID Airtime and MCS Index tools. The "RF math and the link budget" and "From bits to rate" sequences.

Left out on purpose: The sensitivities are conformance floors, the least a radio must manage; real radios do several dB better, so real rings are larger. The path-loss exponent is a chosen value, and walls, people and furniture make real rings uneven. The width is held fixed, where a real AP may send a narrower transmission to a far client. The beacon bar counts beacons only: no probe responses, contention or other APs. The beacon rate is the lowest basic rate, a common vendor choice; the standard asks only for one of the basic rates. Only the legacy rate set is modeled: no Basic HT-MCS, VHT or HE MCS sets.

6 GHz Power and PSD

Plots the most EIRP each 6 GHz device class may radiate, and the SNR it leaves at a distance, against channel width from 20 to 320 MHz, one line per class. A second view draws one channel as a flat block of power per MHz, so you can see the block get wider while its height holds or drops.

Why it's here. 6 GHz power is limited per MHz (power spectral density, PSD) as well as in total. While the PSD limit binds, a wider channel adds 3 dB of EIRP per doubling, which exactly offsets the 3 dB higher noise floor, so SNR holds. Once a class reaches its cap, every doubling costs 3 dB of SNR. The tool also shows that the client 6 dB rule applies as a relative limit only to Standard Power and GVP clients.

How to use

1. Pick US or EU, then turn classes on or off in the controls (below the chart on a phone, in the panel on the right on a wide screen). Each power class has its own color, and every class has its own line style and marker, so you can tell them apart without color.
2. Switch the chart between EIRP, SNR and Spectrum. Tap a column on the chart or use the Width slider to pick 20, 40, 80, 160 or 320 MHz.

3. Read the numbers for each class at that width under the chart: EIRP, received power, SNR, the power per MHz actually radiated, the typical highest MCS, and a line saying whether the class is PSD-limited or cap-limited.
4. Set the distance, extra loss for walls and the receiver noise figure. For Standard Power and GVP, set the AP's authorized power to see its clients follow it 6 dB lower.
5. On a computer or tablet, Present opens the chart full screen for a projector: Up and Down change the channel width, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Region	US or EU	US (47 CFR 15.407) or EU (ETSI EN 303 687); default US
Classes	on / off	US: Standard Power AP, fixed client, SP client, LPI AP, subordinate, LPI client, GVP AP, GVP client, VLP. EU: LPI, VLP. Default US: SP AP, LPI AP, VLP
Channel width	MHz	20, 40, 80, 160 or 320; default 80
Distance	m	1 to 100; default 10
Extra loss	dB	0 to 40; default 0
Noise figure	dB	3 to 12; default 7
AP authorized power	dBm EIRP	Standard Power AP 0 to 36 (default 36); GVP AP 0 to 24 (default 24); US only

How it works. EIRP = min(PSD limit + 10 log10(BW in MHz), max EIRP). SP and GVP clients are also held to their AP's authorized power minus 6 dB, and an SP or GVP AP to its own authorized power. US limits (PSD dBm/MHz, max dBm): SP AP and fixed client 23 and 36, SP client 17 and 30, LPI AP and subordinate 5 and 30, LPI client -1 and 24, GVP AP 11 and 24, GVP client 5 and 18, VLP -5 and 14. EU: LPI 10 and 23, VLP 1 and 14. Noise floor = -174 dBm/Hz + 10 log10(BW in Hz) + noise figure. Received power = EIRP - free-space path loss at 6105 MHz - extra loss, with

a 0 dBi receive antenna. SNR = received power - noise floor. Channel counts are non-overlapping channels that fit wholly inside the class's sub-bands.

Example. At 10 m with no extra loss, free-space loss at 6105 MHz is 68.2 dB. An LPI AP radiates 18, 21, 24, 27 and 30 dBm at 20, 40, 80, 160 and 320 MHz, and its SNR stays at 43.8 dB from 20 to 160 MHz because the noise floor rises by the same 3 dB (-94.0 dBm at 20 MHz, -88.0 dBm at 80 MHz). A Standard Power AP holds 36 dBm at every width, so its SNR falls from 61.8 dB at 20 MHz to 49.8 dB at 320 MHz. An SP client whose AP is authorized at 30 dBm is limited to 24 dBm.

Field notes

- These are regulatory maximums. A device may transmit less than its limit.
- Standard Power and GVP may use only U-NII-5 and U-NII-7 in the US, which leaves 41, 20, 9, 4 and 1 channels of 20 to 320 MHz. LPI and VLP may use the whole 5925 to 7125 MHz band: 59, 29, 14, 7 and 3. The EU band is 5945 to 6425 MHz.
- The LPI client limit of 24 dBm and -1 dBm/MHz is a flat number, not tied to the AP. It only happens to sit 6 dB under the LPI AP. Fixed client devices are exempt from the 6 dB rule, VLP has no client class, and the EU has no client offset.
- Path loss is taken at one frequency for every width so that only the width changes between columns. The MCS shown is the typical figure from this app's Signal Thresholds table, not a guarantee.
- The PSD rule applies to any 1 MHz of the real emission. An HE 20 MHz transmission fills about 18.9 MHz, so a device exactly at the PSD limit radiates about 0.2 dB less than shown. That difference is too small to change the lesson.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: 6 GHz power is limited per MHz as well as in total, so a PSD-limited class keeps its SNR as the channel widens while a capped class loses 3 dB per doubling, and the client 6 dB rule applies only to Standard Power and GVP clients.

Control	What it does	Why it's there
Region: US (FCC) / EU (ETSI) (default US)	Switches the rule set.	The EU has no client offset at all, which the US rules make easy to forget.
Classes to show (checkboxes; US: nine classes, EU: two; default Standard Power AP, Low Power Indoor AP, Very Low Power device)	Turn class lines on and off. Each class has its own color, line style and marker.	Compare classes on one chart; each can be told apart without color.
Chart: EIRP / SNR / Spectrum (default EIRP)	Switches the view. Spectrum draws one channel as a flat block of power per MHz.	EIRP shows the cap and the PSD limit; SNR shows the consequence; Spectrum shows the block getting wider while its height holds or drops.
Class drawn in the spectrum view (picker, Spectrum with more than one class)	Picks the class the block is drawn for.	The spectrum view draws one class at a time.
Width (slider, 20, 40, 80, 160, 320 MHz, default 80); tap a column on the chart	Picks the channel width the readouts describe.	The x axis of the lesson; the readout line says whether each class is PSD-limited or cap-limited at that width.
Distance (1 to 100 m, logarithmic, default 10 m (33 ft))	Sets where SNR is read.	SNR at a chosen distance, with free-space loss at one frequency for every width so only the width changes between columns.
Extra loss (walls) (0 to 40 dB, default 0)	Subtracts a fixed loss from every class.	An optional extra-loss slider for walls.
Noise figure (3 to 12 dB, default 7)	Raises or lowers the noise floor.	Moves every class's SNR by the same amount, so it never changes which class holds and which falls.
AP authorized power: Standard Power AP (AFC) (0 to 36 dBm), GVP AP (geofencing) (0 to 24 dBm); US only, active when an SP or GVP class is shown	Sets each AP's grant. Its clients follow 6 dB lower.	The client -6 dB rule is a relative limit only for Standard Power and GVP clients; the LPI client is a flat number that happens to sit 6 dB under the LPI AP.

A demo that works (about 3 minutes)

1. Set up: US, the default classes, SNR view, 10 m (33 ft), width 20 MHz.
2. Ask the room: "Double the channel to 40, then 80, then 160 MHz. Which class keeps its SNR?"
3. Reveal: press Up three times. The LPI AP holds 43.8 dB from 20 to 160 MHz: its EIRP rises 3 dB per doubling (18, 21, 24, 27 dBm) and the noise floor rises the same 3 dB. The Standard Power AP is already at its 36 dBm cap, so it loses 3 dB per doubling (61.8 dB at 20 MHz, 49.8 dB at 320).
4. Optional follow-up: switch to EIRP and turn on Standard Power client; set the Standard Power AP to 30 dBm and the client drops to 24 dBm. Then switch to EU and point out there is no client offset.

Pairs with: FSPL Simulator and Wi-Fi Through a Wall (the "Why 6 GHz behaves differently" sequence), Uplink vs Downlink (the client rules in a link), Rate vs Range, the 6 GHz channel card in Course Handouts.

Left out on purpose: These are regulatory maximums; a device may transmit less. Path loss is taken at one frequency (6105 MHz) for every width. The MCS shown is the typical figure from the Signal Thresholds table, not a guarantee. The PSD rule applies to any 1 MHz of the real emission, and an HE 20 MHz transmission fills about 18.9 MHz, a 0.2 dB difference too small to change the lesson.

Uplink vs Downlink

Shows both directions of one Wi-Fi link at once, from above. The downlink runs from the AP to the client and the uplink from the client to the AP. A solid ring marks how far the client can still decode the AP, a dashed ring how far the AP can still decode the client, and the shaded band between them is the asymmetry zone. Two arrows between the AP and the client carry each direction's received level and its MCS (modulation and coding scheme, the data-rate step).

Why it's here. Every Wi-Fi link is two links, each with its own transmitter and receiver. The client usually transmits less power than the AP and usually has the worse antenna, so there is a zone where the client hears the AP well enough to stay connected while the AP barely hears the client. That asymmetry is normal, not a fault. Turning the AP down to "match" the client shrinks the cell on the side that was already stronger and does nothing for the uplink: the client is the weak end of the link and needs the AP's extra power.

How to use

1. Drag or tap on the floor to move the client, or use the Client slider under the view. The readout gives the level, MCS and signal-to-noise ratio (SNR) each way, the imbalance in dB and the width of the asymmetry zone in meters.
2. Set the AP and client transmit powers and both antenna gains in the controls (below the view on a phone, in the panel on the right on a wide screen). The values are illustrative. Watch the uplink move with the client's power and the AP's antenna, and never with the AP's power.
3. Pick a regulatory rule. US 6 GHz Standard Power and GVP (geofenced variable power) hold the client 6 dB below the AP's authorized power, so the client follows the AP as you move it. US 6 GHz LPI (low power indoor) gives the client a flat limit, and the EU gives AP and client the same limit. Custom applies no rule. Every rule is a 6 GHz rule, so another band returns to Custom.
4. Press Turn AP down to match to set the AP's transmit power to the client's. The solid ring shrinks, the dashed ring does not move, and the readout says what changed and what did not. Press it again to put the AP back.
5. Predict, then reveal asks: the client shows four bars; can the AP hear it? Reveal the answer, then put the client in the zone to see it.
6. On a computer or tablet, Present opens the tool full screen for a projector: Up and Down move the client out and in, R resets, P reveals the answer, M turns the AP down to match and back, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Regulatory rule	preset	Custom, US 6 GHz Standard Power, US 6 GHz GVP, US 6 GHz LPI or EU 6 GHz LPI; default Custom
Band	GHz	2.4, 5 or 6; default 5. Path loss is taken at 2437, 5500 or 6135 MHz
Channel width	MHz	20 or 40 at 2.4 GHz, up to 160 at 5 GHz, up to 320 at 6 GHz; default 20
AP transmit power	dBm	0 to 30; default 20 (illustrative)
Client transmit power	dBm	0 to 24; default 14 (illustrative). Under a rule, the rule sets it

Input	Unit	Range
AP antenna gain	dBi	0 to 8; default 4 (illustrative)
Client antenna gain	dBi	-5 to 3; default -2 (illustrative, a phone-class antenna)
Path-loss exponent	n	2.0 (free space) to 4.0; default 3.0

How it works. Downlink level = AP transmit power + AP antenna gain - path loss + client antenna gain. Uplink level = client transmit power + client antenna gain - path loss + AP antenna gain. Path loss = free-space loss at 1 m + 10 n log10(d), the same both ways, so downlink minus uplink always equals AP transmit power minus client transmit power. Each direction's MCS is the highest whose minimum receiver sensitivity the level meets (MCS 0 needs -82 dBm at 20 MHz, 3 dB more per doubling of width), and each ring is where that direction falls to the MCS 0 floor. Noise floor = -174 dBm/Hz + 10 log10(BW in Hz) + a 7 dB noise figure at both ends. Rules, in equivalent isotropically radiated power (EIRP, transmit power plus antenna gain): US Standard Power, AP up to 36 dBm and client up to 6 dB below the AP's authorized power; US GVP, AP up to 24 dBm and client 6 dB below; US LPI, AP 5 dBm per MHz up to 30 dBm and client -1 dBm per MHz up to 24 dBm; EU LPI, AP and client both 10 dBm per MHz up to 23 dBm. Under a rule the client transmits at its limit.

Example. At 5 GHz (5500 MHz), 20 MHz and n = 3, with the illustrative AP at 20 dBm and 4 dBi and the client at 14 dBm and -2 dBi, a client 20 m away hears the AP at -64.3 dBm (MCS 6) while the AP hears the client at -70.3 dBm (MCS 3): a 6 dB imbalance. The client can decode the AP out to 78 m, but the AP can decode the client only out to 49 m, so the asymmetry zone is 29 m wide. Turning the AP down to 14 dBm to match pulls the solid ring in from 78 m to 49 m, and the uplink stays at -70.3 dBm.

Field notes

- The imbalance is set by the two transmit powers alone. Each antenna's gain counts in both directions, so the AP's better antenna helps it hear the client as much as it helps the client hear it.
- Under a rule the client is set to the most it may transmit. The rule is a ceiling: many clients transmit less, which widens the gap.
- The US LPI numbers sit 6 dB apart, but that is a coincidence of two flat limits. Only Standard Power and GVP tie the client to its AP, and they tie it to the AP's authorized power: turning the AP's transmitter down does not change its grant, so the client's limit stays put.

- Both ends use the same receiver floors, the conformance minimums for each MCS. Real radios do better, and many APs gain more on receive from extra antennas and receive chains, which this model leaves out.
- A client's bars come from the downlink only. They cannot tell you whether the AP hears the client.
- For one direction in detail, with cable losses and fade margin, use the Link Budget tool.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: every Wi-Fi link is two links, the client usually transmits less than the AP, so there is a zone where the client hears the AP but the AP cannot hear the client, and turning the AP down to "match" fixes nothing on the uplink.

Control	What it does	Why it's there
Drag or tap on the floor; Client (slider, default 20 m (66 ft))	Moves the client. The readout gives each direction's level, MCS and SNR, the imbalance and the width of the asymmetry zone.	Put the client inside the shaded zone and read both directions at once.
Regulatory rule (picker): Custom (no rule applied), US 6 GHz Standard Power, US 6 GHz Geofenced Variable Power (GVP), US 6 GHz Low Power Indoor (LPI), EU 6 GHz Low Power Indoor (LPI); default Custom	Applies a rule to the client's power. Every rule is a 6 GHz rule, so picking another band returns to Custom.	Regulations set the client relative to the AP (Standard Power, GVP), as a flat number (US LPI) or the same as the AP (EU). Standard Power and GVP are separate entries because their AP caps differ.
Band (2.4 / 5 / 6 GHz, default 5)	Sets the frequency path loss is taken at.	Path loss is the same both ways on any band; the imbalance comes from the radios.
Channel width (MHz) (default 20)	Moves each direction's MCS thresholds 3 dB per doubling, and the per-MHz limits under the LPI rules.	Under US LPI the limits are per MHz, so at 20 MHz the AP is 18 dBm and its client 12; the flat 30 and 24 appear only at 320 MHz.

Control	What it does	Why it's there
AP transmit power (0 to 30 dBm, default 20, illustrative)	Moves the downlink only.	The uplink never moves with the AP's power.
Client transmit power (0 to 24 dBm, default 14, illustrative); reads "set by the rule" under a rule	Moves the uplink only. Under a rule the client transmits at its limit.	The client is the weak end of the link; its power sets the uplink. The rule is a ceiling, and many clients transmit less.
AP antenna gain (0 to 8 dBi, default 4), Client antenna gain (-5 to 3 dBi, default -2, a phone-class antenna)	Add each antenna's gain.	Each antenna counts in both directions, so the AP's better antenna helps it hear the client as much as it helps the client hear it. The imbalance is set by the two transmit powers alone.
Path-loss exponent (model) (2.0 to 4.0, default 3.0)	Sets how fast the signal falls with distance.	Stretches or shrinks both rings without changing the dB imbalance between them.
Turn AP down to match / Put the AP back to (power)	Sets the AP's transmit power to the client's, and back.	AP and client power rarely match and should not be made to. The solid ring shrinks, the dashed ring does not move, and the readout says what changed and what did not. The tool never tells you to equalize the two.
Reveal the answer / Hide the answer; Put the client in the zone	Answers "The client shows four bars. Can the AP hear it?" and moves the client into the asymmetry zone.	Bars come from the downlink only and cannot tell you whether the AP hears the client.

A demo that works (about 3 minutes)

1. Set up: the defaults: Custom, 5 GHz, 20 MHz, AP 20 dBm with 4 dBi, client 14 dBm with -2 dBi, client at 20 m (66 ft).
2. Ask the room the question on screen: the client shows four bars. Can the AP hear it?
3. Reveal: press P. At 20 m the client hears the AP at -64.3 dBm (MCS 6) and the AP hears the client at -70.3 dBm (MCS 3), a 6 dB imbalance. Press Put the client in the zone: between 49 m (161 ft) and 78 m (256 ft) the client still decodes the AP while the AP cannot decode the client.

4. Optional follow-up: press M. The solid ring pulls in from 78 m to 49 m and the uplink stays at -70.3 dBm: the AP gave up its reach and the uplink gained nothing. Then pick US 6 GHz Standard Power on 6 GHz and move the AP slider: the client follows the AP's authorized power, 6 dB lower.

Pairs with: Rate vs Range (the same thresholds and path loss), 6 GHz Power and PSD (the same client rules), Body Loss, Link Budget. The "RF math and the link budget" and "Why clients misbehave" sequences.

Left out on purpose: All radio values are illustrative. Both ends use the same conformance receiver floors, where many APs hear better with extra antennas and receive chains. Under a rule the client is set to the most it may transmit. No walls or bodies; path loss is the log-distance model. For one direction in detail, with cable losses and fade margin, use Link Budget (the tool has a button).

Body Loss

Shows what people cost a Wi-Fi signal, from above. An auditorium floor holds one AP on the front wall, a person holding a device (the holder) and up to 50 other people. A straight line runs from the AP to the device, and every person that line passes through is numbered. The readouts give the received level, the loss from the holder's own body, the loss from the crowd, the difference between the empty and the occupied room, and the MCS (modulation and coding scheme, the data-rate step) the level supports. Every body-loss value is an illustrative setting, not a measurement.

Why it's here. A person between the device and the AP costs decibels, and the person holding the device often costs the most, because their body sits right next to its antenna. Facing the AP with the device in front costs little; turning your back puts your body in the path. Crowds add up: a room full of people is a room full of obstacles that are mostly water, which is why a survey of the empty building reads better than the same building in use. Higher bands generally lose more to the body.

How to use

1. Turn the holder with the Holder facing slider, the Face the AP and Back to the AP buttons, or by tapping the floor: the holder turns toward the spot you tap. Watch the loss from the holder rise from nothing to the full value as the back turns toward the AP.
2. Drag the holder to move them, or use the Holder slider under the view. Drag any person to place them on or off the line.

3. Set the body losses in the controls (below the view on a phone, in the panel on the right on a wide screen). Each is labeled illustrative: set them to match what you measure.
4. Switch the building between Empty and Occupied, change how many people are in the room, and press Scatter the crowd for a new arrangement. The same arrangement always comes back for the same seed.
5. Predict, then reveal asks: you surveyed the auditorium empty on Saturday; what happens Monday at 9 a.m.? Reveal the answer, then switch between Saturday and Monday.
6. On a computer or tablet, Present opens the tool full screen for a projector: Left and Right turn the holder, Space empties or fills the room, Up and Down change the crowd size, R resets, P reveals the answer, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Band	GHz	2.4, 5 or 6; default 5. Path loss is taken at 2437, 5500 or 6135 MHz
Holder facing	degrees	0 to 360, 0 up the page and 90 to the right; default facing the AP
Holder body loss at 2.4 GHz	dB	0 to 20; default 8 (illustrative)
Loss per person on the line at 2.4 GHz	dB	0 to 10; default 4 (illustrative)
5 GHz body-loss multiplier	x	1.0 to 2.0; default 1.2 (illustrative)
6 GHz body-loss multiplier	x	1.0 to 2.0; default 1.3 (illustrative)
Building	state	Empty or Occupied; default Occupied
People in the room	people	0 to 50; default 30

How it works. Received level = 20 dBm radiated by the AP - path loss - holder loss - crowd loss, with a 0 dBi device antenna (decibels over an isotropic antenna). Path loss = free-space path loss at 1 m + 10 x 3 x log10(d), the log-distance model at the Roaming Walk default exponent, with d the distance from the AP to the device. Holder loss = the holder setting x the band multiplier x a turn share: 0 while the holder faces anywhere within 120 degrees of the AP, rising along a raised-cosine curve over the last 60 degrees to 1 with the back square to the AP (illustrative). Crowd loss = the per-person setting x the band multiplier x the number of people the straight line from the AP to the device passes through, each body taken as 0.5 m wide seen from above; it is zero while the building is empty, and empty vs occupied changes nothing else. The band multipliers are 1.0 at 2.4 GHz and, by default, 1.2 at 5 GHz and 1.3 at 6 GHz, an illustrative trend, not a measured table. The MCS is the highest whose minimum receiver sensitivity the level meets at 20 MHz (MCS 0 needs -82 dBm). The device is held 0.3 m in front of the body. The crowd is placed at random from a seed, at least 0.7 m apart.

Example. At 5 GHz with the illustrative defaults, the device is 14 m from the AP and the holder faces the AP. Empty, it reads -61.8 dBm (MCS 7). With 30 people in the room, 2 stand on the line, each costing 4 x 1.2 = 4.8 dB, so the occupied room reads -71.4 dBm (MCS 3): 9.6 dB worse than the survey of the empty room. Turn the holder's back to the AP and the body adds 8 x 1.2 = 9.6 dB more: -81.6 dBm, MCS 0.

Field notes

- Every body-loss value in this tool is an illustrative, adjustable setting. No measured body-loss figure stands behind the defaults, so treat the numbers as a way to see the shape of the effect, not as a design value.
- The holder counts first. A device held against the body loses the most when the back turns toward the AP, which is why a reading can change just by turning around.
- A survey walked in an empty building measures a building nobody uses that way. Leave margin for the people who will fill it.
- Only people on the straight line count here. Real signals also arrive by reflections around a crowd, which this model leaves out, so it shows the direct path only.
- For walls and building materials rather than people, use RF Attenuation (RF: radio frequency) and Wi-Fi Through a Wall.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: people cost decibels, the person holding the device often costs the most when they turn their back to the AP, and a crowd on the line is why an empty-building survey reads better than the building in use.

Control	What it does	Why it's there
Band (2.4 / 5 / 6 GHz, default 5)	Sets the frequency and applies that band's body-loss multiplier.	Higher bands generally lose more to the body, shown as an illustrative trend.
Holder facing (slider, 0 to 360 degrees, 0 up the page and 90 to the right); Face the AP, Back to the AP (buttons); tap the floor to turn the holder toward that spot	Turns the holder. The holder loss stays at zero until the holder has turned more than 120 degrees away from the AP, then ramps in to the full value with the back square to it.	Orientation matters: facing the AP with the device in front costs little; turning your back puts your body in the path.
Drag the holder; Holder (slider, distance from the AP)	Moves the holder.	Changes the path loss and which people stand on the line, so the class can find a clear spot.
Drag any person	Places a person on or off the line.	Shows that only the people on the straight line count in this model.
Holder body loss at 2.4 GHz (illustrative) (0 to 20 dB, default 8)	Sets the holder's full loss with the back square to the AP.	No primary source was read for any body-loss figure, so every loss is a labeled, adjustable setting; set it to match what you measure.
Loss per person on the line at 2.4 GHz (illustrative) (0 to 10 dB, default 4)	Sets what each person on the line costs.	Same reason: illustrative, adjustable, labeled so.
5 GHz body-loss multiplier (illustrative) (1.0 to 2.0, default 1.2), 6 GHz body-loss multiplier (illustrative) (1.0 to 2.0, default 1.3)	Scale the losses at 5 and 6 GHz.	The band trend applies to the crowd as well as the holder, because a crowd is bodies.

Control	What it does	Why it's there
Building: Empty / Occupied (default Occupied)	Removes or restores the crowd. Nothing else changes.	The Saturday survey against the Monday room.
People in the room (0 to 50, default 30)	Sets the crowd size.	Crowds add up, one body on the line at a time.
Scatter the crowd	Places the people in a new arrangement from a new seed.	The same number of people costs more or less depending on how many land on the line.
Reveal the answer / Hide the answer; Show Saturday, empty / Show Monday, occupied	Answers "You surveyed the auditorium empty on Saturday. What happens Monday at 9 a.m.?" and switches between the two.	Predict, then reveal, on the tool's question.

A demo that works (about 3 minutes)

1. Set up: the defaults: 5 GHz, Occupied with 30 people, the holder 14 m (46 ft) from the AP, facing it. Two people stand on the line.
2. Ask the room the question on screen: surveyed empty on Saturday, what happens Monday at 9 a.m.?
3. Reveal: press P, then Space to empty the room: -61.8 dBm (MCS 7). Space again to fill it: -71.4 dBm (MCS 3), 9.6 dB worse, from two people at 4.8 dB each.
4. Optional follow-up: press Right one step at a time. The first eight presses change nothing; after that the holder loss ramps in, and with the back square to the AP the level is -81.6 dBm, MCS 0. Then compare 2.4 and 6 GHz.

Pairs with: Why Two Devices Disagree (body as one part of a reading), Uplink vs Downlink, Wi-Fi Through a Wall and RF Attenuation for walls rather than people.

Left out on purpose: Every body-loss value is an illustrative setting with no measured figure behind the defaults, so use it for the shape of the effect, not as a design value. Only people on the straight line count; real signals also reach around a crowd by reflection. Bodies are 0.5 m wide

seen from above, and the device is held 0.3 m in front of the holder. Path loss is the log-distance model at n = 3.

Signals and PHY (7)

Modulation Simulator

Shows how bits become a radio wave. Each group of bits picks one point on the I/Q plane, and that point sets the amplitude and phase of the carrier for one symbol. Add noise and watch the receiver decide which point was sent, measure the EVM, and count the symbol and bit errors.

Why it's here. Every Wi-Fi data rate is a modulation plus a coding rate. Seeing BPSK next to 4096-QAM makes it plain why higher orders carry more bits per symbol and why they need a much cleaner signal to work.

How to use

1. Pick a modulation, from BPSK (1 bit per symbol) to 4096-QAM (12 bits per symbol).
2. Press Step to send one symbol, or Play to keep sending at the chosen speed. The simulator always opens paused. Use +100 to send a batch at once and Reset to clear the points and counts.
3. Move the SNR slider (0 to 45 dB) to add or remove noise. Received points spread around the ideal points, and a point that crosses a decision boundary is decided wrong and marked with an X.
4. Choose Random bits, or Your text to send a message as UTF-8 bits and read back what the receiver decoded.
5. Up to 16-QAM every point shows its bits. Above that, tap or hover a point to see them.
6. Read the waveform strip: the thin solid trace is I cos, the thin dashed trace is -Q sin, and the bold trace is their sum, the carrier actually sent. The bits for each symbol sit above it.
7. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses, the Right arrow steps, R resets, Up and Down move the SNR, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Modulation	BPSK, QPSK, 16, 64, 256, 1024 or 4096-QAM	7 choices
SNR	dB	0 to 45 in 0.5 dB steps
Speed	symbols per second	1, 4 or 20
Bits to send	random or typed text	text up to 120 characters

How it works. Square QAM with M points puts half the bits on I and half on Q. Each axis takes the odd levels -(sqrt(M)-1) to +(sqrt(M)-1) and is Gray coded, so neighboring points differ by one bit. BPSK uses I only, at -1 and +1. Every constellation is scaled by the 802.11 factor 1/sqrt(2(M-1)/3) so average symbol energy is 1. The carrier for a symbol is I cos(wt) - Q sin(wt), which equals A cos(wt + phase) with A = sqrt(I squared + Q squared) and phase = atan2(Q, I). Noise is Gaussian on each axis with standard deviation sqrt(1 / (2 x SNR)), SNR as a linear ratio. The receiver picks the nearest ideal point. With noise alone, RMS EVM is 1 / sqrt(SNR), so EVM in dB equals minus the SNR in dB; the measured value comes from the simulated points.

Example. 16-QAM at 25 dB SNR gives an EVM of -25 dB, about 5.6%. That meets both 16-QAM transmit limits (-16 and -19 dB) and the 64-QAM rate 3/4 limit of -25 dB, but misses 64-QAM rate 5/6 at -27 dB. Typing the letter A sends 8 bits; in 64-QAM that is 2 symbols, with 4 zero bits padding the second.

Field notes

- The drawing is slowed down. It shows 3 carrier cycles per symbol; a real Wi-Fi carrier runs billions of cycles per second.
- The required EVM table is the 802.11 transmitter accuracy requirement: how cleanly a radio must build each point. It is not a receiver sensitivity threshold. Meets or misses compares the limit with the EVM the current SNR produces.

- Changing the modulation, the SNR, the bit source or the message clears the received points and counts, so every reading belongs to one setting.
- Measured EVM from a handful of symbols wanders. Send a few hundred and it settles close to the theoretical value.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a group of bits picks one point on the I/Q plane, higher orders pack more bits into points that sit closer together, and so they need a cleaner signal to decode.

Control	What it does	Why it's there
Modulation (select: BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM, 4096-QAM, each with its bits per symbol)	Picks the constellation. Default 16-QAM. The line under it gives the point count, bits per symbol and the amendment that first used it.	Higher orders carry more bits per symbol and need a cleaner signal. Seeing BPSK next to 4096-QAM makes that plain (help, `whyHere`).
Play / **Pause**	Sends one symbol per tick at the chosen speed. The simulator always opens paused.	Paused by default so nothing moves until you ask. Play lets the cloud build up while you talk.
Step	Sends exactly one symbol.	One symbol is one point, one amplitude and one phase on the carrier strip. Step is how you show that mapping slowly, and it is the suggested path when reduced motion is on.
+100	Sends 100 symbols at once.	Measured EVM from a handful of symbols wanders, and a few hundred settle close to theory (help, `fieldNotes`). +100 gets you there in two presses.
Reset	Clears the received points and counts. Disabled when nothing has been sent.	Starts a fresh reading. Every other change clears the points too, so each reading belongs to one setting (help, `fieldNotes`).

Control	What it does	Why it's there
Signal-to-noise ratio (SNR) slider, 0 to 45 dB in 0.5 dB steps, default 25 dB	Sets the Gaussian noise added to each received point.	The one noise control: the received cloud spreads, points cross decision boundaries and are marked wrong.
Speed (symbols per second): 1/s, 4/s, 20/s. Default 4/s.	Sets how fast Play sends.	1/s is slow enough to narrate each symbol on the carrier strip; 20/s fills the constellation fast enough to see the cloud's shape.
Bits to send: Random, Your text	Random bits, or the UTF-8 bits of a typed message.	The tool calls for both. Typed text makes the errors land in something the class can read, and shows the zero-padding of the last symbol.
Message (text field, up to 120 characters, default "Hello, Wi-Fi"; appears with Your text)	The message to send. Shows bytes, bits, symbols and padding, then Progress, Received so far and Last full copy.	Lets the room watch a readable message break as the SNR drops.
Tap or hover a point on the constellation (64-QAM and above)	Shows that point's bits.	Up to 16-QAM every point is labeled. Above that the labels would overlap, so labels show on hover or tap only.
Present (title bar)	Opens the presenter layout over the same state.	See the shelf intro.

A demo that works (about 3 minutes)

1. Set up: 16-QAM, SNR 25 dB, Random bits. Press +100 twice. The cloud sits tight on each point, EVM measured is close to -25 dB, and there are no symbol errors.
2. Ask the room: "Same SNR, but I switch to 256-QAM. How many errors?"
3. Reveal: switch to 256-QAM and press +100. Errors appear at once, because the same noise now crosses boundaries that are four times closer on each axis. Press Up and watch the errors thin out; by our arithmetic they become rare somewhere past 30 dB.

4. Optional follow-up: switch to Your text, drop the SNR until "Hello, Wi-Fi" arrives garbled, then open the EVM card. 16-QAM at 25 dB meets both 16-QAM limits and the 64-QAM rate 3/4 limit of -25 dB, and misses 64-QAM rate 5/6 at -27 dB (help, `example`).

Pairs with: Rate vs Range (where each MCS stops working), the MCS Index card in Course Handouts, and Fourier and FFT mode 4, which puts these same constellations on every subcarrier.

Left out on purpose: no coding (the coding rates appear only in the EVM limits table), no OFDM (one carrier, one symbol at a time), and only white Gaussian noise: no phase noise, fading or interference. The carrier strip draws 3 cycles per symbol; a real carrier runs billions per second. EVM limits are transmitter accuracy, not receiver sensitivity.

Fourier and FFT

Shows that any signal is a sum of sines, and how a spectrum analyzer and a Wi-Fi radio use the FFT. Waves mode builds a signal from up to five sines and draws it over time and as a spectrum. FFT mode samples that signal, applies a window and runs an FFT, with the bin spacing, capture time, ENBW and RBW worked out on screen. Swept vs FFT race puts a swept analyzer and a real-time FFT analyzer on the same simulated 2.4 GHz band and counts the Bluetooth hops and microwave pulses each one catches. OFDM is an inverse FFT builds a Wi-Fi symbol from subcarriers and shows the receiver's FFT getting every constellation point back.

Why it's here. A Wi-Fi receiver runs an FFT on every symbol, and many spectrum analyzers compute their display the same way. Knowing what the bins, the window and the FFT size do explains why a weak signal can hide next to a strong one, why a peak can read a few dB low, why a swept analyzer misses hopping and bursty interferers that a real-time analyzer shows, and why legacy subcarriers are exactly 312.5 kHz apart.

How to use

1. In Waves, move a sine's frequency, amplitude and phase sliders and watch the time trace and the spectrum change together. Add up to five sines, or start from a preset: one sine, a square wave built from odd harmonics, two close tones, or a tone with a weak neighbor 40 dB down. Play sound to hear the sum.
2. In FFT, the same signal is sampled at Fs with N points. Pick the window, the sample rate and N, or use Halve N and Double N, and read the bin spacing, capture time, ENBW, RBW and the level of the strongest bin against the true level.

3. Pick a lesson to set everything up: a weak neighbor that the rectangular window hides and Blackman-Harris reveals, a tone between bins that reads low until you switch to flat top, and two close tones that merge when you halve N.
4. In Swept vs FFT race, pick the span, the RBW and the run length and read the sweep time. Both waterfalls show the same simulated scene with time running up; the swept analyzer only sees the cells along its lime diagonal. Compare how many Bluetooth hops and microwave pulses each analyzer caught. Run the race replays the run slowed down. The Scene settings change the mains frequency, how far the microwave oven sweeps and how wide the video camera is, and New scene draws new hops.
5. In OFDM is an inverse FFT, tap subcarriers on and off and pick the modulation. Watch the time-domain symbol with its cyclic prefix, each subcarrier's sinc with zeros at every other subcarrier's center, and the constellation the receiver recovers. Switch Legacy and HE to see the subcarriers move 4 times closer together and the symbol last 4 times longer, and switch to Real Wi-Fi for a full 20 MHz channel.
6. On a computer or tablet, Present opens this tool full screen for a projector: 1 to 4 switch modes, Space plays the sound or runs and pauses the race, the Right arrow steps the race or the OFDM highlight, R resets the mode, Up and Down move each mode's main setting, W changes the FFT window, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Sines	count	1 to 5
Frequency	Hz	100 to 5,000 in 10 Hz steps
Amplitude	linear, 1.00 = 0 dB full scale	0 to 1.00
Phase	degrees	-180 to 180
Sample rate (Fs)	kHz	12.8, 25.6, 51.2 or 102.4
FFT size (N)	points	64 to 4,096, powers of two
Window	type	Rectangular, Hann, Hamming, Blackman-Harris (4-term), Flat top

Input	Unit	Range
Span (race)	MHz	20, 50 or 100, centered on 2450
RBW (race)	Hz	3 MHz, 1 MHz, 300 kHz, 100 kHz, 30 kHz or 10 kHz
Run length (race)	ms	50, 200 or 1,000
Mains frequency (race)	Hz	60 or 50
Microwave sweep width (race)	MHz	near CW, 10, 20 or 40
Video camera bandwidth (race)	MHz	4, 8 or 18
Subcarriers (OFDM)	index	-8 to +7 in the teaching view; 52 (legacy) or 242 (HE) in Real Wi-Fi
Numerology (OFDM)	type	Legacy 312.5 kHz, 3.2 µs, GI 0.8 µs; HE 78.125 kHz, 12.8 µs, GI 0.8, 1.6 or 3.2 µs
Modulation (OFDM)	type	BPSK, QPSK, 16-QAM, 64-QAM, 256-QAM, 1024-QAM

How it works. The signal is x(t) = sum of A cos(2 pi f t + phase). FFT mode takes N samples x[n] = x(n / Fs), multiplies them by the window w[n], and computes the DFT X[k] = sum of x[n] w[n] e^(-j 2 pi k n / N) with a radix-2 FFT. Bin spacing = Fs / N and capture time = N / Fs, so their product is always 1. ENBW is computed from the window itself as N x sum(w^2) / (sum w)^2, in bins, and RBW = ENBW x Fs / N. Levels are dB full scale, scaled so a sine of amplitude 1 exactly on a bin reads 0 dB with any window. The highest-sidelobe figure shown for each window is the published one, with the value measured from this window at the current N beside it. Swept vs FFT race: the swept analyzer's sweep time is ST = k x Span / RBW^2 with k = 2.5 (the classic analog figure is 2 to 3). Its tuned frequency climbs across the span once per sweep, and it catches a signal only if, at some instant, the signal is within half an RBW of where it is tuned; the tool solves that exactly for every hop and pulse. The FFT analyzer is ideal and gapless: it transforms the whole span every frame of about 1/RBW, so it catches everything at least one frame long. OFDM is an inverse FFT: each active subcarrier k carries a constellation point X_k, and the symbol is s(t) = sum of X_k e^(j 2 pi k df t) for one useful symbol time T = 1/df. The transmitter makes N samples of it with an N-point inverse FFT, copies the last GI to the front as the cyclic prefix, and

the receiver drops the prefix and runs an N-point FFT to get X_k back. Each subcarrier's spectrum over T is a sinc, sin(pi x) / (pi x) with x = (f - k df) / df, which is zero at every other subcarrier's center.

Example. At Fs = 25.6 kHz and N = 256 the bins are 100 Hz apart and the capture lasts 10 ms. A Hann window has an ENBW of 1.50 bins, so the RBW is 150 Hz. A 20 MHz Wi-Fi receiver sampling at 20 MS/s with N = 64 gets bins exactly 312.5 kHz apart and a 3.2 µs frame; HE uses N = 256 for 78.125 kHz and 12.8 µs. With k = 2.5 and a 100 MHz span, a swept analyzer takes 0.25 ms per sweep at 1 MHz RBW, 25 ms at 100 kHz and 2.5 s at 10 kHz. At 100 kHz over 200 ms, the swept analyzer in the default scene catches 14 of 320 Bluetooth hops and the FFT analyzer catches all 320.

Field notes

- The windows use the periodic form used for spectral analysis. That is why Hann's ENBW comes out exactly 1.50 bins.
- Flat top's published highest sidelobe is -88 dB. The flat top used here measures about -88 dB at N = 64 and about -93 dB from N = 256 up, and the screen shows both.
- Frequencies are real audio frequencies, so the sound plays what the plots show. The sound does not follow the phase sliders.
- Every tone stays below half the lowest sample rate, so nothing aliases in this tool.
- The race scene is simulated from published timings, not captured. The microwave oven is on for half of each mains cycle (8.3 of 16.7 ms at 60 Hz, 10 of 20 ms at 50 Hz); sources disagree on how far it sweeps, so that is a setting. Bluetooth hops are drawn uniformly at random; the real sequence is pseudo-random. Wi-Fi uses 1500-byte frames at 54 Mbps (244 µs) with DIFS and a random backoff between them.
- Real analyzers with digital RBW filters, or FFT-assisted sweeps, sweep faster than the formula. A real FFT analyzer also has a real-time bandwidth limit; the one here covers the whole span with no gaps.
- The waterfalls use the analyzer color scale from dark blue at the noise floor (-95 dBm) through green, yellow and red to white at -30 dBm, with the scale drawn under them.
- In the OFDM teaching view the same 16 subcarriers are kept when you switch to HE, so they span a quarter of the width. Real HE fills the same 20 MHz channel with 4 times as many subcarriers. The carrier frequency never changes.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: any signal is a sum of sines, an analyzer computes its spectrum from a finite chunk of samples so its settings change what you see, and a Wi-Fi receiver is itself an FFT analyzer whose bins are the subcarriers.

Control	What it does	Why it's there
Mode (select): Waves, FFT, Swept vs FFT race, OFDM is an inverse FFT	Switches between the four modes. Default Waves.	The lesson builds in that order: signal, measurement, analyzer, Wi-Fi. The selector became a dropdown at four modes.
Start from (select): One sine, Square wave (odd harmonics), Two close tones, Tone plus weak neighbor. Your own mix appears once you edit.	Loads a preset signal.	The presets. The square wave shows a shape built from sines; the last two set up the FFT lessons.
Show each sine behind the sum (switch)	Draws each component under the summed trace.	Lets the class see the sum is only its parts.
Sine 1 to **Sine 5** chips, **Add a sine**, remove (per sine)	Picks the sine you are editing, adds one (up to five), removes one.	"Add up to five sines; each gets a row with the three knobs."
Frequency slider, 100 to 5,000 Hz in 10 Hz steps	Frequency of the picked sine.	A sine has three knobs. The time trace and the spectrum move together.
Amplitude slider, 0 to 1.00 (1.00 = 0 dB full scale)	Amplitude of the picked sine.	Same, the second knob.
Phase slider, -180 to 180 degrees	Phase of the picked sine.	Same, the third knob. The time trace moves and the magnitude spectrum does not, which is a point worth making aloud.
Play sound / **Stop sound**	Plays the sum through the speaker.	The tool allows optional sound at audio frequencies. The frequencies are real, so the sound is what the plots show. It does not follow the phase sliders.

Control	What it does	Why it's there
Lesson (select): Free play, Weak neighbor 40 dB down, Tone between bins, Halve N	Sets the signal and analyzer for one lesson.	The three lessons: rectangular leakage hides the weak tone and Blackman-Harris reveals it; a tone between bins reads low until flat top; two tones merge when you halve N.
Window (select): Rectangular, Hann, Hamming, Blackman-Harris (4-term), Flat top. Default Hann.	Applies the window before the FFT.	The window trades a clean floor against how sharp each peak is.
Sample rate (Fs) (select): 12.8, 25.6, 51.2, 102.4 kHz. Default 25.6 kHz.	Sets Fs.	Bin spacing is Fs / N and capture time N / Fs, so their product is always 1 (help).
FFT size (N) (select) 64 to 4,096, powers of two, default 256; **Halve N**, **Double N**	Sets N.	More samples give finer bins and a longer capture. The two buttons make it a one-press comparison.
Disclosures: **ENBW, RBW, sidelobes, scalloping** and **The Wi-Fi receiver is an FFT analyzer**	Extra readouts and the closing card.	The "Wi-Fi bridge": 20 MS/s with N = 64 gives bins exactly 312.5 kHz apart and a 3.2 µs frame.
Span (select): 20, 50, 100 MHz, centered on 2450 MHz. Default 100.	Width the swept analyzer covers.	Sweep time is k x Span / RBW squared (Keysight AN 150).
RBW (select): 3 MHz, 1 MHz, 300 kHz, 100 kHz, 30 kHz, 10 kHz. Default 100 kHz.	Resolution bandwidth.	The one knob that makes the swept analyzer slow: at 100 MHz, 0.25 ms at 1 MHz RBW, 25 ms at 100 kHz, 2.5 s at 10 kHz.
Run length (select): 50, 200, 1,000 ms. Default 200.	How much time the race covers.	Longer runs give more hops and pulses to count, so the caught/missed contrast holds up.
Run the race / **Pause** / **Resume**, **Step**	Replays the run slowed down, or steps it one fortieth at a time.	The tool calls this mode "the highest-value lesson". Replaying it slowly lets the room watch the lime diagonal miss hops.

Control	What it does	Why it's there
Scene: mains, oven sweep, camera disclosure: **Mains frequency** (60 Hz, 50 Hz), **Microwave sweep width** (near CW, 10, 20, 40 MHz; default 20), **Video camera bandwidth** (4, 8, 18 MHz; default 8), **New scene**	Changes the synthetic scene and draws new hops.	Sources conflict on how far an oven sweeps, and the camera width is not pinned, so both are settings. 50 Hz is for classes outside North America.
View: Teaching (16 subcarriers), Real Wi-Fi (20 MHz)	Small N for visibility, then the full channel.	A small N such as 16 to start, then real Wi-Fi.
Subcarrier chips (-8 to +7), **All on**, **All off** (Teaching view)	Turns subcarriers on and off. Default 1, 2 and 3 on.	The student builds the symbol one subcarrier at a time.
Modulation on every subcarrier (select): BPSK to 1024-QAM. Default QPSK.	The constellation each subcarrier carries.	Reuses the Modulation Simulator's math, so the two tools agree.
New data	Picks new random points for every subcarrier.	Shows the recovered constellation matches whatever was sent, not one lucky pattern.
Highlighted subcarrier slider	Picks the subcarrier whose sinc is drawn bold.	The orthogonality lesson: every sinc is zero at every other subcarrier's center.
Numerology: Legacy, HE; **Guard interval** (select, HE: 0.8, 1.6, 3.2 µs)	Switches legacy 312.5 kHz / 3.2 µs to HE 78.125 kHz / 12.8 µs.	The symbol stretches four times, and the screen must say plainly that the carrier frequency does not change.
Same time scale for legacy and HE (switch, under **Time scale and the receiver**)	Keeps both numerologies on one time axis. Default on.	So the 4x longer HE symbol looks 4x longer instead of being rescaled to fit.

A demo that works (about 3 minutes)

1. Set up: FFT mode, Lesson "Weak neighbor 40 dB down". The rectangular window is on and the spectrum shows one tone.

2. Ask the room: "There are two tones in this signal. Where is the second one?"
3. Reveal: press W (or pick Blackman-Harris). The leakage skirt drops and the weak neighbor appears 40 dB down.
4. Optional follow-up: press 3 for the race at its defaults (100 MHz span, 100 kHz RBW, 25 ms sweep). Ask which analyzer will catch more Bluetooth hops, then Space to run it and read the Caught and Missed counts.

Pairs with: the Spectrum Analysis guided lesson (the race reuses its interferer fingerprint wording), Modulation Simulator for mode 4, and PHY Preamble Reference for where the symbols sit in a PPDU. The Teacher's Guide lists a Fourier and FFT lab under "Spectrum analysis". It is not built yet.

Left out on purpose: nothing aliases, because every tone stays below half the lowest sample rate. The race scene is synthetic, from published timings: Bluetooth hops are uniform random (the real sequence is pseudo-random), and the FFT analyzer is ideal and gapless, where a real one has a real-time bandwidth limit. Real swept analyzers with digital RBW filters sweep faster than the formula. In the OFDM teaching view the same 16 subcarriers are kept when you switch to HE, so they span a quarter of the width. Real HE fills the channel with four times as many.

OFDMA Resource Units

Shows how 802.11ax OFDMA splits one channel into resource units (RUs) so an access point can talk to several clients in one transmission. You place clients into RUs across the channel, then compare the airtime of one frame per client sent as separate single-user transmissions with one downlink OFDMA and one uplink OFDMA transmission, all drawn to the same microsecond scale.

Why it's here. OFDMA is often sold as a faster PHY. It is not. For small frames the win comes from paying for contention, the preamble and the acknowledgment once for everyone instead of once per client. Seeing the timelines side by side shows where the time goes, and why one client with a large frame gains nothing.

How to use

1. Read the channel strip. Each block is one client's RU, with its letter and color. Hatched slots are center 26-tone RUs, which only a 26-tone RU can use.

2. Move a client by dragging it along the channel, or tap it and then tap one of the dashed outlines: those are the free spots the tone plan offers for its RU size. With a keyboard, focus a client and use the arrow keys.
3. Set the channel width, the number of clients, the frame size and the MCS. Under Resource units, pick a client (or All clients) and its RU size. Largest equal RUs gives every client the biggest RU that fits; Re-pack moves them all to the low edge.
4. Read the three timelines. SU is one full transmit opportunity per client. DL OFDMA is one transmission from the access point. UL OFDMA starts with a trigger frame and ends with one Multi-STA BlockAck. Each client's color follows its data.
5. Direction picks which OFDMA timeline the readouts compare with SU. The readouts give each total, the SU / OFDMA ratio, and a line naming what was saved: contention, preambles, or triggers, acks and SIFS.
6. Tap Show the arithmetic for every segment's formula. Values that are estimates carry an Assumption tag, and the Assumptions card lists them all.
7. On a computer or tablet, Present opens this tool full screen for a projector: the Right arrow selects the next client, Up and Down change the number of clients, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Direction	-	Downlink (AP to clients) or Uplink (clients to AP)
Channel width	MHz	20, 40, 80 or 160
Clients	count	1 to 9 at 20 MHz; up to 18 at 40 MHz and wider
Frame size per client	bytes	64 to 1,500; default 200
MCS	HE MCS	0 to 11, the same for every client; MCS 10 and 11 need a 242-tone RU or larger
RU size	tones	26, 52, 106, 242, 484, 996 or 2x996, where the channel is wide enough

How it works. RUs per width: 26-tone 9/18/37/74, 52-tone 4/8/16/32, 106-tone 2/4/8/16, 242-tone 1/2/4/8, 484-tone -/1/2/4, 996-tone -/-/1/2, 2x996 only at 160 MHz; data tones 24, 48, 102, 234, 468, 980, 1960. SU: each client gets one HE single-user TXOP from the Airtime Anatomy model (AIFS 43 + average backoff 67.5 + preamble + data + SIFS 16 + a 32-byte Block Ack at 24 Mbps). DL OFDMA: one AIFS and backoff, an HE MU preamble (20 + RL-SIG 4 + HE-SIG-A 8 + HE-SIG-B + HE-STF 4 + HE-LTF 7.2), data set by the slowest RU at 13.6 µs per symbol, SIFS, then every client's Block Ack together in one HE TB PPDU. UL OFDMA: one AIFS and backoff, a Basic trigger frame (16 + 8 + 6 per user + 4 bytes at 24 Mbps), SIFS, one HE TB PPDU (preamble 48 µs with an 8 µs HE-STF, data at 14.4 µs per symbol), SIFS, and a Multi-STA BlockAck. Data bits per symbol on an RU = data tones x bits per subcarrier x coding rate, rounded down; symbols = ceil((16 + 8 x subframe bytes) / that).

Example. Four clients, one 200-byte frame each, 20 MHz, MCS 7, one stream. Sent one at a time, each TXOP is 228.9 µs, so four take 915.6 µs. With four 52-tone RUs, DL OFDMA takes 403.3 µs and UL OFDMA 408.1 µs: about 2.2 times less airtime. The saving is three fewer contentions, three fewer preambles and fewer acknowledgments; the data itself takes 13.6 µs longer on the narrower RUs.

Field notes

- A teaching estimate: one frame per client, no collisions or retries, the average backoff, best-effort access, one spatial stream, control frames at 24 Mbps.
- Estimates, each labeled on screen: the HE-SIG-B length (computed from assumed field sizes), the rate of the downlink block acks, the trigger frame size, the Multi-STA BlockAck size, and the Block Ack after each single-user frame.
- The channel strip draws RUs in tone-plan order with every 26-tone slot the same width. Guard, DC and leftover tones are not drawn, and positions are logical, not exact subcarrier numbers.
- Real access points also schedule, send buffer status reports and sound the channel. None of that is modeled here.
- With one client on a full-channel RU, OFDMA is slightly slower than single user: the longer preamble and the triggered acknowledgment cost time and there is no one to share them with.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: OFDMA splits one channel into resource units so the AP serves several clients in one transmission, and the saving for small frames comes from paying for contention, the preamble and the acknowledgment once, not from a faster PHY.

Control	What it does	Why it's there
Direction: Downlink, Uplink. Default Downlink.	Picks which OFDMA timeline the readouts compare with SU.	DL OFDMA and UL OFDMA cost different overhead (UL adds a trigger frame and a Multi-STA BlockAck).
Channel width (select): 20, 40, 80, 160 MHz. Default 20.	Sets how many RUs fit.	The RU table: counts per width.
Clients (select): 1 to 9 at 20 MHz, up to 18 at 40 MHz and wider. Default 4.	Number of clients, each with one frame.	Comparing clients needs several; the single-client case needs one.
Frame size per client (select): 64, 100, 200, 500, 1,000, 1,500 bytes. Default 200.	The frame each client gets.	Small frames are where OFDMA wins; large frames to one client gain little.
MCS (every client) (select): MCS 0 to 11 with modulation and coding rate. Default MCS 7.	One MCS for every client.	The defaults reproduce the worked example in the help. MCS 10 and 11 need a 242-tone RU or larger.
Client (select: All clients, or Client A, B ...) and **RU size** (select: 26, 52, 106, 242, 484, 996, 2x996 tones, with data + pilot and how many fit). Default four 52-tone RUs.	Sets one client's RU, or every client's.	RU size per client, with the tone plan enforcing valid placements.
Largest equal RUs	Gives every client the biggest RU that fits.	A one-press fair split, so the room is not placing RUs by hand.
Re-pack	Moves every RU to the low edge.	Tidies the strip after drags.
Move left, **Move right**	Moves the selected client's RU along the channel.	The keyboard and button route for the RU puzzle.

Control	What it does	Why it's there
Drag a client on the channel strip, or tap it then tap a dashed outline	Places an RU. The dashed outlines are the free spots the tone plan allows.	"The user drags clients into RUs across the channel; the tone plan enforces valid placements".
Show the arithmetic / **Hide the arithmetic**	Shows every segment's formula. Estimates carry an Assumption tag.	Every INF estimate is labeled on screen, never shown as a standard value.

A demo that works (about 3 minutes)

1. Set up: the defaults. Four clients, 200 bytes each, 20 MHz, MCS 7, four 52-tone RUs.
2. Ask the room: "OFDMA is sold as faster. Is the data itself going faster?"
3. Reveal: read the timelines. SU takes 915.6 µs for four TXOPs; DL OFDMA takes 403.3 µs, about 2.2 times less. The data takes 13.6 µs *longer* on the narrower RUs. The saving is three fewer contentions, three fewer preambles and fewer acks (help, `example`).
4. Optional follow-up: press Down to 1 client, pick 1,500 bytes and a 242-tone RU. OFDMA comes out slightly slower than SU: the longer preamble and triggered ack cost time and there is no one to share them with.

Pairs with: Airtime Anatomy (the SU timing comes from its service), Medium Access Simulator (what contention costs), then Spatial Reuse and Multi-Link Operation in the "Wi-Fi 6 and 7 features" sequence. The Teacher's Guide lists an OFDMA lab there; it is not built yet.

Left out on purpose: one frame per client, no collisions or retries, the average backoff, best effort, one spatial stream, control frames at 24 Mbps. The HE-SIG-B length, the block-ack rates, the trigger frame size and the Multi-STA BlockAck size are estimates, each labeled. The strip is in tone-plan order with every 26-tone slot the same width, without guard, DC or leftover tones. Scheduling, buffer status reports and sounding are not modeled.

MIMO and Beamforming

Shows how many spatial streams an access point and a client can use, what the leftover antenna chains do instead, what beamforming costs in airtime, and why a packet capture taken nearby misses beamformed frames that the client received.

Why it's here. A 4x4 AP does not give a 2x2 phone four streams, and the two spare chains are not wasted either. Knowing which side has the spare chains, and what they add in each direction, explains real link rates. The capture lesson matters to anyone who troubleshoots from a laptop sniffer: the sniffer is not the device the AP is aiming at, so a pile of failed downlink frames in a capture can be a property of the capture, not of the network.

How to use

1. Set the AP and client chains. The streams readout is the smaller of the two, in both directions. Press Swap to flip the counts, for example from a 4x4 AP with a 2x2 client to a 2x2 AP with a 4x4 client.
2. Switch between Downlink and Uplink above the streams picture. Each stream is its own labeled lane, S1 to S4. Spare chains that help are drawn dashed into the lanes; idle ones stand alone. The table below the stage shows both directions side by side.
3. Turn AP transmit beamforming on or off. With it on, the beam pattern swings its main lobe toward the client. Drag the client or the sniffer in the pattern, or use the angle sliders, and read the sniffer's level against the client's.
4. Set the sniffer's receive chains. A sniffer with fewer chains than the frame's streams cannot separate them, and the readout says so.
5. Set the channel width and the sounding interval to see the sounding exchange drawn to scale and its share of airtime for one client.
6. On a computer or tablet, Present opens this simulator full screen for a projector: Up and Down steer the client, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
AP chains	antenna chains	1, 2, 3, 4 or 8; default 4

Input	Unit	Range
Client chains	antenna chains	1 to 4; default 2
Direction	Downlink / Uplink	default Downlink
AP transmit beamforming	On / Off	default On; not available with 1 AP chain
Client angle and sniffer angle	degrees from straight ahead of the AP	80 left to 80 right
Sniffer chains	receive chains	1 to 4; default 1
Channel width	MHz	20, 40, 80 or 160; default 80
Sounding interval	ms	2, 5, 10, 20, 50, 100 or 200; default 10

How it works. Spatial streams Nss = min(AP chains, client chains, streams the client advertises). Ideal transmit beamforming gain = 10 log10(transmit chains / Nss) dB and ideal receive combining gain = 10 log10(receive chains / Nss) dB; both are upper bounds. Beam pattern: a line of AP elements half a wavelength apart, steered at the client, with array factor AF = |sum over n of e^(j n (k d sin(a) + b))|, where a is the angle from straight ahead, k d = pi and the steering phase b = -k d sin(client angle). The sniffer's level is 20 log10(AF at the sniffer / AF at the client). Sounding: NDPA, SIFS 16 µs, NDP, SIFS, compressed beamforming report. The report carries the Givens angles for an Nr x Nc matrix, sum over i = 1 to min(Nc, Nr - 1) of 2 (Nr - i) angles, for every 4th subcarrier (estimated as the full-band RU tones / 4, plus one), at 6 bits per phi and 4 per psi, plus one byte of SNR per column and 35 bytes of headers. NDPA: 25 bytes at 6 Mbps, 20 µs + 4 µs per symbol. NDP: 36 µs of preamble + 8 µs per HE-LTF (1, 2, 4, 4, 6, 6, 8, 8 for 1 to 8 antennas) + 4 µs packet extension. Report: HE single user, 1 stream, MCS 4, 0.8 µs guard interval, 13.6 µs symbols. Share of airtime = exchange time / sounding interval.

Example. A 4x4 AP and a 2x2 client use 2 streams both ways. Downlink: the AP's 2 spare transmit chains give up to +3.0 dB of beamforming gain; with a 1-stream client that becomes up to +6.0 dB. Uplink: the AP's 2 spare receive chains give up to +3.0 dB of combining gain. At 80 MHz the estimated report is about 1.6 kB and the whole exchange about 275 µs, so sounding one client every 10 ms takes about 2.8% of the airtime. A 1-chain sniffer cannot decode any of the 2-stream frames.

Field notes

- The gains are ideal upper bounds. Real beamforming and combining gain is lower and changes with the room, the client's position and how fresh the last sounding is.
- The beam pattern is line of sight, one steered stream, from a straight line of elements. Indoors, reflections spread the energy, so real patterns are less clean, but the sniffer is still not the target.
- The client does not beamform in this model. Clients seldom do: in our capture, 93% of the client's frames were not beamformed.
- The sounding figures are estimates. They leave out the wait for the medium, assume one client, and assume a fixed rate for the report. Several beamformed clients multiply the cost.
- A wider channel makes the report bigger, but the report also travels over the wider channel, so in this estimate its airtime stays the same. More antennas and more streams are what make sounding cost more.
- Our measurement, not a model output: in one 802.11ax capture from a nearby sniffer, beamformed downlink frames failed their frame check 50.8% of the time against 5.1% for frames that were not beamformed, at matched signal strength (14,825 and 6,891 frames). The client received those frames; the sniffer was off the beam.
- A capture on the AP sees more than a capture on a laptop: the AP knows what it sent and is the intended receiver for the uplink.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: streams stop at the smaller side, the spare chains still add gain in each direction, beamforming costs sounding airtime, and a nearby sniffer is not the device the AP is aiming at, so it misses beamformed frames the client received fine.

Control	What it does	Why it's there
AP chains (select): 1, 2, 3, 4, 8. Default 4.	AP antenna chains.	A 4x4 AP and a 2x2 client talk at 2 streams. 8 is an optional large AP.
Client chains (select): 1 to 4. Default 2.	Client antenna chains.	Same. Clients stop at 4.

Control	What it does	Why it's there
Swap	Swaps the two counts.	The reverse case: flip to a 2x2 AP with a 4x4 client, and now the client has the spare chains.
Downlink / **Uplink** (on the stage, above the streams picture)	Shows one direction. Spare chains that help are drawn dashed into the lanes.	Spare AP chains give transmit beamforming gain downlink and receive combining gain uplink. The table under the stage shows both side by side.
AP transmit beamforming: On, Off. Default On. Disabled with 1 AP chain.	Steers the beam pattern's main lobe at the client.	"Beamforming on/off". A 1-chain AP cannot beamform.
Client angle and **Sniffer angle** sliders, 80 degrees left to 80 right (default client 20, sniffer -35); or drag either in the pattern	Moves the client and the sniffer around the AP.	The sniffer's level is the pattern value at its angle compared with the client's.
Sniffer chains (select): 1 to 4. Default 1.	The sniffer's receive chains.	A sniffer with fewer chains than the streams sent cannot separate them.
Channel width and sounding interval disclosure: **Channel width** (select: 20, 40, 80, 160 MHz; default 80) and **Sounding interval** slider (2, 5, 10, 20, 50, 100, 200 ms; default 10)	Draws the NDPA, NDP and report exchange to scale and its share of airtime. Disabled when beamforming is off.	Beamforming is not free. Width is there to show it barely matters here: the report grows, but it also travels over the wider channel (help).
Both directions side by side, **Sounding, in detail**, **Our measurement, in full** disclosures	The full tables and the capture figures.	The capture numbers are shown as a cited measurement, not a model output.

A demo that works (about 3 minutes)

1. Set up: the defaults (4x4 AP, 2x2 client, beamforming on, sniffer at 1 chain).
2. Ask the room: "Four antennas on the AP, two on the phone. How many streams, and are the other two AP antennas wasted?"

3. Reveal: 2 streams both ways. Downlink, the AP's 2 spare chains give up to +3.0 dB of beamforming gain; uplink, the same 2 chains give up to +3.0 dB of combining gain. Press Swap and ask who has the spare chains now.
4. Optional follow-up: press Up a few times to swing the beam toward the client and watch the sniffer's relative level fall. Point at "cannot decode" for the 1-chain sniffer, then open **Our measurement, in full**: beamformed downlink frames failed their FCS 50.8% of the time against 5.1% for non-beamformed frames at matched signal strength.

Pairs with: The Number on the Box vs the Number in Your Hand (the streams ceiling, as a rate), OFDMA Resource Units (sounding is one of the things it does not model).

Left out on purpose: the gains are ideal upper bounds. The beam pattern is line of sight, one steered stream, from a straight line of elements; indoors reflections make real patterns less clean. The client does not beamform (in our capture, 93% of the client's frames were not beamformed). Sounding figures are estimates for one client, without the wait for the medium and at a fixed report rate.

PHY Preamble Reference

Draws the preamble of every Wi-Fi PPDU format to scale in microseconds: Legacy (802.11a/g), HT mixed (802.11n), VHT (802.11ac), HE SU, ER SU, MU and TB (802.11ax), and EHT MU and TB (802.11be). Tap any block to open its bit table, with an evidence tag on every field. A second mode walks a receiver's decision from the first symbols after L-SIG, and a calculator shows the L-SIG LENGTH that makes a legacy radio defer for the whole PPDU.

Why it's here. Every PPDU since 802.11a starts with the same 20 µs legacy preamble (L-STF, L-LTF and L-SIG), so an older radio can hear any newer frame and stay quiet. What follows is where the PHYs differ, and it is where a capture's PHY details come from: the MCS, width, streams, BSS color and TXOP a receiver reads before the data. Seeing the bits, and how sure we are of each one, makes a packet capture easier to read.

How to use

1. Pick a PPDU type under Settings. The bar draws each field to scale: sand is the legacy preamble every PHY sends, blue is what this PHY adds, filled blocks carry bits and outlined blocks are training patterns. BPSK or QBPSK is marked under every SIG symbol.

2. Tap a block, or pick it in the list under the bar, to open its bit table: bit range, field name, width, meaning and evidence tag. Tags with a question mark (one source, inferred, not verified, or no tag in the brief) are not settled; treat them as leads.
3. Change the streams, the LTF size and guard interval (HE and EHT), the HE-SIG-B or EHT-SIG symbol count (HE MU and EHT MU) or the channel width (the VHT-SIG-B layout) to see what moves.
4. Switch Mode to Which PHY? and press Next step to walk the receiver's questions, or press Mystery to hide the PPDU's name, predict, then reveal.
5. In the L-SIG LENGTH calculator, enter the true PPDU duration to see the spoofed LENGTH, its value mod 3, and how long a legacy radio defers.
6. On a computer or tablet, Present opens this reference full screen for a projector: the Right arrow opens the next block (in Which PHY?, asks the next question), R closes it (or starts the walk again), Up and Down change the PPDU type, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
PPDU type	choice	Legacy, HT mixed, VHT, HE SU, HE ER SU, HE MU, HE TB, EHT MU, EHT TB; default HE SU
Spatial streams	streams	1 to 4 for HT, 1 to 8 for VHT, HE and EHT; default 1
LTF size and guard interval	choice	2x with a 0.8 or 1.6 µs GI, 4x with a 0.8 or 3.2 µs GI; HE and EHT only; default 2x, 0.8 µs
HE-SIG-B or EHT-SIG symbols	symbols	1 to 12, set by the AP; HE MU and EHT MU only; default 2
Channel width	MHz	20, 40, 80 or 160; changes only the VHT-SIG-B layout
TXTIME	µs	The true PPDU duration, at least the preamble and up to 5,484 µs (plus the signal extension); default 200
2.4 GHz signal extension	on or off	Adds 6 µs after the data at 2.4 GHz; default off

How it works. Legacy preamble: L-STF 8 µs, L-LTF 8 µs, L-SIG 4 µs. HT mixed adds HT-SIG 8 (two QBPSK symbols), HT-STF 4 and 4 µs per HT-LTF (1, 2, 4, 4 for 1 to 4 streams). VHT adds VHT-SIG-A 8 (BPSK, then QBPSK), VHT-STF 4, 4 µs per VHT-LTF and VHT-SIG-B 4, so its preamble is 36 + 4 x N_VHT-LTF µs. HE adds RL-SIG 4, HE-SIG-A 8 (16 in ER SU), HE-SIG-B 4 per symbol in MU, HE-STF 4 (8 in TB) and one HE-LTF per stream group of 6.4 or 12.8 µs plus the guard interval. EHT adds RL-SIG 4, U-SIG 8, EHT-SIG 4 per symbol in MU, EHT-STF 4 (8 in TB) and the EHT-LTFs. L-SIG LENGTH = ceil((TXTIME - signal extension - 20) / 4) x 3 - 3 - m, with m = 0 for HT, VHT and EHT, 1 for HE MU and ER SU, and 2 for HE SU and TB. A legacy radio at 6 Mb/s counts ceil((16 + 8 x LENGTH + 6) / 24) symbols of 4 µs after L-SIG, which covers the whole PPDU. The receiver's decision: QBPSK right after L-SIG means HT; a BPSK repeat of L-SIG means HE or EHT, split by LENGTH mod 3 (1 = HE SU or TB, 2 = HE MU or ER SU, 0 = EHT or later); otherwise a QBPSK second symbol means VHT, and anything else is non-HT.

Example. HE SU, 2 streams, 2x HE-LTF with a 0.8 µs guard interval: 20 + RL-SIG 4 + HE-SIG-A 8 + HE-STF 4 + 2 HE-LTFs x 7.2 = 50.4 µs. For a 200 µs HE SU PPDU, LENGTH = ceil((200 - 20) / 4) x 3 - 3 - 2 = 130, and 130 mod 3 = 1. A legacy radio counts ceil((16 + 8 x 130 + 6) / 24) = 45 symbols and defers 20 + 4 x 45 = 200 µs. The same 200 µs as HE MU gives LENGTH 131, and 131 mod 3 = 2.

Field notes

- Bit positions come from open-source driver and receiver code (Linux, ns-3, GNU Radio) and from secondary sources; the IEEE 802.11 standard itself was not read. Every field shows its tag: P (code or a document read directly), S2 (two independent sources), S1 (one source) or INF (derived).
- Not settled, and marked so on screen: HE TB HE-SIG-A (one vendor, and its second symbol is inferred), the HE MU reserved bit at HE-SIG-A2 B7, HE-SIG-B Beamformed at B14, EHT-SIG positions (one vendor), VHT-SIG-B at 40 and 80 MHz, the VHT-SIG-A2 CRC and Tail (no tag in the brief), LTF counts above 4 streams, and the EHT-LTF length.
- The BSS color moves with the HE PPDU type (HE-SIG-A1 B8-B13 in SU, B5-B10 in MU, B1-B6 in TB) and is fixed in EHT at U-SIG-1 B7-B12.
- The LTF size and guard interval pairs offered are the brief's EHT list; 1x is not offered. LTF counts follow the Airtime Anatomy table. Durations are per field and do not change with width. The packet extension and the 2.4 GHz signal extension come after the data and are not drawn.
- HE-SIG-B and EHT-SIG symbol counts depend on the users and the SIG MCS, so here they are a setting, not computed.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: every PPDU since 802.11a starts with the same 20 µs legacy preamble so older radios defer; what follows is where each PHY differs, and it is where a capture's PHY details come from.

Control	What it does	Why it's there
Mode: Explore, Which PHY?	Explore opens the fields; Which PHY? walks a receiver's decision.	Explore covers the formats and their fields; Which PHY? covers how a receiver tells them apart.
PPDU type (select): Legacy (non-HT), HT mixed (802.11n), VHT (802.11ac), HE SU, HE ER SU, HE MU, HE TB (802.11ax), EHT MU, EHT TB (802.11be). Default HE SU.	Draws that PPDU's preamble to scale in µs.	The preamble for each PHY as labeled blocks, sand for the legacy part, blue for what the PHY adds.
Blocks on the bar, and the field list under it; **Close**	Opens a block's bit table (bit range, field, width, meaning, evidence tag).	What is inside each SIG field, bit by bit, with the evidence level shown so single-source fields are never taught as settled.
Spatial streams (N_STS) slider: 1 for Legacy, 1 to 4 for HT, 1 to 8 for VHT, HE and EHT. Default 1.	Adds or removes LTFs.	The preamble grows with streams, one LTF per stream group.
LTF size and guard interval (select, HE and EHT only): 2x 0.8 µs GI, 2x 1.6 µs GI, 4x 0.8 µs GI, 4x 3.2 µs GI	Changes each HE or EHT LTF's length.	GI and LTF size for HE. 1x is not offered, because the EHT rate list does not include it (help).
HE-SIG-B / EHT-SIG symbols slider, 1 to 12, set by the AP (HE MU and EHT MU only). Default 2.	Lengthens the SIG-B or EHT-SIG field.	The count depends on the users and the SIG MCS, so it is a setting, not computed (help).
Channel width (select): 20, 40, 80, 160 MHz	Changes only the VHT-SIG-B layout.	Bandwidth "where it changes" the preamble. Durations per field do not change with width, and this shows that.

Control	What it does	Why it's there
Next step / **Walk complete**, **Back**, **Reveal**, **Reset** (Which PHY?)	Asks the receiver's next question, takes one back, shows every question and the verdict, starts over.	Walk the decision tree step by step (BPSK vs QBPSK, the RL-SIG repeat, LENGTH mod 3).
Mystery (Which PHY?)	Picks a PPDU at random and hides its name until the walk ends.	Built for predict, then reveal (help, `howToUse`).
L-SIG LENGTH calculator disclosure: **True PPDU duration, TXTIME (µs)** field (default 200), **2.4 GHz: 6 µs signal extension after the data** switch, **This preamble + 100 µs of data**	Shows the spoofed LENGTH, m, LENGTH mod 3, and how long a legacy radio counts and defers.	How L-SIG LENGTH is set so a legacy radio defers for the whole newer PPDU.
Readouts: L-SIG and BSS color and **Sources and evidence tags** disclosures	L-SIG RATE, LENGTH mod 3, where the BSS color sits for this type; the evidence key.	The BSS color moves with the HE PPDU type and is fixed in EHT (help).

A demo that works (about 3 minutes)

1. Set up: Which PHY?, then press **Mystery**.
2. Ask the room: "The first symbol after L-SIG is a BPSK copy of L-SIG. What have we narrowed it to?"
3. Reveal: press Right through the questions. An RL-SIG repeat means HE or EHT, and LENGTH mod 3 splits them: 1 is HE SU or TB, 2 is HE MU or ER SU, 0 is EHT (help, `algorithm`).
4. Optional follow-up: Explore, HE SU, 2 streams, 2x LTF with 0.8 µs GI: 50.4 µs of preamble. Open the L-SIG LENGTH calculator at 200 µs: LENGTH 130, mod 3 = 1, and a legacy radio defers exactly 200 µs.

Pairs with: Airtime Anatomy (the same preamble durations, in a whole TXOP) and A Frame's Journey (what comes after the preamble).

Left out on purpose: bit positions come from open-source driver and receiver code and secondary sources, not a reading of the standard, and every field carries its tag. Unsettled fields are marked on screen (HE TB HE-SIG-A, EHT-SIG positions, LTF counts above 4 streams and others

listed in the help). The packet extension and the 2.4 GHz signal extension come after the data and are not drawn. SIG-B symbol counts are a setting, not computed.

A Frame's Journey

Follows one frame across one hop, from the sending radio's NIC (network interface card) to the receiving one: the laptop's frame to the AP (access point), the same frame Down the Stack sends on its air hop. The bits become RF (radio frequency) waves, cross the distance and are decoded. A receiver that is capturing puts a radiotap header in front of the frame, which was never sent. The receiver checks the FCS (frame check sequence) and only then, after one SIFS (short interframe space), sends the ACK (acknowledgment). Flip one bit and the check fails, no ACK comes back, and the sender tries again.

Why it's here. An ACK is the sender's only evidence that a frame arrived. The receiver runs the same 32-bit CRC (cyclic redundancy check) the sender ran, and one wrong bit anywhere in the frame, even inside the FCS itself, makes the two disagree. A frame that fails the check is dropped without a reply, so the sender learns about the loss only from the ACK that never comes, and every retry costs airtime. Seeing the radiotap header added on the receiving side, and never on the air, also explains why two receivers capturing the same frame report different signal levels and times.

How to use

1. Press Play to follow the frame one step at a time, or use Back, Step, Reset and the Step slider. The top card says what happens at each step. The drawing shows the sending NIC, the air and the receiving NIC, with a timeline underneath: the frame, the SIFS gap and the ACK, not to scale.
2. Pick the band (2.4, 5 or 6 GHz) and the distance. Distance lowers the wave's height and the signal level; the wavelength drawn never changes, because the frequency does not.
3. Turn on Corrupt one bit on the first attempt and pick the bit with the slider; the tool names the field it falls in. Step through: the FCS check fails, no ACK comes back, and the retry, with the Retry bit set and a new FCS, gets through.
4. Turn Receiver is capturing off: nothing is added, and the frame on the air is the same either way. Change TCP or UDP and the data size to change the frame.

5. Predict, then reveal: Ask the class loads the question (one bit of the frame arrives wrong: what does the receiver send back?) with a bit corrupted; pick an answer, then press Reveal.
6. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses, the Right arrow steps, R resets, Up and Down move the receiver 1 m (about 3 ft) farther or closer, C corrupts a bit or sends the frame clean, B picks the next band, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Band	GHz	2.4, 5 (default) or 6; channel 6 at 2437 MHz, 36 at 5180 MHz, 37 at 6135 MHz
Distance	m or ft	1 to 30 m (about 3 to 98 ft); default 8 m
Corrupt one bit on the first attempt	switch	Off (default) or On
Which bit	bit number	0 to the frame's last bit; default 133, inside Address 3
Receiver is capturing (radiotap)	switch	On (default) or Off
Transport	choice	TCP, Transmission Control Protocol (default), or UDP, User Datagram Protocol
Data	bytes	100, 500 or 1460 (default)

How it works. The frame is built byte by byte: a 26-byte 802.11 QoS (quality of service) Data header with To DS 1 and From DS 0, LLC/SNAP (logical link control and subnetwork access protocol), the IPv4 header with its checksum, the TCP or UDP header and the data. The FCS is the CRC-32 of everything before it, the IEEE 802.3 polynomial that 802.11 uses (clause 9.2.4.8). The receiver recomputes the CRC over what arrived and compares it with the FCS that arrived: equal passes, anything else fails. It sends an ACK only for a frame that passes, back to the radio in the frame's Address 2 (clauses 10.3.2.2 and 10.3.2.11). SIFS is 16 µs at 5 and 6 GHz, and 10 µs at 2.4 GHz after the 6 µs signal extension that follows an OFDM (orthogonal frequency division multiplexing) frame there, so the gap is 16 µs (clauses 18.4.5 and 10.3.8). The retry sets the

Retry bit in Frame Control, which changes the CRC. Signal level = 20 dBm EIRP (effective isotropic radiated power, illustrative) minus the free-space path loss at the channel's center frequency. Travel time = distance / the speed of light.

Example. Defaults (5 GHz, 8 m, TCP, 1460 bytes): the frame is 1538 bytes, 12,304 bits. It arrives at -44.8 dBm after 26.7 ns, the FCS matches, and one SIFS of 16 µs later the AP sends its ACK to the laptop's MAC (media access control) address. The trip across takes 26.7 ns; the SIFS is 16,000 ns. With bit 133 corrupted, inside Address 3, the receiver's CRC differs from the FCS that arrived, no ACK is sent, and the retry gets through.

Field notes

- The FCS is a real CRC-32 over this frame's bytes, so the values shown are what a receiver computes for them. The Duration field, the sequence number, the MAC addresses and the data are illustrative.
- The radiotap timer value and the MCS (modulation and coding scheme) are illustrative; the channel and the signal level follow the band and the distance. Radiotap also exists on the sending side when a tool injects frames, and the driver removes it before the frame goes out.
- Not modeled: the preamble (see PHY Preamble), how bits are coded onto subcarriers, the ACK timeout value, the backoff before the retry, retry limits, block acknowledgment for aggregated frames, and encryption.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: one wrong bit anywhere in a frame fails the FCS, the receiver drops it without a reply, and the sender learns about the loss only from the ACK that never comes. Radiotap is added by the capturing receiver and never travels on the air.

Control	What it does	Why it's there
Play / **Pause**, **Back**, **Step**, **Reset**	Moves through the hop one step at a time. The top card says what happens at each step.	One step per event (bits to RF, the FCS check, SIFS, the ACK, the retry) so you can stop and ask at each one.
Step slider (1 of n)	Jumps to any step.	For going straight back to the FCS check.

Control	What it does	Why it's there
Ask the class, answer buttons **An ACK**, **A request to send it again**, **Nothing**, then **Reveal** (card: **Predict, then reveal**)	Loads the question "One bit of the frame arrives wrong. What does the receiver send back?" with a bit corrupted, records the room's pick, and jumps to just after the FCS check.	The predict prompt. "A request to send it again" is there because 802.11 has no negative acknowledgment to send.
Band: 2.4 GHz, 5 GHz, 6 GHz. Default 5 GHz.	Sets channel 6 (2437 MHz), 36 (5180 MHz) or 37 (6135 MHz).	Channel and signal level in the radiotap header follow the band. The SIFS differs at 2.4 GHz (10 µs plus a 6 µs signal extension) and is 16 µs at 5 and 6 GHz.
Distance slider, 1 to 30 m (3 to 98 ft), default 8 m (26 ft)	Moves the receiver. The wave's height and the signal level drop; the wavelength drawn never changes.	Shows distance changes amplitude, never frequency (help).
Corrupt one bit on the first attempt (switch) and **Which bit** slider, 0 to the frame's last bit, default bit 133	Flips one bit and names the field it lands in.	Any single bit fails the check, even one inside the FCS itself.
Receiver is capturing (radiotap) (switch), default on	Adds or removes the radiotap header on the receiving side.	Radiotap is a pseudo-header added by the capturing driver and not sent over the air. The on-air frame is the same either way.
Transport: TCP, UDP. Default TCP.	Changes the transport header.	Changes the frame's size and bytes, so the FCS changes with it.
Data (select): 100, 500, 1460 bytes. Default 1460.	Sets the payload.	Same, and 1460 bytes is a full TCP segment.
Model notes	What is real and what is illustrative.	Answers "is it really like that?"
Metric / imperial switch (title bar)	Changes the distance units.	

A demo that works (about 3 minutes)

1. Set up: the defaults (5 GHz, 8 m, TCP, 1460 bytes). Step through once, clean: the frame is 1538 bytes, it arrives at -44.8 dBm, the FCS matches, and 16 µs later the ACK goes back.
2. Ask the room: press **Ask the class**. "One bit arrives wrong. What does the receiver send back?" Take a show of hands for each answer and tap the winner.
3. Reveal: press Reveal. "Nothing." The receiver cannot trust any part of a frame that fails its FCS, not even the sender's address, so it has no one to reply to. Step on to the retry: the Retry bit is set, the FCS is new, and it gets through.
4. Optional follow-up: turn **Receiver is capturing** off and ask what changed on the air. Nothing did. Point out the trip across takes 26.7 ns and the SIFS is 16,000 ns.

Pairs with: Down the Stack (the same air frame, built by the same function, opened layer by layer), PHY Preamble Reference (what precedes the frame), and Airtime Anatomy (what each retry costs in airtime).

Left out on purpose: the FCS is a real CRC-32 over this frame's bytes, but the Duration, sequence number, MAC addresses and data are illustrative, as are the radiotap timer and MCS. Not modeled: the preamble, how bits are coded onto subcarriers, the ACK timeout, the backoff before the retry, retry limits, block acknowledgment for aggregated frames, and encryption. The timeline is not to scale.

The Number on the Box vs the Number in Your Hand

Shows why the speed class printed on a router box is not what one device gets. Three bars share one scale. The first is the number on the box of a BE19000-class router: 18,656 Mbps (megabits per second), every radio added at its maximum. The second is the best case for one client, a 2x2 phone by default: one link, two spatial streams. The third is an estimate for that client at a stated distance and channel width.

Why it's here. A router's class name adds up its 6 GHz, 5 GHz and 2.4 GHz radios, each at its widest channel, its top MCS (modulation and coding scheme, the data-rate step) and four spatial streams. A phone uses one of those links with its own two streams, so its best case is under a third of the box number, and only close to the router. Distance lowers it again. The stream count sets the ceiling: a 2x2 laptop and a 2x2 phone reach the same one.

How to use

1. Press Show the next step to walk from the box, to the best case for the client, to the estimate at a distance. Each new bar starts at the length of the bar above it and shrinks. Back a step goes the other way.
2. Choose the client: a 2x2 phone (the default), a 2x2 laptop, or a 4-stream reference client as big as the router. The phone and the laptop give the same bars. The 4-stream reference still reaches only the one fastest link.
3. Set the distance and the 6 GHz channel width for the third step. Far enough out, a narrower channel beats a wider one, because a wider channel needs a stronger signal for the same MCS.
4. On a computer or tablet, Present opens the tool full screen for a projector: Right shows the next step and Left the one before, Space jumps between the box and the hand, Up and Down change the client, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Client	streams	A 2x2 phone (default), a 2x2 laptop, or a 4-stream reference client
Step	step	On the box, best case, at a distance; default on the box
Distance from the router	m or ft	1 to 60 m (3 to 197 ft); default 5 m (illustrative)
Channel width, 6 GHz	MHz	20, 40, 80, 160 or 320; default 320, the best case's own width

How it works. Box = 11,520 + 5,760 + 1,376 = 18,656 Mbps, the published figures for a BE19000-class router's 6 GHz, 5 GHz and 2.4 GHz radios; the class name rounds the sum up to the next thousand. The app's own PHY (physical layer) rate math, the Throughput Calculator's data subcarriers x bits per symbol x streams / symbol time for 802.11be with the 0.8 microsecond guard interval, gives 11,529, 5,765 and 1,376 Mbps for four streams at 320, 160 and 40 MHz and MCS 13: the same figures to within 0.1%. Best case = that formula for the fastest radio (6 GHz, 320 MHz, MCS 13) with the client's streams, never more than the router's four. Step 3: received level = 20 dBm EIRP (effective isotropic radiated power) - path loss, with a 0 dBi client antenna (decibels over an isotropic antenna); path loss = free-space path loss at 1 m + 10 x 3 x log10(d) at 6135 MHz (6 GHz channel 37). The MCS is the highest whose 802.11be minimum receiver sensitivity at the chosen width the level meets (the Rate vs Range table). Estimate = the PHY rate at that MCS x 0.80, the Throughput Calculator factor for 802.11be.

Example. With the defaults, the box says 18,656 Mbps. A 2x2 phone's best case is 5,765 Mbps, 31% of the box. At 5 m on a 320 MHz channel the phone receives -49.2 dBm, which supports MCS 7: a PHY rate of 2,882 Mbps and an estimate of 2,306 Mbps, 12% of the box. At 20 m a 160 MHz channel (MCS 2, estimate 346 Mbps) beats a 320 MHz channel (MCS 0, estimate 231 Mbps).

Field notes

- The class number is a sum of radios, and no single device sees it. Read the per-band figures on the datasheet, then halve the fastest one for a 2x2 client.
- The best case assumes the client supports 320 MHz channels and 4096-QAM (quadrature amplitude modulation). If it supports less, its best case is lower.
- The 0.80 factor is the Throughput Calculator's estimate and is favorable, so a speed test usually reads lower. Other clients sharing the airtime, walls and interference are left out, and the client is assumed to keep all its streams at a distance, so step 3 is an upper limit for this client.
- The 5 m default distance is an illustrative choice. No measured home stands behind it.
- The tool also leaves out Multi-Link Operation, where a Wi-Fi 7 client and router use more than one band at once. The Multi-Link Operation simulator shows it.
- For the rate at every MCS, width and stream count, see the MCS Index and the Throughput Calculator. For how the rate falls with distance, see Rate vs Range.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a router's class number adds every radio at its maximum, one client uses one link with its own streams, and distance takes the rate lower again.

Control	What it does	Why it's there
Show the next step / **Back a step** (on the stage)	Walks the three bars: On the box, Best case for this client, At a stated distance and width. Each new bar starts at the length of the one above and shrinks.	The visible change: a bar shrinks from the box number to what that client can reach at best.
Client: phone and laptop are 2x2: Phone, Laptop, 4-stream. Default Phone.	Picks the client's stream count.	The one control. Throughput defaults to a 2x2 client (Apple and Google publish 2x2 for current phones). Phone and laptop give identical bars on purpose: the stream count sets the ceiling. The 4-stream reference still reaches only the one fastest link.
Distance from the router slider, 1 to 60 m (3 to 197 ft), default 5 m (16 ft)	Sets step 3's distance.	Step 3 of the lesson: the signal weakens, the MCS steps down, the rate follows. The 5 m default is illustrative, and the screen says so.
Channel width, 6 GHz (MHz): 20, 40, 80, 160, 320. Default 320.	Sets step 3's width.	A wider channel needs a stronger signal for the same MCS, so far enough out a narrower one wins (help).
Reset	Back to the box, the phone, 5 m and 320 MHz.	Starts the walk over.
Open Throughput Calculator, **Open MCS Index**, **Open Rate vs Range** (under **Why the numbers differ**)	Opens the tools whose math this one reuses.	Rates come from the Throughput Calculator and level and MCS from Rate vs Range, so the class can check every number there.
Metric / imperial switch (title bar)	Changes the distance units.	Metric first, imperial for US rooms.

A demo that works (about 3 minutes)

1. Set up: the defaults. The top bar reads 18,656 Mbps, the sum of the three radios on a BE19000-class box.

2. Ask the room: "My phone is Wi-Fi 7. How much of that does it get, standing next to the router?"
3. Reveal: Right. The best case for a 2x2 phone is 5,765 Mbps, 31% of the box. Right again: at 5 m on 320 MHz, MCS 7 and an estimate of 2,306 Mbps, 12% of the box (help, `example`).
4. Optional follow-up: slide to 20 m and compare 320 MHz with 160 MHz. The narrower channel wins (help: MCS 2 at 346 Mbps against MCS 0 at 231 Mbps). Then switch to Laptop and ask why nothing moved.

Pairs with: MIMO and Beamforming (why the smaller side sets the streams), Rate vs Range, the MCS Index card in Course Handouts, and Multi-Link Operation.

Left out on purpose: the best case assumes the client supports 320 MHz and 4096-QAM. The 0.80 factor is the Throughput Calculator's favorable estimate, so a speed test usually reads lower. Other clients sharing airtime, walls, interference and Multi-Link Operation are left out, and the client keeps all its streams at a distance, so step 3 is an upper limit. The router vendor is not named.

Airtime and Access (13)

Medium Access Simulator

Shows how Wi-Fi stations share one channel. Each station waits for a quiet medium, counts down a random backoff in 9 µs slots, and transmits when the count reaches zero. You watch it happen slot by slot: collisions, the contention window doubling, EDCA priority, hidden nodes and RTS/CTS.

Why it's here. Wi-Fi is half duplex and shared, so airtime runs out long before the PHY rate does. Watching the waiting and the collisions shows why a 54 Mbps link delivers about 31 Mbps to one station, less as stations are added, and why voice traffic gets through first.

How to use

1. Press Play to run the clock, or Step to move one 9 µs slot at a time. The simulation starts paused. Reset replays the same run from time zero.
2. Read the timeline. The top lane is the medium as the AP hears it, and each station has its own lane. Hatched blocks are DIFS, AIFS or EIFS waits, outlined cells are backoff slots with the count inside, filled gray cells are a frozen count, lime blocks are data frames, green blocks are ACKs and red blocks are frames lost to a collision. Zoom changes the time scale; while paused, drag the timeline sideways to look back.
3. Add up to 10 stations and set each one's offered load. Switch to EDCA to give each station an access category. Turn on Hidden node, then RTS/CTS, to see collisions nobody could hear coming and how the handshake removes most of them. Change the frame size and PHY rate to see airtime change.
4. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses, the Right arrow steps one slot, R resets, Up and Down change the speed, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Stations	count	1 to 10
Access category	Voice / Video / Best effort / Background	EDCA only; Legacy DCF gives every station DIFS and a contention window of 15 to 1023
Offered load	frames per second	always has a frame, or 50 to 2,000 frames/s with random arrivals
Frame size	bytes	100 to 2,304; default 1,500
PHY rate	Mbps	6 to 54 (legacy OFDM); default 54
Speed	air time per real second	0.1 ms to 200 ms

How it works. Timing is 802.11 OFDM at 5 GHz: slot 9 µs, SIFS 16 µs, DIFS = SIFS + 2 slots = 34 µs, AIFS = SIFS + AIFSN x slot. A station waits for the medium to be idle for AIFS, then counts down a backoff drawn at random from 0 to CW, one per idle slot. If the medium goes busy the count freezes, and it resumes later without a new draw. At zero the station transmits. A clean frame gets an ACK after SIFS and CW resets to

CWmin. No ACK means CW = 2(CW + 1) - 1, capped at CWmax, and a new draw; after 7 failed tries the frame is dropped. A station that heard a frame it could not decode waits EIFS (SIFS + DIFS + an ACK at 6 Mbps = 94 µs) in place of DIFS. Frame airtime = 20 µs + 4 µs x ceil((16 + 8 x bytes + 6) / (rate x 4)).

Example. A 1,500-byte frame at 54 Mbps takes 20 + 4 x ceil(12,022 / 216) = 244 µs, and its ACK at 24 Mbps takes 28 µs. One station alone spends 34 µs of DIFS, 67.5 µs of backoff on average, 244 µs of data, 16 µs of SIFS and 28 µs of ACK per frame, about 390 µs, so it delivers about 31 Mbps from a 54 Mbps rate. Each station you add brings collisions that eat into that.

Field notes

- Timing is legacy OFDM (802.11a/g). Newer PHYs (802.11n, ac, ax, be) have longer preambles and send many frames per access by aggregating them, so the numbers differ. The lesson about contention is the same.
- Only uplink is modeled: every station sends to the AP, and the AP answers only with ACK or CTS. Rate adaptation and the NAV reset rule are not modeled.
- RTS/CTS removes most hidden-node data collisions, not all of them. A hidden station whose own RTS overlaps the AP's CTS never hears that CTS, so it can still talk over the data frame that follows.
- Runs are seeded, so the same settings give the same run every time.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: Wi-Fi is half duplex and shared, so stations take turns by counting down a random backoff; collisions double the contention window, more stations mean more waiting, and EDCA lets voice win because it waits less.

Control	What it does	Why it's there
Play / Pause	Runs or freezes the simulated clock. The tool always opens paused at time zero.	It opens paused and keeps Step working (reduced motion), so the instructor decides when the clock starts.

Control	What it does	Why it's there
Step 1 slot	Advances one 9 µs slot.	Backoff counts down one slot at a time and freezes when the medium goes busy. Stepping lets the room call each count and catch the freeze and resume.
Reset	Replays the same run from time zero.	Runs are seeded, so the same settings give the same run. A demo you rehearsed plays the same collisions live.
Speed (air time per real second)	0.1, 0.5, 2, 20 or 200 ms per second; opens at 0.1 ms per second.	The slow end shows single slots; the fast end lets the throughput and collision numbers settle.
Zoom: Slots / Frames / Wide	Changes the timeline's time scale. While paused, drag the timeline sideways to look back.	Slots shows the countdown cells; Wide shows many frames so the pattern of turns is visible.
Add station / Remove	1 to 10 stations; opens with 3. Each station row also has its own remove button.	"More stations mean more waiting and more collisions" is a stated goal of the lesson.
Access category (per station)	Voice, Video, Best effort or Background. Only live in EDCA.	The EDCA table gives Voice AIFSN 2 and CW 3 to 7 against Best effort's AIFSN 3 and CW 15 to 1023: priority traffic wins because it waits less.
Offered load (per station)	Always has a frame, or 50, 200, 500, 1,000 or 2,000 frames/s with random arrivals.	Saturated stations show contention at its worst; light loads show how rarely it bites when the channel is quiet.
Per-station table	CW, backoff, delivered and dropped for every station.	The tool asks for current CW per station, so the class can watch 15 become 31, then 63, after each collision.
Channel access: Legacy DCF / EDCA	DCF gives everyone DIFS and CW 15 to 1023; EDCA applies each station's access category.	The tool builds a DCF mode for everyone as the baseline, then EDCA to show voice getting through first.
Hidden node	Stations A, C, E cannot hear B, D, F; everyone still reaches the AP. Needs two or more stations.	Shows collisions nobody could hear coming: the stations never freeze on each other, so frames overlap at the AP.

Control	What it does	Why it's there
RTS/CTS	Every exchange becomes RTS, CTS, data, ACK; anyone who hears the CTS sets its NAV.	The fix for the hidden node.
Frame size	100, 500, 1,000, 1,500 or 2,304 bytes; default 1,500.	The tool makes frame size and PHY rate user controls so the class sees airtime per frame change. The help says the same.
PHY rate	6 to 54 Mbps (legacy OFDM); default 54.	Same as frame size. The Frame card underneath shows the airtime formula with your numbers in it.

A demo that works (about 3 minutes)

1. Set up: the defaults. Three stations, Legacy DCF, 1,500 bytes at 54 Mbps, speed 0.1 ms per second.
2. Ask the room: "Two stations reach zero in the same slot. What happens to both frames, and what does each station do next?"
3. Reveal: press Right until a red collision block appears, then open the per-station table: both colliding stations' CW has gone from 15 to 31. Press Up a few times to speed up, Play, and read the collision rate and throughput. Add stations up to 10 and watch the collision rate climb and throughput fall. The help puts one station alone at about 31 Mbps delivered on a 54 Mbps link.
4. Optional follow-up: switch to EDCA, set station A to Voice and the rest to Best effort, and compare average access delay. Or turn on Hidden node, watch the collisions, then turn on RTS/CTS.

Pairs with: Voice Priority, End to End (runs on this engine), Airtime Anatomy, Channel Utilization Meter, Rate Adaptation. Teacher's Guide sequences "Why is the Wi-Fi slow?" and "QoS and contention".

Left out on purpose: timing is legacy OFDM (802.11a/g, 9 µs slots, 20 µs preamble); newer PHYs have longer preambles and aggregate frames, and the contention lesson is the same. Uplink only: the AP sends only ACK or CTS. No rate adaptation, no aggregation, no NAV reset rule. RTS/CTS removes most hidden-node collisions, not all: a hidden station whose RTS overlaps the AP's CTS never hears that CTS.

Voice Priority, End to End

Follows one voice packet toward a phone on Wi-Fi and shows which of the AP's four queues it waits in. The caller's app marks the packet EF (Expedited Forwarding), a DSCP (Differentiated Services Code Point) value. The access point (AP) turns the DSCP into an 802.11 user priority (UP), and the UP picks one of the four WMM (Wi-Fi Multimedia) queues, or access categories. The call gets the Voice queue only if EF survives every hop before the AP and the AP maps it to UP 6.

Why it's here. Turning on WMM does not make calls good, and a priority mark set on your own network does not carry across the internet. The AP chooses the queue from the mark it receives, so a tunnel that does not copy the mark outward, a provider that resets it, or an AP that ignores it all put the call in Best effort, behind whatever else is downloading. The Medium Access Simulator shows how the four queues contend for the air; this tool is about whether the mark gets there, and it reuses that simulator's engine for the wait on the Wi-Fi hop.

How to use

1. Read the top card: the queue the call waits in, and one sentence saying why.
2. Pick where the marking is lost: nowhere, at the AP mapping, at the tunnel, or at the internet provider. The path shows the mark each hop passes on, and the hop that loses it says Marking lost here. Press Send the packet, or Next hop, to move the packet toward the phone.
3. Read the AP's four queues and the Wait on the Wi-Fi hop bars. With the download running, the call in Best effort sits behind the download's frames and waits far longer than in Voice. Turn the download off and the queues come out within a tenth of a millisecond of each other.
4. Switch the AP mapping to Top three bits to see the older default: EF (46, binary 101110) becomes UP 5, the Video queue, even when nothing on the path loses the mark. Change how many download frames sit ahead in Best effort (illustrative).
5. Predict, then reveal: Ask the class loads the question (the provider resets the mark while a download runs: which queue does the call wait in?) and hides the answer on the stage; pick an answer, then press Reveal.
6. On a computer or tablet, Present opens this simulator full screen for a projector: Space sends the packet or pauses, the Right arrow moves it one hop, R resets, Up and Down move where the marking is lost, D starts or stops the download, M switches the AP mapping, P asks, reveals and closes the question, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Where the marking is lost	choice	Nowhere (default), at the AP mapping, at the tunnel, or at the internet provider
How the AP maps DSCP to user priority	choice	RFC 8325 table (default, recommended) or top three bits (older default)
A download is running	switch	On (default) or Off
Download frames ahead in Best effort	frames	0 to 256; default 64 (illustrative; used only when the call lands in Best effort while a download runs)

How it works. The packet leaves the caller's app marked EF (46). The tunnel gateway copies the mark to the outer header, or leaves the outer mark at 0 when the loss is at the tunnel, and then no later hop can see EF. The internet provider passes the mark, or resets it to 0 when the loss is there. The home router and switch pass it unchanged. The AP maps the DSCP it sees to a user priority: the RFC 8325 table sends EF to UP 6 and 0 to UP 0; the top-three-bits mapping sends EF to UP 5; when the loss is at the AP mapping, the AP sends every frame as UP 0. UP 6 and 7 are Voice (AC_VO), 4 and 5 Video (AC_VI), 0 and 3 Best effort (AC_BE), 1 and 2 Background (AC_BK). The wait on the Wi-Fi hop comes from the Medium Access Simulator's engine, unchanged: legacy OFDM timing, 1,500-byte frames at 54 Mbps and the default client EDCA (Enhanced Distributed Channel Access) settings, with a voice flow of one packet every 20 ms (illustrative) and, while the download runs, a best-effort flow that always has a frame. Each run is seeded and covers 2 s of air time. In Voice or Video the call has its own queue, so its wait is the engine's mean time from the head of its queue to the ACK. In Best effort the call shares the download's queue, so it waits for the frames ahead to leave, one best-effort frame time each, then for its own turn: wait = (frames ahead + 1) x the mean best-effort frame time.

Example. Defaults (nothing lost, RFC 8325 mapping, download running, 64 frames ahead): the call gets UP 6, Voice, and waits 0.48 ms on the Wi-Fi hop. With the loss at the internet provider, the AP sees 0 and the call lands in Best effort: 65 x 0.398 ms = 25.9 ms, 54 times as long. With nothing lost and the top-three-bits mapping, the call gets UP 5, Video, and waits 0.59 ms. With the download off, the three queues wait 0.30, 0.32 and 0.36 ms.

Field notes

- RFC 8325 notes that the common default of copying the top three DSCP bits puts EF in the Video queue, and recommends mapping EF to UP 6. The DSCP / QoS Markings card holds its full table.

- Whether a tunnel copies the inner mark to its outer header is set by the device and its configuration (RFC 4301), so on a real network it depends on the device. Wi-Fi Calling and VPN (virtual private network) traffic both travel in tunnels.
- A network you do not run may re-mark or ignore the DSCP at its boundary (RFC 2475). RFC 8325 calls resetting untrusted marks bleaching.
- The Wi-Fi hop uses the Medium Access Simulator's simplifications: legacy 54 Mbps timing, one frame size for every frame, no aggregation, and the default client EDCA settings; an AP's own settings can differ. Inside one AP the standard gives the higher queue the win when two of its queues finish counting down together; here the download and the call contend as two transmitters.
- Illustrative: a voice packet every 20 ms, and how many download frames sit ahead in Best effort, which depends on the AP's buffer. Wired hops add delay too, and the tool does not put a number on them.
- Not modeled: the call's upstream direction, where the phone picks the user priority itself, and how the AP maps the upstream frame back to a DSCP.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a call gets the Voice queue on Wi-Fi only if its EF marking survives every hop before the AP and the AP maps it to UP 6; turning on WMM alone does not do it.

Control	What it does	Why it's there
Send the packet / Pause	Moves the voice packet along the path toward the phone.	Lets the class watch the mark each hop passes on.
Next hop	Moves the packet one hop.	Stops the packet at the hop that says "Marking lost here" so you can talk about that device.
Reset	Back to the defaults, packet at the caller, no question showing.	Returns to the starting state between demos.

Control	What it does	Why it's there
Where the marking is lost: Nowhere / At the AP mapping / At the tunnel / At the internet provider	Picks the hop that resets EF to 0. A dropdown on a phone, where four segments truncate.	The one control that matters: the visible change is the call dropping from Voice to Best effort while a download runs. It corrects "turning on WMM makes my calls good" and "QoS works across the internet".
How the AP maps DSCP to user priority: RFC 8325 table (recommended) / Top three bits (older default)	RFC 8325 maps EF to UP 6 (Voice); top three bits turns EF (46) into UP 5 (Video).	RFC 8325 notes that the common default of copying the top three bits puts EF in the Video queue, and recommends UP 6. This shows the call landing in the wrong queue with nothing on the path at fault.
A download is running	On by default.	With the download off, the four queues come out within a tenth of a millisecond of each other (help). Priority only matters when something else is queued.
Download frames ahead in Best effort (illustrative)	0 to 256 in steps of 8; default 64. Live only when the call lands in Best effort with the download on; locked while the class predicts.	The queue depth depends on the AP's buffer, so the help labels it illustrative. The default is 64 frames × 0.398 ms = 25.9 ms against 0.48 ms in Voice.
Ask the class / Voice / Video / Best effort / Reveal / close	Loads the question (the provider resets the mark while a download runs), hides the answer, takes a guess, reveals.	The house "predict, then reveal" pattern, built into the tool.
Open Medium Access Simulator / Open DSCP / QoS Markings	Links to the related tools.	EDCA is not rebuilt here (the Medium Access Simulator shows it), and the DSCP / QoS card holds the RFC 8325 table.

A demo that works (about 3 minutes)

1. Set up: the defaults. Marking lost Nowhere, RFC 8325 mapping, download running.

2. Ask the room: press P. "The phone has WMM on and the app marks its voice EF. Your provider resets the mark to 0. While a download runs, which queue does the call wait in?" Take votes on Voice, Video and Best effort.
3. Reveal: press P again. Best effort, behind the download: about 25.9 ms of waiting at the default 64 frames, against 0.48 ms in Voice.
4. Optional follow-up: press P to close, set the loss back to Nowhere and press M. The call lands in Video with nothing on the path losing the mark. Then press D to stop the download and show the queues come out nearly equal.

Pairs with: Medium Access Simulator, the DSCP / QoS Markings card, Power Save (U-APSD and the voice-call traffic pattern).

Left out on purpose: the Wi-Fi hop uses the Medium Access Simulator's simplifications (legacy 54 Mbps timing, one frame size, no aggregation, default client EDCA settings). The download and the call contend as two transmitters, not two queues inside one AP. Whether a tunnel copies the mark outward depends on the device (RFC 4301). Wired hops add delay the tool does not number. The upstream direction, where the phone picks the UP itself, is not modeled. The voice packet every 20 ms and the download queue depth are illustrative.

Airtime Anatomy

Draws one transmit opportunity (TXOP) to scale, microsecond by microsecond: the wait (AIFS plus backoff), optional RTS/CTS, the preamble, the data, SIFS and the acknowledgment. Two scenarios stack on one axis so you can compare them, and the readouts show the PHY rate, total airtime, throughput and how much of the air carried data. A Structure view opens the PSDU (physical layer service data unit, the bytes the data symbols carry) and draws what is inside it: one MPDU (MAC protocol data unit), an A-MSDU, an A-MPDU, or an A-MPDU of A-MSDUs. Corrupt one MSDU (MAC service data unit, one packet of your data) and it shows what the receiver can report and what the sender has to send again.

Why it's here. A PHY rate is the speed of the data symbols only. Every frame also pays for waiting, a preamble and an acknowledgment, and those costs do not shrink as the rate goes up. Seeing them to scale shows why one 1,500-byte frame at 867 Mbps delivers about 56 Mbps, and why aggregation recovers airtime that a faster MCS cannot. Aggregation is not one thing. An A-MSDU (aggregate MAC service data unit) puts several packets under one MAC header and one FCS (frame check sequence), so one bad bit throws all of them away. An A-MPDU (aggregate MAC protocol data unit) gives every subframe its own header and FCS, so the Block Ack bitmap can name the one that failed and only that one is sent again.

How to use

1. Read the bars. Hatched blocks are waiting (AIFS and the average backoff), the gray block is the preamble, lime is the data symbols, the dashed box is SIFS and the outlined block is the ACK or Block Ack. Very short segments get a label under the bar with a leader line.
2. Tap or hover a segment, or pick a duration in the table below the bars, to see how long it is, its share of the TXOP, and the formula with your numbers in it.
3. Pick a preset or change the inputs. Compare with scenario B stacks a second TXOP on the same scale; Edit scenario chooses which one the inputs change. More settings holds encryption, access category, RTS/CTS, control frame rate and the HE packet extension.
4. Watch the Check line. It flags combinations the standard does not allow, such as VHT at 6 GHz or a VHT MCS, width and stream count with no whole number of data bits per symbol. It also enforces the standard's limits on size as well as duration: a PPDU over 5.484 ms, and an aggregate the Structure view marks over a maximum, such as an HT A-MPDU of 64 frames of 1,500 bytes (99,328 bytes, over HT's 65,535). The line names the size and the limit, and when a scenario breaks both the time limit and a size limit, it names both. A flagged scenario is not drawn.
5. Switch the view to Structure to open the PSDU of the scenario you are editing. Arrangement picks Single MPDU, A-MSDU, A-MPDU or A-MPDU of A-MSDUs; MSDUs per A-MSDU sets how many packets share one header (2, 3 or 4). The A-MPDU arrangement is the same aggregate the time view draws, and the line under the inputs says so, with its bytes and its time.
6. Turn on Corrupt one MSDU, tap a lime block, or use the arrow buttons to move the damage. Each box in the PSDU is one MPDU drawn to scale in bytes (lime is payload, gray is overhead), and the one that fails is outlined and marked with a cross. Under it are the parts of that MPDU, the Block Ack bitmap with a 0 for the failed subframe (or, for a lone MPDU, the ACK that never comes), what is resent in bytes and in time, the limits from the standard for this PHY, the bytes part by part, and all four arrangements side by side. Amber in the limits means the frame fits only a receiver that advertises a larger setting; red means it is over the maximum.
7. On a computer or tablet, Present opens this tool full screen for a projector: the Right arrow walks the TXOP one segment at a time (in the Structure view it moves the damage to the next MSDU), Up and Down change the rate of the scenario you are editing, V switches between Time and Structure, A steps through the arrangements, C corrupts one MSDU or repairs it, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Band	GHz	2.4, 5 or 6. 2.4 GHz uses a 10 µs SIFS plus a 6 µs signal extension; 6 GHz is HE only; VHT is 5 GHz only
PHY	802.11 generation	Legacy (a/g), HT (n), VHT (ac), HE (ax, single-user PPDU)
Channel width	MHz	20, 40, 80 or 160; Legacy is 20 MHz; HT is 20 or 40 MHz
MCS	per stream	0 to 11; HT 0 to 7, VHT 0 to 9, HE 0 to 11. Legacy uses a data rate of 6 to 54 Mbps instead
Spatial streams	count	1 to 8; HT up to 4
Guard interval	µs	HT and VHT 0.4 or 0.8; HE 0.8, 1.6 or 3.2
Payload per frame	bytes	64 to 2,304; default 1,500
Frames aggregated	A-MPDU subframes	1 to 64; Legacy sends 1
Encryption overhead	bytes	0 (open) or 16 (CCMP/GCMP)
Access category	EDCA	Voice, Video, Best effort, Background (default EDCA values for a client)
Control frame rate	Mbps	6, 12 or 24; used for RTS, CTS, ACK and Block Ack
HE packet extension	µs	0, 4, 8, 12 or 16; HE only
Arrangement	Structure view	Single MPDU, A-MSDU, A-MPDU (the time view's aggregate) or A-MPDU of A-MSDUs; Legacy offers only a single MPDU
MSDUs per A-MSDU	count	2, 3 or 4; a teaching choice, not a limit from the standard

Input	Unit	Range
Corrupt one MSDU	MSDU	Off, or any one MSDU in the PSDU

How it works. AIFS = SIFS + AIFSN x 9 µs slot. Average backoff = CWmin / 2 x 9 µs. Data bits per symbol N_DBPS = data subcarriers x bits per subcarrier x coding rate x streams (Legacy: rate x 4); HE rounds it down. Symbol time: Legacy 4 µs, HT/VHT 3.2 µs + GI, HE 12.8 µs + GI. MPDU = payload + 26-byte QoS header + 4-byte FCS + encryption; each A-MPDU subframe adds a 4-byte delimiter and pads to 4 bytes. VHT and HE always send an A-MPDU; Legacy and a single HT frame do not. One frame is acknowledged with an ACK, two or more with a Block Ack. Symbols: Legacy/HT/VHT ceil((16 + 8 x PSDU bytes + 6) / N_DBPS); HE ceil((16 + 8 x PSDU bytes) / N_DBPS). Preamble: Legacy 20 µs; HT 20 + 8 + 4 + 4 per HT-LTF; VHT 20 + 8 + 4 + 4 per VHT-LTF + 4; HE 20 + 4 + 8 + 4 + HE-LTFs x (6.4 + GI), or x 16 µs at a 3.2 µs GI (4x HE-LTF). Data: symbols x 4 µs, or with a 0.4 µs GI 4 x ceil(3.6 x symbols / 4); HE symbols x symbol time + packet extension; plus 6 µs at 2.4 GHz. Control frames: 20 + 4 x ceil((16 + 8 x bytes + 6) / (rate x 4)) µs with ACK 14, Block Ack 32, RTS 20 and CTS 14 bytes. Throughput = payload bits / total airtime. Structure view: an MPDU is the 26-byte MAC header, the encryption bytes, the body and a 4-byte FCS. An A-MSDU body is a run of subframes, each a 14-byte subframe header (destination address 6, source address 6, length 2) and one MSDU, every subframe but the last padded to a 4-byte boundary, all under one MAC header and one FCS. An A-MPDU is a run of subframes, each a 4-byte MPDU delimiter, one whole MPDU and padding to a 4-byte boundary. An A-MPDU of A-MSDUs puts one A-MSDU in each A-MPDU subframe. The PSDU is every byte added up, delimiters, headers, padding and FCS included, and the time view uses the same sizes and the same rule for when the PSDU is an A-MPDU. One bad bit fails the FCS of the MPDU it sits in: in an A-MPDU the Block Ack bitmap marks that subframe 0 and only it is resent; a lone MPDU or A-MSDU gets no ACK and is resent whole. The resend is timed as a new PPDU (physical layer protocol data unit) of exactly the resent bytes, preamble plus data, with the same formulas. Limits checked: an HT A-MSDU is at most 3,839 or 7,935 bytes; an MPDU inside an HT A-MPDU is at most 4,095 bytes, 2^12 - 1, because the delimiter's MPDU Length field has 12 bits (14 in VHT and later), so an unencrypted A-MSDU there is at most 4,065; a VHT MPDU is at most 3,895, 7,991 or 11,454 bytes; an A-MPDU is at most 2^(13 + exponent) - 1 bytes, 65,535 for HT and 1,048,575 for VHT; an HE A-MPDU is at most 6,500,631 bytes, the HE PSDU maximum; and preamble plus data is at most 5.484 ms (aPPDUMaxTime), the longest time a legacy station can read from L-SIG: 4,095 bytes at 6 Mbps.

Example. HE at 6 GHz, 80 MHz, MCS 11, 2 streams, 0.8 µs GI, 32 frames of 1,500 bytes: 16,333 data bits per symbol in 13.6 µs is 1,200.96 Mbps. The TXOP is AIFS 43 + backoff 67.5 + preamble 50.4 + data 340 + SIFS 16 + Block Ack 32 = 548.9 µs, so 384,000 payload bits deliver 699.6 Mbps, 58.3% of the PHY rate. The same 1,500 bytes sent alone over VHT at 866.67 Mbps take 214.5 µs, ACK included, and deliver 55.9 Mbps. In the Structure view the same 32 frames are 32 A-MPDU subframes of 1,552 bytes (delimiter 4 + MAC header 26 + security 16 + MSDU

1,500 + FCS 4 + padding 2), 49,664 bytes in all. Corrupt MSDU 5 and bit 5 of the Block Ack is 0: 1,552 bytes, 3.1% of the PSDU, go again in a 64 µs PPDU. Pack three of those MSDUs into one A-MSDU instead and one bad bit costs all 4,596 bytes. Pack three of them into each A-MSDU of a 64-frame HT A-MPDU and every MPDU is 4,592 bytes, more than the 4,095 an HT delimiter can describe.

Field notes

- One station, no contention, no retries. Backoff is the average of a draw from 0 to CWmin; with other stations contending, waits are longer.
- HT and VHT are modeled with one BCC encoder (6 tail bits). At the highest rates the standard uses more encoders and a few more tail bits, under one symbol.
- HE uses LDPC with pre-FEC padding ignored, so the symbol count can be one short. HE-LTFs are 2x (6.4 µs + GI), or 4x (16 µs) at a 3.2 µs GI. It models the single-user PPDU, not OFDMA resource units.
- Control frames go at the legacy rate you choose, and the slot is the 9 µs short slot of an all-OFDM network.
- The Check does not catch every VHT MCS exclusion (for example 80 MHz, MCS 6, 3 streams). Confirm unusual combinations in the MCS Index tool.
- Airtime on data symbols leaves out the HE packet extension and the 2.4 GHz signal extension: both are padding, not data.
- The Structure view checks the limits two sources agree on: the HT Maximum A-MSDU Length (3,839 or 7,935 bytes, HT Capabilities element), 4,095 bytes for an MPDU inside an HT A-MPDU, the VHT Maximum MPDU Length (3,895, 7,991 or 11,454 bytes, VHT Capabilities element), the A-MPDU maximum (HT 65,535, VHT 1,048,575, HE 6,500,631 bytes) and the 5.484 ms PPDU duration. At most MCS values time runs out before bytes do. The time view's Check applies the same limits to the aggregate it draws, so the two views agree: an aggregate the Structure view marks as over a maximum is refused in the time view too.
- Not checked: the HE Maximum MPDU Length, and the smaller A-MPDU limit an HE receiver can advertise, which the sources pin only in part, and not at all for HE in 2.4 GHz. EHT (Wi-Fi 7) is not modeled; its exact A-MPDU maximum differs between sources.
- The Block Ack bitmap counts MPDUs, one bit each: 64 in the compressed Block Ack since HT, 256 in HE, 512 or 1,024 in EHT. A 1,024-bit bitmap is not new in Wi-Fi 7: the original Basic Block Ack had one, for 64 MSDUs of up to 16 fragments each. What EHT added is 1,024 MPDUs.
- Every A-MPDU subframe pads to 4 bytes, the last one included, as the time view counts it. The security header and MIC (message integrity check) are drawn as one block, the HT Control field is left out, and a damaged delimiter, where the receiver searches for the next one, is not modeled.

- A resend is timed as preamble plus data for exactly the resent bytes. The new wait, the larger contention window after a failure and the acknowledgment are not added.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a PHY rate is only the speed of the data symbols; every transmission also pays for waiting, a preamble and an acknowledgment, and aggregation recovers the airtime that a faster MCS cannot.

Control	What it does	Why it's there
Time / Structure (on the stage)	Switches between the TXOP drawn to scale in µs and the inside of the PSDU.	The help says "aggregation is not one thing": the Structure view shows why an A-MSDU and an A-MPDU fail differently.
Compare with scenario B	Stacks a second TXOP on the same scale. On by default.	"Up to two scenarios stacked for comparison (A and B), because comparing is the lesson."
Edit scenario: A / B	Picks which scenario the inputs change.	Needed once there are two scenarios.
Preset buttons: Legacy 6 Mbps, VHT one frame, VHT 32 aggregated, HE 32 aggregated	Loads one of the four scenarios. Opens with A = Legacy 6 Mbps and B = HE 32 aggregated.	These are the four defaults of the WLAN Pros Airtime Calculator, and the tool reproduces their totals: 2,242.5, 214.5, 666.5 and 548.9 µs.
Band: 2.4 / 5 / 6 GHz	Applies the band rules: 2.4 GHz adds a 6 µs signal extension; 6 GHz is HE only; VHT is 5 GHz only.	Band is one of the things the student should see change the share of air that carries data.
PHY	Legacy (a/g), HT (n), VHT (ac), HE (ax, single-user).	The generations differ in preamble length and in data bits per symbol. The tool lists PHY as a per-scenario control.
Channel width	20, 40, 80 or 160 MHz, limited by PHY.	Part of the lesson's goal.

Control	What it does	Why it's there
Legacy data rate / MCS (per stream)	6 to 54 Mbps for Legacy; MCS 0 to 11 otherwise, limited by PHY.	Part of the lesson's goal: a faster MCS shrinks only the data block.
Spatial streams	1 to 8 (HT up to 4).	More streams shrink the data block and leave the fixed costs alone, the same lesson as MCS.
Guard interval	HT and VHT 0.4 or 0.8 µs; HE 0.8, 1.6 or 3.2 µs.	Part of the lesson's goal.
Payload per frame	64 to 2,304 bytes; default 1,500.	Small frames pay the same fixed costs for less data.
Frames aggregated (A-MPDU)	1 to 64; Legacy sends 1.	Part of the lesson's goal, and the headline of the tool: aggregation recovers airtime.
More settings / Fewer settings	Shows or hides the five less-used settings below.	"Group the less-used ones behind More."
Encryption overhead: 0 / 16 bytes	Adds CCMP/GCMP bytes to every frame.	Ported from the calculator sheet. Shows that security costs a little airtime too.
Access category	Voice, Video, Best effort, Background (default client EDCA values).	Changes AIFS and the average backoff, the wait at the front of every TXOP. The Teacher's Guide QoS sequence uses it.
RTS/CTS protection	Reserves the medium before the data frame.	Shows what the handshake costs on a clean channel.
Control frame rate: 6 / 12 / 24 Mbps	Rate for RTS, CTS, ACK and Block Ack.	Shows how much of the fixed cost is control frames sent at a legacy rate.
HE packet extension	0, 4, 8, 12 or 16 µs, HE only.	Ported from the calculator sheet; shows a padding cost that carries no data.
Tap or hover a segment / duration table	Shows the segment's duration, its share of the TXOP and the formula with your numbers in it.	The tool asks for each segment's duration and the formula that produced it.

Control	What it does	Why it's there
Arrangement (Structure view)	Single MPDU, A-MSDU, A-MPDU or A-MPDU of A-MSDUs; Legacy offers only a single MPDU.	An A-MSDU shares one header and one FCS, so one bad bit loses them all; an A-MPDU gives each subframe its own, so Block Ack names the failed one (help).
MSDUs per A-MSDU: 2 / 3 / 4	How many packets share one header.	A teaching choice, not a limit from the standard (help).
Corrupt one MSDU, the arrow buttons, or tap a lime block	Damages one MSDU and shows the Block Ack bitmap, what is resent in bytes and time, and all four arrangements side by side.	Makes the A-MSDU versus A-MPDU difference visible as a resend cost. (help)

A demo that works (about 3 minutes)

1. Set up: scenario A on "VHT one frame", scenario B on "VHT 32 aggregated", compare on.
2. Ask the room: "One 1,500-byte frame at 867 Mbps. What throughput does it deliver?"
3. Reveal: press Right to walk scenario A one segment at a time. The lime data block is a sliver between the wait, the preamble and the ACK. The help puts it at about 56 Mbps. Scenario B carries 32 frames in 666.5 µs against A's one frame in 214.5 µs.
4. Optional follow-up: press V for Structure, A to reach A-MSDU, C to corrupt one MSDU: the whole A-MSDU is resent. Press A to reach A-MPDU: only the one subframe is resent.

Pairs with: Airtime Fairness, Multicast at the Basic Rate, Legacy Protection Cost and Channel Utilization Meter (all built on this timing), the MCS Index tool for unusual combinations, Multi-Link Operation (its frame airtime option). Teacher's Guide sequence "Why is the Wi-Fi slow?".

Left out on purpose: one station, no contention, no retries; backoff is the average of a draw from 0 to CWmin. HT and VHT use one BCC encoder; HE uses LDPC with pre-FEC padding ignored, so the symbol count can be one short. Single-user HE only, not OFDMA. The Check line misses some VHT MCS exclusions. EHT (Wi-Fi 7) is not modeled. A resend is timed as preamble plus data only, without the new wait or the larger contention window.

Airtime Fairness

Shows why one slow client drags every fast client down under plain Wi-Fi contention, and how airtime fairness fixes it. Build a cell of 1 to 8 clients, then compare packet fairness (every client gets the same number of turns) with airtime fairness (every client gets the same share of time).

Why it's here. A PHY rate is not a throughput. Every turn on the air pays a fixed overhead before any data moves, and under plain contention a 6 Mbps client holds the air for milliseconds each turn while an 867 Mbps client needs well under one. Seeing the airtime share and the throughput side by side explains the performance anomaly and why APs offer airtime fairness.

How to use

1. Pick the sharing rule: Packet, Airtime, or Compare to see both. The line under the switch states the result in numbers and changes as you edit.
2. Read the airtime share bar (who holds the air), the throughput bars (what each client and the whole cell get), and the round (each transmission drawn as wide as its airtime, with the overhead outlined and the frames filled). Replay round runs the animation again.
3. Edit the clients: a PHY rate from the presets or a custom rate, and the number of frames sent per turn (aggregation). Legacy clients send one frame per turn. Add up to 8 clients, and change the payload per frame to see small frames waste the air.
4. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses the round, the Right arrow steps one transmission, R empties the round, Up and Down change the sharing rule, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Clients	count	1 to 8; default three at 867 Mbps with 32 frames per turn plus one legacy client at 6 Mbps
PHY rate	Mbps	6, 24, 54 (legacy); 150, 300 (HT); 433, 867 (VHT); 600, 1201 (HE); or a custom rate from 1 to 10,000 with a legacy or HT-or-newer preamble

Input	Unit	Range
Frames per turn	frames	1 to 64; always 1 for legacy clients
Payload per frame	bytes	64 to 1,500; default 1,500

How it works. Airtime of one turn T = AIFS 43 µs + average backoff 67.5 µs + preamble (20 µs legacy, 40 µs HT or newer) + SIFS 16 µs + ACK 28 µs (or Block Ack 32 µs when frames are aggregated) + 8 x frames x (payload + 34) / rate. Packet fairness: each client gets one turn per round, so throughput = 8 x frames x payload / (sum of every client's T). Airtime fairness: each client gets 1/N of the time, so throughput = (1/N) x 8 x frames x payload / T. The total is the sum over clients.

Example. Defaults: an 867 Mbps client sending 32 frames takes about 651 µs per turn; the 6 Mbps client sending one frame takes about 2.22 ms. Under packet fairness the round lasts about 4.17 ms and the slow client holds 53% of it, so each fast client gets 92.0 Mbps, the slow one 2.87 Mbps, 278.9 Mbps in total. Under airtime fairness each client holds 25% of the air: the fast clients get 147.4 Mbps each, the slow one 1.35 Mbps, 443.4 Mbps in total.

Field notes

- This is a teaching model: one AP sending downlink to every client, every client always has data waiting, no collisions, no retries and no rate adaptation. Airtime Anatomy does the exact per-PHY airtime.
- The 40 µs preamble is one teaching value for HT, VHT and HE. Real preambles vary with the PHY, the number of streams and the frame format.
- The limits on how many frames or bytes one aggregate may carry are not enforced.
- Airtime fairness costs the slow client throughput. That is the point: it stops paying for its slow turns with everyone else's time.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: under plain contention every client gets the same number of turns, not the same amount of time, so one slow client drags every fast client down; airtime fairness gives each client an equal share of time instead.

Control	What it does	Why it's there
Sharing rule: Packet / Airtime / Compare	Packet fairness (equal turns), airtime fairness (equal time), or both side by side. Opens on Compare.	The fairness switch and compare view: the performance anomaly (Heusse et al., 2003) against its fix.
Replay round	Runs the round animation again.	The animated round: each transmission drawn as wide as its airtime, so the slow client's block visibly dominates under packet fairness.
Play / Pause, Step, Reset (Present mode)	Plays the round, steps one transmission, empties it.	Same round, paced by the presenter.
Payload per frame	64 to 1,500 bytes; default 1,500.	The lesson: every transmission pays a fixed overhead, so small frames waste most of the air.
Add client	Up to 8 clients. Opens with three at 867 Mbps and one legacy at 6 Mbps.	The default cell, chosen to show the anomaly on sight.
PHY rate (per client)	Presets 6, 24, 54 (legacy); 150, 300 (HT); 433, 867 (VHT); 600, 1201 (HE); or Custom rate.	The preset list, one per generation, so the class can build a realistic mixed cell.
Custom rate and Preamble: Legacy / HT or newer	Any rate from 1 to 10,000 Mbps with a 20 µs or 40 µs preamble.	The tool uses 20 µs legacy and 40 µs as a single teaching value for HT, VHT and HE.
Frames per turn (aggregation)	1 to 64; legacy clients always 1.	The aggregation count: fast clients recover airtime by sending many frames per turn.
Remove (per client)	Removes that client.	Build the cell down as well as up.

A demo that works (about 3 minutes)

1. Set up: the defaults, then set the sharing rule to Packet.
2. Ask the room: "If one client drops to 6 Mbps, what happens to everyone else?" (the Teacher's Guide's own example question).
3. Reveal: play the round. The 6 Mbps block fills most of the round. Read the takeaway line under the switch, which states the slow client's share of the air and what every client gets. Press Up or Down to switch to Airtime: the fast clients recover, the slow client gets less, and the total rises.
4. Optional follow-up: set every client to 867 Mbps. Packet and airtime fairness give identical results (one of the tool's tests), which shows the anomaly needs mixed rates.

Pairs with: the Airtime Fairness lab, four exercises from "A PHY rate is not a throughput" to "Modern clients, same problem". Also Airtime Anatomy, Legacy Protection Cost (a slow device that costs airtime while silent), Rate Adaptation. Teacher's Guide sequences "Why is the Wi-Fi slow?" and "Troubleshooting slow or unstable Wi-Fi".

Left out on purpose: one AP sending downlink to every client; every client always has data; no collisions, no retries, no rate adaptation. The 40 µs preamble is one teaching value for HT, VHT and HE. Aggregate size limits are not enforced. Airtime Anatomy does the exact per-PHY timing.

OFDMA vs MU-MIMO

Compares the two ways an 802.11ax access point sends to several clients at once. OFDMA splits the channel into resource units, one per client. MU-MIMO gives every client the whole channel on its own spatial stream and aims each stream with zero-forcing, which needs a fresh channel measurement (sounding) first. You move clients around the AP and see which needs less airtime, drawn to one microsecond scale.

Why it's here. MU-MIMO is often sold as always faster. It only pays when the clients sit in different directions, the frames are big enough for the whole channel to matter, and one sounding serves several transmissions. Moving the clients and changing the frame size shows each condition fail in turn.

How to use

1. Pick a scenario: Spread out, big frames; Bunched together; Many tiny frames; or One far client. Any change after that reads Your own layout.

2. Read the room. The AP is at the bottom with one dot per antenna. Each colored lobe is the beam zero-forcing aims at that client, with a null toward every other client. The badges say whether each pair is separable.
3. Drag a client, or pick it under Move client and use Turn left, Turn right, Closer and Farther.
4. Set the AP antennas, the number of clients, the channel width, the frame size and how many exchanges share one sounding. Turn on the side-wall reflection to add a second path for each client.
5. Read Which wins here and the sentence under it: how much faster MU-MIMO sends the data, how much longer its preamble is, and what sounding costs.
6. The two bars share one scale. The bracket under MU-MIMO marks the sounding. Tap Show the arithmetic for every segment's formula.
7. On a computer or tablet, Present opens this tool full screen for a projector: the Right arrow steps to the next scenario, Up and Down change the number of clients, R resets the scenario, and you can drag clients on the stage. F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
AP antennas	count	2 to 8, half a wavelength apart
Clients	count	2 to 4, one stream each
Client position	degrees and meters	90 degrees left to 90 degrees right of straight ahead; 2 to 35 m
Channel width	MHz	20, 40, 80 or 160
Frame size	bytes	100, 500 or 1,500 per client per exchange
Exchanges per sounding	count	1, 2, 4 or 8
Side-wall reflection		Off or On (illustrative)

How it works. Channel: a half-wave linear array of M antennas; a client at angle theta has h = a(theta), a_n = e^(j pi n sin theta). With the reflection on, h = a(theta) - 0.5 a(theta'), theta' the direction of the client's mirror image in a wall 20 m to the right. Zero-forcing: W = H^H (H H^H)^-1, so each client hears only its own stream; client k keeps 1 / ([(H H^H)^-1]_kk x ||h_k||^2) of the signal the whole array would give it alone (1 - |rho|^2 for two clients). A pair is separable when that loss is 3 dB or less. Link: received power as in Rate vs Range (20 dBm EIRP, exponent 3, 5.5 GHz). OFDMA keeps each client's own signal; MU-MIMO adds 10 log10(M / K) and subtracts the zero-forcing loss; MCS from the sensitivity table, up to MCS 11. OFDMA airtime: the OFDMA Resource Units model, unchanged, with K equal RUs at the slowest client's MCS, repeated for every exchange. MU-MIMO airtime: AIFS and backoff, HE NDP Announcement (21 + 4 per client bytes), SIFS, HE sounding NDP (one HE-LTF per AP antenna), SIFS, BFRP Trigger, SIFS, every client's compressed beamforming report at once in an HE TB PPDU, SIFS; then per exchange an HE MU preamble with one HE-LTF per stream, data set by the slowest stream on the whole channel, SIFS and the same block acks as OFDMA. Later exchanges wait for the medium again.

Example. Spread out, big frames: four clients 50 degrees left to 55 degrees right of a four-antenna AP, 1,500-byte frames, 80 MHz, eight exchanges per sounding. Zero-forcing costs each client 0.2 to 0.3 dB. OFDMA gives each client a 242-tone RU and takes 149.6 µs of data per exchange; MU-MIMO gives each the whole channel and takes 40.8 µs. Over eight exchanges that saves more than the 533.6 µs of sounding: OFDMA 3,181.6 µs, MU-MIMO 3,017.6 µs, 5% less airtime for MU-MIMO. With one exchange per sounding, OFDMA wins.

Field notes

- A teaching model. Zero-forcing on a line-of-sight channel shows why MU-MIMO needs clients in different directions; it is not how any one access point picks its groups. Real channels scatter, and real access points regroup clients as they move.
- The reflection is illustrative: half strength, one wall, and a fixed phase so the numbers hold still while you drag.
- Estimates, each labeled on screen: the report size (MU codebook, 9 and 7 bits per angle pair, 4-bit delta SNR, the MIMO and Beamforming tool's subcarrier count), control frames at 24 Mbps, the HE-SIG-B length and block-ack rate carried over from OFDMA Resource Units, and sounding sharing a TXOP with the first transmission.
- OFDMA here keeps the same power per tone as one client alone and is not beamformed. An access point that boosts a small RU, or beamforms inside OFDMA, would change the numbers.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: OFDMA and MU-MIMO are two different ways an 802.11ax AP serves several clients at once. OFDMA splits the channel; MU-MIMO gives every client the whole channel on its own beam, but only after it sounds the channel, and only when the clients sit in different directions. Neither always wins.

Control	What it does	Why it's there
Scenario	Four scenarios: Spread out, big frames; Bunched together; Many tiny frames; One far client. Any edit reads "Your own layout".	One scenario per condition MU-MIMO needs, so the class sees each one fail.
The room (drag a client)	Top-down view of the AP and its clients. Each colored lobe is the zero-forcing beam for that client, with a null toward every other client. Pair badges say separable or not separable.	The picture of why direction matters: a null cannot sit on one client without hitting a client right beside it.
Move client, Turn left / Turn right / Closer / Farther	Moves the chosen client 5 deg around the AP or 2 m in or out.	The keyboard path for the drag, and a clean way to walk a client toward another one step by step.
AP antennas	2 to 8.	Zero-forcing needs at least one antenna per client; more antennas make narrower beams and cheaper nulls.
Clients	2 to 4, one stream each.	More clients mean a longer sounding and a harder zero-forcing problem.
Channel width	20, 40, 80 or 160 MHz.	Wider channels make MU-MIMO's whole-channel data faster and its reports bigger.
Frame size	100, 500 or 1,500 bytes.	Small frames leave MU-MIMO nothing to save, so sounding decides the result.
Exchanges per sounding	1, 2, 4 or 8.	Sounding is paid once and shared; with one exchange per sounding, OFDMA wins every scenario.

Control	What it does	Why it's there
Side-wall reflection	Off / On.	An illustrative second path per client: the channel is no longer just a direction, and the separability changes.

A demo that works (about 3 minutes)

1. Set up: Spread out, big frames. Read the verdict: MU-MIMO wins by 164 µs, about 5%.
2. Ask the room: "If MU-MIMO sends four clients at once over the whole channel, why is it only 5% faster?" Point at the sounding bracket (533.6 µs) and at the per-exchange line: the data is 108.8 µs faster, the preamble 21.6 µs longer.
3. Set Exchanges per sounding to 1. OFDMA now wins: sounding is paid for one exchange.
4. Back to the scenario, then Bunched together: every pair reads not separable, zero-forcing costs over 20 dB, and MU-MIMO cannot decode at all.
5. Optional: Many tiny frames (OFDMA wins at any exchange count), then One far client (MU-MIMO wins by 27%, because the far client gets the whole channel instead of one RU at MCS 3).

Pairs with: OFDMA Resource Units (its timelines are the OFDMA side here, unchanged) and MIMO and Beamforming (sounding for one client). Airtime Anatomy for the single-frame timing underneath both.

Left out on purpose: the channel is line of sight plus one optional reflection; zero-forcing is the textbook precoder, not any AP's scheduler. OFDMA keeps the same power per tone as one client alone and is not beamformed. Control frames at 24 Mbps; the report size, HE-SIG-B length and block-ack rate are estimates, each labeled on screen. No retries or collisions.

Rate Adaptation

Runs one Wi-Fi link frame by frame under Minstrel-style rate control, the kind Linux uses, and shows it learning. A strip draws every attempt of the last 5 ms to scale, colored and numbered by MCS, with failures marked. A chart shows the chosen rate over the last 20 s against the rate the

SNR supports, and a table shows what the radio has learned about each rate, updated every 50 ms.

Why it's here. A radio does not know the best rate. It learns it by trying rates and keeping statistics, and every retry costs more airtime than the first try: the contention window doubles, and a retry at a lower rate takes longer to send. Walking the client away and back shows the chosen rate step down as the signal falls and climb again as sampling finds the higher rates.

How to use

1. Press Play, or Step to send one frame with all its retries. The link opens paused with every rate untried. Restart starts a new link that has learned nothing.
2. Pick a path: Walk (away to 60 m and back to 2 m every 20 s), Steady (4 m) or Fading (12 m, with the signal swelling and fading by up to about 8 dB). The SNR offset slider adds or removes signal on top of the path.
3. Watch the table. Each rate shows its smoothed success, its throughput estimate and its place in the retry chain: 1st and 2nd are the best and second-best throughput, P is the best probability, low is the lowest rate.
4. Change the sampling share, the EWMA weight, the retry chain, the retry limit and the ACK timeout while it runs. Set sampling to 0% and Restart to see a link that never finds its higher rates.
5. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses, the Right arrow steps one frame, R restarts, Up and Down move the SNR offset, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Path	preset	Walk (away and back), Steady or Fading; default Walk
SNR offset	dB	-20 to +20; default 0
Sampling share	% of frames	0 to 30; default 10

Input	Unit	Range
EWMA weight on history	%	50 to 95; default 75
Retry chain	on or off	On, or Off (every attempt stays at the first rate); default On
Retry limit	attempts	7 (short) or 4 (long); default 7
ACK timeout	µs	45 or 50; default 45
Speed	air time per real second	1/50, 1/10, 1/4 or real time; default 1/4

How it works. The link is fixed: 5 GHz, 802.11ax, 20 MHz, one spatial stream, 0.8 µs guard interval, 1500-byte frames, best effort, ACK at 24 Mbps, MCS 0 to 11. SNR comes from the path with the Rate vs Range math (20 dBm EIRP, path-loss exponent 3, 5500 MHz) over a -94.0 dBm noise floor. The SNR an MCS needs is its minimum sensitivity minus that noise floor, 12.0 dB for MCS 0 up to 42.0 dB for MCS 11. Each attempt succeeds with a chance from a teaching curve, 90% at that SNR and 10% about 4.4 dB lower. Every 50 ms each rate's success ratio folds into an EWMA (75% on history by default). Throughput estimate = min(success, 90%) x 12,000 bits / the time of one attempt, and rates under 10% success are ignored. About 10% of frames sample a random other rate. Each frame follows a retry chain: best throughput, second best, best probability, lowest rate; a sample frame puts the sample rate first, or second when it is slower than the best rate. Attempts are split over the chain as evenly as they go (7 = 2, 2, 2, 1). Each attempt costs AIFS (43 µs) + mean backoff (CW / 2 x 9 µs) + TXTIME from the Airtime Anatomy calculator + SIFS and ACK, or the ACK timeout. CW is 15 on the first try and doubles per retry (31, 63, ...) up to 1023.

Example. With the client steady at 4 m the SNR is 48.7 dB, above the 42.0 dB MCS 11 needs, and the rate control settles on MCS 11 (143.4 Mbps) within a second or two, with an estimate of 36.9 Mbps: one first try there takes 292.9 µs of air. At MCS 7 a first try that is delivered takes 347.3 µs. If it fails (348.3 µs with a 45 µs timeout) and the retry goes out at MCS 6, the retry alone takes 432.9 µs, because the mean backoff has grown from 67.5 to 139.5 µs and MCS 6 sends for longer.

Field notes

- The success curve is a teaching model, not a measured error curve. The sensitivity table it is anchored to is a set of conformance floors; real radios do better.
- This follows the published outline of Minstrel. Real drivers differ in detail: minstrel_ht samples on a timer with typed sample sets and sizes each chain stage by a time budget rather than a count.
- Delivered throughput is far below the PHY rate here because every frame waits, sends a preamble and waits for its ACK on its own. Aggregation (A-MPDU) and Block Ack, which real links use, are left out.
- There is one station, so there are no collisions, and backoff is the mean rather than a random draw. The Medium Access Simulator shows contention.
- The ACK timeout is SIFS + slot + a receive delay of about 20 to 25 µs. The sources disagree on the exact constant, so it is a setting.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a radio does not know its best rate; it learns it by trying rates and keeping statistics, and every retry costs more airtime than the first try.

Control	What it does	Why it's there
Play / Pause	Runs the link frame by frame. Opens paused with every rate untried.	The play control; opening untried shows the learning from zero.
Step	Sends one frame with all its retries.	Shows one retry chain in full: which rates it tried and in what order.
Restart	Starts a new link that has learned nothing.	The help pairs it with sampling at 0%: restart to see a link that never finds its higher rates.
Speed (air time per second)	1/50 (watch single frames), 1/10, 1/4, Real time; default 1/4.	The tool lists speed with play and step. The slow end shows single frames, the fast end the trend.

Control	What it does	Why it's there
Path: Walk / Steady / Fading	Walk goes out to 60 m and back to 2 m every 20 s; Steady holds 4 m; Fading holds 12 m with up to about 8 dB of swell and fade.	The lesson: as the client walks away the rate steps down, and on the way back sampling finds the higher rates again.
SNR offset	-20 to +20 dB on top of the path; default 0.	The SNR slider: move the whole link up or down the rate table.
Sampling share	0 to 30% of frames; default 10%.	A small share of frames sample other rates. At 0% the link never discovers a better rate.
EWMA weight on history	50 to 95%; default 75%.	The smoothing, 75% weight on history: lower it and the table reacts faster and noisier.
Retry chain: On / Off, same rate	On uses best throughput, second best, best probability, then lowest rate; Off keeps every attempt at the first rate.	The retry chain order. Off shows why falling back to a safer rate matters.
Retry limit (attempts): 7 (short) / 4 (long)	Attempts before the frame is dropped.	The two retry limits.
ACK timeout: 45 µs / 50 µs	How long the sender waits for an ACK that never comes.	The sources disagree on the exact constant, so the help makes it a setting.

A demo that works (about 3 minutes)

1. Set up: the defaults. Walk path, 10% sampling, speed 1/4.
2. Ask the room: "This link has never sent a frame. How does it find the right rate, and when the client walks to 60 m, does the rate slide down smoothly or in steps?"
3. Reveal: Play. The chart shows the chosen rate stepping down against the rate the SNR supports, and the table shows what the radio has learned about each rate, updated every 50 ms. On the walk back, sampling finds the higher rates again.

4. Optional follow-up: set Sampling share to 0% and press Restart. The link stays low. Or press Down several times to cut the SNR and watch retries pile up.

Pairs with: Medium Access Simulator (contention and the doubling CW), Airtime Anatomy (why a lower rate takes longer), What an Interferer Costs (failed frames that rate control can mistake for a weak link). Teacher's Guide sequence "Why is the Wi-Fi slow?".

Left out on purpose: the success curve is a teaching model anchored to the conformance-floor sensitivity table, not a measured error curve; real radios do better. It follows Minstrel's published outline; minstrel_ht samples on a timer and sizes chain stages by a time budget. No A-MPDU or Block Ack, so delivered throughput sits far below the PHY rate. One station, no collisions, mean backoff instead of a random draw.

What an Interferer Costs (and how a NIC hears the air)

Shows how a Wi-Fi radio, the network interface card (NIC) in a laptop or an access point (AP), decides the air is busy, and what each kind of interferer costs it. A level meter draws the two clear channel assessment (CCA) thresholds: preamble detect at -82 dBm, where the radio holds off for another Wi-Fi transmitter it can decode, and energy detect at -62 dBm, where it holds off for anything else. The 20 dB between them is 100 times the power. Bars show how far away the same transmitter still makes you wait, as Wi-Fi and as anything else, and a 60 ms sample of your channel shows when your radio waits and when it sends into the interferer. Four sources are side by side: a neighbor's AP on your channel, a microwave oven, Bluetooth, and an analog video sender.

Why it's here. Many people assume the harm comes from non-Wi-Fi devices. Usually the bigger cost is how readily your own radio backs off for other Wi-Fi on its channel. A neighbor's AP at -80 dBm makes your radio wait every time it transmits; a microwave oven at -70 dBm in your channel does not, because it is under energy detect, and at that level it barely touches your frames. The same 20 dB means a Wi-Fi neighbor defers you from 10 times farther away in free space, and 4.6 times farther indoors (path-loss exponent 3). Non-Wi-Fi interference can still do real harm: a continuous analog video sender stays under energy detect, so nobody waits for it, and ruins most frames anyway. That is the tool's counterexample.

How to use

1. Start with the question at the top: a microwave oven and a neighbor's AP, both at -70 dBm. Pick an answer, then press Reveal to load that scene and read why.
2. Pick a source: Wi-Fi AP (a neighbor on your channel), Microwave, Bluetooth or Video. Drag its level slider under the meter and watch the marker cross the thresholds. The meter's scale never moves.
3. Read the verdict and the two costs: airtime lost waiting (deferral) and airtime lost to corrupted frames. The corrupted-frame shares are illustrative. The table underneath shows every source at its current level, side by side.
4. Pick your AP's channel. The oven's peak lands in channel 11; on channel 6 and channel 1 it is weaker (illustrative), and on 5 GHz there is no oven and no Bluetooth. For the oven, pick 60 Hz or 50 Hz mains and watch its ON bursts in the 60 ms sample.
5. On 5 GHz, pick a channel width of 20, 40, 80 or 160 MHz and watch preamble detect move from -82 to -79, -76 and -73 dBm while energy detect stays at -62, so the gap narrows from 20 dB to 17, 14 and 11. This widening applies to 5 GHz. 6 GHz works differently and is not modeled here.
6. Set the path-loss exponent for the distance bars: 2.0 for free space, 3.0 for a typical indoor space, 3.5 for dense walls. Set the neighbor's airtime if you like; its default is illustrative.
7. On a computer or tablet, Present opens the tool full screen for a projector: Up and Down move the selected source's level 1 dB, N picks the next source, C the next channel, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Source	-	Wi-Fi AP on your channel (default), microwave oven, Bluetooth or analog video sender
Source level	dBm	-100 to -30. Defaults, all illustrative: Wi-Fi -78; oven -50 at its strongest frequency; Bluetooth -42 (a 4 dBm Class 2 radio 2 m away in free space, calculated); video sender -70
Your AP's channel	-	Channel 1, 6 or 11 in 2.4 GHz, or 5 GHz channel 36; default 11

Input	Unit	Range
Channel width	MHz	20, 40, 80 or 160 on 5 GHz; 20 on 2.4 GHz; default 20. The width scaling is the 5 GHz rule; 6 GHz is not offered. 320 MHz is left out: its threshold is not published in the sources used
Mains power	Hz	60 (default) or 50; the oven only
Neighbor airtime	%	10 to 80; default 30, illustrative
Path-loss exponent	n	2.0, 3.0 (default) or 3.5; the distances only

How it works. Preamble detect is -82 dBm for a 20 MHz physical layer protocol data unit (PPDU). On 5 GHz it is -79 for 40 MHz, -76 for 80 MHz and -73 for 160 MHz: 3 dB more per doubling, which is the same -82 dBm in each 20 MHz. This widening applies to 5 GHz; on 2.4 GHz the channel stays at 20 MHz. Energy detect is -62 dBm on the primary 20 MHz at every width. A Wi-Fi source is heard at or above preamble detect; any other source at or above energy detect. When a source is heard, your radio waits for all of its time on the air in your channel: that is the airtime lost waiting. When it is not heard, your radio sends into it, and the airtime lost to corrupted frames is its time on the air times the chance an overlapped frame fails. That chance is a logistic curve in the source's level, 1 / (1 + e^-((level - midpoint) / spread)), with midpoint and spread per source: Wi-Fi -86 and 2.5 dB, oven -55 and 2.5, Bluetooth -70 and 3, video -75 and 2.5, all illustrative. Time on the air in your channel: the neighbor's airtime; the oven 50%, one burst per mains cycle, 8.33 of 16.67 ms at 60 Hz and 10 of 20 ms at 50 Hz; Bluetooth 2 of 6 slots (illustrative) times 20 of 79 hop channels landing in your 20 MHz, 8.4%; the video sender 100%. The oven sits 0 dB below its peak on channel 11, 15 dB on channel 6 and 30 dB on channel 1 (both illustrative). Distances: the level of a 20 dBm transmitter (illustrative) is 20 - free-space loss at 1 m - 10 n log10(d); each distance is where that level meets a threshold. Their ratio is 10^(gap / 10n), independent of the transmitter.

Example. Defaults, channel 11, n = 3.0: a 20 dBm transmitter reaches -82 dBm at 114 m (375 ft) and -62 dBm at 25 m (81 ft), 4.6 times the distance and 21.5 times the area. In free space the ratio is 10. The question's scene, both at -70 dBm: the neighbor is 12 dB above preamble detect, so your radio waits its whole 30% airtime; the oven is 8 dB under energy detect, so your radio never waits and loses about 0.1% of its airtime to corrupted frames (illustrative). The oven at -50 dBm is above energy detect and costs 50%, the ON half of every mains cycle. The video sender at -70 dBm costs no waiting and 88% to corrupted frames (illustrative).

Field notes

- The -82 and -62 dBm thresholds are the standard's minimum requirements (IEEE 802.11-2020, 17.3.10.6), not what chips do. A radio must call the air busy at or above them. Real chips often detect preambles lower, near -91 dBm by one vendor's account (about 4 dB above a -95 dBm noise floor), which makes the real gap wider than 20 dB.
- The 40, 80 and 160 MHz preamble-detect values come from one source (Bejarano, Knightly and Park, IEEE Communications Magazine, 2013), and the tool applies them on 5 GHz only. The -82 and -62 values are confirmed by two. 6 GHz works differently and is not modeled here, so the tool offers no 6 GHz channel.
- The oven's timing and duty come from Airshark (Rayanchu, Patro and Banerjee, ACM IMC 2011), which measured a 16.66 ms cycle and a duty of 0.5, and agree with two older sources. Its peak, mostly 2.45 to 2.47 GHz, was measured on six ovens. How much weaker it is on channels 6 and 1 was not found in any source; those values are illustrative.
- The corrupted-frame shares are illustrative. No per-rate curve of frame loss against interference level was found. The curve is shaped to agree with what Airshark measured on a good link: an oven cost almost nothing below about -60 dBm, Bluetooth never cost more than about a tenth of throughput, and an analog video camera cost 80% or more at -70 dBm.
- Keith's field observation: an AP sitting on top of a running microwave oven, sending at high rates to its clients with no retries. That fits an AP on channel 1 or on 5 GHz, where the oven is weak or absent, and clients across the room hear the oven far weaker than the AP does.
- Waiting is decided at the sender and a frame's survival at its receiver. This tool puts both at your radio, so one level drives both.
- Left out: collisions between two Wi-Fi radios that start at the same moment, the hidden-node case beyond the illustrative curve, Bluetooth's adaptive hopping (which avoids busy channels), rate adaptation lowering the rate after failed frames, secondary channels, and the 802.11ax per-20 MHz rules. The Spatial Reuse tool shows how BSS coloring can relax -82 dBm.
- To see these interferers on a screen, open the Spectrum Analysis lesson (its interferer gallery) and the Swept vs FFT race in Fourier and FFT, which shows how a swept analyzer and a real-time fast Fourier transform (FFT) analyzer catch an oven's bursts and Bluetooth's hops.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: for the common interferers, the bigger cost is usually waiting, not lost frames: your radio defers to Wi-Fi it can decode at -82 dBm but to anything else only at -62 dBm, 20 dB (100 times the power) higher.

Control	What it does	Why it's there
Opening question: The oven / The neighbor's AP / The same, then Reveal, then Ask again	"A microwave oven and a neighbor's AP on your channel, both at -70 dBm. Which one costs your Wi-Fi more airtime?" Reveal loads that scene into the tool.	The lesson: waiting is the cost, not frame loss.
Source: Wi-Fi AP / Microwave / Bluetooth / Video	Picks the interferer on the meter. A dropdown on a phone.	Four sources side by side. The analog video sender is the counterexample: it stays under energy detect, so nobody waits, and it ruins most frames anyway (Airshark).
Level slider (under the meter)	-100 to -30 dBm in 1 dB steps; the label names the source ("Neighbor AP at your radio", "Oven at its strongest frequency"). The meter's scale never moves.	Watch the marker cross preamble detect and energy detect. Defaults are illustrative: Wi-Fi -78, oven -50, Bluetooth -42, video -70.
Mains power: 60 Hz / 50 Hz	The oven's on-off cycle in the 60 ms sample. Oven only.	Airshark measured the oven's cycle at 16.66 ms and a duty of 0.5. The switch lets a 50 Hz country see its own cycle.
Your AP's channel: Ch 1 / Ch 6 / Ch 11 / 5 GHz	The oven peaks in channel 11 and is weaker on 6 and 1 (illustrative); 5 GHz has no oven and no Bluetooth. Default 11.	A common field observation, an AP on a running oven with high rates and no retries, fits an AP on channel 1 or on 5 GHz.
Channel width (MHz): 20 / 40 / 80 / 160	On 5 GHz, preamble detect moves to -79, -76, -73 dBm while energy detect stays at -62.	Only preamble detect scales with width, so the 20 dB gap narrows to 17, 14 and 11 dB: the same -82 per 20 MHz.
Neighbor airtime (illustrative)	10 to 80%; default 30%.	An illustrative default for a co-channel neighbor: 30% airtime at -78 dBm, above -82, so you defer every time it talks.
Path-loss exponent, for the distances: 2.0 free / 3.0 indoor / 3.5 walls	Changes only the distance bars.	Turns 20 dB into distance: a Wi-Fi neighbor defers you from 10 times farther in free space and 4.6 times farther indoors (help).
Reset to defaults	Restores every setting.	Back to the start between demos.

A demo that works (about 3 minutes)

1. Set up: open the tool fresh. The question sits at the top.
2. Ask the room: the question as written. Oven or neighbor's AP, both at -70 dBm? Take a show of hands for The oven, The neighbor's AP, The same.
3. Reveal: pick the room's answer and press Reveal. The neighbor's AP: at -70 it is above -82, so your radio waits every time it transmits. The oven at -70 is under -62, so your radio sends into it, and at that level it barely touches your frames.
4. Optional follow-up: press N to reach Video at -70. Nobody waits and most frames die, so non-Wi-Fi interference can do real harm. Then press C to reach 5 GHz, open the width and pick 160 to show the gap narrow to 11 dB.

Pairs with: Spatial Reuse (BSS coloring relaxes the -82 dBm rule taught here), the Spectrum Analysis lesson's interferer gallery, the Swept vs FFT race in Fourier and FFT, Medium Access Simulator.

Left out on purpose: -82 and -62 dBm are the standard's minimum requirements, not what chips do; real chips often detect preambles near -91 dBm. The 40, 80 and 160 MHz values come from one source, and 6 GHz is not modeled. The corrupted-frame shares and the oven's weaker levels on channels 6 and 1 are illustrative. Waiting and frame survival are both computed at your radio. Left out: simultaneous-start collisions, Bluetooth's adaptive hopping, rate adaptation after failures, secondary channels, the 802.11ax per-20 MHz rules.

Spatial Reuse

Two BSSs share one channel along a line: AP A with client A, AP B with client B. AP A is already sending, and AP B has a frame ready. The tool shows whether AP B waits or sends at the same time, which rule decided it, how much transmit power AP B gives up when it reuses the air, each client's SINR, the highest MCS that SINR supports, whether each link still holds the MCS you pick, and how much airtime the pair uses.

Why it's here. A radio waits for any Wi-Fi frame it hears at -82 dBm or more, so two APs on one channel take turns even when they barely bother each other's clients. 802.11ax adds a BSS color to every frame so a radio can tell a neighbor from its own BSS, and lets it ignore a neighbor below a higher threshold, OBSS_PD, if it turns its own power down by the same amount. This shows the trade: spatial reuse buys concurrency with range.

How to use

1. Start with the default: AP B hears AP A at about -77.7 dBm, above -82, so it waits and the pair needs two frame-times.
2. Raise OBSS_PD. Once it passes the level AP B hears, AP B ignores the neighbor, both frames go at once, and AP B's power drops 1 dB for every 1 dB you raised the threshold. Watch each client's SINR and MCS fall.
3. Give both BSSs the same color number. AP B can no longer tell AP A's frame from its own BSS, so -82 dBm applies again whatever OBSS_PD is. Turn BSS coloring off to see legacy radios.
4. Drag a radio along the line, or use the position sliders. Move the APs closer until energy detect (-62 dBm) fires, which no color or threshold gets past.
5. Pick the MCS each link must hold (default MCS 7). Each link card says whether it still holds that MCS, and the highest MCS its SINR supports.
6. Change the channel width, AP power, TX_PWRref class and path-loss exponent to see how each moves the numbers. Copy puts the whole scenario on the clipboard.
7. On a computer or tablet, Present opens this simulator full screen for a projector: Up and Down move OBSS_PD, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
BSS coloring	on or off	on (802.11ax) by default; off is legacy behavior
BSS color, each BSS	number	1 to 63; defaults 6 and 26
OBSS_PD threshold	dBm per 20 MHz	-82 to -62, default -82
TX_PWRref	dBm	21 (client, or AP with 1 or 2 streams) or 25 (AP with 3 or more)
MCS each link must hold	MCS index	0 to 13, default 7 for both links

Input	Unit	Range
Channel width	MHz	20, 40, 80 or 160, default 20
AP transmit power	dBm	5 to 25, default 20, both APs
Path-loss exponent n	-	2.0 to 4.0, default 3.0
Positions	m	0 to 60 on a line; defaults AP A 5, client A 15, client B 45, AP B 55

How it works. Received level = AP power - (free-space loss at 1 m + 10 n log10(d)) at 5180 MHz with 0 dBi antennas, then minus 10 log10(width / 20) to get the level in each 20 MHz, which is what the thresholds are defined against. AP B decides in this order: energy at or above -62 dBm defers (energy detect); below -82 dBm nothing is detected; with coloring off any Wi-Fi preamble at or above -82 defers; with the same color on both BSSs the frame is intra-BSS and -82 applies; with a neighbor's color AP B defers at or above OBSS_PD, and below it AP B sends with its power capped at TX_PWRmax = TX_PWRref - (OBSS_PD - (-82)). A whole wide frame is compared with thresholds raised 3 dB per doubling of width, which is the same test. SINR at each client = signal - (other AP's level power-summed with the noise floor, -174 + 10 log10(20 MHz) + a 7 dB noise figure = -94.0 dBm). The SNR an MCS needs is its minimum sensitivity at the channel width minus the noise floor at that width, the way Rate vs Range reads SNR: about 12.0 dB for MCS 0 and 30.0 dB for MCS 7. A link holds an MCS when its SINR is at least that; the highest such MCS is the one its SINR supports. Airtime counts one frame each: 1 frame-time together, 2 in turn.

Example. Defaults: the APs are 50 m apart at n = 3, so AP B hears AP A at -77.7 dBm and waits (OBSS_PD at -82). Each link has 37.3 dB SINR, enough for MCS 9, so both hold MCS 7, one after the other. Raise OBSS_PD to -72: AP B sends at once, capped at 21 - 10 = 11 dBm, and the pair needs 1 frame-time instead of 2. Client A's SINR drops to 26.7 dB, which supports MCS 4, not 7. Client B's drops to 9.0 dB, short of the 12.0 dB MCS 0 needs. At -62 AP B is down to 1 dBm and client B's SINR is -1.0 dB: the neighbor is louder than its own AP.

Field notes

- TX_PWRref (21 or 25 dBm) and the 3 dB per doubling of width each come from one secondary source. The -82, -62 and OBSS_PD range have two.
- SRG (spatial reuse group) thresholds and parameterized spatial reuse (PSR) are out of scope.

- The MCS sensitivities are 802.11 minimum-sensitivity conformance floors (10% packet error rate, 4096-byte frames). Real radios beat them by several dB, so a link that fails here may still work; read the verdicts as the worst case the standard allows. The 7 dB noise figure is the Rate vs Range default.
- Only AP B decides. AP A already holds the air, and uplink frames and clients' own deferral are left out.
- Airtime ignores backoff, ACKs and rate changes; it counts one equal-length frame per BSS.
- The -82 and -62 dBm thresholds and the per-20 MHz comparison match the Channel Planner, so the two tools agree.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: two BSSs on one channel take turns whenever they hear each other above -82 dBm; 802.11ax BSS coloring and OBSS_PD let a radio ignore a neighbor, but every 1 dB of threshold costs 1 dB of its own transmit power, so spatial reuse buys concurrency with range.

Control	What it does	Why it's there
BSS coloring on (off = legacy radios); in Present, "BSS coloring (802.11ax)"	On by default. Off applies legacy rules and disables the color and OBSS_PD controls.	The color on/off toggle, where off equals legacy behavior.
BSS A color / BSS B color	1 to 63; defaults 6 and 26.	A 6-bit color lets a radio tell its own BSS from a neighbor. Give both the same number and -82 dBm applies again whatever OBSS_PD is (help).
OBSS_PD threshold, per 20 MHz	-82 to -62 dBm in 1 dB steps; default -82.	The lesson: raise the threshold and AP B may ignore the neighbor, at TX_PWRref minus the amount raised.
TX_PWRref (single source): 21 dBm / 25 dBm	Client or AP with 1 or 2 streams, or AP with 3 or more.	The two reference classes, labeled single-source.
Reset to the starting line	Restores the defaults.	Back to the start between demos.

Control	What it does	Why it's there
Link A MCS / Link B MCS	MCS 0 to 13 each must hold; default MCS 7.	Each link card says whether it still holds that MCS, so the cost of reuse shows up as a verdict, not just a dB number. (help)
Channel width, both BSSs	20, 40, 80 or 160 MHz.	OBSS_PD is per 20 MHz and rises 3 dB per width doubling (single-source).
AP transmit power, both	5 to 25 dBm; default 20.	Sets the starting power the restriction is taken from.
Path-loss exponent n	2.0 to 4.0; default 3.0.	The log-distance model; lets you match the room to the building.
Positions (AP A, Client A, Client B, AP B), or drag a radio on the stage	0 to 60 m on a line; defaults 5, 15, 45 and 55 m.	The draggable positions. The help: move the APs closer until energy detect (-62 dBm) fires, which no color or threshold gets past.

A demo that works (about 3 minutes)

1. Set up: the defaults. AP B hears AP A at about -77.7 dBm, above -82, so it waits and the pair needs two frame-times.
2. Ask the room: "How far do we have to raise OBSS_PD before AP B can send at the same time, and what does AP B give up?"
3. Reveal: press Up once per dB. At -77 dBm AP B stops waiting, both frames go at once, and AP B's power has dropped 5 dB. Watch each client's SINR and MCS verdict fall as you keep going.
4. Optional follow-up: set both colors to the same number and show AP B waiting again at any OBSS_PD. Or drag the APs together until energy detect fires.

Pairs with: What an Interferer Costs (the -82 and -62 dBm rules this tool relaxes), Channel Planner (same thresholds), OFDMA Resource Units and Multi-Link Operation (Teacher's Guide sequence "Wi-Fi 6 and 7 features").

Left out on purpose: only AP B decides; AP A already holds the air, and uplink and clients' own deferral are left out. SRG thresholds and parameterized spatial reuse are out of scope. The MCS sensitivities are conformance floors, so a link that fails here may still work. Airtime counts one equal-length frame per BSS, with no backoff or ACKs. TX_PWRref and the 3 dB per width doubling come from one source.

Power Save

Puts a client and its AP on a timeline and shows how the client saves battery: beacons every beacon interval, a DTIM every few beacons, the frames the AP holds while the client dozes, and the client waking under four modes (always awake, legacy PS with PS-Poll, U-APSD, and Target Wake Time). Two modes side by side show that every saving costs latency.

Why it's here. Battery drain, a sensor that misses broadcasts, and a call that stutters on a phone that just went quiet all come back to power save. A sleeping client wakes for beacons and reads the TIM to learn whether the AP holds frames for it. The DTIM period decides how often broadcast and multicast go out. PS-Poll fetches one frame per poll, U-APSD fetches a service period's worth per trigger, and TWT lets the client sleep through beacons entirely. Each one trades time awake for time waiting.

How to use

1. Pick a mode and a mode to compare it with. Both run the same traffic, beacons and currents.
2. Read the timeline: beacons on top (DTIM beacons taller), then for each mode the frames the AP is holding for the client, a dot where the TIM bit is set, and the client's radio: a thin line while it dozes and a colored bar while it is awake, colored by what it is doing. Switch the span between 0.3 s and 60 s and slide the window along the run.
3. Change the beacon interval, DTIM period and listen interval, the U-APSD access categories and Max SP Length, or the TWT mantissa, exponent and wake duration. The QoS Info byte and the TWT wake interval are worked out as you go.
4. Choose a traffic pattern (phone idle, sensor, voice call, web page) or set your own, and set the currents and battery. The readouts give time awake, average current, a battery estimate, and mean and worst latency.
5. On a computer or tablet, Present opens this simulator full screen for a projector: the Right arrow slides the window along the run, R returns it to the start, Up and Down change the DTIM period, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Mode, Compare with	mode	Always awake, Legacy PS (PS-Poll), U-APSD, or TWT; default Legacy PS compared with TWT
Beacon interval	TU (1 TU = 1024 µs)	50, 100 (default, 102.4 ms), 200, 300, 500 or 1000
DTIM period	beacons	1 to 10; default 3
Listen interval	beacons	1 to 10; default 3
U-APSD access categories	flags	Voice, Video, Best effort, Background; all on by default
Max SP Length	code	0 = all buffered (default), 1 = 2, 2 = 4, 3 = 6 frames
TWT agreement	type	Individual (default) or Broadcast
Wake interval mantissa	16-bit field	1 to 65535; default 62500
Wake interval exponent	5-bit field	0 to 31; default 4 (62500 x 2^4 µs = 1 s). Intervals over 5 minutes are refused
Minimum wake duration	256 µs or 1 TU	1 to 255; default 16 x 256 µs = 4.1 ms
Beacons (TWT)	choice	Sleep through (default) or Wake for DTIM
Traffic	frames	downlink bursts (none to 50 per second) of 1, 4 or 16 frames; uplink (none to 50 per second); broadcast and multicast (none to 20 per second); access category; regular or random arrivals

Input	Unit	Range
Awake and doze current	mA, µA	awake 20, 50 (default), 100 or 200 mA; doze 5, 20 (default), 50 or 200 µA
Battery	mAh	250, 1000 (default), 3000 or 10000

How it works. Time is kept in microseconds; 1 TU = 1024 µs. Beacons go out every beacon interval and every Nth is a DTIM, after which the AP sends the broadcast and multicast frames it held. In legacy PS and U-APSD the client wakes on every listen-interval beacon and on every DTIM beacon; if its TIM bit is set it retrieves the held frames: one PS-Poll per frame, or on a U-APSD access category one trigger per service period of up to Max SP Length frames (an uplink frame on that access category is itself the trigger). In TWT the wake interval is mantissa x 2^exponent µs and the client wakes for the minimum wake duration, during which the AP delivers held frames; broadcast TWT here starts each service period at a beacon. The run lasts 60 s, or four TWT intervals, up to 20 minutes. Time awake is the union of every awake stretch. Average current = awake share x awake current + doze share x doze current. Battery life = capacity / average current. Latency is from a frame reaching the AP (or being ready to send) to its delivery.

Example. Defaults: beacon 100 TU, DTIM 3, listen interval 3, phone-idle traffic. Legacy PS wakes for every third beacon (about 3.3 times a second) plus once for each uplink frame, and a downlink frame waits up to about 306 ms. TWT at 1 s wakes once a second and sleeps through beacons, so it is awake far less, but a downlink frame can wait up to a second and broadcast frames go by while it dozes. Set the voice-call pattern and compare Legacy PS with U-APSD: U-APSD collects each frame with the next uplink frame, within 20 ms.

Field notes

- 100 TU (102.4 ms) is the common beacon interval, not a mandated value.
- A longer DTIM saves the client wakes and makes broadcast and multicast wait longer. When the listen interval and DTIM period do not line up (3 and 4, say), this model's client wakes for both, so a longer DTIM can add wakes; make one a multiple of the other.
- The battery figures in the Currents card are one vendor example, not a general truth: a Silicon Labs SiWx917 tested by Novus Labs in a study Silicon Labs commissioned averaged about 90 µA at DTIM 10 and 49 µA with a 30 s TWT. The simulator does not use them.

- The frame-exchange durations, the traffic patterns and the currents are illustrative. Real clients also stay awake a while after traffic, retry, scan and roam, and APs limit how long they hold frames; none of that is modeled.
- The TIM here is set whenever any frame waits for the client; the finer U-APSD rules for which access categories the TIM covers are not modeled.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a sleeping client wakes for beacons and reads the TIM to learn whether the AP holds frames for it; each power save mode trades time awake for time waiting, so every saving costs latency.

Control	What it does	Why it's there
Mode	Always awake, Legacy PS (PS-Poll), U-APSD or Target Wake Time (TWT); default Legacy PS.	The four modes: PS-Poll fetches one frame per poll, U-APSD a service period per trigger, TWT lets the client sleep through beacons.
Compare with	A second mode, or Nothing; default TWT.	Two modes side by side show that every saving costs latency.
Timeline span: 0.3 s / 1 s / 10 s / 60 s	Zooms the timeline.	0.3 s shows single beacons and polls; 60 s shows the long TWT sleeps.
Window start slider	Slides the view along the run.	Lets you find a moment worth pointing at.
Beacon interval	50, 100, 200, 300, 500 or 1,000 TU; default 100 TU (102.4 ms).	100 TU is the common interval, not a mandated value.
DTIM period	1 to 10 beacons; default 3.	The lesson: DTIM decides how often broadcast and multicast go out; longer saves battery and adds latency.

Control	What it does	Why it's there
Listen interval	1 to 10 beacons; default 3.	The legacy PS wake rule. When it does not line up with DTIM, the tool says the client wakes for both.
U-APSD access categories (Voice, Video, Best effort, Background checkboxes)	Which ACs are trigger-enabled; all on by default. The QoS Info byte is worked out as you go.	The U-APSD flags (QoS Info B0 to B3).
Max SP Length	All buffered, 2, 4 or 6 frames per service period.	The codes 0 to 3 (QoS Info B5 and B6).
Agreement: Individual / Broadcast	TWT agreement type.	Part of the TWT model.
Wake interval mantissa (text field)	1 to 65,535; default 62,500.	The mantissa is a 16-bit field, and the interval is mantissa × 2^exponent µs.
Wake interval exponent	0 to 31; default 4 (62,500 × 2^4 µs = 1 s). Intervals over 5 minutes are refused.	Same formula as the mantissa.
1 s / 30 s buttons	Set 62,500 × 2^4 or 58,594 × 2^9.	Quick jumps to a sensible interval and to the 30 s TWT in the one vendor battery example the tool cites.
Minimum wake duration	1 to 255 units; default 16 × 256 µs = 4.1 ms.	The wake duration field.
Duration unit: 256 µs / 1 TU (1024 µs)	Unit for the wake duration.	The two units.
Beacons: Sleep through / Wake for DTIM	Whether a TWT client still wakes for DTIM beacons.	The lesson: TWT lets the client sleep through beacons entirely.
Traffic pattern	Phone, mostly idle; Sensor; Voice call; Web page; Custom.	The tool lists traffic pattern as a control; each preset carries its own one-line description.

Control	What it does	Why it's there
Downlink bursts, Frames per burst, Uplink frames, Broadcast and multicast, Access category	The traffic detail behind the pattern.	The downlink rate and burstiness plus uplink traffic.
Arrivals: Regular / Random, and New random pattern	Evenly spaced or random arrivals, and a new draw.	Random arrivals show worst-case latency that regular ones hide.
Awake current, Doze current, Battery	Awake 20 to 200 mA, doze 5 to 200 µA, battery 250 to 10,000 mAh.	Energy is awake time × awake current plus doze time × doze current, with currents as labeled parameters.

A demo that works (about 3 minutes)

1. Set up: the defaults. Legacy PS compared with TWT (1 s wake interval, sleeping through beacons), traffic "Phone, mostly idle".
2. Ask the room: "The TWT client skips every beacon and wakes once a second. What does that buy in battery, and what does it cost?"
3. Reveal: read the readouts for both modes: time awake, average current, battery estimate, mean and worst latency. The TWT client is awake far less and waits far longer for its frames.
4. Optional follow-up: switch traffic to "Voice call" and compare Legacy PS with U-APSD, the case U-APSD was made for. Or press Up to lengthen DTIM and watch broadcast latency grow. Keep the listen interval a factor or multiple of DTIM (3 with 6 or 9), because when they do not line up the client wakes for both.

Pairs with: Multicast at the Basic Rate (DTIM holding multicast), Voice Priority, End to End, Medium Access Simulator. Teacher's Guide sequence "QoS and contention".

Left out on purpose: frame-exchange durations, traffic patterns and currents are illustrative. Real clients stay awake after traffic, retry, scan and roam, and APs limit how long they hold frames. The TIM is set whenever any frame waits; the finer U-APSD rules for which ACs the TIM covers are not modeled. The Silicon Labs battery figures in the Currents card are one vendor-commissioned example and the simulator does not use them.

Multicast at the Basic Rate

Shows what multicast and broadcast data cost the channel. They go at a basic rate so every client can decode them, with no acknowledgment and no retry, so the same bytes take many times the airtime of a unicast frame. A one-second timeline draws the frames, the beacons and the DTIM (delivery traffic indication message) beacons, and compares the multicast stream with the same stream converted to one unicast copy per listening client.

Why it's here. One multicast video stream, or a lot of service-discovery chatter, can slow a whole cell. A unicast frame goes at the best rate its one receiver supports and is acknowledged and retried. A multicast or broadcast frame goes at a basic rate, one every client must support, and nobody acknowledges it. If any client on the AP is in power save, the AP also holds multicast until just after the next DTIM beacon, so it arrives late and in bursts. Converting multicast to unicast often uses less airtime when few clients listen, and more when many do; raising the basic rate helps too, at the cost of range for multicast. SSID Airtime shows what beacons cost; this tool is about the data. Airtime Anatomy shows one frame's timing in detail.

How to use

1. Read the headline: the share of the channel the multicast stream uses, and the share the same stream would use as unicast copies. Over 100% means the stream does not fit: the queue grows until packets are dropped.
2. Read the one-second timeline. Thin lines are beacons every 102.4 ms, thick lines are DTIM beacons. Lime blocks are busy air. Play the second sweeps it six times slower than real time; Next beacon moves one beacon at a time; Reset clears it.
3. Read the close-up: one packet drawn to scale, as one multicast frame (waiting, preamble, data, and no acknowledgment) and as unicast copies, each with its own wait, SIFS (short interframe space) and ACK (acknowledgment).
4. Change the stream (a preset, or the bit rate and packet size), the band and the basic rate, how many clients listen and their rates, and whether multicast, unicast or both are shown. Turn on A client is in power save and change the DTIM period to see multicast held and released in bursts.
5. Predict, then reveal: Ask the class loads the question (a 4 Mb/s video stream on a Wi-Fi 6 AP) and hides the airtime answers; pick a range, then press Reveal.
6. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses the second, the Right arrow moves to the next beacon, R resets it, Up and Down change the basic rate, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Stream preset	choice	Voice paging (0.064 Mb/s, 200-byte packets), Video (4 Mb/s, 1,316-byte packets), Service discovery chatter (0.1 Mb/s, 400-byte packets); all illustrative
Stream bit rate	Mb/s	0.064 to 20; default 4 (illustrative)
Packet size	bytes	200, 400, 576, 1,000, 1,316 or 1,500; default 1,316 (illustrative)
Band	GHz	2.4 or 5; default 5
Basic rate	Mb/s	1, 2, 5.5 or 11 (802.11b, 2.4 GHz only), 6, 12 or 24 (802.11a/g); default 6
Show	choice	Multicast, Unicast, or Both (default)
Listening clients	count	1 to 30; default 5
Listener rates	MCS (modulation and coding scheme)	Spread (MCS 9, 5, 7, 3, 11, 1, repeating; default), All close (MCS 9), All far (MCS 1); all illustrative
DTIM period	beacons	1 to 10; default 3. Beacons every 102.4 ms
A client is in power save	switch	Off (default) or On

How it works. Packets per second = bit rate / (8 x packet size). Each packet becomes a frame of packet + 26-byte header + 4-byte checksum + 16 bytes of encryption, as in Airtime Anatomy. Multicast airtime per packet = DIFS (distributed interframe space, SIFS + 2 slots) + average backoff (half the minimum contention window, in slots) + preamble + data, with no acknowledgment. At 802.11a/g rates the preamble and data come from Airtime Anatomy's timing (9 µs slot, minimum contention window 15). At 802.11b rates the preamble is the long one, 192 µs, the data takes 8 x frame bytes / rate microseconds, the slot is 20 µs, SIFS 10 µs and the minimum contention window 31. Each unicast copy is a full

Airtime Anatomy exchange: wait (AIFS, arbitration interframe space, plus average backoff, video access category), preamble, data, SIFS and ACK, as a Wi-Fi 6 frame at 20 MHz, 2 spatial streams, 0.8 µs guard interval, at the listener's MCS, with the ACK at 24 Mb/s. Airtime share = airtime per packet x packets per second / 1 s. The break-even listener count is the smallest count at which the copies take at least as long as the one multicast frame. With a client in power save, a packet that reaches the AP is sent right after the next DTIM beacon (every DTIM period x 102.4 ms), so the added delay is up to one DTIM interval and half of it on average. Frames go on the air in order, each after the one before it.

Example. Defaults: 4 Mb/s video in 1,316-byte packets is about 380 packets per second. At a 6 Mb/s basic rate on 5 GHz each multicast frame takes 1,941.5 µs, so the stream uses 73.8% of the channel. The same stream converted to unicast for five listeners spread across the MCS range takes 1,207.5 µs per packet, 45.9% of the channel, and the break-even is 8 listeners. Raise the basic rate to 24 Mb/s and multicast drops to 21.9%. On 2.4 GHz at 1 Mb/s it would need 435% of the channel: it does not fit.

Field notes

- RFC 9119 (Request for Comments 9119, from the IETF, the Internet Engineering Task Force, 2021) describes these multicast problems on Wi-Fi. The values in this tool are computed from the airtime model, not taken from the RFC.
- The stream presets, the default bit rate and packet size, the listeners' rates and the unicast radio settings are illustrative, chosen to show the effect, not measured.
- One AP, no other traffic, no collisions and no retries. Beacons are drawn as markers; their own airtime is what SSID Airtime shows.
- A unicast copy to a dozing client waits for that client to wake, which this tool does not draw: the power save switch holds multicast only.
- The 802.11b timing (192 µs long preamble, 20 µs slot) uses the standard's well-known values; IEEE 802.11 itself was not re-read for this tool.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: multicast and broadcast go at a basic rate so every client can decode them, with no acknowledgment and no retry, so the same bytes can take many times the airtime of unicast; converting to unicast often wins when few clients listen.

Control	What it does	Why it's there
Play the second / Pause	Sweeps the one-second timeline six times slower than real time.	Shows the stream's busy blocks, the beacons and the DTIM beacons arriving in time.
Next beacon	Moves one beacon at a time.	Walks the DTIM release one beacon at a time with power save on.
Reset	Clears the sweep.	Back to the start of the second.
Presets (illustrative): Voice paging / Video / Service discovery chatter	Loads a stream's bit rate and packet size.	The preset list: a paging stream, a video stream and mDNS-style chatter.
Stream bit rate (illustrative)	0.064 to 20 Mb/s in fixed steps; default 4.	The slider, labeled illustrative.
Packet size (illustrative)	200, 400, 576, 1,000, 1,316 or 1,500 bytes; default 1,316.	The packet size, labeled illustrative.
Band: 2.4 GHz / 5 GHz	Default 5 GHz. The 802.11b rates exist only on 2.4 GHz.	Lets you show the 1 Mb/s worst case, which 5 GHz cannot have.
Basic rate	1, 2, 5.5 or 11 (802.11b, 2.4 GHz only), 6, 12 or 24 Mb/s; default 6.	The lesson: raising the basic rate helps, at the cost of range for multicast.
Show: Multicast / Converted to unicast / Compare both	Default Compare both.	The tool compares the stream with the same stream sent as one unicast copy per listener.
Listening clients	1 to 30; default 5.	Unicast conversion grows with the listener count, and the readout reports the break-even count.
Listener rates (illustrative)	Spread, MCS 1 to 11; All close, MCS 9; All far, MCS 1.	Unicast copies go at each listener's own rate, so far listeners make conversion expensive.

Control	What it does	Why it's there
DTIM period	1 to 10 beacons; default 3.	The lesson: with a client in power save, multicast waits for the DTIM beacon.
A client is in power save	Off by default.	Same lesson: turn it on to see multicast held and released in bursts.
Ask the class / Under 10% / 10 to 30% / 30 to 60% / Over 60% / Reveal / close	Loads the question and hides the airtime answers until Reveal.	The predict-then-reveal question: "A 4 Mb/s video stream on a Wi-Fi 6 AP. How much of the channel does it use?"

A demo that works (about 3 minutes)

1. Set up: the defaults. 4 Mb/s video, 1,316-byte packets, 5 GHz, 6 Mb/s basic rate, five listeners.
2. Ask the room: press Ask the class. "A 4 Mb/s video stream on a Wi-Fi 6 AP. How much of the channel does it use?" Take votes on the four ranges.
3. Reveal: Over 60%. At a 6 Mb/s basic rate it uses 73.8% of the channel (the tool's own screen test). Press Up to raise the basic rate to 24 Mb/s: 21.9%.
4. Optional follow-up: raise Listening clients past the break-even readout (8 listeners at the defaults, per the same test) and watch unicast become the more expensive choice. Or turn on "A client is in power save" and play the second to see the stream held until each DTIM beacon.

Pairs with: SSID Airtime (beacon cost), Airtime Anatomy, Power Save (DTIM). Teacher's Guide sequences "Where the airtime goes" and "Troubleshooting slow or unstable Wi-Fi".

Left out on purpose: one AP, no other traffic, no collisions, no retries. Beacons are drawn as markers; their own airtime is SSID Airtime's job. The stream presets, default rates and listener rates are illustrative. A unicast copy to a dozing client would wait for that client; the power save switch holds multicast only. The 802.11b timing uses the standard's well-known values, not a fresh read of IEEE 802.11. RFC 9119 describes the problem; the numbers come from the airtime model.

Channel Utilization Meter

Shows the number an AP reports as channel utilization, and what it does and does not mean. The AP counts the share of time its radio found the channel busy, averaged over several seconds, and puts it in every beacon as one byte from 0 to 255. A strip of channel time shows the frames and the waits between them, a bracket shows the averaging window sliding over the beacon intervals, and a split bar shows how much of that time was your data, overhead, collisions, other networks, waiting the protocol requires, and truly spare time.

Why it's here. Channel utilization is the busy number in the BSS Load element (BSS: basic service set, one AP's network; IEEE, the Institute of Electrical and Electronics Engineers, 802.11-2024 clause 9.4.2.26, first added by 802.11e as the QBSS Load element, a QBSS being a quality of service basic service set). Busy counts everything the AP hears: its own network, neighbor networks, collisions, retries, and non-Wi-Fi energy loud enough to make the channel read busy. Busy is not the same as used for your data: preambles, acknowledgments and collisions are all busy time. Idle is not the same as available: every frame waits a DIFS (distributed interframe space) and a random backoff first, so one sender at full speed fills a 54 Mb/s channel at about 73% busy as the access point reports it (76% seen by a device outside the exchange), never 100%. The Station Count beside it counts associated stations, not active ones. Clients read these numbers when choosing an AP, so a busy-looking channel pushes clients away whether or not the busy time is the AP's own. Airtime Anatomy shows one frame exchange in detail; the Medium Access Simulator shows contention and backoff; the BSS Load tool (macOS) reads the real element from the AP you are connected to.

How to use

1. Read the headline: channel utilization as a percent and as the byte (0 to 255), the Station Count, and the share of time that was truly spare.
2. Run the channel. The strip scrolls the last few milliseconds: gray preambles and lime data are this network's frames, outlined blocks are ACKs (acknowledgments), dashed gaps are the SIFS (short interframe space) before each ACK, hatched gaps are the DIFS and backoff every sender must wait, striped blocks are another network, crossed blocks are collisions and zigzag blocks are non-Wi-Fi energy. Every kind of block is named in the legend.
3. Watch the averaging window: each bar is one beacon interval (100 TU, time units of 1024 µs, so 102.4 ms), and the bracket spans the window. The meter is the average over the bracket, so a one-second burst barely moves a 5-second average. Next beacon interval runs one interval, Skip one window fills a whole window at once, Add a 1-second non-Wi-Fi burst adds a burst now, and Reset starts over.
4. Change the traffic: senders (1 to 50), offered load, data rate, basic rates (they set the ACK rate), frame payload, and stations that are associated but idle. Change the measurement: the window, and Listener view (counts reservations). Add a neighbor network or non-Wi-Fi

bursts.

5. The meter shows what the access point itself reports: the time its radio heard a signal or was transmitting. The short gap between a data frame and its acknowledgment counts as idle here, because a device never sets its reservation timer (the network allocation vector, NAV) from a frame addressed to it, and the access point is one end of every exchange in its own network. Turn on Listener view to see the channel from a nearby device outside the exchange: it honors the reservation, so the gap counts as busy and the same traffic reads about 76% instead of 73%.
6. Read the split bar and the readouts: payload, overhead, collisions, other network, non-Wi-Fi, required idle and truly spare. The formula card shows the byte computed with the live numbers.
7. Many senders: with every sender saturated, busy stays near its ceiling while your data's share falls, because collisions are busy time. The tool's own run is drawn beside points from a published paper.
8. Predict, then reveal: Ask the class loads one laptop sending as fast as it can and hides the meter; pick an answer, then press Reveal.
9. On a computer or tablet, Present opens this simulator full screen for a projector: Space runs or pauses the channel, the Right arrow runs one beacon interval, R resets, Up and Down change the number of senders, B adds a 1-second burst, W skips ahead one window, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Senders	count	1 to 50; default 1
Offered load, each	%	5 to 100 in steps of 5, a share of what one sender alone can carry; 100 means saturated, a frame always waiting (default)
Data rate	Mb/s	6, 9, 12, 18, 24, 36, 48 or 54 (802.11g, 2.4 GHz); default 54
Basic rates	choice	6, 12, 24 Mb/s (default, the mandatory OFDM rates); 1, 2, 5.5, 11 Mb/s; 6 Mb/s only; 24 Mb/s only; or all eight, 6 to 54 Mb/s. It sets the ACK rate

Input	Unit	Range
Frame payload	bytes	100, 500, 1,000, 1,500 or 2,304; default 1,500
Associated but idle	count	0 to 60; default 0
Averaging window	beacon intervals	1 to 255; default 50 (5.12 s), the standard's default according to one secondary source
Listener view (counts reservations)	switch	Off (default): what the access point reports. On: what a device outside the exchange sees
A neighbor network on this channel	switch	Off (default) or On
Neighbor's airtime	%	5 to 60; default 20 (illustrative)
Non-Wi-Fi bursts	switch	Off (default) or On: 4 ms bursts about 20% of the time (illustrative)
Speed	choice	1 ms per second, 20 ms per second, Real time (default), or 5 s per second

How it works. Channel Utilization = floor(255 × busy µs ÷ (window × beacon period in TU × 1024)), with the beacon period fixed at 100 TU. Busy, as the access point reports it, is physical carrier sense: the radio heard a signal (CCA, clear channel assessment) or was transmitting. In Listener view busy is physical or virtual carrier sense: it adds the NAV that a frame's Duration field sets, here the SIFS between a data frame and its ACK, and the same for the neighbor network's exchanges. Frame timing comes from Airtime Anatomy at 2.4 GHz with 802.11g rates: 9 µs slot, 10 µs SIFS, DIFS = SIFS + 2 slots = 28 µs, 6 µs signal extension, and the ACK at the highest basic rate that is not above the data rate; if no basic rate fits, at the fastest of the mandatory 6, 12 or 24 Mb/s that is not above it (IEEE 802.11-2024 10.6.6.5.2). Only the OFDM rates in the basic set count, because the ACK keeps the modulation class of the frame it answers; the standard does not say how to settle the two rules against each other, and filtering by class first is the model's reading. Each frame is the payload plus a 26-byte header and a 4-byte checksum, with no encryption bytes. Contention follows the Medium Access Simulator's rules for legacy DCF (distributed coordination function): a contention window of 15 slots at first and at most 1023, a random backoff of 0 to the window counted down only while the channel has been idle for a DIFS,

and frozen while it is busy. Two senders reaching zero on the same slot collide, double their window and try again, up to 7 attempts. Senders below saturation get frames at random (Poisson) times. The run is recorded per beacon interval, and the meter averages the last window of them (fewer while the window is still filling). The split bar sorts every microsecond: payload (the payload bits at the data rate), overhead (preamble, header, signal extension, the ACK, and the SIFS in Listener view), collisions, other network, non-Wi-Fi, required idle (DIFS, backoff, the ACK timeout after a collision, and the SIFS in the access point's view) and truly spare (idle with nobody waiting to send).

Example. One sender at 54 Mb/s with 1500-byte payloads: the data frame is 57 OFDM (orthogonal frequency-division multiplexing) symbols = 228 µs, plus 20 µs of preamble and SIGNAL field and 6 µs of signal extension = 254 µs. One cycle is DIFS 28 + mean backoff 7.5 × 9 = 67.5 + data 254 + SIFS 10 + ACK 34 = 393.5 µs. As the access point reports it, busy is 254 + 34 = 288 µs, 73.2% of the cycle; in Listener view, which counts the reserved SIFS, it is 298 µs, 75.7%. The payload itself is 12,000 bits at 54 Mb/s = 222.2 µs, 56.5%. Over a full window the AP reports about 186 of 255 (72.9%). That channel is full there: a meter reading 50% on it means about two-thirds of what it can carry is in use, not half.

Field notes

- The window's default of 50 beacon intervals is the standard's default according to one secondary source; the standard's own definition of it was not read. It is labeled so on screen.
- The formula is the definition, not any product's code. Real access points may compute it differently: one open-source implementation averages the radio driver's busy counters over a period the operator sets. In the two open-source drivers checked, those counters add up airtime (signals heard plus the radio's own transmissions) with no reservation term, which matches the access point view here; closed firmware was not checked.
- Only the primary 20 MHz channel is modeled. Whether the basic field covers only the primary channel is not verified; extended load elements for wider channels exist and are not shown.
- Available Admission Capacity is the medium time left under explicit admission control, in units of 32 µs per second. Most networks do not run admission control, so it is not a general capacity gauge; this tool shows it as not in use.
- The many-sender points are from Bianchi (2000), IEEE Journal on Selected Areas in Communications 18(3), read off the paper's figure: about 0.8 of the channel at 5 stations and 0.55 at 50 for basic access, and about 0.83 with RTS/CTS (request to send / clear to send). Those use 1 Mbps parameters: the shape transfers, the exact numbers do not.

- The ACK rate follows the basic rate set: with 6 Mb/s the only basic rate every ACK goes at 6 and takes 50 µs; with all eight rates basic the ACK goes at the data rate. The lowest basic rate is not a floor for ACKs: with only 24 basic, a frame at 12 is acknowledged at 12. Beacons are left out here, so the basic set changes only the ACKs.
- Simplifications: every station hears every other (no hidden nodes); after a collision every station waits the ACK timeout instead of the extended interframe space; a new frame always draws a backoff; the neighbor network sends the same frames at the same rate; non-Wi-Fi bursts never land on top of a frame; beacons' own airtime is left out (SSID Airtime shows it).
- The neighbor's airtime and the non-Wi-Fi burst pattern are illustrative, chosen to show the effect, not measured.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: the channel utilization an AP puts in its beacon is the share of time its radio found the channel busy, averaged over seconds; busy is not the same as used for your data, and idle is not the same as available.

Control	What it does	Why it's there
Run the channel / Pause	Scrolls the strip of channel time.	The play control.
Next beacon interval	Runs one 102.4 ms interval.	Each bar under the meter is one beacon interval.
Skip one window	Fills a whole averaging window at once.	Saves waiting 5 seconds for the average to settle in front of a class.
Add a 1-second non-Wi-Fi burst	Adds a burst now.	For example: one 1-second burst in a 5.12 s window moves the reading by at most about 20%, so a spike barely moves the average.
Reset	Starts over.	Back to an empty window.
Senders	1 to 50; default 1.	The many-sender lesson: with every sender saturated, busy stays near its ceiling while your data's share falls, drawn beside Bianchi's 2000 figures.

Control	What it does	Why it's there
Offered load, each	5 to 100% in steps of 5; default 100 (saturated).	A lightly loaded channel reads low for the right reason.
Data rate	6 to 54 Mb/s (802.11g); default 54.	The rate control.
Basic rates	6, 12, 24 Mb/s (default); 1, 2, 5.5, 11; 6 only; 24 only; all eight.	Sets the ACK rate: the highest basic rate not faster than the data frame, else the fastest mandatory rate not faster. "6 only" slows every ACK to 50 us; "all eight" sends the ACK at the data rate. The default keeps every worked number.
Frame payload	100, 500, 1,000, 1,500 or 2,304 bytes; default 1,500.	The frame size control.
Associated but idle	0 to 60 stations; default 0.	The lesson: Station Count counts associated stations, not active ones.
Averaging window, beacon intervals (default 50: one secondary source)	1 to 255; default 50 (5.12 s).	The default is the standard's according to one secondary source, and the label says so.
Listener view (counts reservations)	Off: what the AP reports. On: what a device outside the exchange sees, counting the NAV across the SIFS.	The physical-versus-virtual carrier sense toggle. The same traffic reads 73% as the AP reports it and 76% from outside.
A neighbor network on this channel, and Neighbor's airtime (illustrative)	Adds another network's busy time, 5 to 60%; default 20%.	Busy time the AP's own station count cannot explain, and it pushes clients away whether or not it is the AP's own.
Non-Wi-Fi bursts (illustrative)	Adds 4 ms bursts about 20% of the time.	Busy counts non-Wi-Fi energy loud enough to read busy.

Control	What it does	Why it's there
Speed	1 ms per second, 20 ms per second, Real time or 5 s per second; default Real time.	The slow end shows single frames and gaps in the strip; the fast end fills the averaging window.
Ask the class / About 25% / About 50% / About 75% / 100% / Reveal / close	Loads one saturated laptop and hides the meter.	The question: "One laptop sends as fast as it can. What will the meter read?"

A demo that works (about 3 minutes)

1. Set up: the defaults (one saturated sender at 54 Mb/s, 1,500 bytes).
2. Ask the room: press Ask the class. "One laptop sends as fast as it can. What will the meter read?" Most rooms say 100%.
3. Reveal: About 75%. The meter reads 73%, because every frame waits a DIFS and a random backoff first, and that required silence is idle time. Only about 56% of the cycle is payload.
4. Optional follow-up: press B during a run and show the 5-second average barely move. Or raise "Associated but idle" to 40: Station Count jumps and utilization does not move. Then turn on the neighbor network: utilization rises and Station Count does not.

Pairs with: Airtime Anatomy, Medium Access Simulator, the BSS Load tool (macOS, reads the real element), SSID Airtime. Teacher's Guide sequences "Troubleshooting slow or unstable Wi-Fi" and "Where the airtime goes".

Left out on purpose: the formula is the definition, not any product's code; real APs may compute it differently. Only the primary 20 MHz channel is modeled. Every station hears every other (no hidden nodes); after a collision everyone waits the ACK timeout instead of EIFS; beacons' own airtime is left out. The neighbor's airtime and the burst pattern are illustrative. Bianchi's points use 1 Mbps parameters: the shape transfers, the exact numbers do not. Available Admission Capacity shows as not in use.

Legacy Protection Cost

Shows what one old 802.11b device costs a modern network while it sends nothing. An 802.11b device cannot decode OFDM (orthogonal frequency-division multiplexing), which 802.11g and later use, so each modern frame is first protected by a short, slow frame the old device can decode, and an associated 802.11b device also forces a longer slot time. The tool draws both costs on a timeline beside a network of modern devices only, and a beacon inspector shows the ERP (Extended Rate PHY, the 802.11g physical layer) bits and the HT (High Throughput, 802.11n) Protection value that switch it on.

Why it's here. It is easy to think an old device only slows the network when it talks. Protection costs airtime even when it is silent. The protection frame is a CTS (Clear to Send) addressed to the sender itself, called CTS-to-self, or an RTS (Request to Send) and CTS pair, sent at a DSSS or CCK rate (direct sequence spread spectrum, complementary code keying) the old device can decode. Its Duration field makes every listener hold off. At 1 Mb/s with the long preamble, a 14-byte CTS takes 304 µs, longer than the 1,500-byte, 254 µs frame at 54 Mb/s it protects. Add the 20 µs slot instead of 9 µs and one 54 Mb/s laptop's ceiling falls from about 30.5 to about 14.8 Mb/s. Airtime Fairness shows the cost of a slow client's own traffic; Airtime Anatomy shows one frame's timing in detail.

How to use

1. Read the headline: the laptop's payload rate with this network's settings, what it would be with modern devices only, and the share lost.
2. Read the channel: two lanes over a few milliseconds, modern devices only above and this network below. The lime blocks are data. The warning-colored blocks are the two costs, each with its own pattern and label: vertical bars for the protection frame, hatching for the added wait from the long slot. Play sweeps the window about two thousand times slower than real time; Next cycle moves one send cycle at a time; Reset clears it.
3. Read one send cycle drawn to scale, part by part: DIFS (distributed interframe space) and backoff, the protection frame, preamble, data, SIFS (short interframe space) and ACK (acknowledgment). The durations are also written out under each bar.
4. Turn on 802.11b device associated or 802.11b network heard nearby and watch the beacon inspector: associated sets NonERP_Present and Use_Protection and forces the long slot; heard sets Use_Protection only. For a heard network, choose whether it is on this AP's channel or another one, and whether this AP reacts to its own channel only or to adjacent channels too.
5. Pick the protection frame: CTS-to-self or RTS/CTS, then a rate and preamble from the grid; each cell shows what it costs. 1 Mb/s with short preamble is grayed out because 802.11b does not define it. The 6 Mb/s OFDM contrast shows that most of the cost is the old preamble, not

protection itself.

6. Predict, then reveal: Ask the class loads the question (an old 802.11b scanner in a drawer, associated but silent) and hides the answers; pick a range, from Nothing to Over 60%, then press Reveal.
7. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses, the Right arrow moves to the next send cycle, R resets, Up and Down make the protection frame faster or slower, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
802.11b device associated	switch	On (default) or Off
The 802.11b device can use short preamble	switch	Off (default, illustrative: an old scanner is assumed long-preamble only) or On. Off sets Barker_Preamble_Mode while the device is associated
802.11b network heard nearby	switch	Off (default) or On
The old network is	choice	On our channel (default) or On another channel
This AP reacts to	choice	Own channel only (default) or Adjacent channels too. Sources disagree: one open-source access point reacts only to its own channel; a 2004 lab report saw protection spread between channels 1 and 11. The standard allows both.
Protection frame	choice	CTS-to-self (default) or RTS/CTS
Protection rate and preamble	Mb/s	1 (long only), 2, 5.5 or 11, long or short preamble; or 6 Mb/s OFDM for contrast. Default 1 Mb/s long preamble, the worst case the standard allows
Frame size	bytes	64, 256, 512, 1,000 or 1,500; default 1,500
Data rate	Mb/s	6, 9, 12, 18, 24, 36, 48 or 54 (802.11g); default 54

Input	Unit	Range
CWmin with an 802.11b device associated	slots	15 (default) or 31. CWmin is the minimum contention window; 31 comes from one unverified source

How it works. One send cycle = DIFS + mean backoff + protection + data frame + SIFS + ACK. DIFS = SIFS + 2 slots. Mean backoff = CWmin / 2 slots. The slot is 9 µs, or 20 µs while an 802.11b device is associated. The data frame and ACK come from Airtime Anatomy's timing: 802.11g at 2.4 GHz, frame = payload + 26-byte header + 4-byte checksum (an open network, no encryption bytes), a 20 µs preamble, 4 µs symbols, 6 µs signal extension, SIFS 10 µs, and the 14-byte ACK at 24 Mb/s (34 µs). A protection frame at an 802.11b rate takes the 802.11b transmit time: 192 µs (long preamble and header) or 96 µs (short), plus 8 x bytes / rate, rounded up to a whole microsecond. A CTS is 14 bytes and an RTS 20 bytes. CTS-to-self adds CTS + SIFS; RTS/CTS adds RTS + SIFS + CTS + SIFS. The 6 Mb/s OFDM contrast CTS takes 20 + 4 x ceil((16 + 8 x 14 + 6) / 24) + 6 = 50 µs. Payload rate = payload bits / cycle time. The loss compares it with the same frames sent with a 9 µs slot, CWmin 15 and no protection. ERP bits: associated sets NonERP_Present and Use_Protection; a heard network sets Use_Protection only, always on this AP's channel and on another channel only when the AP reacts to adjacent channels; Barker_Preamble_Mode is set when the associated device cannot use short preamble. Use_Protection turns the protection frame on. HT Protection reads 3 (non-HT mixed) with an older station associated, 1 (nonmember) with an older network heard, else 0.

Example. Defaults: 1,500 bytes at 54 Mb/s. With modern devices only, a cycle takes DIFS 28 + backoff 67.5 + data 254 + SIFS 10 + ACK 34 = 393.5 µs, 30.5 Mb/s of payload. The long slot alone makes it 498 µs, 24.1 Mb/s. Add CTS-to-self at 11 Mb/s short preamble (107 + 10 µs) and it is 615 µs, 19.5 Mb/s; at 11 Mb/s long (203 + 10 µs), 711 µs, 16.9 Mb/s; at 1 Mb/s long (304 + 10 µs), 812 µs, 14.8 Mb/s, about half lost. With CWmin 31 as well, 972 µs, 12.3 Mb/s. RTS/CTS at 2 Mb/s adds 540 µs with long preamble and 348 µs with short.

Field notes

- Everything here is 2.4 GHz. There is no DSSS at 5 GHz, so none of these numbers carry over there.
- Protection must turn on when an 802.11b station is associated, and may turn on when one is only heard. Which heard networks an access point reacts to differs between implementations.
- Which rate an access point picks for its protection frames is up to the implementation; 1 Mb/s long preamble is the slowest legal case.

- The HT Protection value is a readout only, not animated: which 802.11n frames need a protection frame in mode 3 is not verified. Value 2 (a 20 MHz-only 802.11n station in a 20/40 MHz network) is not modeled.
- CWmin 31 while an 802.11b station is present comes from one unverified source, so it is a setting, applied only while one is associated.
- The ERP element is element 42 in 802.11g-2003, clause 7.3.2.13; its clause number in the current edition was not verified and is not shown.
- One sender, no collisions, no retries, and the old device sends nothing. Its own slow traffic is what Airtime Fairness shows.
- The network is open (no encryption bytes) so the data frame matches the 254 µs worked example; with 16 bytes of encryption it would take one more symbol, 258 µs.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: one associated 802.11b device costs a 2.4 GHz network airtime even while it sends nothing: a slow protection frame in front of every modern frame, and a longer slot time.

Control	What it does	Why it's there
Play / Pause	Sweeps the two-lane window about two thousand times slower than real time.	Shows modern-only and this network side by side, repeating send cycles.
Next cycle	Moves one send cycle.	Lets you stop on one cycle and point at each cost.
Reset	Clears the sweep.	Back to the start.
802.11b device associated	On by default. Sets NonERP_Present and Use_Protection and forces the 20 µs slot.	Protection must turn on when an 802.11b station is associated, and the long slot is the second cost.
802.11b network heard nearby	Off by default. Sets Use_Protection only, never the long slot.	Protection may turn on when an old network is only heard.

Control	What it does	Why it's there
The old network is: On our channel / On another channel	Where the heard network sits.	The "own channel only / adjacent channels too" question, labeled because sources disagree.
This AP reacts to: Own channel only / Adjacent channels too	Whether a heard network on another channel triggers protection.	One open-source AP reacts only to its own channel; a 2004 lab report saw protection spread between channels 1 and 11. The standard allows both.
The 802.11b device can use short preamble	Off by default, which sets Barker_Preamble_Mode.	The third ERP bit. Off is illustrative: an old scanner is assumed long-preamble only.
Protection frame: CTS-to-self / RTS/CTS	The kind of protection frame.	The picker: CTS-to-self costs one slow frame, RTS/CTS costs two.
Rate and preamble grid	1 (long only), 2, 5.5 or 11 Mb/s, long or short preamble; each cell shows its cost. 1 Mb/s short is grayed out. Default 1 Mb/s long.	The default is the worst case the standard allows. 1 Mb/s short is disabled because 802.11b defines short preamble only for 2, 5.5 and 11 Mb/s.
Contrast: 6 Mb/s OFDM	A protection frame at 6 Mb/s OFDM, which an 802.11b device cannot decode.	The contrast option: most of the cost is the old preamble, not protection itself.
CWmin with an 802.11b device associated: 15 / 31 (unverified)	Minimum contention window while the device is associated.	"31 when an 802.11b station is present" comes from one unverified source, so it is a labeled setting.
Frame size	64, 256, 512, 1,000 or 1,500 bytes; default 1,500.	The sender. Smaller frames make the fixed protection cost a bigger share.
Data rate	6 to 54 Mb/s (802.11g); default 54.	The sender.
Ask the class / Nothing / 5 to 25% / 25 to 40% / 40 to 60% / Over 60% / Reveal / close	Loads the silent-scanner question and hides the answers.	The question: "An old 802.11b scanner sits in a drawer, associated but silent. How much does it cost the 54 Mb/s laptop?"

A demo that works (about 3 minutes)

1. Set up: the defaults. 802.11b device associated, CTS-to-self at 1 Mb/s long preamble, 1,500 bytes at 54 Mb/s.
2. Ask the room: press Ask the class. The scanner in the drawer, associated but silent: how much does it cost the laptop? Take votes on the five ranges.
3. Reveal: 40 to 60%. The laptop's ceiling falls from about 30.5 to about 14.8 Mb/s. Point at the two costs in the lower lane: the 304 µs CTS, longer than the 254 µs data frame it protects, and the stretched slots.
4. Optional follow-up: press Up until the protection frame reaches 11 Mb/s short (19.5 Mb/s in the ceiling table). Pick the 6 Mb/s OFDM contrast. Then turn "associated" off and "heard nearby" on: the protection frame stays and the long slot goes.

Pairs with: Airtime Fairness (the cost of a slow device's own traffic), Airtime Anatomy. Teacher's Guide sequences "Where the airtime goes" and "Troubleshooting slow or unstable Wi-Fi".

Left out on purpose: everything is 2.4 GHz; there is no DSSS at 5 GHz. Which rate an AP uses for protection is up to the implementation; 1 Mb/s long is the slowest legal case. The HT Protection value is a readout only; mode 2 is not modeled. One sender, no collisions, no retries, and the old device sends nothing. The network is open; with 16 bytes of encryption the data frame takes one more symbol.

Multi-Link Operation

Compares Wi-Fi 7 Multi-Link Operation (MLO) modes on the same random traffic. A client holds links on two or three bands, other networks keep each band busy part of the time, and a stream of frames arrives. Single link, STR, NSTR and EMLSR each send the frames their own way, and the tool shows each mode's mean and 99th-percentile latency, a latency histogram, and which links carried the frames.

Why it's here. MLO is often presented as a latency cure. The win is real when the links are about equally busy, because a frame can take whichever link frees up first. It shrinks when one link is much busier, and it can turn into a loss: a slow link that happens to be free first, or EMLSR paying its switch delays on every exchange. Real drivers also switch EMLSR off often, so a Wi-Fi 7 laptop is frequently on one link.

How to use

1. Pick a lesson, or set your own links, traffic and modes. Every change reruns all the modes on the same traffic; New random traffic draws a fresh sample with the same settings.
2. Read the lanes: one row per link, other networks' busy time shaded, and each frame placed where the chosen mode sent it, numbered in arrival order. EMLSR's lead-in and NSTR's padding are drawn hollow. Slide the view along the run.
3. Compare the histograms and the latency table. The verdict line says whether each mode beats the best single link or is worse than it.
4. Turn on EMLSR disabled by driver to see the fallback to one link, and raise the padding and transition delays to see EMLSR's switch cost.
5. On a computer or tablet, Present opens this simulator full screen for a projector: R goes back to the lesson, Up and Down slide the lanes along the run, N draws new random traffic, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Links	band	2.4, 5 and 6 GHz, each on or off; at least one stays on. Default 5 and 6 GHz
Busy	% of the time	0 to 95 per link, in steps of 5; default 50
Mean busy period	µs	200, 500, 1,000 (default), 2,000 or 5,000 per link
Frames per second	frames/s	100 to 3,000 in steps of 100; default 1,000. 2,000 frames per run
Frame airtime	µs	Fixed: 100 to 2,000, default 300, the same on every link. Or from Airtime Anatomy: HE MCS 0 to 11 (default 7) and 1, 8 or 32 frames per A-MPDU
Modes to compare	on/off	STR, NSTR and EMLSR; all on by default. Single link is always shown
EMLSR padding delay	µs	0, 32, 64 (default), 128 or 256
EMLSR transition delay	µs	0, 16, 32, 64 (default), 128 or 256

Input	Unit	Range
EMLSR disabled by driver	on/off	off by default; needs two or more links

How it works. Frames arrive at random (exponential gaps). Each link carries other networks' traffic: busy periods and idle gaps of random, exponentially distributed length, set by the busy fraction and mean busy period, seeded so every mode sees the same traffic. A frame starts the moment its link is idle; anyone due to start while it is on the air waits and goes right after it. Single link: first in, first out on one link. STR: each link sends on its own, and a frame takes whichever link is free first. NSTR: frames start together and end together; frames already waiting join on other idle links and are padded to the longest airtime. EMLSR: one exchange at a time on whichever link is free first; each opens with an initial control frame (sized as an RTS and CTS at 24 Mbps, a lower bound) plus the padding delay, and the next waits out the transition delay. Latency is from arrival to the end of the frame's own airtime. A mode is overloaded when more than 5% of frames have not started by the time the last one arrives. Airtime Anatomy mode uses one TXOP (AIFS, average backoff, preamble, data, SIFS, Block Ack) at HE, 2 streams, 1500-byte frames, 20 MHz on 2.4 GHz, 80 MHz on 5 GHz and 160 MHz on 6 GHz.

Example. The Two equal links lesson (5 and 6 GHz, each 50% busy, 300 µs frames, 1,000 per second, first traffic sample): single link means are 2.01 ms on 5 GHz and 1.74 ms on 6 GHz; STR averages 583 µs, NSTR 621 µs and EMLSR 1.49 ms. In the Slow 2.4 GHz link lesson (8-frame A-MPDUs, 794 µs on 2.4 GHz and 291 µs on 6 GHz), STR averages 462 µs, worse than 428 µs on 6 GHz alone, because some frames take the slow link just because it was free first.

Field notes

- The one published MLO latency study (Carrascosa, Geraci, Knightly and Bellalta, IEEE ICC 2022) is a model built on real traffic, not a field measurement: it replays measured 5 GHz channel occupancy. It found an order-of-magnitude latency win when the links are equally busy, and cases where MLO is worse when they are not. We found no independent latency measurement on Wi-Fi 7 hardware (research as of September 2026).
- Intel's Linux driver leaves EMLSR for reasons that include low signal, Bluetooth coexistence, channel load, link usage and missed beacons, and after some of them it will not re-enter for 300 or 600 seconds.
- EMLMR (moving extra radio chains to one link) is described but not simulated.

- Not modeled: collisions, retries, uplink traffic, more than one frame or A-MPDU per exchange, and EHT (320 MHz, 4096-QAM). HE timing stands in for EHT. With EMLSR switched off, the model keeps the least busy link; a real driver picks by its own rules.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: MLO's latency win comes from sending on whichever link frees up first, so it is real when links are about equally busy, shrinks when one is much busier, and can become a loss; EMLSR pays a switch cost and real drivers often turn it off.

Control	What it does	Why it's there
Lesson	Two equal links: MLO wins; One link much busier: the win shrinks; A slow 2.4 GHz link: MLO can be worse; EMLSR switch cost: one clean link. Shows "Your own settings" once you edit.	Each lesson sets up one learning point in one pick.
New random traffic	Draws a fresh sample with the same settings.	Shows that a result holds across samples, not just one lucky draw.
2.4 GHz / 5 GHz / 6 GHz link switches	Each link on or off; at least one stays on. Default 5 and 6 GHz.	A client holds links on two or three bands.
busy	0 to 95% per link in steps of 5; default 50%.	The busy fraction per link: the lesson turns on whether the links are equally busy.
mean busy period	200, 500, 1,000, 2,000 or 5,000 µs; default 1,000.	The busy-period length per link.
Frames per second	100 to 3,000 in steps of 100; default 1,000.	The arrival rate.
Frame airtime: Fixed / Airtime Anatomy	Fixed µs per frame, or computed from HE MCS and aggregation.	Airtime fixed per frame, or from the Airtime Anatomy service at a chosen MCS.
Airtime per frame, every link	100 to 2,000 µs; default 300.	Fixed-airtime option.

Control	What it does	Why it's there
HE MCS and Frames per A-MPDU (1, 8, 32)	Airtime Anatomy option; default MCS 7.	Lets a slow 2.4 GHz link cost real airtime, as in the "slow link" lesson.
STR / NSTR / EMLSR switches	Which modes to compare; all on. Single link is always shown.	Compare single link, STR, NSTR and EMLSR on the same random traffic.
Padding delay / Transition delay	EMLSR switch delays, 0 to 256 µs; defaults 64 and 64.	The tool takes these values from the Linux kernel's list. EMLSR pays them before each exchange on a newly chosen link.
EMLSR disabled by driver	Falls back to one link. Needs two or more links.	Real drivers may disable EMLSR. Intel's Linux driver leaves it for many reasons, some for 300 or 600 s (help).
Show the lanes for, Time shown (2 / 5 / 20 ms), View starts at	Which mode's frame placement the lanes draw, how much time, and where.	Lets you put one mode's choices on the lanes at a scale where individual frames read.

A demo that works (about 3 minutes)

1. Set up: Lesson "Two equal links: MLO wins".
2. Ask the room: "Wi-Fi 7 adds a second link. Does latency always drop? What if one link is much busier, or slow?"
3. Reveal: read the verdict line and the histograms. With equal links, STR beats the best single link. Switch the lesson to "One link much busier: the win shrinks", then to "A slow 2.4 GHz link: MLO can be worse", and watch the verdict line say so.
4. Optional follow-up: pick "EMLSR switch cost: one clean link" to see EMLSR pay its delays, then turn on "EMLSR disabled by driver". Press N between lessons to show the result is not one lucky draw.

Pairs with: Airtime Anatomy (frame airtime), Spatial Reuse and OFDMA Resource Units (Teacher's Guide sequence "Wi-Fi 6 and 7 features").

Left out on purpose: the one published MLO latency study (Carrascosa et al., IEEE ICC 2022) is a model driven by real traffic traces, not a field measurement, and no independent Wi-Fi 7 hardware measurement was found. EMLMR is described but not simulated. No collisions, retries or uplink; one frame or A-MPDU per exchange; HE timing stands in for EHT. With EMLSR off the model keeps the least busy link, while a real driver picks by its own rules.

Network Design and Security (17)

Channel Planner

Place 2 to 12 access points on a floor, give each a channel and width, and see which ones share airtime. APs whose channels overlap and that hear each other above the clear channel assessment thresholds defer to each other; the largest group that all defer to each other is outlined as the contention domain, and each AP in it gets about 1/N of the airtime.

Why it's here. Channel planning is usually taught as a rule of thumb: use 1, 6 and 11, and keep channels narrow. This shows where the rules come from. Six APs on one floor fit on nine 20 MHz channels without sharing, but at 80 MHz only two channels exist without DFS in the US, so three APs share each one. In 2.4 GHz the transmit mask shows why 1, 6 and 11 beat 1, 4, 8 and 11.

How to use

1. Pick the band, region and width. In 5 GHz, turn DFS and (in the US) U-NII-4 on or off and watch the channel count for each width change. In 2.4 GHz, pick the OFDM or DSSS transmit mask.
2. Tap an AP to select it and drag it to move it, or pick it from the Selected AP list and use its position sliders. Choose its channel and width from the lists. Add up to 12 APs.
3. Turn on the wall tool and drag on the floor to draw a wall, or add a wall straight across from the controls. Every wall costs the same loss, which you set.
4. Solid lines join APs that defer to each other, labeled with the received level and the rule that fired. Dashed lines mean only one side defers. The dashed rings mark the largest contention domain. The strip under the floor shows which channels are in use across the band.

5. Press Auto-plan to spread the APs across the channels available at the chosen width, then change the width and plan again.
6. On a computer or tablet, Present opens the planner full screen for a projector: R resets the floor, Up and Down change the channel width for every AP, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Band and region	2.4 or 5 GHz; US or EU	US 2.4 GHz channels 1 to 11, EU 1 to 13; US 5 GHz 36 to 165 plus U-NII-4 169 to 177, EU 36 to 140
Channel width	MHz	20 or 40 in 2.4 GHz (20 only with DSSS); 20, 40, 80 or 160 in 5 GHz
DFS and U-NII-4	on or off	DFS: 52 to 64 and 100 to 144 (140 in the EU); U-NII-4 in the US only
Access points	count and position	2 to 12 on a floor 20 to 100 m wide and 10 to 60 m deep; default 50 x 30 m
AP EIRP	dBm	0 to 30, default 20
Path-loss exponent n	-	2.0 to 4.0, default 3.0
Loss per wall	dB	0 to 30, default 10

How it works. Received level between two APs = EIRP - (free-space loss at 1 m + 10 n log10(d)) - walls crossed x wall loss, with a 0 dBi receive antenna, computed at 2437 MHz in 2.4 GHz and 5500 MHz in 5 GHz. A wide transmission spreads its power, so each 20 MHz piece carries the total minus 10 log10(width / 20). An AP defers when a neighbor's channel overlaps its own and: the neighbor's preamble lands on its primary 20 MHz at -82 dBm or more (PD); or, in 5 GHz, a signal lands on its secondary 20 MHz at -72 dBm or more (SD); or the energy in any overlapping 20 MHz is -62 dBm or more (ED). Channels that do not overlap never contend. In 2.4 GHz the energy from a partly overlapping neighbor comes from its transmit mask (OFDM: 0 dBr to 9 MHz, -20 at 11, -28 at 20, -40 at 30 and beyond; DSSS: -30 dBr from 11 to 22 MHz, -50 beyond), interpolated in dB and summed over the victim's 20 MHz. A contention domain is the largest set of APs that all defer to each other; each gets 1/N

of the airtime. Auto-plan is greedy: APs that hear the most neighbors choose first, each taking the channel that keeps the largest domain smallest, then every AP gets up to three chances to change its pick.

Example. The starting floor has six APs on 36 at 40 MHz in the US with DFS off, so all six hear each other and share one channel at about 17% each. Auto-plan uses all four 40 MHz channels and leaves pairs 32 m apart sharing (-72.4 dBm, preamble detect), so the largest domain is 2. At 20 MHz nobody shares; at 80 MHz, with two channels, three APs share each one. In 2.4 GHz the mask is -34 dBr at 25 MHz (1 and 6) but -23.6 dBr at 15 MHz (1 and 4).

Field notes

- The transmit mask is a ceiling, not what a radio emits. The adjacent-channel figures are the worst case the mask allows; real radios run below it.
- Auto-plan is a teaching heuristic, not any vendor's RRM or DCA algorithm. It only knows the APs on this floor, not the neighbors, clients or interference a real network hears.
- The -82, -72 and -62 dBm thresholds are the commonly cited 802.11 values from vendor documentation. The EU writes energy detect into law as -75 dBm/MHz, which is -62 dBm over 20 MHz.
- On a secondary channel, deferring means the AP sends on fewer 20 MHz pieces rather than waiting. The Lab counts it as sharing to keep the picture simple.
- 1/N airtime is an even split among APs that all hear each other. Clients, retries, traffic load and APs that hear only part of the group change the real share.
- One wall type with one loss. For real materials use the RF Attenuation tool.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: APs on overlapping channels that hear each other share one channel's airtime, and a wider channel leaves fewer channels to reuse, so more APs end up sharing.

Control	What it does	Why it's there
Band: 2.4 GHz / 5 GHz	Switches the floor between the two bands. A new band starts with every AP on one channel.	The two planning problems differ: 2.4 GHz is about masks and 1, 6 and 11; 5 GHz is about how many channels survive each width. 6 GHz planning is left to the 6 GHz Power tool (code comment).
Region: US / EU	Swaps the channel plan. The EU plan has no U-NII-4 and stops at 140.	Channel counts differ by region (US 25/12/6/2 at 20/40/80/160 MHz, EU 19/9/4/2), and the class should see that the count is a rule, not physics.
DFS channels (5 GHz)	Adds 52 to 64 and 100 to 144.	Shows how much DFS buys: without it the US has 9 channels at 20 MHz and none at 160 MHz.
U-NII-4 (US) (5 GHz)	Adds 169 to 177. Disabled in the EU.	Lets the class see what the newest US channels add at each width.
Transmit mask: OFDM (g/n/ax) / DSSS (802.11b) (2.4 GHz only)	Picks the mask used for adjacent-channel energy. DSSS has no 40 MHz channel.	This is where 1, 6 and 11 comes from: the OFDM mask is about -34 dBr at 25 MHz spacing but -23.6 dBr at 15 MHz (1 and 4).
Channel width, every AP (Up/Down in Present)	Sets every AP's width at once.	The core lesson: at 20 MHz six APs share nothing; at 80 MHz only two non-DFS channels exist, so three APs share each.
Auto-plan	Greedy assignment that shrinks the largest contention domain at the current width.	The floor opens with all six APs on one channel, "the plan Auto-plan is there to fix" (code comment). It is labeled a teaching heuristic, not any vendor's RRM or DCA.
Reset to the starting floor (R)	Six APs on 36 at 40 MHz, US, DFS off.	Returns to the worst case so the fix can be shown again.
Selected AP, Add AP, Remove AP	Picks an AP; adds or removes one (2 to 12).	More APs on the same floor means more sharing at the same width.
AP n channel, AP n width	Sets one AP's primary channel and width.	Lets you put one AP on a wide channel and show it colliding with narrower neighbors.

Control	What it does	Why it's there
AP n across, AP n down (in "Move AP n by the numbers")	Moves the selected AP by the numbers. Tap and drag on the floor do the same.	A keyboard way to do what dragging does.
Wall tool: drag on the floor, Add wall top to bottom, Add wall side to side, Remove wall n, Clear walls	Draws or removes walls.	A wall can drop a neighbor below -82 dBm and split one contention domain into two.
Loss per wall (0 to 30 dB, default 10)	One loss for every wall.	Kept to one wall type and one number on purpose; real materials live in the RF Attenuation tool.
AP EIRP (0 to 30 dBm, default 20), Path-loss exponent n (2.0 to 4.0, default 3.0)	Sets how far each AP is heard.	Contention is decided by what each AP hears, so lower power or a lossier building shrinks the groups.
Floor width, Floor depth	20 to 100 m (66 to 328 ft) by 10 to 60 m (33 to 197 ft).	The same APs on a smaller floor hear each other more.

A demo that works (about 3 minutes)

1. Set up: press R. Six APs, 5 GHz, US, DFS off, all on 36 at 40 MHz: one domain of six, about 17% airtime each.
2. Ask the room: "Press Auto-plan at 40 MHz. How many APs still share a channel? Now what happens at 80 MHz?"
3. Reveal: Auto-plan at 40 MHz uses all four channels and leaves pairs sharing (largest domain 2). Press Up to 80 MHz and Auto-plan again: two channels, three APs on each. Press Down to 20 MHz and plan: nobody shares.
4. Optional follow-up: switch DFS on at 80 MHz and plan again, or switch to 2.4 GHz and open "Why they defer, and why 1, 6 and 11" to compare 1 and 6 against 1 and 4.

Pairs with: Adjacent Channels and AP Stacking (why a clean channel still hurts up close), DFS and Radar, the 2.4 and 5 GHz channel cards in Course Handouts. Lesson sequences "Channels and the rules" and "Designing a cell plan".

Left out on purpose: 6 GHz. The mask is a ceiling, not what a radio emits, so the 2.4 GHz figures are the worst case the mask allows. The -82, -72 and -62 dBm thresholds are the commonly cited values. Deferring on a secondary channel is counted as sharing to keep the picture simple. 1/N airtime assumes an even split among APs that all hear each other; clients, retries and load change the real share. Auto-plan knows only the APs on this floor.

Adjacent Channels and AP Stacking

Shows why a channel that does not overlap yours still hurts when the other transmitter is close. A spectrum strip draws your 20 MHz channel, the wanted signal in it, and a neighbor's transmit mask reaching into it, shaded where its energy lands in your channel. Below it, your receiver, the radio you want to hear, and the neighbor sit on a line. Move the neighbor closer or farther and watch the leakage, the signal quality and the highest usable rate change.

Why it's here. Channel Planner shows which channels overlap on paper. This tool is the other half: a transmitter's energy does not stop at its channel edge. The standard's spectral mask limits how much spills into the next channels, and that leakage lands inside your channel, where no receive filter can remove it. Your receiver's filter can only push down the neighbor's own channel, and only so far. That is why two APs on clean channels stacked on one ceiling, or a client sitting next to someone else's AP, still interfere, and why distance matters as much as channel choice.

How to use

1. Start with the question at the top: two APs on channels 36 and 44, 30 cm apart. Pick an answer, then press Reveal to load that scene and read what happens.
2. Drag the Neighbor distance slider under the floor line. The neighbor's curve and the shaded leakage rise and fall; the channel centers never move.
3. Pick the band, the neighbor's transmit mask and width, and the channel separation. In 2.4 GHz the separations are the 1/6/11 pair (25 MHz) and the 1/4/8/11 pairs (20 and 15 MHz).
4. Set who is listening (your client or your AP), the neighbor's power, and the power and distance of the radio you want to hear.

5. Set the receiver selectivity: how far your receiver's filter pushes down the neighbor's own channel. There is one value for the next channel and one for one gap or more, the same at every rate. These values are illustrative: real receivers vary, and no measured selectivity was read.
6. Read the verdicts: the highest MCS (modulation and coding scheme) without and with the neighbor, the SINR (signal to interference plus noise ratio) and SIR (signal to interference ratio), the leakage and the total interference in your 20 MHz, and whether CCA (clear channel assessment) energy detect calls the air busy.
7. On a computer or tablet, Present opens the tool full screen for a projector: Up and Down move the neighbor away and closer, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Band	GHz	2.4, 5 or 6; default 5
Neighbor transmit mask	-	OFDM (orthogonal frequency-division multiplexing, 802.11a/g, 20 MHz only, not in 6 GHz) or HE/EHT (High Efficiency and Extremely High Throughput, 802.11ax/be); default HE/EHT
Neighbor width	MHz	20 in 2.4 GHz; 20 to 160 in 5 GHz; 20 to 320 in 6 GHz; default 20. You always listen on 20 MHz
Channel separation	-	5 and 6 GHz: next channel, one channel gap, two channel gaps (default one gap: 36 and 44). 2.4 GHz: 1 and 6, 4 and 8, 1 and 4
Who is listening	-	Your client (downlink) or your AP (uplink); labels only; default your AP
Neighbor distance	m	0.3 to 30; default 3
Neighbor power	dBm	0 to 30, antenna gain included; default 20
Wanted transmitter power	dBm	0 to 30, antenna gain included; default 20

Input	Unit	Range
Wanted transmitter distance	m	1 to 60; default 10
Path-loss exponent	n	2.0 to 4.0; default 3.0
Receiver selectivity (illustrative)	dB	20 to 60, one value per separation; defaults 35 (next channel) and 51 (one gap or more)
CCA energy-detect threshold	dBm	-82 to -52; default -62

How it works. Received power = transmit power - path loss, with path loss = free-space loss at 1 m + 10 n log10(d), and free space (n = 2) below 1 m. The neighbor's transmit mask is taken in dBr (decibels relative to its in-channel level), linear in dB between its points: OFDM 20 MHz is 0 dBr to 9 MHz from center, -20 at 11, -28 at 20 and -40 at 30 and beyond; HE/EHT 20 MHz is 0 to 9.75, -20 at 10.5, -28 at 20 and -40 at 30, and the wider HE/EHT masks scale the same way (40 MHz: 19.5, 20.5, 40, 60). Leakage L = the mask added up in linear power across your 20 MHz, divided by the mask added up across the neighbor's own channel. Interference = neighbor power + 10 log10(10^(L/10) + 10^(-S/10)), where S is the receiver selectivity: the leakage is already inside your channel, and the filter acts only on the neighbor's own channel, so the two add in milliwatts. SIR = wanted - interference. SINR = wanted - (interference + noise, added in milliwatts), with noise = -174 dBm/Hz + 10 log10(20 MHz) + a 7 dB noise figure = -94.0 dBm. An MCS is usable when the SINR is at least its minimum sensitivity minus the noise floor, using the 20 MHz sensitivities from Rate vs Range; that is the only place the rate enters. CCA energy detect calls the air busy when the interference is at or above the threshold.

Example. Two APs on channels 36 and 44 (one channel gap, 40 MHz center to center), 30 cm apart, each at 20 dBm, HE/EHT mask. The neighbor on 36 arrives at -16.8 dBm. Its mask at 44's center is -40 dBr, and added up across 44's 20 MHz it is -40.0 dBr, so -56.7 dBm leaks into channel 44. With 51 dB of illustrative selectivity, -67.8 dBm of channel 36 gets past the filter, and the total interference is -56.4 dBm. That is above the -62 dBm energy-detect threshold, so the AP on 44 hears the air as busy and waits. Its own client 10 m away arrives at -57.3 dBm, which alone supports MCS 8. With the neighbor the SINR is -0.8 dB, short of the 12 dB MCS 0 needs, so the link is lost. For 2.4 GHz channels 1 and 6 with the OFDM mask, the mask at channel 6's center is -34 dBr but the leakage across its 20 MHz is -29.8 dBr.

Field notes

- The mask is a ceiling on the transmitter, not a measurement. Real radios run at or below it, so the leakage here is the worst case the rule allows.
- The mask value at your center frequency is not the interference level. What matters is the neighbor's energy added up across your whole 20 MHz, which is shown next to it.
- The receiver's rejection does not change with rate. The standard's minimum adjacent-channel rejection (ACR) plus minimum sensitivity is -66 dBm at every MCS (via a 2024 802.11be test white paper): what changes with rate is only how much interference each MCS can absorb. That is why the tool has one selectivity value, not one per rate.
- The selectivity values are illustrative. Real receivers vary, and no measured selectivity was read.
- For a 20 MHz neighbor, the mask is flat at -40 dBr past 30 MHz (1.5 channel widths) from its center, so a second empty channel buys nothing more from the mask. A wider neighbor's mask keeps falling until 1.5 times its width from its center, so there each extra empty channel still helps. Distance always helps.
- Below 1 m the path is taken as free space. At 30 cm the antennas are only a few wavelengths apart, so treat those numbers as rough.
- Walls, antenna patterns and fading are left out. For which channels overlap on paper and who shares airtime, open Channel Planner.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a neighbor on a channel that does not overlap yours still hurts when it is close, because its transmit mask leaks energy into your 20 MHz where no receive filter can remove it, and distance is what saves you.

Control	What it does	Why it's there
Opening question: Yes, a problem / No, they do not overlap; Reveal; Ask again	Records the class's answer; Reveal loads the scene (36 and 44, 30 cm apart) and explains it.	The predict-then-reveal: "Two APs on channels 36 and 44, 30 cm apart on the same ceiling. Any problem?"

Control	What it does	Why it's there
Band: 2.4 / 5 / 6 GHz (default 5)	Changes the channel plan and the separations offered.	2.4 GHz brings the 1/6/11 against 1/4/8/11 comparison; 5 and 6 GHz show gaps between 20 MHz channels.
Neighbor transmit mask: OFDM (802.11a/g) / HE/EHT (802.11ax/be)	Picks the neighbor's mask shape. Shown only when both apply (not in 6 GHz).	Lets the class compare the older and newer masks' skirts.
Neighbor width (MHz)	20 in 2.4 GHz, up to 160 in 5 GHz and 320 in 6 GHz. You always listen on 20 MHz.	A 20 MHz neighbor's mask is flat past 1.5 channel widths, but a wider neighbor's keeps falling, so an extra empty channel helps there.
Channel separation: Next channel, One channel gap, Two channel gaps; in 2.4 GHz 1 and 6 (25 MHz), 4 and 8 (20 MHz), 1 and 4 (15 MHz)	Moves the neighbor's channel. Center frequencies never move on screen.	Separation and distance both matter; for a 20 MHz neighbor the second empty channel buys nothing more from the mask.
Neighbor distance (slider under the floor line, 0.3 to 30 m (1 to 98 ft), default 3 m; Up/Down in Present)	Moves the neighbor along the line.	The main control: leakage falls with distance at the log-distance slope, and this is what separates "stacked on one ceiling" from "down the hall".
Who is listening: Your client (downlink) / Your AP (uplink)	Relabels the receiver and the wanted transmitter. Default your AP.	Covers both cases the help names: two APs stacked on a ceiling, and a client sitting next to someone else's AP.
Neighbor power (0 to 30 dBm, default 20)	The neighbor's power, antenna gain included.	Leakage rises dB for dB with the neighbor's power.
Your AP power / Your client power; Your AP distance / Your client distance (1 to 60 m, default 10)	The wanted signal's power and distance (label follows Who is listening).	A stronger wanted signal survives more leakage, so the same neighbor can be harmless to a close client and fatal to a far one.
Path-loss exponent (model) (2.0 to 4.0, default 3.0)	Sets the propagation slope.	How fast both signals fall with distance.

Control	What it does	Why it's there
Selectivity, next channel / one gap or more (illustrative) (20 to 60 dB; defaults 35 and 51)	How far your receiver's filter pushes down the neighbor's own channel. One slider, for the current separation.	Rejection does not change with rate (ACR plus minimum sensitivity is -66 dBm at every MCS), so one value is enough. Illustrative: no measured selectivity was read.
CCA energy-detect threshold (-82 to -52 dBm, default -62)	The level at which the receiver calls the air busy.	Leakage can stop your AP from transmitting at all, not just lower its rate.
Reset to defaults (R)	Back to the opening defaults.	Returns to the question's starting point.

A demo that works (about 3 minutes)

1. Set up: press R, then read the opening question aloud.
2. Ask the room: "Channels 36 and 44 don't overlap. 30 cm (12 in) apart on one ceiling: problem or not?" Click the room's answer.
3. Reveal: press Reveal. From the help's worked example: -56.7 dBm leaks into channel 44, above the -62 dBm energy-detect threshold, so the AP hears the air as busy; its client 10 m (33 ft) away would get MCS 8 alone, but with the neighbor the SINR is -0.8 dB and the link is lost. Then press Up a few times and watch the link come back as the neighbor moves away.
4. Optional follow-up: switch to Two channel gaps and show that a 20 MHz neighbor gains almost nothing over one gap; then 2.4 GHz and compare 1 and 6 with 1 and 4.

Pairs with: Channel Planner (which channels overlap on paper, who shares airtime). Lesson sequence "Channels and the rules".

Left out on purpose: the mask is the worst case the rule allows; real radios run at or below it. Selectivity values are illustrative. Below 1 m the path is taken as free space, so the 30 cm numbers are rough. No walls, antenna patterns or fading.

Roaming Walk

Walks a client across a floor of access points and shows when it roams and what each roam costs. The client, not the AP, decides: it holds its AP until the signal drops below its trigger, then moves only to an AP that is at least delta stronger. Every roam leaves a gap with no AP, and most of that gap is scanning.

Why it's here. Sticky clients and clients that flip between two APs are two of the most common roaming complaints, and both come from the same two numbers: the trigger and the delta. Apple publishes them for its phones and laptops. Seeing a client walk past a closer, stronger AP because it is still above its trigger explains a lot of field behavior, and shows why a -67 dBm design still leaves a phone on its AP down to -70 dBm.

How to use

1. Pick a client. The phone and laptop presets use Apple's published trigger and delta; the sticky and jumpy clients are illustrative, because other client platforms publish no roam thresholds. Or set the trigger and delta yourself.
2. Press Play, or use Step and the walk-time slider. On the floor plan the lime line joins the client to its AP. Below it, every AP's signal is drawn in that AP's color, the lime line is the AP the client is using, the dashed line is the trigger, and each roam is marked with its gap in milliseconds (PP marks a ping-pong).
3. Turn on 802.11k, PMK caching or 802.11r (FT) and watch the gap per roam change. Drag the APs or pick another path, including one you draw, and add shadowing to see a jumpy client ping-pong between two APs of similar strength.
4. On a computer or tablet, Present opens the walk full screen for a projector: Space plays or pauses, the Right arrow steps 1 s, R restarts the walk, Up and Down move the roam trigger, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Client	trigger dBm, delta dB	phone transmitting -70/8, phone idle -70/12, laptop -75/12, illustrative sticky -85/12 and jumpy -65/2, or custom (trigger -90 to -55, delta 0 to 20)

Input	Unit	Range
802.11k, PMK caching, 802.11r (FT)	on or off	off by default
Illustrative timings	ms and channels	dwell 5 to 100 ms per channel (default 10), 1 to 40 channels in a full scan (default 25), 1 to 10 ms per authentication frame (default 3), 0 to 200 ms of authentication server time (default 50)
Floor	m	60 m x 20 m, 2 to 6 APs you can drag; paths along the corridor, under the APs, corner to corner, across at 20 m, down and back, or drawn by you
Band, AP power, path-loss exponent, shadowing	GHz, dBm, n, dB	2.4, 5 or 6 GHz; 5 to 30 dBm EIRP (default 14); n 2.0 to 4.0 (default 3.0); shadowing 0 to 6 dB (default 2)

How it works. Signal from each AP: RSSI = EIRP - FSPL(1 m) - 10 x n x log10(d) + S, with d in meters (at least 1 m), FSPL(1 m) = 20 x log10(4 x pi x f / c), and S a seeded Gaussian shadowing term that is correlated along the path over about 5 m. The client samples every 100 ms while walking at 1.4 m/s. While the serving RSSI is below the trigger, it roams to the strongest other AP if that AP is at least delta stronger. Roam gap = scan time + authentication time. Scan = dwell x channels, where 802.11k limits the channels to the neighbors, at most 6. Authentication = frames x time per frame, plus server time for full 802.1X. Full 802.1X is 17 frames here (2 Open authentication, 2 reassociation, 2 EAP identity, 6 EAP method frames as an illustrative count that really depends on the EAP type, 1 EAP-Success, 4 EAPOL-Key); PMK caching is 8 and applies only to an AP the client already joined; FT over the air is 4. A ping-pong is a roam back to the previous AP within 5 s.

Example. Default floor with shadowing off, 3 APs 20 m apart and a walk along the corridor: the transmitting phone roams twice, at 17.7 s and 32.0 s, and spends 0.2 s below -70 dBm. The laptop values give 11.8 s below -70 dBm, and the illustrative sticky client never roams and spends 23.7 s there. Each full 802.1X roam costs 351 ms (250 ms scanning, 101 ms authenticating); with 802.11k and FT it costs 32 ms.

Field notes

- This is a teaching model: APs radiate equally in every direction, there are no walls, and the contours on the floor show the average signal without shadowing.

- Every timing default is illustrative. For context, Mishra, Shin and Arbaugh (2003) measured handoffs of 58.74 to 396.76 ms on 802.11b with open authentication, with scanning over 90% of the time, and one FT lab capture read 14 ms. Neither is used by the model.
- Because scanning dominates, 802.11k shortens a slow roam more than FT does. FT and PMK caching shorten only the authentication.
- 802.11v lets the AP suggest a better AP; the client still decides. The model does not simulate it.
- Real clients average their readings, scan on their own schedule and weigh more than signal strength.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: the client decides when to roam: it holds its AP until the signal falls below its trigger, then moves only to an AP at least delta stronger, and most of each roam's gap is scanning.

Control	What it does	Why it's there
Play / Pause / Play again, Step (Step 1 s), Restart (R), Walk time, Speed: 1x (walking pace), 2x, 5x, 10x	Runs, steps and scrubs the walk.	Lets you stop the client just before a roam and ask the room what happens next.
Client: Phone, transmitting (-70/8); Phone, idle (-70/12); Laptop (-75/12); Illustrative sticky (-85/12); Illustrative jumpy (-65/2); Custom	Loads a trigger (dBm) and delta (dB).	The phone and laptop values are Apple's published rules; sticky and jumpy are illustrative because other client platforms publish no roam thresholds. Generic names on screen, the real source in the source line.
Trigger (-90 to -55 dBm; Up/Down in Present)	Where the client starts looking.	Set too low, the client is sticky and clings to a distant AP at a poor rate.
Delta (how much stronger a new AP must be) (0 to 20 dB)	The hysteresis.	Set too small, the client ping-pongs between two APs of similar strength.

Control	What it does	Why it's there
802.11k neighbor list, PMK caching, 802.11r fast transition (FT)	Shorten the scan (k) or the authentication (PMK caching, FT).	Scanning dominates a slow roam, so 802.11k shortens it more than FT does.
Illustrative timings: Dwell per channel, Channels in a full scan, Time per auth frame, Authentication server time (802.1X)	Set what a roam costs.	Every timing is a labeled parameter: the only measured figures (Mishra et al. 2003, one FT capture of 14 ms) are shown as context, not used.
Path: Along the corridor, Under the APs, Corner to corner, Across at 20 m, Down and back, Draw your own (Use this path, Undo, Cancel)	Picks or draws the walk.	Different paths put the client between APs of similar strength, where ping-pong shows.
APs (2 to 6), Move, AP n across, AP n down (or drag on the floor)	Sets AP count and positions. Changing the count spaces them evenly.	Closer APs mean more overlap and earlier candidates.
Band: 2.4 / 5 / 6 GHz; AP power (EIRP, illustrative) (5 to 30 dBm, default 14); Path-loss exponent n	Sets the signal model. The -67 and -70 dBm reach readouts update.	Apple's -70 dBm trigger means a -67 dBm overlap design still leaves a phone on its AP down to -70 dBm.
Shadowing (sigma) (0 to 6 dB, default 2), New shadowing	Adds smooth, seeded shadow fading along the path.	Shadowing is what makes a jumpy client ping-pong between two similar APs.
Show all / Show the last N (roam log)	Expands the roam log.	Keeps the log short on screen.

A demo that works (about 3 minutes)

1. Set up: Restart, Client "Phone, transmitting", three APs, Along the corridor.
2. Ask the room: "The phone walks right past AP 2. Does it roam to it the moment AP 2 is stronger?"

3. Reveal: play it. The phone holds its AP until it falls below -70 dBm, then roams. In the help's example (shadowing off) it roams twice and spends 0.2 s below -70 dBm; the Laptop preset spends 11.8 s there, and Illustrative sticky never roams.
4. Optional follow-up: pick Illustrative jumpy with shadowing on and count the PP (ping-pong) marks; then turn on 802.11k and FT and watch a 351 ms full 802.1X roam drop to 32 ms (help example).

Pairs with: Band Steering (bands on one AP, where this is APs on one band), 802.1X and EAP Ladder in Roam mode (the frames of a roam), Association, Frame by Frame. Lesson sequences "Association and roaming" and "Designing a cell plan".

Left out on purpose: APs radiate equally in every direction and there are no walls; contours show the average signal without shadowing. Every timing is illustrative. 802.11v is not simulated. Real clients average their readings, scan on their own schedule and weigh more than signal strength.

Band Steering

Shows why a dual-band client so often stays on 2.4 GHz, and what an AP can and cannot do about it. The client chooses the band. The AP can only hide (stop answering broadcast probe requests on 2.4 GHz), refuse (reject authentication on 2.4 GHz), suggest (send a BSS, basic service set, Transition Management request naming the 5 GHz network) or force the client off (send a deauthentication on 2.4 GHz). None of the four stops the 2.4 GHz beacons, and after a deauthentication the client still chooses where it goes back. No clause of the IEEE (Institute of Electrical and Electronics Engineers) 802.11 standard defines band steering; the term is industry usage for those four tools.

Why it's here. Band steering is sold as the AP moving clients to 5 GHz. It does not move anyone: even a deauthentication only disconnects the client, which then chooses again. The same AP sounds weaker on 5 GHz: free-space loss alone is 7.1 dB higher at 5.5 GHz than at 2.437 GHz, and walls usually add more, so near the edge of coverage 5 GHz drops below a client's entry level first. Once a client is on 2.4 GHz with a good signal, its own published rules say it does not look again, and walking toward the AP only makes that signal better. Roaming Walk shows a client choosing between APs on one band; this tool is the choice between bands on one AP, with the same log-distance model.

How to use

1. Read the top card: which band the client is on, how far it is from the AP, and one sentence saying why it did what it did.

2. Press Walk to move the client along its path one meter at a time, or use Step 1 m, Reset and the Walked slider. Pick Edge to AP or AP to edge. On the floor the solid ring is where 2.4 GHz can still be heard and the dashed ring is 5 GHz; the line from the AP to the client is the band it is on, in the same pattern. When the client scans, a probe arrow per band points at the AP: two heads when the AP answered, a cross when it did not.
3. Pick the AP's steering mode: off, probe suppression, authentication refusal, a transition request (802.11v amendment), or deauthentication. Pick Client A, B or C and read its rule set; the gauges draw its thresholds as lines. Turn on Client uses a random address while scanning to see the AP lose track of it. Change the extra 5 GHz wall loss; under authentication refusal, how many refusals a client tolerates; under deauthentication, the client's rescan time and the AP's retry interval. Under deauthentication a dashed arrow labeled Deauthentication runs from the AP to the client, the client reads Deauthenticated: no traffic while it rescans, and the top card counts the deauthentications, the returns to 2.4 GHz and the time without traffic.
4. Predict, then reveal: Ask the class loads the question (the client walks from the edge of coverage right up to the AP: does it move to 5 GHz?); pick an answer, then press Reveal to walk it in.
5. On a computer or tablet, Present opens this simulator full screen for a projector: Space walks or pauses, the Right arrow steps 1 m, R resets the walk, Up and Down move the client 5 m along the path, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
AP steering	choice	Off (default), probe suppression on 2.4 GHz, authentication refusal on 2.4 GHz, a BSS Transition Management request (802.11v amendment), or deauthentication on 2.4 GHz
Client	choice	Client A, Client B or Client C (default A); each follows one published rule set, or none, as described on screen
Client uses a random address while scanning	switch	Off (default) or On
Extra 5 GHz wall loss	dB	0 to 10; default 3 (illustrative)
Refusals a client tolerates	count	1 to 10; default 3 (illustrative; used only with authentication refusal)

Input	Unit	Range
Client rescan time	s	1 to 10; default 3 (illustrative; used only with deauthentication): how long the client is off the air, with zero traffic, after each deauthentication
AP retry interval	s	1 to 30; default 5 (illustrative; used only with deauthentication): how long the AP lets the client stay on 2.4 GHz before it deauthenticates it
Walking path	choice	Edge to AP (55 m to 2 m, default) or AP to edge
Client C's driver supports transition requests	switch	On (default) or Off; shown for Client C only

How it works. Signal on each band: RSSI = EIRP - [FSPL(1 m) + 10 x n x log10(d)] - extra loss, where EIRP is the effective isotropic radiated power, FSPL(1 m) the free-space path loss at 1 m and n the path-loss exponent, with EIRP 14 dBm on both bands and n = 3.0 (both illustrative, the same model and defaults as Roaming Walk), 2437 MHz for 2.4 GHz and 5500 MHz for 5 GHz, and the extra wall loss on 5 GHz only. The frequency term makes 5 GHz 20 x log10(5500 / 2437) = 7.1 dB weaker at every distance. Nothing is heard below -82 dBm (illustrative), and the AP hears the client wherever the client hears the AP. Each ring is where its band reaches -82 dBm. The client walks one meter per step. When it is not connected, or its own rule says to look, it scans: one broadcast probe request per band, 5 GHz first. The AP matches a dual-band client by seeing the same MAC (media access control) address in probe requests on both radios, and remembers it for the walk; with the random-address switch on, every probe comes from a random address and the AP never matches it. Client A looks below -70 dBm and moves only to a band at least 8 dB stronger (12 dB when idle; this tool uses 8); at join it takes 5 GHz when 5 GHz is at or above -70 dBm, else the stronger band (a model choice). Client B joins only above -80 dBm on 2.4 GHz or -77 dBm on 5 GHz, re-runs selection only when its link is not above -73 dBm on 2.4 GHz or -70 dBm on 5 GHz, and picks the higher throughput estimate: the Shannon limit of a 20 MHz (2.4 GHz) or 80 MHz (5 GHz) channel at the signal capped at -73 or -70 dBm, over thermal noise plus a 7 dB noise figure, with the current network's score times 1.2 (all illustrative). Client C joins the strongest band and has no rule for leaving. Under probe suppression the AP does not answer a broadcast probe on 2.4 GHz from a matched client; a probe naming the network is answered and beacons still go out. Under authentication refusal the AP refuses a matched client on 2.4 GHz until it has refused it the tolerated number of times, then lets it in; a refused client tries its other acceptable band, and with none it tries 2.4 GHz again at the next step. A transition request goes out right after the client joins 2.4 GHz and every 5 m walked after that (illustrative); Client A accepts when 5 GHz is at or above -70 dBm (a model choice), Client B when 5 GHz passes its -77 dBm entry level, Client C when it can

hear 5 GHz and its driver supports the request. A decline gives status 7, no suitable candidates. Under deauthentication the AP sends a deauthentication to a matched client once it has been on 2.4 GHz for the retry interval (default 5 s, illustrative). The client is off the air, with zero traffic, for the rescan time (default 3 s, illustrative), then joins by its own rules; if they put it back on 2.4 GHz, the AP deauthenticates it again after the retry interval, a loop. Time counts the walk at 1 m each second (illustrative).

Example. Defaults (Client A, steering off, edge to AP, 3 dB extra 5 GHz loss): at 55 m 2.4 GHz arrives at -78.4 dBm and 5 GHz at -88.5 dBm, below the floor, so the client joins 2.4 GHz. Walking in, 2.4 GHz rises above -70 dBm at about 29 m, and from there Client A does not look again; at 2 m it is still on 2.4 GHz at -35.2 dBm, with 5 GHz at -45.3 dBm. Client B stays on 2.4 GHz the same way. With a transition request, Client B declines (status 7, no suitable candidates) until 5 GHz reaches -77 dBm, then accepts and moves at 20 m. With deauthentication (5 s retry, 3 s rescan), Client A walking in is first deauthenticated at 33 m, comes back to 2.4 GHz three times while 5 GHz is below -70 dBm, and lands on 5 GHz at 6 m after 4 deauthentications and 12 s without traffic.

Field notes

- Probe suppression and authentication refusal are modeled on one open-source access point implementation, whose own configuration notes warn that both can cause connection problems and slow down finding and joining the AP. How commercial APs do it, and their thresholds, are not published in anything read for this tool.
- Clients A, B and C follow rules published by the makers of three client platforms, on their own support and developer pages. Device makers can change every number, so each is one published rule set, not how every device behaves. Client C's maker publishes support for transition requests and no band rule.
- The transition request's answer codes are from the standard's list: 0 accepts; 7 means no suitable candidates.
- Deauthentication is a standard frame, but using it to steer is vendor behavior, not a steering mechanism the 802.11 standard defines. No published source gives AP deauthentication steering timings, so the rescan time, the retry interval and the 1 m per second walk they count on are illustrative.
- A client using PMF (Protected Management Frames, 802.11w) ignores an unprotected deauthentication from anyone but its AP. The AP's own deauthentication is protected and still works, so the tool applies it to every client.
- The AP deauthenticates only a client it has matched on 5 GHz, the same test probe suppression and authentication refusal use. A client that stops scanning before it can hear 5 GHz is never matched, so it is never deauthenticated.

- Platforms document scanning from random MAC (media access control) addresses when not connected; that is what the random-address switch models.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: the client chooses the band; the AP can only hide, refuse, suggest or disconnect, and the same AP sounds weaker on 5 GHz, so a client that joined 2.4 GHz at the edge stays there by its own published rules.

Control	What it does	Why it's there
Walk / Pause, Step 1 m, Reset, Walked slider (on the stage)	Moves the client along its path 1 m (3 ft) at a time.	Walking toward the AP only makes the 2.4 GHz signal better, and the class needs to see that the client still does not move.
Walking path: Edge to AP / AP to edge	Direction of the walk (55 m to 2 m by default).	Edge to AP is the question's scene: joining 2.4 GHz at the edge, then never looking again.
Predict, then reveal: Ask the class; Yes, it moves to 5 GHz / No, it stays on 2.4 GHz; Reveal; Close the question	Loads the question (steering off, edge to AP), then walks it in.	The question: "The client walks from the edge of coverage right up to the AP. Does it move to 5 GHz?" Under Client A's and Client B's rules, no.
AP steering: Off; Probe suppression on 2.4 GHz; Authentication refusal on 2.4 GHz; Transition request (802.11v amendment); Deauthentication on 2.4 GHz	Picks what the AP does to a client it has heard on 5 GHz.	The AP's tools, weakest to strongest, and none of them hides the 2.4 GHz beacons. Deauthentication, the fourth, only disconnects the client, which then chooses again.
Client: Client A / Client B / Client C	Loads one published rule set, drawn as threshold lines on the gauges.	Each is one platform owner's published rules, with no vendor or OS name on screen. Client C has no published band rule.
Client uses a random address while scanning	The AP's tracker loses the client.	The AP recognizes a dual-band client by its MAC on both radios; a random scan address stops probe suppression and refusal from applying.

Control	What it does	Why it's there
Client C's driver supports transition requests (Client C only)	Whether Client C answers an 802.11v request.	Client C's maker publishes transition-request support and no band rule.
Extra 5 GHz wall loss (illustrative) (0 to 10 dB, default 3)	Adds loss on 5 GHz only.	Free space alone makes 5 GHz 7.1 dB weaker than 2.4 GHz, and walls usually add more, so 5 GHz drops below the entry level first at the edge.
Refusals a client tolerates (illustrative) (1 to 10, default 3)	Used only with authentication refusal.	How many refusals a client tolerates is not published, so it is a labeled slider.
Client rescan time (illustrative) (1 to 10 s, default 3); AP retry interval (illustrative) (1 to 30 s, default 5)	Used only with deauthentication.	No source gives deauthentication timings, so 3 s and 5 s are illustrative. Only a 5 GHz-capable client on 2.4 GHz is deauthenticated.

A demo that works (about 3 minutes)

1. Set up: Reset, Client A, steering Off, Edge to AP.
2. Ask the room: press Ask the class and read the question.
3. Reveal: press Reveal and let it walk. At 55 m (180 ft) 5 GHz is below the floor, so it joins 2.4 GHz; above -70 dBm on 2.4 GHz (about 29 m) Client A stops looking, and at 2 m it is still on 2.4 GHz with 5 GHz at -45.3 dBm (help example).
4. Optional follow-up: pick Deauthentication and walk again. Client A is thrown off at 33 m, comes back to 2.4 GHz three times, and lands on 5 GHz at 6 m after 4 deauthentications and 12 s without traffic (help example).

Pairs with: Roaming Walk (APs on one band), Uplink vs Downlink, Why Two Devices Disagree. Lesson sequence "Why clients misbehave".

Left out on purpose: probe suppression and refusal follow one open-source AP's documented behavior; commercial thresholds are not published. Client rules are one published set each; device makers can change every number. Deauthentication steering is vendor behavior, not a mechanism the standard defines, and its timings are illustrative. A PMF client ignores an unprotected deauthentication from anyone but its AP.

Survey Walk

Walks a site survey down a corridor-and-rooms floor and shows why walking speed matters even though radio waves travel at the speed of light. A scanner visits one channel at a time, so each channel is measured once per revisit, and its samples land walking speed x revisit time apart. The tool shows that spacing against the guess range, the white stretches with no data, and how the number of network interface cards (NICs), the channel list, the dwell and the hopping algorithm change it.

Why it's here. Walking never distorts a single measurement: at 1.4 m/s the channel holds still for 14 to 37 ms, and the longest Wi-Fi frame lasts 5.484 ms. What walking speed changes is how far apart the samples of each channel land, and how far a pause or a timestamp shortcut moves them from where they were really taken. A laptop's one radio scanning 28 channels at 250 ms needs 7.0 s to come back to a channel, which is 9.8 m at a normal walk, twice a 5 m guess range. Seeing that on the floor explains why survey guidance asks for a steady, slower pace and why more radios help.

How to use

1. Answer the opening question, then press Walk, or use Step and the walk-time slider. The strip under the floor lights the channel each NIC is on right now.
2. Pick a channel to show. Its samples are placed along the path where the survey app puts them; the band around the path is the guess range, and any stretch with no sample within half of it turns white, meaning no data, never no coverage. Pick All channels to see every sample, shaped by band (circle 2.4 GHz, square 5 GHz, diamond 6 GHz).
3. Change the data-collection device from 1 NIC (a laptop's built-in radio) to 2, 3 or 4 NICs, the channels scanned, the dwell and the hopping algorithm, and watch the revisit time, the spacing and the Rule 4 verdict change. Rule 4: the longest revisit allowed is the guess range divided by the walking pace.
4. Turn on Pause at the door, or switch timestamps to Per cycle, and watch placed samples slide away from the rings that mark where they were really taken. Try Stop and go and Line capture, and the Active and Hybrid survey types.
5. Turn on Signal along the path to see each sample's level: a moving scanner draws a new fade at every sample, while standing still repeats one fade.
6. On a computer or tablet, Present opens the walk full screen for a projector: Space plays or pauses, the Right arrow steps 1 s, R restarts the walk, Up and Down change the walking pace, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Survey type	passive, active or hybrid	passive by default; hybrid needs at least 2 NICs, because one holds the active connection
Capture method	continuous, line or stop and go	continuous by default; stop and go completes two full scan cycles at each stop, 2 to 20 m apart (default 5 m)
Data-collection device	NICs	1 NIC (a laptop's built-in radio), 2, 3 or 4 NICs, on the same channels and dwell
Channels scanned	channels	2.4 GHz 1, 6, 11 (3); plus US 5 GHz (28, default); plus the 15 6 GHz Preferred Scanning Channels, PSCs (43); every US 20 MHz channel (95); 6 GHz only (59); or the first 1 to 95 of the full list
Dwell and channel switch	ms	dwell 20 to 500 ms per channel (default 250, a common survey default); switch 0 to 50 ms per hop (default 0, illustrative)
Hopping algorithm	sequential, band per NIC or priority	priority channels default to 1, 6, 11, 36 and 149, visited every k-th slot, k 2 to 6 (default 3)
Walking pace and guess range	m/s, m	pace 0.5 to 2.0 m/s (default 1.4); guess range, also called accuracy distance, 1 to 20 m (default 5)
Timestamps and the door	per channel or per cycle; s	per channel by default; a pause of 1 to 10 s (default 3) halfway along the first leg

How it works. Each scanning NIC spends one slot of (dwell + switch) on a channel and the visit is stamped at the end of the slot. Sequential, shared: the NICs take the next channel in turn, so revisit = ceil(channels / NICs) x slot. Band per NIC: each NIC owns whole bands and spare NICs split the slowest band, so each band has its own revisit. Priority: every k-th slot goes to the next priority channel and the others go round the rest. Spacing = pace x revisit. Rule 4: longest allowed revisit = guess range / pace, checked against the longest revisit of any channel. Continuous capture places each sample by spreading time evenly between clicks (start, turns, stop); Line does the same per straight segment

and does not record the 2 s pause between segments; Stop and go places every sample at its stop. Per-cycle timestamps give every sample of a cycle the time the cycle completes. Signal: effective isotropic radiated power (EIRP) 14 dBm - free-space path loss at 1 m (FSPL) - 10 x 3.0 x log10(d), plus Rayleigh fading from 24 seeded scatterers evaluated at the sample's true position. The active client uses the strongest of our APs and tests once a second (illustrative).

Example. Straight corridor, 58 m at 1.4 m/s, 28 channels at 250 ms, 5 m guess range. 1 NIC: revisit 7.0 s, spacing 9.8 m, and Rule 4 fails (3.57 s allowed). 2 NICs: 3.5 s and 4.9 m, which only just passes. 3 NICs: 2.5 s and 3.5 m. 4 NICs: 1.75 s and 2.45 m. A 3 s pause at the door in continuous mode misplaces samples by up to 2.0 m; per-cycle timestamps misplace them by up to 9.45 m.

Field notes

- White on a survey map means no data, not no coverage. The fix is to go get the data, not to widen the guess range.
- An active survey shows where one client, at one point in time, on one walking path, with one set of other traffic, roams. It does not show where or how other clients will connect, and it sees only a single AP at a time. Always do a passive survey as well.
- Whether real survey apps stamp each channel or each cycle is not published, and neither is how they place samples between clicks.
- Counting 20 MHz channels in the US, 2.4 + 5 + 6 GHz is 11 + 25 + 59 = 95. At 250 ms per channel on one radio that is almost 24 s per cycle.
- This is a teaching model: the walls are drawn but not modeled, and every timing that is not a common default is marked illustrative.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: walking never distorts a measurement, but walking speed sets how far apart each channel's samples land, through the scanner's revisit time, and the survey rule says the longest revisit allowed is the guess range divided by the pace.

Control	What it does	Why it's there
Predict, then walk: Yes, it matters / No, light is too fast; Reveal now; Ask again	Opening question; the answer appears after the first full walk.	The opening card: "RF travels at the speed of light, so does walking speed matter?" Both answers are half right.
Walk / Pause / Walk again, Step, Back to the start (R), Walk time, Speed (1x to 10x)	Runs and scrubs the walk.	Lets you stop and point at a white gap as it forms.
Survey type: Passive / Active / Hybrid	Active shows its five limits and greys out every AP but the associated one. Hybrid needs 2 NICs.	Active surveys are taught for their limits: one client, one point in time, one path, one set of traffic, a single AP.
Data-collection device, by network interface cards (NICs): 1 NIC (a laptop's built-in radio), 2, 3 or 4 NICs	Same channels and dwell on 1 to 4 radios.	A generic device, no product name, 1 to 4 NICs. At the defaults 1 NIC fails the revisit rule and 2 NICs only just pass.
Channels scanned: 2.4 GHz 1, 6, 11 (3); + US 5 GHz (28); + 6 GHz PSCs (43); every 20 MHz channel, US (95); 6 GHz only (59); Custom count (Channel count slider)	The channel list the scanner cycles.	Revisit time grows with the list: 95 channels at 250 ms is almost 24 s per cycle on one radio.
Dwell per channel (20 to 500 ms, default 250); Channel switch per hop (illustrative) (0 to 50 ms)	Time on each channel, and switching overhead.	250 ms is labeled a common survey default; dwell times the channel count is the cycle.
Hopping algorithm: Sequential / Band per NIC / Priority; Channels in use (priority) chips; Priority every k-th slot (2 to 6, default 3)	How NICs share the list.	Pace works in tandem with how many NICs there are and how the scanner changes channel. Priority channels get short spacing and the rest get longer.
Capture method: Continuous / Line / Stop and go; Stop every (2 to 20 m, Stop and go only)	Where samples land between clicks.	Continuous spreads samples evenly in time, so a pause slides them; stop and go stacks them at a point in one fade.

Control	What it does	Why it's there
Timestamps: Per channel / Per cycle	One stamp per channel or per cycle.	Per-cycle stamping moves early channels up to one revisit distance off; which one products use is not published.
Pause at the door; Pause (1 to 10 s, default 3)	Stops the walker halfway along the first leg with no click.	Shows placed samples sliding away from where they were taken (rings mark the true spots).
Walk path: Straight corridor, Loop, Perimeter, Draw your own	Picks or draws the walk.	Longer paths and turns add clicks and pauses.
Walking pace (0.5 to 2.0 m/s, default 1.4; Up/Down in Present)	Walker speed.	Spacing = pace x revisit time. Comfortable gait is 1.27 to 1.46 m/s (4.2 to 4.8 ft/s).
Guess range (accuracy distance) (1 to 20 m, default 5)	How far the map may guess from a sample.	A good indoor choice is 5 m. Any stretch with no sample within half the guess range turns white: no data, never no coverage.
Show this channel's samples: All channels (shape = band) or one channel	Filters the dots.	Each channel has its own spacing, so one channel's gaps are the lesson.
Signal along the path; New fading	Shows each sample's level, with a new fading draw.	A moving scanner draws a new fade at every sample; a stationary one repeats one fade.

A demo that works (about 3 minutes)

1. Set up: Back to the start. Passive, 1 NIC, 28 channels at 250 ms, 1.4 m/s (4.6 ft/s), guess range 5 m (16 ft), Straight corridor.
2. Ask the room: the opening question. Record the answer.
3. Reveal: walk it. 1 NIC revisits a channel every 7.0 s, 9.8 m (32 ft) apart, and Rule 4 fails (3.57 s allowed). Switch to 2 NICs: 3.5 s and 4.9 m, which only just passes (help example).

4. Optional follow-up: turn on Pause at the door in Continuous (samples slide up to 2.0 m), then switch Timestamps to Per cycle (up to 9.45 m).

Pairs with: Heat Map Builder, then Predict, Then Measure. Lesson sequence "Surveys and heat maps".

Left out on purpose: the walls are drawn but not modeled. Whether real survey apps stamp per channel or per cycle, and how they place samples between clicks, is not published. No product names or vendor timings anywhere. No active-survey direction-skew mode.

Heat Map Builder

Builds a heat map from survey samples you place on a floor, so you can see which cells are measurements and which are guesses. Only the dots are measured. Every other cell is filled in by an interpolation method, and beyond the guess range the map should be white: no data, not no coverage.

Why it's here. Every survey heat map is mostly guesses drawn in the same colors as the measurements. Seeing three samples paint a whole floor, a hidden wall the map never saw, and the error shrink as samples get closer explains why sample spacing, averaging and a sensible guess range matter more than the colors. No survey product publishes its heat-map algorithm, so each method here is one documented method from the interpolation literature, not any product's method.

How to use

1. Place samples: tap the floor to add one, or press Grid or Corridor walk and set the spacing (1 to 10 m). Switch the tap to Inspect a cell to see which samples made a cell's value, drawn as lines as thick as their weights.
2. Pick the method. Inverse distance weighting (IDW) averages the nearest samples, weighted by 1 / distance to the power p; raise p from 1 toward 6 and the map turns blocky, like nearest neighbor. Average in dB or in milliwatts (mW) and watch the answer move.
3. Set the guess range (1 to 20 m). Cells farther than that from every sample are white. Beyond it, Flat IDW keeps filling and goes flat, and Path-loss fill uses a log-distance model fitted to each AP's samples.

4. Switch the map between Heat map, Truth and Error map. The readouts give the root mean square error (RMSE), the typical size of the map's error in dB, and the largest error. Reveal the hidden wall to see where the error comes from, and run the Spacing experiment to plot RMSE against grid spacing, with and without noise and averaging.
5. Predict, then reveal: Start: three dots asks whether three samples should paint the whole floor. The next reveal widens the guess range, and the last one shows the hidden wall on the error map.
6. On a computer or tablet, Present opens the tool full screen for a projector: Space takes the samples again with new noise, R resets, Up and Down move the guess range 1 m, P steps the power through 1, 2, 4 and 6, E switches heat map, truth and error map, W shows or hides the hidden wall, N is the next step of the three-dots lesson, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Samples	m	taps (snapped to 0.5 m), a grid or a corridor walk with spacing 1 to 10 m (default 3); up to 1,200 samples
Method	-	IDW with power p 1 to 6 (default 2), or nearest neighbor
Averaging domain	dB or mW	dB by default
Guess range	m	1 to 20 (default 5)
Beyond the guess range	-	Off (white), Flat IDW, or Path-loss fill
Noise and averaging	dB, readings	sigma 0 to 8 dB (default 0, illustrative); 1 to 36 readings averaged per point
Floor	m, n	40 m x 25 m, 1 to 4 APs (default 2) at 17 dBm effective isotropic radiated power (EIRP) on 5 GHz, path-loss exponent n 2.0 to 4.0 (default 3.0), three drawn walls and one hidden 12 dB wall

How it works. Truth at a point: the strongest AP's EIRP, minus the free-space path loss (FSPL) at 1 m for 5.5 GHz, minus 10 x n x log10(d), minus the loss of every wall crossed, with d in meters (at least 1 m). A sample reads every AP at its spot; with noise on, each reading is the mean of N Gaussian draws of standard deviation sigma, so the noise left is sigma / sqrt(N). IDW: z = sum(w x z_i) / sum(w), w = 1 / d_i^p, over the 8 nearest samples within the guess range; a cell on a sample takes its value. In milliwatts each value is converted with mW = 10^(dBm / 10), averaged with the same weights and converted back. Nearest neighbor takes the nearest sample within the guess range. Beyond the guess range: Off leaves the cell with no data; Flat IDW uses the 8 nearest samples at any distance; Path-loss fill fits reading = A - 10 x n x log10(d) to each AP's samples by least squares (n held to 1.6 to 6) and takes the strongest AP. Error = estimate - truth; RMSE is over the cells with data. The spacing experiment re-runs a grid at 1, 2, 3, 5, 7 and 10 m and scores every cell, filling beyond the guess range with flat IDW when extrapolation is off.

Example. A cell 2 m, 4 m and 6 m from samples of -55, -65 and -70 dBm: with p = 1 the weights are 0.500, 0.250 and 0.167 and the cell reads -60.5 dBm; p = 2 gives -58.1 dBm; p = 4 gives -55.8 dBm, almost the nearest sample; p = 2 averaged in milliwatts gives -56.2 dBm. With n = 3, going from 10 m to 15 m from an AP costs 10 x 3 x log10(1.5) = 5.3 dB, while flat IDW past the last sample changes by less than 1 dB; add one unmeasured 12 dB wall and the error on the far side is about 17 dB.

Field notes

- No survey vendor publishes its heat-map algorithm. Each method here is one documented method, and the tool never says which method any product uses. Which domain, dB or milliwatts, products average in is not published either.
- White means no data, not no coverage. A cell farther than the guess range from every sample was never measured.
- IDW can never predict a value outside the range of its samples, so it cannot show signal falling off past the last sample. A path-loss model can, but only for distance: it cannot see a wall no one measured across.
- Closer samples help until noise on single readings dominates. Averaging several readings per point, or moving through the fades while measuring, brings the error back down.
- This is a teaching model: the truth is a log-distance model with wall losses, the noise is illustrative, and real maps also carry antenna, adapter and position errors.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a heat map is mostly guesses drawn in the same colors as the measurements; the interpolation method, the guess range and the sample spacing decide how wrong it is, and white means no data, not no coverage.

Control	What it does	Why it's there
A tap on the floor will: Add a sample / Inspect a cell	A tap adds a sample (snapped to 0.5 m) or shows which samples made a cell, drawn as lines as thick as their weights.	Inspecting a cell shows the map is a weighted average of dots, not a measurement.
Remove the last sample, Clear every sample	Undo and clear.	Lets you rebuild the three-dots scene by hand.
Grid, Corridor walk; Grid and walk spacing (1 to 10 m, default 3)	Places samples on a grid or along a corridor.	Closer samples make the map more accurate, up to the point where noise on single readings dominates.
Method: IDW / Nearest neighbor	Inverse distance weighting, or copy the closest sample.	One documented method from the interpolation literature; no survey vendor publishes its algorithm, so the tool never says which a product uses.
Power (p) (1 to 6, default 2; P steps 1, 2, 4, 6 in Present)	How local the IDW guess is.	High power approaches nearest neighbor and turns the map blocky; the worked example shows -60.5, -58.1 and -55.8 dBm for p = 1, 2 and 4.
Average the samples in: dB / Milliwatts (mW)	Averaging domain.	The answer moves (-58.1 against -56.2 dBm at p = 2), and which domain products use is not published.
Guess range (1 to 20 m, default 5; Up/Down in Present)	Cells farther than this from every sample are white.	No data is not no coverage. The fix is to go get the data, not to widen the range.
Beyond the guess range: Off / Flat IDW / Path-loss fill	What fills past the guess range.	IDW can never go below its lowest sample, so it goes flat past the last point; a path-loss model can extrapolate distance but cannot see an unmeasured wall.

Control	What it does	Why it's there
Map view: Heat map / Truth / Error map (E in Present)	Switches what the floor shows.	The error map is where the hidden wall shows up.
Start: three dots, then Reveal: widen the guess range, Reveal: the hidden wall, End the lesson	The built-in three-step lesson (N in Present).	The question: "Three dots can paint the whole floor green. Should they?"
Reveal wall / Hide wall	Shows or hides the 12 dB wall no sample crossed (W in Present).	A model cannot see a wall no one measured across: about 17 dB of error on the far side.
Spacing experiment	Re-runs the grid at several spacings and plots RMSE against spacing.	Shows accuracy improving with spacing, then flattening once noise dominates.
Noise on one reading, sigma (0 to 8 dB, illustrative); Readings averaged per point (1 to 36); Take the samples again (Space in Present)	Adds fading noise and averaging.	Averaging N readings shrinks the noise by the square root of N.
Inspected cell, across / down	Moves the inspected cell by the numbers.	A keyboard way to inspect a cell.
APs (1 to 4, default 2); Path-loss exponent (n) (2.0 to 4.0)	Sets the truth the samples read.	Changes the floor under the same method so the class sees the method's error, not the building's.

A demo that works (about 3 minutes)

1. Set up: R, then press Start: three dots (N).
2. Ask the room: "Three dots can paint the whole floor green. Should they?"
3. Reveal: N widens the guess range and the floor fills with confident color from three readings; N again reveals the hidden wall on the error map, about 17 dB wrong on its far side.
4. Optional follow-up: press P to step the power and watch the map turn blocky, or run the Spacing experiment.

Pairs with: Survey Walk before it, Predict, Then Measure after it. Lesson sequence "Surveys and heat maps".

Left out on purpose: the truth is a log-distance model with wall losses, and the noise is illustrative; real maps also carry antenna, adapter and position errors. IDW uses the 8 nearest samples: plain IDW over every sample rises past the last point and would fail the "goes flat" lesson.

Predict, Then Measure

Tests a predictive design against the building with an AP on a stick (APoS): a temporary AP placed where the design says, measured on site. Every wall loss in a predictive design is a claim. Walk the floor, compare the Predicted and Measured maps, and see which walls the design got wrong, and which your walk never tested.

Why it's here. A predictive design is a model: walls drawn with assumed losses, APs placed, a heat map computed. It can show good coverage everywhere and still be wrong about a wall. You only learn a wall's real loss by measuring on both sides of it (Keith's Rule 5), so a validation walk that stays on one side of a wall cannot find the error. All wall losses here, including the material defaults, are illustrative, adjustable values, not measured material data.

How to use

1. Pick a scenario (office, school or warehouse; illustrative). The Predicted map is the design. Each wall is labeled with its design loss and 'untested'.
2. Walk the floor: drag on the floor to draw a walk leg (a tap continues the last leg), or press One side only or Both sides of every wall. A sample is taken every meter. Set the noise on each sample (0 to 8 dB, illustrative) to see what real readings do to the estimates.
3. A wall with samples on both sides of it on one leg reads 'tested' with its measured loss. Switch the map between Predicted, Measured (the samples spread with inverse distance weighting (IDW), white where there is no data) and Difference (each sample minus the design at that spot, spread the same way).
4. Update model replaces each tested wall's design loss with its measured loss and recomputes the Predicted map. Untested walls keep what the design said.

5. Predict, then reveal: the design shows green everywhere. What would you check before signing it off? Reveal the truth to see the true wall losses and the Truth map, and which wrong walls your walk missed. Then move the AP, or decide the design needs another AP.
6. On a computer or tablet, Present opens the tool full screen for a projector: Space reveals or hides the truth, M steps through the maps, R resets, U updates the model, C clears the walk, B walks both sides of every wall, O walks one side only, Up and Down change the noise, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Scenario	-	Office (30 m x 20 m, 11 walls), School (32 m x 20 m, 11 walls) or Warehouse (40 m x 26 m, 8 walls); illustrative
Walk	m	drawn legs, or One side only, or Both sides of every wall; a sample every 1 m, up to 600 samples
Noise	dB	0 to 8, standard deviation of each sample (default 0, illustrative)
Wall design loss	dB	0 to 45 per wall; material defaults drywall 3, glass 4, brick 10, concrete 15, elevator shaft 25 (illustrative)
AP position	m	anywhere on the floor, snapped to 0.5 m
Hidden truth	dB	by default two walls 8 to 14 dB worse than the design and one 4 to 8 dB better, picked by a seed; the instructor can set any wall's true loss, 0 to 45 dB (illustrative)

How it works. Signal at a point: an illustrative 20 dBm effective isotropic radiated power (EIRP), minus the free-space path loss (FSPL) at 1 m for 5.5 GHz, minus 10 x n x log10(d), minus the loss of every wall the straight line from the AP crosses, with d in meters (at least 1 m) and n set per scenario (illustrative). The Predicted map uses the design losses, the Truth map the true losses. Each sample reads the truth plus Gaussian noise. Measured map: IDW over the 8 nearest samples within 5 m, weights 1 / d^2; white beyond 5 m. Difference: for each sample, measured minus the design's prediction at that spot, spread with the same IDW. A wall is tested when two neighboring samples on one leg sit on opposite sides of it: the leg crosses it, and the line from the AP crosses it for one sample and not the other, with every other wall the same for both. Its measured loss is S(near) - S(far) - (PL(far) - PL(near)), averaged over every such pair; the other walls cancel. Readouts: walls tested and

untested, the largest difference, the share of the floor where the difference is more than 5 dB, and the share below a -67 dBm design target (illustrative).

Example. Two samples 1 m apart straddle a wall, at 7.5 m and 8.5 m from the AP, with n = 2.8 and no other wall between them and the AP. The near one reads -51.8 dBm and the far one -69.3 dBm. The extra meter costs 10 x 2.8 x log10(8.5 / 7.5) = 1.5 dB, so the wall costs 69.3 - 51.8 - 1.5 = 16.0 dB. If the design said 4 dB, the wall is 12 dB worse than the model.

Field notes

- Wall losses here are illustrative, adjustable values. No primary source for multi-wall model coefficients was used, and real walls of one material vary widely.
- Capture on both sides of what you care about. A sample inside a room tells you nothing about its wall unless another sample sits on the other side of it.
- A design that looks right everywhere is still a model. An AP-on-a-stick walk tests the walls it crosses and nothing else.
- This is a teaching model: straight-line wall losses only, no doors, reflections or floors above and below.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: every wall loss in a predictive design is a claim, and only an AP-on-a-stick walk with samples on both sides of a wall can test it.

Control	What it does	Why it's there
Scenario (illustrative): Office / School / Warehouse	Loads a floor, walls and a hidden truth.	Different buildings, same lesson.
Reset scenario	Resets the design, the AP, the walk and the reveal; keeps the hidden truth.	Lets you walk the same building again.

Control	What it does	Why it's there
A tap or drag on the floor will: Draw the walk / Move the AP	Drag draws a walk leg (a tap continues it); or moves the AP on a stick.	The student designs the validation walk.
One side only; Both sides of every wall (O and B in Present)	Loads a corridor walk that stays on one side, or short legs across every wall.	The Rule 5 contrast: one side leaves walls untested; both sides tests them.
Remove the last leg of the walk; Clear the walk (C in Present)	Undo and clear.	Rebuild the walk.
Noise on each sample (illustrative) (0 to 8 dB; Up/Down in Present)	Adds noise to each reading.	Shows what real readings do to the wall estimates.
Map: Predicted / Measured / Difference / Truth (Truth only after reveal; M in Present)	Switches the map.	The Difference map is each sample minus the design at that spot, which is where a wrong wall shows.
Update model (U in Present)	Replaces each tested wall's design loss with its measured loss.	After correcting the model, the design may need an AP moved, added or removed. Untested walls keep what the design said.
Reveal the truth / Hide the truth (Space in Present)	Shows the true wall losses and the Truth map.	The reveal for "The design shows green everywhere. What would you check before signing it off?"
Wall, "Wall n design loss" (0 to 45 dB), " default, X dB"	Edits a wall's predicted loss; resets it to its material default.	Every loss is illustrative: drywall 3, glass 4, brick 10, concrete 15, elevator shaft 25 dB.
AP across, AP down	Moves the AP on a stick.	Moving the AP walks the same path again from the new spot.
Instructor: the hidden truth: "Wall n true loss", New hidden truth	Sets any wall's true loss, or draws a new seeded truth.	The instructor can plant the wrong wall where the lesson needs it. By default two walls are much worse than the design and one is better.

A demo that works (about 3 minutes)

1. Set up: Office, Reset scenario. Show the Predicted map: green, every wall "untested".
2. Ask the room: "The design shows green everywhere. What would you check before signing it off?"
3. Reveal: press O (one side only) and show the walls still untested. Press B (both sides): walls turn "tested" with measured losses, and the Difference map lights up the wrong ones. Press Space to reveal the truth and show which wrong walls the first walk missed.
4. Optional follow-up: press U and show the Predicted map change, then ask whether the AP needs to move.

Pairs with: Heat Map Builder (the same sample-and-map method), Survey Walk, Room Propagation and Wi-Fi Through a Wall for how one wall attenuates. Lesson sequence "Surveys and heat maps".

Left out on purpose: straight-line wall losses only: no doors, reflections or floors above and below. Wall losses are illustrative values, not material data; no primary source for multi-wall coefficients was used. An AP-on-a-stick walk tests the walls it crosses and nothing else.

Where Am I? Signal Strength vs Round-Trip Timing

Finds a device on a floor of 3 to 6 APs two ways and shows how far off each one is. Signal strength estimates each AP's distance from the level heard; fine timing measurement (FTM), from 802.11mc, estimates it from the round-trip time (RTT) of frames. Three or more distances give a position by trilateration, drawn as circles around the APs and the point that best fits them.

Why it's here. Signal strength does fall with distance, so it can estimate distance, but a few dB of shadowing becomes meters of error, and the error grows the farther away the AP is. Timing measures something light does at about 30 cm per nanosecond, so it gives meter-class distances. Seeing both scatters side by side, and a blocked direct path pulling a timing position off, explains why indoor location needs more than signal strength.

How to use

1. Drag the device across the floor, or use the device position sliders. Each AP's circle is the distance the method estimated; the lime marker is the position that best fits all the circles, and the dotted line to the true position is the position error.

2. Pick the method: Signal strength (diamond), FTM timing (triangle) or Both, side by side. The small marks are 50 repeated estimates; Re-sample draws new measurement errors.
3. For signal strength, set the path-loss exponent n (2 to 4) and the shadowing sigma (0 to 10 dB, illustrative). The readout gives the one-sigma distance factor, 10^(sigma / 10n), and what it does to a device 10 m away.
4. For timing, set the timing error (0.5 to 3 m; a vendor developer document gives 1 to 2 m), and block the direct path to any AP. A blocked AP reads long by the extra distance you set (illustrative), and its circle is drawn dashed.
5. When two circles cannot meet, the tool says so: no point fits every distance, and the marker is the best compromise.
6. Predict, then reveal: your phone sees an AP at -70 dBm. How far away is it? Say a number, then reveal what the model says and how wide one sigma of shadowing makes it.
7. On a computer or tablet, Present opens the tool full screen for a projector: Space re-samples, Tab switches signal strength, FTM and both while no control has focus (Shift+Tab moves into the controls, where Tab moves between them as usual), M also switches the method, R resets, Up and Down move the shadowing sigma 1 dB, B blocks or clears AP 1's direct path, P steps the -70 dBm question, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Method	-	Signal strength, FTM timing, or Both (default Both)
Path-loss exponent n	-	2.0 to 4.0 (default 3.0)
Shadowing sigma	dB	0 to 10 (default 6, illustrative)
Timing error	m	0.5 to 3 one standard deviation (default 1.5; vendor-documented 1 to 2 m)
Blocked direct path	m	per AP, on or off; extra distance 0 to 10 (default 4, illustrative)
Floor	m	30 m x 20 m, 3 to 6 APs (default 4), each radiating 17 dBm on 5 GHz (illustrative)

How it works. Signal strength: the level heard from an AP d meters away is the power it radiates, minus the free-space path loss at 1 m for 5.5 GHz, minus 10 x n x log10(d), plus shadowing (sigma times a seeded standard normal draw). The device inverts the same model without the shadowing: d_est = 10^((power - loss at 1 m - level) / (10 n)). So d_est = d x 10^(-sigma x draw / (10 n)), and a one-sigma error multiplies the distance by 10^(sigma / (10 n)). Timing: distance = speed of light x RTT / 2, plus a seeded Gaussian error with the chosen standard deviation, plus the blocked-path extra distance for a blocked AP, never below zero. Trilateration: the position that minimizes the sum of (distance from the point to AP i minus the estimated distance to AP i) squared, started from a linear solve and refined by Gauss-Newton. The whole measurement is repeated 50 times with new draws; the spread radius is the root mean square distance of those 50 positions from their average. Two circles cannot meet when their APs are farther apart than the two radii added up, or when one circle sits inside the other.

Example. With sigma = 6 dB and n = 3 the factor is 10^(6 / 30) = 1.585, so a device 10 m away reads 6.3 m to 15.8 m, one sigma either way. The lesson's -70 dBm, with each AP radiating 17 dBm on 5 GHz and n = 3, inverts to 21.1 m, and one sigma of shadowing makes it 13.3 m to 33.5 m. The round trip for 10 m takes 66.7 ns, and 1 ns of RTT is 15 cm of distance.

Field notes

- The 1 to 2 m timing accuracy is from a vendor developer document, not a measurement in this tool. Real results depend on the devices and the building.
- A signal-strength distance error is a factor, not a fixed number of meters: at n = 3 the same one sigma of 6 dB that stretches a 5 m reading to 7.9 m stretches a 20 m reading to 31.7 m.
- When the direct path is blocked, the first signal to arrive at the device is a reflection that traveled farther, so timing reads long, never short.
- This is a teaching model: the log-distance model, the shadowing, the power each AP radiates and the blocked-path distance are illustrative, and real location systems also face AP position errors, clock offsets and floors above and below.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: signal strength gives distance badly, as a factor that grows with distance, while fine timing measurement (FTM) gives meter-class distance, and a blocked direct path makes timing read long.

Control	What it does	Why it's there
Method: Signal strength / Timing / Both (Tab or M in Present)	Shows one method's circles and scatter, or both side by side. Default Both.	Side-by-side scatters make the difference visible at a glance.
Re-sample (Space in Present)	Draws new measurement errors.	The 50 repeated estimates show the spread, not one lucky answer.
Ask: -70 dBm, how far?; Reveal the answer; End the question (P in Present)	The built-in question.	The question: "Your phone sees an AP at -70 dBm. How far away is it?"
Path-loss exponent (n) (2.0 to 4.0, default 3.0)	The model the distance is inverted with.	The one-sigma error factor is 10^(sigma / 10n).
Shadowing sigma (illustrative) (0 to 10 dB, default 6; Up/Down in Present)	Random loss on each signal reading.	A few dB of shadowing becomes meters: at 6 dB and n = 3 the factor is 1.585, so 10 m reads 6.3 to 15.8 m.
Timing error, one standard deviation (vendor-documented 1 to 2 m) (0.5 to 3 m, default 1.5)	FTM ranging noise.	Keeps the timing claim tied to its source, a vendor developer document.
Blocked AP chips (AP 1 to AP n; B toggles AP 1 in Present); Blocked-path extra distance (illustrative) (0 to 10 m, default 4)	Blocks an AP's direct path; its circle is dashed and reads long.	The first path to arrive is a reflection, so timing reads long, never short.
APs (3 to 6, default 4); Device position, across / down (or drag the device)	Sets the geometry.	Three or more distances give a position; bad distances give a bad, and sometimes impossible, position.

A demo that works (about 3 minutes)

1. Set up: R. Method Both, four APs, sigma 6 dB.

2. Ask the room: press P. "Your phone sees an AP at -70 dBm. How far away is it?" Take numbers.
3. Reveal: press P again. The model says 21.1 m (69 ft), and one sigma of shadowing makes it 13.3 to 33.5 m (help example). Point at the two scatters: timing is tight, signal strength is wide.
4. Optional follow-up: press B to block AP 1 and watch the timing position pull off while its circle turns dashed.

Pairs with: the Find My, Explained lesson (where the dot on the map comes from), Roaming Walk for the same log-distance model.

Left out on purpose: the log-distance model, shadowing, 17 dBm per AP and the blocked-path distance are illustrative. The 1 to 2 m timing figure is a vendor document, not a measurement here. Real systems also face AP position errors, clock offsets and floors above and below.

Repeaters and Mesh Backhaul

Shows what relaying costs. A corridor holds the root AP, one to three relays (a repeater, extender or mesh node) and a client. Each hop is drawn with its MCS (modulation and coding scheme, the data-rate step) and the throughput it would carry alone, and an animation shows frames crossing the hops: taking turns when the relays use one radio on one channel, all at once when they have a dedicated backhaul radio. The readouts compare the end-to-end throughput with the client connected straight to the AP from the same spot.

Why it's here. A repeater or mesh node with one radio on one channel has to receive every frame and then send it again on that same channel, so its two hops share the same air and take turns. For one radio relaying on one channel the end-to-end throughput T follows 1/T = 1/T1 + 1/T2, where T1 and T2 are what each hop carries alone: two equal hops give half, and a slow hop drags the total toward itself. That is why a repeater can show full bars to a laptop and still be slower than the AP was. A dedicated backhaul radio on another channel, or a cable, lets the hops run at once, and where the relay stands matters more than where the client is: a weak hop back to the AP limits everything behind it.

How to use

1. Read the headline: the end-to-end throughput, the straight-to-the-AP throughput from the same spot, and the forwarding delay. The corridor shows each hop's MCS and throughput; the slowest hop is marked.

2. Drag a relay or the client along the corridor, or use the position sliders. Move the relay toward the AP and watch the hop back to the AP get faster.
3. Pick the number of relays (1 to 3; the relays are spaced evenly) and the backhaul: Same channel, one radio; Dedicated backhaul radio, another channel; or Wired backhaul. Change the band, the efficiency and the forwarding delay; the efficiency and the delay are illustrative.
4. Play the frames runs the animation: on one channel a single frame crosses one hop at a time, and the Air in use lane shows each hop's share of the air back to back. With a dedicated backhaul every hop has its own lane and all send at once; the slowest hop is busy all the time. Reset returns to the opening scene.
5. Predict, then reveal: Ask the class loads the question ("Your repeater shows full bars to the laptop. Why is it slower than before?") with a repeater 30 m down the corridor and the laptop 2 m from it, and hides the throughputs; pick a reason, then press Reveal.
6. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses the frames, Up and Down add or remove a relay, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Relays	count	1 to 3; default 1. Changing the count spaces them evenly between the AP and the client
Backhaul	choice	Same channel, one radio (default); Dedicated backhaul radio, another channel; Wired backhaul
Band	GHz	2.4, 5 or 6; default 5. Path loss is taken at 2437, 5500 or 6525 MHz
Relay and client positions	m	Along a 60 m corridor from the AP, in order, at least 2 m apart; default relay at 18 m, client at 36 m
Efficiency	factor	0.40 to 0.80; default 0.60 (illustrative)
Forwarding delay per hop	ms	0 to 10; default 1 (illustrative)

How it works. Each hop's received level = 20 dBm EIRP (equivalent isotropically radiated power) - path loss, with path loss = free-space loss at 1 m + 10 x 3.0 x log10(d), the model Roaming Walk uses. The hop's MCS is the highest whose minimum receiver sensitivity the level meets (the

Rate vs Range floors: MCS 0 needs -82 dBm at 20 MHz, 3 dB more per doubling of width), held at MCS 11. Its PHY (physical layer) rate is Airtime Anatomy's 802.11ax figure for that MCS at 2 spatial streams and a 0.8 µs guard interval, on 80 MHz at 5 and 6 GHz and 20 MHz at 2.4 GHz. Hop throughput Ti = PHY rate x efficiency. End to end: same channel, one radio, 1/T = 1/T1 + 1/T2 + … over every hop; dedicated backhaul radio, T = the smallest Ti; wired backhaul, T = the last hop alone. A hop with no link (below MCS 0) that the chain needs gives 0. Each hop's share of the air = T / Ti. Delay = forwarding delay x number of hops; straight to the AP is one hop. The straight-to-the-AP figure is the same calculation for one hop from the AP to the client.

Example. Defaults: 5 GHz, one relay at 18 m, the client at 36 m, one channel, efficiency 0.6. Each 18 m hop receives -64.9 dBm, MCS 3, a PHY rate of 288.2 Mb/s, so it carries 172.9 Mb/s alone. On one channel the two take turns: 1/T = 1/172.9 + 1/172.9, so T = 86.5 Mb/s, exactly half. A dedicated backhaul radio gives 172.9 Mb/s. Straight to the AP from 36 m the client gets MCS 0, 43.2 Mb/s. The question's scene: the repeater at 30 m hears the AP at -71.6 dBm (MCS 1, 86.5 Mb/s) and the laptop 2 m away at MCS 11 (720.6 Mb/s): full bars, yet 1/T = 1/86.5 + 1/720.6 gives 77.2 Mb/s, less than the 86.5 Mb/s the laptop got straight from the AP at 32 m. Move the repeater to 12 m and the total rises to 115.3 Mb/s.

Field notes

- The formula 1/T = 1/T1 + 1/T2 is for one radio relaying on one channel. It is worked out from airtime; no published multihop measurement was checked for this tool.
- Adding relays shortens the hops, which raises each hop's rate, but on one channel every hop costs its own airtime. At the default spot two relays beat one, and three lose to two.
- The dedicated backhaul gives every hop a channel of its own. With two or more relays sharing one backhaul channel, those backhaul hops would take turns again.
- The efficiency factor, the forwarding delay, the 20 dBm EIRP, the path-loss exponent and the channel widths are illustrative, chosen to show the effect, not measured. Traffic flows one way, with no neighbors, collisions or retries.
- The delay counts only the forwarding delay at each hop, not the time a frame waits for the shared channel.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a relay with one radio on one channel makes every frame cross the air twice, so the hops take turns (1/T = 1/T1 + 1/T2); a dedicated backhaul radio or a cable lets them run at once, and a weak hop back to the AP limits everything behind it.

Control	What it does	Why it's there
Predict, then reveal: Ask the class; three reasons; Reveal; Done / Close the question	Loads the question's scene (repeater 30 m down the corridor, laptop 2 m from it) and hides the throughputs.	The question: "Your repeater shows full bars to the laptop. Why is it slower than before?"
Relays: 1 / 2 / 3 (Up/Down in Present)	Number of relays, spaced evenly.	Each extra same-channel hop costs airtime again and adds delay. At the default spot two relays beat one, and three lose to two.
Backhaul: Same channel, one radio; Dedicated backhaul radio, another channel; Wired backhaul	How the relays reach the AP.	The core contrast: turns on one channel against the slowest hop (dedicated) against the last hop alone (wired).
Band: 2.4 / 5 / 6 GHz	Changes path loss and channel width.	Shows the same relay lesson at each band's range.
Efficiency (illustrative) (0.40 to 0.80, default 0.60)	Turns PHY rate into what a hop carries.	Labeled illustrative so no one reads the Mb/s as a measurement.
Forwarding delay per hop (illustrative) (0 to 10 ms, default 1)	Adds per-hop delay.	Relaying adds delay as well as cutting throughput.
Relay position / Client position (or drag on the corridor)	Moves a node along the 60 m (197 ft) corridor.	The point: where the relay stands matters more than where the client is. The sliders are the keyboard way to do what dragging does (code comment).
Play the frames / Pause, Reset	Animates frames taking turns, or running in parallel with a dedicated backhaul.	Makes the shared air visible.

A demo that works (about 3 minutes)

1. Set up: Reset. One relay, same channel, 5 GHz.
2. Ask the room: press Ask the class and read the question. Pick the class's reason.
3. Reveal: press Reveal. The laptop has MCS 11 to the repeater, but the repeater hears the AP at MCS 1: 77.2 Mb/s through the repeater against 86.5 Mb/s the laptop got straight from the AP (help example).
4. Optional follow-up: drag the repeater toward the AP (at 12 m the total rises to 115.3 Mb/s), then switch Backhaul to Dedicated and press Space to show both hops sending at once.

Pairs with: Airtime Fairness, the Mesh, Extenders and Wired Access Points explainer.

Left out on purpose: the formula is for one radio relaying on one channel, worked out from airtime; no published multihop measurement was checked. With several relays sharing one dedicated backhaul channel, those hops would take turns again. Traffic flows one way, with no neighbors, collisions or retries. The delay counts only forwarding, not waiting for the channel.

Why a Busy Line Lags

Shows why a video call stutters when someone else in the house uploads. A chart follows the call's round-trip delay over a 20-second run on a home line: idle, then while an upload runs, then after it stops. One switch turns smart queue management (SQM) off or on. Off, the call's delay climbs from the idle figure to hundreds of milliseconds, following the measurements of the FCC (Federal Communications Commission). On, it stays near idle. A picture of the queue at the home line shows why: the call's packet waiting behind the upload, or going in its turn.

Why it's here. Lag is delay, not speed, and the usual fix people reach for, a faster plan, does not cure it. An upload sends as fast as the line allows, so it fills the line's queue on any plan, and every call packet waits behind whatever is in that queue. A faster plan drains the queue faster, but the call still waits. The FCC's 13th Measuring Broadband America report found latency under load significantly higher than idle latency, most of all on DSL (digital subscriber line). What brings the delay down is keeping the queue short, which is the job of active queue management (AQM) such as FQ-CoDel (flow queue controlled delay), which some home routers offer as a smart queue management setting.

How to use

1. Read the headline: the call's delay at the playhead, its delay while the upload runs in the current setting (and how many times idle that is), and the idle figure with the FCC's measured range.
2. Switch Smart queue management (SQM) between Off and On. The solid line is the current setting; the dashed line is the other one, kept for comparison. The bracket marked FCC range is the spread the FCC measured across providers.
3. Pick the home line: Fiber, Cable (the default) or DSL. Each follows the FCC's figures for that technology, and the chart's scale changes with it.
4. Play the call runs the 20 seconds in real time, 1 s on steps a second, and the Time into the run slider or a drag across the chart moves the playhead. The queue picture below the chart shows what the call's packet is waiting behind at that moment. Reset returns to Cable, SQM off, the whole run drawn.
5. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses, Right steps one second, Up turns SQM on and Down turns it off, Q switches it, R resets, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Smart queue management	switch	Off (default) or On
The home line	choice	Fiber, Cable (default) or DSL
Time into the run	s	0 to 20; the upload runs from 4 s to 16 s (illustrative timing)

How it works. Delay = idle + queue. Idle is the middle of the FCC's measured idle range for the technology: fiber 7 to 14 ms, cable 12 to 24 ms, DSL 23 to 34 ms. With SQM off, the queue fills toward (busy - idle), where busy is the median provider's latency under upload load read from the FCC's Chart 7: about 30 ms on fiber, 225 ms on cable and 665 ms on DSL. With SQM on, the queue fills toward 5 ms, the default target of FQ-CoDel in RFC 8290 (RFC: Request for Comments). The queue fills as 1 - e^(-t / 0.8 s) after the upload starts. When the upload stops, a queue holding D ms of data drains in D ms. The chart samples every 100 ms, the rate the FCC sampled latency under load (10 packets a second).

Example. Defaults: a cable line, SQM off. Idle, the call's delay is 18 ms. While the upload runs it climbs to 225 ms, 12.5 times idle, in line with the FCC's cable providers (about 85 to 250 ms under upload load). Switch SQM on and the same upload leaves the call at 23 ms, 1.3 times idle (illustrative). On DSL, SQM off, the delay goes from 29 ms to 665 ms, 23.3 times idle.

Field notes

- The SQM-on curve is illustrative. No published consumer measurement of it was found, so it sits at idle plus FQ-CoDel's 5 ms default target, a stand-in, not a measurement. The FCC notes that some providers have since added queue management of their own.
- The busy figures are read by eye from the bars of the FCC's chart and are approximate. The FCC measured in its 2022 test period and published in 2024. Under download load its bars differ, and one cable provider reached about 380 ms there.
- Left out: the Wi-Fi hop, queues inside the provider's network and at the far end, jitter, and packet loss. One upload, one call, one line. The 20-second timing and the 0.8 s fill time are illustrative.
- To measure your own line while it is busy, use Network Quality (its responsiveness figure measures round-trip time during a download) or Test My Connection (its loaded responsiveness check). The tool's About card opens both.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: lag is delay, not speed: an upload fills the line's queue on any plan and the call waits behind it, and smart queue management keeps the queue short where a faster plan does not.

Control	What it does	Why it's there
Smart queue management (SQM): Off / On (Up turns on, Down turns off, Q switches in Present)	Switches the router's queue management. The other setting stays as a dashed line.	The one control: the latency trace drops from hundreds of milliseconds back toward idle. It corrects "more Mbps fixes lag".
The home line: Fiber / Cable / DSL (default Cable)	Loads the FCC's figures for that technology and rescales the chart.	The picker sits beside the one switch so the class sees DSL suffer most.

Control	What it does	Why it's there
Play the call / Pause, 1 s on, Reset	Runs the 20 s in real time or steps it.	Lets you stop while the upload runs and read the delay.
Time into the run (0 to 20 s), or drag across the chart	Moves the playhead; the queue picture shows what the call's packet waits behind.	The queue picture is the mechanism behind the number.
Open Network Quality, Open Test My Connection	Opens the two tools that measure a real line under load.	Moves the class from the model to their own line.

A demo that works (about 3 minutes)

1. Set up: Reset (Cable, SQM off, whole run drawn).
2. Ask the room: "Someone starts a big upload during your video call. Does a faster plan fix the lag?"
3. Reveal: play it. The call goes from 18 ms idle to 225 ms, 12.5 times idle. Press Up: with SQM on the same upload leaves it at 23 ms (help example).
4. Optional follow-up: switch to DSL with SQM off: 29 ms to 665 ms.

Pairs with: Test My Connection and Network Quality, the Home Internet explainer.

Left out on purpose: the SQM-on curve is illustrative (idle plus FQ-CoDel's 5 ms target); no consumer measurement of it was found. The busy figures are read by eye from the FCC's chart. Left out: the Wi-Fi hop, queues inside the provider and at the far end, jitter and loss. There is no built-in predict question on this tool; ask it yourself.

DFS and Radar

Runs one access point on a simulated one-hour clock and shows what Dynamic Frequency Selection (DFS) costs. On a DFS channel the AP must listen for radar before it transmits (the channel availability check, or CAC). When radar appears it stops its traffic, leaves the channel within 10 seconds and may not return for 30 minutes. You press Radar now and watch the outage.

Why it's here. DFS is why an AP can go quiet for a minute after a reboot, why clients sometimes lose an AP for no visible reason, and why some channels sit unused on a busy plan. The listening time, the move and the 30-minute block are fixed by regulation, so the only real choice is what the AP does next: a non-DFS channel is available at once, another DFS channel means another CAC, and in the EU channels 120, 124 and 128 need 10 minutes.

How to use

1. Pick a region, a width and a starting channel. The channel list says which are DFS and how long each CAC takes.
2. Press Play, or use Step and the clock slider. The channel strip shows every 20 MHz channel: in use, listening (CAC), blocked after radar with its countdown, or not in the plan. A bar under a channel marks it as DFS.
3. Press Radar now while the AP is on a DFS channel. The timeline shows the AP and three clients: CAC, service, the radar, the move and any gap with no service, plus a row for blocked channels. Switch the timeline to 3 minutes to see a short move.
4. Choose what the AP does after radar: prefer a non-DFS channel, or take another DFS channel and pay its CAC. Add random radar to see repeated moves.
5. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses, the Right arrow steps 10 s, D is Radar now, R restarts, Up and Down move the clock a minute, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Region	US or EU	US (FCC 47 CFR 15.407): 36 to 165, DFS 52 to 64 and 100 to 144. EU (ETSI EN 301 893): 36 to 64 and 100 to 140, DFS 52 to 64 and 100 to 140. Default US
Channel width	MHz	20, 40, 80 or 160; default 20
Starting channel	channel	any channel of the plan at that width; default 100

Input	Unit	Range
After radar	policy	prefer non-DFS (default) or another DFS channel
Random radar	events per hour	off (default), 2, 6 or 12
Clock speed	x	1, 10 (default), 60 or 300 times real time
AP EIRP	dBm	17, 20, 23 (default), 27 or 30; for the detection threshold readout only

How it works. CAC: 60 s on a DFS channel; in the EU, 600 s for any channel partly or fully in 5600 to 5650 MHz (20 MHz channels 120, 124 and 128). A wide channel is DFS if any of its 20 MHz channels is. On radar while serving, normal traffic stops within the closing time (US 200 ms, EU 1 s of transmissions in aggregate) and the AP leaves 1 s after the radar, inside the 10 s move time. On radar during a CAC it leaves at once. Every 20 MHz channel it was using is blocked for 30 minutes. The new channel follows the policy; if no channel is free at the width, the AP narrows. Clients follow the AP straight to a non-DFS channel; if the AP must run a CAC first, they drop and rejoin 1 to 2.5 s after it returns. Outage is radar to resumed service. Detection threshold: FCC -64 dBm at 200 mW EIRP or more, -62 dBm below; ETSI -62 + 10 - PSD + G dBm with a floor of -64 dBm, G = 0 dBi here.

Example. US, channel 100 at 20 MHz: the AP listens for 60 s and first transmits at 1:00. Radar at 5:00: it moves to channel 36, which is not DFS, so the outage is 1.0 s and channel 100 is blocked until 35:00. With Another DFS it moves to 104 instead and waits a 60 s CAC, a 61.0 s outage. In the EU, the same move from 116 lands on 120 and waits 10 minutes, a 10 min 1 s outage.

Field notes

- The 60 ms of control signals the FCC allows after the 200 ms closing time comes from one secondhand source, a test lab's transcription of FCC KDB 905462 D02; the FCC document itself was not read.
- The 10-minute CAC is ETSI only. Under the FCC every DFS channel takes 60 seconds.
- Channel 144 is DFS in the US and is not in the EU plan: it crosses the 5725 MHz edge.

- The AP's 1 s move and the clients' rejoin delays are illustrative. Real APs differ, and some block only the part of a wide channel that saw the radar.
- Not modeled: off-channel CAC in the EU, a second radio that listens while the first serves, and radar signal levels. The detection threshold is shown for context only.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: before transmitting on a DFS channel an AP must listen for radar (60 s, or 10 minutes on EU channels 120 to 128); after radar it must leave within 10 s and stay off for 30 minutes, and the only real choice is where it goes next.

Control	What it does	Why it's there
Play / Pause / Play again, Step 10 s, Restart, Clock, Clock speed: 1x (real time), 10x, 60x (1 min per second), 300x (5 min per second)	Runs the one-hour clock. Default 10x.	A 30-minute block and a 10-minute CAC need a fast clock to fit a class.
Radar now (D in Present)	The AP detects radar now; unavailable on a non-DFS channel.	The radar-event button: shows the closing, the move and the 30-minute block.
Region: US (FCC) / EU (ETSI)	Swaps the channel plan and the rules.	The 10-minute CAC is ETSI only; channel 144 is DFS in the US and not in the EU plan.
Width (20, 40, 80, 160 MHz)	Channel width. Default 20.	Wider channels touch more DFS channels.
Starting channel (default 100)	Where the AP starts; the list says which are DFS and how long each CAC takes.	Starting on DFS shows the CAC before the first frame.
After radar: Prefer non-DFS / Another DFS	Where the AP moves.	The only real choice: a non-DFS channel is available at once; another DFS channel means another CAC.

Control	What it does	Why it's there
Random radar: Off, 2, 6 or 12 per hour; New random pattern	Adds seeded random radar.	Shows repeated moves over an hour.
Timeline: Whole hour / 3 minutes	Zooms the timeline.	A short move is invisible at the hour scale.
AP EIRP (17 to 30 dBm, default 23)	Feeds the detection-threshold readout only.	Detection thresholds are shown for context, not modeled.

A demo that works (about 3 minutes)

1. Set up: Restart. US, 20 MHz, channel 100, Prefer non-DFS.
2. Ask the room: "The AP reboots on channel 100. When does it send its first frame? And if radar hits at 5:00, how long are clients without service?"
3. Reveal: play. First transmission at 1:00. Press D near 5:00: it moves to 36, a 1.0 s outage, and 100 is blocked until 35:00 (help example).
4. Optional follow-up: switch After radar to Another DFS (104, a 61.0 s outage), then EU, starting on 116: the move lands on 120 and waits 10 minutes.

Pairs with: Channel Planner (DFS on or off), the 5 GHz channel card in Course Handouts. Lesson sequences "Channels and the rules" and "Designing a cell plan".

Left out on purpose: the FCC's 60 ms of control signals comes from one secondhand source. The AP's 1 s move and client rejoin delays are illustrative. Not modeled: EU off-channel CAC, a second radio listening while the first serves, radar signal levels. Some real APs block only the part of a wide channel that saw radar.

Down the Stack, Across the Air, Up the Other Side

Animates the protocol stack the way it is taught in a classroom. A laptop on Wi-Fi sends data to a wired server through an AP (access point) and a router. Going down the laptop's stack the data gains a port (the TCP, Transmission Control Protocol, or UDP, User Datagram Protocol, header), IP (Internet Protocol) addresses and MAC (media access control) addresses, then becomes bits carried by RF (radio frequency) waves. It crosses the air, and each device climbs only as far as it needs: the AP to the data link layer, the router to the network layer, the server all the way up. A second view shows the four combinations of the To DS and From DS bits (DS, the distribution system) and what Addresses 1 to 4 hold in each.

Why it's here. The IP addresses stay the same from the laptop to the server while the MAC addresses change on every hop, and watching each header get added, read and replaced makes that stick. The frame on the air carries a third address, and in the laptop-to-server case it is the router's MAC, not the server's: the server is on another subnet, so the laptop sends to its default gateway, and the server's address appears only in the IP header. A fourth address exists for the case where both radios on a hop are relays, such as a mesh or WDS (wireless distribution system) backhaul, so the frame needs two more fields for where it started and where it is going.

How to use

1. Press Play to move the data one step at a time, or use Back, Step, Reset and the Journey step slider. The top card says where the data is and what that layer did. In the drawing each column is a device and each box a layer; the lime line is the path so far and the dot is where the data is now. Layers a device does not use are dashed.
2. The next card shows what is on the wire or in the air: each header is a labeled piece with its size in bytes, and a new or changed piece is outlined and marked new. The card below lists the ports, the IP addresses and the TTL (time to live), and the layer 2 addresses in force: 802.11 Addresses 1 to 3 with their roles on the air, or the Ethernet source and destination on a cable.
3. Change the scene: To the server or The reply, and a server on Another subnet (through the router) or the Same subnet (no router, and Address 3 becomes the server). Change the frame: TCP or UDP, 100, 500 or 1460 bytes of data, and a QoS (quality of service) Data header of 26 bytes or the textbook 24-byte Data header.
4. Show Address fields, or press one of the four To DS / From DS buttons (To DS first). Each case shows a worked scene, who plays which role on the air, and Addresses 1 to 4 with the device and MAC address in each.
5. Predict, then reveal: Ask the class loads the question (the laptop sends to a server on another subnet: whose MAC address is in Address 3?); pick an answer, then press Reveal to jump to the step where the laptop fills in the 802.11 header.

6. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses, the Right arrow steps, R resets, Up and Down move one step forward or back, D shows the next To DS / From DS case, V switches between the journey and the address fields, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Direction	choice	To the server (default) or The reply
Server	choice	Another subnet, through the router (default), or the laptop's own subnet
Transport	choice	TCP (default, 20-byte header) or UDP (8-byte header)
Data	bytes	100, 500 or 1460 (default; fills a 1500-byte IPv4 packet with TCP)
QoS Data header	switch	On (default, 26 bytes) or Off (the textbook 24-byte Data header)
To DS / From DS	choice	0 0, 0 1, 1 0 or 1 1 (To DS first); follows the journey: 1 0 going to the server, 0 1 on the reply

How it works. Sizes: TCP header 20 bytes (RFC 9293), UDP header 8 (RFC 768), IPv4 (Internet Protocol version 4) header 20 (RFC 791), LLC/SNAP (logical link control and subnetwork access protocol) 8 (RFC 1042), 802.11 MAC header 24 bytes with three addresses and 30 with four, plus 2 for QoS Control, and a 4-byte FCS (frame check sequence), IEEE 802.11-2020 clause 9.2.4.8. Ethernet adds a 14-byte header and a 4-byte FCS. Addresses by To DS / From DS follow clause 9.3.2.1: 0 0 gives RA = DA, TA = SA, BSSID; 0 1 gives RA = DA, TA = BSSID, SA; 1 0 gives RA = BSSID, TA = SA, DA; 1 1 gives RA, TA, DA, SA. RA is the receiver address, TA the transmitter address, DA the destination address, SA the source address and the BSSID (basic service set identifier) the AP radio's MAC for the network. The laptop compares the server's IP address with its own under a /24 mask; off its subnet, it sends to its default gateway's MAC, found by ARP (Address Resolution Protocol). The AP is a bridge: it removes the 802.11 header and LLC/SNAP and forwards on Ethernet with the laptop's MAC as the source (RFC 1042). The router removes the Ethernet header, lowers the TTL by 1, recomputes the IPv4 header checksum (RFC 1812) and builds a new Ethernet header from its outgoing interface to the next device.

Example. Defaults, laptop to server, TCP, 1460 bytes: 1460 + 20 = 1480 bytes, + 20 = 1500 (the IPv4 packet), + 8 LLC/SNAP = 1508, + 26 header + 4 FCS = 1538 bytes, or 12,304 bits, on the air. There Address 1 is the AP's BSSID, Address 2 the laptop and Address 3 the router's LAN interface. On the cable to the router the same packet is 14 + 1500 + 4 = 1518 bytes, from the laptop's MAC to the router's. The router lowers the TTL from 64 to 63 and sends a new frame from its server-side MAC to the server's. Source 192.0.2.10 and destination 198.51.100.20 never change.

Field notes

- No NAT (network address translation). Most home routers translate the source address, and then the source IP changes at the router. Without NAT the IP addresses stay the same end to end, and only the TTL and the header checksum change.
- Every MAC address, the laptop's port 51000 and the starting TTL of 64 are illustrative. The IP addresses come from the ranges RFC 5737 reserves for documentation.
- The drawing shows the OSI (Open Systems Interconnection) layers 7, 4, 3, 2 and 1 and leaves out 5 and 6.
- Left out: the ARP exchange itself, the ACK (acknowledgment), the PHY (physical layer) preamble, encryption, aggregation, VLAN (virtual LAN) tags and IPv6. A Frame's Journey opens up the air hop: the FCS check, SIFS (short interframe space) and the ACK.
- A mesh network built on 802.11s can carry up to six addresses using its Mesh Control field; the four-address case here is the plain one. Frames that carry several packets at once (an A-MSDU, aggregate MAC service data unit) fill Addresses 3 and 4 differently.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: data gains a port, an IP address and a MAC address on the way down; the IP addresses stay the same end to end while the MACs change on every hop, and Address 3 on the air is the router's MAC, not the server's.

Control	What it does	Why it's there
Predict, then reveal: Ask the class; The server's / The router's / The AP's; Reveal; Close the question	Loads laptop to a server on another subnet; Reveal jumps to where the laptop fills in the 802.11 header.	The predict prompt "Whose MAC is in Address 3?"
Play / Pause, Back, Step, Reset, Journey step (Up/Down in Present)	Moves the data one layer or hop at a time.	Each header added, read and replaced is one step, so the class can watch each change.
Show: The journey / Address fields (V in Present)	Switches the drawing and the address table.	Separates the path from the four address fields.
To DS / From DS buttons: 0 0, 0 1, 1 0, 1 1 (D in Present)	Shows each case's worked scene and Addresses 1 to 4.	The fourth address exists for a hop where both radios are relays, such as a mesh or WDS backhaul.
Direction: To the server / The reply	Reverses the journey.	Shows the MACs change on every hop in both directions.
Server: Another subnet / Same subnet	Adds or removes the router.	On the same subnet there is no router and Address 3 becomes the server.
Transport: TCP / UDP	20-byte or 8-byte header.	Shows the transport header's size in the stack.
Data: 100, 500 or 1460 bytes (fills 1500 with TCP)	Payload size.	1460 bytes fills a 1500-byte IPv4 packet, so the class can add up 1538 bytes on the air.
QoS (quality of service) Data header, 26 bytes	26-byte QoS Data header or the textbook 24-byte header.	Nearly every data frame on a modern network is a QoS Data frame (on-screen note).

A demo that works (about 3 minutes)

1. Set up: Reset. To the server, Another subnet, TCP, 1460 bytes.

2. Ask the room: press Ask the class. "On the air, whose MAC address is in Address 3?"
3. Reveal: Reveal jumps to the 802.11 header: Address 1 is the AP's BSSID, Address 2 the laptop, Address 3 the router's LAN interface. Step on and show the router lower the TTL from 64 to 63 and build a new frame, while 192.0.2.10 and 198.51.100.20 never change (help example).
4. Optional follow-up: switch Server to Same subnet and show Address 3 become the server; press D through the four DS cases.

Pairs with: A Frame's Journey (on Signals and PHY, the air hop, FCS, SIFS and the ACK), Association, Frame by Frame.

Left out on purpose: no NAT (most home routers change the source IP). MACs, port 51000 and TTL 64 are illustrative; IPs come from the RFC 5737 documentation ranges. Layers 5 and 6 are not drawn. Left out: ARP, the ACK, the PHY preamble, encryption, aggregation, VLAN tags, IPv6, and 802.11s six-address mesh frames.

Conference Wi-Fi Runs Out of Addresses

Follows the DHCP (Dynamic Host Configuration Protocol) address pool through a conference morning, 07:00 to 13:00. People arrive, put devices on the Wi-Fi, stay a while and leave. Each device borrows an IP (Internet Protocol) address for a lease time. Set the lease and watch the pool: with a long lease it runs dry mid-morning while the hall is half empty, and with a short one it holds.

Why it's here. When an event network runs out of addresses, devices still associate and show full bars, and nothing loads, so the call is usually that the Wi-Fi is down and more APs are needed. The radio is fine. A device that walks out sends nothing, so its address stays taken until its lease runs out, and the pool has to cover everyone who arrived within one lease time, not just the people in the room. Devices that rotate their private MAC (media access control) address on open networks come back looking like new devices and take a second address. Association, Frame by Frame shows one DHCP exchange; this tool is what happens to the pool when hundreds of them pile up.

How to use

1. Read the top card: whether the pool ran dry and when, and one sentence saying why.

2. Press Play the morning, +10 min, or drag Time to move through the morning. On the chart the solid line is addresses taken (in use or still held), the dashed line is devices in the hall, the dotted line is the pool, and the shaded area is devices with no address. The pool grid shows every address at that moment: solid in use, hatched still held for a device that left, cross-hatched still held for an old private address, outlined free.
3. Change the lease time and watch the dry time move or disappear. Turn on private-address rotation and set the two assumptions: the share of devices that rotate and how often a rotating device comes back as new.
4. Change the pool (subnet /24 to /19, and the reserved addresses) and the crowd (people over the morning, devices each, average stay). All crowd numbers are illustrative; set them to your event.
5. Predict, then reveal: Ask the class loads the defaults and stops at the first device with no address (the Wi-Fi seems down mid-morning: what fixes it?); pick an answer, then press Reveal.
6. On a computer or tablet, Present opens this simulator full screen for a projector: Space plays or pauses the morning, the Right arrow moves 10 minutes, R resets, Up and Down change the lease time, M turns private-address rotation on or off, P asks, reveals and closes the question, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Lease time	choice	5 min to 8 days; default 1 day
Devices rotate their private address on this open network	switch	Off (default) or On
Share of devices that rotate	%	0 to 100; default 50 (assumption; used only with rotation on)
A rotating device comes back as new every	minutes	15 to 240; default 60 (assumption; used only with rotation on)
Subnet	prefix	/24 (default), /23, /22, /21, /20 or /19
Reserved: gateway, servers, printers	addresses	0 to 100; default 10 (illustrative)

Input	Unit	Range
People over the morning	count	50 to 5,000; default 250 (illustrative)
Devices each person puts on the Wi-Fi	devices	1.0 to 3.0; default 1.5 (illustrative)
Average stay	minutes	15 to 360; default 90 (illustrative)

How it works. Pool = 2^(32 - prefix) - 2 usable addresses, minus the reserved ones. Devices = people x devices each, arriving in half-hour shares of 4, 8, 14, 16, 10, 8, 8, 8, 7, 6, 6 and 5% from 07:00 (illustrative: a rush before a 09:00 keynote, then a trickle), spread evenly inside each half hour. Each device stays between half and one and a half times the average stay, on a fixed sequence rather than random draws, so the same settings give the same morning. Every minute: leases that ran out go back to the pool; devices that left, or rotated to a new address, stop renewing; new clients ask for an address and get the lowest free one, first come first served, or wait with none. RFC 2131 lease rules: a device that stays renews at T1, half the lease, so it keeps its address; the server keys each lease to the client's hardware address, so a rotated address is a new client; a device that leaves sends nothing, so its address is free one lease after its last renewal. With rotation on, the chosen share of devices comes back as a new client every chosen interval while in the hall, the first time somewhere inside the first interval.

Example. Defaults (/24 with 10 reserved = 244 addresses, 1-day lease, 250 people with 1.5 devices each = 375 devices, 90-minute average stay): the pool runs dry at 10:18. At that moment only 113 devices are in the hall; 112 addresses are in use and 132 are still held for devices that already left. Up to 74 devices go without an address. With a 30-minute lease the same morning holds, at most 170 of 244 addresses taken. A 2-hour lease holds (peak 232) until rotation is turned on, and then it runs dry at 09:21. A /23 holds even with the 1-day lease.

Field notes

- A device with no address still associates and shows full signal. Adding APs or a faster internet line adds no addresses.
- No device maker publishes how often a device rotates its private address, so the share and interval here are assumptions. One platform maker documents rotating as the default on open and captive-portal networks.
- Some devices do send a release when they leave a network on purpose; walking out of range sends nothing. This tool assumes nothing is released early.

- Not modeled: servers that give short leases to new clients or a different lease per network, captive sign-in pages, and IPv6 (Internet Protocol version 6) addressing, which does not draw from this pool.
- Illustrative: the crowd, the arrival shape, the stay and the reserved addresses. Set them to your event.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: the DHCP pool has to cover everyone who arrived within one lease time, not the people in the room, so an event network runs dry with full bars, and adding APs adds no addresses.

Control	What it does	Why it's there
Predict, then reveal: Ask the class; Add more APs / A shorter lease or a bigger pool / A faster internet line; Reveal; Close the question (P in Present)	Loads the defaults and stops at the first device with no address.	Corrects "the Wi-Fi is down, add APs".
Play the morning / Pause, +10 min, Reset, Time	Runs 07:00 to 13:00.	The dry time is a moment in the morning, so the class watches it arrive.
Lease time (5 min to 8 days, default 1 day; Up/Down in Present)	The server's lease.	The one control: the pool runs dry mid-morning with long leases and holds with short ones.
Devices rotate their private address on this open network (M in Present)	Rotating devices come back as new clients.	Phones that rotate on open networks take a second address. One platform maker documents rotating as the default on open networks.
Share of devices that rotate (assumption) (0 to 100%, default 50); A rotating device comes back as new every (assumption) (15 to 240 min, default 60)	The rotation assumptions.	No device maker publishes a rotation interval, so both are labeled assumptions.
Subnet: /24 to /19	Pool size.	The other fix: a /23 holds even with the 1-day lease.

Control	What it does	Why it's there
Reserved: gateway, servers, printers (illustrative) (0 to 100, default 10)	Addresses taken off the top.	Set to your event.
People over the morning (50 to 5,000, default 250); Devices each person puts on the Wi-Fi (1.0 to 3.0, default 1.5); Average stay (15 to 360 min, default 90)	The crowd. All illustrative.	The defaults are an example; set them to your event.
Open Association, Frame by Frame; Open Subnet Planner	Links out.	One DHCP exchange frame by frame, and subnet sizing.

A demo that works (about 3 minutes)

1. Set up: Reset, then P (Ask the class).
2. Ask the room: "Mid-morning, full bars, nothing loads. What fixes it?"
3. Reveal: P again. The pool (244 addresses) ran dry at 10:18 with only 113 devices in the hall: 132 addresses are still held for devices that left (help example). Press Down until the lease reads 30 min and the same morning holds.
4. Optional follow-up: set a 2-hour lease (it holds), then press M for rotation: it runs dry at 09:21.

Pairs with: Association, Frame by Frame (one DHCP exchange), Subnet Planner.

Left out on purpose: nothing is released early, though some devices do send a release when they leave on purpose. Not modeled: short leases for new clients, per-network leases, captive portals, IPv6. The crowd, arrival shape and stay are illustrative.

Association, Frame by Frame

Plays every frame a client sends and receives to associate with a Wi-Fi network, from the first scan to the first useful packet: the scan, Open System or SAE authentication, association, the EAP exchange for 802.1X, the 4-way handshake, DHCP, the address check, then ARP and DNS. A channel strip above the ladder shows the scan channel by channel, and a timeline under it shows how long each phase takes. It teaches the

frames; to actually connect this device to a network, use Join a Network. A second mode, Why won't it associate?, pairs a client with a network and shows whether they associate, and if not, the exact frame where it stops.

Why it's here. Open System authentication and association are bookkeeping, not security, and the network is not usable after association: the keys come in the 4-way handshake, and the client still has no IP address until DHCP and an address check finish. Seeing each phase to scale shows where a slow association spends its time, which is often the scan and the address check, not the Wi-Fi security.

How to use

1. Pick the security (Open, OWE, WPA2-Personal, WPA3-Personal or 802.1X), the band and the scan type, then press Play, or use Step and Back to go one frame at a time. Blue arrows cross the air; sand, dashed arrows cross the wire. DHCP, ARP and DNS arrows are bridged: solid to the AP, dashed beyond it.
2. Tap any sent message to see what it carries. The Association Request lists the SSID, listen interval, capability information, supported rates, the RSN element (key management, ciphers and the PMF capable and required bits) and the HT, VHT, HE and EHT capabilities; the response carries the status code and the association ID.
3. Each step number has a mark beside it: M for a management frame, D for a data frame. The EAPOL-Key frames of the 4-way handshake are data frames. A lock marks data frames encrypted after message 4; a shield marks a management frame protected by PMF (802.11w), which only happens once keys exist.
4. Watch the channel strip: in a passive scan the AP beacons every 102.4 ms, and a beacon sent while the radio is on another channel is missed (hollow). Lower the passive dwell below 51 ms to see the scan miss the AP altogether. Switch to 6 GHz to see probes go only to the 15 PSCs, or pick Known via RNR to see a 2.4 or 5 GHz beacon name the 6 GHz AP.
5. In 6 GHz, set 6 GHz view to Compare all four to race four ways of finding the AP on one clock: listen on all 59 channels, probe the 15 PSCs, go straight to the channel an RNR named, or listen 20 TU on each PSC for FILS Discovery. Each strip is one method's scan to scale, and each ends with the time it first heard the AP. Play sweeps the clock; Step jumps to the next finding. RNR time: Include counts the 5 GHz scan in which the client heard the RNR; Leave out shows the 6 GHz part alone.
6. Address check: pick ACD, Address Conflict Detection (RFC 5227), in which the client checks that no one else already has its new address with three ARP probes 1 to 2 s apart and then an announcement, or DNAv4, Detecting Network Attachment (RFC 4436), one unicast ARP to the gateway it remembers from before. The timeline shows Address Conflict Detection as the long bar it is.

7. Mode: pick Why won't it associate? to pair a client with a network. Choose a client (Older laptop, WPA2 only; Printer, no PMF; Phone, WPA3 capable; New laptop, 6 GHz; Phone, Wi-Fi 7) and the network's security, PMF, band, Wi-Fi 7 AP and H2E only. The verdict above the ladder says what happens and why, with its source; the ladder plays up to the frame where it stops, marked with a red X, and ends with Stopped here and what the help desk sees. Open Client capabilities to switch each key management type, cipher, band, H2E and Wi-Fi 7 on or off; the client then reads Custom.
8. Four outcomes. Associates: the key management (AKM), cipher and PMF both sides agree on. Never tries to associate: nothing in the AP's RSN element fits, so the client sends no Authentication or Association Request at all and the AP has nothing in its log; this is the common case. Refused: a device older than PMF ignores a network that requires PMF, tries, and gets status 31 in the Association Response. Not offered: 6 GHz allows only OWE, WPA3-Personal only and WPA3-Enterprise only.
9. On a computer or tablet, Present opens the tool full screen for a projector: Space plays or pauses, the Right arrow sends the next frame and the Left arrow takes one back, R resets, Up and Down change the passive dwell by 10 ms (in Why won't it associate?, they step the network security and C cycles the client), S turns the 6 GHz race on or off in Play the association (switching to 6 GHz; the race is not offered in Why won't it associate?), F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Security	choice	Open, OWE (Enhanced Open), WPA2-Personal (PSK), WPA3-Personal (SAE), 802.1X (EAP-TLS, PEAP or EAP-TTLS); default WPA2-Personal. In 6 GHz, Open is drawn as OWE and WPA2-Personal as WPA3-Personal
PMF (802.11w)	choice	Off, Optional or Required, for WPA2-Personal and 802.1X outside 6 GHz; OWE, WPA3 and 6 GHz require it; Open has no keys
Band	choice	2.4 GHz (channels 1 to 11), 5 GHz (25 US channels), 6 GHz (59 channels, 15 PSCs); default 5 GHz. The AP is on channel 6, 44 or 37
Scan	choice	Active (probe requests; DFS channels are only listened to) or Passive (listen for beacons); default active
6 GHz discovery	choice	PSC scan, or Known via RNR (a 2.4 or 5 GHz beacon already named the 6 GHz AP)

Input	Unit	Range
6 GHz view	choice	One method (the ladder) or Compare all four (the race); 6 GHz only; default One method
RNR time	choice	Include the 5 GHz scan that heard the RNR, or Leave out; race only; default Include
Address check	choice	ACD, Address Conflict Detection (RFC 5227), or DNAv4, Detecting Network Attachment (RFC 4436); default ACD
Timing	ms, illustrative	active dwell 10 to 100 (30), passive dwell 20 to 250 (111), frame time 0.5 to 5 (1), RADIUS round trip 1 to 200 (10), crypto 0 to 500 (50), wired LAN round trip 1 to 50 (5), Address Conflict Detection wait before the first probe 0 to 1000 (500) and time between probes 1000 to 2000 (1500)
Mode	choice	Play the association (default), or Why won't it associate?
Client	choice	Older laptop, WPA2 only; Printer, no PMF; Phone, WPA3 capable; New laptop, 6 GHz; Phone, Wi-Fi 7 (generic and illustrative), or Custom: AKMs PSK (2), SAE (8), SAE (24), OWE (18), 802.1X SHA-1 (1), 802.1X SHA-256 (5), 802.1X 192-bit (12); ciphers TKIP, CCMP-128, GCMP-256; PMF older than PMF, not capable, capable or required; H2E; 2.4, 5 and 6 GHz radios; Wi-Fi 7
Network security (Why won't it associate?)	choice	Open, OWE, WPA2-Personal, WPA3-Personal transition, WPA3-Personal only, WPA2-Enterprise, WPA3-Enterprise transition, WPA3-Enterprise only, WPA3-Enterprise 192-bit; default WPA3-Personal only. PMF is a choice only for the two WPA2 modes
Wi-Fi 7 AP, H2E only	on or off	A Wi-Fi 7 AP also enables SAE (24) and GCMP-256; H2E only sets BSS membership selector 123 in the WPA3-Personal modes

How it works. Scan: each channel of the band is visited once for its dwell (active: a probe request, and the AP answers on its own channel; passive, and DFS channels in an active 5 GHz scan: listen only). The AP beacons every 102.4 ms, the first one 51.2 ms into the dwell on its channel, so a passive dwell shorter than that misses it. In 6 GHz an active scan probes only the 15 PSCs, and the AP also sends a FILS Discovery frame every 20 TU. Then Open System authentication (2 frames) or SAE commit and confirm (4 frames, before association), association (2 frames), for 802.1X the EAP exchange from the 802.1X and EAP Ladder, and the 4-way handshake (4 EAPOL-Key data frames). IP: DHCP Discover, Offer, Request, Ack (RFC 2131); Address Conflict Detection (RFC 5227: wait 0 to 1 s, three ARP probes 1 to 2 s apart, then 2 s before

the announcement, 4 to 7 s in all) or Detecting Network Attachment (RFC 4436: one unicast ARP to the remembered gateway and its reply); then ARP for the gateway (not needed after DNAv4) and a DNS query. Time = scan dwells + frames x frame time + EAP round trips x RADIUS time + crypto time (TLS, SAE or OWE) + wired LAN round trips + the Address Conflict Detection waits. A 6 GHz PSC probe waits until the channel has been idle 7 ms (dot11MinPSCProbeDelay, its default in IEEE 802.11-2024); a probe for an AP an RNR named goes at once. The race: listen on all 59 channels at the passive dwell; probe the 15 PSCs at the active dwell; for RNR, a 5 GHz active scan (when included), then one directed probe on channel 37; or listen 20 TU (20.48 ms) on each PSC. Each frame's wait is the average, half its interval. Why won't it associate?, in order: not offered if the band is 6 GHz and the security is not OWE, WPA3-Personal only or WPA3-Enterprise only; never tries if the client has no radio in the band, no AKM in common, no cipher in common, or (for a client that reads the PMF bits) PMF required on one side and not possible on the other (Table 12-5); SAE and OWE only with PMF on, and SAE only with H2E where H2E is required; refused with status 31 when a client older than PMF meets PMF required; otherwise it associates. The client picks SAE (24), SAE (8), then PSK for personal networks (WPA3 order), 192-bit, SHA-256, then SHA-1 802.1X for enterprise, and the strongest cipher both offer (the model's reading). A Wi-Fi 7 connection needs both sides Wi-Fi 7, PMF on, an AKM other than PSK, SAE (8) or 802.1X SHA-1, and GCMP-256 on both sides; otherwise a Wi-Fi 7 client associates as Wi-Fi 6.

Example. WPA2-Personal, 5 GHz active scan, Address Conflict Detection, defaults: 24 frames over the air (8 management, 16 data), 7.59 s in all: scan 2.05 s, authentication 2 ms, association 2 ms, 4-way handshake 4 ms, DHCP 14 ms, Address Conflict Detection 5.50 s, ARP and DNS 14 ms. With DNAv4 (Detecting Network Attachment) instead: 2.08 s, the address check 7 ms. A passive 5 GHz scan takes 2.77 s; an active 2.4 GHz scan 330 ms; an active 6 GHz scan of the 15 PSCs 450 ms; a passive 6 GHz scan of all 59 channels 6.55 s. At a 40 ms passive dwell the AP is not found. The race at the defaults: listen on all 59 finds the AP at 1.01 s, probe the 15 PSCs at 69 ms, RNR at 2.05 s counting the 5 GHz scan (3 ms without it), and listening 20 TU on each PSC at 51 ms, first. At a 10 ms active dwell, probing the PSCs finds it at 29 ms and wins. Why won't it associate?: Older laptop, WPA2 only on WPA3-Personal only: never tries to associate, stopped at the Probe Response. Printer, no PMF on WPA2-Personal with PMF required: refused, status 31. Phone, Wi-Fi 7 on a Wi-Fi 7 WPA3-Personal AP: a Wi-Fi 7 connection with SAE (24), GCMP-256, PMF on; on WPA2-Personal it associates as Wi-Fi 6 with PSK.

Field notes

- Every phase time is a setting: no published measurement breaks a typical association down by phase. The scan dwell defaults are one real example (Linux mac80211, about 30 ms active and 111 ms passive); many drivers scan in firmware with their own dwell. The beacon interval of 102.4 ms is a default, not a mandate.

- OWE (Enhanced Open) encrypts without authenticating: the Diffie-Hellman keys in the association frames give both sides the PMK. 6 GHz requires WPA3 or OWE, with PMF.
- The race's AP sends a FILS Discovery frame every 20 TU. IEEE 802.11-2024 requires that of a 6 GHz-only AP that wants to be found (or one sending unsolicited Probe Responses); an AP with a 2.4 or 5 GHz radio may use a longer interval, and then listening 20 TU per PSC can miss it. The race assumes the client finishes its 5 GHz scan before it tunes to the channel an RNR named.
- PMF (802.11w) protects deauthentication, disassociation and robust Action frames once keys exist. Beacons, probes, authentication and association frames are never protected. The ladder ends with a Block Ack setup (ADDBA) after the association to show the shield; it is not counted in the association time.
- With Detecting Network Attachment (DNAv4), a real client often skips the four DHCP messages and asks only to keep its old address. The ladder keeps them so the two address checks compare side by side.
- Not drawn: channel changes, retries, the DHCP client's own start-up delay, IPv6, and 802.11ai (FILS) fast initial link setup. The DNS server is drawn on the wired LAN lane with the DHCP server and the gateway.
- Roaming between APs, frame by frame, is in the 802.1X and EAP Ladder: switch its Mode to Roam.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.
- Why won't it associate? draws the security each pair actually negotiates. In 6 GHz it does not swap WPA2-Personal for WPA3 the way Play the association does: the network is simply not offered there.
- Status 31 is rare in practice: a client that knows PMF reads the network's bits and never tries. Statuses 42, 43 and 46 (invalid cipher, invalid AKMP, cipher rejected by policy) need a client that ignores the AP's advertisement, so they are not drawn.
- H2E only in 6 GHz is a Wi-Fi Alliance rule (WPA3 Specification v3.5 section 11.2); IEEE 802.11-2024 does not mention H2E there. H2E is a password method, not a key-management type.
- Not modeled: WPA3-Personal compatibility mode (RSN overriding), Transition Disable, FT key management, PSK SHA-256 and SAE-PK. Real devices vary by model, driver and operating system version.

Teaching it

Teaches: Open System authentication and association are bookkeeping, not security; the keys come in the 4-way handshake, and the client still has no IP address until DHCP and an address check finish, which is often where a slow association spends its time.

Control	What it does	Why it's there
Play / Pause / Play again, Back, Step, Reset, Show all, Speed: Slow / Normal / Fast	Sends one frame at a time, takes one back, or draws the whole ladder.	One frame at a time is how the class reads a capture.
Tap any sent message	Shows what it carries; the Association Request lists SSID, rates, RSN element with the PMF bits, and HT to EHT capabilities.	Association is where client and AP agree on capabilities.
Security: Open, OWE (Enhanced Open), WPA2-Personal (PSK), WPA3-Personal (SAE), 802.1X (Enterprise) (default WPA2-Personal)	Changes the middle of the ladder.	SAE replaces Open System authentication before association; 802.1X hands off to the EAP ladder's middle.
EAP method, Inner method (EAP-TTLS), Certificate size (802.1X only)	The EAP middle, shared with the 802.1X ladder.	Uses the same engine as the 802.1X and EAP Ladder, so the two tools always agree.
PMF (802.11w): Off / Optional / Required	The RSN element's MFPC and MFPR bits; forced where the standard requires it.	PMF protects management frames only once keys exist; beacons, probes, authentication and association are never protected.
Band: 2.4 / 5 / 6 GHz (default 5)	The AP is on 6, 44 or 37. In 6 GHz, Open is drawn as OWE and WPA2 as WPA3.	6 GHz requires WPA3 or OWE with PMF.
Scan: Active / Passive	Probe requests, or listening for beacons.	A passive scan waits for beacons every 102.4 ms, a default, not a mandate.
6 GHz discovery: PSC scan / Known via RNR (6 GHz only)	Probe the 15 PSCs, or learn the AP from a 2.4 or 5 GHz beacon.	The point: 6 GHz discovery differs.

Control	What it does	Why it's there
Address check: ACD / DNAv4	Address Conflict Detection (three ARP probes 1 to 2 s apart) or Detecting Network Attachment (one unicast ARP).	Both options side by side; ACD is drawn as the long bar it is.
Timing (illustrative): Active dwell per channel (default 30 ms), Passive dwell per channel (default 111 ms; Up/Down in Present), Time per frame over the air, RADIUS round-trip time, Crypto time (public-key work), Wired LAN round trip (DHCP, gateway, DNS), ACD wait before the first probe, ACD time between probes	Every phase time.	No published measurement breaks an association down by phase, so every phase is a labeled setting; the dwell defaults are Linux mac80211 as one real example.

A demo that works (about 3 minutes)

1. Set up: Reset. WPA2-Personal, 5 GHz, Active, ACD.
2. Ask the room: "After association, is the client on the network?" Then: "Of the 7.6 seconds it takes to associate, where does the time go?"
3. Reveal: Step through: keys arrive only in the 4-way handshake, and the timeline shows scan 2.05 s and ACD 5.50 s against a few milliseconds of Wi-Fi security (help example). Switch Address check to DNAv4: the total time to associate drops to 2.08 s.
4. Optional follow-up: switch Scan to Passive and press Down until the dwell is below 51 ms: the scan misses the beacon and the AP is not found (at 40 ms per the help).

Pairs with: 802.1X and EAP Ladder (Roam mode for the roam), Why a Long Wi-Fi Password Matters More on WPA2 (links in to the 4-way handshake and SAE), Conference Wi-Fi Runs Out of Addresses. Lesson sequences "Association and roaming" and "Wi-Fi security modes".

Left out on purpose: every phase time is a setting. Not drawn: channel changes, retries, the DHCP client's start-up delay, IPv6, 802.11ai (FILS). With DNAv4 a real client often skips the four DHCP messages; the ladder keeps them so the two checks compare side by side. The Block Ack setup at the end is there to show the PMF shield and is not counted.

Teaching it: the 6 GHz race

Teaches: a 6 GHz client can't just probe every channel, so how it finds the AP decides how long it takes. Listening 20 TU on each PSC beats probing them, and RNR is the fastest of all only if you don't count the 2.4 or 5 GHz scan that heard it.

Control	What it does	Why it's there
6 GHz view: One method / Compare all four (6 GHz only)	Swaps the ladder for four channel strips on one time axis: Listen on all 59, Probe the 15 PSCs, Known via RNR, Listen 20 TU on each PSC. Each strip ends "found at" a time; the first is flagged.	The single ladder shows one method at a time. Side by side on one clock, the class sees the difference instead of remembering it.
RNR time: count the 5 GHz scan that heard it: Include / Leave out (default Include)	Include starts the RNR strip after a full 5 GHz scan (2.05 s at the defaults); Leave out shows the 6 GHz part only (3 ms).	A toggle, default on. Both answers are true, and the argument between them is the lesson.
Play / Pause, Step, Back, Reset, Show all, Speed	Play sweeps a clock across the axis; Step jumps to the next finding and Back to the previous; Show all draws the result.	Step is the one to use in front of a room: each press is one "found", in order.
Active dwell, Passive dwell (Timing)	Sets the dwell for the probe strip and the listen-on-all-59 strip.	Drop the active dwell to 10 ms and probing wins (29 ms against 51 ms). The race is a model with inputs, not a ranking of real clients.
Scan, 6 GHz discovery	Disabled in the race, each with a sentence.	Each strip sets its own scan type.

A demo that works (about 3 minutes)

1. Set up: 6 GHz, then Compare all four (S in Present). RNR time on Include. Reset.
2. Ask the room: "Four ways to find the same AP on channel 37. Which one wins, and which one loses?" Most people pick RNR to win and listening on all 59 to lose.
3. Reveal: Step four times. Listen 20 TU on each PSC finds it at 51 ms, Probe the 15 PSCs at 69 ms, listen on all 59 at 1.01 s, and RNR comes last at 2.05 s because the client first had to scan 5 GHz to hear the RNR. Then switch RNR time to Leave out: RNR finds it in 3 ms.

4. Optional follow-up: ask why probing loses to listening. On a PSC the client may not probe until the channel has been idle 7 ms (dot11MinPSCProbeDelay in 802.11-2024), while the AP's FILS Discovery frame arrives on average 10 ms in. Then drop the active dwell to 10 ms and watch probing win.

Pairs with: the Association, Frame by Frame ladder in 6 GHz (one method, frame by frame), Survey Walk (scan cycle time), Band Steering. Lesson sequence "Association and roaming".

Left out on purpose: the AP sends a FILS Discovery frame every 20 TU. 802.11-2024 requires that only of a 6 GHz-only AP that wants to be found; an AP with a 2.4 or 5 GHz radio may not, and then a 20 TU listen can miss it. Each frame's wait is the average, half its interval. The client finishes its 5 GHz scan before it tunes to the channel RNR named, and the 2.4 GHz scan is left out. One AP, US channels, no retries or channel-change time.

Teaching it: Why won't it associate?

Teaches: most security mismatches produce no error at all. The client reads the AP's RSN element, finds nothing it can use, and never sends an Authentication or Association Request, so the AP has nothing in its log. A refusal with a status code is the rare case: status 31 for a device older than PMF that ignores a network requiring PMF.

Control	What it does	Why it's there
Mode: Play the association / Why won't it associate?	Swaps the tool into pairing a client with a network. Play the association is the tool as before.	The ladder is the same frames, stopped where it fails.
Client: Older laptop, WPA2 only; Printer, no PMF; Phone, WPA3 capable; New laptop, 6 GHz; Phone, Wi-Fi 7 (C in Present)	Picks what the client can do. Generic and illustrative.	Each preset reaches a different outcome against the same network.

Control	What it does	Why it's there
Client capabilities (fold): one switch per key management type (PSK 2, SAE 8, SAE 24, OWE 18, 802.1X 1, 5, 12), cipher (TKIP, CCMP-128, GCMP-256), H2E, band and Wi-Fi 7; PMF select (older than PMF, not capable, capable, required)	Builds a custom client; the preset reads Custom.	Lets the room ask "what if it had SAE?" and see the answer.
Network security: Open, OWE, WPA2-Personal, WPA3-Personal transition, WPA3-Personal only, WPA2-Enterprise, WPA3-Enterprise transition, WPA3-Enterprise only, WPA3-Enterprise 192-bit (Up/Down in Present)	What the AP advertises in its RSN element.	Transition modes exist so old and new clients both associate.
PMF (WPA2 modes only), Band, Wi-Fi 7 AP, H2E only	PMF bits; band; a Wi-Fi 7 AP also enables SAE 24 and GCMP-256; H2E only sets selector 123.	PMF required is what turns the printer away; 6 GHz is where WPA2 is not offered at all.
Verdict band (above the ladder)	Associates / Never tries to associate / Refused / Not offered, with the reason and the source.	The answer first, then the frames that prove it.

A demo that works (about 3 minutes)

1. Set up: Mode Why won't it associate?. Client Older laptop, WPA2 only. Network WPA3-Personal only, 5 GHz.
2. Ask the room: "The laptop can't connect. What error does the AP log?"
3. Reveal: Show all. The ladder stops at the Probe Response with an X: never tries to associate, no key management in common. No Authentication, no Association Request, nothing in the AP's log. Switch the network to WPA3-Personal transition: the laptop associates with PSK (2); press C twice to the Phone, WPA3 capable: it picks SAE (8).
4. Optional follow-up: the printer. Network WPA2-Personal, PMF Required, Client Printer, no PMF: refused, status 31 at the Association Response. Then switch Band to 6 GHz with WPA2-Personal: not offered, because 6 GHz allows only WPA3 or OWE. For Wi-Fi 7: Phone, Wi-Fi 7 on WPA3-Personal only with Wi-Fi 7 AP on gives a Wi-Fi 7 connection with SAE (24); on WPA2-Personal it associates as Wi-Fi 6.

Pairs with: 802.1X and EAP Ladder (Break it, for failures after association), Why a Long Wi-Fi Password Matters More on WPA2. Lesson sequence "Wi-Fi security modes".

Left out on purpose: statuses 42, 43 and 46 (they need a client that ignores the AP's advertisement; no standard trigger was found). WPA3-Personal compatibility mode (RSN overriding), Transition Disable, FT key management, PSK SHA-256 and SAE-PK. Which AKM and cipher a client picks when several fit follows the WPA3 order for personal networks; the enterprise order and the cipher choice are the model's reading. H2E only in 6 GHz is a Wi-Fi Alliance rule, not IEEE, and the source line says so.

802.1X and EAP Ladder

Plays an 802.1X connection one message at a time on a three-lane ladder: the client (supplicant), the AP (authenticator) and the RADIUS server (authentication server). EAP rides in EAPOL frames over the air and inside RADIUS Access-Request, Access-Challenge and Access-Accept packets on the wire. Pick EAP-TLS, PEAP or EAP-TTLS to change the middle of the ladder, or PSK and SAE to see a join with no RADIUS at all, and pick a roam mode to see what PMK caching and 802.11r remove. Switch Mode to Roam for a four-lane ladder (client, current AP, target AP, RADIUS server) that plays a whole roam: the scan, Reassociation, and five methods side by side: full 802.1X, PMK caching, OKC, FT over the air and FT over the DS. Break it picks what goes wrong (an untrusted server certificate, a wrong password, a wrong RADIUS shared secret, a wrong passphrase or SAE password) and shows the message where the exchange stops.

Why it's here. The AP does not authenticate anyone in 802.1X; it relays. Seeing every EAP message cross the air and then cross the wire makes that plain, and shows where time goes: each RADIUS round trip, each certificate fragment, and the inner password exchange in PEAP. It also shows the order that matters when troubleshooting: the server sends the PMK to the AP in the Access-Accept, and nothing is encrypted over the air until the 4-way handshake completes.

How to use

1. Pick a method and a roam mode, then press Play, or use Step and Back to go one message at a time. Show all draws the whole ladder at once. Blue arrows cross the air; sand, dashed arrows cross the wire. The caption under the ladder names the frame, the leg it crossed and why it is there.

2. Watch for the two bands: Keys available (for 802.1X, after the Access-Accept) and Traffic protected (after message 4 of the 4-way handshake). Content in braces with a lock travels inside the TLS tunnel, where the AP cannot read it.
3. Raise the certificate size to see a large certificate chain split into fragments, each costing a round trip. Raise the RADIUS round-trip time to see a distant server slow the connection. Switch between methods with the whole ladder shown to compare them.
4. Break it: pick a fault and play the ladder. A red X marks the message that fails or is refused; a dashed arrow that stops short with a clock is one that is never answered. The Stopped here band after the last message says what failed, and the caption adds what the help desk sees. Only the faults that can happen with the chosen method are listed, and only for a full authentication. For the faults that retry, Retries and Wait before a retry are illustrative settings.
5. Roam mode: pick a roam method, a band and a scan type, then play. The Reassociation Request names the current AP. FT over the DS sends its first two messages as Action frames to the current AP, which forwards them to the target AP over the wire (dashed). The timeline bar under the ladder splits the roam into scan, authentication and key handshake: switch between methods and watch FT shrink the middle while the scan bar stays. Tap any sent message to see what it carries. The Timing settings set the scan dwell, frame time, RADIUS round-trip time, crypto time and AP to AP time.
6. On a computer or tablet, Present opens the ladder full screen for a projector: Space plays or pauses, the Right arrow sends the next message and the Left arrow takes one back, R resets, Up and Down change the certificate size, B cycles the Break it fault (in Authenticate; Roam has no Break it), F switches full screen, ? lists the keys and Esc exits. The Mode toggle is at the top of the panel.

Inputs

Input	Unit	Range
Mode	choice	Authenticate (the three-lane ladder) or Roam (four lanes); default Authenticate
Method	choice	EAP-TLS, PEAP (MSCHAPv2), EAP-TTLS, WPA2-Personal (PSK), WPA3-Personal (SAE); default EAP-TLS
Inner method	choice	MSCHAPv2 or PAP, for EAP-TTLS only; PEAP here always carries MSCHAPv2
Roam mode	choice	Full authentication, PMK caching, or 802.11r FT over the air; default full

Input	Unit	Range
Certificate size	fragments per certificate message	1 to 6 (default 1); used only when a certificate is sent
RADIUS round-trip time	ms, illustrative	1 to 200 (default 10); used only when RADIUS is
Break it	choice	None (default); Server certificate not trusted, Wrong password (PEAP and EAP-TTLS), Wrong RADIUS shared secret (the 802.1X methods), Wrong passphrase (PSK), Wrong SAE password (SAE); full authentication only
Retries (Break it)	count, illustrative	1 to 5 (default 2); wrong shared secret, wrong passphrase and wrong SAE password only
Wait before a retry (Break it)	s, illustrative	1 to 10 (default 3)
Roam method (Roam mode)	choice	Full 802.1X, PMK caching, OKC, FT over the air, FT over the DS; default full 802.1X
Band and scan (Roam mode)	choice	2.4, 5 or 6 GHz; active or passive; default 5 GHz active
PMF (802.11w) (Roam mode)	choice	Off, Optional or Required; required in 6 GHz
Timing (Roam mode)	ms, illustrative	active dwell 10 to 100 (30), passive dwell 20 to 250 (111), frame time 0.5 to 5 (1), RADIUS round trip 1 to 200 (10), crypto 0 to 500 (50), AP to AP over the DS 1 to 50 (5)

How it works. Each ladder starts with one probe request and response. A full authentication adds Open System authentication (2 frames) or SAE commit and confirm (4 frames), association (2), for 802.1X the EAP exchange, and the 4-way handshake (4 EAPOL-Key frames). EAP: the AP sends EAP-Request/Identity itself; every later EAP-Request comes from the server in an Access-Challenge and every EAP-Response goes back in an Access-Request, one round trip each, until the Access-Accept carries EAP-Success and the PMK. A certificate message split into n

fragments costs n - 1 extra round trips, because each fragment is acknowledged by an empty message. At one fragment the round trips are: EAP-TLS 4, EAP-TTLS with PAP 4, EAP-TTLS with MSCHAPv2 5, PEAP with MSCHAPv2 8. EAP-TLS fragments both the server's and the client's certificate messages. PMK caching replaces authentication with a PMKID in the Reassociation Request, then the 4-way handshake. FT over the air is 4 frames: two FT authentication frames and the reassociation pair, with the 4-way handshake folded in. Estimated time after the scan = over-the-air frames x 1 ms (illustrative) + RADIUS round trips x the round-trip time. Roam mode: the scan visits every channel of the band (5 GHz: 9 channels probed for the active dwell, the 16 DFS channels listened to for the passive dwell; 6 GHz active: only the 15 PSCs). Full 802.1X: Open System authentication (2 frames), Reassociation (2), the same EAP exchange, the 4-way handshake (4). PMK caching and OKC: authentication, Reassociation with the PMKID, the 4-way handshake; the same frames, with OKC's PMKID computed for the target AP. FT over the air: 4 frames. FT over the DS: FT Action Request to the current AP, forwarded to the target and back over the DS, FT Action Response, then Reassociation (6 messages, 2 of them on the wire). Authentication time = EAP round trips x RADIUS round-trip time + crypto time + frames x frame time; the key handshake is the 4-way (or FT's Reassociation pair). No method changes the scan. Break it cuts a full authentication where the fault stops it. Untrusted server certificate: the client ACKs every certificate fragment but the last, answers the last with a TLS alert (unknown_ca), and the server ends EAP with EAP-Failure in an Access-Reject; the AP relays it and deauthenticates the client (reason 23). Wrong password: PEAP's server sends an MSCHAPv2 Failure (error 691) inside the tunnel, the client acknowledges, both exchange a failure result, then Access-Reject, EAP-Failure and reason 23; EAP-TTLS with MSCHAPv2 sends the error in an Access-Challenge; EAP-TTLS with PAP gets an Access-Reject straight after the password. Wrong RADIUS shared secret: the server silently discards each Access-Request (its Message-Authenticator does not match), so the AP sends it again after each wait and nothing comes back. Wrong passphrase: the AP discards message 2 of the 4-way handshake (its MIC does not verify), sends message 1 again after each wait, then deauthenticates with reason 15. Wrong SAE password: the AP cannot verify the client's SAE Confirm and never sends its own; the client repeats its Confirm, then gives up. The time shown is until the last message, including the waits. These are typical sequences: the alert is optional, and devices differ in who sends the last frame.

Example. EAP-TLS, full authentication, 1 fragment, 10 ms RADIUS: 19 frames over the air, 8 RADIUS messages on the wire, 4 round trips, 57 ms after the scan. With 3 fragments: 27 frames, 16 RADIUS messages, 8 round trips, 105 ms. PEAP at 1 fragment also takes 8 round trips and 105 ms. PMK caching drops the whole EAP exchange (17 fewer messages, 8 ms); FT over the air is 4 frames after the scan, with no RADIUS (4 ms). Roam mode at the defaults (5 GHz active scan 2.05 s, EAP-TLS): full 802.1X spends 103 ms on authentication and 4 ms on the key handshake after the 2.05 s scan; PMK caching and OKC 4 ms and 4 ms (17 fewer messages); FT over the air 2 ms and 2 ms (4 frames); FT over the DS 7 ms and 2 ms. The scan is 2.05 s for all five. Break it, EAP-TLS at 1 fragment with an untrusted certificate: the ladder stops at message 17 of 20, the client's TLS alert, after 3 round trips; no Access-Accept, no 4-way handshake, 42 ms. Wrong RADIUS shared secret with 2 retries 3 s apart: 3 Access-Requests, none answered, about 6 s until the last one.

Field notes

- This is a teaching model. The TLS messages follow the TLS 1.2 shape drawn in RFC 5216 and RFC 5281; TLS 1.3 and session resumption change the count and are not modeled.
- How many fragments a certificate needs depends on the chain and the server's fragment size, so it is a setting, not a fixed count. Real round-trip counts also vary a little between servers and clients.
- The TLS tunnel in PEAP and EAP-TTLS protects the inner exchange between client and RADIUS server, not the Wi-Fi link. A client that skips server-certificate validation can hand its inner credentials to an evil-twin server; see the 802.1X / EAP Types reference.
- Break it draws a typical sequence for each fault, not the only one. RFC 5216 lets a client send a TLS alert or simply stop; RFC 5281 says an EAP-TTLS client abandons after an MS-CHAP-Error, and many answer with an empty response first. Retry counts and timers are set by each client, AP and RADIUS server, so they are settings here. A wrong shared secret fails every 802.1X user on that AP at once and leaves no reject in the RADIUS log, which is its signature.
- In Authenticate mode the scan for the next AP is not counted; Roam mode counts it. In measured handoffs (802.11b, open authentication; Mishra, Shin and Arbaugh 2003) scanning was over 90% of the delay. 802.11k shortens the scan; PMK caching, OKC and FT shorten only the authentication.
- OKC is a vendor extension that shares a cached PMK between the APs of one controller; it is not part of IEEE 802.11, and Apple's own key caching is not compatible with it. FT needs one full first connection in its mobility domain.
- Published roam figures, context only: RFC 5169 says an EAP-TLS run needs at least 3, typically 4 or more, round trips. One lab capture each measured FT over the air at 14 ms and FT over the DS at 88 ms (single captures, not typical figures). Vendor roam-time ranges are marketing, not measurements, and are not shown.
- Capturing a roam yourself: put one capture adapter on each AP channel the roam involves, so every frame lands on some radio; a single adapter misses the frames on the other channel. It usually works, but sometimes the roam has to be repeated a few times to catch every frame.
- Caution with FT: clients whose drivers do not understand the FT AKMs can fail to join a network that offers FT at all. Check the client mix before enabling it, and know whether the WLAN offers FT only or mixed (FT and non-FT AKMs side by side).
- PMF (802.11w) protects deauthentication, disassociation and robust Action frames (such as the FT Action frames over the DS) once keys exist. Beacons, probes, authentication and association frames are never protected. 6 GHz requires WPA3 or OWE with PMF.

- Joining a network frame by frame, from the scan through DHCP and the address check, is its own tool: Association, Frame by Frame.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: in 802.1X the AP relays and the RADIUS server authenticates; EAP rides in EAPOL over the air and inside RADIUS on the wire, the PMK arrives in the Access-Accept, and nothing is encrypted until the 4-way handshake completes.

Control	What it does	Why it's there
Mode: Authenticate / Roam	Three-lane ladder, or four lanes with a target AP.	Roam mode compares a full 802.1X reconnect with the fast roams, and puts the scan back in the total.
Play / Pause / Play again, Back, Step, Reset, Show all, Speed: Slow / Normal / Fast	Sends one message at a time, takes one back, or draws the whole ladder.	Show all lets you switch methods with the whole ladder drawn and compare them (help).
Method: EAP-TLS, PEAP (MSCHAPv2), EAP-TTLS, WPA2-Personal (PSK), WPA3-Personal (SAE) (Authenticate)	Changes the middle of the ladder.	The method changes the middle of the ladder. PSK and SAE are there for contrast: no RADIUS at all.
Inner method: MSCHAPv2 / PAP (EAP-TTLS only)	The method inside the TTLS tunnel.	Only EAP-TTLS has a choice; PEAP here always carries MSCHAPv2.
Roam mode: Full / PMK caching / FT (802.11r) (Authenticate)	What a return or roam skips.	PMK caching skips EAP; FT folds the handshake into four frames.
Certificate size (fragments per certificate message) (1 to 6; Up/Down in Present)	Splits certificates into more EAP messages.	Round trips depend on certificate size and fragmentation, so it is a setting, not a fixed count.

Control	What it does	Why it's there
RADIUS round-trip time (illustrative) (1 to 200 ms, default 10)	Time per trip to the server.	A distant server slows every round trip.
Break it: None: it works, Server certificate not trusted, Wrong password, Wrong RADIUS shared secret, Wrong passphrase, Wrong SAE password (Authenticate, Full only; B in Present)	Cuts the ladder at the message where that fault stops it: a red X on the message that fails or is refused, a dashed arrow with a clock for one that is never answered, a Stopped here band, and "What the help desk sees" in the caption. Only the faults that can happen with the chosen method are listed.	Where a failed connection stops tells you which of three parties to look at: the client (it refused the certificate), the RADIUS server (it refused the password), or the link between them (no answer at all). Two failures are loud, ending in EAP-Failure; a wrong shared secret is silent (RFC 5216 §2.1.3, RFC 3579 §3.2).
Retries (illustrative) (1 to 5, default 2), Wait before a retry (illustrative) (1 to 10 s, default 3) (Break it: wrong shared secret, wrong passphrase, wrong SAE password)	How many times the AP or client tries again, and how long it waits first.	Clients, APs and servers set their own timers, so they are inputs, not claims (RFC 2865 §2.5 and RFC 3579 §2.3 leave them to the implementation).
Roam method: Full 802.1X, PMK caching, OKC, FT over the air, FT over the DS (Roam)	Which roam to play.	OKC is labeled a vendor extension, not IEEE; FT over the DS starts with an Action frame to the current AP.
Band, Scan, PMF (802.11w), 6 GHz discovery, EAP method, Timing sliders (Roam)	The same settings as Association, Frame by Frame, plus AP to AP round trip, over the DS.	The timeline splits scan, authentication and key handshake: FT shrinks the middle while the scan bar stays.

A demo that works (about 3 minutes)

1. Set up: Reset. Authenticate, EAP-TLS, Full, 1 fragment.
2. Ask the room: "Who checks the client's certificate, the AP or the server? And when is the air encrypted?"
3. Reveal: step through: every EAP message crosses the air and then the wire; Keys available lights after the Access-Accept, Traffic protected only after message 4. 19 frames over the air, 8 RADIUS messages, 4 round trips (help example).

4. Optional follow-up: switch Mode to Roam with the full ladder shown, then pick FT over the air: authentication drops from 103 ms to 2 ms while the 2.05 s scan does not move.
5. Break it follow-up: PEAP, Full, Show all. Press B twice for Wrong password: the X lands on the server's MSCHAPv2 Failure (error 691), inside the tunnel, and the RADIUS log names the user. Ask: "Now every user on one AP says the password is wrong. What will the RADIUS log show?" Press B once more for Wrong RADIUS shared secret. Reveal: nothing. The Access-Requests stop short with a clock and the server never answers, so there is no reject to find.

Pairs with: Association, Frame by Frame, the EAP Types reference, Roaming Walk. Lesson sequences "Association and roaming" and "Wi-Fi security modes".

Left out on purpose: TLS follows the TLS 1.2 shape; TLS 1.3 and session resumption are not modeled. Break it draws a typical sequence for each fault: the client's TLS alert is optional under RFC 5216, devices differ in who sends the last frame, and what the AP and client do after a silent RADIUS server is not drawn. The retry counts and waits are settings. Authenticate mode does not count the scan; Roam mode does. The TLS tunnel protects the inner exchange, not the Wi-Fi link. The FT captures (14 ms and 88 ms) are single captures, shown as context only; vendor roam-time ranges are not shown.

PoE: Why the New AP Runs at Half Strength

Shows what a new Wi-Fi 7 AP (access point) does on each kind of PoE (Power over Ethernet) switch port. One generic tri-band AP, with three radios (2.4, 5 and 6 GHz) of four spatial streams each and a need of about 29 W for full function, is plugged into an 802.3af, 802.3at or 802.3bt port. The drawing shows the power that reaches the AP against what it needs, which radios and streams stay live, and the power light, which is on every time.

Why it's here. It corrects the belief that the light is on, so power is fine. A Wi-Fi 7 AP on a port that cannot supply its full power still boots and still lights up, then quietly runs with fewer streams or fewer radios. On an 802.3at port the AP modeled here gives up half its spatial streams, which is how a new AP ends up slower than the old one it replaced, with nothing on the front of it to say so.

How to use

1. Pick the switch port type: 802.3af, 802.3at or 802.3bt. Watch the stream marks on each radio and the power bar under the AP. The power light stays on in all three.
2. On 802.3at, choose what the AP gives up: all three radios at 2x2 (6 of its 12 streams), or two radios at 4x4 with the third turned off (8 of 12). The vendor guide this model follows lists both.
3. Read the numbers beside the drawing: streams live of 12, radios live of 3, and the power at the AP against the about 29 W it needs.
4. Predict, then reveal asks: the new Wi-Fi 7 AP goes on last year's 802.3at switch and its power light comes on. Is it running at full strength?
5. To add up the power for a whole switch, open PoE Budget; for every PoE class and type, open PoE Reference. Both are linked under the explainer.
6. On a computer or tablet, Present opens the tool full screen for a projector: Up and Down or the Right arrow change the port type, 1, 2 and 3 pick 802.3af, 802.3at and 802.3bt, Space switches what the AP gives up on 802.3at, R resets, P reveals the answer, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Switch port type	choice	802.3af (PoE), 802.3at (PoE+) or 802.3bt (PoE++, Type 3); default 802.3at
What the AP gives up on 802.3at	choice	All three radios at 2x2, or two radios at 4x4 with 2.4 GHz off; default all three at 2x2. Applies only on 802.3at

How it works. Power at the AP is the most IEEE 802.3 guarantees at the powered device after the cable: 12.95 W for 802.3af (15.4 W from the switch), 25.5 W for 802.3at (30.0 W) and 51.0 W for 802.3bt Type 3 (60.0 W), the same figures as PoE Reference. The AP needs about 29 W for full function. At 29 W or more, every radio runs 4x4:4, 12 streams. On 802.3at the AP runs either three radios at 2x2:2 (6 streams) or two radios at 4x4:4 with 2.4 GHz off (8 streams). On 802.3af, which the vendor guides do not cover, the AP still boots with less: this model runs 2.4 GHz at 1x1 and 5 GHz at 2x2 with 6 GHz off (3 streams), at lower transmit power, with the USB (Universal Serial Bus) port and second Ethernet port off. That set is illustrative (vendors differ; this is one example), chosen to fit the 12.95 W at the AP. The power light is on in every case. Nothing is computed from the watts beyond this comparison: the radio states come from the published behavior on 802.3at and 802.3bt and from the illustrative set on 802.3af, not from a per-radio power figure.

Example. On an 802.3bt port, 51.0 W reaches the AP and all 12 streams run. Move it to an 802.3at port: 25.5 W reaches it, 3.5 W short of the about 29 W it needs, and it runs three radios at 2x2, 6 of 12 streams, half strength, with the power light still on. Choose two radios at 4x4 instead and it keeps 8 of 12, with 2.4 GHz off.

Field notes

- Vendors cut different things at lower power. This tool models one generic AP on one published guide, so read your own AP's data sheet for what it does.
- What a Wi-Fi 7 AP does on 802.3af is not in either guide. This tool shows one reduced set as an illustrative case, not a measured one: fewer spatial streams, one band off, lower transmit power, and the USB port and second Ethernet port off. Vendors differ; this is one example.
- A 2x2 client uses at most two streams, so one phone alone may notice little. What the AP loses is the streams it could share among many clients at once and, with a radio off, a whole band.
- A light on the AP proves power, not enough power. Check the AP's own power status and the power the switch has granted the port.
- Some APs with two Ethernet ports can combine two 802.3at feeds for full function; the guide this model follows says so of its AP. This tool leaves that out, along with IoT (Internet of Things), Bluetooth and scanning radios, and cable shorter than the 100 m the standard's figures assume.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: a Wi-Fi 7 AP on a port that cannot supply its full power still boots and lights up, then quietly runs with fewer streams or fewer radios.

Control	What it does	Why it's there
Switch port type: 802.3af / 802.3at / 802.3bt (Up, Down, Right, or 1, 2, 3 in Present; default 802.3at)	Changes the power reaching the AP; the stream marks and power bar update. The power light stays on in all three.	The one control, which corrects "the light is on, so power is fine".

Control	What it does	Why it's there
What the AP gives up on 802.3at: Three at 2x2 / Two at 4x4 (Space in Present)	6 of 12 streams, or 8 of 12 with 2.4 GHz off. Applies only on 802.3at.	The vendor guide the model follows lists both; vendors cut different things.
Reveal the answer / Hide the answer (P in Present), Show it on 802.3at	The built-in question, then a jump to the at port.	"The new Wi-Fi 7 AP goes on last year's 802.3at switch and its power light comes on. Is it running at full strength?"
Reset (R)	Back to the defaults.	Returns to the question's port.
Open PoE Budget, Open PoE Reference	Links out.	For adding up a whole switch and every PoE class.

A demo that works (about 3 minutes)

1. Set up: R, then press 3 (802.3bt): 51.0 W reaches the AP and all 12 streams run.
2. Ask the room: read the question. "The light is on. Full strength?"
3. Reveal: press 2 (802.3at): 25.5 W, 3.5 W short of the about 29 W it needs, and it runs 6 of 12 streams with the light still on (help example). Press Space for the 8-of-12 choice with 2.4 GHz off.
4. Optional follow-up: press 1 (802.3af). It boots with fewer capabilities rather than not at all; the tool shows one illustrative reduced set.

Pairs with: PoE Budget, PoE Reference.

Left out on purpose: one generic AP on one published guide; read your own AP's data sheet. The 802.3af behavior has no source and is labeled illustrative. A 2x2 client alone may notice little; the loss is in streams shared among many clients. Dual-port 802.3at feeds, IoT and scanning radios, and short cables are left out.

Why a Long Wi-Fi Password Matters More on WPA2

Shows where a password guess gets checked on three kinds of home or small-office Wi-Fi network, all with the same short password: WPA2-Personal (WPA: Wi-Fi Protected Access, here with a pre-shared key, PSK), WPA3-Personal with SAE (Simultaneous Authentication of Equals), and WPA3 transition mode, which lets WPA2-only devices associate with the same password. A drawing puts a device, your AP (access point) and an attacker's computer side by side, and highlights the path each guess takes. Readouts give the number of possible passwords for the length and characters you choose, where each guess is checked, what sets the pace, whether the AP sees the guessing, and what happens to recorded traffic if the password is learned later. It never shows a time to crack.

Why it's here. It corrects two beliefs: that any password is fine because Wi-Fi is encrypted, and that WPA3 is just a new label. On WPA2-Personal, anyone in range who records one association can test guesses on their own computer, limited only by that computer, and the AP never sees a guess, so the password's length is the whole defense. On WPA3-Personal, a recording gives nothing to check a guess against: each guess has to be a live exchange with the AP, and each wrong one is a failed authentication the AP can notice and limit. Transition mode keeps the WPA2 door open, so a short password stays exposed to offline guessing for as long as it runs.

How to use

1. Pick the network security: WPA2, WPA3 or Transition. The password stays the same on all three, so the only thing that changes is where a guess can be checked.
2. Press Try one guess, or Keep guessing, and watch the dot. On WPA2 and in transition mode it loops inside the attacker's computer and the AP's count of failed attempts stays at 0. On WPA3 it travels to the AP and back, and the AP logs every one. The pace on screen is slowed down so you can follow it; it is not a speed.
3. Change the password's length (8 to 63 characters) and the characters it is drawn from. The number of possible passwords is characters to the power of length: each extra character multiplies it by the number of characters in play.
4. Predict, then reveal uses the Wi-Fi Alliance's own example: a password that is one of 5,000, picked at random, with the attacker holding the list. How much of the list can they try on WPA2, and on WPA3?
5. To see the frames themselves, open Association, Frame by Frame on the WPA2 4-way handshake or the WPA3 SAE exchange from the links under the explainer.

6. On a computer or tablet, Present opens the tool full screen for a projector: Space keeps guessing or stops, the Right arrow tries one guess, 1, 2 and 3 choose WPA2, WPA3 or transition mode, Up and Down change the password length, R resets, P reveals the answer, F switches full screen, ? lists the keys and Esc exits.

Inputs

Input	Unit	Range
Network security	choice	WPA2-Personal, WPA3-Personal (SAE) or WPA3 transition mode; default WPA2-Personal. Changing it starts the counts over
Password length	characters	8 to 63, the limits for a WPA password; default 8
Characters	choice	Lowercase letters (26), lowercase letters and digits (36), upper and lowercase letters and digits (62), or any keyboard character (95); default lowercase letters

How it works. Possible passwords = characters in play ^ length, counted exactly (shown in full up to 18 digits, then as about m x 10^n with the digit count). Where a guess is checked follows IEEE 802.11 and the Wi-Fi Alliance: on WPA2-Personal the keys come from the password, the network name and values sent in the clear during the 4-way handshake, so one recorded association lets each guess be checked offline; SAE is resistant to offline dictionary attacks, so on WPA3-Personal each guess is one live exchange with the AP; in transition mode the password is also used by WPA2-only devices, so it can be found offline from one of their associations. Failed attempts the AP logs = guesses tried on WPA3, and 0 otherwise. In the prediction, the chance of having found a password picked at random after g different guesses out of N is g / N. No time or rate is computed anywhere.

Example. A password of 8 lowercase letters has 26 ^ 8 = 208,827,064,576 possibilities. One more letter makes it 5,429,503,678,976, 26 times as many; at 12 letters it is 95,428,956,661,682,176. Using any keyboard character at 8 characters gives 6,634,204,312,890,625. In the Wi-Fi Alliance's example of 5,000 possible passwords, WPA2 lets the attacker test all 5,000 offline and find it for certain, while on WPA3 the chance reaches 50% only after 2,500 live attempts at the AP.

Field notes

- No time to crack appears anywhere in this tool. How fast offline guessing runs depends entirely on the attacker's hardware, so any figure would be invented.
- The count of possible passwords assumes the password was picked at random. A word, a name, a date or a keyboard pattern is on every guess list and falls long before the count runs out.
- WPA3 does not make the password irrelevant. The Wi-Fi Alliance still says it should be complex enough not to be easily guessed, because live guessing is still possible; it just has to go through the AP.
- Transition mode is for moving a network from WPA2 to WPA3. Once every device supports WPA3, turning transition mode off closes the WPA2 door. If transition mode does not meet a network's needs, the Wi-Fi Alliance recommends separate network names (SSIDs, service set identifiers) with different passwords, on separated network segments.
- Forward secrecy: on WPA3, traffic recorded earlier stays unreadable even if the password is learned later. On WPA2-Personal, the password and a recorded 4-way handshake give the keys to that session.
- This tool leaves out WPA2-Enterprise and WPA3-Enterprise (802.1X), which do not use a shared password; the 802.1X and EAP Ladder covers them. It also leaves out attacks on particular products, and it names no tools and gives no steps.
- The Wireless Classroom is designed for tablets and computers, and on a phone some views are cramped.

Teaching it

Teaches: on WPA2-Personal one recorded join lets an attacker test guesses offline, so length is the whole defense; on WPA3 every guess is a live exchange the AP can see, and transition mode keeps the WPA2 door open.

Control	What it does	Why it's there
Network security: WPA2 / WPA3 / Transition (1, 2, 3 in Present)	Changes where a guess is checked. The password stays the same; the counts start over.	The one control, with the same short password on all three.

Control	What it does	Why it's there
Length (8 to 63; Up/Down in Present)	Password length.	Each extra character multiplies the possibilities by the characters in play.
Characters: Lowercase letters (26), Lowercase letters and digits (36), Upper and lowercase letters and digits (62), Any keyboard character (95)	The character set.	Possible passwords = characters ^ length.
Keep guessing / Stop guessing (Space), Try one guess (Right), Reset (R)	Runs the guessing dot. On WPA2 and transition it loops inside the attacker's computer; on WPA3 it travels to the AP, which logs it.	The visible difference: the AP's failed-attempt count stays at 0 on WPA2. The pace is slowed to follow and is not a speed.
Reveal the answer / Hide the answer (P in Present)	The built-in question, from the Wi-Fi Alliance's example of 5,000 possible passwords.	Uses the source's own example rather than invented crack times.
The WPA2 4-way handshake, The WPA3 SAE exchange, Open WPA Security	Links to the frames.	Shows the frames behind the guess.

A demo that works (about 3 minutes)

1. Set up: R. WPA2, 8 lowercase letters.
2. Ask the room: "The password is one of 5,000 and the attacker has the list. How much of the list can they try on WPA2? On WPA3?"
3. Reveal: P. On WPA2 they test all 5,000 offline and find it for certain; on WPA3 the chance reaches 50% only after 2,500 live attempts at the AP (help example). Press Space on WPA2, then 2 and Space on WPA3, and watch the AP's count.
4. Optional follow-up: press 3 for transition mode and show the dot back inside the attacker's computer.

Pairs with: Association, Frame by Frame (the 4-way handshake and SAE), 802.1X and EAP Ladder, WPA Security reference. Lesson sequence "Wi-Fi security modes".

Left out on purpose: no time to crack anywhere: it depends entirely on the attacker's hardware (so no crack-time numbers are given). The count assumes a random password; words and patterns fall long before it runs out. Enterprise (802.1X) is left out, along with attacks on particular products, tools and steps.

Course Handouts (11)

Every handout on this shelf opens in the shared PDF viewer, which carries a permanent "Share or download" button in its top bar, so any handout can be saved, printed, or AirDropped as a full-resolution PDF for the class.

2.4 GHz Channel Allocations

A built-in, offline, zoomable copy of Keith's published 2.4 GHz channel layout and allocations reference card.

Why it's here. You want the WLAN Pros 2.4 GHz channel map exactly as printed, ready to view and zoom on the phone in your pocket, with no signal needed.

How to use

1. Open the card and pinch or double-tap to zoom. It opens fit to the screen.
2. This is a printed reference image, not a live data table, so a screen reader can't read the values inside it.

Field notes

- The card is a printed image, so a screen reader can't read the values inside it. The card title and the pinch-to-zoom gesture are announced.
- This is Keith's own published WLAN Pros laminated reference card, saved as a PDF and carried inside the app so it works offline on every device.

Teaching it

Teaches: the 2.4 GHz channel map: channels 1 to 14, with 1 to 13 spaced 5 MHz apart, the 22 MHz 802.11b and 20 MHz OFDM channel widths, why 1, 6 and 11 are the three non-overlapping 20 MHz channels, the non-US four-channel plan, and the 802.15.4/Zigbee channels on the same band.

Pairs with: Channel Planner, Adjacent Channels and AP Stacking, Legacy Protection Cost, Fourier and FFT's Swept vs FFT race (the same band, with Bluetooth and a microwave oven in it).

5 GHz Channel Allocations

A built-in, offline, zoomable copy of Keith's published 5 GHz channel layout and allocations reference card.

Why it's here. You want the WLAN Pros 5 GHz channel map exactly as printed, ready to view and zoom on the phone in your pocket, with no signal needed.

How to use

1. Open the card and pinch or double-tap to zoom. It opens fit to the screen.
2. This is a printed reference image, not a live data table, so a screen reader can't read the values inside it.

Field notes

- The card is a printed image, so a screen reader can't read the values inside it. The card title and the pinch-to-zoom gesture are announced.
- This is Keith's own published WLAN Pros laminated reference card, saved as a PDF and carried inside the app so it works offline on every device.

Teaching it

Teaches: the 5 GHz channel map by U-NII band: 28 20 MHz, 14 40 MHz, 7 80 MHz and 3 160 MHz channels, the DFS channels, the TDWR channels, channel 144 and U-NII-4, and the power and indoor/outdoor rules for the US, Canada and Australia.

Pairs with: DFS and Radar, Channel Planner, Adjacent Channels and AP Stacking.

6 GHz Channel Allocations

A built-in, offline, zoomable copy of Keith's published 6 GHz channel layout and allocations reference card.

Why it's here. You want the WLAN Pros 6 GHz channel map exactly as printed, ready to view and zoom on the phone in your pocket, with no signal needed.

How to use

1. Open the card and pinch or double-tap to zoom. It opens fit to the screen.
2. This is a printed reference image, not a live data table, so a screen reader can't read the values inside it.

Field notes

- The card is a printed image, so a screen reader can't read the values inside it. The card title and the pinch-to-zoom gesture are announced.
- This is Keith's own published WLAN Pros laminated reference card, saved as a PDF and carried inside the app so it works offline on every device.

Teaching it

Teaches: the US 6 GHz channel map, 1.2 GHz of spectrum across U-NII-5 to U-NII-8: 59 20 MHz, 29 40 MHz, 14 80 MHz, 7 160 MHz and 3 320 MHz channels, the Preferred Scanning Channels, and the three power classes most people know.

Pairs with: 6 GHz Channel Allocations with GVP, 6 GHz Power and PSD, Channel Planner.

6 GHz Channel Allocations with GVP

Keith's 6 GHz channel chart showing all four US power classes, including Geofenced Variable Power.

Why it's here. The 6 GHz card most people know shows three power classes. A fourth, Geofenced Variable Power, entered the rules effective 27 April 2026. This is the same chart with that class added, so you can see where it is allowed and where it is not.

How to use

1. Open the card and pinch or double-tap to zoom. It opens fit to the screen.
2. Read the Geofenced Variable Power row against Standard Power directly above it. They are authorized in the same two sub-bands, so GVP inherits the same gap in the middle of the band.
3. If you want the familiar three-class layout, the plain 6 GHz Channel Allocations card is still here and unchanged.

Field notes

- Geofenced Variable Power is rules on paper as of this release. No geofencing system has been approved and no GVP access point or client has been certified, so nothing on the shelf uses it yet.
- GVP is authorized in the same two sub-bands as Standard Power, 5.925 to 6.425 GHz and 6.525 to 6.875 GHz. It does not reach U-NII-6 or U-NII-8.
- The card is a printed image, so a screen reader cannot read the values inside it. The card title and the pinch-to-zoom gesture are announced.

Teaching it

Teaches: the same US 6 GHz chart with all four power classes, including Geofenced Variable Power, which entered the rules effective 27 April 2026.

Pairs with: 6 GHz Channel Allocations, 6 GHz Power and PSD.

Wireless LAN Troubleshooting Causes

A bundled, offline, pinch-to-zoom copy of Keith's published "common causes to check when troubleshooting" laminated reference card.

Why it's here. You want the WLAN Pros troubleshooting-causes card exactly as printed, viewable and zoomable on the phone you already have, with no network needed.

How to use

1. Open the card and pinch or double-tap to zoom; it opens fit-to-screen.
2. This is a flat print PDF, not an in-app data table, so a screen reader can't read the content inside it.

Field notes

- Ships as a PDF card on purpose: this is print-layout artwork, separate from the equivalent in-app data tables.
- A screen reader can't read the content inside the card (it's a flat image PDF). The card title and the "pinch to zoom" gesture are announced.
- Source: Keith's own published WLAN Pros laminated reference card, saved to PDF and bundled in the app. It displays offline, with no network needed.

Teaching it

Teaches: the numbered list of places a Wi-Fi problem can come from, from the end user's skills and perceptions, through the client device, the RF medium, contention, the MCS decision, DFS, association, 802.11k/v/r, authentication and encryption, to the upper layers, the controlled port and the fixed access configuration.

Pairs with: Bubble Diagram, Channel Utilization Meter, Rate Adaptation, Why Two Devices Disagree.

WLAN Pros Bubble Diagram

A bundled, offline, pinch-to-zoom copy of Keith's published "Wi-Fi design decision bubble diagram" laminated reference card.

Why it's here. You want the WLAN Pros bubble-diagram design-decision flow exactly as printed, viewable and zoomable on the phone you already have, with no network needed.

How to use

1. Open the card and pinch or double-tap to zoom; it opens fit-to-screen.
2. This is a flat print PDF, not an in-app data table, so a screen reader can't read the content inside it. The screen honestly announces the card name and the zoom gesture instead.

Field notes

- Ships as a PDF card on purpose: this is print-layout artwork, separate from the equivalent in-app data tables.
- A screen reader can't read the content inside the card (it's a flat image PDF). The card title and the "pinch to zoom" gesture are announced.
- This is the only card printed in landscape. The viewer was set up specially so it displays landscape and undistorted; this was fixed and verified on 2026-06-01 and is not an open issue.
- Source: Keith's own published WLAN Pros laminated reference card, saved to PDF and bundled in the app. It displays offline, with no network needed.

Teaching it

Teaches: every piece a Wi-Fi connection depends on, drawn as one map: the client and end user, the RF medium processes (DFS, RRM, MCS, contention, overhead), association, authentication and encryption paths, the AP, and the wired side out to DHCP, DNS, the controller, captive portal and the authentication database.

Pairs with: Troubleshooting Causes; Association, Frame by Frame; 802.1X and EAP Ladder.

Top 20 Wi-Fi Checklist

A bundled, offline, pinch-to-zoom copy of Keith's published "Top 20 Wi-Fi design checklist" laminated reference card (in PDF card form).

Why it's here. You want the WLAN Pros Top 20 checklist card exactly as printed, viewable and zoomable on the phone you already have, with no network needed.

How to use

1. Open the card and pinch or double-tap to zoom; it opens fit-to-screen.
2. This is a flat print PDF, not an in-app data table, so a screen reader can't read the content inside it.

Field notes

- Deliberate split: this Top 20 card is the PDF version, separate from any tappable checklist.
- A screen reader can't read the content inside the card (it's a flat image PDF). The card title and the "pinch to zoom" gesture are announced.
- Source: Keith's own published WLAN Pros laminated reference card, saved to PDF and bundled in the app. It displays offline, with no network needed.

Teaching it

Teaches: the 20 settings and measurements to check first when evaluating a Wi-Fi installation against best practice, per band: channel plan and widths, co-channel interference, minimum basic rates, SSIDs across bands, primary and secondary RSSI, AP transmit power, WPA2/WPA3, 802.11k/r/v, channel utilization and clients per radio.

Pairs with: Extended Checklist, Channel Planner, Rate vs Range, Uplink vs Downlink.

Extended Wi-Fi Checklist

A bundled, offline, pinch-to-zoom copy of Keith's published "extended design checklist items" laminated reference card.

Why it's here. You want the WLAN Pros extended checklist card exactly as printed, viewable and zoomable on the phone you already have, with no network needed.

How to use

1. Open the card and pinch or double-tap to zoom; it opens fit-to-screen.
2. This is a flat print PDF, not an in-app data table, so a screen reader can't read the content inside it.

Field notes

- A screen reader can't read the content inside the card (it's a flat image PDF). The card title and the "pinch to zoom" gesture are announced.
- Source: Keith's own published WLAN Pros laminated reference card, saved to PDF and bundled in the app. It displays offline, with no network needed.

Teaching it

Teaches: items 21 to 50, the next layer after the Top 20: beacon and DTIM intervals, legacy security and 802.11b rates off, 160 and 320 MHz widths, channels 144 and 165, 6 GHz discovery and power classes, hidden SSIDs and SSID count, Wi-Fi 7 MLO, 802.11d/e/h/i/w/u, MU-MIMO, transmit beamforming and OFDMA.

Pairs with: Top 20 Checklist, MIMO and Beamforming, OFDMA Resource Units, Multi-Link Operation, Power Save.

Extended Checklist (Non-Advertised Items)

A bundled, offline, pinch-to-zoom copy of Keith's published "extended checklist, non-advertised items" laminated reference card.

Why it's here. You want the WLAN Pros extended (non-advertised items) checklist card exactly as printed, viewable and zoomable on the phone you already have, with no network needed.

How to use

1. Open the card and pinch or double-tap to zoom; it opens fit-to-screen.
2. This is a flat print PDF, not an in-app data table, so a screen reader can't read the content inside it.

Field notes

- A screen reader can't read the content inside the card (it's a flat image PDF). The card title and the "pinch to zoom" gesture are announced.
- Source: Keith's own published WLAN Pros laminated reference card, saved to PDF and bundled in the app. It displays offline, with no network needed.

Teaching it

Teaches: items 51 to 66, the settings that are not in the beacons, so a scan cannot see them and you must ask the network administrator: automatic channels and widths, client isolation, airtime fairness, DFS events, mesh, band steering, load balancing, auto RF, PoE budget, non-802.11 interference and firmware.

Pairs with: Band Steering, Airtime Fairness, Repeaters and Mesh Backhaul, DFS and Radar.

Wi-Fi Connection Checklist

A bundled, offline, pinch-to-zoom copy of Keith's published "client connection sequence checklist" laminated reference card.

Why it's here. You want the WLAN Pros connection-sequence checklist card exactly as printed, viewable and zoomable on the phone you already have, with no network needed.

How to use

1. Open the card and pinch or double-tap to zoom; it opens fit-to-screen.
2. This is a flat print PDF, not an in-app data table, so a screen reader can't read the content inside it.

Field notes

- A screen reader can't read the content inside the card (it's a flat image PDF). The card title and the "pinch to zoom" gesture are announced.
- Source: Keith's own published WLAN Pros laminated reference card, saved to PDF and bundled in the app. It displays offline, with no network needed.

Teaching it

Teaches: the 14 checks to run from a client, in order: see the SSIDs, Wi-Fi 7 MLO and the MLD MAC, associate, authenticate, get an address, gateway and DNS from DHCP, ping the gateway, DNS, a remote IP and a remote name, check the client's MCS and transmit rate, then a speed test.

Pairs with: Association, Frame by Frame; the Connected, No Internet: Captive Portals lesson; The Number on the Box vs the Number in Your Hand for step 14.

Modulation and Coding Schemes (MCS Index)

A built-in, offline, zoomable copy of Keith's published MCS index, rates, and modulation reference card.

Why it's here. You want the WLAN Pros MCS index card exactly as printed, ready to view and zoom on the phone in your pocket, with no signal needed.

How to use

1. Open the card and pinch or double-tap to zoom. It opens fit to the screen.
2. This is a printed reference image, not a live data table, so a screen reader can't read the values inside it.

Field notes

- This printed MCS card is separate from the interactive MCS Index tool, which lets you look up rates in a live table.
- The card is a printed image, so a screen reader can't read the values inside it. The card title and the pinch-to-zoom gesture are announced.
- This is Keith's own published WLAN Pros laminated reference card, saved as a PDF and carried inside the app so it works offline on every device.

Teaching it

Teaches: the HT, VHT and HE MCS index side by side: modulation, coding rate and the PHY rate for each channel width and guard interval.

Pairs with: Modulation Simulator, Rate vs Range, The Number on the Box vs the Number in Your Hand, the MCS Index tool.